Structure Analysis

Complete source-code archive in PDF form

Repository: Anshu-Bhalani/Structure-Analysis
Source revision: main branch at workflow execution time
Included tracked files: 47
Every tracked text file is included in path order. Long source lines are visually wrapped to fit the page; the source itself is not edited.

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```

01. .github/workflows/deploy-pages.yml

```yaml
name: Deploy GitHub Pages

on:
  push:
    branches:
      - main
  workflow_dispatch:

permissions:
  contents: read

concurrency:
  group: github-pages
  cancel-in-progress: false

jobs:
  build:
    runs-on: ubuntu-24.04
    permissions:
      contents: read
      pages: write
    steps:
      - name: Checkout repository
        uses: actions/checkout@v6

      - name: Configure GitHub Pages
        uses: actions/configure-pages@v6

      - name: Build site with Jekyll
        uses: actions/jekyll-build-pages@v1
        with:
          source: ./
          destination: ./_site

      - name: Upload Pages artifact
        uses: actions/upload-pages-artifact@v5
        with:
          path: ./_site

  deploy:
    environment:
      name: github-pages
      url: ${{ steps.deployment.outputs.page_url }}
    runs-on: ubuntu-24.04
    needs: build
    permissions:
      actions: read
      pages: write
      id-token: write
    steps:
      - name: Deploy to GitHub Pages
        id: deployment
        uses: actions/deploy-pages@v5.0.1
```

02. .github/workflows/javascript-regression-tests.yml

```
name: JavaScript Regression Tests

on:
  push:
    branches:
      - main
  pull_request:
    branches:
      - main
  workflow_dispatch:

permissions:
  contents: read

jobs:
  test:
    runs-on: ubuntu-24.04
    steps:
      - name: Checkout repository
        uses: actions/checkout@v6
      - name: Set up Node.js
        uses: actions/setup-node@v5
        with:
          node-version: "22"
      - name: Check application JavaScript syntax
        run: find assets/js modules -name '*.js' -print0 | xargs -0 -n 1 node --check
      - name: Run non-calculation regression tests
        run: node --test tests/*.test.cjs
```

03. README.md

````
# Structural Analysis – modular project

Open `index.html` in a browser (works from file:// — no build step, no server needed).

```
index.html                      ← only HTML file; links all CSS + JS
assets/
  css/main.css                  ← common styles
  js/main.js                    ← common JS (shell markup, state, helpers, persistence, navigation, bootstrap)
  vendor/                       ← jsPDF + svg2pdf + embedded fonts (vector PDF export)
modules/
  home/ dashboard/ nodes/ elements/ supports/ load-case/ load/ moment-case/ moment/   (*.js, + *.css where needed)
  results/
    results.js / results.css    ← Results frame, summary, equilibrium, orchestration, PDF export
    step-1 … step-8/            ← one folder per calculation part (step-8 = final result tables)
    diagrams/                   ← part 9: all diagrams (live model, reactions, AFD/SFD/BMD/displacement, DoF, full-screen)
```
Page markup lives inside each module's `.js` and is injected by `mountModuleHTML()` (defined in `main.js`), so `index.html` stays a pure link file. Script order in `index.html` matters.


## Offline use (hosted site)

When the GitHub Pages site is first opened while online, its Service Worker caches the application shell, local JavaScript/CSS modules, PDF libraries, and manifest/icon. Wait for the
**Offline ready** status before disconnecting. Afterward, reopen the hosted site on the same browser/device to use the cached app without a network connection.

This requires the hosted HTTPS site; browsers do not allow Service Worker registration from `file://`. Model data continues to be stored locally by the app's existing browser-storage logic.
Clearing site data can remove both saved models and offline cache. Publishing updates requires internet access.
````

04. assets/css/main.css

```
/* ========================================================================
 * main.css
 * Global styles shared by the whole application: reset, layout (sidebar / topbar /
 * container), cards, grids, form controls, buttons, tables, badges, sortable rows,
 * empty states, the analysis loading overlay and the global responsive rules.
 * ======================================================================== */

* {
    box-sizing: border-box;
    margin: 0;
    padding: 0;
}

:root {
    --bg: #f4f7fb;
    --panel: #ffffff;
    --border: #dce3ec;
    --text: #182230;
    --muted: #697586;
    --primary: #2563eb;
    --primary-dark: #1d4ed8;
    --soft-blue: #eff6ff;
    --shadow: 0 8px 25px rgba(15, 23, 42, 0.07);
    --radius: 14px;
}

html {
    overflow-x: hidden;
}

body {
    font-family: Inter, Arial, Helvetica, sans-serif;
    background: var(--bg);
    color: var(--text);
    min-height: 100vh;
    overflow-x: hidden;
    max-width: 100vw;
}

button,
input,
select {
    font: inherit;
}

button {
    cursor: pointer;
}

.app {
    display: flex;
    min-height: 100vh;
}

/* SIDEBAR */
.sidebar {
    width: 260px;
    background: #111827;
    color: white;
    padding: 22px 14px;
    position: sticky;
    top: 0;
    height: 100vh;
    overflow-y: auto;
}

.brand {
    padding: 5px 12px 25px;
    border-bottom: 1px solid rgba(255, 255, 255, 0.1);
    margin-bottom: 18px;
}

.brand h1 {
    font-size: 20px;
    font-weight: 800;
    letter-spacing: -0.4px;
}

.brand p {
    font-size: 12px;
    color: #9ca3af;
    margin-top: 5px;
}

.nav-title {
```

```
    font-size: 10px;
    text-transform: uppercase;
    letter-spacing: 1.2px;
    color: #6b7280;
    padding: 0 12px 8px;
}

.nav-btn {
    width: 100%;
    border: none;
    background: transparent;
    color: #d1d5db;
    display: flex;
    align-items: center;
    gap: 10px;
    padding: 11px 12px;
    border-radius: 9px;
    text-align: left;
    margin-bottom: 4px;
    transition: 0.2s;
}

.nav-btn:hover,
.nav-btn.active {
    background: #1f2937;
    color: white;
}

.nav-icon {
    width: 23px;
    text-align: center;
    font-size: 15px;
}

.sidebar-bottom {
    margin-top: 25px;
    padding: 12px;
    border-top: 1px solid rgba(255, 255, 255, 0.1);
}

.sidebar-btn {
    width: 100%;
    margin-top: 8px;
    padding: 9px;
    border-radius: 8px;
    border: 1px solid #374151;
    background: #1f2937;
    color: #e5e7eb;
}

.sidebar-btn:hover {
    background: #374151;
}

/* MAIN */
.main {
    flex: 1;
    min-width: 0;
}

.topbar {
    background: white;
    border-bottom: 1px solid var(--border);
    padding: 18px 28px;
    display: flex;
    justify-content: space-between;
    align-items: center;
    gap: 20px;
    position: sticky;
    top: 0;
    z-index: 10;
}

.page-title h2 {
    font-size: 22px;
    margin-bottom: 4px;
}

.page-title p {
    color: var(--muted);
    font-size: 13px;
}

.top-actions {
    display: flex;
    gap: 8px;
}
```

```
.model-name-input {
    width: 190px;
    height: 42px;
}

.container {
    padding: 26px 28px 60px;
    max-width: 1500px;
    margin: auto;
}

/* COMMON */
.section {
    display: none;
    animation: fade 0.2s ease;
}

.section.active {
    display: block;
}

@keyframes fade {
    from {
        opacity: 0;
        transform: translateY(3px);
    }
    to {
        opacity: 1;
        transform: translateY(0);
    }
}

.card {
    background: var(--panel);
    border: 1px solid var(--border);
    border-radius: var(--radius);
    box-shadow: var(--shadow);
    margin-bottom: 20px;
    overflow: hidden;
}

.card-header {
    padding: 18px 20px;
    border-bottom: 1px solid var(--border);
    display: flex;
    justify-content: space-between;
    align-items: center;
    gap: 12px;
}

.card-header h3 {
    font-size: 16px;
}

.card-header p {
    color: var(--muted);
    font-size: 12px;
    margin-top: 4px;
}

.card-body {
    padding: 20px;
}

.grid {
    display: grid;
    gap: 14px;
}

.grid-2 {
    grid-template-columns: repeat(2, minmax(0, 1fr));
}

.grid-3 {
    grid-template-columns: repeat(3, minmax(0, 1fr));
}

.grid-4 {
    grid-template-columns: repeat(4, minmax(0, 1fr));
}

.field {
    display: flex;
    flex-direction: column;
    gap: 7px;
}
.field label {
```

```
    font-size: 12px;
    font-weight: 700;
    color: #374151;
}

.field small {
    color: var(--muted);
    font-size: 11px;
}

.input-wrap {
    position: relative;
}

.input-unit {
    position: absolute;
    right: 11px;
    top: 50%;
    transform: translateY(-50%);
    font-size: 11px;
    color: var(--muted);
    pointer-events: none;
}

input,
select {
    width: 100%;
    height: 42px;
    border: 1px solid #cfd7e3;
    border-radius: 9px;
    background: white;
    padding: 0 12px;
    outline: none;
    color: var(--text);
    transition: 0.15s;
}

.has-unit input {
    padding-right: 52px;
}

input:focus,
select:focus {
    border-color: var(--primary);
    box-shadow: 0 0 0 3px rgba(37, 99, 235, 0.1);
}

.btn {
    border: none;
    border-radius: 8px;
    padding: 10px 14px;
    font-weight: 700;
    font-size: 12px;
    transition: 0.2s;
}

.btn-primary {
    background: var(--primary);
    color: white;
}

.btn-primary:hover {
    background: var(--primary-dark);
}

.btn-secondary {
    background: #eef2f7;
    color: #334155;
    border: 1px solid #d6dee9;
}

.btn-secondary:hover {
    background: #e2e8f0;
}

.actions {
    display: flex;
    gap: 8px;
    flex-wrap: wrap;
}

.form-footer {
    display: flex;
    justify-content: flex-end;
    gap: 8px;
    margin-top: 20px;
    padding-top: 18px;
```

```
    border-top: 1px solid var(--border);
}

/* RESTRAINT MATRIX */
.restraint-box {
    border: 1px solid var(--border);
    border-radius: 10px;
    padding: 14px;
    background: #fafcff;
}

.restraint-title {
    font-size: 12px;
    font-weight: 800;
    margin-bottom: 12px;
}

.check-grid {
    display: grid;
    grid-template-columns: repeat(3, 1fr);
    gap: 8px;
}

.check-item {
    border: 1px solid var(--border);
    border-radius: 8px;
    padding: 10px;
    display: flex;
    align-items: center;
    gap: 8px;
    background: white;
}

.check-item input {
    width: auto;
    height: auto;
}

.check-item span {
    font-size: 11px;
    font-weight: 700;
}

/* TABLE */
.table-wrap {
    overflow-x: auto;
}

table {
    width: 100%;
    border-collapse: collapse;
    min-width: 850px;
}

th,
td {
    border-bottom: 1px solid #e5eaf0;
    padding: 11px 10px;
    font-size: 12px;
    text-align: left;
    white-space: nowrap;
}

th {
    background: #f8fafc;
    color: #4b5563;
    font-size: 11px;
    text-transform: uppercase;
    letter-spacing: 0.3px;
}

tr:hover td {
    background: #fbfdff;
}

.badge {
    display: inline-flex;
    align-items: center;
    padding: 4px 8px;
    border-radius: 999px;
    font-size: 10px;
    font-weight: 800;
}

.badge-blue {
    background: #eff6ff;
    color: #1d4ed8;
```

```
}

.badge-green {
    background: #f0fdf4;
    color: #15803d;
}

.badge-orange {
    background: #fff7ed;
    color: #c2410c;
}

.badge-gray {
    background: #f3f4f6;
    color: #4b5563;
}

.table-actions {
    display: flex;
    gap: 5px;
}

/* Sortable input-table rows */
.sort-handle {
    display: inline-flex;
    align-items: center;
    justify-content: center;
    width: 20px;
    height: 24px;
    margin-right: 6px;
    color: #94a3b8;
    font-size: 15px;
    font-weight: 800;
    line-height: 1;
    cursor: grab;
    user-select: none;
    -webkit-user-select: none;
    touch-action: none;
    vertical-align: middle;
    border-radius: 6px;
    transition: color 0.16s ease, background 0.16s ease, transform 0.16s ease, box-shadow 0.16s ease;
}

.sort-handle:active {
    cursor: grabbing;
}

.sort-handle.sort-holding {
    color: var(--primary);
    background: #eff6ff;
    box-shadow: 0 0 0 2px rgba(37, 99, 235, 0.10);
    animation: sortHoldPulse 0.75s ease-in-out infinite;
}

.sort-handle.sort-ready {
    color: var(--primary);
    background: #dbeafe;
    box-shadow: 0 0 0 3px rgba(37, 99, 235, 0.14), 0 4px 10px rgba(37, 99, 235, 0.16);
    transform: scale(1.08);
    animation: sortReadyPulse 0.55s ease-out 1;
}

.sortable-row.sort-dragging td {
    opacity: 1;
    background: #eff6ff;
    transition: background 0.16s ease, box-shadow 0.16s ease, opacity 0.16s ease;
    box-shadow: 0 2px 0 rgba(37, 99, 235, 0.10);
}

.sortable-row.sort-drop-target td {
    background: #f0fdf4;
    transition: background 0.16s ease;
}

@keyframes sortHoldPulse {
    0%, 100% {
        transform: scale(1);
    }
    50% {
        transform: scale(1.08);
    }
}

@keyframes sortReadyPulse {
    0% {
        transform: scale(1);
    }
```

```
    55% {
        transform: scale(1.18);
    }
    100% {
        transform: scale(1.08);
    }
}

.icon-btn {
    border: 1px solid var(--border);
    background: white;
    width: 30px;
    height: 30px;
    border-radius: 7px;
    font-size: 12px;
}

.icon-btn:hover {
    background: #f1f5f9;
}

.length-chip-row {
    display: flex;
    gap: 6px;
    margin-top: 8px;
    flex-wrap: wrap;
}

.length-chip {
    border: 1px solid var(--border);
    background: var(--soft-blue);
    color: var(--primary);
    font-size: 12px;
    font-weight: 700;
    padding: 6px 12px;
    border-radius: 999px;
    cursor: pointer;
}

.length-chip:hover {
    background: #dbeafe;
}

.search-box {
    width: 230px;
}

/* EMPTY */
.empty {
    padding: 45px 20px;
    text-align: center;
    color: var(--muted);
}

.empty-icon {
    font-size: 32px;
    margin-bottom: 10px;
}

.empty h4 {
    color: #374151;
    margin-bottom: 5px;
}

.scientific-shortcut-row {
    margin-top: 6px;
}

.scientific-chip {
    border: 1px solid var(--border);
    background: #f8fafc;
    color: #475569;
    font-size: 12px;
    font-weight: 800;
    min-width: 30px;
    padding: 6px 10px;
    border-radius: 999px;
    cursor: pointer;
}

.scientific-chip:hover {
    background: #e2e8f0;
}

/* RESPONSIVE */
@media (max-width: 1100px) {
    .sidebar {
```

```
        width: 220px;
    }
    .grid-4 {
        grid-template-columns: repeat(2, 1fr);
    }
}

@media (max-width: 800px) {
    .app {
        display: block;
    }
    .sidebar {
        width: 100%;
        height: auto;
        position: relative;
    }
    .nav-buttons {
        display: grid;
        grid-template-columns: repeat(2, 1fr);
    }
    .sidebar-bottom {
        display: block;
        margin-top: 16px;
    }
    .topbar {
        position: relative;
        padding: 15px;
        flex-direction: column;
        align-items: stretch;
        gap: 12px;
    }
    .top-actions {
        display: flex;
        width: 100%;
    }
    .top-actions .btn {
        flex: 1;
    }
    .container {
        padding: 16px;
    }
    .grid-2,
    .grid-3,
    .grid-4 {
        grid-template-columns: 1fr;
    }
}

@media (max-width: 500px) {
    .nav-buttons {
        grid-template-columns: 1fr;
    }
    .check-grid {
        grid-template-columns: repeat(2, 1fr);
    }
    .page-title h2 {
        font-size: 18px;
    }
}

/* ANALYSIS LOADING OVERLAY
   Shown only while Run Analysis is preparing/calculating large models.
   It does not alter the existing page layout or diagram styles. */
.analysis-loading-overlay {
    display: none;
    position: fixed;
    inset: 0;
    z-index: 20000;
    background: rgba(244, 247, 251, 0.94);
    align-items: center;
    justify-content: center;
    padding: 24px;
}

.analysis-loading-overlay.active {
    display: flex;
}

.analysis-loading-card {
    width: min(440px, 100%);
    background: #ffffff;
    border: 1px solid #dce3ec;
    border-radius: 14px;
    box-shadow: 0 12px 35px rgba(15, 23, 42, 0.12);
    padding: 24px;
    text-align: center;
}
```

```
.analysis-loading-title {
    font-size: 18px;
    font-weight: 800;
    color: #182230;
}

.analysis-loading-text {
    margin-top: 8px;
    font-size: 12px;
    line-height: 1.5;
    color: #697586;
}

.analysis-loading-spinner {
    width: 34px;
    height: 34px;
    margin: 0 auto 14px;
    border: 3px solid #dbeafe;
    border-top-color: #2563eb;
    border-radius: 50%;
    animation: analysisSpin 0.8s linear infinite;
}

@keyframes analysisSpin {
    to { transform: rotate(360deg); }
}

/* Keep shared model actions visible on narrow screens. The model-type badge
   and model-name field occupy full rows; Export + Save remain side by side. */
@media (max-width: 800px) {
    .top-actions {
        display: grid;
        grid-template-columns: repeat(2, minmax(0, 1fr));
        width: 100%;
        gap: 8px;
    }

    .top-actions > #topbarHomeButton,
    .top-actions > #currentModelTypeBadge,
    .top-actions > #modelName {
        grid-column: 1 / -1;
        width: 100%;
        min-width: 0 !important;
        max-width: none !important;
    }

    .top-actions > .btn {
        flex: initial;
        width: 100%;
        min-width: 0;
        white-space: nowrap;
    }
}


/* Create-model dialog: select BTF/Grid once, before any model data is entered. */
.model-create-modal {
    position: fixed;
    inset: 0;
    z-index: 30000;
    display: flex;
    align-items: center;
    justify-content: center;
    padding: 18px;
}

.model-create-modal[hidden] {
    display: none !important;
}

.model-create-backdrop {
    position: absolute;
    inset: 0;
    background: rgba(15, 23, 42, 0.56);
}

.model-create-dialog {
    position: relative;
    width: min(100%, 520px);
    max-height: min(90vh, 720px);
    overflow: auto;
    padding: 24px;
    border: 1px solid #e5eaf0;
    border-radius: 16px;
    background: #ffffff;
    box-shadow: 0 24px 70px rgba(15, 23, 42, 0.25);
}
```

```
.model-create-heading h2 {
    margin: 0 0 7px;
    color: #182230;
    font-size: 20px;
}

.model-create-heading p {
    margin: 0 0 20px;
    color: #697586;
    font-size: 13px;
    line-height: 1.5;
}

.model-create-field,
.model-type-options legend {
    display: block;
    margin-bottom: 7px;
    color: #344054;
    font-size: 12px;
    font-weight: 800;
}

.model-create-name {
    width: 100%;
    min-width: 0;
    max-width: none;
    margin-bottom: 20px;
}

.model-type-options {
    display: grid;
    gap: 10px;
    padding: 0;
    margin: 0;
    border: 0;
}

.model-type-options legend {
    padding: 0;
}

.model-type-option {
    display: grid;
    grid-template-columns: 18px minmax(0, 1fr);
    align-items: center;
    gap: 11px;
    width: 100%;
    min-width: 0;
    padding: 13px 14px;
    border: 1px solid #d9e1ec;
    border-radius: 10px;
    cursor: pointer;
    background: #ffffff;
}

.model-type-option:has(input:checked) {
    border-color: #3b82f6;
    background: #eff6ff;
    box-shadow: 0 0 0 1px #3b82f6 inset;
}

.model-type-option input[type="radio"] {
    /* Override the global full-width text-input rules for this radio control. */
    display: block;
    width: 18px;
    min-width: 18px;
    max-width: 18px;
    height: 18px;
    min-height: 18px;
    max-height: 18px;
    margin: 0;
    padding: 0;
    border: 0;
    border-radius: 50%;
    accent-color: #2563eb;
    box-shadow: none;
    flex: 0 0 18px;
}

.model-type-option span {
    display: grid;
    gap: 4px;
    width: 100%;
    min-width: 0;
}

.model-type-option strong {
```

```
    color: #1f2937;
    font-size: 13px;
}

.model-type-option small {
    color: #697586;
    font-size: 11px;
    line-height: 1.4;
}

.model-create-error {
    margin: 12px 0 0;
    color: #b42318;
    font-size: 12px;
    line-height: 1.4;
}

.model-create-error[hidden] {
    display: none;
}

.model-create-actions {
    display: flex;
    justify-content: flex-end;
    gap: 9px;
    margin-top: 22px;
}

.model-type-indicator {
    min-height: 42px;
    justify-content: center;
    white-space: nowrap;
    flex: 0 0 auto;
    font-size: 11px;
    padding: 0 12px;
}

@media (max-width: 480px) {
    .model-create-dialog {
        padding: 19px;
    }

    .model-create-actions {
        display: grid;
        grid-template-columns: 1fr 1fr;
    }
}
```

05. assets/icons/structure-analysis.svg

```
<svg xmlns="http://www.w3.org/2000/svg" width="512" height="512" viewBox="0 0 512 512">
  <rect width="512" height="512" rx="104" fill="#111827"/>
  <path d="M92 372H420M128 372V152M384 372V152M128 176H384M128 248H384M128 320H384" fill="none" stroke="#60A5FA" stroke-width="22" stroke-linecap="round" stroke-linejoin="round"/>
  <path d="M128 176L256 248L384 176M128 248L256 320L384 248M128 320L256 372L384 320" fill="none" stroke="#E5E7EB" stroke-width="14" stroke-linecap="round" stroke-linejoin="round"/>
  <g fill="#F9FAFB" stroke="#2563EB" stroke-width="10">
    <circle cx="128" cy="176" r="19"/><circle cx="256" cy="176" r="19"/><circle cx="384" cy="176" r="19"/>
    <circle cx="128" cy="248" r="19"/><circle cx="256" cy="248" r="19"/><circle cx="384" cy="248" r="19"/>
    <circle cx="128" cy="320" r="19"/><circle cx="256" cy="320" r="19"/><circle cx="384" cy="320" r="19"/>
  </g>
</svg>
```

06. assets/js/main.js

```
/* ========================================================================
 * main.js — global / common JavaScript
 * Shared by every page module: app configuration, application state, generic
 * helpers, unit system, input validation, matrix utilities, persistence
 * (save / load / export / import), sidebar navigation, sortable-table drag,
 * common select/delete helpers, refreshAll() and the application bootstrap.
 *
 * Load order (see index.html): vendor/jspdf + svg2pdf + pdf-fonts -> main.js -> page modules
 * (home ... moment) -> results core -> results steps 1-8 -> diagrams.
 * ======================================================================== */

/* Injects a module's markup into the page.
 * target   : CSS selector (or Element) the markup is attached to
 * position : insertAdjacentHTML position ("beforeend", "afterbegin", "afterend", ...)
 * html     : markup string
 * Every page module / results part calls this once when its script loads. */
function mountModuleHTML(target, position, html) {
    const el = typeof target === "string" ? document.querySelector(target) : target;
    if (!el) {
        console.error("[mountModuleHTML] mount point not found:", target);
        return null;
    }
    el.insertAdjacentHTML(position, html);
    return el;
}

/* ---- Application shell (loading overlay, sidebar, top bar, page container) ---- */
document.body.insertAdjacentHTML("afterbegin", `
<div id="analysisLoadingOverlay" class="analysis-loading-overlay" aria-hidden="true">
    <div class="analysis-loading-card">
        <div class="analysis-loading-spinner"></div>
        <div class="analysis-loading-title">Preparing Structural Analysis</div>
        <div class="analysis-loading-text" id="analysisLoadingText">
            Loading nodes, elements, loads, moments and calculation data…
        </div>
    </div>
</div>



<div class="app home-mode">

    <!-- SIDEBAR -->
    <aside class="sidebar">
        <div class="brand">
            <h1>Structure Analysis</h1>
            <p>Input Model Builder</p>
        </div>
        <div class="nav-title">Model Input</div>
        <div class="nav-buttons">
            <button class="nav-btn active" data-section="home"><span class="nav-icon">■</span>Home</button>
            <button class="nav-btn" data-section="dashboard"><span class="nav-icon">■</span>Dashboard</button>
            <button class="nav-btn" data-section="nodes"><span class="nav-icon">●</span>Nodes</button>
            <button class="nav-btn" data-section="elements"><span class="nav-icon">■</span>Elements</button>
            <button class="nav-btn" data-section="supports"><span class="nav-icon">■</span>Supports</button>
            <button class="nav-btn" data-section="loadCases"><span class="nav-icon">■</span>Load Cases</button>
            <button class="nav-btn" data-section="loads"><span class="nav-icon">↓</span>Loads</button>
            <button class="nav-btn" data-section="momentCases"><span class="nav-icon">■</span>Moment Cases</button>
            <button class="nav-btn" data-section="moments"><span class="nav-icon">■</span>Moments</button>
            <button class="nav-btn" data-section="results"><span class="nav-icon">■</span>Results</button>
        </div>
        <div class="sidebar-bottom">
            <button class="sidebar-btn" onclick="exportModel()">Export JSON</button>
            <button class="sidebar-btn" onclick="document.getElementById('jsonImport').click()">Import JSON</button>
            <input type="file" id="jsonImport" accept=".json" style="display:none" onchange="importModel(event)">
            <button class="sidebar-btn" onclick="clearModel()">Clear Model</button>
        </div>
    </aside>

    <!-- MAIN -->
    <main class="main">

        <header class="topbar">
            <div class="page-title">
                <h2 id="pageTitle">Model Home</h2>
                <p id="pageSubtitle">Manage your structural analysis models.</p>
            </div>
            <div class="top-actions">
                <button id="topbarHomeButton" type="button" class="btn btn-secondary" onclick="goHome()">■ Home</button>
                <span id="currentModelTypeBadge" class="badge badge-blue model-type-indicator" role="status" aria-label="Current model type" title="Model type is selected when creating a
model and cannot be changed afterward">Beam / Truss / Frame Model</span>
                <input id="modelName" class="model-name-input" type="text" maxlength="120" placeholder="Model Name" aria-label="Model Name" value="Structural Model"
oninput="saveModelNamePreference()">
                <button class="btn btn-secondary" onclick="exportModel()">Export</button>
                <button class="btn btn-primary" onclick="saveModel()">Save Model</button>
```

```
            </div>
        </header>

        <div class="container">
        </div>
    </main>
</div>

<div id="newModelModal" class="model-create-modal" hidden aria-hidden="true">
    <div class="model-create-backdrop" onclick="closeNewModelDialog()" aria-hidden="true"></div>
    <form class="model-create-dialog" role="dialog" aria-modal="true" aria-labelledby="newModelDialogTitle" onsubmit="createNewModelFromDialog(event)">
        <div class="model-create-heading">
            <h2 id="newModelDialogTitle">Create New Structural Model</h2>
            <p>Choose the structure type now. It will remain fixed for this model.</p>
        </div>
        <label class="model-create-field" for="newModelNameInput">Model name</label>
        <input id="newModelNameInput" class="model-name-input model-create-name" type="text" maxlength="120" autocomplete="off" placeholder="Model name">
        <fieldset class="model-type-options">
            <legend>Structure type</legend>
            <label class="model-type-option">
                <input type="radio" name="newModelAnalysisMode" value="plane" checked>
                <span><strong>Beam / Truss / Frame</strong><small>Standard 2D structural analysis</small></span>
            </label>
            <label class="model-type-option">
                <input type="radio" name="newModelAnalysisMode" value="grid">
                <span><strong>Grid</strong><small>Grid bending and torsional analysis</small></span>
            </label>
        </fieldset>
        <p id="newModelValidationMessage" class="model-create-error" role="alert" hidden></p>
        <div class="model-create-actions">
            <button type="button" class="btn btn-secondary" onclick="closeNewModelDialog()">Cancel</button>
            <button type="submit" class="btn btn-primary">Create Model</button>
        </div>
    </form>
</div>
`);

// ============================================================================
// 1. APP CONFIGURATION
// Application-wide constants: default values, diagram/rendering constants,
// persistence keys, and UI copy. Nothing in here performs calculations —
// it is pure static configuration consumed by the engines/controllers below.
// ============================================================================
const AppConfig = {
    // Built-in plane-frame support templates. Always available; cannot be
    // deleted. Custom supports remain fully editable through the form.
    builtInSupportTemplates: [
        {
            id: "FIXED",
            description: "Fixed Support",
            type: "fixed",
            builtIn: true,
            restraints: { RFx: true, RFy: true, RFz: true, RMx: true, RMy: true, RMz: true },
            displacement: { dx: 0, dy: 0, dz: 0, tx: 0, ty: 0, tz: 0 },
        },
        {
            id: "ROLLER",
            description: "Roller Support",
            type: "roller",
            builtIn: true,
            restraints: { RFx: false, RFy: true, RFz: false, RMx: false, RMy: false, RMz: false },
            displacement: { dx: 0, dy: 0, dz: 0, tx: 0, ty: 0, tz: 0 },
        },
        {
            id: "HINGED",
            description: "Hinged Support",
            type: "hinged",
            builtIn: true,
            restraints: { RFx: true, RFy: true, RFz: false, RMx: false, RMy: false, RMz: false },
            displacement: { dx: 0, dy: 0, dz: 0, tx: 0, ty: 0, tz: 0 },
        },
    ],

    // Only these input-model keys are persisted to localStorage / export.
    // Solver-derived analysis state is never saved.
    persistedModelKeys: [
        "nodes",
        "elementTemplates",
        "elements",
        "supportTemplates",
        "supportAssignments",
        "loadCases",
        "loads",
        "momentCases",
        "moments",
    ],
    // Shared-baseline diagram scaling constants (see calculateAutoFit /
```

```
    // drawDiagramSVG): the same physical length always maps to the same
    // pixel length across the Structure Model, Reactions, AFD, SFD, BMD and
    // Displacement diagrams, keeping the whole visual system proportional.
    sharedDiagramBaselinePx: 200,
    modelDiagramBaselinePx: 200,

    // Page title/subtitle text shown in the topbar for each sidebar section.
    sectionTitles: {
        home: ["Model Home", "Manage multiple saved structural analysis models."],
        dashboard: ["Model Dashboard", "Define the structural analysis input model."],
        nodes: ["Nodes", "Define global node coordinates."],
        elements: ["Elements", "Define reusable element properties and node connectivity."],
        supports: ["Supports", "Define reusable support conditions."],
        loadCases: ["Load Cases", "Define reusable force load cases."],
        loads: ["Loads", "Define nodal and element force inputs."],
        momentCases: ["Moment Cases", "Define reusable moment cases."],
        moments: ["Moments", "Define nodal and element moment inputs."],
        results: ["Analysis Results", "Stiffness Method output."],
    },
};

// ============================================================================
// 2. APPLICATION STATE
// ============================================================================
const model = {
    nodes: [],
    elementTemplates: [],
    elements: [],
    supportTemplates: [],
    supportAssignments: [],
    loadCases: [],
    loads: [],
    momentCases: [],
    moments: [],
    dofSettings: { skipAxial: false, skipMoment: false },
    analysisMode: "plane",
};

// Named model library. Each saved model stores only the true input model,
// never solver-derived analysis state.
// Storage can be disabled by privacy settings, sandboxing, or browser policy.
// Keep the application usable in-memory if accessing localStorage throws.
function safeStorageGetItem(key) {
    try { return window.localStorage.getItem(key); }
    catch (error) { return null; }
}
function safeStorageSetItem(key, value) {
    try { window.localStorage.setItem(key, String(value)); return true; }
    catch (error) { console.warn(`[Storage] Could not save "${key}"; changes remain in memory only.`, error); return false; }
}
function safeStorageRemoveItem(key) {
    try { window.localStorage.removeItem(key); return true; }
    catch (error) { console.warn(`[Storage] Could not remove "${key}".`, error); return false; }
}

let currentModelLibraryId = safeStorageGetItem("structuralCurrentModelId") || "";
let lastModelLibraryWriteSucceeded = true;
let lastModelSaveStorageFailed = false;
let lastModelLibraryReadFailed = false;

function cloneBuiltInSupportTemplate(template) {
    return JSON.parse(JSON.stringify(template));
}

function ensureBuiltInSupportTemplates() {
    const customTemplates = (model.supportTemplates || []).filter((s) => !s.builtIn && !["FIXED", "ROLLER", "HINGED"].includes(s.id));
    model.supportTemplates = [
        ...AppConfig.builtInSupportTemplates.map(cloneBuiltInSupportTemplate),
        ...customTemplates.map((s) => ({ ...s, type: s.type || "custom", builtIn: false })),
    ];
}

// Active analysis case selections. These are UI state, not part of the persisted input model.
let activeLoadCaseId = "";
let activeMomentCaseId = "";

// Which diagram mode is currently shown in the fullscreen modal (see section 10).
let currentFSMode = "model";

// Analysis result lifecycle: not-run -> valid -> invalidated.
const analysisState = { status: "not-run" };

// Snapshot-style reference to the two central state objects, for anyone
// exploring this file for the first time. This is documentation, not a new
// state container: AppState.model and AppState.analysis are the exact same
// objects as `model` and `analysisState` above, and every function in this
// file continues to read/write `model` / `analysisState` directly.
```

```
const AppState = { model, analysis: analysisState };

function isGridAnalysisMode() {
    return model.analysisMode === "grid";
}

// Keep both diagram selectors aligned with the current solver mode. This can
// also be called immediately before opening the fullscreen picker, so its labels
// do not depend solely on the earlier model-load synchronization.
function syncDiagramModeLabels() {
    const diagramLabels = isGridAnalysisMode()
        ? { afd: "Torsion (Mx)", sfd: "Vertical Shear (Fz)", bmd: "Torsion (My)" }
        : { afd: "AFD", sfd: "SFD", bmd: "BMD" };
    ["resultsDiagramMode", "fsModeSelect"].forEach((selectId) => {
        const select = document.getElementById(selectId);
        if (!select) return;
        Object.entries(diagramLabels).forEach(([value, label]) => {
            const option = Array.from(select.options || []).find((item) => item.value === value);
            if (option) option.textContent = label;
        });
    });
}

function syncAnalysisModeUI() {
    const gridMode = isGridAnalysisMode();
    const typeBadge = document.getElementById("currentModelTypeBadge");
    if (typeBadge) {
        typeBadge.textContent = gridMode ? "Grid Model" : "Beam / Truss / Frame Model";
        typeBadge.title = gridMode
            ? "Grid model — type is fixed for this model"
            : "Beam / Truss / Frame model — type is fixed for this model";
        typeBadge.className = `badge ${gridMode ? "badge-green" : "badge-blue"} model-type-indicator`;
    }
    const gridFields = document.getElementById("gridElementProperties");
    if (gridFields) gridFields.style.display = gridMode ? "grid" : "none";
    // Per-element self-weight values are configured on a Self Weight load case.
    document.querySelectorAll(".grid-only-property").forEach((node) => {
        node.style.display = gridMode ? "table-cell" : "none";
    });
    document.querySelectorAll(".plane-only-property").forEach((node) => {
        const isTableCell = node.tagName === "TH" || node.tagName === "TD";
        node.style.display = gridMode ? "none" : (isTableCell ? "table-cell" : "");
    });
    const dofBox = document.getElementById("planeDofNumberingBox");
    if (dofBox) dofBox.style.display = isGridAnalysisMode() ? "none" : "block";
    document.querySelectorAll(".plane-support-movement").forEach((field) => {
        field.style.display = gridMode ? "none" : "";
    });
    document.querySelectorAll(".grid-support-movement").forEach((field) => {
        field.style.display = gridMode ? "" : "none";
    });
    const modeNote = document.getElementById("analysisModeNote");
    if (modeNote) {
        modeNote.textContent = isGridAnalysisMode()
            ? "Grid geometry lies in the XY plane; node Z coordinates are unused. Grid DOFs: θX, θY and ΔZ. Grid templates require positive E, G, I and J. Loads use Fz; torsion uses Mx (+
right/− left) and My (+ up/− down)."
            : "Plane analysis uses the selected Beam, Truss or Frame type. Thermal loads are defined per member; uniform temperature changes create axial thermal actions and temperature
gradients create bending actions.";
    }
    syncDiagramModeLabels();

    const setLabel = (id, value) => { const el = document.getElementById(id); if (el) el.textContent = value; };
    const setSupportDescription = (id, html) => { const el = document.getElementById(id); if (el) el.innerHTML = html; };
    if (isGridAnalysisMode()) {
        setSupportDescription("supportFixedDescription", "θX + θY + ΔZ restrained");
        setSupportDescription("supportRollerDescription", "ΔZ restrained only<br><span style=\"color:#64748b;\">Grid rotations θX and θY free</span>");
        setSupportDescription("supportHingedDescription", "ΔZ restrained only<br><span style=\"color:#64748b;\">Grid rotations θX and θY free</span>");
        setLabel("supportRFxLabel", "θX restrained"); setLabel("supportRFyLabel", "θY restrained");
        setLabel("supportRFzLabel", "RFz (unused by Grid)"); setLabel("supportRMxLabel", "RMx (unused by Grid)");
        setLabel("supportRMyLabel", "RMy (unused by Grid)"); setLabel("supportRMzLabel", "ΔZ restrained");
        setLabel("supportDispXLabel", "Δx (unused by Grid)"); setLabel("supportDispYLabel", "Δy (unused by Grid)");
        setLabel("supportDispZLabel", "ΔZ prescribed"); setLabel("supportRotXLabel", "θX prescribed");
        setLabel("supportRotYLabel", "θY prescribed"); setLabel("supportRotZLabel", "θz (unused by Grid)");
    } else {
        setSupportDescription("supportFixedDescription", "RFx + RFy + RFz + RMx + RMy + RMz restrained");
        setSupportDescription("supportRollerDescription", "RFy restrained only<br><span style=\"color:#64748b;\">All other 2D restraints free</span>");
        setSupportDescription("supportHingedDescription", "RFx + RFy restrained<br><span style=\"color:#64748b;\">Rotation RZ remains free</span>");
        setLabel("supportRFxLabel", "RFx"); setLabel("supportRFyLabel", "RFy");
        setLabel("supportRFzLabel", "RFz (Ignore)"); setLabel("supportRMxLabel", "RMx (Ignore)");
        setLabel("supportRMyLabel", "RMy (Ignore)"); setLabel("supportRMzLabel", "RMz");
        setLabel("supportDispXLabel", "Δx"); setLabel("supportDispYLabel", "Δy");
        setLabel("supportDispZLabel", "Δz"); setLabel("supportRotXLabel", "θx");
        setLabel("supportRotYLabel", "θy"); setLabel("supportRotZLabel", "θz");
    }
    const loadType = document.getElementById("loadType")?.value || "Nodal Load";
    const isUDL = loadType === "Element UDL";
    if (isGridAnalysisMode()) {
```

```
        setLabel("lblLoadFx", isUDL ? "Start Fx (not used by Grid)" : "Fx (not used by Grid)");
        setLabel("lblLoadFy", isUDL ? "Start Fy (not used by Grid)" : "Fy (not used by Grid)");
        setLabel("lblLoadFz", isUDL ? "Start Fz" : "Fz");
        setLabel("lblLoadFxEnd", "End Fx (not used by Grid)");
        setLabel("lblLoadFyEnd", "End Fy (not used by Grid)");
        setLabel("lblLoadFzEnd", "End Fz");
        setLabel("lblMomentX", "Mx — Right / Left (+ right, − left)");
        setLabel("lblMomentY", "My — Up / Down (+ up, − down)");
        setLabel("lblMomentZ", "Mz (not used by Grid)");
    } else {
        setLabel("lblLoadFz", "Fz (Ignore)");
        setLabel("lblLoadFzEnd", "End Fz (Ignore)");
        setLabel("lblMomentX", "Mx (Ignore)");
        setLabel("lblMomentY", "My (Ignore)");
        setLabel("lblMomentZ", "Mz (Rotation)");
    }

    // User-facing Grid terminology is torsion; the stored model fields remain
    // compatible with existing moment-case and moment solver data.
    const gridNavLabels = [
        ["momentCases", gridMode ? "Torsion Cases" : "Moment Cases"],
        ["moments", gridMode ? "Torsions" : "Moments"],
    ];
    gridNavLabels.forEach(([section, label]) => {
        const button = document.querySelector(`.nav-btn[data-section="${section}"]`);
        if (button) {
            const icon = button.querySelector(".nav-icon");
            button.innerHTML = `${icon ? icon.outerHTML : ""}${label}`;
        }
    });
    const pageTitles = AppConfig.sectionTitles;
    pageTitles.momentCases = gridMode
        ? ["Torsion Cases", "Define reusable Grid torsion cases."]
        : ["Moment Cases", "Define reusable moment cases."];
    pageTitles.moments = gridMode
        ? ["Torsions", "Define nodal and element Grid torsions."]
        : ["Moments", "Define nodal and element moment inputs."];
    const activeSection = document.querySelector(".section.active")?.id;
    if (activeSection === "moments" || activeSection === "momentCases") {
        const title = document.getElementById("pageTitle");
        const subtitle = document.getElementById("pageSubtitle");
        const content = pageTitles[activeSection];
        if (content && title) title.textContent = content[0];
        if (content && subtitle) subtitle.textContent = content[1];
    }
    if (typeof syncLoadInputModeUI === "function") syncLoadInputModeUI();
    if (typeof syncMomentInputModeUI === "function") syncMomentInputModeUI();
    if (typeof syncMomentCaseInputModeUI === "function") syncMomentCaseInputModeUI();
    if (typeof syncElementTemplateTypeUI === "function") syncElementTemplateTypeUI();
}

function setAnalysisMode(value) {
    // A model's solver type is fixed when it is created. Loading/importing a
    // model restores its saved type through applyPersistedInputModel(); this
    // function intentionally cannot switch an existing model's solver.
    const requested = value === "grid" ? "grid" : "plane";
    if (requested !== (model.analysisMode === "grid" ? "grid" : "plane")) {
        console.warn("Model type cannot be changed after creation. Create a new model to use another solver.");
    }
    syncAnalysisModeUI();
    return requested === (model.analysisMode === "grid" ? "grid" : "plane");
}


// Page title/subtitle text shown in the topbar for each sidebar section.

// ============================================================================
// 3. GENERIC HELPERS
// ============================================================================
function getNextId(currentId) {
    const current = String(currentId ?? "");
    const match = current.match(/^(.*?)(\d+)$/);
    if (match) {
        const next = String(parseInt(match[2], 10) + 1).padStart(match[2].length, "0");
        return match[1] + next;
    }
    return current + "1";
}

function val(id) {
    return document.getElementById(id).value.trim();
}
function num(id) {
    const value = document.getElementById(id).value.trim();
    if (value === "") return "";
    const parsed = Number(value);
    return Number.isFinite(parsed) ? parsed : "";
```

```
}

// Mobile-friendly scientific notation helper. Tap "e" after entering the
// mantissa, then enter the exponent, e.g. 2e5, 7.5e-4, 1e-6.
function insertScientificExponent(id) {
    const input = document.getElementById(id);
    if (!input) return;
    input.focus();
    const start = typeof input.selectionStart === "number" ? input.selectionStart : input.value.length;
    const end = typeof input.selectionEnd === "number" ? input.selectionEnd : input.value.length;
    const before = input.value.slice(0, start);
    const after = input.value.slice(end);
    if (/[eE]/.test(input.value)) return;
    input.value = before + "e" + after;
    const pos = start + 1;
    if (typeof input.setSelectionRange === "function") input.setSelectionRange(pos, pos);
}

function escapeHtml(value) {
    return String(value ?? "")
        .replaceAll("&", "&amp;")
        .replaceAll("<", "&lt;")
        .replaceAll(">", "&gt;")
        .replaceAll('"', "&quot;")
        .replaceAll("'", "&#039;");
}

// Safe for values embedded inside single-quoted inline JavaScript arguments.
// encodeURIComponent alone leaves apostrophes unchanged, allowing quote breakout.
function encodeInlineArgument(value) {
    return encodeURIComponent(String(value ?? "")).replace(/'/g, "%27");
}

function formatNum(num) {
    if (Math.abs(num) < 1e-10) return "0.000";
    if (Math.abs(num) > 1e6 || (Math.abs(num) > 0 && Math.abs(num) < 1e-3))
        return Number(num).toExponential(3);
    return Number(num).toFixed(3);
}

// ============================================================================
// UNIT SYSTEM
// Internal analysis units: kN, m, kN/m, kN·m, kN/m².
// Young's modulus E is entered/stored in kN/m². Grid shear modulus G remains entered in GPa.
// ============================================================================
function getElementEInternal(tmpl) {
    if (!tmpl) return 0;
    const hasValue = (value) => value !== undefined && value !== null && value !== "" && Number.isFinite(Number(value));
    // Canonical material unit: kN/m². Legacy fields remain readable during import/migration.
    if (hasValue(tmpl.E_kN_m2)) return Number(tmpl.E_kN_m2);
    if (hasValue(tmpl.E_GPa)) return Number(tmpl.E_GPa) * 1e6;
    // Legacy v14 stored E in N/m².
    if (hasValue(tmpl.E)) return Number(tmpl.E) / 1000;
    return 0;
}

function migrateUnitSystem() {
    model.elementTemplates = (model.elementTemplates || []).map((tmpl) => {
        const normalized = { ...tmpl };
        const hasValue = (value) => value !== undefined && value !== null && value !== "" && Number.isFinite(Number(value));
        if (!hasValue(normalized.E_kN_m2)) {
            if (hasValue(normalized.E_GPa)) {
                // Legacy saved models stored E in GPa; 1 GPa = 10■ kN/m².
                normalized.E_kN_m2 = Number(normalized.E_GPa) * 1e6;
            } else if (hasValue(normalized.E)) {
                // Older models stored E in N/m².
                normalized.E_kN_m2 = Number(normalized.E) / 1000;
            }
        }
        delete normalized.E_GPa;
        delete normalized.E;
        return normalized;
    });
}



// ============================================================================
// INPUT VALIDATION HELPERS
// Centralized validation keeps all model entities consistent before mutation.
// ============================================================================
function isFiniteNumber(value) {
    return typeof value === "number" && Number.isFinite(value);
}

function requireFinite(value, label, options = {}) {
    if (!isFiniteNumber(value)) return `${label} must be a finite number.`;
```

```
    if (options.min !== undefined && value < options.min) return `${label} must be at least ${options.min}.`;
    if (options.minExclusive !== undefined && value <= options.minExclusive) return `${label} must be greater than ${options.minExclusive}.`;
    return null;
}

// Migrate template-level self-weight values from older model files into
// per-element values on Self Weight load cases. Current models store no weight
// on element templates: every element has its own kN/m value on the case.
function normalizeLegacySelfWeightModel(source) {
    if (!source || typeof source !== "object" || Array.isArray(source) || !Array.isArray(source.elementTemplates)) return source;

    const grid = source.analysisMode === "grid";
    const legacyByTemplate = new Map();
    const elementTemplates = source.elementTemplates.map((original, index) => {
        if (!original || typeof original !== "object" || Array.isArray(original)) return original;
        const template = { ...original };
        let legacyValue;
        let hasLegacyValue = false;

        if (Object.prototype.hasOwnProperty.call(template, "selfWeight")) {
            legacyValue = template.selfWeight;
            hasLegacyValue = true;
        } else if (Object.prototype.hasOwnProperty.call(template, "density") ||
                   Object.prototype.hasOwnProperty.call(template, "selfWeightType")) {
            const raw = template.density === undefined || template.density === null || template.density === ""
                ? 0 : Number(template.density);
            if (!Number.isFinite(raw) || raw < 0) {
                legacyValue = template.density;
                hasLegacyValue = true;
            } else {
                const oldType = template.selfWeightType || "density";
                if (oldType === "lineWeight") {
                    legacyValue = raw;
                } else if (oldType === "density") {
                    if (raw > 0) {
                        if (grid) {
                            throw new Error(`Element template '${template.id ?? index + 1}' has an older density-based weight that cannot be converted safely for Grid. Re-enter self-weight
for each element in a Self Weight load case.`);
                        }
                        const area = Number(template.A);
                        if (area === 0 && Number(template.I) > 0) {
                            throw new Error(`Beam template '${template.id ?? index + 1}' has an older density-based weight but A = 0. Re-enter self-weight for each element in a Self Weight
load case.`);
                        }
                        legacyValue = raw * area * 9.81 / 1000;
                    } else {
                        legacyValue = 0;
                    }
                } else {
                    throw new Error(`Element template '${template.id ?? index + 1}' has an unsupported old self-weight format.`);
                }
                hasLegacyValue = true;
            }
        }

        if (hasLegacyValue) {
            if (legacyValue === undefined || legacyValue === null || legacyValue === "" ||
                !Number.isFinite(Number(legacyValue)) || Number(legacyValue) < 0) {
                throw new Error(`Element template '${template.id ?? index + 1}' has an invalid legacy self-weight. Re-enter its weight on a Self Weight load case.`);
            }
            if (template.id !== undefined && template.id !== null) {
                legacyByTemplate.set(String(template.id), Number(legacyValue));
            }
        }
        delete template.selfWeight;
        delete template.density;
        delete template.selfWeightType;
        return template;
    });

    const elements = Array.isArray(source.elements) ? source.elements : [];
    const legacyByElement = Object.create(null);
    elements.forEach((element) => {
        const templateId = String(element?.templateId ?? "");
        if (legacyByTemplate.has(templateId)) legacyByElement[String(element.id)] = legacyByTemplate.get(templateId);
    });

    const normalized = { ...source, elementTemplates };
    const hasPositiveLegacyWeight = Object.values(legacyByElement).some((value) => Number(value) > 0);
    const loadCases = Array.isArray(source.loadCases) ? source.loadCases.map((item) => ({ ...item })) : [];
    const selfWeightCaseExists = loadCases.some((item) => item.type === "Self Weight");

    if (Object.keys(legacyByElement).length && selfWeightCaseExists) {
        normalized.loadCases = loadCases.map((item) => {
            if (item.type !== "Self Weight") return item;
            const existing = item.memberSelfWeights && typeof item.memberSelfWeights === "object" && !Array.isArray(item.memberSelfWeights)
                ? item.memberSelfWeights : {};
```

```
            return { ...item, memberSelfWeights: { ...legacyByElement, ...existing } };
        });
    } else if (hasPositiveLegacyWeight && !selfWeightCaseExists) {
        let id = "SW1", suffix = 2;
        while (loadCases.some((item) => String(item.id) === id)) id = `SW${suffix++}`;
        loadCases.push({
            id,
            type: "Self Weight",
            description: "Migrated member self-weight",
            memberSelfWeights: { ...legacyByElement },
        });
        normalized.loadCases = loadCases;
    } else if (Array.isArray(source.loadCases)) {
        normalized.loadCases = loadCases;
    } else if (source.loadCases === undefined && hasPositiveLegacyWeight) {
        normalized.loadCases = [{ id: "SW1", type: "Self Weight", description: "Migrated member self-weight", memberSelfWeights: { ...legacyByElement } }];
    }

    return normalized;
}
function hasId(items, id, editingId = null) {
    // "__proto__" is special on ordinary object maps used by the UI/solver.
    if (String(id) === "__proto__") return true;
    return Array.isArray(items) && items.some((item) => String(item.id) === String(id) && String(item.id) !== String(editingId));
}

function requireReference(items, id, label) {
    if (!id) return `${label} is required.`;
    if (!items.some((item) => String(item.id) === String(id))) return `${label} "${id}" does not exist.`;
    return null;
}

function getElementLength(elementId) {
    const el = model.elements.find((item) => item.id === elementId);
    if (!el) return null;
    const n1 = model.nodes.find((n) => n.id === el.startNode);
    const n2 = model.nodes.find((n) => n.id === el.endNode);
    if (!n1 || !n2) return null;
    return Math.hypot(Number(n2.x) - Number(n1.x), Number(n2.y) - Number(n1.y));
}

function getElementStructuralType(element) {
    const tmpl = element ? model.elementTemplates.find((t) => t.id === element.templateId) : null;
    if (!tmpl) return null;
    const explicitType = String(tmpl.memberType || "").toLowerCase();
    if (["beam", "truss", "frame"].includes(explicitType)) return explicitType;
    // Backward compatibility for existing models that classified members
    // implicitly from A and I before the explicit member-type field existed.
    const A = Number(tmpl.A);
    const I = Number(tmpl.I);
    if (!Number.isFinite(A) || !Number.isFinite(I)) return null;
    if (A === 0 && I > 0) return "beam";
    if (I === 0 && A > 0) return "truss";
    if (A > 0 && I > 0) return "frame";
    return null;
}

function detectDefaultGlobalDofNumberingConfig() {
    // The Neglect X / Neglect Moment M controls are manual. With both
    // controls unselected, no DOF is omitted just because an element has
    // A=0 or I=0.
    return { skipAxial: false, skipMoment: false };
}

function ensureGlobalDofNumberingConfig() {
    const existing = model.dofSettings;
    if (existing && typeof existing === "object" && typeof existing.skipAxial === "boolean" && typeof existing.skipMoment === "boolean") {
        return existing;
    }

    // Backward compatibility: models created before manual switches existed
    // retain their previous automatic behaviour the first time they are opened.
    model.dofSettings = detectDefaultGlobalDofNumberingConfig();
    return model.dofSettings;
}

function getGlobalDofNumberingConfig() {
    const config = ensureGlobalDofNumberingConfig();
    return {
        skipAxial: !!config.skipAxial,
        skipMoment: !!config.skipMoment,
    };
}

function setGlobalDofNumberingOption(option, checked) {
    if (option !== "skipAxial" && option !== "skipMoment") return;
    const config = ensureGlobalDofNumberingConfig();
```

```
    config[option] = !!checked;
    model.dofSettings = config;

    // Changing a numbering option changes the global matrix numbering, so any
    // previous analysis result must be discarded before another run.
    markAnalysisInvalidated();
    updateGlobalDofNumberingUI();
}

function getNodeStructuralDofRelevance(nodeId) {
    const connected = model.elements.filter((el) => el.startNode === nodeId || el.endNode === nodeId);
    if (isGridAnalysisMode()) {
        const active = connected.length > 0;
        return { ux: active, uy: active, rz: active };
    }
    let axial = false;
    let bending = false;
    const skipAxialForBeam = getGlobalDofNumberingConfig().skipAxial;
    connected.forEach((el) => {
        const tmpl = model.elementTemplates.find((t) => t.id === el.templateId);
        if (!tmpl) return;
        const memberType = getElementStructuralType(el);
        // New Beam templates have an area but their axial DOFs can be excluded
        // by Ignore X. Truss and Frame axial stiffness retain their prior behavior.
        axial = axial || (Number(tmpl.A) > 0 && !(memberType === "beam" && skipAxialForBeam));
        bending = bending || Number(tmpl.I) > 0;
    });
    return { ux: axial, uy: connected.length > 0, rz: bending };
}

function parseLengthInputStrict(rawValue, elementId, label = "Distance") {
    const str = rawValue === undefined || rawValue === null ? "" : String(rawValue).trim();
    if (str === "") return { ok: false, message: `${label} is required.` };
    const L = getElementLength(elementId);
    if (!isFiniteNumber(L) || L <= 0) return { ok: false, message: "The selected element must have a valid positive length before a distance can be assigned." };

    const formulaMatch = str.match(/^(\d*\.?\d*)\s*\*?\s*[lL]\s*(?:\/\s*(\d+\.?\d*))?$/);
    if (formulaMatch) {
        const multiplier = formulaMatch[1] === "" ? 1 : Number(formulaMatch[1]);
        const divisor = formulaMatch[2] ? Number(formulaMatch[2]) : 1;
        if (!Number.isFinite(multiplier) || !Number.isFinite(divisor) || divisor <= 0) return { ok: false, message: `${label} contains an invalid L expression.` };
        return { ok: true, value: (multiplier * L) / divisor, length: L };
    }

    const value = Number(str);
    if (!Number.isFinite(value)) return { ok: false, message: `${label} must be a finite number or a valid L expression such as L/2.` };
    return { ok: true, value, length: L };
}

function validateNodeInput(id, x, y, z, editingId = null) {
    if (!id) return "Node ID is required.";
    if (hasId(model.nodes, id, editingId)) return "Node ID already exists.";
    for (const [value, label] of [[x, "X Coordinate"], [y, "Y Coordinate"], [z, "Z Coordinate"]]) {
        const error = requireFinite(value, label);
        if (error) return error;
    }
    const coincidentNode = model.nodes.find((node) =>
        String(node.id) !== String(editingId) && Number(node.x) === Number(x) && Number(node.y) === Number(y));
    if (coincidentNode) {
        return `Node '${coincidentNode.id}' already occupies (${x}, ${y}). Each node must have unique X/Y coordinates.`;
    }
    return null;
}

// ============================================================================
// 4. MATRIX ALGEBRA UTILITIES
// Pure matrix helpers shared by the stiffness method solver below.
// ============================================================================
function isNumericallyZeroStiffnessDof(matrix, index, tolerance = Number.EPSILON * 100) {
    const row = Array.isArray(matrix?.[index]) ? matrix[index] : [];
    const rowScale = row.reduce((maximum, value) => Math.max(maximum, Math.abs(Number(value) || 0)), 0);
    if (rowScale === 0) return true;
    return Math.abs(Number(row[index]) || 0) <= rowScale * tolerance;
}

const Matrix = {
    zeros: (r, c) => Array.from({ length: r }, () => new Array(c).fill(0)),
    add: (A, B) => A.map((row, i) => row.map((val, j) => val + B[i][j])),
    sub: (A, B) => A.map((row, i) => row.map((val, j) => val - B[i][j])),
    mul: (A, B) =>
        A.map((row) => B[0].map((_, j) => row.reduce((sum, val, k) => sum + val * B[k][j], 0))),
    transpose: (A) => A[0].map((_, c) => A.map((r) => r[c])),
    invert: (M) => {
        let n = M.length;
        let I = Matrix.zeros(n, n);
        for (let i = 0; i < n; i++) I[i][i] = 1;
        let C = M.map((r, i) => [...r, ...I[i]]);
```

```
        // Keep each row's original scale aligned with its row during pivoting.
        const rowScales = M.map((row) => row.reduce((maximum, value) => Math.max(maximum, Math.abs(value)), 0));
        for (let i = 0; i < n; i++) {
            // Scaled partial pivoting avoids treating every small-valued
            // stiffness matrix as singular merely because its units/scale
            // produce pivots below a fixed absolute threshold.
            let maxRow = i;
            let bestRatio = -1;
            for (let k = i; k < n; k++) {
                const ratio = rowScales[k] === 0 ? 0 : Math.abs(C[k][i]) / rowScales[k];
                if (ratio > bestRatio) {
                    bestRatio = ratio;
                    maxRow = k;
                }
            }
            if (bestRatio < 1e-12) return null; // Singular or numerically dependent row
            let tmp = C[maxRow];
            C[maxRow] = C[i];
            C[i] = tmp;
            const scaleTmp = rowScales[maxRow];
            rowScales[maxRow] = rowScales[i];
            rowScales[i] = scaleTmp;
            let pivot = C[i][i];
            for (let j = 0; j < 2 * n; j++) C[i][j] /= pivot;
            for (let k = 0; k < n; k++) {
                if (k !== i) {
                    let factor = C[k][i];
                    for (let j = 0; j < 2 * n; j++) C[k][j] -= factor * C[i][j];
                }
            }
        }
        return C.map((r) => r.slice(n));
    },
};

// ============================================================================
// 5. MATRIX / EQUATION HTML RENDERING HELPERS
// Render [Matrix] and [A] = [B] x [C] style equations as bracketed HTML tables.
// ============================================================================
function matrixToHTML(M, title = "", topLabels = null, rightLabels = null) {
    if (!M || M.length === 0) return "";

    /*
     * Matrix reduction is handled here at the presentation boundary only.
     *
     * The stiffness/load calculations continue to use their existing
     * formulations. When a local element DOF was removed by the global
     * numbering configuration, its label is represented by "—". Those
     * placeholder positions must not be rendered as real matrix rows/columns.
     *
     * Example:
     *   [—, 1, 2, —, 3, 4]  ->  [1, 2, 3, 4]
     *   6×6 matrix            ->  4×4 matrix
     *
     * This keeps the existing calculation untouched while ensuring every
     * displayed matrix is dense and consistently re-indexed.
     */
    const isRemovedDofLabel = (label) =>
        label === "—" || label === "–" || label === "-" || label === null || label === undefined;

    const isVector = !Array.isArray(M[0]);
    const originalRows = isVector ? M.map((val) => [val]) : M;
    const originalRowCount = originalRows.length;
    const originalColCount = isVector ? 1 : originalRows[0].length;

    const topHasRemoved = Array.isArray(topLabels) &&
        topLabels.length === originalColCount &&
        topLabels.some(isRemovedDofLabel);

    const rightHasRemoved = Array.isArray(rightLabels) &&
        rightLabels.length === originalRowCount &&
        rightLabels.some(isRemovedDofLabel);

    let rowIndices = Array.from({ length: originalRowCount }, (_, i) => i);
    let colIndices = Array.from({ length: originalColCount }, (_, i) => i);

    if (rightHasRemoved) {
        rowIndices = rowIndices.filter((i) => !isRemovedDofLabel(rightLabels[i]));
    }

    if (topHasRemoved && !isVector) {
        colIndices = colIndices.filter((i) => !isRemovedDofLabel(topLabels[i]));
    }

    const rows = rowIndices.map((r) =>
        colIndices.map((c) => originalRows[r][c])
    );
```

```
    const displayTopLabels = topHasRemoved
        ? colIndices.map((_, i) => i + 1)
        : topLabels;
    const displayRightLabels = rightHasRemoved
        ? rowIndices.map((_, i) => i + 1)
        : rightLabels;

    let html = `<div class="matrix-wrapper">`;

    if (title) {
        html += `<div class="matrix-title">${escapeHtml(title)}</div>`;
    } else {
        html += `<div></div>`;
    }

    const colsCount = isVector ? 1 : rows[0]?.length || 0;

    if (displayTopLabels && !isVector) {
        html += `<div style="display: flex; width: 100%; padding: 0 8px;">`;
        for (let c = 0; c < colsCount; c++) {
            let lbl = displayTopLabels[c] !== undefined ? displayTopLabels[c] : "";
            html += `<div style="flex: 1; text-align: center; font-size: 11px; font-weight: bold; color: #dc2626;">${escapeHtml(lbl)}</div>`;
        }
        html += `</div>`;
    } else {
        html += `<div></div>`;
    }

    html += `<div style="display: flex; align-items: stretch; gap: 2px;">`;

    // Left Bracket
    html += `<div style="width: 6px; border-left: 2px solid #1e293b; border-top: 2px solid #1e293b; border-bottom: 2px solid #1e293b;"></div>`;

    // Values Table
    html += `<table class="matrix-table">`;

    rows.forEach((row) => {
        html += `<tr>`;
        row.forEach((val) => {
            html += `<td>${typeof val === "number" ? formatNum(val) : escapeHtml(val)}</td>`;
        });
        html += `</tr>`;
    });

    html += `</table>`;

    // Right Bracket
    html += `<div style="width: 6px; border-right: 2px solid #1e293b; border-top: 2px solid #1e293b; border-bottom: 2px solid #1e293b;"></div>`;

    // Right Labels
    if (displayRightLabels) {
        html += `<div style="display: flex; flex-direction: column; justify-content: space-around; margin-left: 6px; padding: 6px 0;">`;
        rows.forEach((_, i) => {
            let lbl = displayRightLabels[i] !== undefined ? displayRightLabels[i] : "";
            html += `<div style="font-size: 11px; font-weight: bold; color: #dc2626; line-height: 1;">${escapeHtml(lbl)}</div>`;
        });
        html += `</div>`;
    }

    html += `</div>`; // Close stretch flex
    html += `</div>`; // Close matrix-wrapper
    return html;
}

function matrixEqHTML(matrices, operators) {
    // Blank operators separate independent matrices into different rows.
    // Non-blank operators connect matrices into the same calculation row.
    let html = `<div class="eq-wrap">`;
    let row = ``;

    for (let i = 0; i < matrices.length; i++) {
        row += matrixToHTML(matrices[i].m, matrices[i].t, matrices[i].top, matrices[i].right);

        const op = i < operators.length ? String(operators[i]) : "";
        if (op.trim() !== "") {
            row += `<div style="display: inline-flex; align-items: center; justify-content: center;">
                        <div class="math-operator" style="margin: 0 10px;">${op}</div>
                    </div>`;
        } else {
            html += `<div class="eq-row">${row}</div>`;
            row = ``;
        }
    }

    if (row) {
        html += `<div class="eq-row">${row}</div>`;
    }
```

```
    html += `</div>`;
    return html;
}

function showEmpty(tbodyId, colspan, title = "No records yet") {
    document.getElementById(tbodyId).innerHTML =
        `<tr><td colspan="${colspan}"><div class="empty"><div class="empty-icon">■</div><h4>${title}</h4><div>Create an item using the form above.</div></div></td></tr>`;
}

// Per-assignment prescribed movements use the same six-component schema as the
// legacy support-template fields, but values now belong to a single node:
// plane: dx, dy, tz; Grid: tx, ty, dz.
const SUPPORT_MOVEMENT_FIELDS = ["dx", "dy", "dz", "tx", "ty", "tz"];
const ZERO_SUPPORT_MOVEMENTS = Object.freeze({ dx: 0, dy: 0, dz: 0, tx: 0, ty: 0, tz: 0 });

function supportTemplateRestrainsMovement(template, field, gridMode = isGridAnalysisMode()) {
    if (!template) return false;
    // The Grid solver defines its own effective built-in supports independently
    // of the plane-model checkbox fields stored in the common UI templates.
    if (gridMode && ["FIXED", "ROLLER", "HINGED"].includes(String(template.id))) {
        const gridBuiltIns = {
            FIXED: { tx: true, ty: true, dz: true },
            ROLLER: { tx: false, ty: false, dz: true },
            HINGED: { tx: false, ty: false, dz: true },
        };
        if (Object.prototype.hasOwnProperty.call(gridBuiltIns, template.id)) {
            return !!gridBuiltIns[template.id][field];
        }
    }
    const restraintKey = gridMode
        ? ({ tx: "RFx", ty: "RFy", dz: "RMz" })[field]
        : ({ dx: "RFx", dy: "RFy", tz: "RMz" })[field];
    // Components unused by the selected analysis mode are not boundary inputs.
    if (!restraintKey) return true;
    return !!template.restraints?.[restraintKey];
}

function getSupportAssignmentPrescribedDisplacements(assignment, template) {
    const input = assignment?.prescribedDisplacements;
    const hasAssignmentValues = input && typeof input === "object" && !Array.isArray(input) &&
        SUPPORT_MOVEMENT_FIELDS.some((field) => Object.prototype.hasOwnProperty.call(input, field));
    const source = hasAssignmentValues ? input : (template?.displacement || ZERO_SUPPORT_MOVEMENTS);
    const result = {};
    SUPPORT_MOVEMENT_FIELDS.forEach((field) => {
        const raw = source?.[field];
        if (raw === undefined || raw === null || raw === "") {
            result[field] = 0;
            return;
        }
        const value = Number(raw);
        result[field] = Number.isFinite(value) ? value : raw;
    });
    return result;
}

// Migrate old projects that stored settlement/rotation on a reusable support
// template. The old value is copied to each assignment using that template, and
// the template field is zeroed only after validation succeeds.
function normalizeLegacySupportMovementAssignments(source, { clearTemplateValues = false } = {}) {
    if (!source || typeof source !== "object" || Array.isArray(source)) return source;
    const templates = Array.isArray(source.supportTemplates) ? source.supportTemplates : [];
    const assignments = Array.isArray(source.supportAssignments) ? source.supportAssignments : [];
    const templateById = new Map(templates.map((template) => [String(template.id), template]));
    const gridMode = source.analysisMode === "grid";
    const normalizedAssignments = assignments.map((assignment) => {
        const template = templateById.get(String(assignment.supportId)) ||
            AppConfig.builtInSupportTemplates.find((item) => String(item.id) === String(assignment.supportId));
        const raw = assignment.prescribedDisplacements;
        const hasAssignmentValues = raw && typeof raw === "object" && !Array.isArray(raw) &&
            SUPPORT_MOVEMENT_FIELDS.some((field) => Object.prototype.hasOwnProperty.call(raw, field));
        const legacyValues = template?.displacement || ZERO_SUPPORT_MOVEMENTS;
        const movements = {};
        SUPPORT_MOVEMENT_FIELDS.forEach((field) => {
            const valueSource = hasAssignmentValues ? raw : legacyValues;
            const value = valueSource?.[field];
            if (value === undefined || value === null || value === "") {
                movements[field] = 0;
                return;
            }
            const number = Number(value);
            movements[field] = Number.isFinite(number) ? number : value;
        });

        // Old solver behavior ignored prescribed movements on unrestrained DOFs.
        // Preserve that behavior while migrating legacy template-owned values.
        if (!hasAssignmentValues) {
            SUPPORT_MOVEMENT_FIELDS.forEach((field) => {
```

```
                if (!supportTemplateRestrainsMovement(template, field, gridMode)) movements[field] = 0;
            });
        }
        return { ...assignment, prescribedDisplacements: movements };
    });

    const normalized = { ...source, supportAssignments: normalizedAssignments };
    if (clearTemplateValues && Array.isArray(source.supportTemplates)) {
        normalized.supportTemplates = templates.map((template) => ({
            ...template,
            displacement: { ...ZERO_SUPPORT_MOVEMENTS },
        }));
    }
    return normalized;
}

// ========================================================================
// 6. PERSISTENCE
// Save/load the input model to localStorage, export/import as JSON, clear all data.
// Only input data is persisted. Solver-derived analysis state is never saved.
// ========================================================================
function getPersistedInputModel() {
    const persisted = {};

    // Persist only true user-input fields. Analysis mutates element and
    // support-assignment objects with calculated matrices, loads and reactions;
    // copying those objects directly would make exported/saved files enormous.
    const INPUT_FIELD_KEYS = {
        nodes: ["id", "x", "y", "z"],
        elementTemplates: ["id", "memberType", "I", "E_kN_m2", "G_GPa", "J", "A", "alpha", "thermalDepth"],
        elements: ["id", "templateId", "startNode", "endNode"],
        supportTemplates: ["id", "description", "type", "builtIn", "restraints", "displacement"],
        supportAssignments: ["id", "supportId", "nodeId", "prescribedDisplacements"],
        loadCases: ["id", "type", "description", "memberSelfWeights"],
        loads: ["id", "loadCaseId", "type", "nodeId", "elementId", "axisSystem", "angle", "distance", "distanceEnd", "Fx", "Fy", "Fz", "FxEnd", "FyEnd", "FzEnd", "temperatureDelta",
"temperatureT1", "temperatureT2"],
        momentCases: ["id", "type", "description"],
        moments: ["id", "momentCaseId", "type", "nodeId", "elementId", "distance", "Mx", "My", "Mz"],
    };

    AppConfig.persistedModelKeys.forEach((key) => {
        const items = Array.isArray(model[key]) ? model[key] : [];
        const fields = INPUT_FIELD_KEYS[key];
        persisted[key] = fields
            ? items.map((item) => {
                const clean = {};
                fields.forEach((field) => {
                    if (Object.prototype.hasOwnProperty.call(item || {}, field)) clean[field] = item[field];
                });
                return clean;
              })
            : [];
    });

    // Persist the manual global DoF switches with the model so each saved
    // model keeps its own numbering configuration.
    persisted.dofSettings = { ...getGlobalDofNumberingConfig() };
    persisted.analysisMode = model.analysisMode === "grid" ? "grid" : "plane";

    return persisted;
}

function validatePersistedInputModel(source) {
    if (!source || typeof source !== "object" || Array.isArray(source)) {
        throw new Error("The file must contain a model JSON object.");
    }
    if (source.analysisMode !== undefined && !["plane", "grid"].includes(source.analysisMode)) {
        throw new Error("Model analysisMode must be either 'plane' or 'grid'.");
    }
    source = normalizeLegacySelfWeightModel(normalizeLegacySupportMovementAssignments(source));

    const requiredCollections = ["nodes", "elementTemplates", "elements"];
    requiredCollections.forEach((key) => {
        if (!Array.isArray(source[key])) {
            throw new Error(`The model is missing the required '${key}' array.`);
        }
    });

    const collections = {};
    AppConfig.persistedModelKeys.forEach((key) => {
        const list = source[key];
        if (list !== undefined && !Array.isArray(list)) {
            throw new Error(`Model field '${key}' must be an array.`);
        }
        collections[key] = list === undefined ? [] : list;
        collections[key].forEach((item, index) => {
            if (!item || typeof item !== "object" || Array.isArray(item)) {
```

```
            throw new Error(`'${key}[${index}]' must be an object.`);
        }
    });
});

const idSets = {};
Object.entries(collections).forEach(([key, list]) => {
    const seen = new Set();
    list.forEach((item, index) => {
        const rawId = item.id;
        if (typeof rawId !== "string" || rawId.trim() === "") {
            throw new Error(`'${key}[${index}]' must have a non-empty string ID.`);
        }
        if (rawId === "__proto__") {
            throw new Error(`'${key}[${index}]' uses the reserved ID "__proto__". Choose a different ID.`);
        }
        if (rawId !== rawId.trim()) {
            throw new Error(`'${key}[${index}]' ID must not contain leading or trailing whitespace.`);
        }
        const id = rawId;
        if (seen.has(id)) throw new Error(`Duplicate ID '${id}' in '${key}'.`);
        seen.add(id);
    });
    idSets[key] = new Set(list.map((item) => String(item.id)));
});

const requireReference = (value, targetSet, description) => {
    if (typeof value !== "string" || value === "" || !targetSet.has(value)) {
        throw new Error(`Invalid or missing ${description}: '${value ?? ""}'. References must match an existing string ID exactly.`);
    }
};
const requireFiniteField = (record, field, description, { min = -Infinity, exclusiveMin = -Infinity, optional = false } = {}) => {
    const value = record[field];
    if (optional && (value === undefined || value === null || value === "")) return;
    if (value === undefined || value === null || value === "" || !Number.isFinite(Number(value))) {
        throw new Error(`${description} must be a finite number.`);
    }
    const number = Number(value);
    if (number < min || number <= exclusiveMin) {
        const limit = Number.isFinite(exclusiveMin) ? `greater than ${exclusiveMin}` : `at least ${min}`;
        throw new Error(`${description} must be ${limit}.`);
    }
};

const nodeIds = idSets.nodes;
const templateIds = idSets.elementTemplates;
const elementIds = idSets.elements;
const loadCaseIds = idSets.loadCases;
const momentCaseIds = idSets.momentCases;
const supportIds = new Set([...idSets.supportTemplates, "FIXED", "ROLLER", "HINGED"]);

collections.loadCases.forEach((loadCase) => {
    if (!["Dead", "Live", "Self Weight"].includes(loadCase.type)) {
        throw new Error(`Load case '${loadCase.id}' has unsupported type '${loadCase.type}'.`);
    }
    const memberWeights = loadCase.memberSelfWeights;
    if (memberWeights === undefined) return;
    if (!memberWeights || typeof memberWeights !== "object" || Array.isArray(memberWeights)) {
        throw new Error(`Load case '${loadCase.id}' memberSelfWeights must be an object keyed by element ID.`);
    }
    Object.entries(memberWeights).forEach(([elementId, weight]) => {
        if (!elementIds.has(String(elementId))) {
            throw new Error(`Load case '${loadCase.id}' self-weight references unknown element '${elementId}'.`);
        }
        requireFiniteField({ weight }, "weight", `Load case '${loadCase.id}' self-weight for element '${elementId}' (kN/m)`, { min: 0 });
    });
});

const occupiedNodeCoordinates = new Map();
collections.nodes.forEach((node) => {
    requireFiniteField(node, "x", `Node '${node.id}' X coordinate`);
    requireFiniteField(node, "y", `Node '${node.id}' Y coordinate`);
    requireFiniteField(node, "z", `Node '${node.id}' Z coordinate`, { optional: true });
    const coordinateKey = JSON.stringify([Number(node.x), Number(node.y)]);
    const existingNodeId = occupiedNodeCoordinates.get(coordinateKey);
    if (existingNodeId !== undefined) {
        throw new Error(`Nodes '${existingNodeId}' and '${node.id}' share the same X/Y coordinates. Every node must have unique coordinates in the analysis plane.`);
    }
    occupiedNodeCoordinates.set(coordinateKey, node.id);
});

collections.elementTemplates.forEach((template) => {
    const grid = source.analysisMode === "grid";
    const materialE = template.E_kN_m2 !== undefined && template.E_kN_m2 !== null && template.E_kN_m2 !== ""
        ? template.E_kN_m2
        : (template.E_GPa !== undefined && template.E_GPa !== null && template.E_GPa !== ""
            ? template.E_GPa : template.E);
```

```
        requireFiniteField({ E: materialE }, "E", `Element template '${template.id}' E`, { exclusiveMin: 0 });
        requireFiniteField(template, "I", `Element template '${template.id}' I`, { min: 0 });
        if (grid) {
            const legacyG = template.G_GPa === undefined || template.G_GPa === null || template.G_GPa === ""
                ? template.G : template.G_GPa;
            requireFiniteField({ G: legacyG }, "G", `Grid element template '${template.id}' G`, { exclusiveMin: 0 });
            requireFiniteField(template, "J", `Grid element template '${template.id}' J`, { exclusiveMin: 0 });
            requireFiniteField(template, "I", `Grid element template '${template.id}' I`, { exclusiveMin: 0 });
        } else {
            requireFiniteField(template, "A", `Element template '${template.id}' A`, { min: 0 });
            if (Number(template.A) === 0 && Number(template.I) === 0) {
                throw new Error(`Element template '${template.id}' cannot have both A and I equal to zero.`);
            }
            const memberType = template.memberType === undefined || template.memberType === null || template.memberType === ""
                ? null : String(template.memberType).toLowerCase();
            if (memberType !== null) {
                if (!["beam", "truss", "frame"].includes(memberType)) {
                    throw new Error(`Element template '${template.id}' has unsupported member type '${template.memberType}'.`);
                }
                requireFiniteField(template, "A", `Element template '${template.id}' A`, { exclusiveMin: 0 });
                if (memberType === "truss") {
                    if (Number(template.I) !== 0) {
                        throw new Error(`Truss template '${template.id}' must have I = 0.`);
                    }
                } else {
                    requireFiniteField(template, "I", `Element template '${template.id}' I`, { exclusiveMin: 0 });
                }
            }
            requireFiniteField(template, "alpha", `Element template '${template.id}' thermal expansion coefficient alpha`, { min: 0, optional: true });
            requireFiniteField(template, "thermalDepth", `Element template '${template.id}' thermal section depth`, { min: 0, optional: true });
        }
    });

    collections.supportTemplates.forEach((support) => {
        const canonicalSupport = AppConfig.builtInSupportTemplates.find((item) => item.id === support.id);
        if (canonicalSupport && support.builtIn === false) {
            throw new Error(`Support ID '${support.id}' is reserved for a built-in support. Rename the custom support before importing.`);
        }
        if (!canonicalSupport && support.builtIn === true) {
            throw new Error(`Custom support '${support.id}' cannot be marked as built-in.`);
        }
        const restraints = support.restraints;
        if (!restraints || typeof restraints !== "object" || Array.isArray(restraints)) {
            throw new Error(`Support template '${support.id}' is missing its restraints object.`);
        }
        ["RFx", "RFy", "RFz", "RMx", "RMy", "RMz"].forEach((field) => {
            if (typeof restraints[field] !== "boolean") {
                throw new Error(`Support template '${support.id}' restraint '${field}' must be true or false.`);
            }
        });
        const displacement = support.displacement;
        if (!displacement || typeof displacement !== "object" || Array.isArray(displacement)) {
            throw new Error(`Support template '${support.id}' is missing its prescribed-displacement object.`);
        }
        ["dx", "dy", "dz", "tx", "ty", "tz"].forEach((field) => {
            requireFiniteField(displacement, field, `Support template '${support.id}' displacement '${field}'`);
        });
        if (canonicalSupport) {
            const restraintMismatch = ["RFx", "RFy", "RFz", "RMx", "RMy", "RMz"]
                .some((field) => restraints[field] !== canonicalSupport.restraints[field]);
            const displacementMismatch = ["dx", "dy", "dz", "tx", "ty", "tz"]
                .some((field) => Number(displacement[field]) !== Number(canonicalSupport.displacement[field]));
            if (restraintMismatch || displacementMismatch) {
                throw new Error(`Built-in support '${support.id}' does not match the application's standard definition. Restore its built-in restraints and prescribed displacements before
importing.`);
            }
        }
    });

    collections.elements.forEach((element) => {
        requireReference(element.templateId, templateIds, `element template reference on element '${element.id}'`);
        requireReference(element.startNode, nodeIds, `start-node reference on element '${element.id}'`);
        requireReference(element.endNode, nodeIds, `end-node reference on element '${element.id}'`);
        if (String(element.startNode) === String(element.endNode)) {
            throw new Error(`Element '${element.id}' cannot start and end at the same node.`);
        }
        const start = collections.nodes.find((node) => String(node.id) === String(element.startNode));
        const end = collections.nodes.find((node) => String(node.id) === String(element.endNode));
        const length = Math.hypot(Number(end.x) - Number(start.x), Number(end.y) - Number(start.y));
        if (!(length > 0) || !Number.isFinite(length)) {
            throw new Error(`Element '${element.id}' must have non-zero length in the XY plane.`);
        }
    });

    const gridModel = source.analysisMode === "grid";
    const assignedSupportNodes = new Set();
    collections.supportAssignments.forEach((assignment) => {
```

```
        requireReference(assignment.nodeId, nodeIds, `node reference on support assignment '${assignment.id}'`);
        requireReference(assignment.supportId, supportIds, `support-template reference on support assignment '${assignment.id}'`);
        const nodeKey = String(assignment.nodeId);
        if (assignedSupportNodes.has(nodeKey)) {
            throw new Error(`Node '${nodeKey}' has multiple support assignments. Combine the required restraints into one support template before importing.`);
        }
        assignedSupportNodes.add(nodeKey);

        if (assignment.prescribedDisplacements !== undefined) {
            const movements = assignment.prescribedDisplacements;
            if (!movements || typeof movements !== "object" || Array.isArray(movements)) {
                throw new Error(`Support assignment '${assignment.id}' prescribedDisplacements must be an object.`);
            }
            SUPPORT_MOVEMENT_FIELDS.forEach((field) => {
                requireFiniteField(movements, field, `Support assignment '${assignment.id}' movement '${field}'`, { optional: true });
            });
            const supportTemplate = collections.supportTemplates.find((item) => String(item.id) === String(assignment.supportId)) ||
                AppConfig.builtInSupportTemplates.find((item) => String(item.id) === String(assignment.supportId));
            const activeMovementFields = gridModel ? ["tx", "ty", "dz"] : ["dx", "dy", "tz"];
            activeMovementFields.forEach((field) => {
                const value = movements[field] === undefined || movements[field] === null || movements[field] === "" ? 0 : Number(movements[field]);
                if (Number.isFinite(value) && value !== 0 && !supportTemplateRestrainsMovement(supportTemplate, field, gridModel)) {
                    throw new Error(`Support assignment '${assignment.id}' prescribes ${field}=${value} at node '${nodeKey}', but support '${assignment.supportId}' does not restrain that
degree of freedom.`);
                }
            });
        }
    });
    const getElementLength = (elementId, description) => {
        const element = collections.elements.find((candidate) => String(candidate.id) === String(elementId));
        if (!element) throw new Error(`${description} references unknown element '${elementId}'.`);
        const start = collections.nodes.find((node) => String(node.id) === String(element.startNode));
        const end = collections.nodes.find((node) => String(node.id) === String(element.endNode));
        const length = Math.hypot(Number(end?.x) - Number(start?.x), Number(end?.y) - Number(start?.y));
        if (!(length > 0) || !Number.isFinite(length)) throw new Error(`${description} requires an element with positive length.`);
        return length;
    };

    collections.loads.forEach((load) => {
        const axisSystem = load.axisSystem;
        if (axisSystem !== undefined && axisSystem !== null && axisSystem !== "" &&
            !["Global", "Local", "Angled"].includes(axisSystem)) {
            throw new Error(`Load '${load.id}' has unsupported axis system '${axisSystem}'.`);
        }
        if (load.type === "Nodal Load" && axisSystem === "Local") {
            throw new Error(`Nodal load '${load.id}' cannot use the Local axis system.`);
        }
        requireReference(load.loadCaseId, loadCaseIds, `load-case reference on load '${load.id}'`);
        const temperatureLoad = ["Uniform Temperature Change", "Temperature Gradient"].includes(load.type);
        // "Element Load" is a legacy alias still handled by both solver paths.
        if (!["Nodal Load", "Element Point Load", "Element Load", "Element UDL", "Uniform Temperature Change", "Temperature Gradient"].includes(load.type)) {
            throw new Error(`Load '${load.id}' has unsupported type '${load.type}'.`);
        }
        if (temperatureLoad) {
            if (gridModel) throw new Error(`Temperature load '${load.id}' is only supported for Beam / Truss / Frame models.`);
            requireReference(load.elementId, elementIds, `element reference on temperature load '${load.id}'`);
            const targetElement = collections.elements.find((item) => item.id === load.elementId);
            const targetTemplate = collections.elementTemplates.find((item) => item.id === targetElement.templateId);
            requireFiniteField(targetTemplate, "alpha", `Temperature load '${load.id}' element template alpha`, { exclusiveMin: 0 });
            if (load.type === "Uniform Temperature Change") {
                requireFiniteField(load, "temperatureDelta", `Temperature load '${load.id}' temperature change ΔT`);
                requireFiniteField(targetTemplate, "A", `Temperature load '${load.id}' element area A`, { exclusiveMin: 0 });
            } else {
                requireFiniteField(load, "temperatureT1", `Temperature gradient load '${load.id}' T1`);
                requireFiniteField(load, "temperatureT2", `Temperature gradient load '${load.id}' T2`);
                requireFiniteField(targetTemplate, "I", `Temperature gradient load '${load.id}' element inertia I`, { exclusiveMin: 0 });
                requireFiniteField(targetTemplate, "thermalDepth", `Temperature gradient load '${load.id}' section depth d`, { exclusiveMin: 0 });
            }
            return;
        }
        if (load.type === "Nodal Load") {
            requireReference(load.nodeId, nodeIds, `node reference on load '${load.id}'`);
        } else {
            requireReference(load.elementId, elementIds, `element reference on load '${load.id}'`);
        }

        const startForceFields = gridModel ? ["Fz"] : ["Fx", "Fy"];
        startForceFields.forEach((field) => requireFiniteField(load, field, `Load '${load.id}' ${field}`));
        if (load.type === "Element Point Load" || load.type === "Element Load" || load.type === "Element UDL") {
            const length = getElementLength(load.elementId, `Load '${load.id}'`);
            requireFiniteField(load, "distance", `Load '${load.id}' distance`, { min: 0 });
            if (Number(load.distance) > length) {
                throw new Error(`Load '${load.id}' distance must not exceed element length ${length}.`);
            }
            if (load.type === "Element UDL") {
                const rawEndDistance = load.distanceEnd;
                const endDistance = rawEndDistance === undefined || rawEndDistance === null || rawEndDistance === ""
```

```
                    ? length : Number(rawEndDistance);
                requireFiniteField({ endDistance }, "endDistance", `Load '${load.id}' end distance`, { min: 0 });
                if (!(Number(load.distance) < endDistance) || endDistance > length) {
                    throw new Error(`Load '${load.id}' UDL interval must satisfy 0 ≤ start distance < end distance ≤ element length ${length}.`);
                }
                // Missing end intensities default to their start intensity, as in the UI/solver.
                const endForceFields = gridModel ? ["FzEnd"] : ["FxEnd", "FyEnd"];
                endForceFields.forEach((field) => requireFiniteField(load, field, `Load '${load.id}' ${field}`, { optional: true }));
            }
        }
        ["Fx", "Fy", "Fz", "FxEnd", "FyEnd", "FzEnd", "angle"].forEach((field) => {
            requireFiniteField(load, field, `Load '${load.id}' ${field}`, { optional: true });
        });
    });
    collections.momentCases.forEach((momentCase) => {
        if (!["Dead", "Live", "Self Weight"].includes(momentCase.type)) {
            throw new Error(`Moment/torsion case '${momentCase.id}' has unsupported type '${momentCase.type}'.`);
        }
    });

    collections.moments.forEach((moment) => {
        requireReference(moment.momentCaseId, momentCaseIds, `moment-case reference on moment '${moment.id}'`);
        if (!["Nodal Moment", "Element Moment"].includes(moment.type)) {
            throw new Error(`Moment '${moment.id}' has unsupported type '${moment.type}'.`);
        }
        if (moment.type === "Nodal Moment") {
            requireReference(moment.nodeId, nodeIds, `node reference on moment '${moment.id}'`);
        } else {
            requireReference(moment.elementId, elementIds, `element reference on moment '${moment.id}'`);
            const length = getElementLength(moment.elementId, `Moment '${moment.id}'`);
            requireFiniteField(moment, "distance", `Moment '${moment.id}' distance`, { min: 0 });
            if (Number(moment.distance) > length) {
                throw new Error(`Moment '${moment.id}' distance must not exceed element length ${length}.`);
            }
        }
        (gridModel ? ["Mx", "My"] : ["Mz"]).forEach((field) => {
            requireFiniteField(moment, field, `Moment '${moment.id}' ${field}`);
        });
    });

    return true;
}

function resetAllEditForms() {
    // Edit IDs belong to the model currently loaded in the workspace.
    const resetters = [
        () => typeof resetNodeForm === "function" && resetNodeForm(),
        () => typeof resetElementTemplateForm === "function" && resetElementTemplateForm(),
        () => typeof resetElementAssignmentForm === "function" && resetElementAssignmentForm(),
        () => typeof resetSupportForm === "function" && resetSupportForm(),
        () => typeof resetSupportAssignmentForm === "function" && resetSupportAssignmentForm(),
        () => typeof resetLoadCaseForm === "function" && resetLoadCaseForm(),
        () => typeof resetLoadForm === "function" && resetLoadForm(),
        () => typeof resetMomentCaseForm === "function" && resetMomentCaseForm(),
        () => typeof resetMomentForm === "function" && resetMomentForm(),
    ];
    resetters.forEach((reset) => {
        try { reset(); }
        catch (error) { console.warn("[Model switch] Could not reset an edit form.", error); }
    });
}

function applyPersistedInputModel(source) {
    // Migrate legacy template-owned settlements to assignments, validate the
    // complete input, and only then clear the old template movement values.
    const assignmentNormalized = normalizeLegacySupportMovementAssignments(normalizeLegacySelfWeightModel(source));
    validatePersistedInputModel(assignmentNormalized);
    const normalizedSource = normalizeLegacySupportMovementAssignments(assignmentNormalized, { clearTemplateValues: true });
    resetAllEditForms();
    AppConfig.persistedModelKeys.forEach((key) => {
        model[key] = Array.isArray(normalizedSource?.[key]) ? normalizedSource[key] : [];
    });
    model.analysisMode = normalizedSource?.analysisMode === "grid" ? "grid" : "plane";
    syncAnalysisModeUI();

    if (normalizedSource?.dofSettings && typeof normalizedSource.dofSettings === "object" &&
        typeof normalizedSource.dofSettings.skipAxial === "boolean" &&
        typeof normalizedSource.dofSettings.skipMoment === "boolean") {
        model.dofSettings = {
            skipAxial: normalizedSource.dofSettings.skipAxial,
            skipMoment: normalizedSource.dofSettings.skipMoment,
        };
    } else {
        delete model.dofSettings;
        ensureGlobalDofNumberingConfig();
    }
    // Never carry solver-derived state across a save/load/import boundary.
```

```
    delete model.nodeDisplacements;
    model.elements.forEach((el) => {
        if (!el) return;
        delete el.plotData;
        delete el.L;
        delete el.cx;
        delete el.cy;
        delete el.SM;
        delete el.RT;
        delete el.RTT;
        delete el.SMS;
        delete el.AML;
        delete el.localwx;
        delete el.localwy;
        delete el.localPointLoads;
        delete el.localUDLs;
        delete el.localPointMoments;
    });
    model.supportAssignments.forEach((sa) => {
        if (!sa) return;
        delete sa.rx;
        delete sa.ry;
        delete sa.rz;
    });
}

function getMissingSelfWeightElement(loadCase, elements = model.elements) {
    if (!loadCase || loadCase.type !== "Self Weight") return null;
    const weights = loadCase.memberSelfWeights && typeof loadCase.memberSelfWeights === "object"
        ? loadCase.memberSelfWeights : {};
    return elements.find((element) => {
        const id = String(element.id);
        if (!Object.prototype.hasOwnProperty.call(weights, id)) return true;
        const weight = weights[id];
        return weight === undefined || weight === null || weight === "" ||
            !Number.isFinite(Number(weight)) || Number(weight) < 0;
    }) || null;
}

function getModelName() {
    const input = document.getElementById("modelName");
    return (input?.value || "Structural Model").trim() || "Structural Model";
}

function saveModelNamePreference() {
    safeStorageSetItem("structuralModelName", getModelName());
}

function sanitizeFileName(name) {
    return String(name || "Structural Model")
        .trim()
        .replace(/[<>:"/\\|?*\x00-\x1F]/g, "")
        .replace(/[. ]+$/g, "")
        .replace(/\s+/g, " ") || "Structural Model";
}

function hasModelInputData(data = getPersistedInputModel()) {
    return AppConfig.persistedModelKeys.some((key) => {
        let items = Array.isArray(data?.[key]) ? data[key] : [];
        // Built-in support templates are always available in the UI; they do
        // not mean the user has created a model that needs saving.
        if (key === "supportTemplates") {
            items = items.filter((item) => item && !item.builtIn && !["FIXED", "ROLLER", "HINGED"].includes(String(item.id)));
        }
        return items.length > 0;
    });
}

function createModelLibraryId() {
    return `model-${Date.now()}-${Math.random().toString(36).slice(2, 8)}`;
}

function normalizeModelLibraryRecords(source) {
    const normalized = Object.create(null);
    const seenNames = new Set();
    let changed = false;

    Object.entries(source || {}).forEach(([storageId, record]) => {
        if (!record || typeof record !== "object" || Array.isArray(record)) {
            changed = true;
            return;
        }

        const id = String(storageId);
        const baseName = String(record.name ?? "").trim() || "Structural Model";
        let name = baseName;
        let suffix = 2;
```

```
        // Names are compared case-insensitively after trimming only the ends.
        // Internal spaces remain significant ("My Model" != "MyModel").
        while (seenNames.has(modelNameKey(name))) {
            name = `${baseName} (${suffix++})`;
        }
        seenNames.add(modelNameKey(name));

        if (record.name !== name || String(record.id || storageId) !== id) changed = true;
        normalized[id] = { ...record, id, name };
    });

    if (Object.keys(normalized).length !== Object.keys(source || {}).length) changed = true;
    return { library: normalized, changed };
}

function getModelLibrary() {
    lastModelLibraryReadFailed = false;
    let raw = null;
    try {
        raw = safeStorageGetItem("structuralModels");
        const saved = JSON.parse(raw || "{}");
        if (!saved || typeof saved !== "object" || Array.isArray(saved)) {
            lastModelLibraryReadFailed = raw !== null && raw !== "";
            return Object.create(null);
        }
        // Preserve rather than silently discard malformed records from otherwise valid JSON.
        if (Object.values(saved).some((record) => !record || typeof record !== "object" || Array.isArray(record))) {
            lastModelLibraryReadFailed = true;
            return Object.create(null);
        }

        const result = normalizeModelLibraryRecords(saved);
        let libraryChanged = result.changed;
        // Migrate older saved models to per-element weights on Self Weight cases
        // when reading. Records whose old value cannot be converted safely are kept
        // intact so the user can see the error when opening/exporting them.
        Object.values(result.library).forEach((record) => {
            if (!record || !record.data || typeof record.data !== "object") return;
            try {
                const normalizedData = normalizeLegacySelfWeightModel(record.data);
                if (JSON.stringify(normalizedData) !== JSON.stringify(record.data)) {
                    record.data = normalizedData;
                    libraryChanged = true;
                }
            } catch (error) {
                // Preserve the legacy record; opening/exporting it will surface
                // the explicit conversion error instead of dropping its load.
            }
        });
        if (libraryChanged) {
            safeStorageSetItem("structuralModels", JSON.stringify(result.library));

            // If an existing duplicate had the current model ID, keep its
            // displayed name synchronized with the automatically repaired name.
            const currentId = safeStorageGetItem("structuralCurrentModelId") || "";
            const oldName = saved[currentId]?.name;
            const repairedName = result.library[currentId]?.name;
            if (repairedName && String(oldName ?? "") !== repairedName) {
                safeStorageSetItem("structuralModelName", repairedName);
                const nameInput = document.getElementById("modelName");
                if (nameInput) nameInput.value = repairedName;
            }
        }
        return result.library;
    } catch (error) {
        lastModelLibraryReadFailed = true;
        return Object.create(null);
    }
}


function getModelLibraryReadFailureMessage() {
    return "The saved-model library could not be read from browser storage. The stored value has been left untouched to avoid overwriting potentially recoverable data. Export the current
workspace as JSON before continuing.";
}
function setModelLibrary(library) {
    const result = normalizeModelLibraryRecords(library);
    lastModelLibraryWriteSucceeded = safeStorageSetItem("structuralModels", JSON.stringify(result.library));
    if (!lastModelLibraryWriteSucceeded) return result.library;

    const activeId = String(currentModelLibraryId || safeStorageGetItem("structuralCurrentModelId") || "");
    const activeName = result.library[activeId]?.name;
    const requestedName = library?.[activeId]?.name;
    if (activeName && requestedName !== activeName) {
        safeStorageSetItem("structuralModelName", activeName);
        const nameInput = document.getElementById("modelName");
        if (nameInput) nameInput.value = activeName;
```

```
    }
    return result.library;
}

function formatModelSavedTime(value) {
    if (!value) return "—";
    const date = new Date(value);
    if (Number.isNaN(date.getTime())) return "—";
    return date.toLocaleString();
}

// ---- Duplicate-model protection ----------------------------------------
// A model counts as a duplicate of a saved one when it has the same name, or
// when its input data is identical to a saved model's data.
function modelNameKey(name) {
    // Ignore all whitespace and letter case when comparing names.
    // Displayed names retain their original internal spacing; whitespace is
    // removed only for duplicate detection (e.g. "Bridge Model" = "bridgemodel").
    return String(name ?? "").replace(/\s+/g, "").toLowerCase();
}

function getAvailableModelName(requestedName, library = getModelLibrary(), ignoreId = "") {
    const baseName = String(requestedName ?? "").trim() || "Structural Model";
    const used = new Set(Object.entries(library).filter(([id]) => String(id) !== String(ignoreId))
        .map(([, item]) => modelNameKey(item?.name)));
    if (!used.has(modelNameKey(baseName))) return baseName;

    let suffix = 2;
    let candidate = `${baseName} (${suffix})`;
    while (used.has(modelNameKey(candidate))) candidate = `${baseName} (${++suffix})`;
    return candidate;
}

function canonicalJSON(value) {
    if (Array.isArray(value)) return "[" + value.map(canonicalJSON).join(",") + "]";
    if (value && typeof value === "object") {
        return "{" + Object.keys(value).sort().map((k) => JSON.stringify(k) + ":" + canonicalJSON(value[k])).join(",") + "}";
    }
    return JSON.stringify(value === undefined ? null : value);
}

function modelDataFingerprint(data) {
    const norm = {};
    AppConfig.persistedModelKeys.forEach((key) => {
        let items = Array.isArray(data?.[key]) ? data[key] : [];
        if (key === "supportTemplates") {
            items = items.filter((t) => t && !t.builtIn && !["FIXED", "ROLLER", "HINGED"].includes(t.id));
        }
        norm[key] = items;
    });
    const hasInput = Object.values(norm).some((items) => items.length > 0);
    if (!hasInput) return "";
    const d = data?.dofSettings;
    norm.dofSettings = { skipAxial: !!(d && d.skipAxial), skipMoment: !!(d && d.skipMoment) };
    norm.analysisMode = data?.analysisMode === "grid" ? "grid" : "plane";
    return canonicalJSON(norm);
}

// Returns { type: "content" | "name", item } for the first saved model (other
// than ignoreId) that duplicates the given name/data, or null.
function findDuplicateModel(name, data, ignoreId, library = getModelLibrary()) {
    const key = modelNameKey(name);
    const fp = modelDataFingerprint(data);
    const others = Object.values(library).filter((m) => m && m.id !== ignoreId);
    if (fp) {
        const same = others.find((m) => modelDataFingerprint(m.data) === fp);
        if (same) return { type: "content", item: same };
    }
    const sameName = others.find((m) => modelNameKey(m.name) === key);
    if (sameName) return { type: "name", item: sameName };
    return null;
}

function duplicateModelMessage(dup) {
    return dup.type === "content"
        ? `Duplicate model: this model is identical to the saved model "${dup.item.name}". It was not saved again.`
        : `Duplicate model: a saved model named "${dup.item.name}" already exists. Rename this model before saving.`;
}

function saveCurrentModelToLibrary(silent = false) {
    lastModelSaveStorageFailed = false;
    const library = getModelLibrary();
    if (lastModelLibraryReadFailed) {
        lastModelSaveStorageFailed = true;
        alert(getModelLibraryReadFailureMessage());
        return false;
    }
```

```
    const data = getPersistedInputModel();
    let modelName = getModelName();

    const dup = findDuplicateModel(modelName, data, currentModelLibraryId, library);
    if (dup) {
        if (!silent) {
            alert(duplicateModelMessage(dup));
            return false;
        }
        // If this is a never-saved workspace, an identical saved model already
        // preserves its content. An existing current record, however, must still
        // be updated so autosave does not silently leave that record stale.
        if (dup.type === "content" && !currentModelLibraryId) return false;
        modelName = getAvailableModelName(modelName, library, currentModelLibraryId);
        // Silent autosaves may need a unique library name, but must not
        // rewrite the user's visible name field without an explicit action.
    }

    const previousCurrentId = currentModelLibraryId;
    const storageKeys = ["structuralModels", "structuralCurrentModelId", "structuralInputModel", "structuralModelName"];
    const previousStorage = Object.fromEntries(storageKeys.map((key) => [key, safeStorageGetItem(key)]));
    if (!currentModelLibraryId) currentModelLibraryId = createModelLibraryId();

    library[currentModelLibraryId] = {
        id: currentModelLibraryId,
        name: modelName,
        data,
        updatedAt: new Date().toISOString(),
    };

    // The named library is now the source of truth. Remove the legacy full-model
    // snapshot before writing so it cannot consume quota alongside the library.
    safeStorageRemoveItem("structuralInputModel");
    setModelLibrary(library);
    const persisted = lastModelLibraryWriteSucceeded &&
        safeStorageSetItem("structuralCurrentModelId", currentModelLibraryId) &&
        safeStorageSetItem("structuralModelName", modelName);

    if (!persisted) {
        lastModelSaveStorageFailed = true;
        storageKeys.forEach((key) => {
            const previous = previousStorage[key];
            if (previous === null) safeStorageRemoveItem(key);
            else safeStorageSetItem(key, previous);
        });
        currentModelLibraryId = previousCurrentId;
        if (typeof renderModelLibrary === "function") renderModelLibrary();
        alert("Model was NOT confirmed saved: browser storage rejected one or more writes (for example, storage may be full or blocked). Your current workspace remains open. Export the
model as JSON before leaving this page.");
        return false;
    }

    renderModelLibrary();
    if (!silent) alert(`Model "${modelName}" saved in browser storage.`);
    return true;
}

function saveModel() {
    // Save preferences only after the library accepts the name; a rejected
    // duplicate must not become the next startup's preferred model name.
    saveCurrentModelToLibrary(false);
}

function loadModel() {
    const savedName = safeStorageGetItem("structuralModelName");
    const nameInput = document.getElementById("modelName");
    if (nameInput && savedName) nameInput.value = savedName;

    // Preferred: restore the currently selected model from the named library.
    const library = getModelLibrary();
    if (lastModelLibraryReadFailed) {
        if (typeof renderModelLibrary === "function") renderModelLibrary();
        return;
    }
    const storedCurrentId = safeStorageGetItem("structuralCurrentModelId") || "";
    if (storedCurrentId && library[storedCurrentId]?.data) {
        try {
            validatePersistedInputModel(library[storedCurrentId].data);
            applyPersistedInputModel(library[storedCurrentId].data);
            currentModelLibraryId = storedCurrentId;
            if (nameInput) nameInput.value = library[storedCurrentId].name || nameInput.value || "Structural Model";
            ensureBuiltInSupportTemplates();
            markAnalysisInvalidated();
            renderModelLibrary();
        } catch (error) {
            // A stale/corrupt library record must not abort application startup.
            console.error("Unable to restore the current saved model:", error);
```

```
            currentModelLibraryId = "";
            safeStorageRemoveItem("structuralCurrentModelId");
            renderModelLibrary();
            alert(`The previously active saved model could not be loaded: ${error && error.message ? error.message : "invalid model data"}. It remains in your saved-model library for
recovery.`);
        }
        return;
    }

    // Backward compatibility: migrate the previous single-model storage into
    // the new named library the first time the updated file is opened.
    const saved = safeStorageGetItem("structuralInputModel");
    if (!saved) {
        renderModelLibrary();
        return;
    }
    try {
        const parsed = JSON.parse(saved);
        applyPersistedInputModel(parsed);
        ensureBuiltInSupportTemplates();
        markAnalysisInvalidated();

        if (!currentModelLibraryId) {
            currentModelLibraryId = createModelLibraryId();
            library[currentModelLibraryId] = {
                id: currentModelLibraryId,
                name: getModelName(),
                data: parsed,
                updatedAt: new Date().toISOString(),
            };
            setModelLibrary(library);
            safeStorageSetItem("structuralCurrentModelId", currentModelLibraryId);
        }
        renderModelLibrary();
    } catch (error) {
        console.error(error);
        renderModelLibrary();
    }
}

function resetAnalysisForModelChange() {
    activeLoadCaseId = "";
    activeMomentCaseId = "";
    analysisState.status = "not-run";
    const resultsContent = document.getElementById("resultsContent");
    if (resultsContent) resultsContent.style.display = "none";
    updateAnalysisStatusUI();
}

function openSavedModel(modelId) {
    const library = getModelLibrary();
    if (lastModelLibraryReadFailed) {
        alert(getModelLibraryReadFailureMessage());
        return;
    }
    const savedModel = library[modelId];
    if (!savedModel?.data) return;

    // Validate the target before changing the current model or saving over its
    // working state. Corrupt library entries must fail without side effects.
    try {
        validatePersistedInputModel(savedModel.data);
    } catch (error) {
        alert(`This saved model cannot be opened: ${error && error.message ? error.message : "invalid model data"}. Your current workspace has not been changed.`);
        return;
    }

    // The saved snapshot above is stale once the current workspace is saved.
    // If Open is clicked on the already-active model, save and stay on that live copy.
    if (modelId === currentModelLibraryId) {
        if (hasModelInputData()) saveCurrentModelToLibrary(true);
        renderModelLibrary();
        const dashboardBtn = document.querySelector('.nav-btn[data-section="dashboard"]');
        switchSection("dashboard", dashboardBtn);
        return;
    }

    // Protect the current work from accidental loss by saving it first when it
    // contains input data. This does not alter the active solver state.
    if (hasModelInputData()) {
        saveCurrentModelToLibrary(true);
        if (lastModelSaveStorageFailed) return;
    }

    currentModelLibraryId = modelId;
    safeStorageSetItem("structuralCurrentModelId", modelId);
    safeStorageSetItem("structuralModelName", savedModel.name || "Structural Model");
```

```
    const nameInput = document.getElementById("modelName");
    if (nameInput) nameInput.value = savedModel.name || "Structural Model";

    applyPersistedInputModel(savedModel.data);
    migrateUnitSystem();
    ensureBuiltInSupportTemplates();
    resetAnalysisForModelChange();
    syncAnalysisModeUI();
    refreshAll();
    renderModelLibrary();

    const dashboardBtn = document.querySelector('.nav-btn[data-section="dashboard"]');
    switchSection("dashboard", dashboardBtn);
}

function newModel() {
    const library = getModelLibrary();
    if (lastModelLibraryReadFailed) {
        alert(getModelLibraryReadFailureMessage());
        return;
    }
    const existingNames = new Set(Object.values(library).map((m) => modelNameKey(m && m.name)));
    if (hasModelInputData()) existingNames.add(modelNameKey(getModelName()));
    let defaultNumber = Object.keys(library).length + 1;
    while (existingNames.has(modelNameKey(`Model ${defaultNumber}`))) defaultNumber++;

    const modal = document.getElementById("newModelModal");
    const nameInput = document.getElementById("newModelNameInput");
    const validation = document.getElementById("newModelValidationMessage");
    if (!modal || !nameInput) {
        // Defensive fallback for an unexpected shell/rendering failure.
        const proposedName = prompt("Enter the new model name:", `Model ${defaultNumber}`);
        if (proposedName === null) return;
        const normalizedName = String(proposedName).trim() || `Model ${defaultNumber}`;
        const names = new Set(Object.values(getModelLibrary()).map((m) => modelNameKey(m && m.name)));
        if (hasModelInputData()) names.add(modelNameKey(getModelName()));
        if (names.has(modelNameKey(normalizedName))) {
            alert(`A model named "${normalizedName}" already exists or is about to be saved. Choose a different name.`);
            return;
        }
        createNewModelRecord(normalizedName, "plane");
        return;
    }

    nameInput.value = `Model ${defaultNumber}`;
    modal.dataset.defaultName = `Model ${defaultNumber}`;
    const btfOption = modal.querySelector('input[name="newModelAnalysisMode"][value="plane"]');
    if (btfOption) btfOption.checked = true;
    if (validation) { validation.textContent = ""; validation.hidden = true; }
    modal.hidden = false;
    modal.setAttribute("aria-hidden", "false");
    requestAnimationFrame(() => nameInput.focus());
}

function closeNewModelDialog() {
    const modal = document.getElementById("newModelModal");
    if (!modal) return;
    modal.hidden = true;
    modal.setAttribute("aria-hidden", "true");
}

function createNewModelFromDialog(event) {
    if (event) event.preventDefault();
    const modal = document.getElementById("newModelModal");
    const nameInput = document.getElementById("newModelNameInput");
    const validation = document.getElementById("newModelValidationMessage");
    if (!modal || !nameInput) return;

    const defaultName = modal.dataset.defaultName || "New Model";
    const modelName = nameInput.value.trim() || defaultName;
    const library = getModelLibrary();
    if (lastModelLibraryReadFailed) {
        if (validation) {
            validation.textContent = getModelLibraryReadFailureMessage();
            validation.hidden = false;
        } else {
            alert(getModelLibraryReadFailureMessage());
        }
        return;
    }
    const existingNames = new Set(Object.values(library).map((m) => modelNameKey(m && m.name)));
    if (hasModelInputData()) existingNames.add(modelNameKey(getModelName()));
    if (existingNames.has(modelNameKey(modelName))) {
        if (validation) {
            validation.textContent = `A saved model named "${modelName}" already exists. Choose a different name.`;
            validation.hidden = false;
        } else {
```

```
            alert(`Duplicate model: a saved model named "${modelName}" already exists. Choose a different name.`);
        }
        nameInput.focus();
        return;
    }

    const selectedMode = modal.querySelector('input[name="newModelAnalysisMode"]:checked')?.value === "grid" ? "grid" : "plane";
    closeNewModelDialog();
    createNewModelRecord(modelName, selectedMode);
}

function createNewModelRecord(modelName, analysisMode) {
    if (hasModelInputData()) {
        saveCurrentModelToLibrary(true);
        if (lastModelSaveStorageFailed) return;
    }
    resetAllEditForms();
    // Re-check after autosaving the previous model to guard every creation
    // path, including the prompt fallback and future callers.
    const library = getModelLibrary();
    if (lastModelLibraryReadFailed) {
        alert(getModelLibraryReadFailureMessage());
        return;
    }
    modelName = getAvailableModelName(modelName, library);

    AppConfig.persistedModelKeys.forEach((key) => {
        model[key] = [];
    });
    model.dofSettings = { skipAxial: false, skipMoment: false };
    // The solver type is chosen once at model creation and is never changed by
    // the in-workspace UI. Saved/imported models restore their own saved type.
    model.analysisMode = analysisMode === "grid" ? "grid" : "plane";
    syncAnalysisModeUI();
    ensureBuiltInSupportTemplates();
    currentModelLibraryId = "";
    safeStorageRemoveItem("structuralCurrentModelId");
    // This legacy snapshot may belong to the model that was open before this new workspace.
    safeStorageRemoveItem("structuralInputModel");
    safeStorageSetItem("structuralModelName", modelName);

    const nameInput = document.getElementById("modelName");
    if (nameInput) nameInput.value = modelName;

    resetAnalysisForModelChange();
    refreshAll();
    renderModelLibrary();
    const dashboardBtn = document.querySelector('.nav-btn[data-section="dashboard"]');
    switchSection("dashboard", dashboardBtn);
}
function deleteSavedModel(modelId) {
    const library = getModelLibrary();
    if (lastModelLibraryReadFailed) {
        alert(getModelLibraryReadFailureMessage());
        return;
    }
    const savedModel = library[modelId];
    if (!savedModel) return;

    if (!confirm(`Delete saved model "${savedModel.name}"? This cannot be undone.`)) return;

    const deletedCurrentModel = currentModelLibraryId === modelId;
    delete library[modelId];
    setModelLibrary(library);
    if (!lastModelLibraryWriteSucceeded) {
        alert("The model was not deleted because browser storage rejected the library update. No deletion is confirmed.");
        return;
    }

    if (deletedCurrentModel || Object.keys(library).length === 0) {
        // Deleting the active model (or the last remaining saved model) also
        // clears its live and persisted working copy to avoid a ghost model.
        resetActiveModelAfterSavedModelDeletion();
    }
    renderModelLibrary();
}

function downloadJSONFile(filename, payload) {
    const blob = new Blob([JSON.stringify(payload, null, 4)], { type: "application/json" });
    const url = URL.createObjectURL(blob);
    const a = document.createElement("a");
    a.href = url;
    a.download = filename;
    document.body.appendChild(a);
    a.click();
    a.remove();
    setTimeout(() => URL.revokeObjectURL(url), 0);
```

```
}

function exportModel() {
    saveModelNamePreference();
    const data = getPersistedInputModel();
    downloadJSONFile(`${sanitizeFileName(getModelName())}.json`, data);
}

function exportSavedModel(modelId) {
    const library = getModelLibrary();
    if (lastModelLibraryReadFailed) {
        alert(getModelLibraryReadFailureMessage());
        return;
    }
    const savedModel = library[modelId];
    if (!savedModel?.data) return;
    try {
        const data = normalizeLegacySelfWeightModel(savedModel.data);
        validatePersistedInputModel(data);
        downloadJSONFile(`${sanitizeFileName(savedModel.name)}.json`, data);
    } catch (error) {
        alert(error.message || "This saved model's self-weight data cannot be exported safely. Open and review the model first.");
    }
}

function exportAllModels() {
    const library = getModelLibrary();
    if (lastModelLibraryReadFailed) {
        alert(getModelLibraryReadFailureMessage());
        return;
    }
    let models = Object.values(library).sort((a, b) => String(a.name).localeCompare(String(b.name)));
    try {
        models = models.map((item) => ({ ...item, data: normalizeLegacySelfWeightModel(item.data) }));
        models.forEach((item) => validatePersistedInputModel(item.data));
    } catch (error) {
        alert(error.message || "At least one saved model has self-weight data that cannot be converted safely. Open and correct that model before exporting all models.");
        return;
    }

    // Always export the current in-memory version of the open model. This means
    // recent edits are included even when the model was saved earlier.
    if (hasModelInputData()) {
        const currentData = getPersistedInputModel();
        const currentName = getModelName();
        const currentIndex = currentModelLibraryId
            ? models.findIndex((item) => item.id === currentModelLibraryId)
            : -1;

        const currentSnapshot = {
            id: currentModelLibraryId || "current-unsaved",
            name: currentName,
            data: currentData,
            updatedAt: new Date().toISOString(),
        };

        if (currentIndex >= 0) {
            models[currentIndex] = currentSnapshot;
        } else {
            models.push(currentSnapshot);
        }
    }

    if (models.length === 0) {
        alert("No saved models available to export.");
        return;
    }

    const bundle = {
        type: "StructuralAnalysisModelCollection",
        version: 1,
        exportedAt: new Date().toISOString(),
        models: models.map((item) => ({
            id: item.id,
            name: item.name,
            updatedAt: item.updatedAt || null,
            data: item.data,
        })),
    };

    const defaultExportFileName =
        safeStorageGetItem("structuralExportAllFileName") || "Structural_Analysis_All_Models";
    const enteredExportFileName = prompt(
        "Enter export file name:",
        defaultExportFileName,
    );
    if (enteredExportFileName === null) return;
```

```
    const safeExportFileName = sanitizeFileName(enteredExportFileName) || defaultExportFileName;
    safeStorageSetItem("structuralExportAllFileName", safeExportFileName);
    downloadJSONFile(`${safeExportFileName}.json`, bundle);
}

function importModel(event) {
    const file = event.target.files[0];
    if (!file) return;

    const reader = new FileReader();
    reader.onload = function (e) {
        try {
            const imported = JSON.parse(e.target.result);

            // Collection import: merge all named models into the local library.
            if (imported?.type === "StructuralAnalysisModelCollection" && Array.isArray(imported.models)) {
                const library = getModelLibrary();
                if (lastModelLibraryReadFailed) {
                    alert(getModelLibraryReadFailureMessage());
                    return;
                }
                let importedCount = 0;
                let duplicateCount = 0;

                imported.models.forEach((item, index) => {
                    if (!item || typeof item !== "object" || !item.data || typeof item.data !== "object") {
                        throw new Error(`Model collection entry ${index + 1} is missing a valid model payload.`);
                    }
                    item.data = normalizeLegacySupportMovementAssignments(normalizeLegacySelfWeightModel(item.data));
                    validatePersistedInputModel(item.data);
                    item.data = normalizeLegacySupportMovementAssignments(item.data, { clearTemplateValues: true });
                    let importName = String(item.name || `Imported Model ${importedCount + 1}`);
                    const fpImport = modelDataFingerprint(item.data);
                    if (fpImport && Object.values(library).some((m) => modelDataFingerprint(m.data) === fpImport)) {
                        duplicateCount++;
                        return;
                    }
                    if (Object.values(library).some((m) => modelNameKey(m.name) === modelNameKey(importName))) {
                        let n = 2;
                        while (Object.values(library).some((m) => modelNameKey(m.name) === modelNameKey(`${importName} (${n})`))) n++;
                        importName = `${importName} (${n})`;
                    }
                    let id = String(item.id || createModelLibraryId());
                    if (library[id]) id = createModelLibraryId();
                    library[id] = {
                        id,
                        name: importName,
                        data: item.data,
                        updatedAt: item.updatedAt || new Date().toISOString(),
                    };
                    importedCount++;
                });

                setModelLibrary(library);
                if (!lastModelLibraryWriteSucceeded) {
                    alert("The models were not imported because browser storage rejected the library update. No import is confirmed.");
                    return;
                }
                renderModelLibrary();
                alert(`${importedCount} model(s) imported into the model library.` +
                    (duplicateCount ? ` ${duplicateCount} duplicate model(s) already saved on this device were skipped.` : ""));
            } else {
                // Validate before prompting or saving the current workspace. An
                // invalid import should have no persistent side effects.
                validatePersistedInputModel(imported);
                // Single-model JSON: load it as the current unsaved model.
                if (hasModelInputData()) {
                    const saveBeforeReplace = confirm(
                        "Importing this file will replace the current workspace model. Choose OK to save the current model first, or Cancel to replace it without saving.",
                    );
                    if (saveBeforeReplace) {
                        saveCurrentModelToLibrary(true);
                        if (lastModelSaveStorageFailed) return;
                    }
                }
                applyPersistedInputModel(imported);
                migrateUnitSystem();
                ensureBuiltInSupportTemplates();
                currentModelLibraryId = "";
                safeStorageRemoveItem("structuralCurrentModelId");
                // Avoid restoring the pre-import workspace as if it were the imported model after refresh.
                safeStorageRemoveItem("structuralInputModel");

                const importedBaseName = file.name.replace(/\.json$/i, "").trim();
                const libraryForName = getModelLibrary();
                const importedName = importedBaseName
                    ? (lastModelLibraryReadFailed ? importedBaseName : getAvailableModelName(importedBaseName, libraryForName))
```

```
                    : "";
                if (lastModelLibraryReadFailed) alert(getModelLibraryReadFailureMessage());
                const nameInput = document.getElementById("modelName");
                if (nameInput && importedName) {
                    nameInput.value = importedName;
                    saveModelNamePreference();
                }

                resetAnalysisForModelChange();
                refreshAll();
                renderModelLibrary();
                alert("Structural input model imported as the current unsaved model.");
            }
        } catch (error) {
            alert(`Invalid model file: ${error && error.message ? error.message : "Invalid JSON."}`);
        } finally {
            event.target.value = "";
        }
    };
    reader.readAsText(file);
}

function resetActiveModelAfterSavedModelDeletion() {
    resetAllEditForms();
    AppConfig.persistedModelKeys.forEach((key) => { model[key] = []; });
    model.dofSettings = { skipAxial: false, skipMoment: false };
    model.analysisMode = "plane";
    currentModelLibraryId = "";
    ["structuralCurrentModelId", "structuralInputModel", "structuralModelName"].forEach((key) => safeStorageRemoveItem(key));

    const nameInput = document.getElementById("modelName");
    if (nameInput) nameInput.value = "Structural Model";
    ensureBuiltInSupportTemplates();
    resetAnalysisForModelChange();
    syncAnalysisModeUI();
    const resultsContent = document.getElementById("resultsContent");
    if (resultsContent) resultsContent.style.display = "none";
    refreshAll();
}

function clearModel() {
    if (!confirm("Delete the entire current input model? Saved models in the Home library will not be deleted.")) return;

    resetAllEditForms();
    AppConfig.persistedModelKeys.forEach((key) => {
        model[key] = [];
    });
    model.dofSettings = { skipAxial: false, skipMoment: false };
    model.analysisMode = "plane";
    ensureBuiltInSupportTemplates();
    currentModelLibraryId = "";
    safeStorageRemoveItem("structuralCurrentModelId");
    safeStorageRemoveItem("structuralInputModel");
    safeStorageRemoveItem("structuralModelName");

    const nameInput = document.getElementById("modelName");
    if (nameInput) nameInput.value = "Structural Model";

    resetAnalysisForModelChange();
    renderModelLibrary();
    document.getElementById("resultsContent").style.display = "none";
    refreshAll();
}

// ============================================================================
// 7. NAVIGATION
// Sidebar section switching.
// ============================================================================
function updatePageTitle(id) {
    document.getElementById("pageTitle").textContent = AppConfig.sectionTitles[id][0];
}

function updatePageSubtitle(id) {
    document.getElementById("pageSubtitle").textContent = AppConfig.sectionTitles[id][1];
}

function switchSection(id, btnEl) {
    document.querySelectorAll(".nav-btn").forEach((b) => b.classList.remove("active"));
    document.querySelectorAll(".section").forEach((s) => s.classList.remove("active"));
    if (btnEl) btnEl.classList.add("active");
    const target = document.getElementById(id);
    if (target) target.classList.add("active");

    const appRoot = document.querySelector(".app");
    if (appRoot) appRoot.classList.toggle("home-mode", id === "home");

    const liveModelDiagramCard = document.getElementById("liveModelDiagramCard");
```

```
    if (liveModelDiagramCard) liveModelDiagramCard.style.display = id === "home" ? "none" : "block";

    const homeButton = document.getElementById("topbarHomeButton");
    if (homeButton) homeButton.style.display = id === "home" ? "none" : "inline-flex";

    updatePageTitle(id);
    updatePageSubtitle(id);
}

function goHome() {
    const homeBtn = document.querySelector('.nav-btn[data-section="home"]');
    switchSection("home", homeBtn);
}

document.querySelectorAll(".nav-btn").forEach((btn) => {
    btn.addEventListener("click", () => {
        switchSection(btn.dataset.section, btn);
    });
});

// ========================================================================
// 8. MODEL CRUD & TABLE RENDERING
// Each subsection: create-form submit handler, form reset, table render, (edit/delete).
// ========================================================================

// ---- Reorderable Input Tables ----
// Drag the grip beside any input ID to change that collection's sequence.
// Pointer events are used so the same control works with mouse and touch.
const sortDragState = {
    row: null,
    handle: null,
    tbody: null,
    collection: null,
    pointerId: null,
    startX: 0,
    startY: 0,
    holdTimer: null,
    active: false,
};

const SORT_HOLD_DELAY_MS = 450;
const SORT_MOVE_CANCEL_PX = 10;

function reorderVisibleCollection(collectionName, orderedEncodedIds) {
    const collection = model[collectionName];
    if (!Array.isArray(collection)) return false;

    const orderedIds = [];
    orderedEncodedIds.forEach((encodedId) => {
        try {
            const id = decodeURIComponent(encodedId);
            if (!orderedIds.includes(id)) orderedIds.push(id);
        } catch (error) {
            // Ignore malformed DOM data; the underlying model remains intact.
        }
    });

    const idSet = new Set(orderedIds);
    const reorderedVisibleItems = orderedIds
        .map((id) => collection.find((item) => String(item?.id) === String(id)))
        .filter(Boolean);

    if (!reorderedVisibleItems.length) return false;

    const visiblePositions = [];
    collection.forEach((item, index) => {
        if (idSet.has(String(item?.id))) visiblePositions.push(index);
    });

    if (visiblePositions.length !== reorderedVisibleItems.length) return false;

    visiblePositions.forEach((position, index) => {
        collection[position] = reorderedVisibleItems[index];
    });

    return true;
}

function activateSortDrag() {
    if (!sortDragState.row || !sortDragState.handle) return;
    if (!sortDragState.row.isConnected) return;

    sortDragState.active = true;
    sortDragState.handle.classList.remove("sort-holding");
    sortDragState.handle.classList.add("sort-ready");
    sortDragState.row.classList.add("sort-dragging");
    if (typeof navigator !== "undefined" && typeof navigator.vibrate === "function") {
```

```
        try { navigator.vibrate([28, 22, 38]); } catch (error) {}
    }
}

function finishSortDrag(commit = true) {
    const row = sortDragState.row;
    const tbody = sortDragState.tbody;
    const collection = sortDragState.collection;
    const wasActive = sortDragState.active;

    if (sortDragState.holdTimer) {
        clearTimeout(sortDragState.holdTimer);
        sortDragState.holdTimer = null;
    }

    if (commit && wasActive && tbody && collection) {
        const orderedIds = Array.from(tbody.querySelectorAll("tr[data-sort-collection]"))
            .filter((tr) => tr.dataset.sortCollection === collection)
            .map((tr) => tr.dataset.sortId);

        if (reorderVisibleCollection(collection, orderedIds)) {
            markAnalysisInvalidated();
            refreshAll();
        }
    }

    if (row) row.classList.remove("sort-dragging", "sort-drop-target");
    if (sortDragState.handle) sortDragState.handle.classList.remove("sort-holding", "sort-ready");

    sortDragState.row = null;
    sortDragState.handle = null;
    sortDragState.tbody = null;
    sortDragState.collection = null;
    sortDragState.pointerId = null;
    sortDragState.startX = 0;
    sortDragState.startY = 0;
    sortDragState.active = false;
}

function handleSortPointerDown(e) {
    const handle = e.target.closest(".sort-handle");
    if (!handle) return;
    if (e.button !== undefined && e.button !== 0) return;

    const row = handle.closest("tr[data-sort-collection][data-sort-id]");
    const tbody = row?.parentElement;
    if (!row || !tbody) return;

    e.preventDefault();

    finishSortDrag(false);

    sortDragState.row = row;
    sortDragState.handle = handle;
    sortDragState.tbody = tbody;
    sortDragState.collection = row.dataset.sortCollection;
    sortDragState.pointerId = e.pointerId;
    sortDragState.startX = e.clientX;
    sortDragState.startY = e.clientY;
    sortDragState.active = false;

    handle.classList.add("sort-holding");

    sortDragState.holdTimer = setTimeout(() => {
        activateSortDrag();
    }, SORT_HOLD_DELAY_MS);

    if (handle.setPointerCapture) {
        try { handle.setPointerCapture(e.pointerId); } catch (error) {}
    }
}

function handleSortPointerMove(e) {
    if (!sortDragState.row || e.pointerId !== sortDragState.pointerId) return;

    const dx = e.clientX - sortDragState.startX;
    const dy = e.clientY - sortDragState.startY;
    const movement = Math.hypot(dx, dy);

    if (!sortDragState.active) {
        if (movement >= SORT_MOVE_CANCEL_PX) {
            finishSortDrag(false);
        }
        return;
    }

    e.preventDefault();
```

```
    const elementAtPointer = document.elementFromPoint(e.clientX, e.clientY);
    const targetRow = elementAtPointer?.closest("tr[data-sort-collection][data-sort-id]");
    if (!targetRow || targetRow.parentElement !== sortDragState.tbody) return;
    if (targetRow.dataset.sortCollection !== sortDragState.collection) return;
    if (targetRow === sortDragState.row) return;

    const currentTarget = sortDragState.tbody.querySelector("tr.sort-drop-target");
    if (currentTarget && currentTarget !== targetRow) currentTarget.classList.remove("sort-drop-target");
    targetRow.classList.add("sort-drop-target");

    const rect = targetRow.getBoundingClientRect();
    const insertBefore = e.clientY < rect.top + rect.height / 2;
    if (insertBefore) {
        sortDragState.tbody.insertBefore(sortDragState.row, targetRow);
    } else {
        sortDragState.tbody.insertBefore(sortDragState.row, targetRow.nextSibling);
    }
}

document.addEventListener("pointerdown", handleSortPointerDown, { passive: false });
document.addEventListener("pointermove", handleSortPointerMove, { passive: false });
document.addEventListener("pointerup", () => finishSortDrag(true));
document.addEventListener("pointercancel", () => finishSortDrag(false));

function handleSortKeyboardReorder(event) {
    if (event.key !== "ArrowUp" && event.key !== "ArrowDown") return;
    const handle = event.target?.closest?.(".sort-handle");
    if (!handle) return;
    const row = handle.closest("tr[data-sort-collection][data-sort-id]");
    const tbody = row?.parentElement;
    if (!row || !tbody) return;

    const collection = row.dataset.sortCollection;
    const rows = Array.from(tbody.querySelectorAll("tr[data-sort-collection][data-sort-id]"))
        .filter((candidate) => candidate.dataset.sortCollection === collection);
    const index = rows.indexOf(row);
    const nextIndex = index + (event.key === "ArrowUp" ? -1 : 1);
    if (index < 0 || nextIndex < 0 || nextIndex >= rows.length) return;
    event.preventDefault();

    const targetRow = rows[nextIndex];
    if (event.key === "ArrowUp") tbody.insertBefore(row, targetRow);
    else tbody.insertBefore(row, targetRow.nextSibling);

    const orderedIds = Array.from(tbody.querySelectorAll("tr[data-sort-collection]"))
        .filter((candidate) => candidate.dataset.sortCollection === collection)
        .map((candidate) => candidate.dataset.sortId);
    const movedId = row.dataset.sortId;
    if (reorderVisibleCollection(collection, orderedIds)) {
        markAnalysisInvalidated();
        refreshAll();
        const movedRow = Array.from(document.querySelectorAll('tr[data-sort-collection][data-sort-id]'))
            .find((candidate) => candidate.dataset.sortCollection === collection && candidate.dataset.sortId === movedId);
        movedRow?.querySelector(".sort-handle")?.focus();
    }
}
document.addEventListener("keydown", handleSortKeyboardReorder);

function sortHandle(id) {
    const safeId = escapeHtml(id);
    return `<span class="sort-handle" role="group" tabindex="0" title="Drag to reorder; use Arrow Up or Arrow Down when focused" aria-keyshortcuts="ArrowUp ArrowDown" aria-label="Reorder ${
safeId}. Use Arrow Up or Arrow Down.">■■</span>`;
}

function setOptions(selectId, items, placeholder, valueKey = "id") {
    const select = document.getElementById(selectId);
    if (!select) return;
    const current = select.value;
    select.innerHTML =
        `<option value="">${placeholder}</option>` +
        items
            .map((item) => {
                const value = typeof item === "string" ? item : item[valueKey];
                return `<option value="${escapeHtml(value)}">${escapeHtml(value)}</option>`;
            })
            .join("");
    if (
        items.some((item) => {
            const value = typeof item === "string" ? item : item[valueKey];
            return String(value) === current;
        })
    )
        select.value = current;
}

function updateSelects() {
    setOptions("elementTemplateSelect", model.elementTemplates, "Select template");
```

```
    setOptions("elementStartNode", model.nodes, "Select node");
    setOptions("elementEndNode", model.nodes, "Select node");
    setOptions("supportTemplateSelect", model.supportTemplates, "Select support");
    setOptions("supportNodeSelect", model.nodes, "Select node");
    setOptions("loadCaseSelect", model.loadCases, "Select load case");
    setOptions("loadNode", model.nodes, "Select node");
    setOptions("loadElement", model.elements, "Select element");
    setOptions("momentCaseSelect", model.momentCases, "Select moment case");
    setOptions("momentNode", model.nodes, "Select node");
    setOptions("momentElement", model.elements, "Select element");
    syncActiveAnalysisCaseSelectors();
}

function getDeletionDependencies(collection, value, property) {
    const refValue = String(value);
    const deps = [];

    const addDependency = (label, items) => {
        if (items.length) deps.push(`${items.length} ${label}${items.length === 1 ? "" : "s"}`);
    };

    switch (collection) {
        case "nodes":
            addDependency("element", model.elements.filter((e) => String(e.startNode) === refValue || String(e.endNode) === refValue));
            addDependency("support assignment", model.supportAssignments.filter((s) => String(s.nodeId) === refValue));
            addDependency("load", model.loads.filter((l) => String(l.nodeId) === refValue));
            addDependency("moment", model.moments.filter((m) => String(m.nodeId) === refValue));
            break;
        case "elementTemplates":
            addDependency("element", model.elements.filter((e) => String(e.templateId) === refValue));
            break;
        case "elements":
            addDependency("load", model.loads.filter((l) => String(l.elementId) === refValue));
            addDependency("moment", model.moments.filter((m) => String(m.elementId) === refValue));
            break;
        case "supportTemplates":
            addDependency("support assignment", model.supportAssignments.filter((s) => String(s.supportId) === refValue));
            break;
        case "loadCases":
            addDependency("load", model.loads.filter((l) => String(l.loadCaseId) === refValue));
            break;
        case "momentCases":
            addDependency("moment", model.moments.filter((m) => String(m.momentCaseId) === refValue));
            break;
    }

    return deps;
}

function deleteItem(collection, encodedValue, property) {
    const value = decodeURIComponent(encodedValue);

    if (collection === "supportTemplates") {
        const support = model.supportTemplates.find((item) => String(item[property]) === String(value));
        if (support?.builtIn) return alert("Built-in support templates cannot be deleted.");
    }

    const dependencies = getDeletionDependencies(collection, value, property);
    if (dependencies.length) {
        const entityNames = {
            nodes: "Node",
            elementTemplates: "Element Template",
            elements: "Element",
            supportTemplates: "Support Template",
            supportAssignments: "Support Assignment",
            loadCases: "Load Case",
            loads: "Load",
            momentCases: "Moment Case",
            moments: "Moment",
        };
        return alert(`Cannot delete ${entityNames[collection] || collection} ${value} because it is referenced by ${dependencies.join(", ")}. Delete or reassign those dependent items first.
`);
    }

    if (!confirm(`Delete ${value}?`)) return;

    if (property === "id") {
        const editResetters = {
            nodes: () => { if (typeof editingNodeId !== "undefined" && String(editingNodeId) === String(value)) resetNodeForm(); },
            elementTemplates: () => { if (typeof editingElementTemplateId !== "undefined" && String(editingElementTemplateId) === String(value)) resetElementTemplateForm(); },
            elements: () => { if (typeof editingElementId !== "undefined" && String(editingElementId) === String(value)) resetElementAssignmentForm(); },
            supportTemplates: () => { if (typeof editingSupportTemplateId !== "undefined" && String(editingSupportTemplateId) === String(value)) resetSupportForm(); },
            supportAssignments: () => { if (typeof editingSupportAssignmentId !== "undefined" && String(editingSupportAssignmentId) === String(value)) resetSupportAssignmentForm(); },
            loadCases: () => { if (typeof editingLoadCaseId !== "undefined" && String(editingLoadCaseId) === String(value)) resetLoadCaseForm(); },
            loads: () => { if (typeof editingLoadId !== "undefined" && String(editingLoadId) === String(value)) resetLoadForm(); },
            momentCases: () => { if (typeof editingMomentCaseId !== "undefined" && String(editingMomentCaseId) === String(value)) resetMomentCaseForm(); },
            moments: () => { if (typeof editingMomentId !== "undefined" && String(editingMomentId) === String(value)) resetMomentForm(); },
```

```
        };
        if (editResetters[collection]) editResetters[collection]();
    }

    markAnalysisInvalidated();
    model[collection] = model[collection].filter(
        (item) => String(item[property]) !== String(value),
    );
    if (collection === "elements") {
        model.loadCases.forEach((loadCase) => {
            const weights = loadCase.memberSelfWeights;
            if (weights && typeof weights === "object" && !Array.isArray(weights)) delete weights[String(value)];
        });
    }
    refreshAll();
}

function refreshAll() {
    syncAnalysisModeUI();
    if (typeof syncLoadInputModeUI === "function") syncLoadInputModeUI();
    if (typeof syncMomentInputModeUI === "function") syncMomentInputModeUI();
    if (typeof syncMomentCaseInputModeUI === "function") syncMomentCaseInputModeUI();
    updateGlobalDofNumberingUI();
    renderModelLibrary();
    renderNodes();
    renderElementTemplates();
    renderElements();
    renderSupportTemplates();
    renderSupportAssignments();
    renderLoadCases();
    renderLoads();
    renderMomentCases();
    renderMoments();
    updateSelects();
    // updateSelects rebuilds the shared case dropdowns, so reapply mode labels
    // and placeholders afterward for Grid torsion terminology.
    if (typeof syncLoadInputModeUI === "function") syncLoadInputModeUI();
    if (typeof syncMomentInputModeUI === "function") syncMomentInputModeUI();
    if (typeof syncMomentCaseInputModeUI === "function") syncMomentCaseInputModeUI();
    updateStats();
    scheduleInlineModelDiagramRedraw();
}

/* ============================================================================
 * APPLICATION BOOTSTRAP
 * Runs once every module script has loaded and registered its markup + logic
 * (index.html loads main.js first and the page / results modules afterwards).
 * ============================================================================ */
document.addEventListener("DOMContentLoaded", function initApplication() {
    // Establish the Home view before restoring saved data. Startup operations
    // can touch old or malformed browser data; they must never leave the shell
    // in dashboard styling while the Home section is active.
    const homeNavBtn = document.querySelector('.nav-btn[data-section="home"]');
    const forceHomeView = () => {
        document.querySelectorAll(".section").forEach((section) => {
            section.classList.toggle("active", section.id === "home");
        });
        document.querySelectorAll(".nav-btn").forEach((button) => {
            button.classList.toggle("active", button === homeNavBtn);
        });
        const appRoot = document.querySelector(".app");
        if (appRoot) appRoot.classList.add("home-mode");
        const liveModelDiagramCard = document.getElementById("liveModelDiagramCard");
        if (liveModelDiagramCard) liveModelDiagramCard.style.display = "none";
        const homeButton = document.getElementById("topbarHomeButton");
        if (homeButton) homeButton.style.display = "none";
        updatePageTitle("home");
        updatePageSubtitle("home");
    };

    forceHomeView();
    try {
        loadModel();
        migrateUnitSystem();
        ensureBuiltInSupportTemplates();
        analysisState.status = "not-run";
        updateAnalysisStatusUI();
        syncAnalysisModeUI();
        toggleLoadTarget();
        toggleMomentTarget();
        refreshAll();
    } catch (error) {
        console.error("[startup] Initialization failed; keeping the Home screen visible.", error);
    } finally {
        // Reassert Home after initialization so no intermediate UI update or
        // partial failure can determine which screen users see on page open.
        forceHomeView();
```

```
        try {
            renderModelLibrary();
        } catch (error) {
            console.error("[startup] Could not refresh the saved-model list.", error);
        }
    }
});


/* ---- Supplied Grid calculation engine (Steps 1–3, embedded in an existing file) ---- */
/*
 * Grid Structure Analysis — Steps 1–3 only.
 * No CSS, UI, DOM rendering, persistence, diagrams, or event handlers.
 *
 * API:
 *   const result = GridSolver.calculateSteps1To3(model, {
 *     loadCaseId: "LC1",
 *     momentCaseId: "MC1"
 *   });
 *
 * This file intentionally stops after member-property and member-stiffness
 * calculations. It does not calculate global equilibrium, displacements,
 * support reactions, or final member end actions.
 */
(function (global) {
  "use strict";

  const BUILT_IN_SUPPORT_TEMPLATES = [
    {
      id: "FIXED", description: "Fixed Grid Support", type: "fixed", builtIn: true,
      restraints: { RFx: true, RFy: true, RFz: false, RMx: false, RMy: false, RMz: true },
      displacement: { dx: 0, dy: 0, dz: 0, tx: 0, ty: 0, tz: 0 }
    },
    {
      id: "ROLLER", description: "Grid Vertical Support", type: "roller", builtIn: true,
      restraints: { RFx: false, RFy: false, RFz: false, RMx: false, RMy: false, RMz: true },
      displacement: { dx: 0, dy: 0, dz: 0, tx: 0, ty: 0, tz: 0 }
    },
    {
      id: "HINGED", description: "Grid Hinge (RFz only)", type: "hinged", builtIn: true,
      restraints: { RFx: false, RFy: false, RFz: false, RMx: false, RMy: false, RMz: true },
      displacement: { dx: 0, dy: 0, dz: 0, tx: 0, ty: 0, tz: 0 }
    }
  ];

  const EMPTY_MODEL = () => ({
    nodes: [], elementTemplates: [], elements: [], supportTemplates: [],
    supportAssignments: [], loadCases: [], loads: [], momentCases: [], moments: []
  });

  const Matrix = Object.freeze({
    zeros: (rows, cols) => Array.from({ length: rows }, () => new Array(cols).fill(0)),
    mul: (A, B) => {
      if (!A.length || !B.length || !B[0].length) return [];
      if (A[0].length !== B.length) throw new Error("Matrix dimensions do not match for multiplication.");
      return A.map(row => B[0].map((_, col) =>
        row.reduce((sum, value, k) => sum + value * B[k][col], 0)
      ));
    },
    transpose: A => A.length ? A[0].map((_, col) => A.map(row => row[col])) : []
  });

  function clone(value) {
    return JSON.parse(JSON.stringify(value));
  }

  function normalizeModel(input) {
    const source = input && input.model ? input.model : input;
    if (!source || typeof source !== "object" || Array.isArray(source)) {
      throw new TypeError("GridSolver.calculateSteps1To3 expects a model object.");
    }

    const normalized = EMPTY_MODEL();
    for (const key of Object.keys(normalized)) {
      if (source[key] !== undefined) {
        if (!Array.isArray(source[key])) throw new TypeError(`Model field '${key}' must be an array.`);
        normalized[key] = clone(source[key]);
      }
    }

    // Normalize legacy element modulus fields into the canonical solver unit (kN/m²).
    normalized.elementTemplates = normalized.elementTemplates.map(template => {
      const item = { ...template };
      const hasValue = value => value !== undefined && value !== null && value !== "" && Number.isFinite(Number(value));
      if (!hasValue(item.E_kN_m2)) {
        if (hasValue(item.E_GPa)) item.E_kN_m2 = Number(item.E_GPa) * 1e6;
        else if (hasValue(item.E)) item.E_kN_m2 = Number(item.E) / 1000;
```

```
    }
    delete item.E_GPa;
    delete item.E;
    if (!hasValue(item.G_GPa) && hasValue(item.G)) {
      item.G_GPa = Number(item.G) / 1e9;
      delete item.G;
    }
    // J is an independent torsional property. Do not fabricate it from I:
    // missing/invalid J is rejected by the member validation below.
    return item;
  });

  const customTemplates = normalized.supportTemplates.filter(support =>
    !support.builtIn && !["FIXED", "ROLLER", "HINGED"].includes(String(support.id))
  );
  normalized.supportTemplates = [
    ...BUILT_IN_SUPPORT_TEMPLATES.map(clone),
    ...customTemplates.map(support => ({ ...support, type: support.type || "custom", builtIn: false }))
  ];
  return normalized;
}

function getElementEInternal(template) {
  if (!template) return 0;
  const hasValue = value => value !== undefined && value !== null && value !== "" && Number.isFinite(Number(value));
  if (hasValue(template.E_kN_m2)) return Number(template.E_kN_m2);
  if (hasValue(template.E_GPa)) return Number(template.E_GPa) * 1e6;
  if (hasValue(template.E)) return Number(template.E) / 1000;
  return 0;
}

function getElementGInternal(template) {
  if (!template) return 0;
  const hasValue = value => value !== undefined && value !== null && value !== "" && Number.isFinite(Number(value));
  if (hasValue(template.G_GPa)) return Number(template.G_GPa) * 1e6;
  if (hasValue(template.G)) return Number(template.G) / 1000;
  return 0;
}

function gridElementMatrices(n1, n2, template) {
  if (!n1 || !n2 || !template) return null;
  const dx = Number(n2.x) - Number(n1.x);
  const dy = Number(n2.y) - Number(n1.y);
  const L = Math.hypot(dx, dy);
  if (!(L > 0)) return null;

  const cx = dx / L;
  const cy = dy / L;
  const E = getElementEInternal(template);
  const G = getElementGInternal(template);
  const I = Number(template.I);
  const J = Number(template.J);
  if (!(E > 0 && G > 0 && I > 0 && J > 0)) return null;

  const a = 12 * E * I / Math.pow(L, 3);
  const b = 6 * E * I / Math.pow(L, 2);
  const c = 4 * E * I / L;
  const m = G * J / L;
  const p = m * cx * cx + c * cy * cy;
  const q = (m - c) * cx * cy;
  const ss = m * cy * cy + c * cx * cx;
  const r = b * cy;
  const t = -b * cx;
  const p1 = -m * cx * cx + 0.5 * c * cy * cy;
  const q1 = (-m - 0.5 * c) * cx * cy;
  const r1 = -b * cy;
  const s1 = -m * cy * cy + 0.5 * c * cx * cx;
  const t1 = b * cx;

  const SM = [
    [m, 0, 0, -m, 0, 0],
    [0, c, -b, 0, c / 2, b],
    [0, -b, a, 0, -b, -a],
    [-m, 0, 0, m, 0, 0],
    [0, c / 2, -b, 0, c, b],
    [0, b, -a, 0, b, a]
  ];
  const RT = [
    [cx, cy, 0, 0, 0, 0],
    [-cy, cx, 0, 0, 0, 0],
    [0, 0, 1, 0, 0, 0],
    [0, 0, 0, cx, cy, 0],
    [0, 0, 0, -cy, cx, 0],
    [0, 0, 0, 0, 0, 1]
  ];
  const RTT = Matrix.transpose(RT);
  const SMS = Matrix.mul(Matrix.mul(RTT, SM), RT);
```

```
  return {
    L, cx, cy, E, G, I, J, A: Number(template.A) || 0,
    a, b, c, m, p, q, s: ss, r, t, p1, q1, r1, s1, t1,
    SM, RT, RTT, SMS
  };
}

function gridPointLoadVector(P, a, L) {
  const x = Math.max(0, Math.min(Number(a) || 0, L));
  const xi = x / L;
  const N1 = 1 - 3 * xi * xi + 2 * xi * xi * xi;
  const N2 = L * (xi - 2 * xi * xi + xi * xi * xi);
  const N3 = 3 * xi * xi - 2 * xi * xi * xi;
  const N4 = L * (-xi * xi + xi * xi * xi);
  const load = Number(P) || 0;
  return [0, -load * N2, load * N1, 0, -load * N4, load * N3];
}

function gridPointTorsionFEM(T, a, L) {
  const x = Math.max(0, Math.min(Number(a) || 0, L));
  const load = Number(T) || 0;
  return [-load * (L - x) / L, 0, 0, -load * x / L, 0, 0];
}

function gridPointMomentVector(M, a, L) {
  const x = Math.max(0, Math.min(Number(a) || 0, L));
  const xi = x / L;
  const dN1 = (-6 * xi + 6 * xi * xi) / L;
  const dN2 = 1 - 4 * xi + 3 * xi * xi;
  const dN3 = (6 * xi - 6 * xi * xi) / L;
  const dN4 = -2 * xi + 3 * xi * xi;
  return [0, -M * dN2, M * dN1, 0, -M * dN4, M * dN3];
}

function gridUDLLoadVector(q1, q2, a, b, L) {
  const lo = Math.max(0, Math.min(Number(a) || 0, L));
  const hi = Math.max(0, Math.min(Number(b) || 0, L));
  const out = Matrix.zeros(6, 1);
  if (!(hi > lo)) return out;
  const span = hi - lo;
  const gp = [
    [-0.9815606342467191, 0.0471753363865118], [-0.9041172563704749, 0.1069393259953184],
    [-0.7699026741943047, 0.1600783285433464], [-0.5873179542866175, 0.2031674267230659],
    [-0.3678314989981802, 0.2334925365383548], [-0.1252334085114689, 0.2491470458134029],
    [0.1252334085114689, 0.2491470458134029], [0.3678314989981802, 0.2334925365383548],
    [0.5873179542866175, 0.2031674267230659], [0.7699026741943047, 0.1600783285433464],
    [0.9041172563704749, 0.1069393259953184], [0.9815606342467191, 0.0471753363865118]
  ];
  // 12-point Gauss-Legendre integration for a linearly varying UDL.
  for (const [z, w] of gp) {
    const x = lo + 0.5 * (z + 1) * span;
    const wt = 0.5 * span * w;
    const xi = x / L;
    const N1 = 1 - 3 * xi * xi + 2 * xi * xi * xi;
    const N2 = L * (xi - 2 * xi * xi + xi * xi * xi);
    const N3 = 3 * xi * xi - 2 * xi * xi * xi;
    const N4 = L * (-xi * xi + xi * xi * xi);
    const qx = Number(q1) + (Number(q2) - Number(q1)) * ((x - lo) / span);
    const v = [0, -N2 * qx, N1 * qx, 0, -N4 * qx, N3 * qx];
    for (let i = 0; i < 6; i++) out[i][0] += wt * v[i];
  }
  return out;
}

function gridLocalMomentComponents(Mx, My, cx, cy) {
  return { Mx: cx * Mx + cy * My, My: -cy * Mx + cx * My };
}

function activeCaseId(options, field, list) {
  const requested = options[field];
  return String(requested ?? (list[0] && list[0].id) ?? "");
}

function calculateSteps1To3(inputModel, options = {}) {
  options = options || {};
  const model = normalizeModel(inputModel);
  if (!model.nodes.length) throw new Error("The model must contain at least one node.");
  if (!model.elements.length) throw new Error("The model must contain at least one element.");

  const nodeIds = new Set();
  model.nodes.forEach(node => {
    const id = String(node.id ?? "");
    if (!id) throw new Error("Every node must have an ID.");
    if (nodeIds.has(id)) throw new Error(`Duplicate node ID '${id}'.`);
    nodeIds.add(id);
  });
  const activeLoadCaseId = activeCaseId(options, "loadCaseId", model.loadCases);
```

```
const activeMomentCaseId = activeCaseId(options, "momentCaseId", model.momentCases);
if (model.loads.length && !activeLoadCaseId) throw new Error("Select a load case before calculating member loads.");
if (model.moments.length && !activeMomentCaseId) throw new Error("Select a moment case before calculating member moments.");
if (activeLoadCaseId && !model.loadCases.some(item => String(item.id) === activeLoadCaseId)) {
  throw new Error(`Load case '${activeLoadCaseId}' does not exist in the model.`);
}
if (activeMomentCaseId && !model.momentCases.some(item => String(item.id) === activeMomentCaseId)) {
  throw new Error(`Moment case '${activeMomentCaseId}' does not exist in the model.`);
}

const isActiveLoadCase = load => !!activeLoadCaseId && String(load.loadCaseId || "") === activeLoadCaseId;
const isActiveMomentCase = moment => !!activeMomentCaseId && String(moment.momentCaseId || "") === activeMomentCaseId;

// STEP 1 — Degree-of-freedom numbering and restraints.
const dofMap = {};
model.nodes.forEach((node, index) => {
  dofMap[node.id] = { x: 3 * index, y: 3 * index + 1, z: 3 * index + 2, num: index };
});
const totalDof = model.nodes.length * 3;
const isRestrained = new Array(totalDof).fill(false);
const prescribedDisp = new Array(totalDof).fill(0);

const gridSupportRestraints = assignment => {
  const template = model.supportTemplates.find(support => support.id === assignment.supportId);
  if (!template) return { rx: false, ry: false, rz: false, dx: 0, dy: 0, dz: 0 };
  const movement = assignment.prescribedDisplacements &&
    typeof assignment.prescribedDisplacements === "object" && !Array.isArray(assignment.prescribedDisplacements)
      ? assignment.prescribedDisplacements
      : (template.displacement || {});
  return {
    rx: !!(template.restraints && template.restraints.RFx),
    ry: !!(template.restraints && template.restraints.RFy),
    rz: !!(template.restraints && template.restraints.RMz),
    dx: Number(movement.tx) || 0,
    dy: Number(movement.ty) || 0,
    dz: Number(movement.dz) || 0
  };
};

model.supportAssignments.forEach(assignment => {
  const dof = dofMap[assignment.nodeId];
  if (!dof) return;
  const restraint = gridSupportRestraints(assignment);
  if (restraint.rx) { isRestrained[dof.x] = true; prescribedDisp[dof.x] = restraint.dx; }
  if (restraint.ry) { isRestrained[dof.y] = true; prescribedDisp[dof.y] = restraint.dy; }
  if (restraint.rz) { isRestrained[dof.z] = true; prescribedDisp[dof.z] = restraint.dz; }
});

model.elements.forEach(element => {
  const n1 = model.nodes.find(node => node.id === element.startNode);
  const n2 = model.nodes.find(node => node.id === element.endNode);
  const template = model.elementTemplates.find(item => item.id === element.templateId);
  if (!n1 || !n2 || !template) throw new Error(`Element '${element.id}' has an invalid node or template reference.`);
  if (!gridElementMatrices(n1, n2, template)) {
    throw new Error(`Element '${element.id}' requires positive I, J, E and G values and a non-zero XY length.`);
  }
});

const step1Nodes = model.nodes.map((node, index) => {
  const dof = dofMap[node.id];
  return {
    nodeId: node.id,
    nodeNumber: index + 1,
    dof: {
      thetaX: { number: dof.x + 1, restrained: isRestrained[dof.x], prescribed: prescribedDisp[dof.x] },
      thetaY: { number: dof.y + 1, restrained: isRestrained[dof.y], prescribed: prescribedDisp[dof.y] },
      deltaZ: { number: dof.z + 1, restrained: isRestrained[dof.z], prescribed: prescribedDisp[dof.z] }
    }
  };
});
const step1 = { title: "Grid Joint DoF Numbering", dofMap, numNodes: model.nodes.length, totalDof, nodes: step1Nodes };

// STEP 2 — Joint coordinates and member geometry/material properties.
const step2Joints = model.nodes.map((node, index) => {
  const dof = dofMap[node.id];
  return {
    jointNumber: index + 1,
    nodeId: node.id,
    x: Number(node.x) || 0,
    y: Number(node.y) || 0,
    z: Number(node.z) || 0,
    dof: {
      thetaX: dof.x + 1, thetaY: dof.y + 1, deltaZ: dof.z + 1
    },
    restrained: {
      thetaX: isRestrained[dof.x], thetaY: isRestrained[dof.y], deltaZ: isRestrained[dof.z]
    }
```

```
  };
});
const step2Members = model.elements.map((element, index) => {
  const n1 = model.nodes.find(node => node.id === element.startNode);
  const n2 = model.nodes.find(node => node.id === element.endNode);
  const template = model.elementTemplates.find(item => item.id === element.templateId);
  const p = gridElementMatrices(n1, n2, template);
  return {
    memberNumber: index + 1,
    elementId: element.id,
    templateId: element.templateId,
    startNode: element.startNode,
    endNode: element.endNode,
    I: p.I, J: p.J, E_kN_m2: p.E, G_GPa: p.G / 1e6, A: p.A,
    length: p.L, cx: p.cx, cy: p.cy,
    m: p.m, a: p.a, b: p.b, c: p.c,
    p: p.p, q: p.q, r: p.r, s: p.s, t: p.t,
    p1: p.p1, q1: p.q1, r1: p.r1, s1: p.s1, t1: p.t1
  };
});
const step2 = { title: "Joint and Member Properties", joints: step2Joints, members: step2Members };

// STEP 3 — Local member stiffness matrices, transformed stiffness matrices,
// and equivalent fixed-end actions for member loads in the selected cases.
const step3Members = model.elements.map(element => {
  const n1 = model.nodes.find(node => node.id === element.startNode);
  const n2 = model.nodes.find(node => node.id === element.endNode);
  const template = model.elementTemplates.find(item => item.id === element.templateId);
  const p = gridElementMatrices(n1, n2, template);
  const AML = Matrix.zeros(6, 1);
  const localPointLoads = [];
  const localUDLs = [];
  const localPointMoments = [];
  const activeLC = model.loadCases.find(item => String(item.id) === activeLoadCaseId);

  if (Number(activeLC?.memberSelfWeights?.[String(element.id)]) > 0 && activeLC?.type === "Self Weight") {
    // Self-weight is a per-element downward line load stored on the case.
    const q = -Number(activeLC.memberSelfWeights[String(element.id)]);
    const fixedEnd = gridUDLLoadVector(q, q, 0, p.L, p.L);
    for (let i = 0; i < 6; i++) AML[i][0] -= fixedEnd[i][0];

    // Keep the same load in the member-load data consumed by Step 8 and
    // the internal-action diagrams. AML already contains the fixed-end
    // action above; this record is for plotting/reporting only and must
    // not be assembled into AML a second time.
    localUDLs.push({ startDistance: 0, endDistance: p.L, q1: q, q2: q });
  }

  model.loads.forEach(load => {
    if (!isActiveLoadCase(load) || load.elementId !== element.id) return;
    if (load.type === "Element Point Load" || load.type === "Element Load") {
      const a = Math.max(0, Math.min(Number(load.distance) || 0, p.L));
      const force = Number(load.Fz) || 0;
      const fixedEnd = gridPointLoadVector(force, a, p.L);
      for (let i = 0; i < 6; i++) AML[i][0] -= fixedEnd[i];
      localPointLoads.push({ distance: a, Fz: force });
    } else if (load.type === "Element UDL") {
      const a = Math.max(0, Math.min(Number(load.distance) || 0, p.L));
      const rawEnd = load.distanceEnd === "" || load.distanceEnd == null ? p.L : Number(load.distanceEnd);
      const b = Math.max(0, Math.min(rawEnd, p.L));
      if (!(b > a)) return;
      const q1 = Number(load.Fz) || 0;
      const q2 = load.FzEnd !== undefined && load.FzEnd !== "" ? (Number(load.FzEnd) || 0) : q1;
      const fixedEnd = gridUDLLoadVector(q1, q2, a, b, p.L);
      for (let i = 0; i < 6; i++) AML[i][0] -= fixedEnd[i][0];
      localUDLs.push({ startDistance: a, endDistance: b, q1, q2 });
    }
  });

  model.moments.forEach(moment => {
    if (!isActiveMomentCase(moment) || moment.elementId !== element.id || moment.type !== "Element Moment") return;
    const a = Math.max(0, Math.min(Number(moment.distance) || 0, p.L));
    const hasMx = Number.isFinite(Number(moment.Mx));
    const hasMy = Number.isFinite(Number(moment.My));
    const globalMx = hasMx ? Number(moment.Mx) : 0;
    const globalMy = hasMy ? Number(moment.My) : 0;
    const local = gridLocalMomentComponents(globalMx, globalMy, p.cx, p.cy);
    const torsion = (Number(moment.Mz) || 0) + local.Mx;
    const bendingVector = gridPointMomentVector(local.My, a, p.L);
    const torsionVector = gridPointTorsionFEM(torsion, a, p.L);
    for (let i = 0; i < 6; i++) AML[i][0] += bendingVector[i] + torsionVector[i];
    localPointMoments.push({ distance: a, Mx: torsion, My: local.My });
  });

  const AMS = Matrix.mul(p.RTT, AML);
  return {
    elementId: element.id,
```

```
        templateId: element.templateId,
        startNode: element.startNode,
        endNode: element.endNode,
        length: p.L,
        directionCosines: { cx: p.cx, cy: p.cy },
        SM: p.SM,
        RT: p.RT,
        RTT: p.RTT,
        SMS: p.SMS,
        AML,
        AMS,
        localPointLoads,
        localUDLs,
        localPointMoments
      };
    });

    return {
      success: true,
      solver: "Grid Structure Analysis — Steps 1–3",
      units: { force: "kN", length: "m", moment: "kN·m", rotation: "rad", translation: "m" },
      activeCases: { loadCaseId: activeLoadCaseId, momentCaseId: activeMomentCaseId },
      dofOrder: ["thetaX", "thetaY", "deltaZ"],
      steps: { step1, step2, step3: { title: "Member Stiffness Matrices and Fixed-End Actions", members: step3Members } }
    };
  }

  global.GridSolver = Object.freeze({
    version: "1.1.0-steps-1-3-only",
    calculateSteps1To3,
    Matrix
  });
})(window);
```

07. assets/js/offline.js

```
"use strict";

/*
 * Register offline support without adding any visible UI.
 * The Service Worker quietly caches the application shell in the background.
 */
(function registerOfflineSupport() {
    if (!("serviceWorker" in navigator)) return;

    if (location.protocol !== "https:" &&
        location.hostname !== "localhost" &&
        location.hostname !== "127.0.0.1") {
        return;
    }

    // Resolve the site root correctly for GitHub Pages project paths.
    const siteRoot = new URL("../../", document.currentScript.src);
    const workerUrl = new URL("service-worker.js", siteRoot);

    navigator.serviceWorker.register(workerUrl.href, {
        scope: siteRoot.pathname,
        updateViaCache: "none"
    }).catch((error) => {
        // Offline support is optional; keep registration failures out of the UI.
        console.warn("[Offline] Background registration failed:", error);
    });
})();
```

08. assets/vendor/driver/LICENSE.txt

```
The MIT License

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LIABILITY, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM,
OUT OF OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN
THE SOFTWARE.
```

09. assets/vendor/driver/README.txt

```
Driver.js v1.3.6 (MIT) — https://driverjs.com
Vendored unmodified from the npm package "driver.js@1.3.6" (dist/driver.js.iife.js, dist/driver.css).
```

10. assets/vendor/driver/driver.css

```
.driver-active .driver-overlay,.driver-active *{pointer-events:none}.driver-active .driver-active-element,.driver-active .driver-active-element *,.driver-popover,.driver-popover *{pointer-
events:auto}@keyframes animate-fade-in{0%{opacity:0}to{opacity:1}}.driver-fade .driver-overlay{animation:animate-fade-in .2s ease-in-out}.driver-fade .driver-popover{animation:animate-fade-
in .2s}.driver-popover{all:unset;box-sizing:border-box;color:#2d2d2d;margin:0;padding:15px;border-radius:5px;min-width:250px;max-width:300px;box-shadow:0 1px 10px #0006;z-index:1000000000;
position:fixed;top:0;right:0;background-color:#fff}.driver-popover *{font-family:Helvetica Neue,Inter,ui-sans-serif,"Apple Color Emoji",Helvetica,Arial,sans-serif}.driver-popover-title{font:
19px/normal sans-serif;font-weight:700;display:block;position:relative;line-height:1.5;zoom:1;margin:0}.driver-popover-close-btn{all:unset;position:absolute;top:0;right:0;width:32px;height:
28px;cursor:pointer;font-size:18px;font-weight:500;color:#d2d2d2;z-index:1;text-align:center;transition:color;transition-duration:.2s}.driver-popover-close-btn:hover,.driver-popover-close-
btn:focus{color:#2d2d2d}.driver-popover-title[style*=block]+.driver-popover-description{margin-top:5px}.driver-popover-description{margin-bottom:0;font:14px/normal sans-serif;line-height:1.
5;font-weight:400;zoom:1}.driver-popover-footer{margin-top:15px;text-align:right;zoom:1;display:flex;align-items:center;justify-content:space-between}.driver-popover-progress-text{font-size:
13px;font-weight:400;color:#727272;zoom:1}.driver-popover-footer button{all:unset;display:inline-block;box-sizing:border-box;padding:3px 7px;text-decoration:none;text-shadow:1px 1px 0 #fff;
background-color:#fff;color:#2d2d2d;font:12px/normal sans-serif;cursor:pointer;outline:0;zoom:1;line-height:1.3;border:1px solid #ccc;border-radius:3px}.driver-popover-footer .driver-
popover-btn-disabled{opacity:.5;pointer-events:none}:not(body):has(>.driver-active-element){overflow:hidden!important}.driver-no-interaction,.driver-no-interaction *{pointer-events:
none!important}.driver-popover-footer button:hover,.driver-popover-footer button:focus{background-color:#f7f7f7}.driver-popover-navigation-btns{display:flex;flex-grow:1;justify-content:flex-
end}.driver-popover-navigation-btns button+button{margin-left:4px}.driver-popover-arrow{content:"";position:absolute;border:5px solid #fff}.driver-popover-arrow-side-over{display:none}.
driver-popover-arrow-side-left{left:100%;border-right-color:transparent;border-bottom-color:transparent;border-top-color:transparent}.driver-popover-arrow-side-right{right:100%;border-left-
color:transparent;border-bottom-color:transparent;border-top-color:transparent}.driver-popover-arrow-side-top{top:100%;border-right-color:transparent;border-bottom-color:transparent;border-
left-color:transparent}.driver-popover-arrow-side-bottom{bottom:100%;border-left-color:transparent;border-top-color:transparent;border-right-color:transparent}.driver-popover-arrow-side-
center{display:none}.driver-popover-arrow-side-left.driver-popover-arrow-align-start,.driver-popover-arrow-side-right.driver-popover-arrow-align-start{top:15px}.driver-popover-arrow-side-
top.driver-popover-arrow-align-start,.driver-popover-arrow-side-bottom.driver-popover-arrow-align-start{left:15px}.driver-popover-arrow-align-end.driver-popover-arrow-side-left,.driver-
popover-arrow-align-end.driver-popover-arrow-side-right{bottom:15px}.driver-popover-arrow-side-top.driver-popover-arrow-align-end,.driver-popover-arrow-side-bottom.driver-popover-arrow-
align-end{right:15px}.driver-popover-arrow-side-left.driver-popover-arrow-align-center,.driver-popover-arrow-side-right.driver-popover-arrow-align-center{top:50%;margin-top:-5px}.driver-
popover-arrow-side-top.driver-popover-arrow-align-center,.driver-popover-arrow-side-bottom.driver-popover-arrow-align-center{left:50%;margin-left:-5px}.driver-popover-arrow-none{display:
none}
```

11. assets/vendor/driver/driver.js.iife.js

```
this.driver=this.driver||{};this.driver.js=function(F){"use strict";let z={},q;function V(e={}){z={animate:!0,allowClose:!0,overlayClickBehavior:"close",overlayOpacity:.7,smoothScroll:!1,
disableActiveInteraction:!1,showProgress:!1,stagePadding:10,stageRadius:5,popoverOffset:10,showButtons:["next","previous","close"],disableButtons:[],overlayColor:"#000",...e}}function s(e){
return e?z[e]:z}function le(e){q=e}function S(){return q}let A={};function N(e,o){A[e]=o}function E(e){var o;(o=A[e])==null||o.call(A)}function de(){A={}}function O(e,o,t,i){return(e/=i/
2)<1?t/2*e*e+o:-t/2*(--e*(e-2)-1)+o}function K(e){const o='a[href]:not([disabled]), button:not([disabled]), textarea:not([disabled]), input[type="text"]:not([disabled]), input[type="radio"]:
not([disabled]), input[type="checkbox"]:not([disabled]), select:not([disabled])';return e.flatMap(t=>{const i=t.matches(o),d=Array.from(t.querySelectorAll(o));return[...i?[t]:[],...d]}).
filter(t=>getComputedStyle(t).pointerEvents!=="none"&&ve(t))}function Y(e){if(!e||ue(e))return;const o=s("smoothScroll"),t=e.offsetHeight>window.innerHeight;e.scrollIntoView({behavior:
!o||pe(e)?"auto":"smooth",inline:"center",block:t?"start":"center"})}function pe(e){if(!e||!e.parentElement)return;const o=e.parentElement;return o.scrollHeight>o.clientHeight}function ue(
e){const o=e.getBoundingClientRect();return o.top>=0&&o.left>=0&&o.bottom<=(window.innerHeight||document.documentElement.clientHeight)&&o.right<=(window.innerWidth||document.documentElement.
clientWidth)}function ve(e){return!!(e.offsetWidth||e.offsetHeight||e.getClientRects().length)}let D={};function k(e,o){D[e]=o}function l(e){return e?D[e]:D}function X(){D={}}function fe(e,
o,t,i){let d=l("__activeStagePosition");const n=d||t.getBoundingClientRect(),f=i.getBoundingClientRect(),w=O(e,n.x,f.x-n.x,o),r=O(e,n.y,f.y-n.y,o),v=O(e,n.width,f.width-n.width,o),g=O(e,n.
height,f.height-n.height,o);d={x:w,y:r,width:v,height:g},Q(d),k("__activeStagePosition",d)}function j(e){if(!e)return;const o=e.getBoundingClientRect(),t={x:o.x,y:o.y,width:o.width,height:o.
height};k("__activeStagePosition",t),Q(t)}function he(){const e=l("__activeStagePosition"),o=l("__overlaySvg");if(!e)return;if(!o){console.warn("No stage svg found.");return}const t=window.
innerWidth,i=window.innerHeight;o.setAttribute("viewBox",`0 0 ${t} ${i}`)}function ge(e){const o=we(e);document.body.appendChild(o),U(o,t=>{t.target.tagName==="path"&&E("overlayClick")}),k(
"__overlaySvg",o)}function Q(e){const o=l("__overlaySvg");if(!o){ge(e);return}const t=o.firstElementChild;if((t==null?void 0:t.tagName)!=="path")throw new Error("no path element found in
stage svg");t.setAttribute("d",Z(e))}function we(e){const o=window.innerWidth,t=window.innerHeight,i=document.createElementNS("http://www.w3.org/2000/svg","svg");i.classList.add("driver-
overlay","driver-overlay-animated"),i.setAttribute("viewBox",`0 0 ${o} ${t}`),i.setAttribute("xmlSpace","preserve"),i.setAttribute("xmlnsXlink","http://www.w3.org/1999/xlink"),i.
setAttribute("version","1.1"),i.setAttribute("preserveAspectRatio","xMinYMin slice"),i.style.fillRule="evenodd",i.style.clipRule="evenodd",i.style.strokeLinejoin="round",i.style.
strokeMiterlimit="2",i.style.zIndex="10000",i.style.position="fixed",i.style.top="0",i.style.left="0",i.style.width="100%",i.style.height="100%";const d=document.createElementNS("http://www.
w3.org/2000/svg","path");return d.setAttribute("d",Z(e)),d.style.fill=s("overlayColor")||"rgb(0,0,0)",d.style.opacity=`${s("overlayOpacity")}`,d.style.pointerEvents="auto",d.style.
cursor="auto",i.appendChild(d),i}function Z(e){const o=window.innerWidth,t=window.innerHeight,i=s("stagePadding")||0,d=s("stageRadius")||0,n=e.width+i*2,f=e.height+i*2,w=Math.min(d,n/2,f/2),
r=Math.floor(Math.max(w,0)),v=e.x-i+r,g=e.y-i,y=n-r*2,a=f-r*2;return`M${o},0L0,0L0,${t}L${o},${t}L${o},0Z
    M${v},${g} h${y} a${r},${r} 0 0 1 ${r},${r} v${a} a${r},${r} 0 0 1 -${r},${r} h-${y} a${r},${r} 0 0 1 -${r},-${r} v-${a} a${r},${r} 0 0 1 ${r},-${r} z`}function me(){const e=l(
"__overlaySvg");e&&e.remove()}function ye(){const e=document.getElementById("driver-dummy-element");if(e)return e;let o=document.createElement("div");return o.id="driver-dummy-element",o.
style.width="0",o.style.height="0",o.style.pointerEvents="none",o.style.opacity="0",o.style.position="fixed",o.style.top="50%",o.style.left="50%";document.body.appendChild(o),o}function G(
e){const{element:o}=e;let t=typeof o=="function"?o():typeof o=="string"?document.querySelector(o):o;t||(t=ye()),xe(t,e)}function be(){const e=l("__activeElement"),o=l("__activeStep");e&&(j(
e),he(),ne(e,o))}function xe(e,o){var C;const i=Date.now(),d=l("__activeStep"),n=l("__activeElement")||e,f=!n||n===e,w=e.id==="driver-dummy-element",r=n.id==="driver-dummy-element",v=s(
"animate"),g=o.onHighlightStarted||s("onHighlightStarted"),y=(o==null?void 0:o.onHighlighted)||s("onHighlighted"),a=(d==null?void 0:d.onDeselected)||s("onDeselected"),p=s(),c=l();!f&&a&&a(
r?void 0:n,d,{config:p,state:c,driver:S()}),g&&g(w?void 0:e,o,{config:p,state:c,driver:S()});const u=!f&&v;let h=!1;_e(),k("previousStep",d),k("previousElement",n),k("activeStep",o),k(
"activeElement",e);const m=()=>{if(l("__transitionCallback")!==m)return;const x=Date.now()-i,T=400-x<=400/2;o.popover&&!h&&u&&(ee(e,o),h=!0),s("animate")&&x<400?fe(x,400,n,e):(j(e),y&&y(
w?void 0:e,o,{config:s(),state:l(),driver:S()})),k("__transitionCallback",void 0),k("__previousStep",d),k("__previousElement",n),k("__activeStep",o),k("__activeElement",e)),window.
requestAnimationFrame(m)};k("__transitionCallback",m),window.requestAnimationFrame(m),Y(e),!u&&o.popover&&ee(e,o),n.classList.remove("driver-active-element","driver-no-interaction"),n.
removeAttribute("aria-haspopup"),n.removeAttribute("aria-expanded"),n.removeAttribute("aria-controls"),((C=o.disableActiveInteraction)!=null?C:s("disableActiveInteraction"))&&e.classList.
add("driver-no-interaction"),e.classList.add("driver-active-element"),e.setAttribute("aria-haspopup","dialog"),e.setAttribute("aria-expanded","true"),e.setAttribute("aria-controls","driver-
popover-content")}function Ce(){var e;(e=document.getElementById("driver-dummy-element"))==null||e.remove(),document.querySelectorAll(".driver-active-element").forEach(o=>{o.classList.
remove("driver-active-element","driver-no-interaction"),o.removeAttribute("aria-haspopup"),o.removeAttribute("aria-expanded"),o.removeAttribute("aria-controls")})}function H(){const e=l(
"__resizeTimeout");e&&window.cancelAnimationFrame(e),k("__resizeTimeout",window.requestAnimationFrame(be))}function Pe(e){var r;if(!l("isInitialized")||!(e.key==="Tab"||e.
keyCode===9))return;const i=l("__activeElement"),d=(r=l("popover"))==null?void 0:r.wrapper,n=K([...d?[d]:[],...i?[i]:[]]),f=n[0],w=n[n.length-1];if(e.preventDefault(),e.shiftKey){const v=n[
n.indexOf(document.activeElement)-1]||w;v==null||v.focus()}else{const v=n[n.indexOf(document.activeElement)+1]||f;v==null||v.focus()}}function J(e){var t;((t=s(
"allowKeyboardControl"))==null||t)&&(e.key==="Escape"?E("escapePress"):e.key==="ArrowRight"?E("arrowRightPress"):e.key==="ArrowLeft"&&E("arrowLeftPress"))}function U(e,o,t){const i=(n,f)=>{
const w=n.target;e.contains(w)&&((!t||t(w))&&(n.preventDefault(),n.stopPropagation(),n.stopImmediatePropagation()),f==null||f(n))};document.addEventListener("pointerdown",i,!0),document.
addEventListener("mousedown",i,!0),document.addEventListener("pointerup",i,!0),document.addEventListener("mouseup",i,!0),document.addEventListener("click",n=>{i(n,o)},!0)}function ke(){
window.addEventListener("keyup",J,!1),window.addEventListener("keydown",Pe,!1),window.addEventListener("resize",H),window.addEventListener("scroll",H)}function Se(){window.
removeEventListener("keyup",J),window.removeEventListener("resize",H),window.removeEventListener("scroll",H)}function _e(){const e=l("popover");e&&(e.wrapper.style.display="none")}function
ee(e,o){var x,P;let t=l("popover");t&&document.body.removeChild(t.wrapper),t=Ee(),document.body.appendChild(t.wrapper);const{title:i,description:d,showButtons:n,disableButtons:f,
showProgress:w,nextBtnText:r=s("nextBtnText")||"Next &rarr;",prevBtnText:v=s("prevBtnText")||"&larr; Previous",progressText:g=s("progressText")||"{current} of {total}"}=o.popover||{};t.
nextButton.innerHTML=r,t.previousButton.innerHTML=v,t.progress.innerHTML=g,i?(t.title.innerHTML=i,t.title.style.display="block"):t.title.style.display="none",d?(t.description.innerHTML=d,t.
description.style.display="block"):t.description.style.display="none";const y=n||s("showButtons"),a=w||s("showProgress")||!1,p=(y==null?void 0:y.includes("next"))||(y==null?void 0:y.
includes("previous"))||a;t.closeButton.style.display=y.includes("close")?"block":"none",p?(t.footer.style.display="flex",t.progress.style.display=a?"block":"none",t.nextButton.style.
display=y.includes("next")?"block":"none",t.previousButton.style.display=y.includes("previous")?"block":"none"):t.footer.style.display="none";const c=f||s("disableButtons")||[];c!=null&&c.
includes("next")&&(t.nextButton.disabled=!0,t.nextButton.classList.add("driver-popover-btn-disabled")),c!=null&&c.includes("previous")&&(t.previousButton.disabled=!0,t.previousButton.
classList.add("driver-popover-btn-disabled")),c!=null&&c.includes("close")&&(t.closeButton.disabled=!0,t.closeButton.classList.add("driver-popover-btn-disabled"));const u=t.wrapper;u.style.
display="block",u.style.left="",u.style.top="",u.style.bottom="",u.style.right="",u.id="driver-popover-content",u.setAttribute("role","dialog"),u.setAttribute("aria-labelledby","driver-
popover-title"),u.setAttribute("aria-describedby","driver-popover-description");const h=t.arrow;h.className="driver-popover-arrow";const m=((x=o.popover)==null?void 0:x.popoverClass)||s(
"popoverClass")||"";u.className=`driver-popover ${m}`.trim(),U(t.wrapper,T=>{var W,I,M;const L=T.target,$=((W=o.popover)==null?void 0:W.onNextClick)||s("onNextClick"),B=((I=o.
popover)==null?void 0:I.onPrevClick)||s("onPrevClick"),R=((M=o.popover)==null?void 0:M.onCloseClick)||s("onCloseClick");if(L.closest(".driver-popover-next-btn"))return $?$(e,o,{config:s(),
state:l(),driver:S()}):E("nextClick");if(L.closest(".driver-popover-prev-btn"))return B?B(e,o,{config:s(),state:l(),driver:S()}):E("prevClick");if(L.closest(".driver-popover-close-
btn"))return R?R(e,o,{config:s(),state:l(),driver:S()}):E("closeClick")},T=>!(t!=null&&t.description.contains(T))&&!(t!=null&&t.title.contains(T))&&typeof T.className=="string"&&T.className.
includes("driver-popover")),k("popover",t);const b=((P=o.popover)==null?void 0:P.onPopoverRender)||s("onPopoverRender");b&&b(t,{config:s(),state:l(),driver:S()}),ne(e,o),Y(u);const C=e.
classList.contains("driver-dummy-element"),_=K([u,...C?[]:[e]]);_.length>0&&_[0].focus()}function te(){const e=l("popover");if(!(e!=null&&e.wrapper))return;const o=e.wrapper.
getBoundingClientRect(),t=s("stagePadding")||0,i=s("popoverOffset")||0;return{width:o.width+t+i,height:o.height+t+i,realWidth:o.width,realHeight:o.height}}function oe(e,o){const{
elementDimensions:t,popoverDimensions:i,popoverPadding:d,popoverArrowDimensions:n}=o;return e==="start"?Math.max(Math.min(t.top-d,window.innerHeight-i.realHeight-n.width),n.width):
e==="end"?Math.max(Math.min(t.top-(i==null?void 0:i.realHeight)+t.height+d,window.innerHeight-(i==null?void 0:i.realHeight)-n.width),n.width):e==="center"?Math.max(Math.min(t.top+t.height/2-
(i==null?void 0:i.realHeight)/2,window.innerHeight-(i==null?void 0:i.realHeight)-n.width),n.width):0}function ie(e,o){const{elementDimensions:t,popoverDimensions:i,popoverPadding:d,
popoverArrowDimensions:n}=o;return e==="start"?Math.max(Math.min(t.left-d,window.innerWidth-i.realWidth-n.width),n.width):e==="end"?Math.max(Math.min(t.left-(i==null?void 0:i.realWidth)+t.
width+d,window.innerWidth-(i==null?void 0:i.realWidth)-n.width),n.width):e==="center"?Math.max(Math.min(t.left+t.width/2-(i==null?void 0:i.realWidth)/2,window.innerWidth-(i==null?void 0:i.
realWidth)-n.width),n.width):0}function ne(e,o){const t=l("popover");if(!t)return;const{align:i="start",side:d="left"}=(o==null?void 0:o.popover)||{},n=i,f=e.id==="driver-dummy-
element"?"over":d,w=s("stagePadding")||0,r=te(),v=t.arrow.getBoundingClientRect(),g=e.getBoundingClientRect(),y=g.top-r.height;let a=y>=0;const p=window.innerHeight-(g.bottom+r.height);let
c=p>=0;const u=g.left-r.width;let h=u>=0;const m=window.innerWidth-(g.right+r.width);let b=m>=0;const C=!a&&!c&&!h&&!b;let _=f;if(f==="top"&&a?b=h=c=!1:f==="bottom"&&c?b=h=a=!1:
f==="left"&&h?b=a=c=!1:f==="right"&&b&&(h=a=c=!1),f==="over"){const x=window.innerWidth/2-r.realWidth/2,P=window.innerHeight/2-r.realHeight/2;t.wrapper.style.left=`${x}px`,t.wrapper.style.
right="auto",t.wrapper.style.top=`${P}px`,t.wrapper.style.bottom="auto"}else if(C){const x=window.innerWidth/2-(r==null?void 0:r.realWidth)/2,P=10;t.wrapper.style.left=`${x}px`,t.wrapper.
style.right="auto",t.wrapper.style.bottom=`${P}px`,t.wrapper.style.top="auto"}else if(h){const x=Math.min(u,window.innerWidth-(r==null?void 0:r.realWidth)-v.width),P=oe(n,{elementDimensions:
g,popoverDimensions:r,popoverPadding:w,popoverArrowDimensions:v});t.wrapper.style.left=`${x}px`,t.wrapper.style.top=`${P}px`,t.wrapper.style.bottom="auto",t.wrapper.style.right="auto",
_="left"}else if(b){const x=Math.min(m,window.innerWidth-(r==null?void 0:r.realWidth)-v.width),P=oe(n,{elementDimensions:g,popoverDimensions:r,popoverPadding:w,popoverArrowDimensions:v});t.
wrapper.style.right=`${x}px`,t.wrapper.style.top=`${P}px`,t.wrapper.style.bottom="auto",t.wrapper.style.left="auto",_="right"}else if(a){const x=Math.min(y,window.innerHeight-r.realHeight-v.
width);let P=ie(n,{elementDimensions:g,popoverDimensions:r,popoverPadding:w,popoverArrowDimensions:v});t.wrapper.style.top=`${x}px`,t.wrapper.style.left=`${P}px`,t.wrapper.style.
bottom="auto",t.wrapper.style.right="auto",_="top"}else if(c){const x=Math.min(p,window.innerHeight-(r==null?void 0:r.realHeight)-v.width);let P=ie(n,{elementDimensions:g,popoverDimensions:
r,popoverPadding:w,popoverArrowDimensions:v});t.wrapper.style.left=`${P}px`,t.wrapper.style.bottom=`${x}px`,t.wrapper.style.top="auto",t.wrapper.style.right="auto",_="bottom"}C?t.arrow.
classList.add("driver-popover-arrow-none"):Te(n,_,e)}function Te(e,o,t){const i=l("popover");if(!i)return;const d=t.getBoundingClientRect(),n=te(),f=i.arrow,w=n.width,r=window.innerWidth,
v=d.width,g=d.left,y=n.height,a=window.innerHeight,p=d.top,c=d.height;f.className="driver-popover-arrow";let u=o,h=e;if(o==="top"?(g+v<=0?(u="right",h="end"):g+v-w<=0&&(u="top",h="start"),
g>=r?(u="left",h="end"):g+w>=r&&(u="top",h="end")):o==="bottom"?(g+v<=0?(u="right",h="start"):g+v-w<=0&&(u="bottom",h="start"),g>=r?(u="left",h="start"):g+w>=r&&(u="bottom",h="end")):
o==="left"?(p+c<=0?(u="bottom",h="end"):p+c-y<=0&&(u="left",h="start"),p>=a?(u="top",h="end"):p+y>=a&&(u="left",h="end")):o==="right"&&(p+c<=0?(u="bottom",h="start"):p+c-y<=0&&(u="right",
h="start"),p>=a?(u="top",h="start"):p+y>=a&&(u="right",h="end")),!u)f.classList.add("driver-popover-arrow-none");else{f.classList.add(`driver-popover-arrow-side-${u}`),f.classList.add(
`driver-popover-arrow-align-${h}`);const m=t.getBoundingClientRect(),b=f.getBoundingClientRect(),C=s("stagePadding")||0,_=m.left-C<window.innerWidth&&m.right+C>0&&m.top-C<window.
innerHeight&&m.bottom+C>0;o==="bottom"&&_&&(b.x>m.x&&b.x+b.width<m.x+m.width?i.wrapper.style.transform="translateY(0)":(f.classList.remove(`driver-popover-arrow-align-${h}`),f.classList.add(
"driver-popover-arrow-none"),i.wrapper.style.transform=`translateY(-${C/2}px)`))}}function Ee(){const e=document.createElement("div");e.classList.add("driver-popover");const o=document.
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createElement("div");o.classList.add("driver-popover-arrow");const t=document.createElement("header");t.id="driver-popover-title",t.classList.add("driver-popover-title"),t.style.
display="none",t.innerText="Popover Title";const i=document.createElement("div");i.id="driver-popover-description",i.classList.add("driver-popover-description"),i.style.display="none",i.
innerText="Popover description is here";const d=document.createElement("button");d.type="button",d.classList.add("driver-popover-close-btn"),d.setAttribute("aria-label","Close"),d.
innerHTML="&times;";const n=document.createElement("footer");n.classList.add("driver-popover-footer");const f=document.createElement("span");f.classList.add("driver-popover-progress-text"),
f.innerText="";const w=document.createElement("span");w.classList.add("driver-popover-navigation-btns");const r=document.createElement("button");r.type="button",r.classList.add("driver-
popover-prev-btn"),r.innerHTML="&larr; Previous";const v=document.createElement("button");return v.type="button",v.classList.add("driver-popover-next-btn"),v.innerHTML="Next &rarr;",w.
appendChild(r),w.appendChild(v),n.appendChild(f),n.appendChild(w),e.appendChild(d),e.appendChild(o),e.appendChild(t),e.appendChild(i),e.appendChild(n),{wrapper:e,arrow:o,title:t,description:
i,footer:n,previousButton:r,nextButton:v,closeButton:d,footerButtons:w,progress:f}}function Le(){var o;const e=l("popover");e&&((o=e.wrapper.parentElement)==null||o.removeChild(e.
wrapper))}const $e="";function Ae(e={}){V(e);function o(){s("allowClose")&&g()}function t(){const a=s("overlayClickBehavior");if(s("allowClose")&&a==="close"){g();return}a==="nextStep"&&i(
)}function i(){const a=l("activeIndex"),p=s("steps")||[];if(typeof a=="undefined")return;const c=a+1;p[c]?v(c):g()}function d(){const a=l("activeIndex"),p=s("steps")||[];if(typeof
a=="undefined")return;const c=a-1;p[c]?v(c):g()}function n(a){(s("steps")||[])[a]?v(a):g()}function f(){var b;if(l("__transitionCallback"))return;const p=l("activeIndex"),c=l(
"__activeStep"),u=l("__activeElement");if(typeof p=="undefined"||typeof c=="undefined"||typeof l("activeIndex")=="undefined")return;const m=((b=c.popover)==null?void 0:b.onPrevClick)||s(
"onPrevClick");if(m)return m(u,c,{config:s(),state:l(),driver:S()});d()}function w(){var m;if(l("__transitionCallback"))return;const p=l("activeIndex"),c=l("__activeStep"),u=l(
"__activeElement");if(typeof p=="undefined"||typeof c=="undefined")return;const h=((m=c.popover)==null?void 0:m.onNextClick)||s("onNextClick");if(h)return h(u,c,{config:s(),state:l(),driver:
S()});i()}function r(){l("isInitialized")||(k("isInitialized",!0),document.body.classList.add("driver-active",s("animate")?"driver-fade":"driver-simple"),ke(),N("overlayClick",t),N(
"escapePress",o),N("arrowLeftPress",f),N("arrowRightPress",w))}function v(a=0){var R,W,I,M,re,se,ae,ce;const p=s("steps");if(!p){console.error("No steps to drive through"),g();return}if(!p[
a]){g();return}k("__activeOnDestroyed",document.activeElement),k("activeIndex",a);const c=p[a],u=p[a+1],h=p[a-1],m=((R=c.popover)==null?void 0:R.doneBtnText)||s("doneBtnText")||"Done",b=s(
"allowClose"),C=typeof((W=c.popover)==null?void 0:W.showProgress)!="undefined"?(I=c.popover)==null?void 0:I.showProgress:s("showProgress"),x=(((M=c.popover)==null?void 0:M.progressText)||s(
"progressText")||"{{current}} of {{total}}").replace("{{current}}",`${a+1}`).replace("{{total}}",`${p.length}`),P=((re=c.popover)==null?void 0:re.showButtons)||s("showButtons"),T=["next",
"previous",...b?["close"]:[]].filter(He=>!(P!=null&&P.length)||P.includes(He)),L=((se=c.popover)==null?void 0:se.onNextClick)||s("onNextClick"),$=((ae=c.popover)==null?void 0:ae.
onPrevClick)||s("onPrevClick"),B=((ce=c.popover)==null?void 0:ce.onCloseClick)||s("onCloseClick");G({...c,popover:{showButtons:T,nextBtnText:u?void 0:m,disableButtons:[...h?[]:["previous"]],
showProgress:C,progressText:x,onNextClick:L||(()=>{u?v(a+1):g()}),onPrevClick:$||(()=>{v(a-1)}),onCloseClick:B||(()=>{g()}),...(c==null?void 0:c.popover)||{}}})}function g(a=!0){const p=l(
"__activeElement"),c=l("__activeStep"),u=l("__activeOnDestroyed"),h=s("onDestroyStarted");if(a&&h){const C=!p||(p==null?void 0:p.id)==="driver-dummy-element";h(C?void 0:p,c,{config:s(),
state:l(),driver:S()});return}const m=(c==null?void 0:c.onDeselected)||s("onDeselected"),b=s("onDestroyed");if(document.body.classList.remove("driver-active","driver-fade","driver-simple"),
Se(),Le(),Ce(),me(),de(),X(),p&&c){const C=p.id==="driver-dummy-element";m&&m(C?void 0:p,c,{config:s(),state:l(),driver:S()}),b&&b(C?void 0:p,c,{config:s(),state:l(),driver:S()})}u&&u.focus(
)}const y={isActive:()=>l("isInitialized")||!1,refresh:H,drive:(a=0)=>{r(),v(a)},setConfig:V,setSteps:a=>{X(),V({...s(),steps:a})},getConfig:s,getState:l,getActiveIndex:()=>l("activeIndex"),
isFirstStep:()=>l("activeIndex")===0,isLastStep:()=>{const a=s("steps")||[],p=l("activeIndex");return p!==void 0&&p===a.length-1},getActiveStep:()=>l("activeStep"),getActiveElement:()=>l(
"activeElement"),getPreviousElement:()=>l("previousElement"),getPreviousStep:()=>l("previousStep"),moveNext:i,movePrevious:d,moveTo:n,hasNextStep:()=>{const a=s("steps")||[],p=l(
"activeIndex");return p!==void 0&&!!a[p+1]},hasPreviousStep:()=>{const a=s("steps")||[],p=l("activeIndex");return p!==void 0&&!!a[p-1]},highlight:a=>{r(),G({...a,popover:a.popover?{
showButtons:[],showProgress:!1,progressText:"",...a.popover}:void 0})},destroy:()=>{g(!1)}};return le(y),y}return F.driver=Ae,Object.defineProperty(F,Symbol.toStringTag,{value:"Module"}),F}(
{});
```

12. assets/vendor/jspdf.umd.min.js

```
/** @license
 *
 * jsPDF - PDF Document creation from JavaScript
 * Version 4.2.1 Built on 2026-03-17T11:11:27.056Z
 *                      CommitID 00000000
 *
 * Copyright (c) 2010-2025 James Hall <james@parall.ax>, https://github.com/MrRio/jsPDF
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 *               2012 Florian Jenett, https://github.com/fjenett
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 *               2013 Stefan Slonevskiy, https://github.com/stefslon
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 *
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 *    kim3er, mfo, alnorth, Flamenco
 */

!function(t,e){"object"==typeof exports&&"undefined"!=typeof module?e(exports):"function"==typeof define&&define.amd?define(["exports"],e):e((t="undefined"!=typeof globalThis?globalThis:
t||self).jspdf={})}(this,function(t){function e(t,e){(null==e||e>t.length)&&(e=t.length);for(var n=0,r=Array(e);n<e;n++)r[n]=t[n];return r}function n(t,n){return function(t){if(Array.
isArray(t))return t}(t)||function(t,e){var n=null==t?null:"undefined"!=typeof Symbol&&t[Symbol.iterator]||t["@@iterator"];if(null!=n){var r,i,a,s,o=[],h=!0,l=!1;try{if(a=(n=n.call(t)).next,
0===e){if(Object(n)!==n)return;h=!1}else for(;!(h=(r=a.call(n)).done)&&(o.push(r.value),o.length!==e);h=!0);}catch(t){l=!0,i=t}finally{try{if(!h&&null!=n.return&&(s=n.return(),Object(
s)!==s))return}finally{if(l)throw i}}return o}}(t,n)||function(t,n){if(t){if("string"==typeof t)return e(t,n);var r={}.toString.call(t).slice(8,-1);return"Object"===r&&t.constructor&&(r=t.
constructor.name),"Map"===r||"Set"===r?Array.from(t):"Arguments"===r||/^(?:Ui|I)nt(?:8|16|32)(?:Clamped)?Array$/.test(r)?e(t,n):void 0}}(t,n)||function(){throw new TypeError("Invalid
attempt to destructure non-iterable instance.\nIn order to be iterable, non-array objects must have a [Symbol.iterator]() method.")}()}function r(t){return r="function"==typeof
Symbol&&"symbol"==typeof Symbol.iterator?function(t){return typeof t}:function(t){return t&&"function"==typeof Symbol&&t.constructor===Symbol&&t!==Symbol.prototype?"symbol":typeof t},r(
t)}var i=function(){return"undefined"!=typeof window?window:"undefined"!=typeof global?global:"undefined"!=typeof self?self:this}();function a(){i.console&&"function"==typeof i.console.
log&&i.console.log.apply(i.console,arguments)}var s={log:a,warn:function(t){i.console&&("function"==typeof i.console.warn?i.console.warn.apply(i.console,arguments):a.call(null,arguments))},
error:function(t){i.console&&("function"==typeof i.console.error?i.console.error.apply(i.console,arguments):a(t))}};function o(t,e,n){var r=new XMLHttpRequest;r.open("GET",t),r.
responseType="blob",r.onload=function(){c(r.response,e,n)},r.onerror=function(){s.error("could not download file")},r.send()}function h(t){var e=new XMLHttpRequest;e.open("HEAD",t,!1);try{e.
send()}catch(n){}return e.status>=200&&e.status<=299}function l(t){try{t.dispatchEvent(new MouseEvent("click"))}catch(n){var e=document.createEvent("MouseEvents");e.initMouseEvent("click",
!0,!0,window,0,0,0,80,20,!1,!1,!1,!1,0,null),t.dispatchEvent(e)}}var c=i.saveAs||("object"!=("undefined"==typeof window?"undefined":r(window))||window!==i?function(){}:"undefined"!=typeof
HTMLAnchorElement&&"download"in HTMLAnchorElement.prototype?function(t,e,n){var r=i.URL||i.webkitURL,a=document.createElement("a");e=e||t.name||"download",a.download=e,a.rel="noopener",
"string"==typeof t?(a.href=t,a.origin!==location.origin?h(a.href)?o(t,e,n):l(a,a.target="_blank"):l(a)):(a.href=r.createObjectURL(t),setTimeout(function(){r.revokeObjectURL(a.href)},4e4),
setTimeout(function(){l(a)},0))}:"msSaveOrOpenBlob"in navigator?function(t,e,n){if(e=e||t.name||"download","string"==typeof t)if(h(t))o(t,e,n);else{var i=document.createElement("a");i.
href=t,i.target="_blank",setTimeout(function(){l(i)})}else navigator.msSaveOrOpenBlob(function(t,e){return void 0===e?e={autoBom:!1}:"object"!==r(e)&&(s.warn("Deprecated: Expected third
argument to be a object"),e={autoBom:!e}),e.autoBom&&/^\s*(?:text\/\S*|application\/xml|\S*\/\S*\+xml)\s*;.*charset\s*=\s*utf-8/i.test(t.type)?new Blob([String.fromCharCode(65279),t],{type:
t.type}):t}(t,n),e)}:function(t,e,n,a){if((a=a||open("","_blank"))&&(a.document.title=a.document.body.innerText="downloading..."),"string"==typeof t)return o(t,e,n);var s="application/octet-
stream"===t.type,h=/constructor/i.test(i.HTMLElement)||i.safari,l=/CriOS\/[\d]+/.test(navigator.userAgent);if((l||s&&h)&&"object"===("undefined"==typeof FileReader?"undefined":r(
FileReader))){var c=new FileReader;c.onloadend=function(){var t=c.result;t=l?t:t.replace(/^data:[^;]*;/,"data:attachment/file;"),a?a.location.href=t:location=t,a=null},c.readAsDataURL(
t)}else{var u=i.URL||i.webkitURL,f=u.createObjectURL(t);a?a.location=f:location.href=f,a=null,setTimeout(function(){u.revokeObjectURL(f)},4e4)}});
/**
   * A class to parse color values
   * @author Stoyan Stefanov <sstoo@gmail.com>
   * {@link   http://www.phpied.com/rgb-color-parser-in-javascript/}
   * @license Use it if you like it
   */function u(t){var e;t=t||"",this.ok=!1,"#"==t.charAt(0)&&(t=t.substr(1,6)),t={aliceblue:"f0f8ff",antiquewhite:"faebd7",aqua:"00ffff",aquamarine:"7fffd4",azure:"f0ffff",beige:"f5f5dc",
bisque:"ffe4c4",black:"000000",blanchedalmond:"ffebcd",blue:"0000ff",blueviolet:"8a2be2",brown:"a52a2a",burlywood:"deb887",cadetblue:"5f9ea0",chartreuse:"7fff00",chocolate:"d2691e",coral:
"ff7f50",cornflowerblue:"6495ed",cornsilk:"fff8dc",crimson:"dc143c",cyan:"00ffff",darkblue:"00008b",darkcyan:"008b8b",darkgoldenrod:"b8860b",darkgray:"a9a9a9",darkgreen:"006400",darkkhaki:
"bdb76b",darkmagenta:"8b008b",darkolivegreen:"556b2f",darkorange:"ff8c00",darkorchid:"9932cc",darkred:"8b0000",darksalmon:"e9967a",darkseagreen:"8fbc8f",darkslateblue:"483d8b",darkslategray:
"2f4f4f",darkturquoise:"00ced1",darkviolet:"9400d3",deeppink:"ff1493",deepskyblue:"00bfff",dimgray:"696969",dodgerblue:"1e90ff",feldspar:"d19275",firebrick:"b22222",floralwhite:"fffaf0",
forestgreen:"228b22",fuchsia:"ff00ff",gainsboro:"dcdcdc",ghostwhite:"f8f8ff",gold:"ffd700",goldenrod:"daa520",gray:"808080",green:"008000",greenyellow:"adff2f",honeydew:"f0fff0",hotpink:
"ff69b4",indianred:"cd5c5c",indigo:"4b0082",ivory:"fffff0",khaki:"f0e68c",lavender:"e6e6fa",lavenderblush:"fff0f5",lawngreen:"7cfc00",lemonchiffon:"fffacd",lightblue:"add8e6",lightcoral:
```

```
"f08080",lightcyan:"e0ffff",lightgoldenrodyellow:"fafad2",lightgrey:"d3d3d3",lightgreen:"90ee90",lightpink:"ffb6c1",lightsalmon:"ffa07a",lightseagreen:"20b2aa",lightskyblue:"87cefa",
lightslateblue:"8470ff",lightslategray:"778899",lightsteelblue:"b0c4de",lightyellow:"ffffe0",lime:"00ff00",limegreen:"32cd32",linen:"faf0e6",magenta:"ff00ff",maroon:"800000",
mediumaquamarine:"66cdaa",mediumblue:"0000cd",mediumorchid:"ba55d3",mediumpurple:"9370d8",mediumseagreen:"3cb371",mediumslateblue:"7b68ee",mediumspringgreen:"00fa9a",mediumturquoise:
"48d1cc",mediumvioletred:"c71585",midnightblue:"191970",mintcream:"f5fffa",mistyrose:"ffe4e1",moccasin:"ffe4b5",navajowhite:"ffdead",navy:"000080",oldlace:"fdf5e6",olive:"808000",olivedrab:
"6b8e23",orange:"ffa500",orangered:"ff4500",orchid:"da70d6",palegoldenrod:"eee8aa",palegreen:"98fb98",paleturquoise:"afeeee",palevioletred:"d87093",papayawhip:"ffefd5",peachpuff:"ffdab9",
peru:"cd853f",pink:"ffc0cb",plum:"dda0dd",powderblue:"b0e0e6",purple:"800080",red:"ff0000",rosybrown:"bc8f8f",royalblue:"4169e1",saddlebrown:"8b4513",salmon:"fa8072",sandybrown:"f4a460",
seagreen:"2e8b57",seashell:"fff5ee",sienna:"a0522d",silver:"c0c0c0",skyblue:"87ceeb",slateblue:"6a5acd",slategray:"708090",snow:"fffafa",springgreen:"00ff7f",steelblue:"4682b4",tan:"d2b48c",
teal:"008080",thistle:"d8bfd8",tomato:"ff6347",turquoise:"40e0d0",violet:"ee82ee",violetred:"d02090",wheat:"f5deb3",white:"ffffff",whitesmoke:"f5f5f5",yellow:"ffff00",yellowgreen:"9acd32"}[
t=(t=t.replace(/ /g,"")).toLowerCase()]||t;for(var n=[{re:/^rgb\((\d{1,3}),\s*(\d{1,3}),\s*(\d{1,3})\)$/,example:["rgb(123, 234, 45)","rgb(255,234,245)"],process:function(t){return[parseInt(
t[1]),parseInt(t[2]),parseInt(t[3])]}},{re:/^(\w{2})(\w{2})(\w{2})$/,example:["#00ff00","336699"],process:function(t){return[parseInt(t[1],16),parseInt(t[2],16),parseInt(t[3],16)]}},{re:/^(\
w{1})(\w{1})(\w{1})$/,example:["#fb0","f0f"],process:function(t){return[parseInt(t[1]+t[1],16),parseInt(t[2]+t[2],16),parseInt(t[3]+t[3],16)]}}],r=0;r<n.length;r++){var i=n[r].re,a=n[r].
process,s=i.exec(t);s&&(e=a(s),this.r=e[0],this.g=e[1],this.b=e[2],this.ok=!0)}this.r=this.r<0||isNaN(this.r)?0:this.r>255?255:this.r,this.g=this.g<0||isNaN(this.g)?0:this.g>255?255:this.g,
this.b=this.b<0||isNaN(this.b)?0:this.b>255?255:this.b,this.toRGB=function(){return"rgb("+this.r+", "+this.g+", "+this.b+")"},this.toHex=function(){var t=this.r.toString(16),e=this.g.
toString(16),n=this.b.toString(16);return 1==t.length&&(t="0"+t),1==e.length&&(e="0"+e),1==n.length&&(n="0"+n),"#"+t+e+n}}var f=i.atob.bind(i),d=i.btoa.bind(i);
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   *
   * Author: Joseph Myers
   * Accessed from: http://www.myersdaily.org/joseph/javascript/md5.js
   *
   * Modified by: Owen Leong
   */
function p(t,e){var n=t[0],r=t[1],i=t[2],a=t[3];n=m(n,r,i,a,e[0],7,-680876936),a=m(a,n,r,i,e[1],12,-389564586),i=m(i,a,n,r,e[2],17,606105819),r=m(r,i,a,n,e[3],22,-1044525330),n=m(n,r,i,a,e[
4],7,-176418897),a=m(a,n,r,i,e[5],12,1200080426),i=m(i,a,n,r,e[6],17,-1473231341),r=m(r,i,a,n,e[7],22,-45705983),n=m(n,r,i,a,e[8],7,1770035416),a=m(a,n,r,i,e[9],12,-1958414417),i=m(i,a,n,r,
e[10],17,-42063),r=m(r,i,a,n,e[11],22,-1990404162),n=m(n,r,i,a,e[12],7,1804603682),a=m(a,n,r,i,e[13],12,-40341101),i=m(i,a,n,r,e[14],17,-1502002290),n=b(n,r=m(r,i,a,n,e[15],22,1236535329),i,
a,e[1],5,-165796510),a=b(a,n,r,i,e[6],9,-1069501632),i=b(i,a,n,r,e[11],14,643717713),r=b(r,i,a,n,e[0],20,-373897302),n=b(n,r,i,a,e[5],5,-701558691),a=b(a,n,r,i,e[10],9,38016083),i=b(i,a,n,r,
e[15],14,-660478335),r=b(r,i,a,n,e[4],20,-405537848),n=b(n,r,i,a,e[9],5,568446438),a=b(a,n,r,i,e[14],9,-1019803690),i=b(i,a,n,r,e[3],14,-187363961),r=b(r,i,a,n,e[8],20,1163531501),n=b(n,r,i,
a,e[13],5,-1444681467),a=b(a,n,r,i,e[2],9,-51403784),i=b(i,a,n,r,e[7],14,1735328473),n=v(n,r=b(r,i,a,n,e[12],20,-1926607734),i,a,e[5],4,-378558),a=v(a,n,r,i,e[8],11,-2022574463),i=v(i,a,n,r,
e[11],16,1839030562),r=v(r,i,a,n,e[14],23,-35309556),n=v(n,r,i,a,e[1],4,-1530992060),a=v(a,n,r,i,e[4],11,1272893353),i=v(i,a,n,r,e[7],16,-155497632),r=v(r,i,a,n,e[10],23,-1094730640),n=v(n,
r,i,a,e[13],4,681279174),a=v(a,n,r,i,e[0],11,-358537222),i=v(i,a,n,r,e[3],16,-722521979),r=v(r,i,a,n,e[6],23,76029189),n=v(n,r,i,a,e[9],4,-640364487),a=v(a,n,r,i,e[12],11,-421815835),i=v(i,
a,n,r,e[15],16,530742520),n=w(n,r=v(r,i,a,n,e[2],23,-995338651),i,a,e[0],6,-198630844),a=w(a,n,r,i,e[7],10,1126891415),i=w(i,a,n,r,e[14],15,-1416354905),r=w(r,i,a,n,e[5],21,-57434055),n=w(n,
r,i,a,e[12],6,1700485571),a=w(a,n,r,i,e[3],10,-1894986606),i=w(i,a,n,r,e[10],15,-1051523),r=w(r,i,a,n,e[1],21,-2054922799),n=w(n,r,i,a,e[8],6,1873313359),a=w(a,n,r,i,e[15],10,-30611744),i=w(
i,a,n,r,e[6],15,-1560198380),r=w(r,i,a,n,e[13],21,1309151649),n=w(n,r,i,a,e[4],6,-145523070),a=w(a,n,r,i,e[11],10,-1120210379),i=w(i,a,n,r,e[2],15,718787259),r=w(r,i,a,n,e[9],21,-343485551),
t[0]=k(n,t[0]),t[1]=k(r,t[1]),t[2]=k(i,t[2]),t[3]=k(a,t[3])}function g(t,e,n,r,i,a){return e=k(k(e,t),k(r,a)),k(e<<i|e>>>32-i,n)}function m(t,e,n,r,i,a,s){return g(e&n|~e&r,t,e,i,a,
s)}function b(t,e,n,r,i,a,s){return g(e&r|n&~r,t,e,i,a,s)}function v(t,e,n,r,i,a,s){return g(e^n^r,t,e,i,a,s)}function w(t,e,n,r,i,a,s){return g(n^(e|~r),t,e,i,a,s)}function y(t){var e,n=t.
length,r=[1732584193,-271733879,-1732584194,271733878];for(e=64;e<=t.length;e+=64)p(r,_(t.substring(e-64,e)));t=t.substring(e-64);var i=[0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0];for(e=0;e<t.length;
e++)i[e>>2]|=t.charCodeAt(e)<<(e%4<<3);if(i[e>>2]|=128<<(e%4<<3),e>55)for(p(r,i),e=0;e<16;e++)i[e]=0;return i[14]=8*n,p(r,i),r}function _(t){var e,n=[];for(e=0;e<64;e+=4)n[e>>2]=t.
charCodeAt(e)+(t.charCodeAt(e+1)<<8)+(t.charCodeAt(e+2)<<16)+(t.charCodeAt(e+3)<<24);return n}var x="0123456789abcdef".split("");function A(t){for(var e="",n=0;n<4;n++)e+=x[t>>8*n+4&15]+x[
t>>8*n&15];return e}function L(t){return String.fromCharCode(255&t,(65280&t)>>8,(16711680&t)>>16,(4278190080&t)>>24)}function N(t){return function(t){return t.map(L).join("")}(y(t))}var
S="5d41402abc4b2a76b9719d911017c592"!=function(t){for(var e=0;e<t.length;e++)t[e]=A(t[e]);return t.join("")}(y("hello"));function k(t,e){if(S){var n=(65535&t)+(65535&e);return(t>>16)+(
e>>16)+(n>>16)<<16|65535&n}return t+e&4294967295}
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   *
   * Reference: http://www.fpdf.org/en/script/script37.php
   */function P(t,e){var n,r,i,a;if(t!==n){for(var s=(i=t,a=1+(256/t.length|0),new Array(a+1).join(i)),o=[],h=0;h<256;h++)o[h]=h;var l=0;for(h=0;h<256;h++){var c=o[h];l=(l+c+s.charCodeAt(
h))%256,o[h]=o[l],o[l]=c}n=t,r=o}else o=r;var u=e.length,f=0,d=0,p="";for(h=0;h<u;h++)d=(d+(c=o[f=(f+1)%256]))%256,o[f]=o[d],o[d]=c,s=o[(o[f]+o[d])%256],p+=String.fromCharCode(e.charCodeAt(
h)^s);return p}
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   * Author: Owen Leong (@owenl131)
   * Date: 15 Oct 2020
   * References:
   * https://www.cs.cmu.edu/~dst/Adobe/Gallery/anon21jul01-pdf-encryption.txt
   * https://github.com/foliojs/pdfkit/blob/master/lib/security.js
   * http://www.fpdf.org/en/script/script37.php
   */var F={print:4,modify:8,copy:16,"annot-forms":32};function I(t,e,n,r){this.v=1,this.r=2;var i=192;t.forEach(function(t){if(void 0!==F.perm)throw new Error("Invalid permission: "+t);
i+=F[t]}),this.padding="(¿N^Nu■Ad\0NVÿü■\b..\0¶Ðh>■/\f©þdSiz";var a=(e+this.padding).substr(0,32),s=(n+this.padding).substr(0,32);this.O=this.processOwnerPassword(a,s),this.P=-(1+(255^i)),
this.encryptionKey=N(a+this.O+this.lsbFirstWord(this.P)+this.hexToBytes(r)).substr(0,5),this.U=P(this.encryptionKey,this.padding)}function C(t){if(/[^\u0000-\u00ff]/.test(t))throw new Error(
"Invalid PDF Name Object: "+t+", Only accept ASCII characters.");for(var e="",n=t.length,r=0;r<n;r++){var i=t.charCodeAt(r);
e+=i<33||35===i||37===i||40===i||41===i||47===i||60===i||62===i||91===i||93===i||123===i||125===i||i>126?"#"+("0"+i.toString(16)).slice(-2):t[r]}return e}function j(t){if("object"!==r(
t))throw new Error("Invalid Context passed to initialize PubSub (jsPDF-module)");var e={};this.subscribe=function(t,n,r){if(r=r||!1,"string"!=typeof t||"function"!=typeof
n||"boolean"!=typeof r)throw new Error("Invalid arguments passed to PubSub.subscribe (jsPDF-module)");e.hasOwnProperty(t)||(e[t]={});var i=Math.random().toString(35);return e[t][i]=[n,!!r],
i},this.unsubscribe=function(t){for(var n in e)if(e[n][t])return delete e[n][t],0===Object.keys(e[n]).length&&delete e[n],!0;return!1},this.publish=function(n){if(e.hasOwnProperty(n)){var
r=Array.prototype.slice.call(arguments,1),a=[];for(var o in e[n]){var h=e[n][o];try{h[0].apply(t,r)}catch(l){i.console&&s.error("jsPDF PubSub Error",l.message,l)}h[1]&&a.push(o)}a.length&&a.
forEach(this.unsubscribe)}},this.getTopics=function(){return e}}function E(t){if(!(this instanceof E))return new E(t);var e="opacity,stroke-opacity".split(",");for(var n in t)t.
hasOwnProperty(n)&&e.indexOf(n)>=0&&(this[n]=t[n]);this.id="",this.objectNumber=-1}function O(t,e){this.gState=t,this.matrix=e,this.id="",this.objectNumber=-1}function B(t,e,n,r,i){if(!(
this instanceof B))return new B(t,e,n,r,i);this.type="axial"===t?2:3,this.coords=e,this.colors=n,O.call(this,r,i)}function M(t,e,n,r,i){if(!(this instanceof M))return new M(t,e,n,r,i);this.
boundingBox=t,this.xStep=e,this.yStep=n,this.stream="",this.cloneIndex=0,O.call(this,r,i)}function R(t){var e,n="string"==typeof arguments[0]?arguments[0]:"p",a=arguments[1],o=arguments[2],
h=arguments[3],l=[],f=1,p=16,g="S",m=null;"object"===r(t=t||{})&&(n=t.orientation,a=t.unit||a,o=t.format||o,h=t.compress||t.compressPdf||h,null!==(m=t.encryption||null)&&(m.userPassword=m.
userPassword||"",m.ownerPassword=m.ownerPassword||"",m.userPermissions=m.userPermissions||[]),f="number"==typeof t.userUnit?Math.abs(t.userUnit):1,void 0!==t.precision&&(e=t.precision),void
0!==t.floatPrecision&&(p=t.floatPrecision),g=t.defaultPathOperation||"S"),l=t.filters||(!0===h?["FlateEncode"]:l),a=a||"mm",n=(""+(n||"P")).toLowerCase();var b=t.putOnlyUsedFonts||!1,v={},
w={internal:{},__private__:{}};w.__private__.PubSub=j;var y="1.3",_=w.__private__.getPdfVersion=function(){return y};w.__private__.setPdfVersion=function(t){y=t};var x={a0:[2383.94,3370.39],
a1:[1683.78,2383.94],a2:[1190.55,1683.78],a3:[841.89,1190.55],a4:[595.28,841.89],a5:[419.53,595.28],a6:[297.64,419.53],a7:[209.76,297.64],a8:[147.4,209.76],a9:[104.88,147.4],a10:[73.7,104.
88],b0:[2834.65,4008.19],b1:[2004.09,2834.65],b2:[1417.32,2004.09],b3:[1000.63,1417.32],b4:[708.66,1000.63],b5:[498.9,708.66],b6:[354.33,498.9],b7:[249.45,354.33],b8:[175.75,249.45],b9:[124.
72,175.75],b10:[87.87,124.72],c0:[2599.37,3676.54],c1:[1836.85,2599.37],c2:[1298.27,1836.85],c3:[918.43,1298.27],c4:[649.13,918.43],c5:[459.21,649.13],c6:[323.15,459.21],c7:[229.61,323.15],
c8:[161.57,229.61],c9:[113.39,161.57],c10:[79.37,113.39],dl:[311.81,623.62],letter:[612,792],"government-letter":[576,756],legal:[612,1008],"junior-legal":[576,360],ledger:[1224,792],
tabloid:[792,1224],"credit-card":[153,243]};w.__private__.getPageFormats=function(){return x};var A=w.__private__.getPageFormat=function(t){return x[t]};o=o||"a4";var L="compat",
N="advanced",S=L;function k(){this.saveGraphicsState(),ct(new Wt(Nt,0,0,-Nt,0,Pn()*Nt).toString()+" cm"),this.setFontSize(this.getFontSize()/Nt),g="n",S=N}function P(){this.
restoreGraphicsState(),g="S",S=L}var F=w.__private__.combineFontStyleAndFontWeight=function(t,e){if(
"bold"==t&&"normal"==e||"bold"==t&&400==e||"normal"==t&&"italic"==e||"bold"==t&&"italic"==e)throw new Error("Invalid Combination of fontweight and fontstyle");return e&&(
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t=400==e||"normal"===e?"italic"===t?"italic":"normal":700!=e&&"bold"!==e||"normal"!==t?(700==e?"bold":e)+""+t:"bold"),t};w.advancedAPI=function(t){var e=S===L;return e&&k.call(this),
"function"!=typeof t||(t(this),e&&P.call(this)),this},w.compatAPI=function(t){var e=S===N;return e&&P.call(this),"function"!=typeof t||(t(this),e&&k.call(this)),this},w.
isAdvancedAPI=function(){return S===N};var O,T=function(t){if(S!==N)throw new Error(t+" is only available in 'advanced' API mode. You need to call advancedAPI() first.")},D=w.
roundToPrecision=w.__private__.roundToPrecision=function(t,n){var r=e||n;if(isNaN(t)||isNaN(r))throw new Error("Invalid argument passed to jsPDF.roundToPrecision");return t.toFixed(r).
replace(/0+$/,"")};O=w.hpf=w.__private__.hpf="number"==typeof p?function(t){if(isNaN(t))throw new Error("Invalid argument passed to jsPDF.hpf");return D(t,p)}:"smart"===p?function(t){if(
isNaN(t))throw new Error("Invalid argument passed to jsPDF.hpf");return D(t,t>-1&&t<1?16:5)}:function(t){if(isNaN(t))throw new Error("Invalid argument passed to jsPDF.hpf");return D(t,16)};
var q=w.f2=w.__private__.f2=function(t){if(isNaN(t))throw new Error("Invalid argument passed to jsPDF.f2");return D(t,2)},z=w.__private__.f3=function(t){if(isNaN(t))throw new Error("Invalid
argument passed to jsPDF.f3");return D(t,3)},U=w.scale=w.__private__.scale=function(t){if(isNaN(t))throw new Error("Invalid argument passed to jsPDF.scale");return S===L?t*Nt:S===N?t:void
0},H=function(t){return U(function(t){return S===L?Pn()-t:S===N?t:void 0}(t))};w.__private__.setPrecision=w.setPrecision=function(t){"number"==typeof parseInt(t,10)&&(e=parseInt(t,10))};var
W,V="00000000000000000000000000000000",G=w.__private__.getFileId=function(){return V},Y=w.__private__.setFileId=function(t){return V=void 0!==t&&/^[a-fA-F0-9]{32}$/.test(t)?t.toUpperCase():
V.split("").map(function(){return"ABCDEF0123456789".charAt(Math.floor(16*Math.random()))}).join(""),null!==m&&(Ee=new I(m.userPermissions,m.userPassword,m.ownerPassword,V)),V};w.
setFileId=function(t){return Y(t),this},w.getFileId=function(){return G()};var Z=w.__private__.convertDateToPDFDate=function(t){var e=t.getTimezoneOffset(),n=e<0?"+":"-",r=Math.floor(Math.
abs(e/60)),i=Math.abs(e%60),a=[n,Q(r),"'",Q(i),"'"].join("");return["D:",t.getFullYear(),Q(t.getMonth()+1),Q(t.getDate()),Q(t.getHours()),Q(t.getMinutes()),Q(t.getSeconds()),a].join("")},
J=w.__private__.convertPDFDateToDate=function(t){var e=parseInt(t.substr(2,4),10),n=parseInt(t.substr(6,2),10)-1,r=parseInt(t.substr(8,2),10),i=parseInt(t.substr(10,2),10),a=parseInt(t.
substr(12,2),10),s=parseInt(t.substr(14,2),10);return new Date(e,n,r,i,a,s,0)},X=w.__private__.setCreationDate=function(t){var e;if(void 0===t&&(t=new Date),t instanceof Date)e=Z(t);else{if(
!/^D:(20[0-2][0-9]|203[0-7]|19[7-9][0-9])(0[0-9]|1[0-2])([0-2][0-9]|3[0-1])(0[0-9]|1[0-9]|2[0-3])(0[0-9]|[1-5][0-9])(0[0-9]|[1-5][0-9])(\+0[0-9]|\+1[0-4]|-0[0-9]|-1[0-1])'(0[0-9]|[1-5][0-
9])'?$/.test(t))throw new Error("Invalid argument passed to jsPDF.setCreationDate");e=t}return W=e},K=w.__private__.getCreationDate=function(t){var e=W;return"jsDate"===t&&(e=J(W)),e};w.
setCreationDate=function(t){return X(t),this},w.getCreationDate=function(t){return K(t)};var $,Q=w.__private__.padd2=function(t){return("0"+parseInt(t)).slice(-2)},tt=w.__private__.
padd2Hex=function(t){return("00"+(t=t.toString())).substr(t.length)},et=0,nt=[],rt=[],it=0,at=[],st=[],ot=!1,ht=rt;w.__private__.setCustomOutputDestination=function(t){ot=!0,ht=t};var
lt=function(t){ot||(ht=t)};w.__private__.resetCustomOutputDestination=function(){ot=!1,ht=rt};var ct=w.__private__.out=function(t){return t=t.toString(),it+=t.length+1,ht.push(t),ht},ut=w.
__private__.write=function(t){return ct(1===arguments.length?t.toString():Array.prototype.join.call(arguments," "))},ft=w.__private__.getArrayBuffer=function(t){for(var e=t.length,n=new
ArrayBuffer(e),r=new Uint8Array(n);e--;)r[e]=t.charCodeAt(e);return n},dt=[["Helvetica","helvetica","normal","WinAnsiEncoding"],["Helvetica-Bold","helvetica","bold","WinAnsiEncoding"],[
"Helvetica-Oblique","helvetica","italic","WinAnsiEncoding"],["Helvetica-BoldOblique","helvetica","bolditalic","WinAnsiEncoding"],["Courier","courier","normal","WinAnsiEncoding"],["Courier-
Bold","courier","bold","WinAnsiEncoding"],["Courier-Oblique","courier","italic","WinAnsiEncoding"],["Courier-BoldOblique","courier","bolditalic","WinAnsiEncoding"],["Times-Roman","times",
"normal","WinAnsiEncoding"],["Times-Bold","times","bold","WinAnsiEncoding"],["Times-Italic","times","italic","WinAnsiEncoding"],["Times-BoldItalic","times","bolditalic","WinAnsiEncoding"],[
"ZapfDingbats","zapfdingbats","normal",null],["Symbol","symbol","normal",null]];w.__private__.getStandardFonts=function(){return dt};var pt=t.fontSize||16;w.__private__.setFontSize=w.
setFontSize=function(t){return pt=S===N?t/Nt:t,this};var gt,mt=w.__private__.getFontSize=function(){return S===L?pt:pt*Nt},bt=t.R2L||!1;w.__private__.setR2L=w.setR2L=function(
t){return bt=t,this},w.__private__.getR2L=w.getR2L=function(){return bt};var vt,wt=w.__private__.setZoomMode=function(t){if(/^(?:\d+\.\d*|\d*\.\d+|\d+)%$/.test(t))gt=t;else if(isNaN(t)){if(-
1===[void 0,null,"fullwidth","fullheight","fullpage","original"].indexOf(t))throw new Error('zoom must be Integer (e.g. 2), a percentage Value (e.g. 300%) or fullwidth, fullheight, fullpage,
 original. "'+t+'" is not recognized.');gt=t}else gt=parseInt(t,10)};w.__private__.getZoomMode=function(){return gt};var yt,_t=w.__private__.setPageMode=function(t){if(-1==[void 0,null,
"UseNone","UseOutlines","UseThumbs","FullScreen"].indexOf(t))throw new Error('Page mode must be one of UseNone, UseOutlines, UseThumbs, or FullScreen. "'+t+'" is not recognized.');vt=t};w.
__private__.getPageMode=function(){return vt};var xt=w.__private__.setLayoutMode=function(t){if(-1==[void 0,null,"continuous","single","twoleft","tworight","two"].indexOf(t))throw new Error(
'Layout mode must be one of continuous, single, twoleft, tworight. "'+t+'" is not recognized.');yt=t};w.__private__.getLayoutMode=function(){return yt},w.__private__.setDisplayMode=w.
setDisplayMode=function(t,e,n){return wt(t),xt(e),_t(n),this};var At={title:"",subject:"",author:"",keywords:"",creator:""};w.__private__.getDocumentProperty=function(t){if(-1===Object.keys(
At).indexOf(t))throw new Error("Invalid argument passed to jsPDF.getDocumentProperty");return At[t]},w.__private__.getDocumentProperties=function(){return At},w.__private__.
setDocumentProperties=w.setProperties=w.setDocumentProperties=function(t){for(var e in At)At.hasOwnProperty(e)&&t[e]&&(At[e]=t[e]);return this},w.__private__.setDocumentProperty=function(t,
e){if(-1===Object.keys(At).indexOf(t))throw new Error("Invalid arguments passed to jsPDF.setDocumentProperty");return At[t]=e};var Lt,Nt,St,kt,Pt,Ft={},It={},Ct=[],jt={},Et={},Ot={},Bt={},
Mt=null,Rt=0,Tt=[],Dt=new j(w),qt=t.hotfixes||[],zt={},Ut={},Ht=[],Wt=function t(e,n,r,i,a,s){if(!(this instanceof t))return new t(e,n,r,i,a,s);isNaN(e)&&(e=1),isNaN(n)&&(n=0),isNaN(r)&&(
r=0),isNaN(i)&&(i=1),isNaN(a)&&(a=0),isNaN(s)&&(s=0),this._matrix=[e,n,r,i,a,s]};Object.defineProperty(Wt.prototype,"sx",{get:function(){return this._matrix[0]},set:function(t){this._matrix[
0]=t}}),Object.defineProperty(Wt.prototype,"shy",{get:function(){return this._matrix[1]},set:function(t){this._matrix[1]=t}}),Object.defineProperty(Wt.prototype,"shx",{get:function(){return
this._matrix[2]},set:function(t){this._matrix[2]=t}}),Object.defineProperty(Wt.prototype,"sy",{get:function(){return this._matrix[3]},set:function(t){this._matrix[3]=t}}),Object.
defineProperty(Wt.prototype,"tx",{get:function(){return this._matrix[4]},set:function(t){this._matrix[4]=t}}),Object.defineProperty(Wt.prototype,"ty",{get:function(){return this._matrix[5]},
set:function(t){this._matrix[5]=t}}),Object.defineProperty(Wt.prototype,"a",{get:function(){return this._matrix[0]},set:function(t){this._matrix[0]=t}}),Object.defineProperty(Wt.prototype,
"b",{get:function(){return this._matrix[1]},set:function(t){this._matrix[1]=t}}),Object.defineProperty(Wt.prototype,"c",{get:function(){return this._matrix[2]},set:function(t){this._matrix[
2]=t}}),Object.defineProperty(Wt.prototype,"d",{get:function(){return this._matrix[3]},set:function(t){this._matrix[3]=t}}),Object.defineProperty(Wt.prototype,"e",{get:function(){return
this._matrix[4]},set:function(t){this._matrix[4]=t}}),Object.defineProperty(Wt.prototype,"f",{get:function(){return this._matrix[5]},set:function(t){this._matrix[5]=t}}),Object.
defineProperty(Wt.prototype,"rotation",{get:function(){return Math.atan2(this.shx,this.sx)}}),Object.defineProperty(Wt.prototype,"scaleX",{get:function(){return this.decompose().scale.sx}}),
Object.defineProperty(Wt.prototype,"scaleY",{get:function(){return this.decompose().scale.sy}}),Object.defineProperty(Wt.prototype,"isIdentity",{get:function(){return 1===this.sx&&0===this.
shy&&0===this.shx&&1===this.sy&&0===this.tx&&0===this.ty}}),Wt.prototype.join=function(t){return[this.sx,this.shy,this.shx,this.sy,this.tx,this.ty].map(O).join(t)},Wt.prototype.
multiply=function(t){var e=t.sx*this.sx+t.shy*this.shx,n=t.sx*this.shy+t.shy*this.sy,r=t.shx*this.sx+t.sy*this.shx,i=t.shx*this.shy+t.sy*this.sy,a=t.tx*this.sx+t.ty*this.shx+this.tx,s=t.
tx*this.shy+t.ty*this.sy+this.ty;return new Wt(e,n,r,i,a,s)},Wt.prototype.decompose=function(){var t=this.sx,e=this.shy,n=this.shx,r=this.sy,i=this.tx,a=this.ty,s=Math.sqrt(t*t+e*e),o=(t/
=s)*n+(e/=s)*r;n-=t*o,r-=e*o;var h=Math.sqrt(n*n+r*r);return o/=h,t*(r/=h)<e*(n/=h)&&(t=-t,e=-e,o=-o,s=-s),{scale:new Wt(s,0,0,h,0,0),translate:new Wt(1,0,0,1,i,a),rotate:new Wt(t,e,-e,t,0,
0),skew:new Wt(1,0,o,1,0,0)}},Wt.prototype.toString=function(t){return this.join(" ")},Wt.prototype.inversed=function(){var t=this.sx,e=this.shy,n=this.shx,r=this.sy,i=this.tx,a=this.ty,s=1/
(t*r-e*n),o=r*s,h=-e*s,l=-n*s,c=t*s;return new Wt(o,h,l,c,-o*i-l*a,-h*i-c*a)},Wt.prototype.applyToPoint=function(t){var e=t.x*this.sx+t.y*this.shx+this.tx,n=t.x*this.shy+t.y*this.sy+this.ty;
return new bn(e,n)},Wt.prototype.applyToRectangle=function(t){var e=this.applyToPoint(t),n=this.applyToPoint(new bn(t.x+t.w,t.y+t.h));return new vn(e.x,e.y,n.x-e.x,n.y-e.y)},Wt.prototype.
clone=function(){var t=this.sx,e=this.shy,n=this.shx,r=this.sy,i=this.tx,a=this.ty;return new Wt(t,e,n,r,i,a)},w.Matrix=Wt;var Vt=w.matrixMult=function(t,e){return e.multiply(t)},Gt=new Wt(
1,0,0,1,0,0);w.unitMatrix=w.identityMatrix=Gt;var Yt=function(t,e){if(!Et[t]){var n=(e instanceof B?"Sh":"P")+(Object.keys(jt).length+1).toString(10);e.id=n,Et[t]=n,jt[n]=e,Dt.publish(
"addPattern",e)}};w.ShadingPattern=B,w.TilingPattern=M,w.addShadingPattern=function(t,e){return T("addShadingPattern()"),Yt(t,e),this},w.beginTilingPattern=function(t){T("beginTilingPattern(
)"),yn(t.boundingBox[0],t.boundingBox[1],t.boundingBox[2]-t.boundingBox[0],t.boundingBox[3]-t.boundingBox[1],t.matrix)},w.endTilingPattern=function(t,e){T("endTilingPattern()"),e.stream=st[
$].join("\n"),Yt(t,e),Dt.publish("endTilingPattern",e),Ht.pop().restore()};var Zt,Jt=w.__private__.newObject=function(){var t=Xt();return Kt(t,!0),t},Xt=w.__private__.
newObjectDeferred=function(){return et++,nt[et]=function(){return it},et},Kt=function(t,e){return e="boolean"==typeof e&&e,nt[t]=it,e&&ct(t+" 0 obj"),t},$t=w.__private__.
newAdditionalObject=function(){var t={objId:Xt(),content:""};return at.push(t),t},Qt=Xt(),te=Xt(),ee=w.__private__.decodeColorString=function(t){var e=t.split(" ");if(2!==e.length||"g"!==e[
1]&&"G"!==e[1])5!==e.length||"k"!==e[4]&&"K"!==e[4]||(e=[(1-e[0])*(1-e[3]),(1-e[1])*(1-e[3]),(1-e[2])*(1-e[3]),"r"]);else{var n=parseFloat(e[0]);e=[n,n,n,"r"]}for(var r="#",i=0;i<3;i++)r+=(
"0"+Math.floor(255*parseFloat(e[i])).toString(16)).slice(-2);return r},ne=w.__private__.encodeColorString=function(t){var e;"string"==typeof t&&(t={ch1:t});var n=t.ch1,i=t.ch2,a=t.ch3,s=t.
ch4,o="draw"===t.pdfColorType?["G","RG","K"]:["g","rg","k"];if("string"==typeof n&&"#"!==n.charAt(0)){var h=new u(n);if(h.ok)n=h.toHex();else if(!/^\d*\.?\d*$/.test(n))throw new Error(
'Invalid color "'+n+'" passed to jsPDF.encodeColorString.')}if("string"==typeof n&&/^#[0-9A-Fa-f]{3}$/.test(n)&&(n="#"+n[1]+n[1]+n[2]+n[2]+n[3]+n[3]),"string"==typeof n&&/^#[0-9A-Fa-f]{6}$/.
test(n)){var l=parseInt(n.substr(1),16);n=l>>16&255,i=l>>8&255,a=255&l}if(void 0===i||void 0===s&&n===i&&i===a)e="string"==typeof n?n+" "+o[0]:2===t.precision?q(n/255)+" "+o[0]:z(n/255)+"
"+o[0];else if(void 0===s||"object"===r(s)){if(s&&!isNaN(s.a)&&0===s.a)return["1.","1.","1.",o[1]].join(" ");e="string"==typeof n?[n,i,a,o[1]].join(" "):2===t.precision?[q(n/255),q(i/255),q(
a/255),o[1]].join(" "):[z(n/255),z(i/255),z(a/255),o[1]].join(" ")}else e="string"==typeof n?[n,i,a,s,o[2]].join(" "):2===t.precision?[q(n),q(i),q(a),q(s),o[2]].join(" "):[z(n),z(i),z(a),z(
s),o[2]].join(" ");return e},re=w.__private__.getFilters=function(){return l},ie=w.__private__.putStream=function(t){var e=(t=t||{}).data||"",n=t.filters||re(),r=t.alreadyAppliedFilters||[],
i=t.addLength1||!1,a=e.length,s=t.objectId,o=function(t){return t};if(null!==m&&void 0===s)throw new Error("ObjectId must be passed to putStream for file encryption");null!==m&&(o=Ee.
encryptor(s,0));var h={};!0===n&&(n=["FlateEncode"]);var l=t.additionalKeyValues||[],c=(h=void 0!==R.API.processDataByFilters?R.API.processDataByFilters(e,n):{data:e,reverseChain:[]}).
reverseChain+(Array.isArray(r)?r.join(" "):r.toString());if(0!==h.data.length&&(l.push({key:"Length",value:h.data.length}),!0===i&&l.push({key:"Length1",value:a})),0!=c.length)if(c.split("/
").length-1==1)l.push({key:"Filter",value:c});else{l.push({key:"Filter",value:"["+c+"]"});for(var u=0;u<l.length;u+=1)if("DecodeParms"===l[u].key){for(var f=[],d=0;d<h.reverseChain.split("/
").length-1;d+=1)f.push("null");f.push(l[u].value),l[u].value="["+f.join(" ")+"]"}}ct("<<");for(var p=0;p<l.length;p++)ct("/"+l[p].key+" "+l[p].value);ct(">>"),0!==h.data.length&&(ct(
"stream"),ct(o(h.data)),ct("endstream"))},ae=w.__private__.putPage=function(t){var e=t.number,n=t.data,r=t.objId,i=t.contentsObjId;Kt(r,!0),ct("<</Type /Page"),ct("/Parent "+t.
rootDictionaryObjId+" 0 R"),ct("/Resources "+t.resourceDictionaryObjId+" 0 R"),ct("/MediaBox ["+parseFloat(O(t.mediaBox.bottomLeftX))+" "+parseFloat(O(t.mediaBox.bottomLeftY))+" "+O(t.
mediaBox.topRightX)+" "+O(t.mediaBox.topRightY)+"]"),null!==t.cropBox&&ct("/CropBox ["+O(t.cropBox.bottomLeftX)+" "+O(t.cropBox.bottomLeftY)+" "+O(t.cropBox.topRightX)+" "+O(t.cropBox.
topRightY)+"]"),null!==t.bleedBox&&ct("/BleedBox ["+O(t.bleedBox.bottomLeftX)+" "+O(t.bleedBox.bottomLeftY)+" "+O(t.bleedBox.topRightX)+" "+O(t.bleedBox.topRightY)+"]"),null!==t.trimBox&&ct(
"/TrimBox ["+O(t.trimBox.bottomLeftX)+" "+O(t.trimBox.bottomLeftY)+" "+O(t.trimBox.topRightX)+" "+O(t.trimBox.topRightY)+"]"),null!==t.artBox&&ct("/ArtBox ["+O(t.artBox.bottomLeftX)+" "+O(t.
artBox.bottomLeftY)+" "+O(t.artBox.topRightX)+" "+O(t.artBox.topRightY)+"]"),"number"==typeof t.userUnit&&1!==t.userUnit&&ct("/UserUnit "+t.userUnit),Dt.publish("putPage",{objId:r,
pageContext:Tt[e],pageNumber:e,page:n}),ct("/Contents "+i+" 0 R"),ct(">>"),ct("endobj");var a=n.join("\n");return S===N&&(a+="\nQ"),Kt(i,!0),ie({data:a,filters:re(),objectId:i}),ct(
"endobj"),r},se=w.__private__.putPages=function(){var t,e,n=[];for(t=1;t<=Rt;t++)Tt[t].objId=Xt(),Tt[t].contentsObjId=Xt();for(t=1;t<=Rt;t++)n.push(ae({number:t,data:st[t],objId:Tt[t].objId,
contentsObjId:Tt[t].contentsObjId,mediaBox:Tt[t].mediaBox,cropBox:Tt[t].cropBox,bleedBox:Tt[t].bleedBox,trimBox:Tt[t].trimBox,artBox:Tt[t].artBox,userUnit:Tt[t].userUnit,rootDictionaryObjId:
Qt,resourceDictionaryObjId:te}));Kt(Qt,!0),ct("<</Type /Pages");var r="/Kids [";for(e=0;e<Rt;e++)r+=n[e]+" 0 R ";ct(r+"]"),ct("/Count "+Rt),ct(">>"),ct("endobj"),Dt.publish("postPutPages")},
oe=function(t){Dt.publish("putFont",{font:t,out:ct,newObject:Jt,putStream:ie}),!0!==t.isAlreadyPutted&&(t.objectNumber=Jt(),ct("<<"),ct("/Type /Font"),ct("/BaseFont /"+C(t.postScriptName)),
ct("/Subtype /Type1"),"string"==typeof t.encoding&&ct("/Encoding /"+t.encoding),ct("/FirstChar 32"),ct("/LastChar 255"),ct(">>"),ct("endobj"))},he=function(t){t.objectNumber=Jt();var e=[];e.
```

```
push({key:"Type",value:"/XObject"}),e.push({key:"Subtype",value:"/Form"}),e.push({key:"BBox",value:"["+[O(t.x),O(t.y),O(t.x+t.width),O(t.y+t.height)].join(" ")+"]"}),e.push({key:"Matrix",
value:"["+t.matrix.toString()+"]"});var n=t.pages[1].join("\n");ie({data:n,additionalKeyValues:e,objectId:t.objectNumber}),ct("endobj")},le=function(t,e){e||(e=21);var n=Jt(),r=function(t,
e){var n,r=[],i=1/(e-1);for(n=0;n<1;n+=i)r.push(n);if(r.push(1),0!=t[0].offset){var a={offset:0,color:t[0].color};t.unshift(a)}if(1!=t[t.length-1].offset){var s={offset:1,color:t[t.length-
1].color};t.push(s)}for(var o="",h=0,l=0;l<r.length;l++){for(n=r[l];n>t[h+1].offset;)h++;var c=t[h].offset,u=(n-c)/(t[h+1].offset-c),f=t[h].color,d=t[h+1].color;o+=tt(Math.round((1-u)*f[
0]+u*d[0]).toString(16))+tt(Math.round((1-u)*f[1]+u*d[1]).toString(16))+tt(Math.round((1-u)*f[2]+u*d[2]).toString(16))}return o.trim()}(t.colors,e),i=[];i.push({key:"FunctionType",value:
"0"}),i.push({key:"Domain",value:"[0.0 1.0]"}),i.push({key:"Size",value:"["+e+"]"}),i.push({key:"BitsPerSample",value:"8"}),i.push({key:"Range",value:"[0.0 1.0 0.0 1.0 0.0 1.0]"}),i.push({
key:"Decode",value:"[0.0 1.0 0.0 1.0 0.0 1.0]"}),ie({data:r,additionalKeyValues:i,alreadyAppliedFilters:["/ASCIIHexDecode"],objectId:n}),ct("endobj"),t.objectNumber=Jt(),ct("<< /ShadingType
"+t.type),ct("/ColorSpace /DeviceRGB");var a="/Coords ["+O(parseFloat(t.coords[0]))+" "+O(parseFloat(t.coords[1]))+" ";2===t.type?a+=O(parseFloat(t.coords[2]))+" "+O(parseFloat(t.coords[
3])):a+=O(parseFloat(t.coords[2]))+" "+O(parseFloat(t.coords[3]))+" "+O(parseFloat(t.coords[4]))+" "+O(parseFloat(t.coords[5])),ct(a+="]"),t.matrix&&ct("/Matrix ["+t.matrix.toString()+"]"),
ct("/Function "+n+" 0 R"),ct("/Extend [true true]"),ct(">>"),ct("endobj")},ce=function(t,e){var n=Xt(),r=Jt();e.push({resourcesOid:n,objectOid:r}),t.objectNumber=r;var i=[];i.push({key:
"Type",value:"/Pattern"}),i.push({key:"PatternType",value:"1"}),i.push({key:"PaintType",value:"1"}),i.push({key:"TilingType",value:"1"}),i.push({key:"BBox",value:"["+t.boundingBox.map(O).
join(" ")+"]"}),i.push({key:"XStep",value:O(t.xStep)}),i.push({key:"YStep",value:O(t.yStep)}),i.push({key:"Resources",value:n+" 0 R"}),t.matrix&&i.push({key:"Matrix",value:"["+t.matrix.
toString()+"]"}),ie({data:t.stream,additionalKeyValues:i,objectId:t.objectNumber}),ct("endobj")},ue=function(t){for(var e in t)switch(e){case"opacity":ct("/ca
"+q(t[e]));break;case"stroke-opacity":ct("/CA "+q(t[e]))}ct(">>"),ct("endobj")},fe=function(t){Kt(t.resourcesOid,!0),ct("<<"),ct("/ProcSet [/PDF /Text /ImageB /ImageC /ImageI]"),function(){
for(var t in ct("/Font <<"),Ft)Ft.hasOwnProperty(t)&&(!1===b||!0===b&&v.hasOwnProperty(t))&&ct("/"+t+" "+Ft[t].objectNumber+" 0 R");ct(">>")}(),function(){if(Object.keys(jt).length>0){for(
var t in ct("/Shading <<"),jt)jt.hasOwnProperty(t)&&jt[t]instanceof B&&jt[t].objectNumber>=0&&ct("/"+t+" "+jt[t].objectNumber+" 0 R");Dt.publish("putShadingPatternDict"),ct(">>")}}(),
function(t){if(Object.keys(jt).length>0){for(var e in ct("/Pattern <<"),jt)jt.hasOwnProperty(e)&&jt[e]instanceof w.TilingPattern&&jt[e].objectNumber>=0&&jt[e].objectNumber<t&&ct("/"+e+"
"+jt[e].objectNumber+" 0 R");Dt.publish("putTilingPatternDict"),ct(">>")}}(t.objectOid),function(){if(Object.keys(Ot).length>0){var t;for(t in ct("/ExtGState <<"),Ot)Ot.hasOwnProperty(
t)&&Ot[t].objectNumber>=0&&ct("/"+t+" "+Ot[t].objectNumber+" 0 R");Dt.publish("putGStateDict"),ct(">>")}}(),function(){for(var t in ct("/XObject <<"),zt)zt.hasOwnProperty(t)&&zt[t].
objectNumber>=0&&ct("/"+t+" "+zt[t].objectNumber+" 0 R");Dt.publish("putXobjectDict"),ct(">>")}(),ct(">>"),ct("endobj")},de=function(t){It[t.fontName]=It[t.fontName]||{},It[t.fontName][t.
fontStyle]=t.id},pe=function(t,e,n,r,i){var a={id:"F"+(Object.keys(Ft).length+1).toString(10),postScriptName:t,fontName:e,fontStyle:n,encoding:r,isStandardFont:i||!1,metadata:{}};return Dt.
publish("addFont",{font:a,instance:this}),Ft[a.id]=a,de(a),a.id},ge=w.__private__.pdfEscape=w.pdfEscape=function(t,e){return function(t,e){var n,r,i,a,s,o,h,l,c;if(i=(e=e||{}).
sourceEncoding||"Unicode",s=e.outputEncoding,(e.autoencode||s)&&Ft[Lt].metadata&&Ft[Lt].metadata[i]&&Ft[Lt].metadata[i].encoding&&(a=Ft[Lt].metadata[i].encoding,!s&&Ft[Lt].encoding&&(s=Ft[
Lt].encoding),!s&&a.codePages&&(s=a.codePages[0]),"string"==typeof s&&(s=a[s]),s)){for(h=!1,o=[],n=0,r=t.length;n<r;n++)(l=s[t.charCodeAt(n)])?o.push(String.fromCharCode(l)):o.push(t[n]),o[
n].charCodeAt(0)>>8&&(h=!0);t=o.join("")}for(n=t.length;void 0===h&&0!==n;)t.charCodeAt(n-1)>>8&&(h=!0),n--;if(!h)return t;for(o=e.noBOM?[]:[254,255],n=0,r=t.length;n<r;n++){if((c=(l=t.
charCodeAt(n))>>8)>>8)throw new Error("Character at position "+n+" of string '"+t+"' exceeds 16bits. Cannot be encoded into UCS-2 BE");o.push(c),o.push(l-(c<<8))}return String.fromCharCode.
apply(void 0,o)}(t,e).replace(/\\/g,"\\\\").replace(/\(/g,"\\(").replace(/\)/g,"\\)")},me=w.__private__.beginPage=function(t){st[++Rt]=[],Tt[Rt]={objId:0,contentsObjId:0,userUnit:Number(f),
artBox:null,bleedBox:null,cropBox:null,trimBox:null,mediaBox:{bottomLeftX:0,bottomLeftY:0,topRightX:Number(t[0]),topRightY:Number(t[1])}},we(Rt),lt(st[$])},be=function(t,e){var r,i,a;switch(
n=e||n,"string"==typeof t&&(r=A(t.toLowerCase()),Array.isArray(r)&&(i=r[0],a=r[1])),Array.isArray(t)&&(i=t[0]*Nt,a=t[1]*Nt),isNaN(i)&&(i=o[0],a=o[1]),(i>14400||a>14400)&&(s.warn("A page in
a PDF can not be wider or taller than 14400 userUnit. jsPDF limits the width/height to 14400"),i=Math.min(14400,i),a=Math.min(14400,a)),o=[i,a],n.substr(0,1)){case"l":a>i&&(o=[a,i]);break;
case"p":i>a&&(o=[a,i])}me(o),Qe(Ke),ct(hn),0!==pn&&ct(pn+" J"),0!==gn&&ct(gn+" j"),Dt.publish("addPage",{pageNumber:Rt})},ve=function(t){t>0&&t<=Rt&&(st.splice(t,1),Tt.splice(t,1),Rt--,
$>Rt&&($=Rt),this.setPage($))},we=function(t){t>0&&t<=Rt&&($=t)},ye=w.__private__.getNumberOfPages=w.getNumberOfPages=function(){return st.length-1},_e=function(t,e,n){var r,i=void 0;return
n=n||{},t=void 0!==t?t:Ft[Lt].fontName,e=void 0!==e?e:Ft[Lt].fontStyle,r=t.toLowerCase(),void 0!==It[r]&&void 0!==It[r][e]?i=It[r][e]:void 0!==It[t]&&void 0!==It[t][e]?i=It[t][e]:!1===n.
disableWarning&&s.warn("Unable to look up font label for font '"+t+"', '"+e+"'. Refer to getFontList() for available fonts."),i||n.noFallback||null==(i=It.times[e])&&(i=It.times.normal),i},
xe=w.__private__.putInfo=function(){var t=Jt(),e=function(t){return t};for(var n in null!==m&&(e=Ee.encryptor(t,0)),ct("<<"),ct("/Producer ("+ge(e("jsPDF "+R.version))+")"),At)At.
hasOwnProperty(n)&&At[n]&&ct("/"+n.substr(0,1).toUpperCase()+n.substr(1)+" ("+ge(e(At[n]))+")");ct("/CreationDate ("+ge(e(W))+")"),ct(">>"),ct("endobj")},Ae=w.__private__.
putCatalog=function(t){var e=(t=t||{}).rootDictionaryObjId||Qt;switch(Jt(),ct("<<"),ct("/Type /Catalog"),ct("/Pages "+e+" 0 R"),gt||(gt="fullwidth"),gt){case"fullwidth":ct("/OpenAction [3 0
R /FitH null]");break;case"fullheight":ct("/OpenAction [3 0 R /FitV null]");break;case"fullpage":ct("/OpenAction [3 0 R /Fit]");break;case"original":ct("/OpenAction [3 0 R /XYZ null null
1]");break;default:var n=""+gt;"%"===n.substr(n.length-1)&&(gt=parseInt(gt)/100),"number"==typeof gt&&ct("/OpenAction [3 0 R /XYZ null null "+q(gt)+"]")}switch(yt||(yt="continuous"),yt){
case"continuous":ct("/PageLayout /OneColumn");break;case"single":ct("/PageLayout /SinglePage");break;case"two":case"twoleft":ct("/PageLayout /TwoColumnLeft");break;case"tworight":ct("/
PageLayout /TwoColumnRight")}vt&&ct("/PageMode /"+vt),Dt.publish("putCatalog"),ct(">>"),ct("endobj")},Le=w.__private__.putTrailer=function(){ct("trailer"),ct("<<"),ct("/Size "+(et+1)),ct("/
Root "+et+" 0 R"),ct("/Info "+(et-1)+" 0 R"),null!==m&&ct("/Encrypt "+Ee.oid+" 0 R"),ct("/ID [ <"+V+"> <"+V+"> ]"),ct(">>")},Ne=w.__private__.putHeader=function(){ct("%PDF-"+y),ct("%"ß¬à")},
Se=w.__private__.putXRef=function(){var t="0000000000";ct("xref"),ct("0 "+(et+1)),ct("0000000000 65535 f ");for(var e=1;e<=et;e++)"function"==typeof nt[e]?ct((t+nt[e])).slice(-10)+" 00000
n "):void 0!==nt[e]?ct((t+nt[e]).slice(-10)+" 00000 n "):ct("0000000000 00000 n ")},ke=w.__private__.buildDocument=function(){var t;et=0,it=0,rt=[],nt=[],at=[],Qt=Xt(),te=Xt(),lt(rt),Dt.
publish("buildDocument"),Ne(),se(),function(){Dt.publish("putAdditionalObjects");for(var t=0;t<at.length;t++){var e=at[t];Kt(e.objId,!0),ct(e.content),ct("endobj")}Dt.publish(
"postPutAdditionalObjects")}(),t=[],function(){for(var t in Ft)Ft.hasOwnProperty(t)&&(!1===b||!0===b&&v.hasOwnProperty(t))&&oe(Ft[t])}(),function(){var t;for(t in Ot)Ot.hasOwnProperty(
t)&&ue(Ot[t])}(),function(){for(var t in zt)zt.hasOwnProperty(t)&&he(zt[t])}(),function(t){var e;for(e in jt)jt.hasOwnProperty(e)&&(jt[e]instanceof B?le(jt[e]):jt[e]instanceof M&&ce(jt[e],
t))}(t),Dt.publish("putResources"),t.forEach(fe),fe({resourcesOid:te,objectOid:Number.MAX_SAFE_INTEGER}),Dt.publish("postPutResources"),null!==m&&(Ee.oid=Jt(),ct("<<"),ct("/Filter /
Standard"),ct("/V "+Ee.v),ct("/R "+Ee.r),ct("/U <"+Ee.toHexString(Ee.U)+">"),ct("/O <"+Ee.toHexString(Ee.O)+">"),ct("/P "+Ee.P),ct(">>"),ct("endobj")),xe(),Ae();var e=it;return Se(),Le(),ct(
"startxref"),ct(""+e),ct("%%EOF"),lt(st[$]),rt.join("\n")},Pe=w.__private__.getBlob=function(t){return new Blob([ft(t)],{type:"application/pdf"})},Fe=function(t){for(;t.firstChild;)t.
removeChild(t.firstChild)},Ie=function(t){var e,n=t.document,r=n.documentElement,i=n.head,a=n.body;return i||(i=n.createElement("head"),r.appendChild(i)),a||(a=n.createElement("body"),r.
appendChild(a)),Fe(i),Fe(a),(e=n.createElement("style")).appendChild(n.createTextNode("html, body { padding: 0; margin: 0; } iframe { width: 100%; height: 100%; border: 0;}")),i.appendChild(
e),{document:n,body:a}},Ce=w.output=w.__private__.output=(Zt=function(t,e){switch("string"==typeof(e=e||{})?e={filename:e}:e.filename=e.filename||"generated.pdf",t){case void 0:return ke();
case"save":w.save(e.filename);break;case"arraybuffer":return ft(ke());case"blob":return Pe(ke());case"bloburi":case"bloburl":if(void 0!==i.URL&&"function"==typeof i.URL.
createObjectURL)return i.URL&&i.URL.createObjectURL(Pe(ke()))||void 0;s.warn("bloburl is not supported by your system, because URL.createObjectURL is not supported by your browser.");break;
case"datauristring":case"dataurlstring":var n="",r=ke();try{n=d(r)}catch(x){n=d(unescape(encodeURIComponent(r)))}return"data:application/pdf;filename="+encodeURIComponent(e.filename)+";
base64,"+n;case"pdfobjectnewwindow":if("[object Window]"===Object.prototype.toString.call(i)){var a="https://cdnjs.cloudflare.com/ajax/libs/pdfobject/2.1.1/pdfobject.min.js",o=!e.
pdfObjectUrl;o||(a=e.pdfObjectUrl);var h=i.open();if(null!==h){var l=Ie(h),c=l.document.createElement("script"),u=this;c.src=a,o&&(c.integrity="sha512-4ze/a9/4jqu+tX9dfOqJYSvyYd5M6qum/
3HpCLr+/Jqf0whc37VUbkpNGHR7/8pSnCFw47T1fmIpwBV7UySh3g==",c.crossOrigin="anonymous"),c.onload=function(){h.PDFObject.embed(u.output("datauristring"),e)},l.body.appendChild(c)}return h}throw
new Error("The option pdfobjectnewwindow just works in a browser-environment.");case"pdfjsnewwindow":if("[object Window]"===Object.prototype.toString.call(i)){var f=e.pdfJsUrl||"examples/
PDF.js/web/viewer.html",p=i.open();if(null!==p){var g=Ie(p),m=g.document.createElement("iframe"),b=-1===f.indexOf("?")?"?":"&";u=this,m.id="pdfViewer",m.width="500px",m.height="400px",m.
src=f+b+"file=&downloadName="+encodeURIComponent(e.filename),m.onload=function(){p.document.title=e.filename,m.contentWindow.PDFViewerApplication.open(u.output("bloburl"))},g.body.
appendChild(m)}return p}throw new Error("The option pdfjsnewwindow just works in a browser-environment.");case"dataurlnewwindow":if("[object Window]"!==Object.prototype.toString.call(
i))throw new Error("The option dataurlnewwindow just works in a browser-environment.");var v=i.open();if(null!==v){var y=Ie(v),_=y.document.createElement("iframe");_.src=this.output(
"datauristring",e),y.body.appendChild(_),v.document.title=e.filename}if(v||"undefined"==typeof safari)return v;break;case"datauri":case"dataurl":return i.document.location.href=this.output(
"datauristring",e);default:return null}},Zt.foo=function(){try{return Zt.apply(this,arguments)}catch(n){var t=n.stack||"";~t.indexOf(" at ")&&(t=t.split(" at ")[1]);var e="Error in function
"+t.split("\n")[0].split("<")[0]+": "+n.message;if(!i.console)throw new Error(e);i.console.error(e,n),i.alert&&alert(e)}},Zt.foo.bar=Zt,Zt.foo),je=function(t){return!0===Array.isArray(
qt)&&qt.indexOf(t)>-1};switch(a){case"pt":Nt=1;break;case"mm":Nt=72/25.4;break;case"cm":Nt=72/2.54;break;case"in":Nt=72;break;case"px":Nt=1==je("px_scaling")?.75:96/72;break;case"pc":
case"em":Nt=12;break;case"ex":Nt=6;break;default:if("number"!=typeof a)throw new Error("Invalid unit: "+a);Nt=a}var Ee=null;X(),Y();var Oe=w.__private__.getPageInfo=w.getPageInfoByObjId=function(
t){if(isNaN(t)||t%1!=0)throw new Error("Invalid argument passed to jsPDF.getPageInfo");return{objId:Tt[t].objId,pageNumber:t,pageContext:Tt[t]}},Be=w.__private__.getPageInfoByObjId=function(
t){if(isNaN(t)||t%1!=0)throw new Error("Invalid argument passed to jsPDF.getPageInfoByObjId");for(var e in Tt)if(Tt[e].objId===t)break;return Oe(e)},Me=w.__private__.getCurrentPageInfo=w.
getCurrentPageInfo=function(){return{objId:Tt[$].objId,pageNumber:$,pageContext:Tt[$]}};w.addPage=function(){return be.apply(this,arguments),this},w.setPage=function(){return we.apply(this,
arguments),lt.call(this,st[$]),this},w.insertPage=function(t){return this.addPage(),this.movePage($,t),this},w.movePage=function(t,e){var n,r;if(t>e){n=st[t],r=Tt[t];for(var i=t;i>e;i--)st[
i]=st[i-1],Tt[i]=Tt[i-1];st[e]=n,Tt[e]=r,this.setPage(e)}else if(t<e){n=st[t],r=Tt[t];for(var a=t;a<e;a++)st[a]=st[a+1],Tt[a]=Tt[a+1];st[e]=n,Tt[e]=r,this.setPage(e)}return this},w.
deletePage=function(){return ve.apply(this,arguments),this},w.__private__.text=w.text=function(t,e,n,i,a){var s,o,h,l,c,u,f,d,p,g=(i=i||{}).scope||this;if("number"==typeof
t&&"number"==typeof e&&("string"==typeof n||Array.isArray(n))){var m=n;n=e,e=t,t=m}if(arguments[3]instanceof Wt==0?(h=arguments[4],l=arguments[5],"object"===r(f=arguments[3])&&null!==f||(
"string"==typeof h&&(l=h,h=null),"string"==typeof f&&(l=f,f=null),"number"==typeof f&&(h=f,f=null),i={flags:f,angle:h,align:l})):(T("The transform parameter of text() with a Matrix value"),
p=a),isNaN(e)||isNaN(n)||null==t)throw new Error("Invalid arguments passed to jsPDF.text");if(0===t.length)return g;var b,w="",y="number"==typeof i.lineHeightFactor?i.lineHeightFactor:Xe,
_=g.internal.scaleFactor;function x(t){return t=t.split("\t").join(Array(i.TabLen||9).join(" ")),ge(t,f)}function A(t){for(var e,n=t.concat(),r=[],i=n.length;i--;)"string"==typeof(e=n.shift(
))?r.push(e):Array.isArray(t)&&(1===e.length||void 0===e[1]&&void 0===e[2])?r.push(e[0]):r.push([e[0],e[1],e[2]]);return r}function L(t,e){var n;if("string"==typeof t)n=e(t)[0];else if(
Array.isArray(t)){for(var r,i,a=t.concat(),s=[],o=a.length;o--;)"string"==typeof(r=a.shift())?s.push(e(r)[0]):Array.isArray(r)&&"string"==typeof r[0]&&(i=e(r[0],r[1],r[2]),s.push([i[0],i[1],
i[2]]));n=s}return n}var k=!1,P=!0;if("string"==typeof t)k=!0;else if(Array.isArray(t)){var F=t.concat();o=[];for(var I,C=F.length;C--;)("string"!=typeof(I=F.shift())||Array.isArray(
I)&&"string"!=typeof I[0])&&(P=!1);k=P}if(!1===k)throw new Error('Type of text must be string or Array. "'+t+'" is not recognized.');"string"==typeof t&&(t=t.match(/[\r?\n]/)?t.split(/\r\n|\
r|\n/g):[t]);var j=pt/g.internal.scaleFactor,E=j*(y-1);switch(i.baseline){case"bottom":n-=E;break;case"top":n+=j-E;break;case"hanging":n+=j-2*E;break;case"middle":n+=j/2-E}if((u=i.
maxWidth||0)>0&&("string"==typeof t?t=g.splitTextToSize(t,u):"[object Array]"===Object.prototype.toString.call(t)&&(t=t.reduce(function(t,e){return t.concat(g.splitTextToSize(e,u))},[]))),
s={text:t,x:e,y:n,options:i,mutex:{pdfEscape:ge,activeFontKey:Lt,fonts:Ft,activeFontSize:pt}},Dt.publish("preProcessText",s),t=s.text,h=(i=s.options).angle,p instanceof
Wt==0&&h&&"number"==typeof h){h*=Math.PI/180,0===i.rotationDirection&&(h=-h),S===N&&(h=-h);var B=Math.cos(h),M=Math.sin(h);p=new Wt(B,M,-M,B,0,0)}else h&&h instanceof Wt&&(p=h);S!==N||p||(
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p=Gt),void 0!==(c=i.charSpace||fn)&&(w+=O(U(c))+" Tc\n",this.setCharSpace(this.getCharSpace()||0)),void 0!==(d=i.horizontalScale)&&(w+=O(100*d)+" Tz\n"),i.lang;var R=-1,D=void 0!==i.
renderingMode?i.renderingMode:i.stroke,q=g.internal.getCurrentPageInfo().pageContext;switch(D){case 0:case1:case"fill":R=0;break;case 1:case!0:case"stroke":R=1;break;case 2:
case"fillThenStroke":R=2;break;case 3:case"invisible":R=3;break;case 4:case"fillAndAddForClipping":R=4;break;case 5:case"strokeAndAddPathForClipping":R=5;break;case 6:
case"fillThenStrokeAndAddToPathForClipping":R=6;break;case 7:case"addToPathForClipping":R=7}var z=void 0!==q.usedRenderingMode?q.usedRenderingMode:-1;-1!==R?w+=R+" Tr\n":-1!==z&&(w+="0 Tr\
n"),-1!==R&&(q.usedRenderingMode=R),l=i.align||"left";var H,W=pt*y,V=g.internal.pageSize.getWidth(),G=Ft[Lt];c=i.charSpace||fn,u=i.maxWidth||0,f=Object.assign({autoencode:!0,noBOM:!0},i.
flags);var Y=[],Z=function(t){return g.getStringUnitWidth(t,{font:G,charSpace:c,fontSize:pt,doKerning:!1})*pt/_};if("[object Array]"===Object.prototype.toString.call(t)){var J;o=A(t),
"left"!==l&&(H=o.map(Z));var X,K=0;if("right"===l){e-=H[0],t=[],C=o.length;for(var $=0;$<C;$++)0===$?(X=rn(e),J=an(n)):(X=U(K-H[$]),J=-W),t.push([o[$],X,J]),K=H[$]}else if("center"===l){e-
=H[0]/2,t=[],C=o.length;for(var Q=0;Q<C;Q++)0===Q?(X=rn(e),J=an(n)):(X=U((K-H[Q])/2),J=-W),t.push([o[Q],X,J]),K=H[Q]}else if("left"===l){t=[],C=o.length;for(var tt=0;tt<C;tt++)t.push(o[
tt])}else if("justify"===l&&"Identity-H"===G.encoding){t=[],C=o.length,u=0!==u?u:V;for(var et=0,nt=0;nt<C;nt++)if(J=0===nt?an(n):-W,X=0===nt?rn(e):et,nt<C-1){var rt=U((u-H[nt])/(o[nt].split(
" ").length-1)),it=o[nt].split(" ");t.push([it[0]+" ",X,J]),et=0;for(var at=1;at<it.length;at++){var st=(Z(it[at-1]+" "+it[at])-Z(it[at]))*_+rt;at==it.length-1?t.push([it[at],st,0]):t.push([
it[at]+" ",st,0]),et-=st}}else t.push([o[nt],X,J]);t.push(["",et,0])}else{if("justify"!==l)throw new Error('Unrecognized alignment option, use "left", "center", "right" or "justify".');for(
t=[],C=o.length,u=0!==u?u:V,nt=0;nt<C;nt++){J=0===nt?an(n):-W,X=0===nt?rn(e):0;var ot=o[nt].split(" ").length-1,ht=ot>0?(u-H[nt])/ot:0;nt<C-1?Y.push(O(U(ht))):Y.push(0),t.push([o[nt],X,
J])}}}!0===("boolean"==typeof i.R2L?i.R2L:bt)&&(t=L(t,function(t,e,n){return[t.split("").reverse().join(""),e,n]})),s={text:t,x:e,y:n,options:i,mutex:{pdfEscape:ge,activeFontKey:Lt,fonts:Ft,
activeFontSize:pt}},Dt.publish("postProcessText",s),t=s.text,b=s.mutex.isHex||!1;var lt=Ft[Lt].encoding;"WinAnsiEncoding"!==lt&&"StandardEncoding"!==lt||(t=L(t,function(t,e,n){return[x(t),e,
n]})),o=A(t),t=[];for(var ut,ft,dt,gt=Array.isArray(o[0])?1:0,mt="",vt=function(t,e,n){var r="";return n instanceof Wt?(n="number"==typeof i.angle?Vt(n,new Wt(1,0,0,1,t,e)):Vt(new Wt(1,0,0,
1,t,e),n),S===N&&(n=Vt(new Wt(1,0,0,-1,0,0),n)),r=n.join(" ")+" Tm\n"):r=O(t)+" "+O(e)+" Td\n",r},wt=0;wt<o.length;wt++){switch(mt="",gt){case 1:dt=(b?"<":"(")+o[wt][0]+(b?">":")");
ut=parseFloat(o[wt][1]),ft=parseFloat(o[wt][2]);break;case 0:dt=(b?"<":"(")+o[wt]+(b?">":")"),ut=rn(e),ft=an(n)}void 0!==Y&&void 0!==Y[wt]&&(mt=Y[wt]+" Tw\n"),0===wt?t.push(mt+vt(ut,ft,
p)+dt):0===gt?t.push(mt+dt):1===gt&&t.push(mt+vt(ut,ft,p)+dt)}t=0===gt?t.join(" Tj\nT* "):t.join(" Tj\n"),t+=" Tj\n";var yt="BT\n/";return yt+=Lt+" "+pt+" Tf\n",yt+=O(pt*y)+" TL\n",yt+=cn+"\
n",yt+=w,yt+=t,ct(yt+="ET"),v[Lt]=!0,g};var Re=w.__private__.clip=w.clip=function(t){return ct("evenodd"===t?"W*":"W"),this};w.clipEvenOdd=function(){return Re("evenodd")},w.__private__.
discardPath=w.discardPath=function(){return ct("n"),this};var Te=w.__private__.isValidStyle=function(t){var e=!1;return-1!==[void 0,null,"S","D","F","DF","FD","f","f*","B","B*","n"].indexOf(
t)&&(e=!0),e};w.__private__.setDefaultPathOperation=w.setDefaultPathOperation=function(t){return Te(t)&&(g=t),this};var De=w.__private__.getStyle=w.getStyle=function(t){var e=g;switch(t){
case"D":case"S":e="S";break;case"F":e="f";break;case"FD":case"DF":e="B";break;case"f":case"f*":case"B":case"B*":e=t}return e},qe=w.close=function(){return ct("h"),this};w.stroke=function(){
return ct("S"),this},w.fill=function(t){return ze("f",t),this},w.fillEvenOdd=function(t){return ze("f*",t),this},w.fillStroke=function(t){return ze("B",t),this},w.fillStrokeEvenOdd=function(
t){return ze("B*",t),this};var ze=function(t,e){"object"===r(e)?We(e,t):ct(t)},Ue=function(t){null===t||S===N&&void 0===t||(t=De(t),ct(t))};function He(t,e,n,r,i){var a=new M(e||this.
boundingBox,n||this.xStep,r||this.yStep,this.gState,i||this.matrix);a.stream=this.stream;var s=t+"$$"+this.cloneIndex+++"$$";return Yt(s,a),a};var We=function(t,e){var n=Et[t.key],r=jt[n];if(
r instanceof B)ct("q"),ct(Ve(e)),r.gState&&w.setGState(r.gState),ct(t.matrix.toString()+" cm"),ct("/"+n+" sh"),ct("Q");else if(r instanceof M){var i=new Wt(1,0,0,-1,0,Pn());t.matrix&&(i=i.
multiply(t.matrix||Gt),n=He.call(r,t.key,t.boundingBox,t.xStep,t.yStep,i).id),ct("q"),ct("/Pattern cs"),ct("/"+n+" scn"),r.gState&&w.setGState(r.gState),ct(e),ct("Q")}},Ve=function(t){
switch(t){case"f":case"F":case"n":return"W n";case"f*":return"W* n";case"B":case"S":return"W S";case"B*":return"W* S"}},Ge=w.moveTo=function(t,e){return ct(O(U(t))+" "+O(H(e))+" m"),this},
Ye=w.lineTo=function(t,e){return ct(O(U(t))+" "+O(H(e))+" l"),this},Ze=w.curveTo=function(t,e,n,r,i,a){return ct([O(U(t)),O(H(e)),O(U(n)),O(H(r)),O(U(i)),O(H(a)),"c"].join(" ")),this};w.
__private__.line=w.line=function(t,e,n,r,i){if(isNaN(t)||isNaN(e)||isNaN(n)||isNaN(r)||!Te(i))throw new Error("Invalid arguments passed to jsPDF.line");return S===L?this.lines([[n-t,r-e]],t,
e,[1,1],i||"S"):this.lines([[n-t,r-e]],t,e,[1,1]).stroke()},w.__private__.lines=w.lines=function(t,e,n,r,i,a){var s,o,h,l,c,u,f,d,p,g,m,b;if("number"==typeof t&&(b=n,n=e,e=t,t=b),r=||[1,1],
a=a||!1,isNaN(e)||isNaN(n)||!Array.isArray(t)||!Array.isArray(r)||!Te(i)||"boolean"!=typeof a)throw new Error("Invalid arguments passed to jsPDF.lines");for(Ge(e,n),s=r[0],o=r[1],l=t.length,
g=e,m=n,h=0;h<l;h++)2===(c=t[h]).length?(g=c[0]*s+g,m=c[1]*o+m,Ye(g,m)):(u=c[0]*s+g,f=c[1]*o+m,d=c[2]*s+g,p=c[3]*o+m,g=c[4]*s+g,m=c[5]*o+m,Ze(u,f,d,p,g,m));return a&&qe(),Ue(i),this},w.
path=function(t){for(var e=0;e<t.length;e++){var n=t[e],r=n.c;switch(n.op){case"m":Ge(r[0],r[1]);break;case"l":Ye(r[0],r[1]);break;case"c":Ze.apply(this,r);break;case"h":qe()}}return this},
w.__private__.rect=w.rect=function(t,e,n,r,i){if(isNaN(t)||isNaN(e)||isNaN(n)||isNaN(r)||!Te(i))throw new Error("Invalid arguments passed to jsPDF.rect");return S===L&&(r=-r),ct([O(U(t)),O(
H(e)),O(U(n)),O(U(r)),"re"].join(" ")),Ue(i),this},w.__private__.triangle=w.triangle=function(t,e,n,r,i,a,s){if(isNaN(t)||isNaN(e)||isNaN(n)||isNaN(r)||isNaN(i)||isNaN(a)||!Te(s))throw new
Error("Invalid arguments passed to jsPDF.triangle");return this.lines([[n-t,r-e],[i-n,a-r],[t-i,e-a]],t,e,[1,1],s,!0),this},w.__private__.roundedRect=w.roundedRect=function(t,e,n,r,i,a,s){
if(isNaN(t)||isNaN(e)||isNaN(n)||isNaN(r)||isNaN(i)||isNaN(a)||!Te(s))throw new Error("Invalid arguments passed to jsPDF.roundedRect");var o=4/3*(Math.SQRT2-1);return i=Math.min(i,.5*n),
a=Math.min(a,.5*r),this.lines([[n-2*i,0],[i*o,0,i,a-a*o,i,a],[0,r-2*a],[0,a*o,-i*o,a,-i,a],[2*i-n,0],[-i*o,0,-i,-a*o,-i,-a],[0,2*a-r],[0,-a*o,i*o,-a,i,-a]],t+i,e,[1,1],s,!0),this},w.
__private__.ellipse=w.ellipse=function(t,e,n,r,i){if(isNaN(t)||isNaN(e)||isNaN(n)||isNaN(r)||!Te(i))throw new Error("Invalid arguments passed to jsPDF.ellipse");var a=4/3*(Math.SQRT2-1)*n,
s=4/3*(Math.SQRT2-1)*r;return Ge(t+n,e),Ze(t+n,e-s,t+a,e-r,t,e-r),Ze(t-a,e-r,t-n,e-s,t-n,e),Ze(t-n,e+s,t-a,e+r,t,e+r),Ze(t+a,e+r,t+n,e+s,t+n,e),Ue(i),this},w.__private__.circle=w.
circle=function(t,e,n,r){if(isNaN(t)||isNaN(e)||isNaN(n)||!Te(r))throw new Error("Invalid arguments passed to jsPDF.circle");return this.ellipse(t,e,n,n,r)},w.setFont=function(t,e,n){return
n&&(e=F(e,n)),Lt=_e(t,e,{disableWarning:!1}),this};var Je=w.__private__.getFont=w.getFont=function(){return Ft[_e.apply(w,arguments)]};w.__private__.getFontList=w.getFontList=function(){var
t,e,n={};for(t in It)if(It.hasOwnProperty(t))for(e in n[t]=[],It[t])It[t].hasOwnProperty(e)&&n[t].push(e);return n},w.addFont=function(t,e,n,r,i){var a=["StandardEncoding",
"MacRomanEncoding","Identity-H","WinAnsiEncoding"];return arguments[3]&&-1!==a.indexOf(arguments[3])?i=arguments[3]:arguments[3]&&-1==a.indexOf(arguments[3])&&(n=F(n,r)),pe.call(this,t,e,n,
i=i||"Identity-H")};var Xe,Ke=t.lineWidth||.200025,$e=w.__private__.getLineWidth=w.getLineWidth=function(){return Ke},Qe=w.__private__.setLineWidth=w.setLineWidth=function(t){return Ke=t,ct(
O(U(t))+" w"),this};w.__private__.setLineDash=R.API.setLineDash=R.API.setLineDashPattern=function(t,e){if(t=t||[],e=e||0,isNaN(e)||!Array.isArray(t))throw new Error("Invalid arguments
passed to jsPDF.setLineDash");return t=t.map(function(t){return O(U(t))}).join(" "),e=O(U(e)),ct("["+t+"] "+e+" d"),this};var tn=w.__private__.getLineHeight=w.getLineHeight=function(){
return pt*Xe};w.__private__.getLineHeight=w.getLineHeight=function(){return pt*Xe};var en=w.__private__.setLineHeightFactor=w.setLineHeightFactor=function(t){return"number"==typeof(t=t||1.
15)&&(Xe=t),this},nn=w.__private__.getLineHeightFactor=w.getLineHeightFactor=function(){return Xe};en(t.lineHeight);var rn=w.__private__.getHorizontalCoordinate=function(t){return U(t)},
an=w.__private__.getVerticalCoordinate=function(t){return S===N?t:Tt[$].mediaBox.topRightY-Tt[$].mediaBox.bottomLeftY-U(t)},sn=w.__private__.getHorizontalCoordinateString=w.
getHorizontalCoordinateString=function(t){return O(rn(t))},on=w.__private__.getVerticalCoordinateString=w.getVerticalCoordinateString=function(t){return O(an(t))},hn=t.strokeColor||"0 G";w.
__private__.getStrokeColor=w.getDrawColor=function(){return ee(hn)},w.__private__.setStrokeColor=w.setDrawColor=function(t,e,n,r){return hn=ne({ch1:t,ch2:e,ch3:n,ch4:r,pdfColorType:"draw",
precision:2}),ct(hn),this};var ln=t.fillColor||"0 g";w.__private__.getFillColor=w.getFillColor=function(){return ee(ln)},w.__private__.setFillColor=w.setFillColor=function(t,e,n,r){return
ln=ne({ch1:t,ch2:e,ch3:n,ch4:r,pdfColorType:"fill",precision:2}),ct(ln),this};var cn=t.textColor||"0 g",un=w.__private__.getTextColor=w.getTextColor=function(){return ee(cn)};w.__private__.
setTextColor=w.setTextColor=function(t,e,n,r){return cn=ne({ch1:t,ch2:e,ch3:n,ch4:r,pdfColorType:"text",precision:3}),this};var fn=t.charSpace,dn=w.__private__.getCharSpace=w.
getCharSpace=function(){return parseFloat(fn||0)};w.__private__.setCharSpace=w.setCharSpace=function(t){if(isNaN(t))throw new Error("Invalid argument passed to jsPDF.setCharSpace");return
fn=t,this};var pn=0;w.CapJoinStyles={0:0,butt:0,but:0,miter:0,1:1,round:1,rounded:1,circle:1,2:2,projecting:2,project:2,square:2,bevel:2},w.__private__.setLineCap=w.setLineCap=function(t){
var e=w.CapJoinStyles[t];if(void 0===e)throw new Error("Line cap style of '"+t+"' is not recognized. See or extend .CapJoinStyles property for valid styles");return pn=e,ct(e+" J"),this};
var gn=0;w.__private__.setLineJoin=w.setLineJoin=function(t){var e=w.CapJoinStyles[t];if(void 0===e)throw new Error("Line join style of '"+t+"' is not recognized. See or extend .
CapJoinStyles property for valid styles");return gn=e,ct(e+" j"),this},w.__private__.setLineMiterLimit=w.__private__.setMiterLimit=w.setLineMiterLimit=w.setMiterLimit=function(t){if(t=t||0,
isNaN(t))throw new Error("Invalid argument passed to jsPDF.setLineMiterLimit");return ct(O(U(t))+" M"),this},w.GState=function(t){{t="string"==typeof t?Ot[Bt[t]]:mn(null,t)).
equals(Mt)||(ct("/"+t.id+" gs"),Mt=t)};var mn=function(t,e){if(!t||!Bt[t]){var n=!1;for(var r in Ot)if(Ot.hasOwnProperty(r)&&Ot[r].equals(e)){n=!0;break}if(n)e=Ot[r];else{var i="GS"+(Object.
keys(Ot).length+1).toString(10);Ot[i]=e,e.id=i}return t&&(Bt[t]=e.id),Dt.publish("addGState",e),e}};w.addGState=function(t,e){return mn(t,e),this},w.saveGraphicsState=function(){return ct(
"q"),Ct.push({key:Lt,size:pt,color:cn}),this},w.restoreGraphicsState=function(){ct("Q");var t=Ct.pop();return Lt=t.key,pt=t.size,cn=t.color,Mt=null,this},w.
setCurrentTransformationMatrix=function(t){return ct(t.toString()+" cm"),this},w.comment=function(t){return ct("#"+t),this};var bn=function(t,e){var n=t||0;Object.defineProperty(this,"x",{
enumerable:!0,get:function(){return n},set:function(t){isNaN(t)||(n=parseFloat(t))}});var r=e||0;Object.defineProperty(this,"y",{enumerable:!0,get:function(){return r},set:function(t){isNaN(
t)||(r=parseFloat(t))}});var i="pt";return Object.defineProperty(this,"type",{enumerable:!0,get:function(){return i},set:function(t){i=t.toString()}}),this},vn=function(t,e,n,r){bn.call(
this,t,e),this.type="rect";var i=n||0;Object.defineProperty(this,"w",{enumerable:!0,get:function(){return i},set:function(t){isNaN(t)||(i=parseFloat(t))}});var a=r||0;return Object.
defineProperty(this,"h",{enumerable:!0,get:function(){return a},set:function(t){isNaN(t)||(a=parseFloat(t))}}),this},wn=function(){this.page=Rt,this.currentPage=$,this.pages=st.slice(0),
this.pagesContext=Tt.slice(0),this.x=St,this.y=kt,this.matrix=Pt,this.width=xn($),this.height=Ln($),this.outputDestination=ht,this.id="",this.objectNumber=-1};wn.prototype.restore=function(
){Rt=this.page,$=this.currentPage,Tt=this.pagesContext,st=this.pages,St=this.x,kt=this.y,Pt=this.matrix,An($,this.width),Nn($,this.height),ht=this.outputDestination};var yn=function(t,e,n,r,
i){Ht.push(new wn),Rt=$=0,st=[],St=t,kt=e,Pt=i,me([n,r])};for(var _n in w.beginFormObject=function(t,e,n,r,i){return yn(t,e,n,r,i),this},w.endFormObject=function(t){return function(t){if(Ut[
t])Ht.pop().restore();else{var e=new wn,n="Xo"+(Object.keys(zt).length+1).toString(10);e.id=n,Ut[t]=n,zt[n]=e,Dt.publish("addFormObject",e),Ht.pop().restore()}}(t),this},w.
doFormObject=function(t,e){var n=zt[Ut[t]];return ct("q"),ct(e.toString()+" cm"),ct("/"+n.id+" Do"),ct("Q"),this},w.getFormObject=function(t){var e=zt[Ut[t]];return{x:e.x,y:e.y,width:e.
width,height:e.height,matrix:e.matrix}},w.save=function(t,e){return t=t||"generated.pdf",(e=e||{}).returnPromise=e.returnPromise||!1,!1===e.returnPromise?(c(Pe(ke()),t),"function"==typeof c.
unload&&i.setTimeout&&setTimeout(c.unload,911),this):new Promise(function(e,n){try{var r=c(Pe(ke()),t);"function"==typeof c.unload&&i.setTimeout&&setTimeout(c.unload,911),e(r)}catch(a){n(a.
message)}})},R.API)R.API.hasOwnProperty(_n)&&("events"===_n&&R.API.events.length?function(t,e){var n,r,i;for(i=e.length-1;-1!==i;i--)n=e[i][0],r=e[i][1],t.subscribe.apply(t,[n].concat(
"function"==typeof r?[r]:r))}(Dt,R.API.events):w[_n]=R.API[_n]);function xn(t){return Tt[t].mediaBox.topRightX-Tt[t].mediaBox.bottomLeftX}function An(t,e){Tt[t].mediaBox.topRightX=e+Tt[t].
mediaBox.bottomLeftX}function Ln(t){return Tt[t].mediaBox.topRightY-Tt[t].mediaBox.bottomLeftY}function Nn(t,e){Tt[t].mediaBox.topRightY=e+Tt[t].mediaBox.bottomLeftY}var Sn=w.
getPageWidth=function(t){return xn(t=t||$)/Nt},kn=w.setPageWidth=function(t,e){An(t,e*Nt)},Pn=w.getPageHeight=function(t){return Ln(t=t||$)/Nt},Fn=w.setPageHeight=function(t,e){Nn(t,e*Nt)};
return w.internal={pdfEscape:ge,getStyle:De,getFont:Je,getFontSize:mt,getCharSpace:dn,getTextColor:un,getLineHeight:tn,getLineHeightFactor:nn,getLineWidth:$e,write:ut,
getHorizontalCoordinate:rn,getVerticalCoordinate:an,getCoordinateString:sn,getVerticalCoordinateString:on,collections:{},newObject:Jt,newAdditionalObject:$t,newObjectDeferred:Xt,
newObjectDeferredBegin:Kt,getFilters:re,putStream:ie,events:Dt,scaleFactor:Nt,pageSize:{getWidth:function(){return Sn($)},setWidth:function(t){kn($,t)},getHeight:function(){return Pn($)},
setHeight:function(t){Fn($,t)}},encryptionOptions:m,encryption:Ee,getEncryptor:function(t){return null!==m?Ee.encryptor(t,0):function(t){return t}},output:Ce,getNumberOfPages:ye,get pages(){
return st},out:ct,f2:q,f3:z,getPageInfo:Oe,getPageInfoByObjId:Be,getCurrentPageInfo:Me,getPDFVersion:_,Point:bn,Rectangle:vn,Matrix:Wt,hasHotfix:je},Object.defineProperty(w.internal.
pageSize,"width",{get:function(){return Sn($)},set:function(t){kn($,t)},enumerable:!0,configurable:!0}),Object.defineProperty(w.internal.pageSize,"height",{get:function(){return Pn($)},set:

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function(t){Fn($,t)},enumerable:!0,configurable:!0}),function(t){for(var e=0,n=dt.length;e<n;e++){var r=pe.call(this,t[e][0],t[e][1],t[e][2],dt[e][3],!0);!1===b&&(v[r]=!0);var i=t[e][0].
split("-");de({id:r,fontName:i[0],fontStyle:i[1]||""})}Dt.publish("addFonts",{fonts:Ft,dictionary:It})}.call(w,dt),Lt="F1",be(o,n),Dt.publish("initialized"),w}I.prototype.
lsbFirstWord=function(t){return String.fromCharCode(255&t,t>>8&255,t>>16&255,t>>24&255)},I.prototype.toHexString=function(t){return t.split("").map(function(t){return("0"+(255&t.charCodeAt(
0)).toString(16)).slice(-2)}).join("")},I.prototype.hexToBytes=function(t){for(var e=[],n=0;n<t.length;n+=2)e.push(String.fromCharCode(parseInt(t.substr(n,2),16)));return e.join("")},I.
prototype.processOwnerPassword=function(t,e){return P(N(e).substr(0,5),t)},I.prototype.encryptor=function(t,e){var n=N(this.encryptionKey+String.fromCharCode(255&t,t>>8&255,t>>16&255,255&e,
e>>8&255)).substr(0,10);return function(t){return t^D},Y=function(t){var e=new ct,n=Lt.internal.getHeight(t)||0,r=Lt.internal.getWidth(t)||0;return e.BBox=[0,0,Number(r),Number(n)],e},Z=T.
__acroform__.setBit=function(t,e){if(t=t||0,e=e||0,isNaN(t)||isNaN(e))throw new Error("Invalid arguments passed to jsPDF.API.__acroform__.setBit");return t|1<<e},J=T.__acroform__.
clearBit=function(t,e){if(t=t||0,e=e||0,isNaN(t)||isNaN(e))throw new Error("Invalid arguments passed to jsPDF.API.__acroform__.clearBit");return t&~(1<<e)},X=T.__acroform__.getBit=function(
t,e){if(isNaN(t)||isNaN(e))throw new Error("Invalid arguments passed to jsPDF.API.__acroform__.getBit");return t&1<<e?1:0},K=T.__acroform__.getBitForPdf=function(t,e){if(isNaN(t)||isNaN(
e))throw new Error("Invalid arguments passed to jsPDF.API.__acroform__.getBitForPdf");return X(t,e-1)},$=T.__acroform__.setBitForPdf=function(t,e){if(isNaN(t)||isNaN(e))throw new Error(
"Invalid arguments passed to jsPDF.API.__acroform__.setBitForPdf");return Z(t,e-1)},Q=T.__acroform__.clearBitForPdf=function(t,e){if(isNaN(t)||isNaN(e))throw new Error("Invalid arguments
passed to jsPDF.API.__acroform__.clearBitForPdf");return J(t,e-1)},tt=T.__acroform__.calculateCoordinates=function(t,e){var n=e.internal.getHorizontalCoordinate,r=e.internal.
getVerticalCoordinate,i=t[0],a=t[1],s=t[2],o=t[3],n={};return h.lowerLeft_X=n(i)||0,h.lowerLeft_Y=r(a+o)||0,h.upperRight_X=n(i+s)||0,h.upperRight_Y=r(a)||0,[Number(H(h.lowerLeft_X)),Number(
H(h.lowerLeft_Y)),Number(H(h.upperRight_X)),Number(H(h.upperRight_Y))]},et=function(t){if(t.appearanceStreamContent)return t.appearanceStreamContent;if(t.V||t.DV){var e=[],n=t._V||t.DV,r=nt(
t,n),i=t.scope.internal.getFont(t.fontName,t.fontStyle).id;e.push("/Tx BMC"),e.push("q"),e.push("BT"),e.push(t.scope.__private__.encodeColorString(t.color)),e.push("/"+i+" "+H(r.fontSize)+"
Tf"),e.push("1 0 0 1 0 0 Tm"),e.push(r.text),e.push("ET"),e.push("Q"),e.push("EMC");var a=Y(t);return a.scope=t.scope,a.stream=e.join("\n"),a}},nt=function(t,e){var n=0===t.fontSize?t.
maxFontSize:t.fontSize,r={text:"",fontSize:""},i=(e=")"==(e="("==e.substr(0,1)?e.substr(1):e).substr(e.length-1)?e.substr(0,e.length-1):e).split(" ");i=t.multiline?i.map(function(t){return
t.split("\n")}):i.map(function(t){return[t]});var a=n,s=Lt.internal.getHeight(t)||0;s=s<0?-s:s;var o=Lt.internal.getWidth(t)||0;o=o<0?-o:o;var h=function(e,n,r){if(e+1<i.length){var a=n+"
"+i[e+1][0];return rt(a,t,r).width<=o-4}return!1};a++;t:for(;a>0;){e="",a--;var l,c,u=rt("3",t,a).height,f=t.multiline?s-a:(s-u)/2,d=f+=2,p=0,g=0,m=0;if(a<=0){e="(...) Tj\n",e+="% Width of
Text: "+rt(e,t,a=12).width+", FieldWidth:"+o+"\n";break}for(var b="",v=0,w=0;w<i.length;w++)if(i.hasOwnProperty(w)){var y=!1;if(1!==i[w].length&&m!==i[w].length-1){if((u+2)*(
v+2)+2>s)continue t;b+=i[w][m],y=!0,g=w,w-->}else{b=" "==(b+=i[w][m]+" ").substr(b.length-1)?b.substr(0,b.length-1):b;var _=parseInt(w),x=h(_,b,a),A=w>=i.length-1;if(x&&!A){b+=" ",m=0;
continue}if(x||A){if(A)g=_;else if(t.multiline&&(u+2)*(v+2)+2>s)continue t}else{if(!t.multiline)continue t;if((u+2)*(v+2)+2>s)continue t;g=_}}for(var L="",N=p;N<=g;N++){var S=i[N];if(t.
multiline){if(N===g){L+=S[m]+" ",m=(m+1)%S.length;continue}if(N===p){L+=S[S.length-1]+" ";continue}}L+=S[0]+" "}switch(L=" "==L.substr(L.length-1)?L.substr(0,L.length-1):L,c=rt(L,t,a).width,
t.textAlign){case"right":l=o-c-2;break;case"center":l=(o-c)/2;break;default:l=2}e+=H(l)+" "+H(d)+" Td\n",e+="("+q(L)+") Tj\n",e+=-H(l)+" 0 Td\n",d=-(a+2),c=0,p=y?g:g+1,v++,b=""}break}return
r.text=e,r.fontSize=a,r},rt=function(t,e,n){var r=e.scope.internal.getFont(e.fontName,e.fontStyle),i=e.scope.getStringUnitWidth(t,{font:r,fontSize:parseFloat(n),charSpace:0})*parseFloat(n);
return{height:e.scope.getStringUnitWidth("3",{font:r,fontSize:parseFloat(n),charSpace:0})*parseFloat(n)*1.5,width:i}},it={fields:[],xForms:[],acroFormDictionaryRoot:null,printedOut:!1,
internal:null,isInitialized:!1},at=function(t,e){var n={type:"reference",object:t};void 0===e.internal.getPageInfo(t.page).pageContext.annotations.find(function(t){return t.type===n.type&&t.
object===n.object})&&e.internal.getPageInfo(t.page).pageContext.annotations.push(n)},st=function(t,e){if(e.scope=t,void 0!==t.internal&&(void 0===t.internal.acroformPlugin||!1===t.internal.
acroformPlugin.isInitialized)){if(ft.FieldNum=0,t.internal.acroformPlugin=JSON.parse(JSON.stringify(it)),t.internal.acroformPlugin.acroFormDictionaryRoot)throw new Error("Exception while
creating AcroformDictionary");D=t.internal.scaleFactor,t.internal.acroformPlugin.acroFormDictionaryRoot=new ut,t.internal.acroformPlugin.acroFormDictionaryRoot.scope=t,t.internal.
acroformPlugin.acroFormDictionaryRoot._eventID=t.internal.events.subscribe("postPutResources",function(){!function(t){t.internal.events.unsubscribe(t.internal.acroformPlugin.
acroFormDictionaryRoot._eventID),delete t.internal.acroformPlugin.acroFormDictionaryRoot._eventID,t.internal.acroformPlugin.printedOut=!0}(t)}),t.internal.events.subscribe("buildDocument",
function(){!function(t){t.internal.acroformPlugin.acroFormDictionaryRoot.objId=void 0;var e=t.internal.acroformPlugin.acroFormDictionaryRoot.Fields;for(var n in e)if(e.hasOwnProperty(n)){
var r=e[n];r.objId=void 0,r.hasAnnotation&&at(r,t)}}(t)}),t.internal.events.subscribe("putCatalog",function(){!function(t){if(void 0===t.internal.acroformPlugin.acroFormDictionaryRoot)throw
new Error("putCatalogCallback: Root missing.");t.internal.write("/AcroForm "+t.internal.acroformPlugin.acroFormDictionaryRoot.objId+" 0 R")}(t)}),t.internal.events.subscribe("postPutPages",
function(e){!function(t,e){var n=!t;for(var i in t||(e.internal.newObjectDeferredBegin(e.internal.acroformPlugin.acroFormDictionaryRoot.objId,!0),e.internal.acroformPlugin.
acroFormDictionaryRoot.putStream()),t=t||e.internal.acroformPlugin.acroFormDictionaryRoot.Kids)if(t.hasOwnProperty(i)){var a=t[i],s=[],o=a.Rect;if(a.Rect&&(a.Rect=tt(a.Rect,e)),e.internal.
newObjectDeferredBegin(a.objId,!0),a.DA=Lt.createDefaultAppearanceStream(a),"object"===r(a)&&"function"==typeof a.getKeyValueListForStream&&(s=a.getKeyValueListForStream()),a.Rect=o,a.
hasAppearanceStream&&!a.appearanceStreamContent){var h=et(a);s.push({key:"AP",value:"<</N "+h+">>"}),e.internal.acroformPlugin.xForms.push(h)}if(a.appearanceStreamContent){var l="";for(var
c in a.appearanceStreamContent)if(a.appearanceStreamContent.hasOwnProperty(c)){var u=a.appearanceStreamContent[c];if(l+="/"+c+" ",l+="<<",Object.keys(u).length>=1||Array.isArray(u)){for(var
i in u)if(u.hasOwnProperty(i)){var f=u[i];"function"==typeof f&&(f=f.call(e,a)),l+="/"+i+" "+f+" ",e.internal.acroformPlugin.xForms.indexOf(f)>=0||e.internal.acroformPlugin.xForms.push(
f)}}else"function"==typeof(f=u)&&(f=f.call(e,a)),l+="/"+i+" "+f,e.internal.acroformPlugin.xForms.indexOf(f)>=0||e.internal.acroformPlugin.xForms.push(f);l+=">>"}s.push({key:"AP",value:"<<\
n"+l+">>"})}e.internal.putStream({additionalKeyValues:s,objectId:a.objId}),e.internal.out("endobj")}n&&function(t,e){for(var n in t)if(t.hasOwnProperty(n)){var i=n,a=t[n];e.internal.
newObjectDeferredBegin(a.objId,!0),"object"===r(a)&&"function"==typeof a.putStream&&a.putStream(),delete t[i]}}(e.internal.acroformPlugin.xForms,e)}(e,t)}),t.internal.acroformPlugin.
isInitialized=!0}},ot=T.__acroform__.arrayToPdfArray=function(t,e,n){var i=function(t){return t};if(Array.isArray(t)){for(var a="[",s=0;s<t.length;s++)switch(0!==s&&(a+=" "),r(t[s])){
case"boolean":case"number":case"object":a+=t[s].toString();break;case"string":"/"===t[s].substr(0,1)?a+="/"+U(t[s].substr(1)):(void 0!==e&&n&&(i=n.internal.getEncryptor(e)),a+="("+q(i(t[s].
toString()))+")")}return a+"]"}throw new Error("Invalid argument passed to jsPDF.__acroform__.arrayToPdfArray")},ht=function(t,e,n){var r=function(t){return t};return void 0!==e&&n&&(r=n.
internal.getEncryptor(e)),(t=t||"").toString(),"("+q(r(t))+")"},lt=function(){this._objId=void 0,this._scope=void 0,Object.defineProperty(this,"objId",{get:function(){if(void 0===this.
_objId){if(void 0===this.scope)return;this._objId=this.scope.internal.newObjectDeferred()}return this._objId},set:function(t){this._objId=t}}),Object.defineProperty(this,"scope",{value:this.
_scope,writable:!0})};lt.prototype.toString=function(){return this.objId+" 0 R"},lt.prototype.putStream=function(){var t=this.getKeyValueListForStream();this.scope.internal.putStream({data:
this.stream,additionalKeyValues:t,objectId:this.objId}),this.scope.internal.out("endobj")},lt.prototype.getKeyValueListForStream=function(){var t=[],e=Object.getOwnPropertyNames(this).
filter(function(t){return"content"!=t&&"appearanceStreamContent"!=t&&"scope"!=t&&"objId"!=t&&"_"!=t.substring(0,1)});for(var n in e)if(!1===Object.getOwnPropertyDescriptor(this,e[n]).
configurable){var r=e[n],i=this[r];i&&(Array.isArray(i)?t.push({key:r,value:ot(i,this.objId,this.scope)}):i instanceof lt?(i.scope=this.scope,t.push({key:r,value:i.objId+" 0 R"})):
"function"!=typeof i&&t.push({key:r,value:i}))}return t};var ct=function(){lt.call(this),Object.defineProperty(this,"Type",{value:"/XObject",configurable:!1,writable:!0}),Object.
defineProperty(this,"Subtype",{value:"/Form",configurable:!1,writable:!0}),Object.defineProperty(this,"FormType",{value:1,configurable:!1,writable:!0});var t,e=[];Object.defineProperty(this,
"BBox",{configurable:!1,get:function(){return e},set:function(t){e=t}}),Object.defineProperty(this,"Resources",{value:"2 0 R",configurable:!1,writable:!0}),Object.defineProperty(this,
"stream",{enumerable:!1,configurable:!0,set:function(e){t=e.trim()},get:function(){return t||null}})};V(ct,lt);var ut=function(){lt.call(this);var t,e=[];Object.defineProperty(this,"Kids",{
enumerable:!1,configurable:!0,get:function(){return e.length>0?e:void 0}}),Object.defineProperty(this,"Fields",{enumerable:!1,configurable:!1,get:function(){return e}}),Object.
defineProperty(this,"DA",{enumerable:!1,configurable:!1,get:function(){if(t){var e=function(t){return t};return this.scope&&(e=this.scope.internal.getEncryptor(this.objId)),"("+q(e(
t))+")"}},set:function(e){t=e}})};V(ut,lt);var ft=function t(){lt.call(this);var e=4;Object.defineProperty(this,"F",{enumerable:!1,configurable:!1,get:function(){return e},set:function(t){
if(isNaN(t))throw new Error('Invalid value "'+t+'" for attribute F supplied.');e=t}}),Object.defineProperty(this,"showWhenPrinted",{enumerable:!0,configurable:!0,get:function(){return
Boolean(K(e,3))},set:function(t){!0===Boolean(t)?this.F=$(e,3):this.F=Q(e,3)}});var n=0;Object.defineProperty(this,"Ff",{enumerable:!1,configurable:!1,get:function(){return n},set:function(
t){if(isNaN(t))throw new Error('Invalid value "'+t+'" for attribute Ff supplied.');n=t}});var r=[];Object.defineProperty(this,"Rect",{enumerable:!1,configurable:!1,get:function(){if(0!==r.
length)return r},set:function(t){r=void 0!==t?t:[]}}),Object.defineProperty(this,"x",{enumerable:!0,configurable:!0,get:function(){return!r||isNaN(r[0])?0:r[0]},set:function(t){r[0]=t}}),
Object.defineProperty(this,"y",{enumerable:!0,configurable:!0,get:function(){return!r||isNaN(r[1])?0:r[1]},set:function(t){r[1]=t}}),Object.defineProperty(this,"width",{enumerable:!0,
configurable:!0,get:function(){return!r||isNaN(r[2])?0:r[2]},set:function(t){r[2]=t}}),Object.defineProperty(this,"height",{enumerable:!0,configurable:!0,get:function(){return!r||isNaN(r[
3])?0:r[3]},set:function(t){r[3]=t}});var i="";Object.defineProperty(this,"FT",{enumerable:!0,configurable:!1,get:function(){return i},set:function(t){switch(t){case"/Btn":case"/Tx":case"/
Ch":case"/Sig":i=t;break;default:throw new Error('Invalid value "'+t+'" for attribute FT supplied.')}}});var a=null;Object.defineProperty(this,"T",{enumerable:!0,configurable:!1,get:
function(){if(!a||a.length<1){if(this instanceof yt)return;a="FieldObject"+t.FieldNum++}var e=function(t){return t};return this.scope&&(e=this.scope.internal.getEncryptor(this.objId)),"("+q(
e(a))+")"},set:function(t){a=t.toString()}}),Object.defineProperty(this,"fieldName",{configurable:!0,enumerable:!0,get:function(){return a},set:function(t){a=t}});var s="helvetica";Object.
defineProperty(this,"fontName",{enumerable:!0,configurable:!0,get:function(){return s},set:function(t){s=t}});var o="normal";Object.defineProperty(this,"fontStyle",{enumerable:!0,
configurable:!0,get:function(){return o},set:function(t){o=t}});var h=0;Object.defineProperty(this,"fontSize",{enumerable:!0,configurable:!0,get:function(){return h},set:function(t){h=t}});
var l=void 0;Object.defineProperty(this,"maxFontSize",{enumerable:!0,configurable:!0,get:function(){return void 0===l?50/D:l},set:function(t){l=t}});var c="black";Object.defineProperty(this,
"color",{enumerable:!0,configurable:!0,get:function(){return c},set:function(t){c=t}});var u="/F1 0 Tf 0 g";Object.defineProperty(this,"DA",{enumerable:!0,configurable:!1,get:function(){if(
!(!u||this instanceof yt||this instanceof xt))return ht(u,this.objId,this.scope)},set:function(t){t=t.toString(),u=t}});var f=null;Object.defineProperty(this,"DV",{enumerable:!1,
configurable:!1,get:function(){if(f)return this instanceof bt==0?ht(f,this.objId,this.scope):f},set:function(t){t=t.toString(),f=this instanceof bt==0?"("===t.substr(0,1)?z(t.substr(1,t.
length-2)):z(t):t}}),Object.defineProperty(this,"defaultValue",{enumerable:!0,configurable:!0,get:function(){return this instanceof bt==1?z(f.substr(1,f.length-1)):f},set:function(t){t=t.
toString(),f=this instanceof bt==1?"/"+U(t):t}});var d=null;Object.defineProperty(this,"_V",{enumerable:!1,configurable:!1,get:function(){if(d)return d},set:function(t){this.V=t}}),Object.
defineProperty(this,"V",{enumerable:!1,configurable:!1,get:function(){if(d)return this instanceof bt==0?ht(d,this.objId,this.scope):d},set:function(t){t=t.toString(),d=this instanceof
bt==0?"("===t.substr(0,1)?z(t.substr(1,t.length-2)):z(t):t}}),Object.defineProperty(this,"value",{enumerable:!0,configurable:!0,get:function(){return this instanceof bt==1?z(d.substr(1,d.
length-1)):d},set:function(t){t=t.toString(),d=this instanceof bt==1?"/"+U(t):t}}),Object.defineProperty(this,"hasAnnotation",{enumerable:!0,configurable:!0,get:function(){return this.
Rect}}),Object.defineProperty(this,"Type",{enumerable:!0,configurable:!1,get:function(){return this.hasAnnotation?"/Annot":null}}),Object.defineProperty(this,"Subtype",{enumerable:!0,
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configurable:!1,get:function(){return this.hasAnnotation?"/Widget":null}});var p,g=!1;Object.defineProperty(this,"hasAppearanceStream",{enumerable:!0,configurable:!0,get:function(){return
g},set:function(t){t=Boolean(t),g=t}}),Object.defineProperty(this,"page",{enumerable:!0,configurable:!0,get:function(){if(p)return p},set:function(t){p=t}}),Object.defineProperty(this,
"readOnly",{enumerable:!0,configurable:!0,get:function(){return Boolean(K(this.Ff,1))},set:function(t){!0===Boolean(t)?this.Ff=$(this.Ff,1):this.Ff=Q(this.Ff,1)}}),Object.defineProperty(
this,"required",{enumerable:!0,configurable:!0,get:function(){return Boolean(K(this.Ff,2))},set:function(t){!0===Boolean(t)?this.Ff=$(this.Ff,2):this.Ff=Q(this.Ff,2)}}),Object.
defineProperty(this,"noExport",{enumerable:!0,configurable:!0,get:function(){return Boolean(K(this.Ff,3))},set:function(t){!0===Boolean(t)?this.Ff=$(this.Ff,3):this.Ff=Q(this.Ff,3)}});var
m=null;Object.defineProperty(this,"Q",{enumerable:!0,configurable:!1,get:function(){if(null!==m)return m},set:function(t){if(-1===[0,1,2].indexOf(t))throw new Error('Invalid value "'+t+'"
for attribute Q supplied.');m=t}}),Object.defineProperty(this,"textAlign",{get:function(){var t;switch(m){case 0:default:t="left";break;case 1:t="center";break;case 2:t="right"}return t},
configurable:!0,enumerable:!0,set:function(t){switch(t){case"right":case 2:m=2;break;case"center":case 1:m=1;break;default:m=0}}})};V(ft,lt);var dt=function(){ft.call(this),this.FT="/Ch",
this.V=""(),this.fontName="zapfdingbats";var t=0;Object.defineProperty(this,"TI",{enumerable:!0,configurable:!1,get:function(){return t},set:function(e){t=e}}),Object.defineProperty(this,
"topIndex",{enumerable:!0,configurable:!0,get:function(){return t},set:function(e){t=e}});var e=[];Object.defineProperty(this,"Opt",{enumerable:!0,configurable:!1,get:function(){return ot(e,
this.objId,this.scope)},set:function(t){var n,r;r=[],"string"==typeof(n=t)&&(r=function(t,e,n){n||(n=1);for(var r,i=[];r=e.exec(t);)i.push(r[n]);return i}(n,/\((.*?)\)/g)),e=r}}),this.
getOptions=function(){return e},this.setOptions=function(t){e=t,this.sort&&e.sort()},this.addOption=function(t){t=(t=t||"").toString(),e.push(t),this.sort&&e.sort()},this.
removeOption=function(t,n){for(n=n||!1,t=(t=t||"").toString();-1!==e.indexOf(t)&&(e.splice(e.indexOf(t),1),!1!==n););}},Object.defineProperty(this,"combo",{enumerable:!0,configurable:!0,get:
function(){return Boolean(K(this.Ff,18))},set:function(t){!0===Boolean(t)?this.Ff=$(this.Ff,18):this.Ff=Q(this.Ff,18)}}),Object.defineProperty(this,"edit",{enumerable:!0,configurable:!0,get:
function(){return Boolean(K(this.Ff,19))},set:function(t){!0===this.combo&&(!0===Boolean(t)?this.Ff=$(this.Ff,19):this.Ff=Q(this.Ff,19))}}),Object.defineProperty(this,"sort",{enumerable:!0,
configurable:!0,get:function(){return Boolean(K(this.Ff,20))},set:function(t){!0===Boolean(t)?(this.Ff=$(this.Ff,20),e.sort()):this.Ff=Q(this.Ff,20)}}),Object.defineProperty(this,
"multiSelect",{enumerable:!0,configurable:!0,get:function(){return Boolean(K(this.Ff,22))},set:function(t){!0===Boolean(t)?this.Ff=$(this.Ff,22):this.Ff=Q(this.Ff,22)}}),Object.
defineProperty(this,"doNotSpellCheck",{enumerable:!0,configurable:!0,get:function(){return Boolean(K(this.Ff,23))},set:function(t){!0===Boolean(t)?this.Ff=$(this.Ff,23):this.Ff=Q(this.Ff,
23)}}),Object.defineProperty(this,"commitOnSelChange",{enumerable:!0,configurable:!0,get:function(){return Boolean(K(this.Ff,27))},set:function(t){!0===Boolean(t)?this.Ff=$(this.Ff,27):this.
Ff=Q(this.Ff,27)}}),this.hasAppearanceStream=!1};V(dt,ft);var pt=function(){dt.call(this),this.fontName="helvetica",this.combo=!1};V(pt,dt);var gt=function(){pt.call(this),this.combo=!0};V(
gt,pt);var mt=function(){gt.call(this),this.edit=!0};V(mt,gt);var bt=function(){ft.call(this),this.FT="/Btn",Object.defineProperty(this,"noToggleToOff",{enumerable:!0,configurable:!0,get:
function(){return Boolean(K(this.Ff,15))},set:function(t){!0===Boolean(t)?this.Ff=$(this.Ff,15):this.Ff=Q(this.Ff,15)}}),Object.defineProperty(this,"radio",{enumerable:!0,configurable:!0,
get:function(){return Boolean(K(this.Ff,16))},set:function(t){!0===Boolean(t)?this.Ff=$(this.Ff,16):this.Ff=Q(this.Ff,16)}}),Object.defineProperty(this,"pushButton",{enumerable:!0,
configurable:!0,get:function(){return Boolean(K(this.Ff,17))},set:function(t){!0===Boolean(t)?this.Ff=$(this.Ff,17):this.Ff=Q(this.Ff,17)}}),Object.defineProperty(this,"radioIsUnison",{
enumerable:!0,configurable:!0,get:function(){return Boolean(K(this.Ff,26))},set:function(t){!0===Boolean(t)?this.Ff=$(this.Ff,26):this.Ff=Q(this.Ff,26)}});var t,e={};Object.defineProperty(
this,"MK",{enumerable:!1,configurable:!1,get:function(){var t=function(t){return t};if(this.scope&&(t=this.scope.internal.getEncryptor(this.objId)),0!==Object.keys(e).length){var n,r=[];for(
n in r.push("<<"),e)r.push("/"+n+" ("+q(t(e[n]))+")");return r.push(">>"),r.join("\n")}},set:function(t){"object"===r(t)&&(e=t)}}),Object.defineProperty(this,"caption",{enumerable:!0,
configurable:!0,get:function(){return e.CA||""},set:function(t){"string"==typeof t&&(e.CA=t)}}),Object.defineProperty(this,"AS",{enumerable:!1,configurable:!1,get:function(){return t},set:
function(e){var n=null==e?"":e.toString();"/"===n.substr(0,1)&&(n=n.substr(1)),t="/"+U(n)}}),Object.defineProperty(this,"appearanceState",{enumerable:!0,configurable:!0,get:function(){
return t.substr(1,t.length-1)},set:function(e){t="/"+U(e)}})};V(bt,ft);var vt=function(){bt.call(this),this.pushButton=!0};V(vt,bt);var wt=function(){bt.call(this),this.radio=!0,this.
pushButton=!1;var t=[];Object.defineProperty(this,"Kids",{enumerable:!0,configurable:!1,get:function(){return t},set:function(e){t=void 0!==e?e:[]}})};V(wt,bt);var yt=function(){var t,e;ft.
call(this),Object.defineProperty(this,"Parent",{enumerable:!1,configurable:!1,get:function(){return t},set:function(e){t=e}}),Object.defineProperty(this,"optionName",{enumerable:!1,
configurable:!0,get:function(){return e},set:function(t){e=t}});var n,i={};Object.defineProperty(this,"MK",{enumerable:!1,configurable:!1,get:function(){var t=function(t){return t};this.
scope&&(t=this.scope.internal.getEncryptor(this.objId));var e,n=[];for(e in n.push("<<"),i)n.push("/"+e+" ("+q(t(i[e]))+")");return n.push(">>"),n.join("\n")},set:function(t){"object"===r(
t)&&(i=t)}}),Object.defineProperty(this,"caption",{enumerable:!0,configurable:!0,get:function(){return i.CA||""},set:function(t){"string"==typeof t&&(i.CA=t)}}),Object.defineProperty(this,
"AS",{enumerable:!1,configurable:!1,get:function(){return n},set:function(t){var e=null==t?"":t.toString();"/"===e.substr(0,1)&&(e=e.substr(1)),n="/"+U(e)}}),Object.defineProperty(this,
"appearanceState",{enumerable:!0,configurable:!0,get:function(){return n.substr(1,n.length-1)},set:function(t){var e=null==t?"":t.toString();"/"===e.substr(0,1)&&(e=e.substr(1)),n="/"+U(
e)}}),this.caption="l",this.appearanceState="Off",this._AppearanceType=Lt.RadioButton.Circle,this.appearanceStreamContent=this._AppearanceType.createAppearanceStream(this.optionName)};V(yt,
ft),wt.prototype.setAppearance=function(t){if(!("createAppearanceStream"in t)||!("getCA"in t))throw new Error("Couldn't assign Appearance to RadioButton. Appearance was Invalid!");for(var e
in this.Kids)if(this.Kids.hasOwnProperty(e)){var n=this.Kids[e];n.appearanceStreamContent=t.createAppearanceStream(n.optionName),n.caption=t.getCA()}},wt.prototype.createOption=function(t){
var e=new yt;return e.Parent=this,e.optionName=t,this.Kids.push(e),Nt.call(this.scope,e),e};var _t=function(){bt.call(this),this.fontName="zapfdingbats",this.caption="3",this.
appearanceState="On",this.value="On",this.textAlign="center",this.appearanceStreamContent=Lt.CheckBox.createAppearanceStream()};V(_t,bt);var xt=function(){ft.call(this),this.FT="/Tx",Object.
defineProperty(this,"multiline",{enumerable:!0,configurable:!0,get:function(){return Boolean(K(this.Ff,13))},set:function(t){!0===Boolean(t)?this.Ff=$(this.Ff,13):this.Ff=Q(this.Ff,13)}}),
Object.defineProperty(this,"fileSelect",{enumerable:!0,configurable:!0,get:function(){return Boolean(K(this.Ff,21))},set:function(t){!0===Boolean(t)?this.Ff=$(this.Ff,21):this.Ff=Q(this.Ff,
21)}}),Object.defineProperty(this,"doNotSpellCheck",{enumerable:!0,configurable:!0,get:function(){return Boolean(K(this.Ff,23))},set:function(t){!0===Boolean(t)?this.Ff=$(this.Ff,23):this.
Ff=Q(this.Ff,23)}}),Object.defineProperty(this,"doNotScroll",{enumerable:!0,configurable:!0,get:function(){return Boolean(K(this.Ff,24))},set:function(t){!0===Boolean(t)?this.Ff=$(this.Ff,
24):this.Ff=Q(this.Ff,24)}}),Object.defineProperty(this,"comb",{enumerable:!0,configurable:!0,get:function(){return Boolean(K(this.Ff,25))},set:function(t){!0===Boolean(t)?this.Ff=$(this.Ff,
25):this.Ff=Q(this.Ff,25)}}),Object.defineProperty(this,"richText",{enumerable:!0,configurable:!0,get:function(){return Boolean(K(this.Ff,26))},set:function(t){!0===Boolean(t)?this.Ff=$(
this.Ff,26):this.Ff=Q(this.Ff,26)}});var t=null;Object.defineProperty(this,"MaxLen",{enumerable:!0,configurable:!1,get:function(){return t},set:function(e){t=e}}),Object.defineProperty(this,
"maxLength",{enumerable:!0,configurable:!0,get:function(){return t},set:function(e){Number.isInteger(e)&&(t=e)}}),Object.defineProperty(this,"hasAppearanceStream",{enumerable:!0,
configurable:!0,get:function(){return this.V||this.DV}})};V(xt,ft);var At=function(){xt.call(this),Object.defineProperty(this,"password",{enumerable:!0,configurable:!0,get:function(){return
Boolean(K(this.Ff,14))},set:function(t){!0===Boolean(t)?this.Ff=$(this.Ff,14):this.Ff=Q(this.Ff,14)}}),this.password=!0};V(At,xt);var Lt={CheckBox:{createAppearanceStream:function(){return{
N:{On:Lt.CheckBox.YesNormal},D:{On:Lt.CheckBox.YesPushDown,Off:Lt.CheckBox.OffPushDown}}},YesPushDown:function(t){var e=Y(t);e.scope=t.scope;var n=[],r=t.scope.internal.getFont(t.fontName,t.
fontStyle).id,i=t.scope.__private__.encodeColorString(t.color),a=nt(t,t.caption);return n.push("0.749023 g"),n.push("0 0 "+H(Lt.internal.getWidth(t))+" "+H(Lt.internal.getHeight(t))+" re"),
n.push("f"),n.push("BMC"),n.push("q"),n.push("0 0 1 rg"),n.push("/"+r+" "+H(a.fontSize)+" Tf "+i),n.push("BT"),n.push(a.text),n.push("ET"),n.push("Q"),n.push("EMC"),e.stream=n.join("\n"),e},
YesNormal:function(t){var e=Y(t);e.scope=t.scope;var n=t.scope.internal.getFont(t.fontName,t.fontStyle).id,r=t.scope.__private__.encodeColorString(t.color),i=[],a=Lt.internal.getHeight(t),
s=Lt.internal.getWidth(t),o=nt(t,t.caption);return i.push("1 g"),i.push("0 0 "+H(s)+" "+H(a)+" re"),i.push("f"),i.push("q"),i.push("0 0 1 rg"),i.push("0 0 "+H(s-1)+" "+H(a-1)+" re"),i.push(
"W"),i.push("n"),i.push("0 g"),i.push("BT"),i.push("/"+n+" "+H(o.fontSize)+" Tf "+r),i.push(o.text),i.push("ET"),i.push("Q"),e.stream=i.join("\n"),e},OffPushDown:function(t){var e=Y(t);e.
scope=t.scope;var n=[];return n.push("0.749023 g"),n.push("0 0 "+H(Lt.internal.getWidth(t))+" "+H(Lt.internal.getHeight(t))+" re"),n.push("f"),e.stream=n.join("\n"),e}},RadioButton:{Circle:{
createAppearanceStream:function(t){var e={D:{Off:Lt.RadioButton.Circle.OffPushDown},N:{}};return e.N[t]=Lt.RadioButton.Circle.YesNormal,e.D[t]=Lt.RadioButton.Circle.YesPushDown,e},getCA:
function(){return"l"},YesNormal:function(t){var e=Y(t);e.scope=t.scope;var n=[],r=Lt.internal.getWidth(t)<=Lt.internal.getHeight(t)?Lt.internal.getWidth(t)/4:Lt.internal.getHeight(t)/4;
r=Number((.9*r).toFixed(5));var i=Lt.internal.Bezier_C,a=Number((r*i).toFixed(5));return n.push("q"),n.push("1 0 0 1 "+W(Lt.internal.getWidth(t)/2)+" "+W(Lt.internal.getHeight(t)/2)+" cm"),
n.push(r+" 0 m"),n.push(r+" "+a+" "+a+" "+r+" 0 "+r+" c"),n.push("-"+a+" "+r+" -"+r+" "+a+" -"+r+" 0 c"),n.push("-"+r+" -"+a+" -"+a+" -"+r+" 0 -"+r+" c"),n.push(a+" -"+r+" "+r+" -"+a+"
"+r+" 0 c"),n.push("f"),n.push("Q"),e.stream=n.join("\n"),e},YesPushDown:function(t){var e=Y(t);e.scope=t.scope;var n=[],r=Lt.internal.getWidth(t)<=Lt.internal.getHeight(t)?Lt.internal.
getWidth(t)/4:Lt.internal.getHeight(t)/4;r=Number((.9*r).toFixed(5));var i=Number((2*r).toFixed(5)),a=Number((i*Lt.internal.Bezier_C).toFixed(5)),s=Number((r*Lt.internal.Bezier_C).toFixed(
5));return n.push("0.749023 g"),n.push("q"),n.push("1 0 0 1 "+W(Lt.internal.getWidth(t)/2)+" "+W(Lt.internal.getHeight(t)/2)+" cm"),n.push(i+" 0 m"),n.push(i+" "+a+" "+a+" "+i+" 0 "+i+" c"),
n.push("-"+a+" "+i+" -"+i+" "+a+" -"+i+" 0 c"),n.push("-"+i+" -"+a+" -"+a+" -"+i+" 0 -"+i+" c"),n.push(a+" -"+i+" "+i+" -"+a+" "+i+" 0 c"),n.push("f"),n.push("Q"),n.push("0 g"),n.push("q"),
n.push("1 0 0 1 "+W(Lt.internal.getWidth(t)/2)+" "+W(Lt.internal.getHeight(t)/2)+" cm"),n.push(r+" 0 m"),n.push(r+" "+s+" "+s+" "+r+" 0 "+r+" c"),n.push("-"+s+" "+r+" -"+r+" "+s+" -"+r+" 0
c"),n.push("-"+r+" -"+s+" -"+s+" -"+r+" 0 -"+r+" c"),n.push(s+" -"+r+" "+r+" -"+s+" "+r+" 0 c"),n.push("f"),n.push("Q"),e.stream=n.join("\n"),e},OffPushDown:function(t){var e=Y(t);e.scope=t.
scope;var n=[],r=Lt.internal.getWidth(t)<=Lt.internal.getHeight(t)?Lt.internal.getWidth(t)/4:Lt.internal.getHeight(t)/4;r=Number((.9*r).toFixed(5));var i=Number((2*r).toFixed(5)),a=Number((
i*Lt.internal.Bezier_C).toFixed(5));return n.push("0.749023 g"),n.push("q"),n.push("1 0 0 1 "+W(Lt.internal.getWidth(t)/2)+" "+W(Lt.internal.getHeight(t)/2)+" cm"),n.push(i+" 0 m"),n.push(
i+" "+a+" "+a+" "+i+" 0 "+i+" c"),n.push("-"+a+" "+i+" -"+i+" "+a+" -"+i+" 0 c"),n.push("-"+i+" -"+a+" -"+a+" -"+i+" 0 -"+i+" c"),n.push(a+" -"+i+" "+i+" -"+a+" "+i+" 0 c"),n.push("f"),n.
push("Q"),e.stream=n.join("\n"),e}},Cross:{createAppearanceStream:function(t){var e={D:{Off:Lt.RadioButton.Cross.OffPushDown},N:{}};return e.N[t]=Lt.RadioButton.Cross.YesNormal,e.D[t]=Lt.
RadioButton.Cross.YesPushDown,e},getCA:function(){return"8"},YesNormal:function(t){var e=Y(t);e.scope=t.scope;var n=[],r=Lt.internal.calculateCross(t);return n.push("q"),n.push("1 1 "+H(Lt.
internal.getWidth(t)-2)+" "+H(Lt.internal.getHeight(t)-2)+" re"),n.push("W"),n.push("n"),n.push(H(r.x1.x)+" "+H(r.x1.y)+" m"),n.push(H(r.x2.x)+" "+H(r.x2.y)+" l"),n.push(H(r.x4.x)+" "+H(r.
x4.y)+" m"),n.push(H(r.x3.x)+" "+H(r.x3.y)+" l"),n.push("s"),n.push("Q"),e.stream=n.join("\n"),e},YesPushDown:function(t){var e=Y(t);e.scope=t.scope;var n=Lt.internal.calculateCross(t),r=[];
return r.push("0.749023 g"),r.push("0 0 "+H(Lt.internal.getWidth(t))+" "+H(Lt.internal.getHeight(t))+" re"),r.push("f"),r.push("q"),r.push("1 1 "+H(Lt.internal.getWidth(t)-2)+" "+H(Lt.
internal.getHeight(t)-2)+" re"),r.push("W"),r.push("n"),r.push(H(n.x1.x)+" "+H(n.x1.y)+" m"),r.push(H(n.x2.x)+" "+H(n.x2.y)+" l"),r.push(H(n.x4.x)+" "+H(n.x4.y)+" m"),r.push(H(n.x3.x)+" "+H(
n.x3.y)+" l"),r.push("s"),r.push("Q"),e.stream=r.join("\n"),e},OffPushDown:function(t){var e=Y(t);e.scope=t.scope;var n=[];return n.push("0.749023 g"),n.push("0 0 "+H(Lt.internal.getWidth(
t))+" "+H(Lt.internal.getHeight(t))+" re"),n.push("f"),e.stream=n.join("\n"),e}}},createDefaultAppearanceStream:function(t){var e=t.scope.internal.getFont(t.fontName,t.fontStyle).id,n=t.
scope.__private__.encodeColorString(t.color);return"/"+e+" "+t.fontSize+" Tf "+n}};Lt.internal={Bezier_C:.551915024494,calculateCross:function(t){var e=Lt.internal.getWidth(t),n=Lt.internal.
getHeight(t),r=Math.min(e,n);return{x1:{x:(e-r)/2,y:(n-r)/2+r},x2:{x:(e-r)/2+r,y:(n-r)/2},x3:{x:(e-r)/2,y:(n-r)/2},x4:{x:(e-r)/2+r,y:(n-r)/2+r}}},Lt.internal.getWidth=function(t){var e=0;
return"object"===r(t)&&(e=G(t.Rect[2])),e},Lt.internal.getHeight=function(t){var e=0;return"object"===r(t)&&(e=G(t.Rect[3])),e};var Nt=T.addField=function(t){if(st(this,t),!(t instanceof
ft))throw new Error("Invalid argument passed to jsPDF.addField.");var e;return(e=t).scope.internal.acroformPlugin.printedOut&&(e.scope.internal.acroformPlugin.printedOut=!1,e.scope.internal.
acroformPlugin.acroFormDictionaryRoot=null),e.scope.internal.acroformPlugin.acroFormDictionaryRoot.Fields.push(e),t.page=t.scope.internal.getCurrentPageInfo().pageNumber,this};T.
AcroFormChoiceField=dt,T.AcroFormListBox=pt,T.AcroFormComboBox=gt,T.AcroFormEditBox=mt,T.AcroFormButton=bt,T.AcroFormPushButton=vt,T.AcroFormRadioButton=wt,T.AcroFormCheckBox=_t,T.
AcroFormTextField=xt,T.AcroFormPasswordField=At,T.AcroFormAppearance=Lt,T.AcroForm={ChoiceField:dt,ListBox:pt,ComboBox:gt,EditBox:mt,Button:bt,PushButton:vt,RadioButton:wt,CheckBox:_t,
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TextField:xt,PasswordField:At,Appearance:Lt},R.AcroForm={ChoiceField:dt,ListBox:pt,ComboBox:gt,EditBox:mt,Button:bt,PushButton:vt,RadioButton:wt,CheckBox:_t,TextField:xt,PasswordField:At,
Appearance:Lt};var St=R.AcroForm;function kt(t){return t.reduce(function(t,e,n){return t[e]=n,t},{})}!function(t){var e="addImage_";t.__addimage__={};var n="UNKNOWN",i={PNG:[[137,80,78,71]],
TIFF:[[77,77,0,42],[73,73,42,0]],JPEG:[[255,216,255,224,void 0,void 0,74,70,73,70,0],[255,216,255,225,void 0,void 0,69,120,105,102,0,0],[255,216,255,219],[255,216,255,238]],JPEG2000:[[0,0,0,
12,106,80,32,32]],GIF87a:[[71,73,70,56,55,97]],GIF89a:[[71,73,70,56,57,97]],WEBP:[[82,73,70,70,void 0,void 0,void 0,void 0,87,69,66,80]],BMP:[[66,77],[66,65],[67,73],[67,80],[73,67],[80,
84]]},a=t.__addimage__.getImageFileTypeByImageData=function(t,e){var r,a,s,o,h,l=n;if("RGBA"===(e=e||n)||void 0!==t.data&&t.data instanceof Uint8ClampedArray&&"height"in t&&"width"in
t)return"RGBA";if(L(t))for(h in i)for(s=i[h],r=0;r<s.length;r+=1){for(o=!0,a=0;a<s[r].length;a+=1)if(void 0!==s[r][a]&&s[r][a]!==t[a]){o=!1;break}if(!0===o){l=h;break}}else for(h in i)for(
s=i[h],r=0;r<s.length;r+=1){for(o=!0,a=0;a<s[r].length;a+=1)if(void 0!==s[r][a]&&s[r][a]!==t.charCodeAt(a)){o=!1;break}if(!0===o){l=h;break}}return l===n&&e!==n&&(l=e),l},s=function t(e){
for(var n=this.internal.write,r=this.internal.putStream,i=(0,this.internal.getFilters)();-1!==i.indexOf("FlateEncode");)i.splice(i.indexOf("FlateEncode"),1);e.objectId=this.internal.
newObject();var a=[];if(a.push({key:"Type",value:"/XObject"}),a.push({key:"Subtype",value:"/Image"}),a.push({key:"Width",value:e.width}),a.push({key:"Height",value:e.height}),e.
colorSpace===w.INDEXED?a.push({key:"ColorSpace",value:"[/Indexed /DeviceRGB "+(e.palette.length/3-1)+" "+("sMask"in e&&void 0!==e.sMask?e.objectId+2:e.objectId+1)+" 0 R]"}):(a.push({key:
"ColorSpace",value:"/"+e.colorSpace}),e.colorSpace===w.DEVICE_CMYK&&a.push({key:"Decode",value:"[1 0 1 0 1 0 1 0]"})),a.push({key:"BitsPerComponent",value:e.bitsPerComponent}),
"decodeParameters"in e&&void 0!==e.decodeParameters&&a.push({key:"DecodeParms",value:"<<"+e.decodeParameters+">>"}),"transparency"in e&&Array.isArray(e.transparency)&&e.transparency.
length>0){for(var s="",o=0,h=e.transparency.length;o<h;o++)s+=e.transparency[o]+" "+e.transparency[o]+" ";a.push({key:"Mask",value:"["+s+"]"})}void 0!==e.sMask&&a.push({key:"SMask",value:e.
objectId+1+" 0 R"});var l=void 0!==e.filter?["/"+e.filter]:void 0;if(r({data:e.data,additionalKeyValues:a,alreadyAppliedFilters:l,objectId:e.objectId}),n("endobj"),"sMask"in e&&void 0!==e.
sMask){var c,u=null!==(c=e.sMaskBitsPerComponent)&&void 0!==c?c:e.bitsPerComponent,f={width:e.width,height:e.height,colorSpace:"DeviceGray",bitsPerComponent:u,data:e.sMask};"filter"in e&&(f.
decodeParameters="/Predictor "+(new t.__addimage__.PNG_Predictor).concat(e.predictor," /Colors 1 /BitsPerComponent ").concat(u," /Columns ").concat(e.width),f.filter=e.filter),t.call(this,f)}if(e.colorSpace===w.INDEXED){var
d=this.internal.newObject();r({data:S(new Uint8Array(e.palette)),objectId:d}),n("endobj")}},o=function(){var t=this.internal.collections[e+"images"];for(var n in t)s.call(this,t[n])},
h=function(){var t,n=this.internal.collections[e+"images"],r=this.internal.write;for(var i in n)r("/I"+(t=n[i]).index,t.objectId,"0","R")},l=function(){this.internal.collections[
e+"images"]||(this.internal.collections[e+"images"]={},this.internal.events.subscribe("putResources",o),this.internal.events.subscribe("putXobjectDict",h))},c=function(){var t=this.internal.
collections[e+"images"];return l.call(this),t},u=function(){return Object.keys(this.internal.collections[e+"images"]).length},d=function(e){return"function"==typeof t["process"+e.
toUpperCase()]},p=function(t){return"object"===r(t)&&1===t.nodeType},g=function(e,n){if("IMG"===e.nodeName&&e.hasAttribute("src")){var r=""+e.getAttribute("src");if(0===r.indexOf("data:
image/"))return f(unescape(r).split("base64,").pop());var i=t.loadFile(r,!0);if(void 0!==i)return i}if("CANVAS"===e.nodeName){if(0===e.width||0===e.height)throw new Error("Given canvas must
have data. Canvas width: "+e.width+", height: "+e.height);var a;switch(n){case"PNG":a="image/png";break;case"WEBP":a="image/webp";break;default:a="image/jpeg"}return f(e.toDataURL(a,1).
split("base64,").pop())}},m=function(t){var n=this.internal.collections[e+"images"];if(n)for(var r in n)if(t===n[r].alias)return n[r]},b=function(t,e,n){return t||e||(t=-96,e=-96),t<0&&(t=-
1*n.width*72/t/this.internal.scaleFactor),e<0&&(e=-1*n.height*72/e/this.internal.scaleFactor),0===t&&(t=e*n.width/n.height),0===e&&(e=t*n.height/n.width),[t,e]},v=function(t,e,n,r,i,a){var
s=b.call(this,n,r,i),o=this.internal.getCoordinateString,h=this.internal.getVerticalCoordinateString,l=c.call(this);if(n=s[0],r=s[1],l[i.index]=i,a){a*=Math.PI/180;var u=Math.cos(a),f=Math.
sin(a),d=function(t){return t.toFixed(4)},p=[d(u),d(f),d(-1*f),d(u),0,0,"cm"]}this.internal.write("q"),a?(this.internal.write([1,"0","0",1,o(t),h(e+r),"cm"].join(" ")),this.internal.write(p.
join(" ")),this.internal.write([o(n),"0","0",o(r),"0","0","cm"].join(" "))):this.internal.write([o(n),"0","0",o(r),o(t),h(e+r),"cm"].join(" ")),this.isAdvancedAPI()&&this.internal.write([1,
0,0,-1,0,0,"cm"].join(" ")),this.internal.write("/I"+i.index+" Do"),this.internal.write("Q")},w=t.color_spaces={DEVICE_RGB:"DeviceRGB",DEVICE_GRAY:"DeviceGray",DEVICE_CMYK:"DeviceCMYK",
CAL_GREY:"CalGray",CAL_RGB:"CalRGB",LAB:"Lab",ICC_BASED:"ICCBased",INDEXED:"Indexed",PATTERN:"Pattern",SEPARATION:"Separation",DEVICE_N:"DeviceN"};t.decode={DCT_DECODE:"DCTDecode",
FLATE_DECODE:"FlateDecode",LZW_DECODE:"LZWDecode",JPX_DECODE:"JPXDecode",JBIG2_DECODE:"JBIG2Decode",ASCII85_DECODE:"ASCII85Decode",ASCII_HEX_DECODE:"ASCIIHexDecode",RUN_LENGTH_DECODE:
"RunLengthDecode",CCITT_FAX_DECODE:"CCITTFaxDecode"};var y=t.image_compression={NONE:"NONE",FAST:"FAST",MEDIUM:"MEDIUM",SLOW:"SLOW"},_=t.__addimage__.sHashCode=function(t){var e,n,r=0;if(
"string"==typeof t)for(n=t.length,e=0;e<n;e++)r=(r<<5)-r+t.charCodeAt(e),r|=0;else if(L(t))for(n=t.byteLength/2,e=0;e<n;e++)r=(r<<5)-r+t[e],r|=0;return r},x=t.__addimage__.
validateStringAsBase64=function(t){(t=t||"").toString().trim();var e=!0;return 0===t.length&&(e=!1),t.length%4!=0&&(e=!1),!1===/^[A-Za-z0-9+/]+$/.test(t.substr(0,t.length-2))&&(e=!1),!1===/
^[A-Za-z0-9/][A-Za-z0-9+/]|[A-Za-z0-9+/]=|==$/.test(t.substr(-2))&&(e=!1),e},A=t.__addimage__.extractImageFromDataUrl=function(t){if(null==t)return null;if(!(t=t.trim()).startsWith("data:
"))return null;var e=t.indexOf(",");return e<0?null:t.substring(0,e).trim().endsWith("base64")?t.substring(e+1):null};t.__addimage__.isArrayBuffer=function(t){return t instanceof
ArrayBuffer};var L=t.__addimage__.isArrayBufferView=function(t){return t instanceof Int8Array||t instanceof Uint8Array||t instanceof Uint8ClampedArray||t instanceof Int16Array||t instanceof
Uint16Array||t instanceof Int32Array||t instanceof Uint32Array||t instanceof Float32Array||t instanceof Float64Array},N=t.__addimage__.binaryStringToUint8Array=function(t){for(var e=t.
length,n=new Uint8Array(e),r=0;r<e;r++)n[r]=t.charCodeAt(r);return n},S=t.__addimage__.arrayBufferToBinaryString=function(t){for(var e="",n=L(t)?t:new Uint8Array(t),r=0;r<n.length;
r+=8192)e+=String.fromCharCode.apply(null,n.subarray(r,r+8192));return e};t.addImage=function(){var t,e,i,a,s,o,h,c,u;if("number"==typeof arguments[1]?(e=n,i=arguments[1],a=arguments[2],
s=arguments[3],o=arguments[4],h=arguments[5],c=arguments[6],u=arguments[7]):(e=arguments[1],i=arguments[2],a=arguments[3],s=arguments[4],o=arguments[5],h=arguments[6],c=arguments[7],
u=arguments[8]),"object"===r(t=arguments[0])&&!p(t)&&"imageData"in t){var f=t;t=f.imageData,e=f.format||e||n,i=f.x||i||0,a=f.y||a||0,s=f.w||f.width||s,o=f.h||f.height||o,h=f.alias||h,c=f.
compression||c,u=f.rotation||f.angle||u}var d=this.internal.getFilters();if(void 0===c&&-1!==d.indexOf("FlateEncode")&&(c="SLOW"),isNaN(i)||isNaN(a))throw new Error("Invalid coordinates
passed to jsPDF.addImage");l.call(this);var g=k.call(this,t,e,h,c);return v.call(this,i,a,s,o,g,u),this};var k=function(e,r,i,s){var o,h,l;if("string"==typeof e&&a(e)===n){e=unescape(e);var
c=P(e,!1);(""!==c||void 0!==(c=t.loadFile(e,!0)))&&(e=c)}if(p(e)&&(e=g(e,r)),r=a(e,r),!d(r))throw new Error("addImage does not support files of type '"+r+"', please ensure that a plugin for
'"+r+"' support is added.");if((null==(l=i)||0===l.length)&&(i=function(t){return"string"==typeof t||L(t)?_(t):L(t.data)?_(t.data):null}(e)),(o=m.call(this,i))||(e instanceof
Uint8Array||"RGBA"===r||(h=e,e=N(e)),o=this["process"+r.toUpperCase()](e,u.call(this),i,function(e){return e&&"string"==typeof e&&(e=e.toUpperCase()),e in t.image_compression?e:y.NONE}(s),
h)),!o)throw new Error("An unknown error occurred whilst processing the image.");return o},P=t.__addimage__.convertBase64ToBinaryString=function(t,e){e="boolean"!=typeof e||e;var n,r="";if(
"string"==typeof t){var i;n=null!==(i=A(t))&&void 0!==i?i:t;try{r=f(n)}catch(a){if(e)throw x(n)?new Error("atob-Error in jsPDF.convertBase64ToBinaryString "+a.message):new Error("Supplied
Data is not a valid base64-String jsPDF.convertBase64ToBinaryString ")}}return r};t.getImageProperties=function(e){var r,i,s="";if(p(e)&&(e=g(e)),"string"==typeof e&&a(e)===n&&(""===(s=P(e,
!1))&&(s=t.loadFile(e)||""),e=s),i=a(e),!d(i))throw new Error("addImage does not support files of type '"+i+"', please ensure that a plugin for '"+i+"' support is added.");if(e instanceof
Uint8Array||(e=N(e)),!(r=this["process"+i.toUpperCase()](e)))throw new Error("An unknown error occurred whilst processing the image");return r.fileType=i,r}}(R.API),
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function(t){var e=function(t){if(void 0!==t&&""!=t)return!0};R.API.events.push(["addPage",function(t){this.internal.getPageInfo(t.pageNumber).pageContext.annotations=[]}]),t.events.push([
"putPage",function(t){for(var n,r,i,a=this.internal.getCoordinateString,s=this.internal.getVerticalCoordinateString,o=this.internal.getPageInfoByObjId(t.objId),h=t.pageContext.annotations,
l=!1,c=0;c<h.length&&!l;c++)switch((n=h[c]).type){case"link":(e(n.options.url)||e(n.options.pageNumber))&&(l=!0);break;case"reference":case"text":case"freetext":l=!0}if(0!=l){this.internal.
write("/Annots [");for(var u=0;u<h.length;u++){n=h[u];var f=this.internal.pdfEscape,d=this.internal.getEncryptor(t.objId);switch(n.type){case"reference":this.internal.write(" "+n.object.
objId+" 0 R ");break;case"text":var p=this.internal.newAdditionalObject(),g=this.internal.newAdditionalObject(),m=this.internal.getEncryptor(p.objId),b=n.title||"Note";i="<</Type /Annot /
Subtype /Text "+(r="/Rect ["+a(n.bounds.x)+" "+s(n.bounds.y+n.bounds.h)+" "+a(n.bounds.x+n.bounds.w)+" "+s(n.bounds.y)+"] ")+"/Contents ("+f(m(n.contents))+")",i+=" /Popup "+g.objId+" 0 R",
i+=" /P "+o.objId+" 0 R",i+=" /T ("+f(m(b))+") >>",p.content=i;var v=p.objId+" 0 R";i="<</Type /Annot /Subtype /Popup "+(r="/Rect ["+a(n.bounds.x+30)+" "+s(n.bounds.y+n.bounds.h)+" "+a(n.
bounds.x+n.bounds.w+30)+" "+s(n.bounds.y)+"] ")+" /Parent "+v,n.open&&(i+=" /Open true"),i+=" >>",g.content=i,this.internal.write(p.objId,"0 R",g.objId,"0 R");break;case"freetext":r="/Rect [
"+a(n.bounds.x)+" "+s(n.bounds.y)+" "+a(n.bounds.x+n.bounds.w)+" "+s(n.bounds.y+n.bounds.h)+"] ";var w="font: Helvetica,sans-serif 12.0pt; text-align:left; color:#"+(n.color||"#000000");
i="<</Type /Annot /Subtype /FreeText "+r+"/Contents ("+f(d(n.contents))+")",i+=" /DS("+f(d(w))+")",i+=" /Border [0 0 0]",i+=" >>",this.internal.write(i);break;case"link":if(n.options.name){
var y=this.annotations._nameMap[n.options.name];n.options.pageNumber=y.page,n.options.top=y.y}else n.options.top||(n.options.top=0);if(r="/Rect ["+n.finalBounds.x+" "+n.finalBounds.y+" "+n.
finalBounds.w+" "+n.finalBounds.h+"] ",i="",n.options.url)i="<</Type /Annot /Subtype /Link "+r+"/Border [0 0 0] /A <</S /URI /URI ("+f(d(n.options.url))+") >>";else if(n.options.
pageNumber)switch(i="<</Type /Annot /Subtype /Link "+r+"/Border [0 0 0] /Dest ["+this.internal.getPageInfo(n.options.pageNumber).objId+" 0 R",n.options.magFactor=n.options.magFactor||"XYZ",
n.options.magFactor){case"Fit":i+=" /Fit]";break;case"FitH":i+=" /FitH "+n.options.top+"]";break;case"FitV":n.options.left=n.options.left||0,i+=" /FitV "+n.options.left+"]";break;default:
var _=s(n.options.top);n.options.left=n.options.left||0,void 0===n.options.zoom&&(n.options.zoom=0),i+=" /XYZ "+n.options.left+" "+_+" "+n.options.zoom+"]"}""!=i&&(i+=" >>",this.internal.
write(i))}}this.internal.write("]")}}),t.createAnnotation=function(t){var e=this.internal.getCurrentPageInfo();switch(t.type){case"link":this.link(t.bounds.x,t.bounds.y,t.bounds.w,t.bounds.
h,t);break;case"text":case"freetext":e.pageContext.annotations.push(t)}},t.link=function(t,e,n,r,i){var a=this.internal.getCurrentPageInfo(),s=this.internal.getCoordinateString,o=this.
internal.getVerticalCoordinateString;a.pageContext.annotations.push({finalBounds:{x:s(t),y:o(e),w:s(t+n),h:o(e+r)},options:i,type:"link"})},t.textWithLink=function(t,e,n,r){var i,a,s=this.
getTextWidth(t),o=this.internal.getLineHeight()/this.internal.scaleFactor;if(void 0!==r.maxWidth){a=r.maxWidth;var h=this.splitTextToSize(t,a).length;i=Math.ceil(o*h)}else a=s,i=o;return
this.text(t,e,n,r),n+=.2*o,"center"===r.align&&(e-=s/2),"right"===r.align&&(e-=s),this.link(e,n-o,a,i,r),s},t.getTextWidth=function(t){var e=this.internal.getFontSize();return this.
getStringUnitWidth(t)*e/this.internal.scaleFactor}}(R.API),
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function(t){var e={1569:[65152],1570:[65153,65154],1571:[65155,65156],1572:[65157,65158],1573:[65159,65160],1574:[65161,65162,65163,65164],1575:[65165,65166],1576:[65167,65168,65169,65170],
1577:[65171,65172],1578:[65173,65174,65175,65176],1579:[65177,65178,65179,65180],1580:[65181,65182,65183,65184],1581:[65185,65186,65187,65188],1582:[65189,65190,65191,65192],1583:[65193,
65194],1584:[65195,65196],1585:[65197,65198],1586:[65199,65200],1587:[65201,65202,65203,65204],1588:[65205,65206,65207,65208],1589:[65209,65210,65211,65212],1590:[65213,65214,65215,65216],
1591:[65217,65218,65219,65220],1592:[65221,65222,65223,65224],1593:[65225,65226,65227,65228],1594:[65229,65230,65231,65232],1601:[65233,65234,65235,65236],1602:[65237,65238,65239,65240],
1603:[65241,65242,65243,65244],1604:[65245,65246,65247,65248],1605:[65249,65250,65251,65252],1606:[65253,65254,65255,65256],1607:[65257,65258,65259,65260],1608:[65261,65262],1609:[65263,
65264,64488,64489],1610:[65265,65266,65267,65268],1649:[64336,64337],1655:[64477],1657:[64358,64359,64360,64361],1658:[64350,64351,64352,64353],1659:[64338,64339,64340,64341],1662:[64342,
64343,64344,64345],1663:[64354,64355,64356,64357],1664:[64346,64347,64348,64349],1667:[64374,64375,64376,64377],1668:[64370,64371,64372,64373],1670:[64378,64379,64380,64381],1671:[64382,
64383,64384,64385],1672:[64392,64393],1676:[64388,64389],1677:[64386,64387],1678:[64390,64391],1681:[64396,64397],1688:[64394,64395],1700:[64362,64363,64364,64365],1702:[64366,64367,64368,
64369],1705:[64398,64399,64400,64401],1709:[64467,64468,64469,64470],1711:[64402,64403,64404,64405],1713:[64410,64411,64412,64413],1715:[64406,64407,64408,64409],1722:[64414,64415],1723:[
64416,64417,64418,64419],1726:[64426,64427,64428,64429],1728:[64420,64421],1729:[64422,64423,64424,64425],1733:[64480,64481],1734:[64473,64474],1735:[64471,64472],1736:[64475,64476],1737:[
64482,64483],1739:[64478,64479],1740:[64508,64509,64510,64511],1744:[64484,64485,64486,64487],1746:[64430,64431],1747:[64432,64433]},n={65247:{65154:65269,65156:65271,65160:65273,65166:
65275},65248:{65154:65270,65156:65272,65160:65274,65166:65276},65165:{65247:{65248:{65258:65010}}},1617:{1612:64606,1613:64607,1614:64608,1615:64609,1616:64610}},r={1612:64606,1613:64607,
1614:64608,1615:64609,1616:64610},i=[1570,1571,1573,1575];t.__arabicParser__={};var a=t.__arabicParser__.isInArabicSubstitutionA=function(t){return void 0!==e[t.charCodeAt(0)]},s=t.
__arabicParser__.isArabicLetter=function(t){return"string"==typeof t&&/^[\u0600-\u06FF\u0750-\u077F\u08A0-\u08FF\uFB50-\uFDFF\uFE70-\uFEFF]+$/.test(t)},o=t.__arabicParser__.
isArabicEndLetter=function(t){return s(t)&&a(t)&&e[t.charCodeAt(0)].length<=2},h=t.__arabicParser__.isArabicAlfLetter=function(t){return s(t)&&i.indexOf(t.charCodeAt(0))>=0};t.
__arabicParser__.arabicLetterHasIsolatedForm=function(t){return s(t)&&a(t)&&e[t.charCodeAt(0)].length>=1};var l=t.__arabicParser__.arabicLetterHasFinalForm=function(t){return s(t)&&a(t)&&e[
t.charCodeAt(0)].length>=2};t.__arabicParser__.arabicLetterHasInitialForm=function(t){return s(t)&&a(t)&&e[t.charCodeAt(0)].length>=3};var c=t.__arabicParser__.
arabicLetterHasMedialForm=function(t){return s(t)&&a(t)&&4==e[t.charCodeAt(0)].length},u=t.__arabicParser__.resolveLigatures=function(t){var e=0,r=n,i="",a=0;for(e=0;e<t.length;e+=1)void
0!==r[t.charCodeAt(e)]?(a++,"number"==typeof(r=r[t.charCodeAt(e)])&&(i+=String.fromCharCode(r),r=n,a=0),e===t.length-1&&(r=n,i+=t.charAt(e-(a-1)),e-=a-1,a=0)):(r=n,i+=t.charAt(e-a),e-=a,
a=0);return i};t.__arabicParser__.isArabicDiacritic=function(t){return void 0!==t&&void 0!==r[t.charCodeAt(0)]};var f=t.__arabicParser__.getCorrectForm=function(t,e,n){return s(t)?1===a(
t)?-1:!l(t)||!s(e)&&!s(n)||!s(n)&&o(e)||o(t)&&!s(e)||o(t)&&h(e)||o(t)&&o(e)?0:c(t)&&s(e)&&!o(e)&&s(n)&&l(n)?3:o(t)||!s(n)?1:2:-1},d=function(t){var n=0,r=0,i=0,a="",o="",h="",l=(t=t||"").
split("\\s+"),c=[];for(n=0;n<l.length;n+=1){for(c.push(""),r=0;r<l[n].length;r+=1)a=l[n][r],o=l[n][r-1],h=l[n][r+1],s(a)?(i=f(a,o,h),c[n]+=-1!==i?String.fromCharCode(e[a.charCodeAt(0)][i]):
a):c[n]+=a;c[n]=u(c[n])}return c.join(" ")},p=t.__arabicParser__.processArabic=t.processArabic=function(){var t,e="string"==typeof arguments[0]?arguments[0]:arguments[0].text,n=[];if(Array.
isArray(e)){var r=0;for(n=[],r=0;r<e.length;r+=1)Array.isArray(e[r])?n.push([d(e[r][0]),e[r][1],e[r][2]]):n.push([d(e[r])]);t=n}else t=d(e);return"string"==typeof arguments[0]?t:(arguments[
0].text=t,arguments[0])};t.events.push(["preProcessText",p])}(R.API),
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function(t){t.autoPrint=function(t){var e;return(t=t||{}).variant=t.variant||"non-conform","javascript"===t.variant?this.addJS("print({});"):(this.internal.events.subscribe(
"postPutResources",function(){e=this.internal.newObject(),this.internal.out("<<"),this.internal.out("/S /Named"),this.internal.out("/Type /Action"),this.internal.out("/N /Print"),this.
internal.out(">>"),this.internal.out("endobj")}),this.internal.events.subscribe("putCatalog",function(){this.internal.out("/OpenAction "+e+" 0 R")})),this}}(R.API),
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function(t){var e=function(){var t=void 0;Object.defineProperty(this,"pdf",{get:function(){return t},set:function(e){t=e}});var e=150;Object.defineProperty(this,"width",{get:function(){
return e},set:function(t){e=isNaN(t)||!1===Number.isInteger(t)||t<0?150:t,this.getContext("2d").pageWrapXEnabled&&(this.getContext("2d").pageWrapX=e+1)}});var n=300;Object.defineProperty(
this,"height",{get:function(){return n},set:function(t){n=isNaN(t)||!1===Number.isInteger(t)||t<0?300:t,this.getContext("2d").pageWrapYEnabled&&(this.getContext("2d").pageWrapY=n+1)}});var
r=[];Object.defineProperty(this,"childNodes",{get:function(){return r},set:function(t){r=t}});var i={};Object.defineProperty(this,"style",{get:function(){return i},set:function(t){i=t}}),
Object.defineProperty(this,"parentNode",{})};e.prototype.getContext=function(t,e){var n;if("2d"!==(t=t||"2d"))return null;for(n in e)this.pdf.context2d.hasOwnProperty(n)&&(this.pdf.
context2d[n]=e[n]);return this.pdf.context2d._canvas=this,this.pdf.context2d},e.prototype.toDataURL=function(){throw new Error("toDataURL is not implemented.")},t.events.push(["initialized",
function(){this.canvas=new e,this.canvas.pdf=this}])}(R.API),function(t){var e={left:0,top:0,bottom:0,right:0},n=!1,i=function(){void 0===this.internal.__cell__&&(this.internal.__cell__={},
this.internal.__cell__.padding=3,this.internal.__cell__.headerFunction=void 0,this.internal.__cell__.margins=Object.assign({},e),this.internal.__cell__.margins.width=this.getPageWidth(),a.
call(this))},a=function(){this.internal.__cell__.lastCell=new s,this.internal.__cell__.pages=1},s=function(){var t=arguments[0];Object.defineProperty(this,"x",{enumerable:!0,get:function(){
return t},set:function(e){t=e}});var e=arguments[1];Object.defineProperty(this,"y",{enumerable:!0,get:function(){return e},set:function(t){e=t}});var n=arguments[2];Object.defineProperty(
this,"width",{enumerable:!0,get:function(){return n},set:function(t){n=t}});var r=arguments[3];Object.defineProperty(this,"height",{enumerable:!0,get:function(){return r},set:function(t){
r=t}});var i=arguments[4];Object.defineProperty(this,"text",{enumerable:!0,get:function(){return i},set:function(t){i=t}});var a=arguments[5];Object.defineProperty(this,"lineNumber",{
enumerable:!0,get:function(){return a},set:function(t){a=t}});var s=arguments[6];return Object.defineProperty(this,"align",{enumerable:!0,get:function(){return s},set:function(t){s=t}}),
this};s.prototype.clone=function(){return new s(this.x,this.y,this.width,this.height,this.text,this.lineNumber,this.align)},s.prototype.toArray=function(){return[this.x,this.y,this.width,
this.height,this.text,this.lineNumber,this.align]},t.setHeaderFunction=function(t){return i.call(this),this.internal.__cell__.headerFunction="function"==typeof t?t:void 0,this},t.
getTextDimensions=function(t,e){i.call(this);var n=(e=e||{}).fontSize||this.getFontSize(),r=e.font||this.getFont(),a=e.scaleFactor||this.internal.scaleFactor,s=0,o=0,h=0,l=this;if(!Array.
isArray(t)&&"string"!=typeof t){if("number"!=typeof t)throw new Error("getTextDimensions expects text-parameter to be of type String or type Number or an Array of Strings.");t=String(t)}var
c=e.maxWidth;c>0?"string"==typeof t?t=this.splitTextToSize(t,c):"[object Array]"===Object.prototype.toString.call(t)&&(t=t.reduce(function(t,e){return t.concat(l.splitTextToSize(e,c))},[])):
t=Array.isArray(t)?t:[t];for(var u=0;u<t.length;u++)s<(h=this.getStringUnitWidth(t[u],{font:r})*n)&&(s=h);return 0!==s&&(o=t.length),{w:s/=a,h:Math.max((o*n*this.getLineHeightFactor()-n*(
this.getLineHeightFactor()-1))/a,0)}},t.cellAddPage=function(){i.call(this),this.addPage();var t=this.internal.__cell__.margins||e;return this.internal.__cell__.lastCell=new s(t.left,t.top,
void 0,void 0),this.internal.__cell__.pages+=1,this};var o=t.cell=function(){var t;t=arguments[0]instanceof s?arguments[0]:new s(arguments[0],arguments[1],arguments[2],arguments[3],
arguments[4],arguments[5],arguments[6]),i.call(this);var r=this.internal.__cell__.lastCell,a=this.internal.__cell__.padding,o=this.internal.__cell__.margins||e,h=this.internal.__cell__.
tableHeaderRow,l=this.internal.__cell__.printHeaders;return void 0!==r.lineNumber&&(r.lineNumber===t.lineNumber?(t.x=(r.x||0)+(r.width||0),t.y=r.y||0):r.y+r.height+t.height+o.bottom>this.
getPageHeight()?(this.cellAddPage(),t.y=o.top,l&&h&&(this.printHeaderRow(t.lineNumber,!0),t.y+=h[0].height)):t.y=r.y+r.height||t.y),void 0!==t.text[0]&&(this.rect(t.x,t.y,t.width,t.height,
!0===n?"FD":void 0),"right"===t.align?this.text(t.text,t.x+t.width-a,t.y+a,{align:"right",baseline:"top"}):"center"===t.align?this.text(t.text,t.x+t.width/2,t.y+a,{align:"center",baseline:
"top",maxWidth:t.width-a-a}):this.text(t.text,t.x+a,t.y+a,{align:"left",baseline:"top",maxWidth:t.width-a-a})),this.internal.__cell__.lastCell=t,this};t.table=function(t,n,l,c,u){if(i.call(
this),!l)throw new Error("No data for PDF table.");var f,d,p,g,m=[],b=[],v=[],w={},y={},_=[],x=[],A=(u=u||{}).autoSize||!1,L=!1!==u.printHeaders,N=u.css&&void 0!==u.css["font-size"]?16*u.
css["font-size"]:u.fontSize||12,S=u.margins||Object.assign({width:this.getPageWidth()},e),k="number"==typeof u.padding?u.padding:3,P=u.headerBackgroundColor||"#c8c8c8",F=u.
headerTextColor||"#000";if(a.call(this),this.internal.__cell__.printHeaders=L,this.internal.__cell__.margins=S,this.internal.__cell__.table_font_size=N,this.internal.__cell__.padding=k,this.
internal.__cell__.headerBackgroundColor=P,this.internal.__cell__.headerTextColor=F,this.setFontSize(N),null==c)b=m=Object.keys(l[0]),v=m.map(function(){return"left"});else if(Array.isArray(
c)&&"object"===r(c[0]))for(m=c.map(function(t){return t.name}),b=c.map(function(t){return t.prompt||t.name||""}),v=c.map(function(t){return t.align||"left"}),f=0;f<c.length;f+=1)y[c[f].
name]=.7499990551181103*c[f].width;else Array.isArray(c)&&"string"==typeof c[0]&&(b=m=c,v=m.map(function(){return"left"}));if(A||Array.isArray(c)&&"string"==typeof c[0])for(f=0;f<m.length;
f+=1){for(w[g=m[f]]=l.map(function(t){return t[g]}),this.setFont(void 0,"bold"),_.push(this.getTextDimensions(b[f],{fontSize:this.internal.__cell__.table_font_size,scaleFactor:this.internal.
scaleFactor}).w),d=w[g],this.setFont(void 0,"normal"),p=0;p<d.length;p+=1)_.push(this.getTextDimensions(d[p],{fontSize:this.internal.__cell__.table_font_size,scaleFactor:this.internal.
scaleFactor}).w);y[g]=Math.max.apply(null,_)+k+k,_=[]}if(L){var I={};for(f=0;f<m.length;f+=1)I[m[f]]={},I[m[f]].text=b[f],I[m[f]].align=v[f];var C=h.call(this,I,y);x=m.map(function(e){
return new s(t,n,y[e],C,I[e].text,void 0,I[e].align)}),this.setTableHeaderRow(x),this.printHeaderRow(1,!1)}var j=c.reduce(function(t,e){return t[e.name]=e.align,t},{});for(f=0;f<l.length;
f+=1){"rowStart"in u&&u.rowStart instanceof Function&&u.rowStart({row:f,data:l[f]},this);var E=h.call(this,l[f],y);for(p=0;p<m.length;p+=1){var O=l[f][m[p]];"cellStart"in u&&u.cellStart
instanceof Function&&u.cellStart({row:f,col:p,data:O},this),o.call(this,new s(t,n,y[m[p]],E,O,f+2,j[m[p]]))}return this.internal.__cell__.table_x=t,this.internal.__cell__.table_y=n,this};
var h=function(t,e){var n=this.internal.__cell__.padding,r=this.internal.__cell__.table_font_size,i=this.internal.scaleFactor;return Object.keys(t).map(function(r){var i=t[r];return this.
splitTextToSize(i.hasOwnProperty("text")?i.text:i,e[r]-n-n)},this).map(function(t){return this.getLineHeightFactor()*t.length*r/i+n+n},this).reduce(function(t,e){return Math.max(t,e)},0)};t.
setTableHeaderRow=function(t){i.call(this),this.internal.__cell__.tableHeaderRow=t},t.printHeaderRow=function(t,e){if(i.call(this),!this.internal.__cell__.tableHeaderRow)throw new Error(
"Property tableHeaderRow does not exist.");var r;if(n=!0,"function"==typeof this.internal.__cell__.headerFunction){var a=this.internal.__cell__.headerFunction(this,this.internal.__cell__.
pages);this.internal.__cell__.lastCell=new s(a[0],a[1],a[2],a[3],void 0,-1)}this.setFont(void 0,"bold");for(var h=[],l=0;l<this.internal.__cell__.tableHeaderRow.length;l+=1){r=this.internal.
__cell__.tableHeaderRow[l].clone(),e&&(r.y=this.internal.__cell__.margins.top||0,h.push(r)),r.lineNumber=t;var c=this.getTextColor();this.setTextColor(this.internal.__cell__.
headerTextColor),this.setFillColor(this.internal.__cell__.headerBackgroundColor),o.call(this,r),this.setTextColor(c)}h.length>0&&this.setTableHeaderRow(h),this.setFont(void 0,"normal"),
n=!1}}(R.API);var Pt={italic:["italic","oblique","normal"],oblique:["oblique","italic","normal"],normal:["normal","oblique","italic"]},Ft=["ultra-condensed","extra-condensed","condensed",
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"semi-condensed","normal","semi-expanded","expanded","extra-expanded","ultra-expanded"],It=kt(Ft),Ct=[100,200,300,400,500,600,700,800,900],jt=kt(Ct);function Et(t){var e=t.family.replace(/
"|'/g,"").toLowerCase(),n=function(t){return Pt[t=t||"normal"]?t:"normal"}(t.style),r=function(t){return t?"number"==typeof t?t>=100&&t<=900&&t%100==0?t:400:/^\d00$/.test(t)?parseInt(t):
"bold"===t?700:400:400}(t.weight),i=function(t){return"number"==typeof It[t=t||"normal"]?t:"normal"}(t.stretch);return{family:e,style:n,weight:r,stretch:i,src:t.src||[],ref:t.ref||{name:e,
style:[i,n,r].join(" ")}}}function Ot(t,e,n,r){var i;for(i=n;i>=0&&i<e.length;i+=r)if(t[e[i]])return t[e[i]];for(i=n;i>=0&&i<e.length;i-=r)if(t[e[i]])return t[e[i]]}var Bt={"sans-serif":
"helvetica",fixed:"courier",monospace:"courier",terminal:"courier",cursive:"times",fantasy:"times",serif:"times"},Mt={caption:"times",icon:"times",menu:"times","message-box":"times","small-
caption":"times","status-bar":"times"};function Rt(t){return[t.stretch,t.style,t.weight,t.family].join(" ")}function Tt(t){return t.trimLeft()}function Dt(t,e){for(var n=0;n<t.length;){if(t.
charAt(n)===e)return[t.substring(0,n),t.substring(n+1)];n+=1}return null}function qt(t){var e=t.match(/^(-[a-z_]|[a-z_])[a-z0-9_-]*/i);return null===e?null:[e[0],t.substring(e[0].
length)]}var zt=["times"];!function(t){var e,n,i,a,o,h,l,c,f,d=function(t){return t=t||{},this.isStrokeTransparent=t.isStrokeTransparent||!1,this.strokeOpacity=t.strokeOpacity||1,this.
strokeStyle=t.strokeStyle||"#000000",this.fillStyle=t.fillStyle||"#000000",this.isFillTransparent=t.isFillTransparent||!1,this.fillOpacity=t.fillOpacity||1,this.font=t.font||"10px sans-
serif",this.textBaseline=t.textBaseline||"alphabetic",this.textAlign=t.textAlign||"left",this.lineWidth=t.lineWidth||1,this.lineJoin=t.lineJoin||"miter",this.lineCap=t.lineCap||"butt",this.
path=t.path||[],this.transform=void 0!==t.transform?t.transform.clone():new c,this.globalCompositeOperation=t.globalCompositeOperation||"normal",this.globalAlpha=t.globalAlpha||1,this.
clip_path=t.clip_path||[],this.currentPoint=t.currentPoint||new h,this.miterLimit=t.miterLimit||10,this.lastPoint=t.lastPoint||new h,this.lineDashOffset=t.lineDashOffset||0,this.lineDash=t.
lineDash||[],this.margin=t.margin||[0,0,0,0],this.prevPageLastElemOffset=t.prevPageLastElemOffset||0,this.ignoreClearRect="boolean"!=typeof t.ignoreClearRect||t.ignoreClearRect,this};t.
events.push(["initialized",function(){this.context2d=new p(this),e=this.internal.f2,n=this.internal.getCoordinateString,i=this.internal.getVerticalCoordinateString,a=this.internal.
getHorizontalCoordinate,o=this.internal.getVerticalCoordinate,h=this.internal.Point,l=this.internal.Rectangle,c=this.internal.Matrix,f=new d}]);var p=function(t){Object.defineProperty(this,
"canvas",{get:function(){return{parentNode:!1,style:!1}}});var e=t;Object.defineProperty(this,"pdf",{get:function(){return e}});var n=!1;Object.defineProperty(this,"pageWrapXEnabled",{get:
function(){return n},set:function(t){n=Boolean(t)}});var r=!1;Object.defineProperty(this,"pageWrapYEnabled",{get:function(){return r},set:function(t){r=Boolean(t)}});var i=0;Object.
defineProperty(this,"posX",{get:function(){return i},set:function(t){isNaN(t)||(i=t)}});var a=0;Object.defineProperty(this,"posY",{get:function(){return a},set:function(t){isNaN(t)||(
a=t)}}),Object.defineProperty(this,"margin",{get:function(){return f.margin},set:function(t){var e;"number"==typeof t?e=[t,t,t,t]:((e=new Array(4))[0]=t[0],e[1]=t.length>=2?t[1]:e[0],e[2]=t.
length>=3?t[2]:e[0],e[3]=t.length>=4?t[3]:e[1]),f.margin=e}});var s=!1;Object.defineProperty(this,"autoPaging",{get:function(){return s},set:function(t){s=t}});var o=0;Object.defineProperty(
this,"lastBreak",{get:function(){return o},set:function(t){o=t}});var h=[];Object.defineProperty(this,"pageBreaks",{get:function(){return h},set:function(t){h=t}}),Object.defineProperty(
this,"ctx",{get:function(){return f},set:function(t){t instanceof d&&(f=t)}}),Object.defineProperty(this,"path",{get:function(){return f.path},set:function(t){f.path=t}});var l=[];Object.
defineProperty(this,"ctxStack",{get:function(){return l},set:function(t){l=t}}),Object.defineProperty(this,"fillStyle",{get:function(){return this.ctx.fillStyle},set:function(t){var e;e=g(
t),this.ctx.fillStyle=e.style,this.ctx.isFillTransparent=0===e.a,this.ctx.fillOpacity=e.a,this.pdf.setFillColor(e.r,e.g,e.b,{a:e.a}),this.pdf.setTextColor(e.r,e.g,e.b,{a:e.a})}}),Object.
defineProperty(this,"strokeStyle",{get:function(){return this.ctx.strokeStyle},set:function(t){var e=g(t);this.ctx.strokeStyle=e.style,this.ctx.isStrokeTransparent=0===e.a,this.ctx.
strokeOpacity=e.a,0===e.a?this.pdf.setDrawColor(255,255,255):(e.a,this.pdf.setDrawColor(e.r,e.g,e.b))}}),Object.defineProperty(this,"lineCap",{get:function(){return this.ctx.lineCap},set:
function(t){-1!==["butt","round","square"].indexOf(t)&&(this.ctx.lineCap=t,this.pdf.setLineCap(t))}}),Object.defineProperty(this,"lineWidth",{get:function(){return this.ctx.lineWidth},set:
function(t){isNaN(t)||(this.ctx.lineWidth=t,this.pdf.setLineWidth(t))}}),Object.defineProperty(this,"lineJoin",{get:function(){return this.ctx.lineJoin},set:function(t){-1!==["bevel",
"round","miter"].indexOf(t)&&(this.ctx.lineJoin=t,this.pdf.setLineJoin(t))}}),Object.defineProperty(this,"miterLimit",{get:function(){return this.ctx.miterLimit},set:function(t){isNaN(t)||(
this.ctx.miterLimit=t,this.pdf.setMiterLimit(t))}}),Object.defineProperty(this,"textBaseline",{get:function(){return this.ctx.textBaseline},set:function(t){this.ctx.textBaseline=t}}),Object.
defineProperty(this,"textAlign",{get:function(){return this.ctx.textAlign},set:function(t){-1!==["right","end","center","left","start"].indexOf(t)&&(this.ctx.textAlign=t)}});var c=null,
u=null;var p=null;Object.defineProperty(this,"fontFaces",{get:function(){return p},set:function(t){c=null,u=null,p=t}}),Object.defineProperty(this,"font",{get:function(){return this.ctx.
font},set:function(t){var e;if(this.ctx.font=t,null!==(e=/^\s*(?=(?:(?:[-a-z]+\s*){0,2}(italic|oblique))?)(?=(?:(?:[-a-z]+\s*){0,2}(small-caps))?)(?=(?:(?:[-a-z]+\s*){0,2}(bold(?:
er)?|lighter|[1-9]00))?)(?:(?:normal|\1|\2|\3)\s*){0,3}((?:xx?-)?(?:small|large)|medium|smaller|larger|[.\d]+(?:\%|in|[cem]m|ex|p[ctx]))(?:\s*\/\s*(normal|[.\d]+(?:\%|in|[cem]m|ex|p[
ctx])))?\s*([-_,\"\'\sa-z0-9]+?)\s*$/i.exec(t))){var n=e[1];e[2];var r=e[3],i=e[4];e[5];var a=e[6],s=/^([.\d]+)((?:%|in|[cem]m|ex|p[ctx]))$/i.exec(i)[2];i="px"===s?Math.floor(parseFloat(
i)*this.pdf.internal.scaleFactor):"em"===s?Math.floor(parseFloat(i)*this.pdf.getFontSize()):Math.floor(parseFloat(i)*this.pdf.internal.scaleFactor),this.pdf.setFontSize(i);var o=function(t){
var e,n,r=[],i=t.trim();if(""===i)return zt;if(i in Mt)return[Mt[i]];for(;""!==i;)switch(e=null,i=(i=Tt(i)).charAt(0)){case"'":case'"':n=Dt(i.substring(1),e);break;default:n=qt(i)}if(
null===n)return zt;if(r.push(n[0]),""!==(i=Tt(n[1]))&&","!==i.charAt(0))return zt;i=i.replace(/^,/,"")}return r}(a);if(this.fontFaces){var h=function(t,e){var n=t.getFontList(),r=JSON.
stringify(n);if(null===c||u!==r){var i=function(t){var e=[];return Object.keys(t).forEach(function(n){t[n].forEach(function(t){var r=null;switch(t){case"bold":r=[family:n,weight:"bold"};
break;case"italic":r={family:n,style:"italic"};break;case"bolditalic":r={family:n,weight:"bold",style:"italic"};break;case"":case"normal":r={family:n}}null!=r&&(r.ref={name:n,style:t},e.
push(r))})}),e}(n);c=function(t){for(var e={},n=0;n<t.length;++n){var r=Et(t[n]),i=r.family,a=r.stretch,s=r.style,o=r.weight;e[i]=e[i]||{},e[i][a]=e[i][a]||{},e[i][a][s]=e[i][a][s]||{},e[i][
a][s][o]=r}return e}(i.concat(e)),u=r}return c}(this.pdf,this.fontFaces),l=o.map(function(t){return{family:t,stretch:"normal",weight:t,style:n}}),f=function(t,e,n){for(var r=(n=n||{}).
defaultFontFamily||"times",i=Object.assign({},Bt,n.genericFontFamilies||{}),a=null,s=null,o=0;o<e.length;++o)if(i[(a=Et(e[o])).family]&&(a.family=i[a.family]),t.hasOwnProperty(a.family)){
s=t[a.family];break}if(!(s=s||t[r]))throw new Error("Could not find a font-family for the rule '"+Rt(a)+"' and default family '"+r+"'.");if(s=function(t,e){if(e[t])return e[t];var n=It[t],
r=n<=It.normal?-1:1,i=Ot(e,Ft,n,r);if(!i)throw new Error("Could not find a matching font-stretch value for "+t);return i}(a.stretch,s),s=function(t,e){if(e[t])return e[t];for(var n=Pt[t],
r=0;r<n.length;++r)if(e[n[r]])return e[n[r]];throw new Error("Could not find a matching font-style for "+t)}(a.style,s),!(s=function(t,e){if(e[t])return e[t];if(400===t&&e[500])return e[
500];if(500===t&&e[400])return e[400];var n=jt[t],r=Ot(e,Ct,n,t<400?-1:1);if(!r)throw new Error("Could not find a matching font-weight for value "+t);return r}(a.weight,s)))throw new Error(
"Failed to resolve a font for the rule '"+Rt(a)+"'.");return s}(h,l);this.pdf.setFont(f.ref.name,f.ref.style)}else{var d="";("bold"===r||parseInt(r,10)>=700||"bold"===n)&&(d="bold"),
"italic"===n&&(d+="italic"),0===d.length&&(d="normal");for(var p="",g={arial:"Helvetica",Arial:"Helvetica",verdana:"Helvetica",Verdana:"Helvetica",helvetica:"Helvetica",Helvetica:
"Helvetica","sans-serif":"Helvetica",fixed:"Courier",monospace:"Courier",terminal:"Courier",cursive:"Times",fantasy:"Times",serif:"Times"},m=0;m<o.length;m++){if(void 0!==this.pdf.internal.
getFont(o[m],d,{noFallback:!0,disableWarning:!0})){p=o[m];break}if("bolditalic"===d&&void 0!==this.pdf.internal.getFont(o[m],"bold",{noFallback:!0,disableWarning:!0}))p=o[m],d="bold";else
if(void 0!==this.pdf.internal.getFont(o[m],"normal",{noFallback:!0,disableWarning:!0}))p=o[m],d="normal";break}if(""===p)for(var b=0;b<o.length;b++)if(g[o[b]]){p=g[o[b]];
break}p=""===p?"Times":p,this.pdf.setFont(p,d)}}}),Object.defineProperty(this,"globalCompositeOperation",{get:function(){return this.ctx.globalCompositeOperation},set:function(t){this.ctx.
globalCompositeOperation=t}}),Object.defineProperty(this,"globalAlpha",{get:function(){return this.ctx.globalAlpha},set:function(t){this.ctx.globalAlpha=t}}),Object.defineProperty(this,
"lineDashOffset",{get:function(){return this.ctx.lineDashOffset},set:function(t){this.ctx.lineDashOffset=t,q.call(this)}}),Object.defineProperty(this,"lineDash",{get:function(){return this.
ctx.lineDash},set:function(t){this.ctx.lineDash=t,q.call(this)}}),Object.defineProperty(this,"ignoreClearRect",{get:function(){return this.ctx.ignoreClearRect},set:function(t){this.ctx.
ignoreClearRect=Boolean(t)}})};p.prototype.setLineDash=function(t){this.lineDash=t},p.prototype.getLineDash=function(){return this.lineDash.length%2?this.lineDash.concat(this.lineDash):this.
lineDash.slice()},p.prototype.fill=function(){A.call(this,"fill",!1)},p.prototype.stroke=function(){A.call(this,"stroke",!1)},p.prototype.beginPath=function(){this.path=[{type:"begin"}]},p.
prototype.moveTo=function(t,e){if(isNaN(t)||isNaN(e))throw s.error("jsPDF.context2d.moveTo: Invalid arguments",arguments),new Error("Invalid arguments passed to jsPDF.context2d.moveTo");var
n=this.ctx.transform.applyToPoint(new h(t,e));this.path.push({type:"mt",x:n.x,y:n.y}),this.ctx.lastPoint=new h(t,e)},p.prototype.closePath=function(){var t=new h(0,0),e=0;for(e=this.path.
length-1;-1!==e;e--)if("begin"===this.path[e].type&&"object"===r(this.path[e+1])&&"number"==typeof this.path[e+1].x){t=new h(this.path[e+1].x,this.path[e+1].y);break}this.path.push({type:
"close"}),this.ctx.lastPoint=new h(t.x,t.y)},p.prototype.lineTo=function(t,e){if(isNaN(t)||isNaN(e))throw s.error("jsPDF.context2d.lineTo: Invalid arguments",arguments),new Error("Invalid
arguments passed to jsPDF.context2d.lineTo");var n=this.ctx.transform.applyToPoint(new h(t,e));this.path.push({type:"lt",x:n.x,y:n.y}),this.ctx.lastPoint=new h(n.x,n.y)},p.prototype.
clip=function(){this.ctx.clip_path=JSON.parse(JSON.stringify(this.path)),A.call(this,null,!0)},p.prototype.quadraticCurveTo=function(t,e,n,r){if(isNaN(n)||isNaN(r)||isNaN(t)||isNaN(e))throw
s.error("jsPDF.context2d.quadraticCurveTo: Invalid arguments",arguments),new Error("Invalid arguments passed to jsPDF.context2d.quadraticCurveTo");var i=this.ctx.transform.applyToPoint(new
h(n,r)),a=this.ctx.transform.applyToPoint(new h(t,e));this.path.push({type:"qct",x1:a.x,y1:a.y,x:i.x,y:i.y}),this.ctx.lastPoint=new h(i.x,i.y)},p.prototype.bezierCurveTo=function(t,e,n,r,i,
a){if(isNaN(i)||isNaN(a)||isNaN(t)||isNaN(e)||isNaN(n)||isNaN(r))throw s.error("jsPDF.context2d.bezierCurveTo: Invalid arguments",arguments),new Error("Invalid arguments passed to jsPDF.
context2d.bezierCurveTo");var o=this.ctx.transform.applyToPoint(new h(i,a)),l=this.ctx.transform.applyToPoint(new h(t,e)),c=this.ctx.transform.applyToPoint(new h(n,r));this.path.push({type:
"bct",x1:l.x,y1:l.y,x2:c.x,y2:c.y,x:o.x,y:o.y}),this.ctx.lastPoint=new h(o.x,o.y)},p.prototype.arc=function(t,e,n,r,i,a){if(isNaN(t)||isNaN(e)||isNaN(n)||isNaN(r)||isNaN(i))throw s.error(
"jsPDF.context2d.arc: Invalid arguments",arguments),new Error("Invalid arguments passed to jsPDF.context2d.arc");if(a=Boolean(a),!this.ctx.transform.isIdentity){var o=this.ctx.transform.
applyToPoint(new h(t,e));t=o.x,e=o.y;var l=this.ctx.transform.applyToPoint(new h(0,n)),c=this.ctx.transform.applyToPoint(new h(0,0));n=Math.sqrt(Math.pow(l.x-c.x,2)+Math.pow(l.y-c.y,
2))}Math.abs(i-r)>=2*Math.PI&&(r=0,i=2*Math.PI),this.path.push({type:"arc",x:t,y:e,radius:n,startAngle:r,endAngle:i,counterclockwise:a})},p.prototype.arcTo=function(t,e,n,r,i){throw new
Error("arcTo not implemented.")},p.prototype.rect=function(t,e,n,r){if(isNaN(t)||isNaN(e)||isNaN(n)||isNaN(r))throw s.error("jsPDF.context2d.rect: Invalid arguments",arguments),new Error(
"Invalid arguments passed to jsPDF.context2d.rect");this.moveTo(t,e),this.lineTo(t+n,e),this.lineTo(t+n,e+r),this.lineTo(t,e+r),this.lineTo(t,e),this.lineTo(t+n,e),this.lineTo(t,e)},p.
prototype.fillRect=function(t,e,n,r){if(isNaN(t)||isNaN(e)||isNaN(n)||isNaN(r))throw s.error("jsPDF.context2d.fillRect: Invalid arguments",arguments),new Error("Invalid arguments passed to
jsPDF.context2d.fillRect");if(!m.call(this)){var i={};"butt"!==this.lineCap&&(i.lineCap=this.lineCap,this.lineCap="butt"),"miter"!==this.lineJoin&&(i.lineJoin=this.lineJoin,this.
lineJoin="miter"),this.beginPath(),this.rect(t,e,n,r),this.fill(),i.hasOwnProperty("lineCap")&&(this.lineCap=i.lineCap),i.hasOwnProperty("lineJoin")&&(this.lineJoin=i.lineJoin)}},p.
prototype.strokeRect=function(t,e,n,r){if(isNaN(t)||isNaN(e)||isNaN(n)||isNaN(r))throw s.error("jsPDF.context2d.strokeRect: Invalid arguments",arguments),new Error("Invalid arguments passed
to jsPDF.context2d.strokeRect");b.call(this)||(this.beginPath(),this.rect(t,e,n,r),this.stroke())},p.prototype.clearRect=function(t,e,n,r){if(isNaN(t)||isNaN(e)||isNaN(n)||isNaN(r))throw s.
error("jsPDF.context2d.clearRect: Invalid arguments",arguments),new Error("Invalid arguments passed to jsPDF.context2d.clearRect");this.ignoreClearRect||(this.fillStyle="#ffffff",this.
fillRect(t,e,n,r))},p.prototype.save=function(t){t="boolean"!=typeof t||t;for(var e=this.pdf.internal.getCurrentPageInfo().pageNumber,n=0;n<this.pdf.internal.getNumberOfPages();n++)this.pdf.
setPage(n+1),this.pdf.internal.out("q");if(this.pdf.setPage(e),t){this.ctx.fontSize=this.pdf.internal.getFontSize();var r=new d(this.ctx);this.ctxStack.push(this.ctx),this.ctx=r}},p.
prototype.restore=function(t){t="boolean"!=typeof t||t;for(var e=this.pdf.internal.getCurrentPageInfo().pageNumber,n=0;n<this.pdf.internal.getNumberOfPages();n++)this.pdf.setPage(n+1),this.
pdf.internal.out("Q");this.pdf.setPage(e),t&&0!==this.ctxStack.length&&(this.ctx=this.ctxStack.pop(),this.fillStyle=this.ctx.fillStyle,this.strokeStyle=this.ctx.strokeStyle,this.font=this.
ctx.font,this.lineCap=this.ctx.lineCap,this.lineWidth=this.ctx.lineWidth,this.lineJoin=this.ctx.lineJoin,this.lineDash=this.ctx.lineDash,this.lineDashOffset=this.ctx.lineDashOffset)},p.
prototype.toDataURL=function(){throw new Error("toDataUrl not implemented.")};var g=function(t){var e,n,r,i;if(!0===t.isCanvasGradient&&(t=t.getColor()),!t)return{r:0,g:0,b:0,a:0,style:t};
if(/transparent|rgba\s*\(\s*(\d+)\s*,\s*(\d+)\s*,\s*(\d+)\s*,\s*0+\s*\)/.test(t))e=0,n=0,r=0,i=0;else{var a=/rgb\s*\(\s*(\d+)\s*,\s*(\d+)\s*,\s*(\d+)\s*\)/.exec(t);if(null!==a)e=parseInt(a[
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1]),n=parseInt(a[2]),r=parseInt(a[3]),i=1;else if(null!==(a=/rgba\s*\(\s*(\d+)\s*,\s*(\d+)\s*,\s*(\d+)\s*,\s*([\d.]+)\s*\)/.exec(t)))e=parseInt(a[1]),n=parseInt(a[2]),r=parseInt(a[3]),
i=parseFloat(a[4]);else{if(i=1,"string"==typeof t&&"#"!==t.charAt(0)){var s=new u(t);t=s.ok?s.toHex():"#000000"}4===t.length?(e=t.substring(1,2),e+=e,n=t.substring(2,3),n+=n,r=t.substring(3,
4),r+=r):(e=t.substring(1,3),n=t.substring(3,5),r=t.substring(5,7)),e=parseInt(e,16),n=parseInt(n,16),r=parseInt(r,16)}}return{r:e,g:n,b:r,a:i,style:t}},m=function(){return this.ctx.
isFillTransparent||0==this.globalAlpha},b=function(){return Boolean(this.ctx.isStrokeTransparent||0==this.globalAlpha)};p.prototype.fillText=function(t,e,n,r){if(isNaN(e)||isNaN(
n)||"string"!=typeof t)throw s.error("jsPDF.context2d.fillText: Invalid arguments",arguments),new Error("Invalid arguments passed to jsPDF.context2d.fillText");if(r=isNaN(r)?void 0:r,!m.
call(this)){var i=R(this.ctx.transform.rotation),a=this.ctx.transform.scaleX;C.call(this,{text:t,x:e,y:n,scale:a,angle:i,align:this.textAlign,maxWidth:r})}},p.prototype.strokeText=function(
t,e,n,r){if(isNaN(e)||isNaN(n)||"string"!=typeof t)throw s.error("jsPDF.context2d.strokeText: Invalid arguments",arguments),new Error("Invalid arguments passed to jsPDF.context2d.
strokeText");if(!b.call(this)){r=isNaN(r)?void 0:r;var i=R(this.ctx.transform.rotation),a=this.ctx.transform.scaleX;C.call(this,{text:t,x:e,y:n,scale:a,renderingMode:"stroke",angle:i,align:
this.textAlign,maxWidth:r})}},p.prototype.measureText=function(t){if("string"!=typeof t)throw s.error("jsPDF.context2d.measureText: Invalid arguments",arguments),new Error("Invalid
arguments passed to jsPDF.context2d.measureText");var e=this.pdf,n=this.pdf.internal.scaleFactor,r=e.internal.getFontSize(),i=e.getStringUnitWidth(t)*r/e.internal.scaleFactor;return new
function(t){var e={t=t||{}).width||0;return Object.defineProperty(this,"width",{get:function(){return e}}),this}({width:i*=Math.round(96*n/72*1e4)/1e4})},p.prototype.scale=function(t,e){if(
isNaN(t)||isNaN(e))throw s.error("jsPDF.context2d.scale: Invalid arguments",arguments),new Error("Invalid arguments passed to jsPDF.context2d.scale");var n=new c(t,0,0,e,0,0);this.ctx.
transform=this.ctx.transform.multiply(n)},p.prototype.rotate=function(t){if(isNaN(t))throw s.error("jsPDF.context2d.rotate: Invalid arguments",arguments),new Error("Invalid arguments passed
to jsPDF.context2d.rotate");var e=new c(Math.cos(t),Math.sin(t),-Math.sin(t),Math.cos(t),0,0);this.ctx.transform=this.ctx.transform.multiply(e)},p.prototype.translate=function(t,e){if(isNaN(
t)||isNaN(e))throw s.error("jsPDF.context2d.translate: Invalid arguments",arguments),new Error("Invalid arguments passed to jsPDF.context2d.translate");var n=new c(1,0,0,1,t,e);this.ctx.
transform=this.ctx.transform.multiply(n)},p.prototype.transform=function(t,e,n,r,i,a){if(isNaN(t)||isNaN(e)||isNaN(n)||isNaN(r)||isNaN(i)||isNaN(a))throw s.error("jsPDF.context2d.transform:
Invalid arguments",arguments),new Error("Invalid arguments passed to jsPDF.context2d.transform");var o=new c(t,e,n,r,i,a);this.ctx.transform=this.ctx.transform.multiply(o)},p.prototype.
setTransform=function(t,e,n,r,i,a){t=isNaN(t)?1:t,e=isNaN(e)?0:e,n=isNaN(n)?0:n,r=isNaN(r)?1:r,i=isNaN(i)?0:i,a=isNaN(a)?0:a,this.ctx.transform=new c(t,e,n,r,i,a)};var v=function(){return
this.margin[0]>0||this.margin[1]>0||this.margin[2]>0||this.margin[3]>0};p.prototype.drawImage=function(t,e,n,r,i,a,s,o,h){var u=this.pdf.getImageProperties(t),f=1,d=1,p=1,g=1;void
0!==r&&void 0!==o&&(p=o/r,g=h/i,f=u.width/r*o/r,d=u.height/i*h/i),void 0===a&&(a=e,s=n,e=0,n=0),void 0!==r&&void 0===o&&(o=r,h=i),void 0===r&&void 0===o&&(o=u.height),u.height);var m=this.
ctx.transform.decompose(),b=R(m.rotate.shx),y=new c,A=(y=(y=(y=y.multiply(m.translate)).multiply(m.skew)).multiply(m.scale)).applyToRectangle(new l(a-e*p,s-n*g,r*f,i*d));if(this.autoPaging){
for(var N,S=w.call(this,A),k=[],P=0;P<S.length;P++=1)-1===k.indexOf(S[P])&&k.push(S[P]);x(k);for(var F=k[0],I=k[k.length-1],C=F;C<I+1;C++){this.pdf.setPage(C);var j=this.pdf.internal.
pageSize.width-this.margin[3]-this.margin[1],E=1===C?this.posY+this.margin[0]:this.margin[0],O=this.pdf.internal.pageSize.height-this.posY-this.margin[0]-this.margin[2],B=this.pdf.internal.
pageSize.height-this.margin[0]-this.margin[2],M=1===C?0:O+(C-2)*B;if(0!==this.ctx.clip_path.length){var T=this.path;N=JSON.parse(JSON.stringify(this.ctx.clip_path)),this.path=_(N,this.
posX+this.margin[3],-M+E+this.ctx.prevPageLastElemOffset),L.call(this,"fill",!0),this.path=T}var D=JSON.parse(JSON.stringify(A));D=_([D],this.posX+this.margin[3],-M+E+this.ctx.
prevPageLastElemOffset)[0];var q=(C>F||C<I)&&v.call(this);q&&(this.pdf.saveGraphicsState(),this.pdf.rect(this.margin[3],this.margin[0],j,B,null).clip().discardPath()),this.pdf.addImage(t,
"JPEG",D.x,D.y,D.w,D.h,null,null,b),q&&this.pdf.restoreGraphicsState()}else this.pdf.addImage(t,"JPEG",A.x,A.y,A.w,A.h,null,null,b)};var w=function(t,e,n){var r=[];e=e||this.pdf.internal.
pageSize.width,n=n||this.pdf.internal.pageSize.height-this.margin[0]-this.margin[2];var i=this.posY+this.ctx.prevPageLastElemOffset;switch(t.type){default:case"mt":case"lt":r.push(Math.
floor((t.y+i)/n)+1);break;case"arc":r.push(Math.floor((t.y+i-t.radius)/n)+1),r.push(Math.floor((t.y+i+t.radius)/n)+1);break;case"qct":var a=T(this.ctx.lastPoint.x,this.ctx.lastPoint.y,t.x1,
t.y1,t.x,t.y);r.push(Math.floor((a.y+i)/n)+1),r.push(Math.floor((a.y+a.h+i)/n)+1);break;case"bct":var s=D(this.ctx.lastPoint.x,this.ctx.lastPoint.y,t.x1,t.y1,t.x2,t.y2,t.x,t.y);r.push(Math.
floor((s.y+i)/n)+1),r.push(Math.floor((s.y+s.h+i)/n)+1);break;case"rect":r.push(Math.floor((t.y+i)/n)+1),r.push(Math.floor((t.y+t.h+i)/n)+1)}for(var o=0;o<r.length;o+=1)for(;this.pdf.
internal.getNumberOfPages()<r[o];)y.call(this);return r},y=function(){var t=this.fillStyle,e=this.strokeStyle,n=this.font,r=this.lineCap,i=this.lineWidth,a=this.lineJoin;this.pdf.addPage(),
this.fillStyle=t,this.strokeStyle=e,this.font=n,this.lineCap=r,this.lineWidth=i,this.lineJoin=a},_=function(t,e,n){for(var r=0;r<t.length;r++)switch(t[r].type){case"bct":t[r].x2+=e,t[r].
y2+=n;case"qct":t[r].x1+=e,t[r].y1+=n;default:t[r].x+=e,t[r].y+=n}return t},x=function(t){return t.sort(function(t,e){return t-e})},A=function(t,e){var n=this.fillStyle,r=this.strokeStyle,
i=this.lineCap,a=this.lineWidth,s=Math.abs(a*this.ctx.transform.scaleX),o=this.lineJoin;if(this.autoPaging){for(var h,l,c=JSON.parse(JSON.stringify(this.path)),u=JSON.parse(JSON.stringify(
this.path)),f=[],d=0;d<u.length;d++)if(void 0!==u[d].x)for(var p=w.call(this,u[d]),g=0;g<p.length;g+=1)-1===f.indexOf(p[g])&&f.push(p[g]);for(var m=0;m<f.length;m++)for(;this.pdf.internal.
getNumberOfPages()<f[m];)y.call(this);x(f);for(var b=f[0],A=f[f.length-1],N=b;N<A+1;N++){this.pdf.setPage(N),this.fillStyle=n,this.strokeStyle=r,this.lineCap=i,this.lineWidth=s,this.
lineJoin=o;var S=this.pdf.internal.pageSize.width-this.margin[3]-this.margin[1],k=1===N?this.posY+this.margin[0]:this.margin[0],P=this.pdf.internal.pageSize.height-this.posY-this.margin[0]-
this.margin[2],F=this.pdf.internal.pageSize.height-this.margin[0]-this.margin[2],I=1===N?0:P+(N-2)*F;if(0!==this.ctx.clip_path.length){var C=this.path;h=JSON.parse(JSON.stringify(this.ctx.
clip_path)),this.path=_(h,this.posX+this.margin[3],-I+k+this.ctx.prevPageLastElemOffset),L.call(this,t,!0),this.path=C}if(l=JSON.parse(JSON.stringify(c)),this.path=_(l,this.posX+this.margin[
3],-I+k+this.ctx.prevPageLastElemOffset),!1===e||0===N){var j=(N>b||N<A)&&v.call(this);j&&(this.pdf.saveGraphicsState(),this.pdf.rect(this.margin[3],this.margin[0],S,F,null).clip().
discardPath()),L.call(this,t,e),j&&this.pdf.restoreGraphicsState()}this.lineWidth=a}this.path=c}else this.lineWidth=s,L.call(this,t,e),this.lineWidth=a},L=function(t,e){if((
"stroke"!==t||e||!b.call(this))&&("stroke"===t||e||!m.call(this))){for(var n,r,i=[],a=this.path,s=0;s<a.length;s++){var o=a[s];switch(o.type){case"begin":i.push({begin:!0});break;
case"close":i.push({close:!0});break;case"mt":i.push({start:o,deltas:[],abs:[]});break;case"lt":var h=i.length;if(a[s-1]&&!isNaN(a[s-1].x)&&(n=[o.x-a[s-1].x,o.y-a[s-1].y],h>0))for(;h>=0;h--
)if(!0!==i[h-1].close&&!0!==i[h-1].begin){i[h-1].deltas.push(n),i[h-1].abs.push(o);break}break;case"bct":n=[o.x1-a[s-1].x,o.y1-a[s-1].y,o.x2-a[s-1].x,o.y2-a[s-1].y,o.x-a[s-1].x,o.y-a[s-1].
y],i[i.length-1].deltas.push(n);break;case"qct":var l=a[s-1].x+2/3*(o.x1-a[s-1].x),c=a[s-1].y+2/3*(o.y1-a[s-1].y),u=o.x+2/3*(o.x1-o.x),f=o.y+2/3*(o.y1-o.y),d=o.x,p=o.y;n=[l-a[s-1].x,c-a[s-
1].y,u-a[s-1].x,f-a[s-1].y,d-a[s-1].x,p-a[s-1].y],i[i.length-1].deltas.push(n);break;case"arc":i.push({deltas:[],abs:[],arc:!0}),Array.isArray(i[i.length-1].abs)&&i[i.length-1].abs.push(
o)}}r=e?null:"stroke"===t?"stroke":"fill";for(var g=!1,v=0;v<i.length;v++)if(i[v].arc)for(var w=i[v].abs,y=0;y<w.length;y++){var _=w[y];"arc"===_.type?k.call(this,_.x,_.y,_.radius,_.
startAngle,_.endAngle,_.counterclockwise,void 0,e,!g):j.call(this,_.x,_.y),g=!0}else if(!0===i[v].close)this.pdf.internal.out("h"),g=!1;else if(!0!==i[v].begin){var x=i[v].start.x,A=i[v].
start.y;E.call(this,i[v].deltas,x,A),g=!0}r&&P.call(this,r),e&&F.call(this)}},N=function(t){var e=this.pdf.internal.getFontSize()/this.pdf.internal.scaleFactor,n=e*(this.pdf.internal.
getLineHeightFactor()-1);switch(this.ctx.textBaseline){case"bottom":return t-n;case"top":return t+e-n;case"hanging":return t+e-2*n;case"middle":return t+e/2-n;default:return t}},S=function(
t){return t+this.pdf.internal.getFontSize()/this.pdf.internal.scaleFactor*(this.pdf.internal.getLineHeightFactor()-1)};p.prototype.createLinearGradient=function(){var t=function(){};return
t.colorStops=[],t.addColorStop=function(t,e){this.colorStops.push([t,e])},t.getColor=function(){return 0===this.colorStops.length?"#000000":this.colorStops[0][1]},t.isCanvasGradient=!0,t},p.
prototype.createPattern=function(){return this.createLinearGradient()},p.prototype.createRadialGradient=function(){return this.createLinearGradient()};var k=function(t,e,n,r,i,a,s,o,h){for(
var l=B.call(this,n,r,i,a),c=0;c<l.length;c++){var u=l[c];0===c&&h?I.call(this,u.x1+t,u.y1+e):j.call(this,u.x1+t,u.y1+e)),O.call(this,t,e,u.x2,u.y2,u.x3,u.y3,u.x4,u.y4)}o?F.call(this):P.
call(this,s)},P=function(t){switch(t){case"stroke":this.pdf.internal.out("S");break;case"fill":this.pdf.internal.out("f")}},F=function(){this.pdf.clip(),this.pdf.discardPath()},I=function(t,
e){this.pdf.internal.out(n(t)+" "+i(e)+" m")},C=function(t){var e;switch(t.align){case"right":case"end":e="right";break;case"center":e="center";break;default:e="left"}var n,r,i,a=this.pdf.
getTextDimensions(t.text),s=N.call(this,t.y),o=S.call(this,s)-a.h,u=this.ctx.transform.applyToPoint(new h(t.x,s));if(this.autoPaging){var f=this.ctx.transform.decompose(),d=new c;d=(d=(d=d.
multiply(f.translate)).multiply(f.skew)).multiply(f.scale);for(var p=this.ctx.transform.applyToRectangle(new l(t.x,s,a.w,a.h)),g=d.applyToRectangle(new l(t.x,o,a.w,a.h)),m=w.call(this,g),b=[
],y=0;y<m.length;y+=1)-1===b.indexOf(m[y])&&b.push(m[y]);x(b);for(var A=b[0],k=b[b.length-1],P=A;P<k+1;P++){this.pdf.setPage(P);var F=1===P?this.posY+this.margin[0]:this.margin[0],I=this.
pdf.internal.pageSize.height-this.posY-this.margin[0]-this.margin[2],C=this.pdf.internal.pageSize.height-this.margin[2],j=C-this.margin[0],E=this.pdf.internal.pageSize.width-this.margin[1],
O=E-this.margin[3],B=1===P?0:I+(P-2)*j;if(0!==this.ctx.clip_path.length){var M=this.path;n=JSON.parse(JSON.stringify(this.ctx.clip_path)),this.path=_(n,this.posX+this.margin[3],-1*B+F),L.
call(this,"fill",!0),this.path=M}var R=_([JSON.parse(JSON.stringify(g))],this.posX+this.margin[3],-B+F+this.ctx.prevPageLastElemOffset)[0];t.scale>=.01&&(r=this.pdf.internal.getFontSize(),
this.pdf.setFontSize(r*t.scale),i=this.lineWidth,this.lineWidth=i*t.scale);var T="text"!==this.autoPaging;if(T||R.y+R.h<=C){if(T||R.y>=F&&R.x<=E){var D=T?t.text:this.pdf.splitTextToSize(t.
text,t.maxWidth||E-R.x)[0],q=_([JSON.parse(JSON.stringify(p))],this.posX+this.margin[3],-B+F+this.ctx.prevPageLastElemOffset)[0],z=T&&(P>A||P<k)&&v.call(this);z&&(this.pdf.saveGraphicsState(
),this.pdf.rect(this.margin[3],this.margin[0],O,j,null).clip().discardPath()),this.pdf.text(D,q.x,q.y,{angle:t.angle,align:e,renderingMode:t.renderingMode}),z&&this.pdf.restoreGraphicsState(
)}}else R.y<C&&(this.ctx.prevPageLastElemOffset+=C-R.y);t.scale>=.01&&(this.pdf.setFontSize(r),this.lineWidth=i)}else t.scale>=.01&&(r=this.pdf.internal.getFontSize(),this.pdf.setFontSize(
r*t.scale),i=this.lineWidth,this.lineWidth=i*t.scale),this.pdf.text(t.text,u.x+this.posX,u.y+this.posY,{angle:t.angle,align:e,renderingMode:t.renderingMode,maxWidth:t.maxWidth}),t.scale>=.
01&&(this.pdf.setFontSize(r),this.lineWidth=i)},j=function(t,e,r,a){r=r||0,a=a||0,this.pdf.internal.out(n(t+r)+" "+i(e+a)+" l")},E=function(t,e,n){return this.pdf.lines(t,e,n,null,null)},
O=function(t,n,r,i,s,h,l,c){this.pdf.internal.out([e(a(r+t)),e(o(i+n)),e(a(s+t)),e(o(h+n)),e(a(l+t)),e(o(c+n)),"c"].join(" "))},B=function(t,e,n,r){for(var i=2*Math.PI,a=Math.PI/2;e>n;)e-=i;
var s=Math.abs(n-e);s<i&&r&&(s=i-s);for(var o=[],h=r?-1:1,l=e;s>1e-5;){var c=l+h*Math.min(s,a);o.push(M.call(this,t,l,c)),s-=Math.abs(c-l),l=c}return o},M=function(t,e,n){var r=(n-e)/2,
i=t*Math.cos(r),a=t*Math.sin(r),s=i,o=-a,h=s*s+o*o,l=h+s*i+o*a,c=4/3*(Math.sqrt(2*h*l)-l)/(s*a-o*i),u=s-c*o,f=o+c*s,d=u,p=-f,g=r+e,m=Math.cos(g),b=Math.sin(g);return{x1:t*Math.cos(e),y1:
t*Math.sin(e),x2:u*m-f*b,y2:u*b+f*m,x3:d*m-p*b,y3:d*b+p*m,x4:t*Math.cos(n),y4:t*Math.sin(n)}},R=function(t){return 180*t/Math.PI},T=function(t,e,n,r,i,a){var s=t+.5*(n-t),o=e+.5*(r-e),h=i+.
5*(n-i),c=a+.5*(r-a),u=Math.min(t,i,s,h),f=Math.max(t,i,s,h),d=Math.min(e,a,o,c),p=Math.max(e,a,o,c);return new l(u,d,f-u,p-d)},D=function(t,e,n,r,i,a,s,o){var h,c,u,f,d,p,g,m,b,v,w,y,_,x,
A=n-t,L=r-e,N=i-n,S=a-r,k=s-i,P=o-a;for(c=0;c<41;c++)b=(g=(u=t+(h=c/40)*A)+h*((d=n+h*N)-u))+h*(d+h*(i+h*k-d)-g),v=(m=(f=e+h*L)+h*((p=r+h*S)-f))+h*(p+h*(a+h*P-p)-m),0==c?(w=b,y=v,_=b,x=v):(
w=Math.min(w,b),y=Math.min(y,v),_=Math.max(_,b),x=Math.max(x,v));return new l(Math.round(w),Math.round(y),Math.round(_-w),Math.round(x-y))},q=function(){if(this.prevLineDash||this.ctx.
lineDash.length||this.ctx.lineDashOffset){var t,e,n=(t=this.ctx.lineDash,e=this.ctx.lineDashOffset,JSON.stringify({lineDash:t,lineDashOffset:e}));this.prevLineDash!==n&&(this.pdf.
setLineDash(this.ctx.lineDash,this.ctx.lineDashOffset),this.prevLineDash=n)}},(R.API);var Ut=Uint8Array,Ht=Uint16Array,Wt=Int32Array,Vt=new Ut([0,0,0,0,0,0,0,0,1,1,1,1,2,2,2,2,3,3,3,3,4,4,4,
4,5,5,5,5,5,0,0,0,0]),Gt=new Ut([0,0,0,0,1,1,2,2,3,3,4,4,5,5,6,6,7,7,8,8,9,9,10,10,11,11,12,12,13,13,0,0]),Yt=new Ut([16,17,18,0,8,7,9,6,10,5,11,4,12,3,13,2,14,1,15]),Zt=function(t,e){for(var
n=new Ht(31),r=0;r<31;++r)n[r]=e+=1<<t[r-1];var i=new Wt(n[30]);for(r=1;r<30;++r)for(var a=n[r];a<n[r+1];++a)i[a]=a-n[r]<<5|r;return{b:n,r:i}},Jt=Zt(Vt,2),Xt=Jt.b,Kt=Jt.r;Xt[28]=258,Kt[
258]=28;for(var $t=Zt(Gt,0).r,Qt=new Ht(32768),te=0;te<32768;++te){var ee=(43690&te)>>>1|(21845&te)<<1;ee=(61680&(ee=(52428&ee)>>>2|(13107&ee)<<2))>>>4|(3855&ee)<<4,Qt[te]=((65280&ee)>>>8|(
255&ee)<<8)>>>1}var ne=function(t,e,n){for(var r=t.length,i=0,a=new Ht(e);i<r;++i)t[i]&&++a[t[i]-1];var s,o=new Ht(e);for(i=1;i<e;++i)o[i]=o[i-1]+a[i-1]<<1;if(n){s=new Ht(1<<e);var h=15-e;
for(i=0;i<r;++i)if(t[i])for(var l=i<<4|t[i],c=e-t[i],u=o[t[i]-1]++<<c,f=u|(1<<c)-1;u<=f;++u)s[Qt[u]>>h]=l}else for(s=new Ht(r),i=0;i<r;++i)t[i]&&(s[i]=Qt[o[t[i]-1]++]>>15-t[i]);return s},
re=new Ut(288);for(te=0;te<144;++te)re[te]=8;for(te=144;te<256;++te)re[te]=9;for(te=256;te<280;++te)re[te]=7;for(te=280;te<288;++te)re[te]=8;var ie=new Ut(32);for(te=0;te<32;++te)ie[te]=5;
var ae=ne(re,9,0),se=ne(ie,5,0),oe=function(t){return(t+7)/8|0},he=function(t,e,n){n<<=7&e;var r=e/8|0;t[r]|=n,t[r+1]|=n>>8},le=function(t,e,n){n<<=7&e;var r=e/8|0;t[r]|=n,t[r+1]|=n>>8,t[
r+2]|=n>>16},ce=function(t,e){for(var n=[],r=0;r<t.length;++r)t[r]&&n.push({s:r,f:t[r]});var i=n.length,a=n.slice();if(!i)return{t:be,l:0};if(1==i){var s=new Ut(n[0].s+1);return s[n[0].s]=1,
{t:s,l:1}}n.sort(function(t,e){return t.f-e.f}),n.push({s:-1,f:25001});var o=n[0],h=n[1],l=0,c=1,u=2;for(n[0]={s:-1,f:o.f+h.f,l:o,r:h};c!=i-1;)o=n[n[l].f<n[u].f?l++:u++],h=n[l!=c&&n[l].f<n[
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u].f?l++:u++],n[c++]={s:-1,f:o.f+h.f,l:o,r:h};var f=a[0].s;for(r=1;r<i;++r)a[r].s>f&&(f=a[r].s);var d=new Ht(f+1),p=ue(n[c-1],d,0);if(p>e){r=0;var g=0,m=p-e,b=1<<m;for(a.sort(function(t,e){
return d[e.s]-d[t.s]||t.f-e.f});r<i;++r){var v=a[r].s;if(!(d[v]>e))break;g+=b-(1<<p-d[v]),d[v]=e}for(g>>=m;g>0;){var w=a[r].s;d[w]<e?g-=1<<e-d[w]++-1:++r}for(;r>=0&&g;--r){var y=a[r].s;d[
y]==e&&(--d[y],++g)}p=e}return{t:new Ut(d),l:p}},ue=function(t,e,n){return-1==t.s?Math.max(ue(t.l,e,n+1),ue(t.r,e,n+1)):e[t.s]=n},fe=function(t){for(var e=t.length;e&&!t[--e];);for(var
n=new Ht(++e),r=0,i=t[0],a=1,s=function(t){n[r++]=t},o=1;o<=e;++o)if(t[o]==i&&o!=e)++a;else{if(!i&&a>2){for(;a>138;a-=138)s(32754);a>2&&(s(a>10?a-11<<5|28690:a-3<<5|12305),a=0)}else if(a>3){
for(s(i),--a;a>6;a-=6)s(8304);a>2&&(s(a-3<<5|8208),a=0)}for(;a--;)s(i);a=1,i=t[o]}return{c:n.subarray(0,r),n:e}},de=function(t,e){for(var n=0,r=0;r<e.length;++r)n+=t[r]*e[r];return n},
pe=function(t,e,n){var r=n.length,i=oe(e+2);t[i]=255&r,t[i+1]=r>>8,t[i+2]=255^t[i],t[i+3]=255^t[i+1];for(var a=0;a<r;++a)t[i+a+4]=n[a];return 8*(i+4+r)},ge=function(t,e,n,r,i,a,s,o,h,l,c){
he(e,c++,n),++i[256];for(var u=ce(i,15),f=u.t,d=u.l,p=ce(a,15),g=p.t,m=p.l,b=fe(f),v=b.c,w=b.n,y=fe(g),_=y.c,x=y.n,A=new Ht(19),L=0;L<v.length;++L)++A[31&v[L]];for(L=0;L<_.length;++L)++A[
31&_[L]];for(var N=ce(A,7),S=N.t,k=N.l,P=19;P>4&&!S[Yt[P-1]];--P);var F,I,C,j,E=l+5<<3,O=de(i,re)+de(a,ie)+s,B=de(i,f)+de(a,g)+s+14+3*P+de(A,S)+2*A[16]+3*A[17]+7*A[18];if(
h>=0&&E<=O&&E<=B)return pe(e,c,t.subarray(h,h+l));if(he(e,c,1+(B<O)),c+=2,B<O){F=ne(f,d,0),I=f,C=ne(g,m,0),j=g;var M=ne(S,k,0);for(he(e,c,w-257),he(e,c+5,x-1),he(e,c+10,P-4),c+=14,L=0;L<P;
++L)he(e,c+3*L,S[Yt[L]]);c+=3*P;for(var R=[v,_],T=0;T<2;++T){var D=R[T];for(L=0;L<D.length;++L){var q=31&D[L];he(e,c,M[q]),c+=S[q],q>15&&(he(e,c,D[L]>>5&127),c+=D[L]>>12)}}}else F=ae,I=re,
C=se,j=ie;for(L=0;L<o;++L){var z=r[L];if(z>255){le(e,c,F[257+(q=z>>18&31)]),c+=I[q+257],q>7&&(he(e,c,z>>23&31),c+=Vt[q]);var U=31&z;le(e,c,C[U]),c+=j[U],U>3&&(le(e,c,z>>5&8191),c+=Gt[
U])}else le(e,c,F[z]),c+=I[z]}return le(e,c,F[256]),c+I[256]},me=new Wt([65540,131080,131088,131104,262176,1048704,1048832,2114560,2117632]),be=new Ut(0),ve=function(){var t=1,e=0;return{p:
function(n){for(var r=t,i=e,a=0|n.length,s=0;s!=a;){for(var o=Math.min(s+2655,a);s<o;++s)i+=r+=n[s];r=(65535&r)+15*(r>>16),i=(65535&i)+15*(i>>16)}t=r,e=i},d:function(){return(255&(
t%=65521))<<24|(65280&t)<<8|(255&(e%=65521))<<8|e>>8}},we=function(t,e,n){for(;n;++e)t[e]=n,n>>>=8};function ye(t,e){e||(e={});var n=ve();n.p(t);var r=function(t,e,n,r,i){if(!i&&(i={l:1},e.
dictionary)){var a=e.dictionary.subarray(-32768),s=new Ut(a.length+t.length);s.set(a),s.set(t,a.length),t=s,i.w=a.length}return function(t,e,n,r,i,a){var s=a.z||t.length,o=new Ut(r+s+5*(
1+Math.ceil(s/7e3))+i),h=o.subarray(r,o.length-i),l=a.l,c=7&(a.r||0);if(e){c&&(h[0]=a.r>>3);for(var u=me[e-1],f=u>>13,d=8191&u,p=(1<<n)-1,g=a.p||new Ht(32768),m=a.h||new Ht(p+1),b=Math.ceil(
n/3),v=2*b,w=function(e){return(t[e]^t[e+1]<<b^t[e+2]<<v)&p},y=new Wt(25e3),_=new Ht(288),x=new Ht(32),A=0,L=0,N=a.i||0,S=0,k=a.w||0,P=0;N+2<s;++N){var F=w(N),I=32767&N,C=m[F];if(g[I]=C,m[
F]=I,k<=N){var j=s-N;if((A>7e3||S>24576)&&(j>423||!l)){c=ge(t,h,0,y,_,x,L,S,P,N-P,c),S=A=L=0,P=N;for(var E=0;E<286;++E)_[E]=0;for(E=0;E<30;++E)x[E]=0}var O=2,B=0,M=d,R=I-C&32767;if(
j>2&&F==w(N-R))for(var T=Math.min(f,j)-1,D=Math.min(32767,N),q=Math.min(258,j);R<=D&&--M&&I!=C;){if(t[N+O]==t[N+O-R]){for(var z=0;z<q&&t[N+z]==t[N+z-R];++z);if(z>O){if(O=z,B=R,z>T)break;var
U=Math.min(R,z-2),H=0;for(E=0;E<U;++E){var W=N-R+E&32767,V=W-g[W]&32767;V>H&&(H=V,C=W)}}}R+=(I=C)-(C=g[I])&32767}if(B){y[S++]=268435456|Kt[O]<<18|$t[B];var G=31&Kt[O],Y=31&$t[B];L+=Vt[G]+Gt[
Y],++_[257+G],++x[Y],k=N+O,++A}else y[S++]=t[N],++_[t[N]]}}for(N=Math.max(N,k);N<s;++N)y[S++]=t[N],++_[t[N]];c=ge(t,h,l,y,_,x,L,S,P,N-P,c),l||(a.r=7&c|h[c/8|0]<<3,c-=7,a.h=m,a.p=g,a.i=N,a.
w=k)}else{for(N=a.w||0;N<s+l;N+=65535){var Z=N+65535;Z>=s&&(h[c/8|0]=l,Z=s),c=pe(h,c+1,t.subarray(N,Z))}a.i=s}return function(t,e,n){return(null==e||e<0)&&(e=0),(null==n||n>t.length)&&(n=t.
length),new Ut(t.subarray(e,n))}(o,0,r+oe(c)+i)}(t,null==e.level?6:e.level,null==e.mem?Math.ceil(1.5*Math.max(8,Math.min(13,Math.log(t.length))))-12+e.mem,n,4,i)}(t,e,e.dictionary?6:2);
return function(t,e){var n=e.level,r=0==n?0:n<6?1:9==n?3:2;if(t[0]=120,t[1]=r<<6|(e.dictionary&&32),t[1]|=31-(t[0]<<8|t[1])%31,e.dictionary){var i=ve();i.p(e.dictionary),we(t,2,i.d())}}(r,
e),we(r,r.length-4,n.d()),r}var _e="undefined"!=typeof TextDecoder&&new TextDecoder;try{_e.decode(be,{stream:!0})}catch(is){}
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   * Licensed under the MIT License.
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   */function xe(t,e="utf8"){return new TextDecoder(e).decode(t)}!function(t){var e=function(t){var e,n,r,i,a,s,o,h,l,c;for(/[^\x00-\xFF]/.test(t),n=[],r=0,i=(t+=e="\0\0\0\0".slice(t.
length%4||4)).length;i>r;r+=4)0!==(a=(t.charCodeAt(r)<<24)+(t.charCodeAt(r+1)<<16)+(t.charCodeAt(r+2)<<8)+t.charCodeAt(r+3))?(s=(a=((a=((a=(a-(c=a%85))/85)-(l=a%85))/85)-(h=a%85))/85)-(
o=a%85))/85)%85,n.push(s+33,o+33,h+33,l+33,c+33)):n.push(122);return function(t,e){for(var n=e;n>0;n--)t.pop()}(n,e.length),String.fromCharCode.apply(String,n)+"~>"},n=function(t){var e,n,r,
i,a,s=String,o="length",h=255,l="charCodeAt",c="slice",u="replace";for(t[c](-2),t=t[c](0,-2)[u](/\s/g,"")[u]("z","!!!!!"),r=[],i=0,a=(t+=e="uuuuu"[c](t[o]%5||5))[o];a>i;i+=5)n=52200625*(t[
l](i-33)+614125*(t[l](i+1)-33)+7225*(t[l](i+2)-33)+85*(t[l](i+3)-33)+(t[l](i+4)-33),r.push(h&n>>24,h&n>>16,h&n>>8,h&n);return function(t,e){for(var n=e;n>0;n--)t.pop()}(r,e[o]),s.
fromCharCode.apply(s,r)},r=function(t){return t.split("").map(function(t){return("0"+t.charCodeAt().toString(16)).slice(-2)}).join("")+">"},i=function(t){var e=new RegExp(/^([0-9A-Fa-f]{
2})+$/);if(-1!==(t=t.replace(/\s/g,"")).indexOf(">")&&(t=t.substr(0,t.indexOf(">"))),t.length%2&&(t+="0"),!1===e.test(t))return"";for(var n="",r=0;r<t.length;r+=2)n+=String.fromCharCode(
"0x"+(t[r]+t[r+1]));return n},a=function(t){for(var e=new Uint8Array(t.length),n=t.length;n--;)e[n]=t.charCodeAt(n);return(e=ye(e)).reduce(function(t,e){return t+String.fromCharCode(e)},
"")};t.processDataByFilters=function(t,s){var o=0,h=t||"",l=[];for("string"==typeof(s=s||[])&&(s=[s]),o=0;o<s.length;o+=1)switch(s[o]){case"ASCII85Decode":case"/ASCII85Decode":h=n(h),l.push(
"/ASCII85Encode");break;case"ASCII85Encode":case"/ASCII85Encode":h=e(h),l.push("/ASCII85Decode");break;case"ASCIIHexDecode":case"/ASCIIHexDecode":h=i(h),l.push("/ASCIIHexEncode");break;
case"ASCIIHexEncode":case"/ASCIIHexEncode":h=r(h),l.push("/ASCIIHexDecode");break;case"FlateEncode":case"/FlateEncode":h=a(h),l.push("/FlateDecode");break;default:throw new Error('The
filter: "'+s[o]+'" is not implemented')}return{data:h,reverseChain:l.reverse().join(" ")}}}(R.API),
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   */
function(t){t.loadFile=function(t,e,n){return function(t,e,n){e=!1!==e,n="function"==typeof n?n:function(){};var r=void 0;try{r=function(t,e,n){var r=new XMLHttpRequest,i=0,a=function(t){
var e=t.length,n=[],r=String.fromCharCode;for(i=0;i<e;i+=1)n.push(r(255&t.charCodeAt(i)));return n.join("")};if(r.open("GET",t,!e),r.overrideMimeType("text/plain; charset=x-user-defined"),
!1===e&&(r.onload=function(){200===r.status?n(a(this.responseText)):n(void 0)}),r.send(null),e&&200===r.status)return a(r.responseText)}(t,e,n)}catch(is){}return r}(t,e,n)},t.
allowFsRead=void 0,t.loadImageFile=t.loadFile}(R.API),function(e){function n(){return(i.html2canvas?Promise.resolve(i.html2canvas):"object"===(void 0===t?"undefined":r(
t))&&"undefined"!=typeof module?new Promise(function(t,e){try{t(require("html2canvas"))}catch(is){e(is)}}):"function"==typeof define&&define.amd?new Promise(function(t,e){try{require([
"html2canvas"],t)}catch(is){e(is)}}):Promise.reject(new Error("Could not load html2canvas"))).catch(function(t){return Promise.reject(new Error("Could not load html2canvas: "+t))}).then(
function(t){return t.default?t.default:t})}function a(){return(i.DOMPurify?Promise.resolve(i.DOMPurify):"object"===(void 0===t?"undefined":r(t))&&"undefined"!=typeof module?new Promise(
function(t,e){try{t(require("dompurify"))}catch(is){e(is)}}):"function"==typeof define&&define.amd?new Promise(function(t,e){try{require(["dompurify"],t)}catch(is){e(is)}}):Promise.reject(
new Error("Could not load dompurify"))).catch(function(t){return Promise.reject(new Error("Could not load dompurify: "+t))}).then(function(t){return t.default?t.default:t})}var s=function(
t){var e=r(t);return"undefined"===e?"undefined":"string"===e||t instanceof String?"string":"number"===e||t instanceof Number?"number":"function"===e||t instanceof Function?"function":t&&t.
constructor===Array?"array":t&&1===t.nodeType?"element":"object"===e?"object":"unknown"},o=function(t,e){var n=document.createElement(t);for(var r in e.className&&(n.className=e.className),
e.innerHTML&&e.dompurify&&(n.innerHTML=e.dompurify.sanitize(e.innerHTML)),e.style)n.style[r]=e.style[r];return n},h=function t(e,n){for(var r=3===e.nodeType?document.createTextNode(e.
nodeValue):e.cloneNode(!1),i=e.firstChild;i;i=i.nextSibling)!0!==n&&1===i.nodeType&&"SCRIPT"===i.nodeName||r.appendChild(t(i,n));return 1===e.nodeType&&("CANVAS"===e.nodeName?(r.width=e.
width,r.height=e.height,r.getContext("2d").drawImage(e,0,0)):"TEXTAREA"!==e.nodeName&&"SELECT"!==e.nodeName||(r.value=e.value),r.addEventListener("load",function(){r.scrollTop=e.scrollTop,r.
scrollLeft=e.scrollLeft},!0)),r},l=function t(e){var n=Object.assign(t.convert(Promise.resolve()),JSON.parse(JSON.stringify(t.template))),r=t.convert(Promise.resolve(),n);return(r=r.
setProgress(1,t,1,[t])).set(e)};(l.prototype=Object.create(Promise.prototype)).constructor=l,l.convert=function(t,e){return t.__proto__=e||l.prototype,t},l.template={prop:{src:null,
container:null,overlay:null,canvas:null,img:null,pdf:null,pageSize:null,callback:function(){}},progress:{val:0,state:null,n:0,stack:[]},opt:{filename:"file.pdf",margin:[0,0,0,0],enableLinks:
!0,x:0,y:0,html2canvas:{},jsPDF:{},backgroundColor:"transparent"}},l.prototype.from=function(t,e){return this.then(function(){switch(e=e||function(t){switch(s(t)){case"string":
return"string";case"element":return"canvas"===t.nodeName.toLowerCase()?"canvas":"element";default:return"unknown"}}(t),e){case"string":return this.then(a).then(function(e){return this.set({
src:o("div",{innerHTML:t,dompurify:e})})});case"element":return this.set({src:t});case"canvas":return this.set({canvas:t});case"img":return this.set({img:t});default:return this.error(
"Unknown source type.")}})},l.prototype.to=function(t){switch(t){case"container":return this.toContainer();case"canvas":return this.toCanvas();case"img":return this.toImg();case"pdf":return
this.toPdf();default:return this.error("Invalid target.")}},l.prototype.toContainer=function(){return this.thenList([function(){return this.prop.src||this.error("Cannot duplicate - no
source HTML.")},function(){return this.prop.pageSize||this.setPageSize()}]).then(function(){var t={position:"relative",display:"inline-block",width:("number"!=typeof this.opt.width||isNaN(
this.opt.width)||"number"!=typeof this.opt.windowWidth||isNaN(this.opt.windowWidth)?Math.max(this.prop.src.clientWidth,this.prop.src.scrollWidth,this.prop.src.offsetWidth):this.opt.
windowWidth)+"px",left:0,right:0,top:0,margin:"auto",backgroundColor:this.opt.backgroundColor},e=h(this.prop.src,this.opt.html2canvas.javascriptEnabled);"BODY"===e.tagName&&(t.height=Math.
max(document.body.scrollHeight,document.body.offsetHeight,document.documentElement.clientHeight,document.documentElement.scrollHeight,document.documentElement.offsetHeight)+"px"),this.prop.
overlay=o("div",{className:"html2pdf__overlay",style:{position:"fixed",overflow:"hidden",zIndex:1e3,left:"-100000px",right:0,bottom:0,top:0}}),this.prop.container=o("div",{className:
"html2pdf__container",style:t}),this.prop.container.appendChild(e),this.prop.container.firstChild.appendChild(o("div",{style:{clear:"both",border:"0 none transparent",margin:0,padding:0,
height:0}})),this.prop.container.style.float="none",this.prop.overlay.appendChild(this.prop.container),document.body.appendChild(this.prop.overlay),this.prop.container.firstChild.style.
position="relative",this.prop.container.height=Math.max(this.prop.container.firstChild.clientHeight,this.prop.container.firstChild.scrollHeight,this.prop.container.firstChild.
offsetHeight)+"px"})},l.prototype.toCanvas=function(){var t=[function(){return document.body.contains(this.prop.container)||this.toContainer()}];return this.thenList(t).then(n).then(
function(t){var e=Object.assign({},this.opt.html2canvas);return delete e.onrendered,t(this.prop.container,e)}).then(function(t){(this.opt.html2canvas.onrendered||function(){})(t),this.prop.
canvas=t,document.body.removeChild(this.prop.overlay)})},l.prototype.toContext2d=function(){var t=[function(){return document.body.contains(this.prop.container)||this.toContainer()}];return
this.thenList(t).then(n).then(function(t){var e=this.opt.jsPDF,n=this.opt.fontFaces,r="number"!=typeof this.opt.width||isNaN(this.opt.width)||"number"!=typeof this.opt.windowWidth||isNaN(
this.opt.windowWidth)?1:this.opt.width/this.opt.windowWidth,i=Object.assign({async:!0,allowTaint:!0,scale:r,scrollX:this.opt.scrollX||0,scrollY:this.opt.scrollY||0,backgroundColor:"#ffffff",
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imageTimeout:15e3,logging:!0,proxy:null,removeContainer:!0,foreignObjectRendering:!1,useCORS:!1},this.opt.html2canvas);if(delete i.onrendered,e.context2d.autoPaging=void 0===this.opt.
autoPaging||this.opt.autoPaging,e.context2d.posX=this.opt.x,e.context2d.posY=this.opt.y,e.context2d.margin=this.opt.margin,e.context2d.fontFaces=n,n)for(var a=0;a<n.length;++a){var s=n[a],
o=s.src.find(function(t){return"truetype"===t.format});o&&e.addFont(o.url,s.ref.name,s.ref.style)}return i.windowHeight=i.windowHeight||0,i.windowHeight=0==i.windowHeight?Math.max(this.prop.
container.clientHeight,this.prop.container.scrollHeight,this.prop.container.offsetHeight):i.windowHeight,e.context2d.save(!0),t(this.prop.container,i)}).then(function(t){this.opt.jsPDF.
context2d.restore(!0),(this.opt.html2canvas.onrendered||function(){})(t),this.prop.canvas=t,document.body.removeChild(this.prop.overlay)})},l.prototype.toImg=function(){return this.thenList(
[function(){return this.prop.canvas||this.toCanvas()}]).then(function(){var t=this.prop.canvas.toDataURL("image/"+this.opt.image.type,this.opt.image.quality);this.prop.img=document.
createElement("img"),this.prop.img.src=t})},l.prototype.toPdf=function(){return this.thenList([function(){return this.toContext2d()}]).then(function(){this.prop.pdf=this.prop.pdf||this.opt.
jsPDF})},l.prototype.output=function(t,e,n){return"img"===(n=n||"pdf").toLowerCase()||"image"===n.toLowerCase()?this.outputImg(t,e):this.outputPdf(t,e)},l.prototype.outputPdf=function(t,e){
return this.thenList([function(){return this.prop.pdf||this.toPdf()}]).then(function(){return this.prop.pdf.output(t,e)})},l.prototype.outputImg=function(t){return this.thenList([function(){
return this.prop.img||this.toImg()}]).then(function(){switch(t){case void 0:case"img":return this.prop.img;case"datauristring":case"dataurlstring":return this.prop.img.src;case"datauri":
case"dataurl":return document.location.href=this.prop.img.src;default:throw'Image output type "'+t+'" is not supported.'}})},l.prototype.save=function(t){return this.thenList([function(){
return this.prop.pdf||this.toPdf()}]).set(t?{filename:t}:null).then(function(){this.prop.pdf.save(this.opt.filename)})},l.prototype.doCallback=function(){return this.thenList([function(){
return this.prop.pdf||this.toPdf()}]).then(function(){this.prop.callback(this.prop.pdf)})},l.prototype.set=function(t){if("object"!==s(t))return this;var e=Object.keys(t||{}).map(function(
e){if(e in l.template.prop)return function(){this.prop[e]=t[e]};switch(e){case"margin":return this.setMargin.bind(this,t.margin);case"jsPDF":return function(){return this.opt.jsPDF=t.jsPDF,
this.setPageSize()};case"pageSize":return this.setPageSize.bind(this,t.pageSize);default:return function(){this.opt[e]=t[e]}}},this);return this.then(function(){return this.thenList(e)})},l.
prototype.get=function(t,e){return this.then(function(){var n=t in l.template.prop?this.prop[t]:this.opt[t];return e?e(n):n})},l.prototype.setMargin=function(t){return this.then(function(){
switch(s(t)){case"number":t=[t,t,t,t];case"array":if(2===t.length&&(t=[t[0],t[1],t[0],t[1]]),4===t.length)break;default:return this.error("Invalid margin array.")}this.opt.margin=t}).then(
this.setPageSize)},l.prototype.setPageSize=function(t){function e(t,e){return Math.floor(t*e/72*96)}return this.then(function(){(t=t||R.getPageSize(this.opt.jsPDF)).hasOwnProperty(
"inner")||(t.inner={width:t.width-this.opt.margin[1]-this.opt.margin[3],height:t.height-this.opt.margin[0]-this.opt.margin[2]},t.inner.px={width:e(t.inner.width,t.k),height:e(t.inner.height,
t.k)},t.inner.ratio=t.inner.height/t.inner.width),this.prop.pageSize=t})},l.prototype.setProgress=function(t,e,n,r){return null!=t&&(this.progress.val=t),null!=e&&(this.progress.state=e),
null!=n&&(this.progress.n=n),null!=r&&(this.progress.stack=r),this.progress.ratio=this.progress.val/this.progress.state,this},l.prototype.updateProgress=function(t,e,n,r){return this.
setProgress(t?this.progress.val+t:null,e||null,n?this.progress.n+n:null,r?this.progress.stack.concat(r):null)},l.prototype.then=function(t,e){var n=this;return this.thenCore(t,e,function(t,
e){return n.updateProgress(null,null,1,[t]),Promise.prototype.then.call(this,function(e){return n.updateProgress(null,t),e}).then(t,e).then(function(t){return n.updateProgress(1),t})})},l.
prototype.thenCore=function(t,e,n){n=n||Promise.prototype.then;var r=this;t&&(t=t.bind(r)),e&&(e=e.bind(r));var i=-1!==Promise.toString().indexOf("[native code]")&&"Promise"===Promise.
name?r:l.convert(Object.assign({},r),Promise.prototype),a=n.call(i,t,e);return l.convert(a,r.__proto__)},l.prototype.thenExternal=function(t,e){return Promise.prototype.then.call(this,t,e)},
l.prototype.thenList=function(t){var e=this;return t.forEach(function(t){e=e.thenCore(t)}),e},l.prototype.catch=function(t){t&&(t=t.bind(this));var e=Promise.prototype.catch.call(this,t);
return l.convert(e,this)},l.prototype.catchExternal=function(t){return Promise.prototype.catch.call(this,t)},l.prototype.error=function(t){return this.then(function(){throw new Error(t)})},
l.prototype.using=l.prototype.set,l.prototype.saveAs=l.prototype.save,l.prototype.export=l.prototype.output,l.prototype.run=l.prototype.then,R.getPageSize=function(t,e,n){if("object"===r(
t)){var i=t;t=i.orientation,e=i.unit||e,n=i.format||n}e=e||"mm",n=n||"a4",t=(""+(t||"P")).toLowerCase();var a,s=(""+n).toLowerCase(),o={a0:[2383.94,3370.39],a1:[1683.78,2383.94],a2:[1190.55,
1683.78],a3:[841.89,1190.55],a4:[595.28,841.89],a5:[419.53,595.28],a6:[297.64,419.53],a7:[209.76,297.64],a8:[147.4,209.76],a9:[104.88,147.4],a10:[73.7,104.88],b0:[2834.65,4008.19],b1:[2004.
09,2834.65],b2:[1417.32,2004.09],b3:[1000.63,1417.32],b4:[708.66,1000.63],b5:[498.9,708.66],b6:[354.33,498.9],b7:[249.45,354.33],b8:[175.75,249.45],b9:[124.72,175.75],b10:[87.87,124.72],c0:[
2599.37,3676.54],c1:[1836.85,2599.37],c2:[1298.27,1836.85],c3:[918.43,1298.27],c4:[649.13,918.43],c5:[459.21,649.13],c6:[323.15,459.21],c7:[229.61,323.15],c8:[161.57,229.61],c9:[113.39,161.
57],c10:[79.37,113.39],dl:[311.81,623.62],letter:[612,792],"government-letter":[576,756],legal:[612,1008],"junior-legal":[576,360],ledger:[1224,792],tabloid:[792,1224],"credit-card":[153,
243]};switch(e){case"pt":a=1;break;case"mm":a=72/25.4;break;case"cm":a=72/2.54;break;case"in":a=72;break;case"px":a=.75;break;case"pc":case"em":a=12;break;case"ex":a=6;break;default:
throw"Invalid unit: "+e}var h,l=0,c=0;if(o.hasOwnProperty(s))l=o[s][1]/a,c=o[s][0]/a;else try{l=n[1],c=n[0]}catch(qn){throw new Error("Invalid format: "+n)}if("p"===t||"portrait"===t)t="p",
c>l&&(h=c,c=l,l=h);else{if("l"!==t&&"landscape"!==t)throw"Invalid orientation: "+t;t="l",l>c&&(h=c,c=l,l=h)}return{width:c,height:l,unit:e,k:a,orientation:t}},e.html=function(t,e){(e=e||{}).
callback=e.callback||function(){},e.html2canvas=e.html2canvas||{},e.html2canvas.canvas=e.html2canvas.canvas||this.canvas,e.jsPDF=e.jsPDF||this,e.fontFaces=e.fontFaces?e.fontFaces.map(Et):
null;var n=new l(e);return e.worker?n:n.from(t).doCallback()}}(R.API),
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   * ====================================================================
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   * OF CONTRACT, TORT OR OTHERWISE, ARISING FROM, OUT OF OR IN CONNECTION
   * WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN THE SOFTWARE.
   * ====================================================================
   */
function(t){t.addJS=function(t){var e,n,r=function(t){for(var e="",n=0;n<t.length;n++){var r=t[n];if("("===r||")"===r){for(var i=0,a=n-1;a>=0&&"\\"===t[a];a--)i++;e+=i%2==0?"\\"+r:r}else
e+=r}return e}(t);return this.internal.events.subscribe("postPutResources",function(){e=this.internal.newObject(),this.internal.out("<<"),this.internal.out("/Names [(EmbeddedJS) "+(e+1)+" 0
R]"),this.internal.out(">>"),this.internal.out("endobj"),n=this.internal.newObject(),this.internal.out("<<"),this.internal.out("/S /JavaScript"),this.internal.out("/JS ("+r+")"),this.
internal.out(">>"),this.internal.out("endobj")}),this.internal.events.subscribe("putCatalog",function(){void 0!==e&&void 0!==n&&this.internal.out("/Names <</JavaScript "+e+" 0 R>>")}),
this}}(R.API),
/**
   * @license
   * Copyright (c) 2014 Steven Spungin (TwelveTone LLC)  steven@twelvetone.tv
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   * Licensed under the MIT License.
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   */
function(t){var e;t.events.push(["postPutResources",function(){var t=this,n=/^(\d+) 0 obj$/;if(this.outline.root.children.length>0)for(var r=t.outline.render().split(/\r\n/),i=0;i<r.length;
i++){var a=r[i],s=n.exec(a);if(null!=s){var o=s[1];t.internal.newObjectDeferredBegin(o,!1)}t.internal.write(a)}if(this.outline.createNamedDestinations){var h=this.internal.pages.length,l=[];
for(i=0;i<h;i++){var c=t.internal.newObject();l.push(c);var u=t.internal.getPageInfo(i+1);t.internal.write("<< /D["+u.objId+" 0 R /XYZ null null null]>> endobj")}var f=t.internal.newObject(
);for(t.internal.write("<< /Names [ "),i=0;i<l.length;i++)t.internal.write("(page_"+(i+1)+")"+l[i]+" 0 R");t.internal.write(" ] >>","endobj"),e=t.internal.newObject(),t.internal.write("<< /
Dests "+f+" 0 R"),t.internal.write(">>","endobj")}}]),t.events.push(["putCatalog",function(){var t=this;t.outline.root.children.length>0&&(t.internal.write("/Outlines",this.outline.makeRef(
this.outline.root)),this.outline.createNamedDestinations&&t.internal.write("/Names "+e+" 0 R"))}]),t.events.push(["initialized",function(){var t=this;t.outline={createNamedDestinations:!1,
root:{children:[]}},t.outline.add=function(t,e,n){var r={title:e,options:n,children:[]};return null==t&&(t=this.root),t.children.push(r),r},t.outline.render=function(){return this.ctx={},
this.ctx.val="",this.ctx.pdf=t,this.genIds_r(this.root),this.renderRoot(this.root),this.renderItems(this.root),this.ctx.val},t.outline.genIds_r=function(e){e.id=t.internal.newObjectDeferred(
);for(var n=0;n<e.children.length;n++)this.genIds_r(e.children[n])},t.outline.renderRoot=function(t){this.objStart(t),this.line("/Type /Outlines"),t.children.length>0&&(this.line("/First
"+this.makeRef(t.children[0])),this.line("/Last "+this.makeRef(t.children[t.children.length-1]))),this.line("/Count "+this.count_r({count:0},t)),this.objEnd()},t.outline.
renderItems=function(e){for(var n=this.ctx.pdf.internal.getVerticalCoordinateString,r=0;r<e.children.length;r++){var i=e.children[r];this.objStart(i),this.line("/Title "+this.makeString(i.
title)),this.line("/Parent "+this.makeRef(e)),r>0&&this.line("/Prev "+this.makeRef(e.children[r-1])),r<e.children.length-1&&this.line("/Next "+this.makeRef(e.children[r+1])),i.children.
```

```
length>0&&(this.line("/First "+this.makeRef(i.children[0])),this.line("/Last "+this.makeRef(i.children[i.children.length-1])));var a=this.count=this.count_r({count:0},i);if(a>0&&this.line("/
Count "+a),i.options&&i.options.pageNumber){var s=t.internal.getPageInfo(i.options.pageNumber);this.line("/Dest ["+s.objId+" 0 R /XYZ 0 "+n(0)+" 0]")}this.objEnd()}for(var o=0;o<e.children.
length;o++)this.renderItems(e.children[o])},t.outline.line=function(t){this.ctx.val+=t+"\r\n"},t.outline.makeRef=function(t){return t.id+" 0 R"},t.outline.makeString=function(e){return"("+t.
internal.pdfEscape(e)+")"},t.outline.objStart=function(t){this.ctx.val+="\r\n"+t.id+" 0 obj\r\n<<\r\n"},t.outline.objEnd=function(){this.ctx.val+=">> \r\nendobj\r\n"},t.outline.
count_r=function(t,e){for(var n=0;n<e.children.length;n++)t.count++,this.count_r(t,e.children[n]);return t.count}}])}(R.API),
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   *
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   */
function(t){var e=[192,193,194,195,196,197,198,199];t.processJPEG=function(t,n,r,i,a,s){var o,h=this.decode.DCT_DECODE,l=null;if("string"==typeof t||this.__addimage__.isArrayBuffer(t)||this.
__addimage__.isArrayBufferView(t)){switch(t=a||t,t=this.__addimage__.isArrayBuffer(t)?new Uint8Array(t):t,o=function(t){for(var n,r=256*t.charCodeAt(4)+t.charCodeAt(5),i=t.length,a={width:0,
height:0,numcomponents:1},s=4;s<i;s+=2){if(s+=r,-1!==e.indexOf(t.charCodeAt(s+1))){n=256*t.charCodeAt(s+5)+t.charCodeAt(s+6),a={width:256*t.charCodeAt(s+7)+t.charCodeAt(s+8),height:n,
numcomponents:t.charCodeAt(s+9)};break}r=256*t.charCodeAt(s+2)+t.charCodeAt(s+3)}return a}(t=this.__addimage__.isArrayBufferView(t)?this.__addimage__.arrayBufferToBinaryString(t):t),o.
numcomponents){case 1:s=this.color_spaces.DEVICE_GRAY;break;case 4:s=this.color_spaces.DEVICE_CMYK;break;case 3:s=this.color_spaces.DEVICE_RGB}l={data:t,width:o.width,height:o.height,
colorSpace:s,bitsPerComponent:8,filter:h,index:n,alias:r}}return l}}(R.API);const Ae=new TextEncoder,Le=(()=>{const t=new Uint8Array(4);return!((new Uint32Array(t.buffer)[0]=1)&t[0])})(),
Ne={int8:globalThis.Int8Array,uint8:globalThis.Uint8Array,int16:globalThis.Int16Array,uint16:globalThis.Uint16Array,int32:globalThis.Int32Array,uint32:globalThis.Uint32Array,uint64:
globalThis.BigUint64Array,int64:globalThis.BigInt64Array,float32:globalThis.Float32Array,float64:globalThis.Float64Array};class Se{buffer;byteLength;byteOffset;length;offset;lastWrittenByte;
littleEndian;_data;_mark;_marks;constructor(t=8192,e={}){let n=!1;"number"==typeof t?t=new ArrayBuffer(t):(n=!0,this.lastWrittenByte=t.byteLength);const r=e.offset?e.offset>>>0:0,i=t.
byteLength-r;let a=r;(ArrayBuffer.isView(t)||t instanceof Se)&&(t.byteLength!==t.buffer.byteLength&&(a=t.byteOffset+r),t=t.buffer),this.lastWrittenByte=n?i:0,this.buffer=t,this.length=i,
this.byteLength=i,this.byteOffset=a,this.offset=0,this.littleEndian=!0,this._data=new DataView(this.buffer,a,i),this._mark=0,this._marks=[]}available(t=1){return this.offset+t<=this.
length}isLittleEndian(){return this.littleEndian}setLittleEndian(){return this.littleEndian=!0,this}isBigEndian(){return!this.littleEndian}setBigEndian(){return this.littleEndian=!1,
this}skip(t=1){return this.offset+=t,this}back(t=1){return this.offset-=t,this}seek(t){return this.offset=t,this}mark(){return this._mark=this.offset,this}reset(){return this.offset=this.
_mark,this}pushMark(){return this._marks.push(this.offset),this}popMark(){const t=this._marks.pop();if(void 0===t)throw new Error("Mark stack empty");return this.seek(t),this}rewind(){
return this.offset=0,this}ensureAvailable(t=1){if(!this.available(t)){const e=2*(this.offset+t),n=new Uint8Array(e);n.set(new Uint8Array(this.buffer)),this.buffer=n.buffer,this.length=e,
this.byteLength=e,this._data=new DataView(this.buffer)}return this}readBoolean(){return 0!==this.readUint8()}readInt8(){return this._data.getInt8(this.offset++)}readUint8(){return this.
_data.getUint8(this.offset++)}readByte(){return this.readUint8()}readBytes(t=1){return this.readArray(t,"uint8")}readArray(t,e){const n=Ne[e].BYTES_PER_ELEMENT*t,r=this.byteOffset+this.
offset,i=this.buffer.slice(r,r+n);if(this.littleEndian===Le&&"uint8"!==e&&"int8"!==e){const t=new Uint8Array(this.buffer.slice(r,r+n));t.reverse();const i=new Ne[e](t.buffer);return this.
offset+=n,i.reverse(),i}const a=new Ne[e](i);return this.offset+=n,a}readInt16(){const t=this._data.getInt16(this.offset,this.littleEndian);return this.offset+=2,t}readUint16(){const t=this.
_data.getUint16(this.offset,this.littleEndian);return this.offset+=2,t}readInt32(){const t=this._data.getInt32(this.offset,this.littleEndian);return this.offset+=4,t}readUint32(){const
t=this._data.getUint32(this.offset,this.littleEndian);return this.offset+=4,t}readFloat32(){const t=this._data.getFloat32(this.offset,this.littleEndian);return this.offset+=4,t}readFloat64(
){const t=this._data.getFloat64(this.offset,this.littleEndian);return this.offset+=8,t}readBigInt64(){const t=this._data.getBigInt64(this.offset,this.littleEndian);return this.offset+=8,
t}readBigUint64(){const t=this._data.getBigUint64(this.offset,this.littleEndian);return this.offset+=8,t}readChar(){return String.fromCharCode(this.readInt8())}readChars(t=1){let e="";for(
let n=0;n<t;n++)e+=this.readChar();return e}readUtf8(t=1){return xe(this.readBytes(t))}decodeText(t=1,e="utf8"){return xe(this.readBytes(t),e)}writeBoolean(t){return this.writeUint8(t?255:
0),this}writeInt8(t){return this.ensureAvailable(1),this._data.setInt8(this.offset++,t),this._updateLastWrittenByte(),this}writeUint8(t){return this.ensureAvailable(1),this._data.setUint8(
this.offset++,t),this._updateLastWrittenByte(),this}writeByte(t){return this.writeUint8(t)}writeBytes(t){this.ensureAvailable(t.length);for(let e=0;e<t.length;e++)this._data.setUint8(this.
offset++,t[e]);return this._updateLastWrittenByte(),this}writeInt16(t){return this.ensureAvailable(2),this._data.setInt16(this.offset,t,this.littleEndian),this.offset+=2,this.
_updateLastWrittenByte(),this}writeUint16(t){return this.ensureAvailable(2),this._data.setUint16(this.offset,t,this.littleEndian),this.offset+=2,this._updateLastWrittenByte(),
this}writeInt32(t){return this.ensureAvailable(4),this._data.setInt32(this.offset,t,this.littleEndian),this.offset+=4,this._updateLastWrittenByte(),this}writeUint32(t){return this.
ensureAvailable(4),this._data.setUint32(this.offset,t,this.littleEndian),this.offset+=4,this._updateLastWrittenByte(),this}writeFloat32(t){return this.ensureAvailable(4),this._data.
setFloat32(this.offset,t,this.littleEndian),this.offset+=4,this._updateLastWrittenByte(),this}writeFloat64(t){return this.ensureAvailable(8),this._data.setFloat64(this.offset,t,this.
littleEndian),this.offset+=8,this._updateLastWrittenByte(),this}writeBigInt64(t){return this.ensureAvailable(8),this._data.setBigInt64(this.offset,t,this.littleEndian),this.offset+=8,this.
_updateLastWrittenByte(),this}writeBigUint64(t){return this.ensureAvailable(8),this._data.setBigUint64(this.offset,t,this.littleEndian),this.offset+=8,this._updateLastWrittenByte(),
this}writeChar(t){return this.writeUint8(t.charCodeAt(0))}writeChars(t){for(let e=0;e<t.length;e++)this.writeUint8(t.charCodeAt(e));return this}writeUtf8(t){return this.writeBytes(function(
t){return Ae.encode(t)}(t))}toArray(){return new Uint8Array(this.buffer,this.byteOffset,this.lastWrittenByte)}getWrittenByteLength(){return this.lastWrittenByte-this.
byteOffset}_updateLastWrittenByte(){this.offset>this.lastWrittenByte&&(this.lastWrittenByte=this.offset)}}
/*! pako 2.1.0 https://github.com/nodeca/pako @license (MIT AND Zlib) */function ke(t){let e=t.length;for(;--e>=0;)t[e]=0}const Pe=new Uint8Array([0,0,0,0,0,0,0,0,1,1,1,1,2,2,2,2,3,3,3,3,4,
4,4,4,5,5,5,5,0]),Fe=new Uint8Array([0,0,0,0,1,1,2,2,3,3,4,4,5,5,6,6,7,7,8,8,9,9,10,10,11,11,12,12,13,13]),Ie=new Uint8Array([0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,1,1,2,2,3,7]),Ce=new Uint8Array([16,
17,18,0,8,7,9,6,10,5,11,4,12,3,13,2,14,1,15]),je=new Array(576);ke(je);const Ee=new Array(60);ke(Ee);const Oe=new Array(512);ke(Oe);const Be=new Array(256);ke(Be);const Me=new Array(29);ke(
Me);const Re=new Array(30);function Te(t,e,n,r,i){this.static_tree=t,this.extra_bits=e,this.extra_base=n,this.elems=r,this.max_length=i,this.has_stree=t&&t.length}let De,qe,ze;function Ue(t,
e){this.dyn_tree=t,this.max_code=0,this.stat_desc=e}ke(Re);const He=t=>t<256?Oe[t]:Oe[256+(t>>>7)],We=(t,e)=>{t.pending_buf[t.pending++]=255&e,t.pending_buf[t.pending++]=e>>>8&255},Ve=(t,e,
n)=>{t.bi_valid>16-n?(t.bi_buf|=e<<t.bi_valid&65535,We(t,t.bi_buf),t.bi_buf=e>>16-t.bi_valid,t.bi_valid+=n-16):(t.bi_buf|=e<<t.bi_valid&65535,t.bi_valid+=n)},Ge=(t,e,n)=>{Ve(t,n[2*e],n[
2*e+1])},Ye=(t,e)=>{let n=0;do{n|=1&t,t>>>=1,n<<=1}while(--e>0);return n>>>1},Ze=(t,e,n)=>{const r=new Array(16);let i,a,s=0;for(i=1;i<=15;i++)s=s+n[i-1]<<1,r[i]=s;for(a=0;a<=e;a++){let e=t[
2*a+1];0!==e&&(t[2*a]=Ye(r[e]++,e))}},Je=t=>{let e;for(e=0;e<286;e++)t.dyn_ltree[2*e]=0;for(e=0;e<30;e++)t.dyn_dtree[2*e]=0;for(e=0;e<19;e++)t.bl_tree[2*e]=0;t.dyn_ltree[512]=1,t.opt_len=t.
static_len=0,t.sym_next=t.matches=0},Xe=t=>{t.bi_valid>8?We(t,t.bi_buf):t.bi_valid>0&&(t.pending_buf[t.pending++]=t.bi_buf),t.bi_buf=0,t.bi_valid=0},Ke=(t,e,n,r)=>{const i=2*e,a=2*n;return
t[i]<t[a]||t[i]===t[a]&&r[e]<=r[n]},$e=(t,e,n)=>{const r=t.heap[n];let i=n<<1;for(;i<=t.heap_len&&(i<t.heap_len&&Ke(e,t.heap[i+1],t.heap[i],t.depth)&&i++,!Ke(e,r,t.heap[i],t.depth));)t.heap[
n]=t.heap[i],n=i,i<<=1;t.heap[n]=r},Qe=(t,e,n)=>{let r,i,a,s,o=0;if(0!==t.sym_next)do{r=255&t.pending_buf[t.sym_buf+o++],r+=(255&t.pending_buf[t.sym_buf+o++])<<8,i=t.pending_buf[t.
sym_buf+o++],0===r?Ge(t,i,e):(a=Be[i],Ge(t,a+256+1,e),s=Pe[a],0!==s&&(i-=Me[a],Ve(t,i,s)),r--,a=He(r),Ge(t,a,n),s=Fe[a],0!==s&&(r-=Re[a],Ve(t,r,s)))}while(o<t.sym_next);Ge(t,256,e)},tn=(t,
e)=>{const n=e.dyn_tree,r=e.stat_desc.static_tree,i=e.stat_desc.has_stree,a=e.stat_desc.elems;let s,o,h,l=-1;for(t.heap_len=0,t.heap_max=573,s=0;s<a;s++)0!==n[2*s]?(t.heap[++t.heap_len]=l=s,
t.depth[s]=0):n[2*s+1]=0;for(;t.heap_len<2;)h=t.heap[++t.heap_len]=l<2?++l:0,n[2*h]=1,t.depth[h]=0,t.opt_len--,i&&(t.static_len-=r[2*h+1]);for(e.max_code=l,s=t.heap_len>>1;s>=1;s--)$e(t,n,
s);h=a;do{s=t.heap[1],t.heap[1]=t.heap[t.heap_len--],$e(t,n,1),o=t.heap[1],t.heap[--t.heap_max]=s,t.heap[--t.heap_max]=o,n[2*h]=n[2*s]+n[2*o],t.depth[h]=(t.depth[s]>=t.depth[o]?t.depth[s]:t.
depth[o])+1,n[2*s+1]=n[2*o+1]=h,t.heap[1]=h++,$e(t,n,1)}while(t.heap_len>=2);t.heap[--t.heap_max]=t.heap[1],((t,e)=>{const n=e.dyn_tree,r=e.max_code,i=e.stat_desc.static_tree,a=e.stat_desc.
has_stree,s=e.stat_desc.extra_bits,o=e.stat_desc.extra_base,h=e.stat_desc.max_length;let l,c,u,f,d,p,g=0;for(f=0;f<=15;f++)t.bl_count[f]=0;for(n[2*t.heap[t.heap_max]+1]=0,l=t.heap_max+1;
l<573;l++)c=t.heap[l],f=n[2*n[2*c+1]+1]+1,f>h&&(f=h,g++),n[2*c+1]=f,c>r||(t.bl_count[f]++,d=0,c>=o&&(d=s[c-o]),p=n[2*c],t.opt_len+=p*(f+d),a&&(t.static_len+=p*(i[2*c+1]+d)));if(0!==g){do{
for(f=h-1;0===t.bl_count[f];)f--;t.bl_count[f]--,t.bl_count[f+1]+=2,t.bl_count[h]--,g-=2}while(g>0);for(f=h;0!==f;f--)for(c=t.bl_count[f];0!==c;)u=t.heap[--l],u>r||(n[2*u+1]!==f&&(t.
opt_len+=(f-n[2*u+1])*n[2*u],n[2*u+1]=f),c--)}})(t,e),Ze(n,l,t.bl_count)},en=(t,e,n)=>{let r,i,a=-1,s=e[1],o=0,h=7,l=4;for(0===s&&(h=138,l=3),e[2*(n+1)+1]=65535,r=0;r<=n;r++)i=s,s=e[2*(
r+1)+1],++o<h&&i===s||(o<l?t.bl_tree[2*i]+=o:0!==i?(i!==a&&t.bl_tree[2*i]++,t.bl_tree[32]++):o<=10?t.bl_tree[34]++:t.bl_tree[36]++,o=0,a=i,0===s?(h=138,l=3):i===s?(h=6,l=3):(h=7,l=4))},nn=(
t,e,n)=>{let r,i,a=-1,s=e[1],o=0,h=7,l=4;for(0===s&&(h=138,l=3),r=0;r<=n;r++)if(i=s,s=e[2*(r+1)+1],!(++o<h&&i===s)){if(o<l)do{Ge(t,i,t.bl_tree)}while(0!==--o);else 0!==i?(i!==a&&(Ge(t,i,t.
bl_tree),o--),Ge(t,16,t.bl_tree),Ve(t,o-3,2)):o<=10?(Ge(t,17,t.bl_tree),Ve(t,o-3,3)):(Ge(t,18,t.bl_tree),Ve(t,o-11,7));o=0,a=i,0===s?(h=138,l=3):i===s?(h=6,l=3):(h=7,l=4)};let rn=!1;const
an=(t,e,n,r)=>{Ve(t,0+(r?1:0),3),Xe(t),We(t,n),We(t,~n),n&&t.pending_buf.set(t.window.subarray(e,e+n),t.pending),t.pending+=n};var sn={_tr_init:t=>{rn||((()=>{let t,e,n,r,i;const a=new
Array(16);for(n=0,r=0;r<28;r++)for(Me[r]=n,t=0;t<1<<Pe[r];t++)Be[n++]=r;for(Be[n-1]=r,i=0,r=0;r<16;r++)for(Re[r]=i,t=0;t<1<<Fe[r];t++)Oe[i++]=r;for(i>>=7;r<30;r++)for(Re[r]=i<<7,t=0;t<1<<Fe[
r]-7;t++)Oe[256+i++]=r;for(e=0;e<=15;e++)a[e]=0;for(t=0;t<=143;)je[2*t+1]=8,t++,a[8]++;for(;t<=255;)je[2*t+1]=9,t++,a[9]++;for(;t<=279;)je[2*t+1]=7,t++,a[7]++;for(;t<=287;)je[2*t+1]=8,t++,a[
8]++;for(Ze(je,287,a),t=0;t<30;t++)Ee[2*t+1]=5,Ee[2*t]=Ye(t,5);De=new Te(je,Pe,257,286,15),qe=new Te(new Array(0),Ie,0,19,7)})(),rn=!0),t.l_desc=new Ue(t.dyn_ltree,
De),t.d_desc=new Ue(t.dyn_dtree,qe),t.bl_desc=new Ue(t.bl_tree,ze),t.bi_buf=0,t.bi_valid=0,Je(t)},_tr_stored_block:an,_tr_flush_block:(t,e,n,r)=>{let i,a,s=0;t.level>0?(2===t.strm.
data_type&&(t.strm.data_type=(t=>{let e,n=4093624447;for(e=0;e<=31;e++,n>>>=1)if(1&n&&0!==t.dyn_ltree[2*e])return 0;if(0!==t.dyn_ltree[18]||0!==t.dyn_ltree[20]||0!==t.dyn_ltree[26])return 1;
for(e=32;e<256;e++)if(0!==t.dyn_ltree[2*e])return 1;return 0})(t)),tn(t,t.l_desc),tn(t,t.d_desc),s=(t=>{let e;for(en(t,t.dyn_ltree,t.l_desc.max_code),en(t,t.dyn_dtree,t.d_desc.max_code),tn(
t,t.bl_desc),e=18;e>=3&&0===t.bl_tree[2*Ce[e]+1];e--);return t.opt_len+=3*(e+1)+5+5+4,e})(t),i=t.opt_len+3+7>>>3,a=t.static_len+3+7>>>3,a<=i&&(i=a)):i=a=n+5,n+4<=i&&-1!==e?an(t,e,n,r):4===t.
strategy||a===i?(Ve(t,2+(r?1:0),3),Qe(t,je,Ee)):(Ve(t,4+(r?1:0),3),((t,e,n,r)=>{let i;for(Ve(t,e-257,5),Ve(t,n-1,5),Ve(t,r-4,4),i=0;i<r;i++)Ve(t,t.bl_tree[2*Ce[i]+1],3);nn(t,t.dyn_ltree,e-
1),nn(t,t.dyn_dtree,n-1)})(t,t.l_desc.max_code+1,t.d_desc.max_code+1,s+1),Qe(t,t.dyn_ltree,t.dyn_dtree)),Je(t),r&&Xe(t)},_tr_tally:(t,e,n)=>(t.pending_buf[t.sym_buf+t.sym_next++]=e,t.
pending_buf[t.sym_buf+t.sym_next++]=e>>8,t.pending_buf[t.sym_buf+t.sym_next++]=n,0===e?t.dyn_ltree[2*n]++:(t.matches++,e--,t.dyn_ltree[2*(Be[n]+256+1)]++,t.dyn_dtree[2*He(e)]++),t.
sym_next===t.sym_end),_tr_align:t=>{Ve(t,2,3),Ge(t,256,je),(t=>{16===t.bi_valid?(We(t,t.bi_buf),t.bi_buf=0,t.bi_valid=0):t.bi_valid>=8&&(t.pending_buf[t.pending++]=255&t.bi_buf,t.bi_buf>>=8,
t.bi_valid-=8)})(t)}},on=(t,e,n,r)=>{let i=65535&t,a=t>>>16&65535,s=0;for(;0!==n;){s=n>2e3?2e3:n,n-=s;do{i=i+e[r++]|0,a=a+i|0}while(--s);i%=65521,a%=65521}return i|a<<16};const hn=new
Uint32Array((()=>{let t,e=[];for(var n=0;n<256;n++){t=n;for(var r=0;r<8;r++)t=1&t?3988292384^t>>>1:t>>>1;e[n]=t}return e})());var ln=(t,e,n,r)=>{const i=hn,a=r+n;t^=-1;for(let s=r;s<a;
s++)t=t>>>8^i[255&(t^e[s])];return-1^t},cn={2:"need dictionary",1:"stream end",0:"","-1":"file error","-2":"stream error","-3":"data error","-4":"insufficient memory","-5":"buffer error","-
6":"incompatible version"},un={Z_NO_FLUSH:0,Z_PARTIAL_FLUSH:1,Z_SYNC_FLUSH:2,Z_FULL_FLUSH:3,Z_FINISH:4,Z_BLOCK:5,Z_TREES:6,Z_OK:0,Z_STREAM_END:1,Z_NEED_DICT:2,Z_ERRNO:-1,Z_STREAM_ERROR:-2,
Z_DATA_ERROR:-3,Z_MEM_ERROR:-4,Z_BUF_ERROR:-5,Z_NO_COMPRESSION:0,Z_BEST_SPEED:1,Z_BEST_COMPRESSION:9,Z_DEFAULT_COMPRESSION:-1,Z_FILTERED:1,Z_HUFFMAN_ONLY:2,Z_RLE:3,Z_FIXED:4,
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Z_DEFAULT_STRATEGY:0,Z_BINARY:0,Z_TEXT:1,Z_UNKNOWN:2,Z_DEFLATED:8};const{_tr_init:fn,_tr_stored_block:dn,_tr_flush_block:pn,_tr_tally:gn,_tr_align:mn}=sn,{Z_NO_FLUSH:bn,Z_PARTIAL_FLUSH:vn,
Z_FULL_FLUSH:wn,Z_FINISH:yn,Z_BLOCK:_n,Z_OK:xn,Z_STREAM_END:An,Z_STREAM_ERROR:Nn,Z_DATA_ERROR:Nn,Z_BUF_ERROR:Sn,Z_DEFAULT_COMPRESSION:kn,Z_FILTERED:Pn,Z_HUFFMAN_ONLY:Fn,Z_RLE:In,Z_FIXED:Cn,
Z_DEFAULT_STRATEGY:jn,Z_UNKNOWN:En,Z_DEFLATED:On}=un,Bn=258,Mn=262,Rn=42,Tn=113,Dn=666,qn=(t,e)=>(t.msg=cn[e],e),zn=t=>2*t-(t>4?9:0),Un=t=>{let e=t.length;for(;--e>=0;)t[e]=0},Hn=t=>{let e,
n,r,i=t.w_size;e=t.hash_size,r=e;do{n=t.head[--r],t.head[r]=n>=i?n-i:0}while(--e);e=i,r=e;do{n=t.prev[--r],t.prev[r]=n>=i?n-i:0}while(--e)};let Wn=(t,e,n)=>(e<<t.hash_shift^n)&t.hash_mask;
const Vn=t=>{const e=t.state;let n=e.pending;n>t.avail_out&&(n=t.avail_out),0!==n&&(t.output.set(e.pending_buf.subarray(e.pending_out,e.pending_out+n),t.next_out),t.next_out+=n,e.
pending_out+=n,t.total_out+=n,t.avail_out-=n,e.pending-=n,0===e.pending&&(e.pending_out=0))},Gn=(t,e)=>{pn(t,t.block_start>=0?t.block_start:-1,t.strstart-t.block_start,e),t.block_start=t.
strstart,Vn(t.strm)},Yn=(t,e)=>{t.pending_buf[t.pending++]=e},Zn=(t,e)=>{t.pending_buf[t.pending++]=e>>>8&255,t.pending_buf[t.pending++]=255&e},Jn=(t,e,n,r)=>{let i=t.avail_in;return i>r&&(
i=r),0===i?0:(t.avail_in-=i,e.set(t.input.subarray(t.next_in,t.next_in+i),n),1===t.state.wrap?t.adler=on(t.adler,e,i,n):2===t.state.wrap&&(t.adler=ln(t.adler,e,i,n)),t.next_in+=i,t.
total_in+=i,i)},Xn=(t,e)=>{let n,r,i=t.max_chain_length,a=t.strstart,s=t.prev_length,o=t.nice_match;const h=t.strstart>t.w_size-Mn?t.strstart-(t.w_size-Mn):0,l=t.window,c=t.w_mask,u=t.prev,
f=t.strstart+Bn;let d=l[a+s-1],p=l[a+s];t.prev_length>=t.good_match&&(i>>=2),o>t.lookahead&&(o=t.lookahead);do{if(n=e,l[n+s]===p&&l[n+s-1]===d&&l[n]===l[a]&&l[++n]===l[a+1]){a+=2,n++;do{
}while(l[++a]===l[++n]&&l[++a]===l[++n]&&l[++a]===l[++n]&&l[++a]===l[++n]&&l[++a]===l[++n]&&l[++a]===l[++n]&&a<f);if(r=Bn-(f-a),a=f-Bn,r>s){if(t.
match_start=e,s=r,r>=o)break;d=l[a+s-1],p=l[a+s]}}}while((e=u[e&c])>h&&0!==--i);return s<=t.lookahead?s:t.lookahead},Kn=t=>{const e=t.w_size;let n,r,i;do{if(r=t.window_size-t.lookahead-t.
strstart,t.strstart>=e+(e-Mn)&&(t.window.set(t.window.subarray(e,e+e-r),0),t.match_start-=e,t.strstart-=e,t.block_start-=e,t.insert>t.strstart&&(t.insert=t.strstart),Mn(t),r+=e),0===t.strm.
avail_in)break;if(n=Jn(t.strm,t.window,t.strstart+t.lookahead,r),t.lookahead+=n,t.lookahead+t.insert>=3)for(i=t.strstart-t.insert,t.ins_h=t.window[i],t.ins_h=Wn(t,t.ins_h,t.window[i+1]);t.
insert&&(t.ins_h=Wn(t,t.ins_h,t.window[i+3-1]),t.prev[i&t.w_mask]=t.head[t.ins_h],t.head[t.ins_h]=i,i++,t.insert--,!(t.lookahead+t.insert<3)););}while(t.lookahead<Mn&&0!==t.strm.avail_in)},
$n=(t,e)=>{let n,r,i,a=t.pending_buf_size-5>t.w_size?t.w_size:t.pending_buf_size-5,s=0,o=t.strm.avail_in;do{if(n=65535,i=t.bi_valid+42>>3,t.strm.avail_out<i)break;if(i=t.strm.avail_out-i,
r=t.strstart-t.block_start,n>r+t.strm.avail_in&&(n=r+t.strm.avail_in),n>i&&(n=i),n<a&&(0===n&&e!==yn||e===bn||n!==r+t.strm.avail_in))break;s=e===yn&&n===r+t.strm.avail_in?1:0,dn(t,0,0,s),t.
pending_buf[t.pending-4]=n,t.pending_buf[t.pending-3]=n>>8,t.pending_buf[t.pending-2]=~n,t.pending_buf[t.pending-1]=~n>>8,Vn(t.strm),r&&(r>n&&(r=n),t.strm.output.set(t.window.subarray(t.
block_start,t.block_start+r),t.strm.next_out),t.strm.next_out+=r,t.strm.avail_out-=r,t.strm.total_out+=r,t.block_start+=r,n-=r),n&&(Jn(t.strm,t.strm.output,t.strm.next_out,n),t.strm.
next_out+=n,t.strm.avail_out-=n,t.strm.total_out+=n)}while(0===s);return o-=t.strm.avail_in,o&&(o>=t.w_size?(t.matches=2,t.window.set(t.strm.input.subarray(t.strm.next_in-t.w_size,t.strm.
next_in),0),t.strstart=t.w_size,t.insert=t.strstart):(t.window_size-t.strstart<=o&&(t.strstart-=t.w_size,t.window.set(t.window.subarray(t.w_size,t.w_size+t.strstart),0),t.matches<2&&t.
matches++,t.insert>t.strstart&&(t.insert=t.strstart)),t.window.set(t.strm.input.subarray(t.strm.next_in-o,t.strm.next_in),t.strstart),t.strstart+=o,t.insert+=o>t.w_size-t.insert?t.w_size-t.
insert:o),t.block_start=t.strstart),t.high_water<t.strstart&&(t.high_water=t.strstart),s?4:e!==bn&&e!==yn&&0===t.strm.avail_in&&t.strstart===t.block_start?2:(i=t.window_size-t.strstart,t.
strm.avail_in>i&&t.block_start>=t.w_size&&(t.block_start-=t.w_size,t.strstart-=t.w_size,t.window.set(t.window.subarray(t.w_size,t.w_size+t.strstart),0),t.matches<2&&t.matches++,i+=t.w_size,
t.insert>t.strstart&&(t.insert=t.strstart)),i>t.strm.avail_in&&(i=t.strm.avail_in),i&&(Jn(t.strm,t.window,t.strstart,i),t.strstart+=i,t.insert+=i>t.w_size-t.insert?t.w_size-t.insert:i),t.
high_water<t.strstart&&(t.high_water=t.strstart),i=t.bi_valid+42>>3,i=t.pending_buf_size-i>65535?65535:t.pending_buf_size-i,a=i>t.w_size?t.w_size:i,r=t.strstart-t.block_start,(r>=a||(
r||e===yn)&&e!==bn&&0===t.strm.avail_in&&r<=i)&&(n=r>i?i:r,s=e===yn&&0===t.strm.avail_in&&n===r?1:0,dn(t,t.block_start,n,s),t.block_start+=n,Vn(t.strm)),s?3:1)},Qn=(t,e)=>{let n,r;for(;;){
if(t.lookahead<Mn){if(Kn(t),t.lookahead<Mn&&e===bn)return 1;if(0===t.lookahead)break}if(n=0,t.lookahead>=3&&(t.ins_h=Wn(t,t.ins_h,t.window[t.strstart+3-1]),n=t.prev[t.strstart&t.w_mask]=t.
head[t.ins_h],t.head[t.ins_h]=t.strstart),0!==n&&t.strstart-n<=t.w_size-Mn&&(t.match_length=Xn(t,n)),t.match_length>=3)if(r=gn(t,t.strstart-t.match_start,t.match_length-3),t.lookahead-=t.
match_length,t.match_length<=t.max_lazy_match&&t.lookahead>=3){t.match_length--;do{t.strstart++,t.ins_h=Wn(t,t.ins_h,t.window[t.strstart+3-1]),n=t.prev[t.strstart&t.w_mask]=t.head[t.ins_h],
t.head[t.ins_h]=t.strstart}while(0!==--t.match_length);t.strstart++}else t.strstart+=t.match_length,t.match_length=0,t.ins_h=t.window[t.strstart],t.ins_h=Wn(t,t.ins_h,t.window[t.
strstart+1]);else r=gn(t,0,t.window[t.strstart]),t.lookahead--,t.strstart++;if(r&&(Gn(t,!1),0===t.strm.avail_out))return 1}return t.insert=t.strstart<2?t.strstart:2,e===yn?(Gn(t,!0),0===t.
strm.avail_out?3:4):t.sym_next&&(Gn(t,!1),0===t.strm.avail_out)?1:2},tr=(t,e)=>{let n,r,i;for(;;){if(t.lookahead<Mn){if(Kn(t),t.lookahead<Mn&&e===bn)return 1;if(0===t.lookahead)break}if(n=0,
t.lookahead>=3&&(t.ins_h=Wn(t,t.ins_h,t.window[t.strstart+3-1]),n=t.prev[t.strstart&t.w_mask]=t.head[t.ins_h],t.head[t.ins_h]=t.strstart),t.prev_length=t.match_length,t.prev_match=t.
match_start,t.match_length=2,0!==n&&t.prev_length<t.max_lazy_match&&t.strstart-n<=t.w_size-Mn&&(t.match_length=Xn(t,n),t.match_length<=5&&(t.strategy===Pn||3===t.match_length&&t.strstart-t.
match_start>4096)&&(t.match_length=2)),t.prev_length>=3&&t.match_length<=t.prev_length){i=t.strstart+t.lookahead-3,r=gn(t,t.strstart-1-t.prev_match,t.prev_length-3),t.lookahead-=t.
prev_length-1,t.prev_length-=2;do{++t.strstart<=i&&(t.ins_h=Wn(t,t.ins_h,t.window[t.strstart+3-1]),n=t.prev[t.strstart&t.w_mask]=t.head[t.ins_h],t.head[t.ins_h]=t.strstart)}while(0!==--t.
prev_length);if(t.match_available=0,t.match_length=2,t.strstart++,r&&(Gn(t,!1),0===t.strm.avail_out))return 1}else if(t.match_available){if(r=gn(t,0,t.window[t.strstart-1]),r&&Gn(t,!1),t.
strstart++,t.lookahead--,0===t.strm.avail_out)return 1}else t.match_available=1,t.strstart++,t.lookahead--}return t.match_available&&(r=gn(t,0,t.window[t.strstart-1]),t.match_available=0),t.
insert=t.strstart<2?t.strstart:2,e===yn?(Gn(t,!0),0===t.strm.avail_out?3:4):t.sym_next&&(Gn(t,!1),0===t.strm.avail_out)?1:2};function er(t,e,n,r,i){this.good_length=t,this.max_lazy=e,this.
nice_length=n,this.max_chain=r,this.func=i}const nr=[new er(0,0,0,0,$n),new er(4,4,8,4,Qn),new er(4,5,16,8,Qn),new er(4,6,32,32,Qn),new er(4,4,16,16,tr),new er(8,16,32,32,tr),new er(8,16,
128,128,tr),new er(8,32,128,256,tr),new er(32,128,258,1024,tr),new er(32,258,258,4096,tr)];function rr(){this.strm=null,this.status=0,this.pending_buf=null,this.pending_buf_size=0,this.
pending_out=0,this.pending=0,this.wrap=0,this.gzhead=null,this.gzindex=0,this.method=On,this.last_flush=-1,this.w_size=0,this.w_bits=0,this.w_mask=0,this.window=null,this.window_size=0,this.
prev=null,this.head=null,this.ins_h=0,this.hash_size=0,this.hash_bits=0,this.hash_mask=0,this.hash_shift=0,this.block_start=0,this.match_length=0,this.prev_match=0,this.match_available=0,
this.strstart=0,this.match_start=0,this.lookahead=0,this.prev_length=0,this.max_chain_length=0,this.max_lazy_match=0,this.level=0,this.strategy=0,this.good_match=0,this.nice_match=0,this.
dyn_ltree=new Uint16Array(1146),this.dyn_dtree=new Uint16Array(122),this.bl_tree=new Uint16Array(78),Un(this.dyn_ltree),Un(this.dyn_dtree),Un(this.bl_tree),this.l_desc=null,this.d_desc=null,
this.bl_desc=null,this.bl_count=new Uint16Array(16),this.heap=new Uint16Array(573),Un(this.heap),this.heap_len=0,this.heap_max=0,this.depth=new Uint16Array(573),Un(this.depth),this.
sym_buf=0,this.lit_bufsize=0,this.sym_next=0,this.sym_end=0,this.opt_len=0,this.static_len=0,this.matches=0,this.insert=0,this.bi_buf=0,this.bi_valid=0}const ir=t=>{if(!t)return 1;const e=t.
state;return!e||e.strm!==t||e.status!==Rn&&57!==e.status&&69!==e.status&&73!==e.status&&91!==e.status&&103!==e.status&&e.status!==Tn&&e.status!==Dn?1:0},ar=t=>{if(ir(t))return qn(t,Ln);t.
total_in=t.total_out=0,t.data_type=En;const e=t.state;return e.pending=0,e.pending_out=0,e.wrap<0&&(e.wrap=-e.wrap),e.status=2===e.wrap?57:e.wrap?Rn:Tn,t.adler=2===e.wrap?0:1,e.last_flush=-
2,fn(e),xn},sr=t=>{const e=ar(t);return e===xn&&((n=t.state).window_size=2*n.w_size,Un(n.head),n.max_lazy_match=nr[n.level].max_lazy,n.good_match=nr[n.level].good_length,n.nice_match=nr[n.
level].nice_length,n.max_chain_length=nr[n.level].max_chain,n.strstart=0,n.block_start=0,n.lookahead=0,n.insert=0,n.match_length=n.prev_length=2,n.match_available=0,n.ins_h=0),e;var n},or=(
t,e,n,r,i,a)=>{if(!t)return Ln;let s=1;if(e===kn&&(e=6),r<0?(s=0,r=-r):r>15&&(s=2,r-=16),i<1||i>9||n!==On||r<8||r>15||e<0||e>9||a<0||a>Cn||8===r&&1!==s)return qn(t,Ln);8===r&&(r=9);const
o=new rr;return t.state=o,o.strm=t,o.status=Rn,o.wrap=s,o.gzhead=null,o.w_bits=r,o.w_size=1<<o.w_bits,o.w_mask=o.w_size-1,o.hash_bits=i+7,o.hash_size=1<<o.hash_bits,o.hash_mask=o.hash_size-
1,o.hash_shift=~~((o.hash_bits+3-1)/3),o.window=new Uint8Array(2*o.w_size),o.head=new Uint16Array(o.hash_size),o.prev=new Uint16Array(o.w_size),o.lit_bufsize=1<<i+6,o.pending_buf_size=4*o.
lit_bufsize,o.pending_buf=new Uint8Array(o.pending_buf_size),o.sym_buf=o.lit_bufsize,o.sym_end=3*(o.lit_bufsize-1),o.level=e,o.strategy=a,o.method=n,sr(t)};var hr=or,lr=(t,e)=>ir(t)||2!==t.
state.wrap?Ln:(t.state.gzhead=e,xn),cr=(t,e)=>{if(ir(t)||e>_n||e<0)return t?qn(t,Ln):Ln;const n=t.state;if(!t.output||0!==t.avail_in&&!t.input||n.status===Dn&&e!==yn)return qn(t,0===t.
avail_out?Sn:Ln);const r=n.last_flush;if(n.last_flush=e,0!==n.pending){if(Vn(t),0===t.avail_out)return n.last_flush=-1,xn}else if(0===t.avail_in&&zn(e)<=zn(r)&&e!==yn)return qn(t,Sn);if(n.
status===Dn&&0!==t.avail_in)return qn(t,Sn);if(n.status===Rn&&0===n.wrap&&(n.status=Tn),n.status===Rn){let e=On+(n.w_bits-8<<4)<<8,r=-1;if(r=n.strategy>=Fn||n.level<2?0:n.level<6?1:6===n.
level?2:3,e|=r<<6,0!==n.strstart&&(e|=32),e+=31-e%31,Zn(n,e),0!==n.strstart&&(Zn(n,t.adler>>>16),Zn(n,65535&t.adler)),t.adler=1,n.status=Tn,Vn(t),0!==n.pending)return n.last_flush=-1,xn}if(
57===n.status)if(t.adler=0,Yn(n,31),Yn(n,139),Yn(n,8),n.gzhead)Yn(n,(n.gzhead.text?1:0)+(n.gzhead.hcrc?2:0)+(n.gzhead.extra?4:0)+(n.gzhead.name?8:0)+(n.gzhead.comment?16:0)),Yn(n,255&n.
gzhead.time),Yn(n,n.gzhead.time>>8&255),Yn(n,n.gzhead.time>>16&255),Yn(n,n.gzhead.time>>24&255),Yn(n,9===n.level?2:n.strategy>=Fn||n.level<2?4:0),Yn(n,255&n.gzhead.os),n.gzhead.extra&&n.
gzhead.extra.length&&(Yn(n,255&n.gzhead.extra.length),Yn(n,n.gzhead.extra.length>>8&255)),n.gzhead.hcrc&&(t.adler=ln(t.adler,n.pending_buf,n.pending,0)),n.gzindex=0,n.status=69;else if(Yn(n,
0),Yn(n,0),Yn(n,0),Yn(n,0),Yn(n,0),Yn(n,0),Yn(n,9===n.level?2:n.strategy>=Fn||n.level<2?4:0),Yn(n,3),n.status=Tn,Vn(t),0!==n.pending)return n.last_flush=-1,xn;if(69===n.status){if(n.gzhead.extra){
let e=n.pending,r=(65535&n.gzhead.extra.length)-n.gzindex;for(;n.pending+r>n.pending_buf_size;){let i=n.pending_buf_size-n.pending;if(n.pending_buf.set(n.gzhead.extra.subarray(n.gzindex,n.
gzindex+i),n.pending),n.pending=n.pending_buf_size,n.gzhead.hcrc&&n.pending>e&&(t.adler=ln(t.adler,n.pending_buf,n.pending-e,e)),n.gzindex+=i,Vn(t),0!==n.pending)return n.last_flush=-1,xn;
e=0,r-=i}let i=new Uint8Array(n.gzhead.extra);n.pending_buf.set(i.subarray(n.gzindex,n.gzindex+r),n.pending),n.pending+=r,n.gzhead.hcrc&&n.pending>e&&(t.adler=ln(t.adler,n.pending_buf,n.
pending-e,e)),n.gzindex=0}n.status=73}if(73===n.status){if(n.gzhead.name){let e,r=n.pending;do{if(n.pending===n.pending_buf_size){if(n.gzhead.hcrc&&n.pending>r&&(t.adler=ln(t.adler,n.
pending_buf,n.pending-r,r)),Vn(t),0!==n.pending)return n.last_flush=-1,xn;r=0}e=n.gzindex<n.gzhead.name.length?255&n.gzhead.name.charCodeAt(n.gzindex++):0,Yn(n,e)}while(0!==e);n.gzhead.
hcrc&&n.pending>r&&(t.adler=ln(t.adler,n.pending_buf,n.pending-r,r)),n.gzindex=0}n.status=91}if(91===n.status){if(n.gzhead.comment){let e,r=n.pending;do{if(n.pending===n.pending_buf_size){
if(n.gzhead.hcrc&&n.pending>r&&(t.adler=ln(t.adler,n.pending_buf,n.pending-r,r)),Vn(t),0!==n.pending)return n.last_flush=-1,xn;r=0}e=n.gzindex<n.gzhead.comment.length?255&n.gzhead.comment.
charCodeAt(n.gzindex++):0,Yn(n,e)}while(0!==e);n.gzhead.hcrc&&n.pending>r&&(t.adler=ln(t.adler,n.pending_buf,n.pending-r,r))}n.status=103}if(103===n.status){if(n.gzhead.hcrc){if(n.
pending+2>n.pending_buf_size&&(Vn(t),0!==n.pending))return n.last_flush=-1,xn;Yn(n,255&t.adler),Yn(n,t.adler>>8&255),t.adler=0}if(n.status=Tn,Vn(t),0!==n.pending)return n.last_flush=-1,
xn}if(0!==t.avail_in||0!==n.lookahead||e!==bn&&n.status!==Dn){let r=0===n.level?2:n.strategy===Fn?((t,e)=>{let n;for(;;){if(0===t.lookahead&&(Kn(t),0===t.lookahead)){if(e===bn)return
1;break}if(t.match_length=0,n=gn(t,0,t.window[t.strstart]),t.lookahead--,t.strstart++,n&&(Gn(t,!1),0===t.strm.avail_out))return 1}return t.insert=0,e===yn?(Gn(t,!0),0===t.strm.avail_out?3:
4):t.sym_next&&(Gn(t,!1),0===t.strm.avail_out)?1:2})(n,e):n.strategy===In?((t,e)=>{let n,r,i,a;const s=t.window;for(;;){if(t.lookahead<=Bn){if(Kn(t),t.lookahead<=Bn&&e===bn)return 1;if(
0===t.lookahead)break}if(t.match_length=0,t.lookahead>=3&&t.strstart>0&&(i=t.strstart-1,r=s[i],r===s[++i]&&r===s[++i]&&r===s[++i])){a=t.strstart+Bn;do{}while(r===s[++i]&&r===s[++i]&&r===s[
++i]&&r===s[++i]&&r===s[++i]&&r===s[++i]&&r===s[++i]&&i<a);t.match_length=Bn-(a-i),t.match_length>t.lookahead&&(t.match_length=t.lookahead)}if(t.match_length>=3?(n=gn(t,1,t.
match_length-3),t.lookahead-=t.match_length,t.strstart+=t.match_length,t.match_length=0):(n=gn(t,0,t.window[t.strstart]),t.lookahead--,t.strstart++),n&&(Gn(t,!1),0===t.strm.
avail_out))return 1}return t.insert=0,e===yn?(Gn(t,!0),0===t.strm.avail_out?3:4):t.sym_next&&(Gn(t,!1),0===t.strm.avail_out)?1:2})(n,e):nr[n.level].func(n,e);if(3!==r&&4!==r||(n.status=Dn),
1===r||3===r)return 0===t.avail_out&&(n.last_flush=-1),xn;if(2===r&&(e===vn?mn(n):e!==_n&&(dn(n,0,0,!1),e===wn&&(Un(n.head),0===n.lookahead&&(n.strstart=0,n.block_start=0,n.insert=0))),Vn(
t),0===t.avail_out))return n.last_flush=-1,xn}return e!==yn?xn:n.wrap<=0?An:(2===n.wrap?(Yn(n,255&t.adler),Yn(n,t.adler>>8&255),Yn(n,t.adler>>16&255),Yn(n,t.adler>>24&255),Yn(n,255&t.
total_in),Yn(n,t.total_in>>8&255),Yn(n,t.total_in>>16&255),Yn(n,t.total_in>>24&255)):(Zn(n,t.adler>>>16),Zn(n,65535&t.adler)),Vn(t),n.wrap>0&&(n.wrap=-n.wrap),0!==n.pending?xn:An)},ur=t=>{
if(ir(t))return Ln;const e=t.state.status;return t.state=null,e===Tn?qn(t,Nn):xn},fr=(t,e)=>{let n=e.length;if(ir(t))return Ln;const r=t.state,i=r.wrap;if(2===i||1===i&&r.status!==Rn||r.
lookahead)return Ln;if(1===i&&(t.adler=on(t.adler,e,n,0)),r.wrap=0,n>=r.w_size){0===i&&(Un(r.head),r.strstart=0,r.block_start=0,r.insert=0);let t=new Uint8Array(r.w_size);t.set(e.subarray(n-
r.w_size,n),0),e=t,n=r.w_size}const a=t.avail_in,s=t.next_in,o=t.input;for(t.avail_in=n,t.next_in=0,t.input=e,Kn(r);r.lookahead>=3;){let t=r.strstart,e=r.lookahead-2;do{r.ins_h=Wn(r,r.ins_h,
r.window[t+3-1]),r.prev[t&r.w_mask]=r.head[r.ins_h],r.head[r.ins_h]=t,t++}while(--e);r.strstart=t,r.lookahead=2,Kn(r)}return r.strstart+=r.lookahead,r.block_start=r.strstart,r.insert=r.
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lookahead,r.lookahead=0,r.match_length=r.prev_length=2,r.match_available=0,t.next_in=s,t.input=o,t.avail_in=a,r.wrap=i,xn};const dr=(t,e)=>Object.prototype.hasOwnProperty.call(t,e);var
pr=function(t){const e=Array.prototype.slice.call(arguments,1);for(;e.length;){const n=e.shift();if(n){if("object"!=typeof n)throw new TypeError(n+"must be non-object");for(const e in n)dr(
n,e)&&(t[e]=n[e])}}return t},gr=t=>{let e=0;for(let r=0,i=t.length;r<i;r++)e+=t[r].length;const n=new Uint8Array(e);for(let r=0,i=0,a=t.length;r<a;r++){let e=t[r];n.set(e,i),i+=e.
length}return n};let mr=!0;try{String.fromCharCode.apply(null,new Uint8Array(1))}catch(as){mr=!1}const br=new Uint8Array(256);for(let ss=0;ss<256;ss++)br[ss]=ss>=252?6:ss>=248?5:ss>=240?4:
ss>=224?3:ss>=192?2:1;br[254]=br[254]=1;var vr=t=>{if("function"==typeof TextEncoder&&TextEncoder.prototype.encode)return(new TextEncoder).encode(t);let e,n,r,i,a,s=t.length,o=0;for(i=0;i<s;
i++)n=t.charCodeAt(i),55296==(64512&n)&&i+1<s&&(r=t.charCodeAt(i+1),56320==(64512&r)&&(n=65536+(n-55296<<10)+(r-56320),i++)),o+=n<128?1:n<2048?2:n<65536?3:4;for(e=new Uint8Array(o),a=0,i=0;
a<o;i++)n=t.charCodeAt(i),55296==(64512&n)&&i+1<s&&(r=t.charCodeAt(i+1),56320==(64512&r)&&(n=65536+(n-55296<<10)+(r-56320),i++)),n<128?e[a++]=n:n<2048?(e[a++]=192|n>>>6,e[a++]=128|63&n):
n<65536?(e[a++]=224|n>>>12,e[a++]=128|n>>>6&63,e[a++]=128|63&n):(e[a++]=240|n>>>18,e[a++]=128|n>>>12&63,e[a++]=128|n>>>6&63,e[a++]=128|63&n);return e},wr=(t,e)=>{const n=e||t.length;if(
"function"==typeof TextDecoder&&TextDecoder.prototype.decode)return(new TextDecoder).decode(t.subarray(0,e));let r,i;const a=new Array(2*n);for(i=0,r=0;r<n;){let e=t[r++];if(e<128){a[i++]=e;
continue}let s=br[e];if(s>4)a[i++]=65533,r+=s-1;else{for(e&=2===s?31:3===s?15:7;s>1&&r<n;)e=e<<6|63&t[r++],s--;s>1?a[i++]=65533:e<65536?a[i++]=e:(e-=65536,a[i++]=55296|e>>>10&1023,a[
i++]=56320|1023&e)}}return((t,e)=>{if(e<65534&&t.subarray&&mr)return String.fromCharCode.apply(null,t.length===e?t:t.subarray(0,e));let n="";for(let r=0;r<e;r++)n+=String.fromCharCode(t[r]);
return n})(a,i)},yr=(t,e)=>{(e=e||t.length)>t.length&&(e=t.length);let n=e-1;for(;n>=0&&128==(192&t[n]);)n--;return n<0||0===n?e:n+br[t[n]]>e?n:e},_r=function(){this.input=null,this.
next_in=0,this.avail_in=0,this.total_in=0,this.output=null,this.next_out=0,this.avail_out=0,this.total_out=0,this.msg="",this.state=null,this.data_type=2,this.adler=0};const xr=Object.
prototype.toString,{Z_NO_FLUSH:Ar,Z_SYNC_FLUSH:Lr,Z_FULL_FLUSH:Nr,Z_FINISH:Sr,Z_OK:kr,Z_STREAM_END:Pr,Z_DEFAULT_COMPRESSION:Fr,Z_DEFAULT_STRATEGY:Ir,Z_DEFLATED:Cr}=un;function jr(t){this.
options=pr({level:Fr,method:Cr,chunkSize:16384,windowBits:15,memLevel:8,strategy:Ir},t||{});let e=this.options;e.raw&&e.windowBits>0?e.windowBits=-e.windowBits:e.gzip&&e.windowBits>0&&e.
windowBits<16&&(e.windowBits+=16),this.err=0,this.msg="",this.ended=!1,this.chunks=[],this.strm=new _r,this.strm.avail_out=0;let n=hr(this.strm,e.level,e.method,e.windowBits,e.memLevel,e.
strategy);if(n!==kr)throw new Error(cn[n]);if(e.header&&lr(this.strm,e.header),e.dictionary){let t;if(t="string"==typeof e.dictionary?vr(e.dictionary):"[object ArrayBuffer]"===xr.call(e.
dictionary)?new Uint8Array(e.dictionary):e.dictionary,n=fr(this.strm,t),n!==kr)throw new Error(cn[n]);this.prototype.push=function(t,e){const n=this.strm,r=this.options.
chunkSize;let i,a;if(this.ended)return!1;for(a=e===~e?e:!0===e?Sr:Ar,"string"==typeof t?n.input=vr(t):"[object ArrayBuffer]"===xr.call(t)?n.input=new Uint8Array(t):n.input=t,n.next_in=0,n.
avail_in=n.input.length;;)if(0===n.avail_out&&(n.output=new Uint8Array(r),n.next_out=0,n.avail_out=r),(a===Lr||a===Nr)&&n.avail_out<=6)this.onData(n.output.subarray(0,n.next_out)),n.
avail_out=0;else{if(i=cr(n,a),i===Pr)return n.next_out>0&&this.onData(n.output.subarray(0,n.next_out)),i=ur(this.strm),this.onEnd(i),this.ended=!0,i===kr;if(0!==n.avail_out){if(a>0&&n.
next_out>0)this.onData(n.output.subarray(0,n.next_out)),n.avail_out=0;else if(0===n.avail_in)break}else this.onData(n.output)}return!0},jr.prototype.onData=function(t){this.chunks.push(t)},
jr.prototype.onEnd=function(t){t===kr&&(this.result=gr(this.chunks)),this.chunks=[],this.err=t,this.msg=this.strm.msg};const Er=16209;var Or=function(t,e){let n,r,i,a,s,o,h,l,c,u,f,d,p,g,m,
b,v,w,y,_,x,A,L,N;const S=t.state;n=t.next_in,L=t.input,r=n+(t.avail_in-5),i=t.next_out,N=t.output,a=i-(e-t.avail_out),s=i+(t.avail_out-257),o=S.dmax,h=S.wsize,l=S.whave,c=S.wnext,u=S.
window,f=S.hold,d=S.bits,p=S.lencode,g=S.distcode,m=(1<<S.lenbits)-1,b=(1<<S.distbits)-1;t:do{d<15&&(f+=L[n++]<<d,d+=8,f+=L[n++]<<d,d+=8),v=p[f&m];e:for(;;){if(w=v>>>24,f>>>=w,d-=w,
w=v>>>16&255,0===w)N[i++]=65535&v;else{if(!(16&w)){if(64&w){if(32&w){S.mode=16191;break t}t.msg="invalid literal/length code",S.mode=Er;break t}v=p[(65535&v)+(f&(1<<w)-1)];continue e}for(
y=65535&v,w&=15,w&&(d<w&&(f+=L[n++]<<d,d+=8),y+=f&(1<<w)-1,f>>>=w,d-=w),d<15&&(f+=L[n++]<<d,d+=8,f+=L[n++]<<d,d+=8),v=g[f&b];;){if(w=v>>>24,f>>>=w,d-=w,w=v>>>16&255,16&w){if(_=65535&v,w&=15,
d<w&&(f+=L[n++]<<d,d+=8,d<w&&(f+=L[n++]<<d,d+=8)),_+=f&(1<<w)-1,_>o){t.msg="invalid distance too far back",S.mode=Er;break t}if(f>>>=w,d-=w,w=i-a,_>w){if(w=_-w,w>l&&S.sane){t.msg="invalid
distance too far back",S.mode=Er;break t}if(x=0,A=u,0===c){if(x+=h-w,w<y){y-=w;do{N[i++]=u[x++]}while(--w);x=i-_,A=N}}else if(c<w){if(x+=h+c-w,w-=c,w<y){y-=w;do{N[i++]=u[x++]}while(--w);if(
x=0,c<y){w=c,y-=w;do{N[i++]=u[x++]}while(--w);x=i-_,A=N}}else if(x+=c-w,w<y){y-=w;do{N[i++]=u[x++]}while(--w);x=i-_,A=N}for(;y>2;)N[i++]=A[x++],N[i++]=A[x++],y-=3;y&&(N[
i++]=A[x++],y>1&&(N[i++]=A[x++]))}else{x=i-_;do{N[i++]=N[x++],N[i++]=N[x++],N[i++]=N[x++],y-=3}while(y>2);y&&(N[i++]=N[x++],y>1&&(N[i++]=N[x++]))}break}if(64&w){t.msg="invalid distance
code",S.mode=Er;break t}v=g[(65535&v)+(f&(1<<w)-1)]}}break}while(n<r&&i<s);y=d>>3,n-=y,d-=y<<3,f&=(1<<d)-1,t.next_in=n,t.next_out=i,t.avail_in=n<r?r-n+5:5-(n-r),t.avail_out=i<s?s-i+257:257-
(i-s),S.hold=f,S.bits=d};const Br=new Uint16Array([3,4,5,6,7,8,9,10,11,13,15,17,19,23,27,31,35,43,51,59,67,83,99,115,131,163,195,227,258,0,0]),Mr=new Uint8Array([16,16,16,16,16,16,16,16,17,
17,17,17,18,18,18,18,19,19,19,19,20,20,20,20,21,21,21,21,16,72,78]),Rr=new Uint16Array([1,2,3,4,5,7,9,13,17,25,33,49,65,97,129,193,257,385,513,769,1025,1537,2049,3073,4097,6145,8193,12289,
16385,24577,0,0]),Tr=new Uint8Array([16,16,16,16,17,17,18,18,19,19,20,20,20,20,21,21,21,21,16,72,78]),Dr=(t,e,n,r,i,a,s,o)=>{const h=o.bits;let l,c,u,f,
d,p,g=0,m=0,b=0,v=0,w=0,y=0,_=0,x=0,A=0,L=0,N=null;const S=new Uint16Array(16),k=new Uint16Array(16);let P,F,I,C=null;for(g=0;g<=15;g++)S[g]=0;for(m=0;m<r;m++)S[e[n+m]]++;for(w=h,v=15;
v>=1&&0===S[v];v--);if(w>v&&(w=v),0===v)return i[a++]=20971520,i[a++]=20971520,o.bits=1,0;for(b=1;b<v&&0===S[b];b++);for(w<b&&(w=b),x=1,g=1;g<=15;g++)if(x<<=1,x-=S[g],x<0)return-1;if(x>0&&(
0===t||1!==v))return-1;for(k[1]=0,g=1;g<15;g++)k[g+1]=k[g]+S[g];for(m=0;m<r;m++)0!==e[n+m]&&(s[k[e[n+m]]++]=m);if(0===t?(N=C=s,p=20):1===t?(N=Br,C=Mr,p=257):(N=Rr,C=Tr,p=0),L=0,m=0,g=b,d=a,
y=w,_=0,u=-1,A=1<<w,f=A-1,1===t&&A>852||2===t&&A>592)return 1;for(;;){P=g-_,s[m]+1<p?(F=0,I=s[m]):s[m]>=p?(F=C[s[m]-p],I=N[s[m]-p]):(F=96,I=0),l=1<<g-_,c=1<<y,b=c;do{c-=l,i[d+(
L>>_)+c]=P<<24|F<<16|I}while(0!==c);for(l=1<<g-1;L&l;)L>>=1;if(0!==l?(L&=l-1,L+=l):L=0,m++,0===--S[g]){if(g===v)break;g=e[n+s[m]]}if(g>w&&(L&f)!==u){for(0===_&&(_=w),d+=b,y=g-_,x=1<<y;
y+_<v&&(x-=S[y+_],!(x<=0));)y++,x<<=1;if(A+=1<<y,1===t&&A>852||2===t&&A>592)return 1;u=L&f,i[u]=w<<24|y<<16|d-a}return 0!==L&&(i[d+L]=g-_<<24|64<<16),o.bits=w,0};const{Z_FINISH:qr,Z_BLOCK:
zr,Z_TREES:Ur,Z_OK:Hr,Z_STREAM_END:Wr,Z_NEED_DICT:Vr,Z_STREAM_ERROR:Gr,Z_DATA_ERROR:Yr,Z_MEM_ERROR:Zr,Z_BUF_ERROR:Jr,Z_DEFLATED:Xr}=un,Kr=16180,$r=16190,Qr=16191,ti=16192,ei=16194,ni=16199,
ri=16200,ii=16206,ai=16209,si=t=>(t>>>24&255)+(t>>>8&65280)+((65280&t)<<8)+((255&t)<<24);function oi(){this.strm=null,this.mode=0,this.last=!1,this.wrap=0,this.havedict=!1,this.flags=0,this.
dmax=0,this.check=0,this.total=0,this.head=null,this.wbits=0,this.wsize=0,this.whave=0,this.wnext=0,this.window=null,this.hold=0,this.bits=0,this.length=0,this.offset=0,this.extra=0,this.
lencode=null,this.distcode=null,this.lenbits=0,this.distbits=0,this.ncode=0,this.nlen=0,this.ndist=0,this.have=0,this.next=null,this.lens=new Uint16Array(320),this.work=new Uint16Array(288),
this.lendyn=null,this.distdyn=null,this.sane=0,this.back=0,this.was=0};const hi=t=>{if(!t)return 1;const e=t.state;return!e||e.strm!==t||e.mode<Kr||e.mode>16211?1:0},li=t=>{if(hi(t))return
Gr;const e=t.state;return t.total_in=t.total_out=e.total=0,t.msg="",e.wrap&&(t.adler=1&e.wrap),e.mode=Kr,e.last=0,e.havedict=0,e.flags=-1,e.dmax=32768,e.head=null,e.hold=0,e.bits=0,e.
lencode=e.lendyn=new Int32Array(852),e.distcode=e.distdyn=new Int32Array(592),e.sane=1,e.back=-1,Hr},ci=t=>{if(hi(t))return Gr;const e=t.state;return e.wsize=0,e.whave=0,e.wnext=0,li(t)},
ui=(t,e)=>{let n;if(hi(t))return Gr;const r=t.state;return e<0?(n=0,e=-e):(n=5+(e>>4),e<48&&(e&=15)),e&&(e<8||e>15)?Gr:(null!==r.window&&r.wbits!==e&&(r.window=null),r.wrap=n,r.wbits=e,ci(
t))},fi=(t,e)=>{if(!t)return Gr;const n=new oi;t.state=n,n.strm=t,n.window=null,n.mode=Kr;const r=ui(t,e);return r!==Hr&&(t.state=null),r};let di,pi,gi=!0;const mi=t=>{if(gi){di=new
Int32Array(512),pi=new Int32Array(32);let e=0;for(;e<144;)t.lens[e++]=8;for(;e<256;)t.lens[e++]=9;for(;e<280;)t.lens[e++]=7;for(;e<288;)t.lens[e++]=8;for(Dr(1,t.lens,0,288,di,0,t.work,{bits:
9}),e=0;e<32;)t.lens[e++]=5;Dr(2,t.lens,0,32,pi,0,t.work,{bits:5}),gi=!1}t.lencode=di,t.lenbits=9,t.distcode=pi,t.distbits=5},bi=(t,e,n,r)=>{let i;const a=t.state;return null===a.window&&(a.
wsize=1<<a.wbits,a.wnext=0,a.whave=0,a.window=new Uint8Array(a.wsize)),r>=a.wsize?(a.window.set(e.subarray(n-a.wsize,n),0),a.wnext=0,a.whave=a.wsize):(i=a.wsize-a.wnext,i>r&&(i=r),a.window.
set(e.subarray(n-r,n-r+i),a.wnext),(r-=i)?(a.window.set(e.subarray(n-r,n),0),a.wnext=r,a.whave=a.wsize):(a.wnext+=i,a.wnext===a.wsize&&(a.wnext=0),a.whave<a.wsize&&(a.whave+=i))),0};var
vi=ci,wi=fi,yi=(t,e)=>{let n,r,i,a,s,o,h,l,c,u,f,d,p,g,m,b,v,w,y,_,x,A,L=0;const N=new Uint8Array(4);let S,k;const P=new Uint8Array([16,17,18,0,8,7,9,6,10,5,11,4,12,3,13,2,14,1,15]);if(hi(
t)||!t.output||!t.input&&0!==t.avail_in)return Gr;n=t.state,n.mode===Qr&&(n.mode=ti),s=t.next_out,i=t.output,h=t.avail_out,a=t.next_in,r=t.input,o=t.avail_in,l=n.hold,c=n.bits,u=o,f=h,A=Hr;
t:for(;;)switch(n.mode){case Kr:if(0===n.wrap){n.mode=ti;break}for(;c<16;){if(0===o)break t;o--,l+=r[a++]<<c,c+=8}if(2&n.wrap&&35615===l){0===n.wbits&&(n.wbits=15),n.check=0,N[0]=255&l,N[
1]=l>>>8&255,n.check=ln(n.check,N,2,0),l=0,c=0,n.mode=16181;break}if(n.head&&(n.head.done=!1),!(1&n.wrap)||(((255&l)<<8)+(l>>8))%31){t.msg="incorrect header check",n.mode=ii;break}if((
15&l)!==Xr){t.msg="unknown compression method",n.mode=ai;break}if(l>>>=4,c-=4,x=8+(15&l),0===n.wbits&&(n.wbits=x),x>15||x>n.wbits){t.msg="invalid window size",n.mode=ai;break}n.dmax=1<<n.
wbits,n.flags=0,t.adler=n.check=1,n.mode=512&l?16189:Qr,l=0,c=0;break;case 16181:for(;c<16;){if(0===o)break t;o--,l+=r[a++]<<c,c+=8}if(n.flags=l,(255&n.flags)!==Xr){t.msg="unknown
compression method",n.mode=ai;break}if(57344&n.flags){t.msg="unknown header flags set",n.mode=ai;break}n.head&&(n.head.text=l>>8&1),512&n.flags&&4&n.wrap&&(N[0]=255&l,N[1]=l>>>8&255,n.
check=ln(n.check,N,2,0)),l=0,c=0,n.mode=16182;case 16182:for(;c<32;){if(0===o)break t;o--,l+=r[a++]<<c,c+=8}n.head&&(n.head.time=l),512&n.flags&&4&n.wrap&&(N[0]=255&l,N[1]=l>>>8&255,N[
2]=l>>>16&255,N[3]=l>>>24&255,n.check=ln(n.check,N,4,0)),l=0,c=0,n.mode=16183;case 16183:for(;c<16;){if(0===o)break t;o--,l+=r[a++]<<c,c+=8}n.head&&(n.head.xflags=255&l,n.head.os=l>>8),
512&n.flags&&4&n.wrap&&(N[0]=255&l,N[1]=l>>>8&255,n.check=ln(n.check,N,2,0)),l=0,c=0,n.mode=16184;case 16184:if(1024&n.flags){for(;c<16;){if(0===o)break t;o--,l+=r[a++]<<c,c+=8}n.length=l,n.
head&&(n.head.extra_len=l),512&n.flags&&4&n.wrap&&(N[0]=255&l,N[1]=l>>>8&255,n.check=ln(n.check,N,2,0)),l=0,c=0}else n.head&&(n.head.extra=null);n.mode=16185;case 16185:if(1024&n.flags&&(
d=n.length,d>o&&(d=o),d&&(n.head&&(x=n.head.extra_len-n.length,n.head.extra||(n.head.extra=new Uint8Array(n.head.extra_len)),n.head.extra.set(r.subarray(a,a+d),x)),512&n.flags&&4&n.wrap&&(n.
check=ln(n.check,r,d,a)),o-=d,a+=d,n.length-=d),n.length))break t;n.length=0,n.mode=16186;case 16186:if(2048&n.flags){if(0===o)break t;d=0;do{x=r[a+d++],n.head&&x&&n.length<65536&&(n.head.
name+=String.fromCharCode(x))}while(x&&d<o);if(512&n.flags&&4&n.wrap&&(n.check=ln(n.check,r,d,a)),o-=d,a+=d,x)break t}else n.head&&(n.head.name=null);n.length=0,n.mode=16187;case 16187:if(
4096&n.flags){if(0===o)break t;d=0;do{x=r[a+d++],n.head&&x&&n.length<65536&&(n.head.comment+=String.fromCharCode(x))}while(x&&d<o);if(512&n.flags&&4&n.wrap&&(n.check=ln(n.check,r,d,a)),o-=d,
a+=d,x)break t}else n.head&&(n.head.comment=null);n.mode=16188;case 16188:if(512&n.flags){for(;c<16;){if(0===o)break t;o--,l+=r[a++]<<c,c+=8}if(4&n.wrap&&l!==(65535&n.check)){t.msg="header
crc mismatch",n.mode=ai;break}l=0,c=0}n.head&&(n.head.hcrc=n.flags>>9&1,n.head.done=!0),t.adler=n.check=0,n.mode=Qr;break;case 16189:for(;c<32;){if(0===o)break t;o--,l+=r[a++]<<c,c+=8}t.
adler=n.check=si(l),l=0,c=0,n.mode=$r;case $r:if(0===n.havedict)return t.next_out=s,t.avail_out=h,t.next_in=a,t.avail_in=o,n.hold=l,n.bits=c,Vr;t.adler=n.check=1,n.mode=Qr;case Qr:if(
e===zr||e===Ur)break t;case ti:if(n.last){l>>>=7&c,c-=7&c,n.mode=ii;break}for(;c<3;){if(0===o)break t;o--,l+=r[a++]<<c,c+=8}switch(n.last=1&l,l>>>=1,c-=1,3&l){case 0:n.mode=16193;break;case
1:if(mi(n),n.mode=ni,e===Ur){l>>>=2,c-=2;break t}break;case 2:n.mode=16196;break;case 3:t.msg="invalid block type",n.mode=ai}l>>>=2,c-=2;break;case 16193:for(l>>>=7&c,c-=7&c;c<32;){if(
0===o)break t;o--,l+=r[a++]<<c,c+=8}if((65535&l)!=(l>>>16^65535)){t.msg="invalid stored block lengths",n.mode=ai;break}if(n.length=65535&l,l=0,c=0,n.mode=ei,e===Ur)break t;case ei:n.
mode=16195;case 16195:if(d=n.length,d){if(d>o&&(d=o),d>h&&(d=h),0===d)break t;i.set(r.subarray(a,a+d),s),o-=d,a+=d,h-=d,s+=d,n.length-=d;break}n.mode=Qr;break;case 16196:for(;c<14;){if(
0===o)break t;o--,l+=r[a++]<<c,c+=8}if(n.nlen=257+(31&l),l>>>=5,c-=5,n.ndist=1+(31&l),l>>>=5,c-=5,n.ncode=4+(15&l),l>>>=4,c-=4,n.nlen>286||n.ndist>30){t.msg="too many length or distance
symbols",n.mode=ai;break}n.have=0,n.mode=16197;case 16197:for(;n.have<n.ncode;){for(;c<3;){if(0===o)break t;o--,l+=r[a++]<<c,c+=8}n.lens[P[n.have++]]=7&l,l>>>=3,c-=3}for(;n.have<19;)n.lens[
P[n.have++]]=0;if(n.lencode=n.lendyn,n.lenbits=7,S={bits:n.lenbits},A=Dr(0,n.lens,0,19,n.lencode,0,n.work,S),n.lenbits=S.bits,A){t.msg="invalid code lengths set",n.mode=ai;break}n.have=0,n.
mode=16198;case 16198:for(;n.have<n.nlen+n.ndist;){for(;L=n.lencode[l&(1<<n.lenbits)-1],m=L>>>24,b=L>>>16&255,v=65535&L,!(m<=c);){if(0===o)break t;o--,l+=r[a++]<<c,c+=8}if(v<16)l>>>=m,c-=m,
n.lens[n.have++]=v;else{if(16===v){for(k=m+2;c<k;){if(0===o)break t;o--,l+=r[a++]<<c,c+=8}if(l>>>=m,c-=m,0===n.have){t.msg="invalid bit length repeat",n.mode=ai;break}x=n.lens[n.have++]}else if(17===v){for(k=m+3;c<k;){if(0===o)break t;o--,l+=r[a++]<<c,c+=8}l>>>=m,c-=m,x=0,d=3+(7&l),l>>>=3}else{for(k=m+7;c<k;){if(0===o)break t;o--,l+=r[a++]<<c,
c+=8}l>>>=m,c-=m,x=0,d=11+(127&l),l>>>=7,c-=7}if(n.have+d>n.nlen+n.ndist){t.msg="invalid bit length repeat",n.mode=ai;break}for(;d--;)n.lens[n.have++]=x}}if(n.mode===ai)break;if(0===n.lens[
256]){t.msg="invalid code -- missing end-of-block",n.mode=ai;break}if(n.lenbits=9,S={bits:n.lenbits},A=Dr(1,n.lens,0,n.nlen,n.lencode,0,n.work,S),n.lenbits=S.bits,A){t.msg="invalid literal/
lengths set",n.mode=ai;break}if(n.distbits=6,n.distcode=n.distdyn,S={bits:n.distbits},A=Dr(2,n.lens,n.nlen,n.ndist,n.distcode,0,n.work,S),n.distbits=S.bits,A){t.msg="invalid distances set",
n.mode=ai;break}if(n.mode=ni,e===Ur)break t;case ni:n.mode=ri;case ri:if(o>=6&&h>=258){t.next_out=s,t.avail_out=h,t.next_in=a,t.avail_in=o,n.hold=l,n.bits=c,Or(t,f),s=t.next_out,i=t.output,
h=t.avail_out,a=t.next_in,r=t.input,o=t.avail_in,l=n.hold,c=n.bits,n.mode===Qr&&(n.back=-1);break}for(n.back=0;L=n.lencode[l&(1<<n.lenbits)-1],m=L>>>24,b=L>>>16&255,v=65535&L,!(m<=c);){if(
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0===o)break t;o--,l+=r[a++]<<c,c+=8}if(b&&!(240&b)){for(w=m,y=b,_=v;L=n.lencode[_+((l&(1<<w+y)-1)>>w)],m=L>>>24,b=L>>>16&255,v=65535&L,!(w+m<=c);){if(0===o)break t;o--,l+=r[a++]<<c,
c+=8}l>>>=w,c-=w,n.back+=w}if(l>>>=m,c-=m,n.back+=m,n.length=v,0===b){n.mode=16205;break}if(32&b){n.back=-1,n.mode=Qr;break}if(64&b){t.msg="invalid literal/length code",n.mode=ai;break}n.
extra=15&b,n.mode=16201;case 16201:if(n.extra){for(k=n.extra;c<k;){if(0===o)break t;o--,l+=r[a++]<<c,c+=8}n.length+=l&(1<<n.extra)-1,l>>>=n.extra,c-=n.extra,n.back+=n.extra}n.was=n.length,n.
mode=16202;case 16202:for(;L=n.distcode[l&(1<<n.distbits)-1],m=L>>>24,b=L>>>16&255,v=65535&L,!(m<=c);){if(0===o)break t;o--,l+=r[a++]<<c,c+=8}if(!(240&b)){for(w=m,y=b,_=v;L=n.distcode[_+((
l&(1<<w+y)-1)>>w)],m=L>>>24,b=L>>>16&255,v=65535&L,!(w+m<=c);){if(0===o)break t;o--,l+=r[a++]<<c,c+=8}l>>>=w,c-=w,n.back+=w}if(l>>>=m,c-=m,n.back+=m,64&b){t.msg="invalid distance code",n.
mode=ai;break}n.offset=v,n.extra=15&b,n.mode=16203;case 16203:if(n.extra){for(k=n.extra;c<k;){if(0===o)break t;o--,l+=r[a++]<<c,c+=8}n.offset+=l&(1<<n.extra)-1,l>>>=n.extra,c-=n.extra,n.
back+=n.extra}if(n.offset>n.dmax){t.msg="invalid distance too far back",n.mode=ai;break}n.mode=16204;case 16204:if(0===h)break t;if(d=f-h,n.offset>d){if(d=n.offset-d,d>n.whave&&n.sane){t.
msg="invalid distance too far back",n.mode=ai;break}d>n.wnext?(d-=n.wnext,p=n.wsize-d):p=n.wnext-d,d>n.length&&(d=n.length),g=n.window}else g=i,p=s-n.offset,d=n.length;d>h&&(d=h),h-=d,n.
length-=d;do{i[s++]=g[p++]}while(--d);0===n.length&&(n.mode=ri);break;case 16205:if(0===h)break t;i[s++]=n.length,h--,n.mode=ri;break;case ii:if(n.wrap){for(;c<32;){if(0===o)break t;o--,
l|=r[a++]<<c,c+=8}if(f-=h,t.total_out+=f,n.total+=f,4&n.wrap&&f&&(t.adler=n.check=n.flags?ln(n.check,i,f,s-f):on(n.check,i,f,s-f)),f=h,4&n.wrap&&(n.flags?l:si(l))!==n.check){t.
msg="incorrect data check",n.mode=ai;break}l=0,c=0}n.mode=16207;case 16207:if(n.wrap&&n.flags){for(;c<32;){if(0===o)break t;o--,l+=r[a++]<<c,c+=8}if(4&n.wrap&&l!==(4294967295&n.total)){t.
msg="incorrect length check",n.mode=ai;break}l=0,c=0}n.mode=16208;case 16208:A=Wr;break t;case ai:A=Yr;break t;case 16210:return Zr;default:return Gr}return t.next_out=s,t.avail_out=h,t.
next_in=a,t.avail_in=o,n.hold=l,n.bits=c,(n.wsize||f!==t.avail_out&&n.mode<ai&&(n.mode<ii||e!==gr))&&bi(t,t.output,t.next_out,f-t.avail_out),u-=t.avail_in,f-=t.avail_out,t.total_in+=u,t.
total_out+=f,n.total+=f,4&n.wrap&&f&&(t.adler=n.check=n.flags?ln(n.check,i,f,t.next_out-f):on(n.check,i,f,t.next_out-f)),t.data_type=n.bits+(n.last?64:0)+(n.mode===Qr?128:0)+(n.mode===ni||n.
mode===ei?256:0),(0===u&&0===f||e===gr)&&A===Hr&&(A=Jr),A},_i=t=>{if(hi(t))return Gr;let e=t.state;return e.window&&(e.window=null),t.state=null,Hr},xi=(t,e)=>{if(hi(t))return Gr;const n=t.
state;return 2&n.wrap?(n.head=e,e.done=!1,Hr):Gr},Ai=(t,e)=>{const n=e.length;let r,i,a;return hi(t)?Gr:(r=t.state,0!==r.wrap&&r.mode!==$r?Gr:r.mode===$r&&(i=1,i=on(i,e,n,0),i!==r.check)?Yr:
(a=bi(t,e,n,n),a?(r.mode=16210,Zr):(r.havedict=1,Hr)))},Li=function(){this.text=0,this.time=0,this.xflags=0,this.os=0,this.extra=null,this.extra_len=0,this.name="",this.comment="",this.
hcrc=0,this.done=!1};const Ni=Object.prototype.toString,{Z_NO_FLUSH:Si,Z_FINISH:ki,Z_OK:Pi,Z_STREAM_END:Fi,Z_NEED_DICT:Ii,Z_STREAM_ERROR:Ci,Z_DATA_ERROR:ji,Z_MEM_ERROR:Ei}=un;function Oi(t){
this.options=pr({chunkSize:65536,windowBits:15,to:""},t||{});const e=this.options;e.raw&&e.windowBits>=0&&e.windowBits<16&&(e.windowBits=-e.windowBits,0===e.windowBits&&(e.windowBits=-15)),
!(e.windowBits>=0&&e.windowBits<16)||t&&t.windowBits||(e.windowBits+=32),e.windowBits>15&&e.windowBits<48&&(15&e.windowBits)||(e.windowBits|=15)),this.err=0,this.msg="",this.ended=!1,this.
chunks=[],this.strm=new _r,this.strm.avail_out=0;let n=wi(this.strm,e.windowBits);if(n!==Pi)throw new Error(cn[n]);if(this.header=new Li,xi(this.strm,this.header),e.dictionary&&(
"string"==typeof e.dictionary?e.dictionary=vr(e.dictionary):"[object ArrayBuffer]"===Ni.call(e.dictionary)&&(e.dictionary=new Uint8Array(e.dictionary)),e.raw&&(n=Ai(this.strm,e.dictionary),
n!==Pi)))throw new Error(cn[n])}function Bi(t,e){const n=new Oi(e);if(n.push(t),n.err)throw n.msg||cn[n.err];return n.result}Oi.prototype.push=function(t,e){const n=this.strm,r=this.options.
chunkSize,i=this.options.dictionary;let a,s,o;if(this.ended)return!1;for(s=e===~~e?e:!0===e?ki:Si,"[object ArrayBuffer]"===Ni.call(t)?n.input=new Uint8Array(t):n.input=t,n.next_in=0,n.
avail_in=n.input.length;;){for(0===n.avail_out&&(n.output=new Uint8Array(r),n.next_out=0,n.avail_out=r),a=yi(n,s),a===Ii&&i&&(a=Ai(n,i),a===Pi?a=yi(n,s):a===ji&&(a=Ii));n.
avail_in>0&&a===Fi&&n.state.wrap>0&&0!==t[n.next_in];)vi(n),a=yi(n,s);switch(a){case Ci:case ji:case Ii:case Ei:return this.onEnd(a),this.ended=!0,!1}if(o=n.avail_out,n.next_out&&(0===n.
avail_out||a===Fi))if("string"===this.options.to){let t=yr(n.output,n.next_out),e=n.next_out-t,i=wr(n.output,t);n.next_out=e,n.avail_out=r-e,e&&n.output.set(n.output.subarray(t,t+e),0),this.
onData(i)}else this.onData(n.output.length===n.next_out?n.output:n.output.subarray(0,n.next_out));if(a!==Pi||0!==o){if(a===Fi)return a=_i(this.strm),this.onEnd(a),this.ended=!0,!0;if(0===n.
avail_in)break}}return!0},Oi.prototype.onData=function(t){this.chunks.push(t)},Oi.prototype.onEnd=function(t){t===Pi&&("string"===this.options.to?this.result=this.chunks.join(""):this.
result=gr(this.chunks)),this.chunks=[],this.err=t,this.msg=this.strm.msg};var Mi={Inflate:Oi,inflateBi,inflateRaw:function(t,e){return(e=e||{}).raw=!0,Bi(t,e)},ungzip:Bi,constants:un};
const{Inflate:Ri,inflate:Ti,inflateRaw:Di,ungzip:qi}=Mi;var zi=Ri,Ui=Ti;const Hi=[];for(let ss=0;ss<256;ss++){let t=ss;for(let e=0;e<8;e++)1&t?t=3988292384^t>>>1:t>>>=1;Hi[ss]=t}const
Wi=4294967295;function Vi(t,e,n){const r=t.readUint32(),i=(a=new Uint8Array(t.buffer,t.byteOffset+t.offset-e-4,e),(function(t,e,n){let r=t;for(let i=0;i<n;i++)r=Hi[255&(r^e[i])]^r>>>8;
return r}(Wi,a,e)^Wi)>>>0);var a;if(i!==r)throw new Error(`CRC mismatch for chunk ${n}. Expected ${r}, found ${i}`)}function Gi(t,e,n){for(let r=0;r<n;r++)e[r]=t[r]}function Yi(t,e,n,r){let
i=0;for(;i<r;i++)e[i]=t[i];for(;i<n;i++)e[i]=t[i]+e[i-r]&255}function Zi(t,e,n,r){let i=0;if(0===n.length)for(;i<r;i++)e[i]=t[i];else for(;i<r;i++)e[i]=t[i]+e[a-i]+n[a]>>1)&255;function Ji(t,e,n,r,i){
let a=0;if(0===n.length){for(;a<i;a++)e[a]=t[a];for(;a<r;a++)e[a]=t[a]+(e[a-i]>>1)&255}else{for(;a<i;a++)e[a]=t[a]+(n[a]>>1)&255;for(;a<r;a++)e[a]=t[a]+(e[a-i]+n[a]>>1)&255}function Xi(t,e,
n,r,i){let a=0;if(0===n.length){for(;a<i;a++)e[a]=t[a];for(;a<r;a++)e[a]=t[a]+e[a-i]&255}else{for(;a<i;a++)e[a]=t[a]+n[a]&255;for(;a<r;a++)e[a]=t[a]+Ki(e[a-i],n[a],n[a-i])&255}}function Ki(
t,e,n){const r=t+e-n,i=Math.abs(r-t),a=Math.abs(r-e),s=Math.abs(r+r-n);return i<=a&&i<=s?t:a<=s?e:n}function $i(t,e,n,r,i,a){switch(t){case 0:Gi(e,n,i);break;case 1:Yi(e,n,i,a);break;case 2:
Zi(e,n,r,i);break;case 3:Ji(e,n,r,i,a);break;case 4:Xi(e,n,r,i,a);break;default:throw new Error(`Unsupported filter: ${t}`)}}const Qi=new Uint16Array([255]),ta=255===new Uint8Array(Qi.
buffer)[0];function ea(t){return(255&t)<<8|t>>8&255}const na=new Uint16Array([255]),ra=255===new Uint8Array(na.buffer)[0],ia=new Uint8Array(0);function aa(t){const{data:e,width:n,height:r,
channels:i,depth:a}=t,s=Math.ceil(a/8)*i,o=Math.ceil(a/8*i*n),h=new Uint8Array(r*o);let l,c,u=ia,f=0;for(let d=0;d<r;d++){switch(l=e.subarray(f+1,f+1+o),c=h.subarray(d*o,(d+1)*o),e[f]){case
0:Gi(l,c,o);break;case 1:Yi(l,c,o,s);break;case 2:Zi(l,c,u,o);break;case 3:Ji(l,c,u,o,s);break;case 4:Xi(l,c,u,o,s);break;default:throw new Error(`Unsupported filter: ${e[f]}`)}u=c,
f+=o+1}if(16===a){const t=new Uint16Array(h.buffer);if(ra)for(let e=0;e<t.length;e++)t[e]=sa(t[e]);return t}return h}function sa(t){return(255&t)<<8|t>>8&255}const oa=Uint8Array.of(137,80,
78,71,13,10,26,10);function ha(t){if(!function(t){if(t.length<oa.length)return!1;for(let e=0;e<oa.length;e++)if(t[e]!==oa[e])return!1;return!0}(t.readBytes(oa.length)))throw new Error(
"wrong PNG signature")}const la=new TextDecoder("latin1");const ca=/^[\u0000-\u00FF]*$/;function ua(t){for(t.mark();0!==t.readByte(););const e=t.offset;t.reset();const n=la.decode(t.
readBytes(e-t.offset-1));return t.skip(1),function(t){if(function(t){if(!ca.test(t))throw new Error("invalid latin1 text")}(t),0===t.length||t.length>79)throw new Error("keyword length must
be between 1 and 79")}(n),n}class fa extends Se{_checkCrc;_inflator;_png;_apng;_end;_hasPalette;_palette;_hasTransparency;_transparency;_compressionMethod;_filterMethod;_interlaceMethod;
_colorType;_isAnimated;_numberOfFrames;_numberOfPlays;_frames;_writingDataChunks;constructor(t,e={}){super(t);const{checkCrc:n=!1}=e;this._checkCrc=n,this._inflator=new zi,this._png={width:-
1,height:-1,channels:-1,data:new Uint8Array(0),depth:1,text:{}},this._apng={width:-1,height:-1,channels:-1,depth:1,numberOfFrames:1,numberOfPlays:0,text:{},frames:[]},this._end=!1,this.
_hasPalette=!1,this._palette=[],this._hasTransparency=!1,this._transparency=new Uint16Array(0),this._compressionMethod=-1,this._filterMethod=-1,this._interlaceMethod=-1,this._colorType=-1,
this._isAnimated=!1,this._numberOfFrames=1,this._numberOfPlays=0,this._frames=[],this._writingDataChunks=!1,this.setBigEndian()}decode(){for(ha(this);!this._end;){const t=this.readUint32(),
e=this.readChars(4);this.decodeChunk(t,e)}return this.decodeImage(),this._png}decodeApng(){for(ha(this);!this._end;){const t=this.readUint32(),e=this.readChars(4);this.decodeApngChunk(t,
e)}return this.decodeApngImage(),this._apng}decodeChunk(t,e){const n=this.offset;switch(e){case"IHDR":this.decodeIHDR();break;case"PLTE":this.decodePLTE(t);break;case"IDAT":this.decodeIDAT(
t);break;case"IEND":this._end=!0;break;case"tRNS":this.decodetRNS(t);break;case"iCCP":this.decodeiCCP(t);break;case"tEXt":!function(t,e,n){const r=ua(e);t[r]=function(t,e){return la.decode(
t.readBytes(e))}(e,n-r.length-1)}(this._png.text,this,t);break;case"pHYs":this.decodepHYs();break;default:this.skip(t)}if(this.offset-n!==t)throw new Error(`Length mismatch while decoding
chunk ${e}`);this._checkCrc?Vi(this,t+4,e):this.skip(4)}decodeApngChunk(t,e){const n=this.offset;switch("fdAT"!==e&&"IDAT"!==e&&this._writingDataChunks&&this.pushDataToFrame(),e){case"acTL":
this.decodeACTL();break;case"fcTL":this.decodeFCTL();break;case"fdAT":this.decodeFDAT(t);break;default:this.decodeChunk(t,e),this.offset=n+t}if(this.offset-n!==t)throw new Error(`Length
mismatch while decoding chunk ${e}`);this._checkCrc?Vi(this,t+4,e):this.skip(4)}decodeIHDR(){const t=this._png;t.width=this.readUint32(),t.height=this.readUint32(),t.depth=function(t){if(
1!==t&&2!==t&&4!==t&&8!==t&&16!==t)throw new Error(`invalid bit depth: ${t}`);return t}(this.readUint8());let n;switch(this._colorType=e,e){case 0:case 3:n=1;break;
case 2:n=3;break;case 4:n=2;break;case 6:n=4;break;default:throw new Error(`Unknown color type: ${e}`)}if(this._png.channels=n,this._compressionMethod=this.readUint8(),0!==this.
_compressionMethod)throw new Error(`Unsupported compression method: ${this._compressionMethod}`);this._filterMethod=this.readUint8(),this._interlaceMethod=this.readUint8()}decodeACTL(){this.
_numberOfFrames=this.readUint32(),this._numberOfPlays=this.readUint32(),this._isAnimated=!0}decodeFCTL(){const t={sequenceNumber:this.readUint32(),width:this.readUint32(),height:this.
readUint32(),xOffset:this.readUint32(),yOffset:this.readUint32(),delayNumber:this.readUint16(),delayDenominator:this.readUint16(),disposeOp:this.readUint8(),blendOp:this.readUint8(),data:
new Uint8Array(0)};this._frames.push(t)}decodePLTE(t){if(t%3!=0)throw new RangeError(`PLTE field length must be a multiple of 3. Got ${t}`);const e=t/3;this._hasPalette=!0;const n=[];this.
_palette=n;for(let r=0;r<e;r++)n.push([this.readUint8(),this.readUint8(),this.readUint8()])}decodeIDAT(t){this._writingDataChunks=!0;const e=t,n=this.offset+this.byteOffset;if(this.
_inflator.push(new Uint8Array(this.buffer,n,e)),this._inflator.err)throw new Error(`Error while decompressing the data: ${this._inflator.err}`);this.skip(t)}decodeFDAT(t){this.
_writingDataChunks=!0;let e=t,n=this.offset+this.byteOffset;if(n+=4,e-=4,this._inflator.push(new Uint8Array(this.buffer,n,e)),this._inflator.err)throw new Error(`Error while decompressing
the data: ${this._inflator.err}`);this.skip(t)}decodetRNS(t){switch(this._colorType){case 0:case 2:if(t%2!=0)throw new RangeError(`tRNS chunk length must be a multiple of 2. Got ${t}`);if(t/
2>this._png.width*this._png.height)throw new Error(`tRNS chunk contains more alpha values than there are pixels (${t/2} vs ${this._png.width*this._png.height})`);this._hasTransparency=!0,
this._transparency=new Uint16Array(t/2);for(let e=0;e<t/2;e++)this._transparency[e]=this.readUint16();break;case 3:{if(t>this._palette.length)throw new Error(`tRNS chunk contains more alpha
values than there are palette colors (${t} vs ${this._palette.length})`);let e=0;for(;e<t;e++){const t=this.readByte();this._palette[e].push(t)}for(;e<this._palette.length;e++)this._palette[
e].push(255);break}default:throw new Error(`tRNS chunk is not supported for color type ${this._colorType}`)}}decodeiCCP(t){const e=ua(this),n=this.readUint8();if(0!==n)throw new Error(
`Unsupported iCCP compression method: ${n}`);const r=this.readBytes(t-e.length-2);this._png.iccEmbeddedProfile={name:e,profile:Ui(r)}}decodepHYs(){const t=this.readUint32(),e=this.
readUint32(),n=this.readByte();this._png.resolution={x:t,y:e,unit:n}}decodeApngImage(){this._apng.width=this._png.width,this._apng.height=this._png.height,this._apng.channels=this._png.
channels,this._apng.depth=this._png.depth,this._apng.numberOfFrames=this._numberOfFrames,this._apng.numberOfPlays=this._numberOfPlays,this._apng.text=this._png.text,this._apng.
resolution=this._png.resolution;for(let t=0;t<this._numberOfFrames;t++){const e={sequenceNumber:this._frames[t].sequenceNumber,delayNumber:this._frames[t].delayNumber,delayDenominator:this.
_frames[t].delayDenominator,data:8===this._apng.depth?new Uint8Array(this._apng.width*this._apng.height*this._apng.channels):new Uint16Array(this._apng.width*this._apng.height*this._apng.
channels)},n=this._frames.at(t);if(n){if(n.data=aa({data:n.data,width:n.width,height:n.height,channels:this._apng.channels,depth:this._apng.depth}),this._hasPalette&&(this._apng.
palette=this._palette),this._hasTransparency&&(this._apng.transparency=this._transparency),0===t||0===n.xOffset&&0===n.yOffset&&n.width===this._png.width&&n.height===this._png.height)e.
data=n.data;else{const r=this._apng.frames.at(t-1);this.disposeFrame(n,r,e),this.addFrameDataToCanvas(e,n)}this._apng.frames.push(e)}}return this.disposeFrame(t,e,n){switch(t.
disposeOp){case 0:break;case 1:for(let e=0;e<this._png.height;e++)for(let r=0;r<this._png.width;r++){const i=(e*t.width+r)*this._png.channels;for(let t=0;t<this._png.channels;t++)n.data[
i+t]=0}break;case 2:n.data.set(e.data);break;default:throw new Error("Unknown disposeOp")}}addFrameDataToCanvas(t,e){const n=1<<this._png.depth,r=(t,n)=>({index:((t+e.yOffset)*this._png.
width+e.xOffset+n)*this._png.channels,frameIndex:(t*e.width+n)*this._png.channels});switch(e.blendOp){case 0:for(let n=0;n<e.height;n++)for(let i=0;i<e.width;i++){const{index:a,frameIndex:
s}=r(n,i);for(let n=0;n<this._png.channels;n++)t.data[a+n]=e.data[s+n]}break;case 1:for(let i=0;i<e.height;i++)for(let a=0;a<e.width;a++){const{index:s,frameIndex:o}=r(i,a);for(let r=0;
r<this._png.channels;r++){const i=e.data[o+this._png.channels-1]/n,a=r%(this._png.channels-1)==0?1:e.data[o+r],h=Math.floor(i*a+(1-i)*t.data[s+r]);t.data[s+r]+=h}}break;default:throw new
Error("Unknown blendOp")}}decodeImage(){if(this._inflator.err)throw new Error(`Error while decompressing the data: ${this._inflator.err}`);const t=this._isAnimated?(this._frames?.at(0)).
data:this._inflator.result;if(0!==this._filterMethod)throw new Error(`Filter method ${this._filterMethod} not supported`);if(0===this._interlaceMethod)this._png.data=aa({data:t,width:this.
_png.width,height:this._png.height,channels:this._png.channels,depth:this._png.depth});else{if(1!==this._interlaceMethod)throw new Error(`Interlace method ${this._interlaceMethod} not
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supported`);this._png.data=function(t){const{data:e,width:n,height:r,channels:i,depth:a}=t,s=[{x:0,y:0,xStep:8,yStep:8},{x:4,y:0,xStep:8,yStep:8},{x:0,y:4,xStep:4,yStep:8},{x:2,y:0,xStep:4,
yStep:4},{x:0,y:2,xStep:2,yStep:4},{x:1,y:0,xStep:2,yStep:2},{x:0,y:1,xStep:1,yStep:2}],o=Math.ceil(a/8)*i,h=new Uint8Array(r*n*o);let l=0;for(let c=0;c<7;c++){const t=s[c],i=Math.ceil((n-t.
x)/t.xStep),a=Math.ceil((r-t.y)/t.yStep);if(i<=0||a<=0)continue;const u=i*o,f=new Uint8Array(u);for(let s=0;s<a;s++){const a=e[l++],c=e.subarray(l,l+u);l+=u;const d=new Uint8Array(u);$i(a,c,
d,f,u,o),f.set(d);for(let e=0;e<i;e++){const i=t.x+e*t.xStep,a=t.y+s*t.yStep;if(!(i>=n||a>=r))for(let t=0;t<o;t++)h[(a*n+i)*o+t]=d[e*o+t]}}}if(16===a){const t=new Uint16Array(h.buffer);if(
ta)for(let e=0;e<t.length;e++)t[e]=ea(t[e]);return t}return h}({data:t,width:this._png.width,height:this._png.height,channels:this._png.channels,depth:this._png.depth})}this._hasPalette&&(
this._png.palette=this._palette),this._hasTransparency&&(this._png.transparency=this._transparency)}pushDataToFrame(){const t=this._inflator.result,e=this._frames.at(-1);e?e.data=t:this.
_frames.push({sequenceNumber:0,width:this._png.width,height:this._png.height,xOffset:0,yOffset:0,delayNumber:0,delayDenominator:0,disposeOp:0,blendOp:0,data:t}),this._inflator=new zi,this.
_writingDataChunks=!1}}var da,pa,ga,ma,ba,va;function wa(t,e,n,r,i){var a=4,s=xa;switch(i){case R.API.image_compression.FAST:a=1,s=_a;break;case R.API.image_compression.MEDIUM:a=6,s=Aa;
break;case R.API.image_compression.SLOW:a=9,s=La}t=function(t,e,n,r){for(var i,a=t.length/e,s=new Uint8Array(t.length+a),o=[ya,_a,xa,Aa,La],h=0;h<a;h+=1){var l=h*e,c=t.subarray(l,l+e);if(
r)s.set(r(c,n,i),l+h);else{for(var u=o.length,f=[],d=0;d<u;d+=1)f[d]=o[d](c,n,i);var p=Sa(f.concat());s.set(f[p],l+h)}i=c}return s}(t,e,Math.ceil(n*r/8),s);var o=ye(t,{level:a});return R.
API.__addimage__.arrayBufferToBinaryString(o)}function ya(t){var e=Array.apply([],t);return e.unshift(0),e}function _a(t,e){var n=t.length,r=[];r[0]=1;for(var i=0;i<n;i+=1){var a=t[i-e]||0;
r[i+1]=t[i]-a+256&255}return r}function xa(t,e,n){var r=t.length,i=[];i[0]=2;for(var a=0;a<r;a+=1){var s=n&&n[a]||0;i[a+1]=t[a]-s+256&255}return i}function Aa(t,e,n){var r=t.length,i=[];i[
0]=3;for(var a=0;a<r;a+=1){var s=t[a-e]||0,o=n&&n[a]||0;i[a+1]=t[a]+256-(s+o>>>1)&255}return i}function La(t,e,n){var r=t.length,i=[];i[0]=4;for(var a=0;a<r;a+=1){var s=Na(t[a-e]||0,n&&n[
a]||0,n&&n[a-e]||0);i[a+1]=t[a]-s+256&255}return i}function Na(t,e,n){if(t===e&&e===n)return t;var r=Math.abs(e-n),i=Math.abs(t-n),a=Math.abs(t+e-n-n);return r<=i&&r<=a?t:i<=a?e:n}function
Sa(t){var e=t.map(function(t){return t.reduce(function(t,e){return t+Math.abs(e)},0)});return e.indexOf(Math.min.apply(null,e))}function ka(t,e,n){var r=e*n,i=Math.floor(r/8),a=16-(r-8*i+n),
s=(1<<n)-1;return Fa(t,i)>>a&s}function Pa(t,e,n,r){var i=n*r,a=Math.floor(i/8),s=16-(i-8*a+r),o=(1<<r)-1,h=(e&o)<<s;!function(t,e,n){if(e+1<t.byteLength)t.setUint16(e,n,!1);else{var
r=n>>8&255;t.setUint8(e,r)}}
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   *
   * https://github.com/deanm/omggif
   *
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   * IN THE SOFTWARE.
   *
   * omggif is a JavaScript implementation of a GIF 89a encoder and decoder,
   * including animation and compression.  It does not rely on any specific
   * underlying system, so should run in the browser, Node, or Plask.
   */(t,a,Fa(t,a)&~(o<<s)&65535|h)}function Fa(t,e){return e+1<t.byteLength?t.getUint16(e,!1):t.getUint8(e)<<8}function Ia(t){var e=0;if(71!==t[e++]||73!==t[e++]||70!==t[e++]||56!==t[
e++]||56!=(t[e++]+1&253)||97!==t[e++])throw new Error("Invalid GIF 87a/89a header.");var n=t[e++]|t[e++]<<8,r=t[e++]|t[e++]<<8,i=t[e++],a=i>>7,s=1<<1+(7&i);t[e++],t[e++];var o=null,h=null;
a&&(o=e,h=s,e+=3*s);var l=!0,c=[],u=0,f=null,d=0,p=null;for(this.width=n,this.height=r;l&&e<t.length;)switch(t[e++]){case 33:switch(t[e++]){case 255:if(11!==t[e]||78==t[e+1]&&69==t[
e+2]&&84==t[e+3]&&83==t[e+4]&&67==t[e+5]&&65==t[e+6]&&80==t[e+7]&&69==t[e+8]&&50==t[e+9]&&46==t[e+10]&&48==t[e+11]&&3==t[e+12]&&1==t[e+13]&&0==t[e+16])e+=14,p=t[e++]|t[e++]<<8,e+=else for(
e+=12;;){if(!((k=t[e++])>=0))throw Error("Invalid block size");if(0===k)break;e+=k}break;case 249:if(4!==t[e++]||0!==t[e+4])throw new Error("Invalid graphics extension block.");var g=t[e++];
u=t[e++]|t[e++]<<8,f=t[e++],1&g||(f=null),d=g>>2&7,e++;break;case 254:for(;;){if(!((k=t[e++])>=0))throw Error("Invalid block size");if(0===k)break;e+=k}break;default:throw new Error(
"Unknown graphic control label: 0x"+t[e-1].toString(16))}break;case 44:var m=t[e++]|t[e++]<<8,b=t[e++]|t[e++]<<8,v=t[e++]|t[e++]<<8,w=t[e++]|t[e++]<<8,y=t[e++],_=y>>6&1,x=1<<1+(7&y),A=o,L=h,
N=!1;y>>7&&(N=!0,A=e,L=x,e+=3*x);var S=e;for(e++;;){var k;if(!((k=t[e++])>=0))throw Error("Invalid block size");if(0===k)break;e+=k}c.push({x:m,y:b,width:v,height:w,has_local_palette:N,
palette_offset:A,palette_size:L,data_offset:S,data_length:e-S,transparent_index:f,interlaced:!!_,delay:u,disposal:d});break;case 59:l=!1;break;default:throw new Error("Unknown gif block:
0x"+t[e-1].toString(16))}this.numFrames=function(){return c.length},this.loopCount=function(){return p},this.frameInfo=function(t){if(t<0||t>=c.length)throw new Error("Frame index out of
range.");return c[t]},this.decodeAndBlitFrameBGRA=function(e,r){var i=this.frameInfo(e),a=i.width*i.height;if(a>536870912)throw new Error("Image dimensions exceed 512MB, which is too large.
");var s=new Uint8Array(a);Ca(t,i.data_offset,s,a);var o=i.palette_offset,h=i.transparent_index;null===h&&(h=256);var l=i.width,c=n-l,u=l,f=4*(i.y*n+i.x),d=4*((i.y+i.height)*n+i.x),p=f,
g=4*c;!0===i.interlaced&&(g+=4*n*7);for(var m=8,b=0,v=s.length;b<v;++b){var w=s[b];if(0===u&&(u=l,(p+=g)>=d&&(g=4*c+4*n*(m-1),p=f+(l+c)*(m<<1),m>>=1)),w===h)p+=4;else{var y=t[o+3*w],_=t[
o+3*w+1],x=t[o+3*w+2];r[p++]=x,r[p++]=_,r[p++]=y,r[p++]=255}--u}},this.decodeAndBlitFrameRGBA=function(e,r){var i=this.frameInfo(e),a=i.width*i.height;if(a>536870912)throw new Error("Image
dimensions exceed 512MB, which is too large.");var s=new Uint8Array(a);Ca(t,i.data_offset,s,a);var o=i.palette_offset,h=i.transparent_index;null===h&&(h=256);var l=i.width,c=n-l,u=l,f=4*(i.
y*n+i.x),d=4*((i.y+i.height)*n+i.x),p=f,g=4*c;!0===i.interlaced&&(g+=4*n*7);for(var m=8,b=0,v=s.length;b<v;++b){var w=s[b];if(0===u&&(u=l,(p+=g)>=d&&(g=4*c+4*n*(m-1),p=f+(l+c)*(m<<1),
m>>=1)),w===h)p+=4;else{var y=t[o+3*w],_=t[o+3*w+1],x=t[o+3*w+2];r[p++]=y,r[p++]=_,r[p++]=x,r[p++]=255}--u}}function Ca(t,e,n,r){for(var i=t[e++],a=1<<i,o=a+1,h=o+1,l=i+1,c=(1<<l)-1,u=0,
f=0,d=0,p=t[e++],g=new Int32Array(4096),m=null;;){for(;u<16&&0!==p;)f|=t[e++]<<u,u+=8,1===p?p=t[e++]:--p;if(u<l)break;var b=f&c;if(f>>=l,u-=l,b!==a){if(b===o)break;for(var v=b<h?b:m,w=0,y=v;
y>a;)y=g[y]>>8,++w;var _=y;if(d+w+(v!==b?1:0)>r)return void s.log("Warning, gif stream longer than expected.");n[d++]=_;var x=d+=w;for(v!==b&&(n[d++]=_),y=v;w--;)y=g[y],n[--x]=255&y,y>>=8;
null!==m&&h<4096&&(g[h++]=m<<8|_,h>=c+1&&l<12&&(++l,c=c<<1|1)),m=b}else h=o+1,c=(1<<(l=i+1))-1,m=null}return d!==r&&s.log("Warning, gif stream shorter than expected.");n}
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    NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS
    SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
  */function ja(t){var e,n,r,i,a,s=Math.floor,o=new Array(64),h=new Array(64),l=new Array(64),c=new Array(64),u=new Array(65535),f=new Array(65535),d=new Array(64),p=new Array(64),g=[],m=0,
b=7,v=new Array(64),w=new Array(64),y=new Array(64),_=new Array(256),x=new Array(2048),A=[0,1,5,6,14,15,27,28,2,4,7,13,16,26,29,42,3,8,12,17,25,30,41,43,9,11,18,24,31,40,44,53,10,19,23,32,
39,45,52,54,20,22,33,38,46,51,55,60,21,34,37,47,50,56,59,61,35,36,48,49,57,58,62,63],L=[0,0,1,5,1,1,1,1,1,1,0,0,0,0,0,0,0,0],N=[0,1,2,3,4,5,6,7,8,9,10,11],S=[0,0,2,1,3,3,2,4,3,5,5,4,4,0,0,1,
125],k=[1,2,3,0,4,17,5,18,33,49,65,6,19,81,97,7,34,113,20,50,129,145,161,8,35,66,177,193,21,82,209,240,36,51,98,114,130,9,10,22,23,24,25,26,37,38,39,40,41,42,52,53,54,55,56,57,58,67,68,69,
70,71,72,73,74,83,84,85,86,87,88,89,90,99,100,101,102,103,104,105,106,115,116,117,118,119,120,121,122,131,132,133,134,135,136,137,138,146,147,148,149,150,151,152,153,154,162,163,164,165,166,
167,168,169,170,178,179,180,181,182,183,184,185,186,194,195,196,197,198,199,200,201,202,210,211,212,213,214,215,216,217,218,225,226,227,228,229,230,231,232,233,234,241,242,243,244,245,246,
247,248,249,250],P=[0,0,3,1,1,1,1,1,1,1,1,1,0,0,0,0,0,0],F=[0,1,2,3,4,5,6,7,8,9,10,11],I=[0,0,2,1,2,4,4,3,4,7,5,4,4,0,1,2,119],C=[0,1,2,3,17,4,5,33,49,6,18,65,81,7,97,113,19,34,50,129,8,20,66,
145,161,177,193,9,35,51,82,240,21,98,114,209,10,22,36,52,225,37,241,23,24,25,26,38,39,40,41,42,53,54,55,56,57,58,67,68,69,70,71,72,73,74,83,84,85,86,87,88,89,90,99,100,101,102,103,104,105,
106,115,116,117,118,119,120,121,122,130,131,132,133,134,135,136,137,138,146,147,148,149,150,151,152,153,154,162,163,164,165,166,167,168,169,170,178,179,180,181,182,183,184,185,186,194,195,
196,197,198,199,200,201,202,210,211,212,213,214,215,216,217,218,226,227,228,229,230,231,232,233,234,242,243,244,245,246,247,248,249,250];function j(t,e){for(var n=0,r=0,i=new Array,a=1;
a<=16;a++){for(var s=1;s<=t[a];s++)i[e[r]]=[],i[e[r]][0]=n,i[e[r]][1]=a,r++,n++;n*=2}return i}function E(t){for(var e=t[0],n=t[1]-1;n>=0;)e&1<<n&&(m|=1<<b),n--,--b<0&&(255==m?(O(255),O(0)):
O(m),b=7,m=0)}function O(t){g.push(t)}function B(t){O(t>>8&255),O(255&t)}function M(t,e,n,r,i){for(var a,s=i[0],o=i[240],h=function(t,e){var n,r,i,a,s,o,h,l,c,u,f=0;for(c=0;c<8;++c){n=t[f],
r=t[f+1],i=t[f+2],a=t[f+3],s=t[f+4],o=t[f+5],h=t[f+6];var p=n+(l=t[f+7]),g=n-l,m=r+h,b=r-h,v=i+o,w=i-o,y=a+s,_=a-s,x=p+y,A=p-y,L=m+v,N=m-v;t[f]=x+L,t[f+4]=x-L;var S=.707106781*(N+A);t[
f+2]=A+S,t[f+6]=A-S;var k=.382683433*((x=_+w)-(N=b+g)),P=.5411961*x+k,F=1.306562965*N+k,I=.707106781*(L=w+b),C=g+I,j=g-I;t[f+5]=j+P,t[f+3]=j-P,t[f+1]=C+F,t[f+7]=C-F,f+=8}for(f=0,c=0;c<8;
++c){n=t[f],r=t[f+8],i=t[f+16],a=t[f+24],s=t[f+32],o=t[f+40],h=t[f+48];var E=n+(l=t[f+56]),O=n-l,B=r+h,M=r-h,R=i+o,T=i-o,D=a+s,q=a-s,z=E+D,U=E-D,H=B+R,W=B-R;t[f]=z+H,t[f+32]=z-H;var V=.
707106781*(W+U);t[f+16]=U+V,t[f+48]=U-V;var G=.382683433*((z=q+T)-(W=M+O)),Y=.5411961*z+G,Z=1.306562965*W+G,J=.707106781*(H=T+M),X=O+J,K=O-J;t[f+40]=K+Y,t[f+24]=K-Y,t[f+8]=X+Z,t[f+56]=X-Z,
f++}for(c=0;c<64;++c)u=t[c]*e[c],d[c]=u>0?u+.5|0:u-.5|0;return d}(t,e),l=0;l<64;++l)p[A[l]]=h[l];var c=p[0]-n;n=p[0],0==c?E(r[0]):(E(r[f[a=32767+c]]),E(u[a]));for(var g=63;g>0&&0==p[g];)g--;
if(0==g)return E(s),n;for(var m,b=1;b<=g;){for(var v=b;0==p[b]&&b<=g;)++b;var w=b-v;if(w>=16){for(var y=1;y<=m;++y)E(o);w&=15}a=32767+p[b],E(i[(w<<4)+f[a]]),E(u[a]),b++}return
63!=g&&E(s),n}function R(t){t=Math.min(Math.max(t,1),100),a!=t&&(function(t){for(var e=[16,11,10,16,24,40,51,61,12,12,14,19,26,58,60,55,14,13,16,24,40,57,69,56,14,17,22,29,51,87,80,62,18,22,
37,56,68,109,103,77,24,35,55,64,81,104,113,92,49,64,78,87,103,121,120,101,72,92,95,98,112,100,103,99],n=0;n<64;n++){var r=s((e[n]*t+50)/100);r=Math.min(Math.max(r,1),255),o[A[n]]=r}for(var
i=[17,18,24,47,99,99,99,99,18,21,26,66,99,99,99,99,24,26,56,99,99,99,99,99,47,66,99,99,99,99,99,99,99,99,99,99,99,99,99,99,99,99,99,99,99,99,99,99,99,99,99,99,99,99,99,99,99,99,99,99,99,99,
99,99],a=0;a<64;a++){var u=s((i[a]*t+50)/100);u=Math.min(Math.max(u,1),255),h[A[a]]=u}for(var f=[1,1.387039845,1.306562965,1.175875602,1,.785694958,.5411961,.275899379],d=0,p=0;p<8;p++)for(
var g=0;g<8;g++)l[d]=1/(o[A[d]]*f[p]*f[g]*8),c[d]=1/(h[A[d]]*f[p]*f[g]*8),d++}(t<50?Math.floor(5e3/t):Math.floor(200-2*t)),a=t)}this.encode=function(t,a){a&&R(a),g=new Array,m=0,b=7,B(
65496),B(65504),B(16),O(74),O(70),O(73),O(70),O(0),O(1),O(1),O(0),B(1),B(1),O(0),O(0),function(){B(65499),B(132),O(0);for(var t=0;t<64;t++)O(o[t]);O(1);for(var e=0;e<64;e++)O(h[e])}(),
function(t,e){B(65472),B(17),O(8),B(e),B(t),O(3),O(1),O(17),O(0),O(2),O(17),O(1),O(3),O(17),O(1)}(t.width,t.height),function(){B(65476),B(418),O(0);for(var t=0;t<16;t++)O(L[t+1]);for(var
e=0;e<=11;e++)O(N[e]);O(16);for(var n=0;n<16;n++)O(S[n+1]);for(var r=0;r<=161;r++)O(k[r]);O(1);for(var i=0;i<16;i++)O(P[i+1]);for(var a=0;a<=11;a++)O(F[a]);O(17);for(var s=0;s<16;s++)O(I[
s+1]);for(var o=0;o<=161;o++)O(C[o])}(),B(65498),B(12),O(3),O(1),O(0),O(2),O(17),O(3),O(17),O(0),O(63),O(0);var s=0,u=0,f=0;m=0,b=7,this.encode.displayName="_encode_";for(var d,p,_,A,j,T,D,
q,z,U=t.data,H=t.width,W=t.height,V=4*H,G=0;G<W;){for(d=0;d<V;){for(j=V*G+d,D=-1,q=0,z=0;z<64;z++)T=j+(q=z>>3)*V+(D=4*(7&z)),G+q>=W&&(T-=V*(G+1+q-W)),d+D>=V&&(T-=d+D-V+4),p=U[T++],_=U[T++],
A=U[T++],v[z]=(x[p]+x[_+256|0]+x[A+512|0]>>16)-128,w[z]=(x[p+768|0]+x[_+1024|0]+x[A+1280|0]>>16)-128,y[z]=(x[p+1280|0]+x[_+1536|0]+x[A+1792|0]>>16)-128;s=M(v,l,s,e,r),u=M(w,c,u,n,i),f=M(y,c,
f,n,i),d+=32}G+=8}if(b>=0){var Y=[];Y[1]=b+1,Y[0]=(1<<b+1)-1,E(Y)}return B(65497),new Uint8Array(g)},t=t||50,function(){for(var t=String.fromCharCode,e=0;e<256;e++)_[e]=t(e)}(),e=j(L,N),n=j(
P,F),r=j(S,k),i=j(I,C),function(){for(var t=1,e=2,n=1;n<=15;n++){for(var r=t;r<e;r++)f[32767+r]=n,u[32767+r]=[],u[32767+r][1]=n,u[32767+r][0]=-(e-1);i<=-t;i++)f[32767+i]=n,u[
32767+i]=[],u[32767+i][1]=n,u[32767+i][0]=e-1+i;t<<=1,e<<=1}}(),function(){for(var t=0;t<256;t++)x[t]=19595*t,x[t+256|0]=38470*t,x[t+512|0]=7471*t+32768,x[t+768|0]=-11059*t,x[t+1024|0]=-
21709*t,x[t+1280|0]=32768*t+8421375,x[t+1536|0]=-27439*t,x[t+1792|0]=-5329*t}(),R(t)}
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   */function Ea(t,e){if(this.pos=0,this.buffer=t,this.datav=new DataView(t.buffer),this.is_with_alpha=!!e,this.bottom_up=!0,this.flag=String.fromCharCode(this.buffer[0])+String.
fromCharCode(this.buffer[1]),this.pos+=2,-1===["BM","BA","CI","CP","IC","PT"].indexOf(this.flag))throw new Error("Invalid BMP File");this.parseHeader(),this.parseBGR()}function Oa(t){
function e(t){if(!t)throw Error("assert :P")}function n(t,e,n){for(var r=0;4>r;r++)if(t[e+r]!=n.charCodeAt(r))return!0;return!1}function r(t,e,n,r,i){for(var a=0;a<i;a++)t[e+a]=n[
r+a]}function i(t,e,n,r){for(var i=0;i<r;i++)t[e+i]=n}function a(t){return new Int32Array(t)}function s(t,e){for(var n=[],r=0;r<t;r++)n.push(new e);return n}function o(t,e){var n=[];return
function t(n,r,i){for(var a=i[r],s=0;s<a&&(n.push(i.length>r+1?[]:new e),!(i.length<r+1));s++)t(n[s],r+1,i)}(n,0,t),n}var h=function(){var t=this;function h(t,e){for(var n=1<<e-1>>>0;t&n;
)n>>>=1;return n?(t&n-1)+n:t}function l(t,n,r,i,a){e(!(i%r));do{t[n+(i-=r)]=a}while(0<i)}function c(t,n,r,i,s){if(e(2328>=s),512>=s)var o=a(512);else if(null==(o=a(s)))return 0;return
function(t,n,r,i,s,o){var c,f,d=n,p=1<<r,g=a(16),m=a(16);for(e(0!=s),e(null!=i),e(null!=t),e(0<r),f=0;f<s;++f){if(15<i[f])return 0;++g[i[f]]}if(g[0]==s)return 0;for(m[1]=0,c=1;15>c;++c){if(
g[c]>1<<c)return 0;m[c+1]=m[c]+g[c]}for(f=0;f<s;++f)c=i[f],0<i[f]&&(o[m[c]++]=f);if(1==m[15])return(i=new u).g=0,i.value=o[0],l(t,d,1,p,i),p;var b,v=-1,w=p-1,y=0,_=1,x=1,A=1<<r;for(f=0,c=1,
s=2;c<=r;++c,s<<=1){if(_+=x<<=1,0>(x-=g[c]))return 0;for(;0<g[c];--g[c])(i=new u).g=c,i.value=o[f++],l(t,d+y,s,A,i),y=h(y,c)}for(c=r+1,s=2;15>=c;++c,s<<=1){if(_+=x<<=1,0>(x-=g[c]))return 0;
for(;0<g[c];--g[c]){if(i=new u,(y&w)!=v){for(d+=A,b=1<<(v=c)-r;15>v&&!(0>=(b-=g[v]));)++v,b<<=1;p+=A=1<<(b=v-r),t[n+(v=y&w)].g=b+r,t[n+v].value=d-n-v}i.g=c-r,i.value=o[f++],l(t,d+(y>>r),s,A,
i),y=h(y,c)}}return _!=2*m[15]-1?0:p}(t,n,r,i,s,o)}function u(){this.value=this.g=0}function f(){this.value=this.g=0}function d(){this.G=s(5,u),this.H=a(5),this.jc=this.Qb=this.nd=0,
this.pd=s(Tn,f)}function p(t,n,r,i){e(null!=t),e(null!=n),e(2147483648>i),t.Ca=254,t.I=0,t.b=-8,t.Ka=0,t.oa=n,t.pa=r,t.Jd=n,t.Yc=r+i,t.Zc=4<=i?r+i-4+1:r,S(t)}function g(t,e){for(var n=0;0<e-
-;)n|=P(t,128)<<e;return n}function m(t,e){var n=g(t,e);return k(t)?-n:n}function b(t,n,r,i){var a,s=0;for(e(null!=t),e(null!=n),e(4294967288>i),t.Sb=i,t.Ra=0,t.u=0,t.h=0,4<i&&(i=4),a=0;a<i;
++a)s+=n[r+a]<<8*a;t.Ra=s,t.bb=i,t.oa=n,t.pa=r}function v(t){for(;8<=t.u&&t.bb<t.Sb;)t.Ra>>>=8,t.Ra+=t.oa[t.pa+t.bb]<<zn-8>>>0,++t.bb,t.u-=8;A(t)&&(t.h=1,t.u=0)}function w(t,n){if(e(0<=n),
!t.h&&n<=qn){var r=x(t)&Dn[n];return t.u+=n,v(t),r}return t.h=1,t.u=0}function y(){this.b=this.Ca=this.I=0,this.oa=[],this.pa=0,this.Jd=[],this.Yc=0,this.Zc=[],this.Ka=0}function _(){this.
Ra=0,this.oa=[],this.h=this.u=this.bb=this.Sb=this.pa=0}function x(t){return t.Ra>>>(t.u&zn-1)>>>0}function A(t){return e(t.bb<=t.Sb),t.h||t.bb==t.Sb&&t.u>zn}function L(t,e){t.u=e,t.h=A(
t)}function N(t){t.u>=Un&&(e(t.u>=Un),v(t))}function S(t){e(null!=t&&null!=t.oa),t.pa<t.Zc?(t.I=(t.oa[t.pa++]|t.I<<8)>>>0,t.b+=8):(e(null!=t&&null!=t.oa),t.pa<t.Yc?(t.b+=8,t.I=t.oa[t.
pa++]|t.I<<8):t.Ka?t.b=0:(t.I<<=8,t.b+=8,t.Ka=1))}function k(t){return g(t,1)}function P(t,e){var n=t.Ca;0>t.b&&S(t);var r=t.b,i=n*e>>>8,a=(t.I>>>r>i)+0;for(a?(n-=i,t.I-=i+1<<r>>>0):n=i+1,
r=n,i=0;256<=r;)i+=8,r>>=8;return r=7^i+Hn[r],t.b-=r,t.Ca=(n<<r)-1,a}function F(t,e,n){t[e+0]=n>>>24&255,t[e+1]=n>>>16&255,t[e+2]=n>>>8&255,t[e+3]=255&n}function I(t,e){return t[e+0]|t[
e+1]<<8}function C(t,e){return I(t,e)|t[e+2]<<16}function j(t,e){return I(t,e)|I(t,e+2)<<16}function E(t,n){var r=1<<n;return e(null!=t),e(0<=n),t.X=a(r),null==t.X?0:(t.Mb=32-n,t.Xa=n,
1)}function O(t,n){e(null!=t),e(null!=n),e(t.Xa==n.Xa),r(n.X,0,t.X,0,1<<n.Xa)}function B(){this.X=[],this.Xa=this.Mb=0}function M(t,n,r,i){e(null!=r),e(null!=i);var a=r[0],s=i[0];return
0==a&&(a=(t*s+n/2)/n),0==s&&(s=(n*a+t/2)/t),0>=a||0>=s?0:(r[0]=a,i[0]=s,1)}function R(t,e){return t+(1<<e)-1>>>e}function T(t,e){return((4278255360&e)+(4278255360&e)>>>0&4278255360)+((
16711935&t)+(16711935&e)>>>0&16711935)>>>0}function D(e,n){t[n]=function(n,r,i,a,s,o,h){var l;for(l=0;l<s;++l){var c=t[e](o[h+l-1],i,a+l);o[h+l]=T(n[r+l],c)}}}function q(){this.ud=this.
hd=this.jd=0}function z(t,e){return((4278124286&(t^e))>>>1)+(t&e)>>>0}function U(t){return 0<=t&&256>t?t:0>t?0:255<t?255:void 0}function H(t,e){return U(t+(t-e+.5>>1))}function W(t,e,n){
return Math.abs(e-n)-Math.abs(t-n)}function V(t,e,n,r,i,a,s){for(r=a[s-1],n=0;n<i;++n)a[s+n]=r=T(t[e+n],r)}function G(t,e,n,r,i){var a;for(a=0;a<n;++a){var s=t[e+a],o=s>>8&255,h=16711935&(
h=(h=16711935&s)+((o<<16)+o));r[i+a]=(4278255360&s)+h>>>0}}function Y(t,e){e.jd=255&t,e.hd=t>>8&255,e.ud=t>>16&255}function Z(t,e,n,r,i,a){var s;for(s=0;s<r;++s){var o=e[n+s],h=o>>>8,l=o,
c=255&(c=(c=o>>>16)+((t.jd<<24>>24)*(h<<24>>24)>>>5));l=255&(l=(l+=(t.hd<<24>>24)*(h<<24>>24)>>>5)+((t.ud<<24>>24)*(c<<24>>24)>>>5)),i[a+s]=(4278255360&o)+(c<<16)+l}}function J(e,n,r,i,a){t[
n]=function(t,e,n,r,s,o,h,l,c){for(r=h;r<l;++r)for(h=0;h<c;++h)s[o++]=a(n[i(t[e++])])},t[e]=function(e,n,s,o,h,l,c){var u=8>>e.b,f=e.Ea,d=e.K[0],p=e.w;if(8>u)for(e=(1<<e.b)-1,p=(1<<u)-1;n<s;
++n){var g,m=0;for(g=0;g<f;++g)g&e||(m=i(o[h++])),l[c++]=a(d[m&p]),m>>=u}else t["VP8LMapColor"+r](o,h,d,p,l,c,n,s,f)}}function X(t,e,n,r,i){for(n=e+n;e<n;){var a=t[e++];r[i++]=a>>16&255,r[
i++]=a>>8&255,r[i++]=255&a}}function K(t,e,n,r,i){for(n=e+n;e<n;){var a=t[e++];r[i++]=a>>16&255,r[i++]=a>>8&255,r[i++]=255&a,s=t[e++];r[i++]=a,r[i++]=s}}function Q(t,e,n,r,i){for(n=e+n;e<n;){var a=(s=t[e++])>>16&248|s>>13&7,s=s>>5&224|s>>3&31;r[i++]=a,r[i++]=s}}function tt(t,e,n,
r,i){for(n=e+n;e<n;){var a=t[e++];r[i++]=255&a,r[i++]=a>>8&255,r[i++]=a>>16&255}}function et(t,e,n,i,a,s){if(0==s)for(n=e+n;e<n;)F(i,((s=t[e++])[0]>>24|s[1]>>8&65280|s[2]<<8&16711680|s[
3]<<24)>>>0),a+=32;else r(i,a,t,e,n)}function nt(e,n){t[n][0]=t[e+"0"],t[n][1]=t[e+"1"],t[n][2]=t[e+"2"],t[n][3]=t[e+"3"],t[n][4]=t[e+"4"],t[n][5]=t[e+"5"],t[n][6]=t[e+"6"],t[n][7]=t[e+"7"],
t[n][8]=t[e+"8"],t[n][9]=t[e+"9"],t[n][10]=t[e+"10"],t[n][11]=t[e+"11"],t[n][12]=t[e+"12"],t[n][13]=t[e+"13"],t[n][14]=t[e+"0"],t[n][15]=t[e+"0"]}function rt(t){return
t==Ur||t===Hr||t===Wr||t==Vr}function it(){this.eb=[],this.size=this.A=this.fb=0}function at(){this.y=[],this.f=[],this.ea=[],this.F=[],this.Tc=this.Ed=this.Cd=this.Fd=this.lb=this.Db=this.
Ab=this.fa=this.J=this.W=this.N=this.O=0}function st(){this.Rd=this.height=this.width=this.S=0,this.f={},this.f.RGBA=new it,this.f.kb=new at,this.sd=null}function ot(){this.width=[0],this.
height=[0],this.Pd=[0],this.Qd=[0],this.format=[0]}function ht(){this.Id=this.fd=this.Md=this.hb=this.ib=this.da=this.bd=this.cd=this.j=this.v=this.Da=this.Sd=this.ob=0}function lt(t){
return alert("todo:WebPSamplerProcessPlane"),t.T}function ct(t,e){var n=t.T,i=e.ba.f.RGBA,a=i.eb,s=i.fb+t.ka*i.A,o=mi[e.ba.S],h=t.y,l=t.O,c=t.f,u=t.N,f=t.ea,d=t.W,p=e.cc,g=e.dc,m=e.Mc,b=e.
Nc,v=t.ka,w=t.ka+t.T,y=t.U,_=y+1>>1;for(0==v?o(h,l,null,null,c,u,f,d,a,s,null,null,y):(o(e.ec,e.fc,h,l,p,g,m,b,c,u,f,d,a,s-i.A,a,s,y),++n);v+2<w;v+=2)p=c,g=u,m=f,b=d,u+=t.Rc,d+=t.Rc,
s+=2*i.A,o(h,(l+=2*t.fa)-t.fa,h,l,p,g,m,b,c,u,f,d,a,s-i.A,a,s,y);return l+=t.fa,t.j+w<t.o?(r(e.ec,e.fc,h,l,y),r(e.cc,e.dc,c,u,_),r(e.Mc,e.Nc,f,d,_),n--):1&w||o(h,l,null,null,c,u,f,d,c,u,f,d,
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a,s+i.A,null,null,y),n}function ut(t,n,r){var i=t.F,a=[t.J];if(null!=i){var s=t.U,o=n.ba.S,h=o==Dr||o==Wr;n=n.ba.f.RGBA;var l=[0],c=t.ka;l[0]=t.T,t.Kb&&(0==c?--l[0]:(--c,a[0]-=t.width),t.
j+t.ka+t.T==t.o&&(l[0]=t.o-t.j-c));var u=n.eb;c=n.fb+c*n.A,t=Lr(i,a[0],t.width,s,l,u,c+(h?0:3),n.A),e(r==l),t&&rt(o)&&xr(u,c,h,s,l,n.A)}return 0}function ft(t){var e=t.ma,n=e.ba.S,r=11>n,
i=n==Mr||n==Tr||n==Dr||n==qr||12==n||rt(n);if(e.memory=null,e.Ib=null,e.Jb=null,e.Nd=null,!Bn(e.Oa,t,i?11:12))return 0;if(i&&rt(n)&&vn(),t.da)alert("todo:use_scaling");else{if(r){if(e.Ib=lt,
t.Kb){if(n=t.U+1>>1,e.memory=a(t.U+2*n),null==e.memory)return 0;e.ec=e.memory,e.fc=0,e.cc=e.ec,e.dc=e.fc+t.U,e.Mc=e.cc,e.Nc=e.dc+n,e.Ib=ct,vn()}}else alert("todo:EmitYUV");i&&(e.Jb=ut,r&&mn(
))}if(r&&!Ii){for(t=0;256>t;++t)Ci[t]=89858*(t-128)+Ni>>Li,Oi[t]=-22014*(t-128)+Ni,Ei[t]=-45773*(t-128),ji[t]=113618*(t-128)+Ni>>Li;for(t=Si;t<ki;++t)e=76283*(t-16)+Ni>>Li,Bi[t-Si]=Vt(e,
255),Mi[t-Si]=Vt(e+8>>4,15);Ii=1}return 1}function dt(t){var n=t.ma,r=t.U,i=t.T;return e(!(1&t.ka)),0>=r||0>=i?0:(r=n.Ib(t,n),null!=n.Jb&&n.Jb(t,n,r),n.Dc+=r,1)}function pt(t){t.ma.
memory=null}function gt(t,e,n,r){return 47!=w(t,8)?0:(e[0]=w(t,14)+1,n[0]=w(t,14)+1,r[0]=w(t,1),0!=w(t,3)?0:!t.h)}function mt(t,e){if(4>t)return t+1;var n=t-2>>1;return(2+(1&t)<<n)+w(e,
n)+1}function bt(t,e){return 120<e?e-120:1<=(n=((n=Kr[e-1])>>4)*t+(8-(15&n)))?n:1;var n}function vt(t,e,n){var r=x(n),i=t[e+=255&r].g-8;return 0<i&&(L(n,n.u+8),r=x(n),e+=t[e].value,e+=r&(
1<<i)-1),L(n,n.u+t[e].g),t[e].value}function wt(t,n,r){return r.g+=t.g,r.value+=t.value<<n>>>0,e(8>=r.g),t.g}function yt(t,n,r){var i=t.xc;return e((n=0==i?0:t.vc[t.md*(r>>i)+(n>>i)])<t.Wb),
t.Ya[n]}function _t(t,n,i,a){var s=t.ab,o=t.c*n,h=t.C;n=h+n;var l=i,c=a;for(a=t.Ta,i=t.Ua;0<s--;){var u=t.gc[s],f=h,d=n,p=l,g=c,m=(c=a,l=i,u.Ea);switch(e(f<d),e(d<=u.nc),u.hc){case 2:Gn(p,g,
(d-f)*m,c,l);break;case 0:var b=f,v=d,w=c,y=l,_=(S=u).Ea;0==b&&(Wn(p,g,null,null,1,w,y),V(p,g+1,0,0,_-1,w,y+1),g+=_,y+=_,++b);for(var x=1<<S.b,A=x-1,L=R(_,S.b),N=S.K,S=S.w+(S.b>>S.b)*k;b<v;){
var k=N,P=S,F=1;for(Vn(p,g,w,y-_,1,w,y);F<_;){var I=(F&~A)+x;I>_&&(I=_),(0,Kn[k[P++]>>8&15])(p,g+ +F,w,y+F-_,I-F,w,y+F),F=I}g+=_,y+=_,++b&A||(S+=L)}d!=u.nc&&r(c,l-m,c,l+(d-f-1)*m,m);break;
case 1:for(m=p,v=g,_=(p=u.Ea)-(y=p&~(w=(g=1<<u.b)-1)),b=R(p,u.b),x=u.K,u=u.w+(f>>u.b)*b;f<d;){for(A=x,L=u,N=new q,S=v+y,k=v+p;v<S;)Y(A[L++],N),$n(N,m,v,g,c,l),v+=_,l+=_},++f&w||(u+=b)}break;case 3:if(p==c&&g==l&&0<u.b){for(v=c,p=m=l+(d-f)*m-(y=(d-f)*R(u.Ea,u.b)),g=c,w=l,b=[],y=(_=y)-1;0<=y;--y)b[y]=g[w+y];for(y=_-1;0<=y;--y)v[p+y]=b[
y];Yn(u,f,d,c,m,c,l)}else Yn(u,f,d,p,g,c,l)}l=a,c=i}c!=i&&r(a,i,l,c,o)}function xt(t,n){var r=t.V,i=t.Ba+t.c*t.C,a=n-t.C;if(e(n<=t.l.o),e(16>=a),0<a){var s=t.l,o=t.Ta,h=t.Ua,l=s.width;if(_t(
t,a,r,i),a=h[h],e((r=t.C)<(i=n)),e(s.v<s.va),i>s.o&&(i=s.o),r<s.j){var c=s.j-r;r=s.j,a[0]+=c*l}if(r>=i?r=0:(a[0]+=4*s.v,s.ka=r-s.j,s.U=s.va-s.v,s.T=i-r,r=1),r){if(h=h[0],11>(r=t.ca).S){var
u=r.f.RGBA,f=(i=r.S,a=s.U,s=s.T,c=u.eb,u.A),d=s;for(u=u.fb+t.Ma*u.A;0<d--;){var p=o,g=h,m=a,b=c,v=u;switch(i){case Br:Qn(p,g,m,b,v);break;case Mr:tr(p,g,m,b,v);break;case Ur:tr(p,g,m,b,v),
xr(b,v,0,m,1,0);break;case Rr:rr(p,g,m,b,v);break;case Tr:et(p,g,m,b,v,1);break;case Hr:et(p,g,m,b,v,1),xr(b,v,0,m,1,0);break;case Dr:et(p,g,m,b,v,0);break;case Wr:et(p,g,m,b,v,0),xr(b,v,1,
m,1,0);break;case qr:er(p,g,m,b,v);break;case Vr:er(p,g,m,b,v),Ar(b,v,m,1,0);break;case zr:nr(p,g,m,b,v);break;default:e(0)}h+=l,u+=f}t.Ma+=s}else alert("todo:EmitRescaledRowsYUVA");e(t.
Ma<=r.height)}}t.C=n,e(t.C<=t.i)}function At(t){var e;if(0<t.ua)return 0;for(e=0;e<t.Wb;++e){var n=t.Ya[e].G,r=t.Ya[e].H;if(0<n[1][r[1]+0].g||0<n[3][r[3]+0].g)return
0}return 1}function Lt(t,n,r,i,a,s){if(0!=t.Z){var o=t.qd,h=t.rd;for(e(null!=gi[t.Z]);n<r;++n)gi[t.Z](o,h,i,a,i,a,s),o=i,h=a,a+=s;t.qd=o,t.rd=h}}function Nt(t,n){var r=t.l.ma,i=0==r.Z||1==r.
Z?t.l.j:t.C;if(i=t.C<i?i:t.C,e(n<=t.l.o),n>i){var a=t.l.width,s=r.ca,o=r.tb+a*i,h=t.V,l=t.Ba+t.c*i,c=t.gc;e(1==t.ab),e(3==c[0].hc),Jn(c[0],i,n,h,l,s,o),Lt(r,i,n,s,o,a)}t.C=t.Ma=n}function
St(t,n,r,i,a,s,o){var h=t.$/i,l=t.$%i,c=t.m,u=t.s,f=r+t.$,d=f;a=r+i*a;var p=r+i*s,g=280+u.ua,m=t.Pb?h:16777216,b=0<u.ua?u.Wa:null,v=u.wc,w=f<p?yt(u,l,h):null;e(t.C<s),e(p<=a);var y=!1;t:for(
;;){for(;y||f<p;){var _=0;if(h>=m){var S=f-r;e((m=t).Pb),m.wd=m.m,m.xd=S,0<m.s.ua&&O(m.s.Wa,m.s.vb),m=h+Qr}if(l&v||(w=yt(u,l,h)),e(null!=w),w.Qb&&(n[f]=w.qb,y=!0),!y)if(N(c),w.jc){_=c,S=n;
var k=f,P=w.pd[x(_)&Tn-1];e(w.jc),256>P.g?(L(_,_.u+P.g),S[k]=P.value,_=0):(L(_,_.u+P.g-256),e(256<=P.value),_=P.value),0==_&&(y=!0)}else _=vt(w.G[0],w.H[0],c);if(c.h)break;if(y||256>_){if(
!y)if(w.nd)n[f]=(w.qb|_<<8)>>>0;else{if(N(c),y=vt(w.G[1],w.H[1],c),N(c),S=vt(w.G[2],w.H[2],c),k=vt(w.G[3],w.H[3],c),c.h)break;n[f]=(k<<24|y<<16|_<<8|S)>>>0}if(y=!1,++f,++l>=i&&(l=0,++h,
null!=o&&h<=s&&!(h%16)&&o(t,h),null!=b))for(;d<f;)_=n[d++],b.X[(506832829*_&4294967295)>>>b.Mb]=_}else if(280>_){if(_=mt(_-256,c),N(c),S=bt(i,S=mt(S,c)),c.h)break;if(y||256>_){if(
r<S||a-f<_)break t;for(k=0;k<_;++k)n[f+k]=n[f+k-S];for(f+=_,l+=_;l>=i;)l-=i,++h,null!=o&&h<=s&&!(h%16)&&o(t,h);if(e(f<=a),l&v&&(w=yt(u,l,h)),null!=b)for(;d<f;)_=n[d++],b.X[(
506832829*_&4294967295)>>>b.Mb]=_}else{if(!(_<g))break t;for(y=_-280,e(null!=b);d<f;)_=n[d++],b.X[(506832829*_&4294967295)>>>b.Mb]=_;_=f,e(!(y>>>(S=b).Xa)),n[_]=S.X[y],y=!0}y||e(c.h==A(
c))}if(t.Pb&&c.h&&f<a)e(t.m.h),t.a=5,t.m=t.wd,t.$=t.xd,0<t.s.ua&&O(t.s.vb,t.s.Wa);else{if(c.h)break t;null!=o&&o(t,h>s?s:h),t.a=0,t.$=f-r}return 1}return t.a=3,0}function kt(t){e(null!=t),t.
vc=null,t.yc=null,t.Ya=null;var n=t.Wa;null!=n&&(n.X=null),t.vb=null,e(null!=t)}function Pt(){{var e=new sn;return null==e?null:(e.a=0,e.xb=pi,nt("Predictor","VP8LPredictors"),nt("Predictor",
"VP8LPredictors_C"),nt("PredictorAdd","VP8LPredictorsAdd"),nt("PredictorAdd","VP8LPredictorsAdd_C"),Gn=G,$n=Z,Qn=X,tr=K,er=$,nr=Q,rr=tt,t.VP8LMapColor32b=Zn,t.VP8LMapColor8b=Xn,e)}function
Ft(t,n,r,o,h){var l=1,f=[t],p=[n],g=o.m,m=o.s,b=null,v=0;t:for(;;){if(r)for(;l&&w(g,1);){var y=f,_=p,A=o,S=1,k=A.m,P=A.gc[A.ab],F=w(k,2);if(A.Oc&1<<F)l=0;else{switch(A.Oc|=1<<F,P.hc=F,P.
Ea=y[0],P.nc=_[0],P.K=[null],++A.ab,e(4>=A.ab),F){case 0:case 1:P.b=w(k,3)+2,S=Ft(R(P.Ea,P.b),R(P.nc,P.b),0,A,P.K),P.K=P.K[0];break;case 3:var I,C=w(k,8)+1,j=16<C?0:4<C?1:2<C?2:3;if(y[0]=R(
P.Ea,j),P.b=j,I=S=Ft(C,1,0,A,P.K))}var O,B=C,M=P,D=1<<(8>>M.b),q=a(D);if(null==q)I=0;else{var z=M.K[0],U=M.w;for(q[0]=M.K[0][0],O=1;O<1*B;++O)q[O]=T(z[U+O],q[O-1]);for(;O<4*D;++O)q[O]=0;M.K[
0]=null,M.K[0]=q,I=1}}S=I;break;case 2:break;default:e(0)}l=S}}if(f=f[0],p=p[0],l&&w(g,1)&&!(l=1<=(v=w(g,4))&&11>=v)){o.a=3;break t}var H;if(H=l)e:{var W,V,G,Y=o,Z=f,J=p,X=v,K=r,$=Y.m,Q=Y.s,
tt=[null],et=1,nt=0,rt=$r[X];n:for(;;){if(K&&w($,1)){var it=w($,3)+2,at=R(Z,it),st=R(J,it),ot=at*st;if(!Ft(at,st,0,Y,tt)||0==W)break n;for(tt=tt[0],Q.xc=it,W=0;W<ot;++W){var ht=tt[W]>>8&65535;tt[
W]=ht,ht>=et&&(et=ht+1)}}if($.h)break n;for(V=0;5>V;++V){var lt=Zr[V];!V&&0<X&&(lt+=1<<X),nt<lt&&(nt=lt)}var ct=s(et*rt,u),ut=et,ft=s(ut,d);if(null==ft)var dt=null;else e(65536>=ut),dt=ft;
var pt=a(nt);if(null==dt||null==pt||null==ct){Y.a=1;break n}var gt=ct;for(W=G=0;W<et;++W){var mt=dt[W],bt=mt.G,vt=mt.H,yt=0,_t=1,xt=0;for(V=0;5>V;++V){lt=Zr[V],0<lt&&(xt|=gt,vt[V]=G,!V&&0<X&&(
lt+=1<<X);r:{var At,Lt=lt,Nt=Y,Pt=pt,It=gt,Ct=G,jt=0,Et=Nt.m,Ot=w(Et,1);if(i(Pt,0,0,Lt),Ot){var Bt=w(Et,1)+1,Mt=w(Et,1),Rt=w(Et,0==Mt?1:8);Pt[Rt]=1,2==Bt&&(Pt[Rt=w(Et,8)]=1);var Tt=1}else{
var Dt=a(19),qt=w(Et,4)+4;if(19<qt){Nt.a=3;var zt=0;break r}for(At=0;At<qt;++At)Dt[Xr[At]]=w(Et,3);var Ut=void 0,Ht=void 0,Wt=Nt,Vt=Dt,Gt=Lt,Yt=Pt,Zt=0,Jt=Wt.m,Xt=8,Kt=s(128,u);i:for(;c(Kt,
0,7,Vt,19);){if(w(Jt,1)){var $t=2+2*w(Jt,3);if((Ut=2+w(Jt,$t))>Gt)break i}else Ut=Gt;for(Ht=0;Ht<Gt&&Ut--;){N(Jt);var Qt=Kt[0+(127&x(Jt))];L(Jt,Jt.u+Qt.g);var te=Qt.value;if(16>te)Yt[
Ht++]=te,0!=te&&(Xt=te);else{var ee=16==te,ne=te-16,re=Yr[ne],ie=w(Jt,Gr[ne])+re;if(Ht+ie>Gt)break i;for(var ae=ee?Xt:0;0<ie--;)Yt[Ht++]=ae}}Zt=1;break i}Zt||(Wt.a=3),Tt=Zt)(Tt=Tt&&!Et.h)&&(
jt=c(It,Ct,8,Pt,Lt)),Tt&&0!=jt?zt=jt:(Nt.a=3,zt=0)}if(0==zt)break n;if(_t&&1==Jr[V]&&(_t=0==gt[G].g),yt+=gt[G].g,G+=zt,3>=V){var se,oe=pt[0];for(;s=1;se<lt;++se)pt[se]>oe&&(oe=pt[se]);
xt+=oe}}if(mt.nd=_t,mt.Qb=0,_t&&(mt.qb=(bt[3][vt[3]+0].value<<24|bt[1][vt[1]+0].value<<16|bt[2][vt[2]+0].value)>>>0,0==yt&&256>bt[0][vt[0]+0].value&&(mt.Qb=1,mt.qb+=bt[0][vt[0]+0].
value<<8)),mt.jc=!mt.Qb&&6>xt,mt.jc){var he,le=mt;for(he=0;he<Tn;++he){var ce=he,ue=le.pd[ce],fe=le.G[0][le.H[0]+ce];256<=fe.value?(ue.g=fe.g+256,ue.value=fe.value):(ue.g=0,ue.value=0,
ce>>=wt(fe,8,ue),ce>>=wt(le.G[1][le.H[1]+ce],16,ue),ce>>=wt(le.G[2][le.H[2]+ce],0,ue),wt(le.G[3][le.H[3]+ce],24,ue))}}}}Q.vc=tt,Q.Wb=et,Q.Ya=dt,Q.yc=ct,H=1;break e}H=0}if(!(l=H)){o.a=3;break
t}if(0<v){if(m.ua=1<<v,!E(m.Wa,v)){o.a=1,l=0;break t}}else m.ua=0;var de=o,pe=f,ge=p,me=de.s,be=me.xc;if(de.c=pe,de.i=ge,me.md=R(pe,be),me.wc=0==be?-1:(1<<be)-1,r){o.xb=di;break t}if(null==(
b=a(f*p))){o.a=1,l=0;break t}l=(l=St(o,b,0,f,p,p,null))&&!g.h;break t}return l?(null!=h?h[0]=b:(e(null==b),e(r)),o.$=0,r||kt(m)):kt(m),l}function It(t,n){var r=t.c*t.i,i=r+n+16*n;return e(t.
c<=n),t.V=a(i),null==t.V?(t.Ta=null,t.Ua=0,t.a=1,0):(t.Ta=t.V,t.Ua=t.Ba+r+n,1)}function Ct(t,n){var r=t.C,i=n-r,a=t.V,s=t.Ba+t.c*r;for(e(n<=t.l.o);0<i;){var o=16<i?16:i,h=t.l.ma,l=t.l.width,
c=l*o,u=h.ca,f=h.tb+l*r,d=t.Ta,p=t.Ua;_t(t,o,a,s),Nr(d,p,u,f,c),Lt(h,r,r+o,u,f,l),i-=o,a+=o*t.c,r+=o}e(r==n),t.C=t.Ma=n}function jt(){this.ub=this.yd=this.td=this.Rb=0}function Et(){this.
Kd=this.Ld=this.Ud=this.Td=this.i=this.c=0}function Ot(){this.Fb=this.Bb=this.Cb=0,this.Zb=a(4),this.Lb=a(4)}function Bt(){this.Yb=function(){var t=[];return function t(e,n,r){for(var i=r[
n],a=0;a<i&&(e.push(r.length>n+1?[]:0),!(r.length<n+1));a++)t(e[a],n+1,r)}(t,0,[3,11]),t}()}function Mt(){this.jb=a(3),this.Wc=o([4,8],Bt),this.Xc=o([4,17],Bt)}function Rt(){this.Pc=this.
wb=this.Tb=this.zd=0,this.vd=new a(4),this.od=new a(4)}function Tt(){this.ld=this.La=this.dd=this.tc=0}function Dt(){this.Na=this.la=0}function qt(){this.Sc=[0,0],this.Eb=[0,0],this.Qc=[0,
0],this.ia=this.lc=0}function zt(){this.ad=a(384),this.Za=0,this.Ob=a(16),this.$b=this.Ad=this.ia=this.Gc=this.Hc=this.Dd=0}function Ut(){this.uc=this.M=this.Nb=0,this.wa=Array(new Tt),this.
Y=0,this.ya=Array(new zt),this.aa=0,this.l=new Gt}function Ht(){this.y=a(16),this.f=a(8),this.ea=a(8)}function Wt(){this.cb=this.a=0,this.sc="",this.m=new y,this.Od=new jt,this.Kc=new Et,
this.ed=new Rt,this.Qa=new Ot,this.Ic=this.$c=this.Aa=0,this.D=new Ut,this.Xb=this.Va=this.Hb=this.zb=this.yb=this.Ub=this.za=0,this.Jc=s(8,y),this.ia=0,this.pb=s(4,qt),this.Pa=new Mt,this.
Bd=this.kc=0,this.Ac=[],this.Bc=0,this.zc=[0,0,0,0],this.Gd=Array(new Ht),this.Hd=0,this.rb=Array(new Dt),this.sb=0,this.wa=Array(new Tt),this.Y=0,this.oc=[],this.pc=0,this.sa=[],this.ta=0,
this.qa=[],this.ra=0,this.Ha=[],this.B=this.R=this.Ia=0,this.Ec=[],this.M=this.ja=this.Vb=this.Fc=0,this.ya=Array(new zt),this.L=this.aa=0,this.gd=o([4,2],Tt),this.ga=null,this.Fa=[],this.
Cc=this.qc=this.P=0,this.Gb=[],this.Uc=0,this.mb=[],this.nb=0,this.rc=[],this.Ga=this.Vc=0}function Vt(t,e){return 0>t?0:t>e?e:t}function Gt(){this.T=this.U=this.ka=this.height=this.width=0,
this.y=[],this.f=[],this.ea=[],this.Rc=this.fa=this.W=this.N=this.O=0,this.ma="void",this.put="VP8IoPutHook",this.ac="VP8IoSetupHook",this.bc="VP8IoTeardownHook",this.ha=this.Kb=0,this.
data=[],this.hb=this.ib=this.da=this.o=this.j=this.va=this.v=this.Da=this.ob=this.w=0,this.F=[],this.J=0}function Yt(){var t=new Wt;return null!=t&&(t.a=0,t.sc="OK",t.cb=0,t.Xb=0,ni||(
ni=Kt)),t}function Zt(t,e,n){return 0==t.a&&(t.a=e,t.sc=n,t.cb=0),0}function Jt(t,e,n){return 3<=n&&157==t[e+0]&&1==t[e+1]&&42==t[e+2]}function Xt(t,n){if(null==t)return 0;if(t.a=0,t.
sc="OK",null==n)return Zt(t,2,"null VP8Io passed to VP8GetHeaders()");var r=n.data,a=n.w,s=n.ha;if(4>s)return Zt(t,7,"Truncated header.");var o=r[a+0]|r[a+1]<<8|r[a+2]<<16,h=t.Od;if(h.Rb=!(
1&o),h.td=o>>1&7,h.yd=o>>4&1,h.ub=o>>5,3<h.td)return Zt(t,3,"Incorrect keyframe parameters.");if(!h.yd)return Zt(t,4,"Frame not displayable.");a+=3,s-=3;var l=t.Kc;if(h.Rb){if(7>s)return Zt(
t,7,"cannot parse picture header");if(!Jt(r,a,s))return Zt(t,3,"Bad code word");l.c=16383&(r[a+4]<<8|r[a+3]),l.Td=r[a+4]>>6,l.i=16383&(r[a+6]<<8|r[a+5]),l.Ud=r[a+6]>>6,a+=7,s-=7,t.za=l.
c+15>>4,t.Ub=l.i+15>>4,n.width=l.c,n.height=l.i,n.Da=0,n.j=0,n.v=0,n.va=n.width,n.o=n.height,n.da=0,n.ib=n.width,n.hb=n.height,n.U=n.width,n.T=n.height,i((o.t=(a+=o.Pa).jb,0,255,o.jb.length),e(
null!=(o=t.Qa)),o.Cb=0,o.Bb=0,o.Fb=1,i(o.Zb,0,0,o.Zb.length),i(o.Lb,0,0,o.Lb))if(h.ub>s)return Zt(t,7,"bad partition length");p(o=t.m,r,a,h.ub),a+=h.ub,s-=h.ub,h.Rb&&(l.Ld=k(o),l.Kd=k(o)),
l=t.Qa;var c,u=t.Pa;if(e(null!=o),e(null!=l),l.Cb=k(o),l.Cb){if(l.Bb=k(o),k(o)){for(l.Fb=k(o),c=0;4>c;++c)l.Zb[c]=k(o)?m(o,7):0;for(c=0;4>c;++c)l.Lb[c]=k(o)?m(o,6):0}if(l.Bb)for(c=0;3>c;
++c)u.jb[c]=k(o)?g(o,8):255}else l.Bb=0;if(o.Ka)return Zt(t,3,"cannot parse segment header");if((l=t.ed).zd=k(o),l.Tb=g(o,6),l.wb=g(o,3),l.Pc=k(o),l.Pc&&k(o)){for(u=0;4>u;++u)k(o)&&(l.vd[
u]=m(o,6));for(u=0;4>u;++u)k(o)&&(l.od[u]=m(o,6))}if(t.L=0==l.Tb?0:l.zd?1:2,o.Ka)return Zt(t,3,"cannot parse filter header");var f=s;if(s=c=a,a=c+f,l=f,t.Xb=(1<<g(t.m,2))-1,f<3*(u=t.Xb))r=7;
else{for(c+=3*u,l-=3*u,f=0;f<u;++f){var d=r[s+0]|r[s+1]<<8|r[s+2]<<16;d>l&&(d=l),p(t.Jc[+f],r,c,d),c+=d,l-=d,s+=3}p(t.Jc[+u],r,c,l),r=c<a?0:5}if(0!=r)return Zt(t,r,"cannot parse
partitions");for(r=g(c=t.m,7),s=k(c)?m(c,4):0,a=k(c)?m(c,4):0,l=k(c)?m(c,4):0,u=k(c)?m(c,4):0,c=k(c)?m(c,4):0,f=t.Qa,d=0;4>d;++d){if(f.Cb){var b=f.Zb[d];f.Fb||(b+=r)}else{if(0<d){t.pb[d]=t.
pb[0];continue}b=r}var v=t.pb[d];v.Sc[0]=ti[Vt(b+s,127)],v.Sc[1]=ei[Vt(b+0,127)],v.Eb[0]=2*ti[Vt(b+a,127)],v.Eb[1]=101581*ei[Vt(b+l,127)]>>16,8>v.Eb[1]&&(v.Eb[1]=8),v.Qc[0]=ti[Vt(b+c,127)],
v.Qc[1]=ei[Vt(b+c,127)],v.lc=b+c}if(!h.Rb)return Zt(t,4,"Not a key frame.");for(k(o),h=t.Pa,r=0;4>r;++r){for(s=0;8>s;++s)for(a=0;3>a;++a)for(l=0;11>l;++l)u=P(o,hi[r][s][a][l])?g(o,8):si[r][
s][a][l],h.Wc[r][s].Yb[a][l]=u;for(s=0;17>s;++s)h.Xc[r][s]=h.Wc[r][li[s]]}return t.kc=k(o),t.kc&&(t.Bd=g(o,8)),t.cb=1}function Kt(t,e,n,r,i,a,s){var o=e[i].Yb[n];for(n=0;16>i;++i){if(!P(t,o[
n+0]))return i;for(;!P(t,o[n+1]);)if(o=e[++i].Yb[0],n=0,16==i)return 16;var h=e[i+1].Yb;if(P(t,o[n+2])){var l=t,c=0;if(P(l,(f=o)[(u=n)+3]))if(P(l,f[u+6]))for(o=0,u=2*(c=P(l,f[u+8]))+(f=P(l,
f[u+9+c])),c=0,f=ri[u];f[o];++o)c+=c+P(l,f[o]);c+=3+(8<<u)}else P(l,f[u+7])?(c=7+2*P(l,165),c+=P(l,145)):c=5+P(l,159);else c=P(l,f[u+4])?3+P(l,f[u+5]):2;o=h[2]}else c=1,o=h[1];h=s+ii[i],0>(
l=t).bā&S(l);var u,f=l.b,d=(u=l.Ca>>1)-(l.I>>f)>>31;--l.b,l.Ca+=d,l.Ca|=1,l.I-=(u+1&d)<<f,a[h]=((c^d)-d)*r[(0<i)+0];return 16}function $t(t){var e=t.rb[t.sb-1];e.la=0,e.Na=0,i(t.oc,0,0,t.zc.
length),t.ja=0}function Qt(t,e,n,r,i){i=t[e+n+32*r]+(i>>3),t[e+n+32*r]=-256&i?0>i?0:255:i}function te(t,e,n,r,i,a){Qt(t,e,0,n,r+i),Qt(t,e,1,n,r+a),Qt(t,e,2,n,r-a),Qt(t,e,3,n,r-i)}function
ee(t){return(20091*t>>16)+t}function ne(t,e,n,r){var i,s=0,o=a(16);for(i=0;4>i;++i){var h=t[e+0]+t[e+8],l=t[e+0]-t[e+8],c=(35468*t[e+4]>>16)-ee(t[e+12]),u=ee(t[e+4])+(35468*t[e+12]>>16);t=s[
s+0]=h+u,o[s+1]=l+c,o[s+2]=l-c,o[s+3]=h-u,s+=4,e++}for(i=s=0;4>i;++i)h=(t=o[s+0]+4)+o[s+8],l=t-o[s+8],c=(35468*o[s+4]>>16)-ee(o[s+12]),Qt(n,r,0,0,h+(u=ee(o[s+4])+(35468*o[s+12]>>16)),Qt(n,
r,1,0,l+c),Qt(n,r,2,0,l-c),Qt(n,r,3,0,h-u),s++,r+=32}function re(t,e,n,r){var i=t[e+0]+4,a=35468*t[e+4]>>16,s=ee(t[e+4]),o=35468*t[e+1]>>16;te(n,r,0,i+s,t=ee(t[e+1]),o),te(n,r,1,i+a,t,o),te(
n,r,2,i-a,t,o),te(n,r,3,i-s,t,o)}function ie(t,e,n,r,i){ne(t,e,n,r),i&&ne(t,e+16,n,r+4)}function ae(t,e,n,r){ar(t,e+0,n,r,1),ar(t,e+32,n,r+128,1)}function se(t,e,n,r){var i;for(t=t[e+0]+4,
i=0;4>i;++i)for(e=0;4>e;++e)Qt(n,r,e,i,t)}function oe(t,e,n,r){t[e+0]&&hr(t,e+0,n,r),t[e+16]&&hr(t,e+16,n,r+4),t[e+32]&&hr(t,e+32,n,r+128),t[e+48]&&hr(t,e+48,n,r+128+4)}function he(t,e,n,r){
var i,s=a(16);for(i=0;4>i;++i){var o=t[e+0+i]+t[e+12+i],h=t[e+4+i]+t[e+8+i],l=t[e+4+i]-t[e+8+i],c=t[e+0+i]-t[e+12+i];s[0+i]=o+h,s[4+i]=c+l,s[12+i]=c-l}for(i=0;4>i;++i)o=t[s[
0+4*i]+3)+s[3+4*i],h=s[1+4*i]+s[2+4*i],l=s[1+4*i]-s[2+4*i],c=t-s[3+4*i],n[r+0]=o+h>>3,n[r+16]=c+l>>3,n[r+32]=o-h>>3,n[r+48]=c-l>>3,r+=64}function le(t,e,n){var r,i=e-32,a=Er,s=255-t[i-1];
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for(r=0;r<n;++r){var o,h=a,l=s+t[e-1];for(o=0;o<n;++o)t[e+o]=h[l+t[i+o]];e+=32}}function ce(t,e){le(t,e,4)}function ue(t,e){le(t,e,8)}function fe(t,e){le(t,e,16)}function de(t,e){var n;for(
n=0;16>n;++n)r(t,e+32*n,t,e-32,16)}function pe(t,e){var n;for(n=16;0<n;--n)i(t,e,t[e-1],16),e+=32}function ge(t,e,n){var r;for(r=0;16>r;++r)i(e,n+32*r,t,16)}function me(t,e){var n,r=16;for(
n=0;16>n;++n)r+=t[e-1+32*n]+t[e+n-32];ge(r>>5,t,e)}function be(t,e){var n,r=8;for(n=0;16>n;++n)r+=t[e-1+32*n];ge(r>>4,t,e)}function ve(t,e){var n,r=8;for(n=0;16>n;++n)r+=t[e+n-32];ge(r>>4,t,
e)}function we(t,e){ge(128,t,e)}function ye(t,e,n){return t+2*e+n+2>>2}function _e(t,e){var n,i=e-32;for(i=new Uint8Array([ye(t[i-1],t[i+0],t[i+1]),ye(t[i+0],t[i+1],t[i+2]),ye(t[i+1],t[i+2],
t[i+3]),ye(t[i+2],t[i+3],t[i+4])]),n=0;4>n;++n)r(t,e+32*n,i,0,i.length)}function xe(t,e){var n=t[e-1],r=t[e-1+32],i=t[e-1+64],a=t[e-1+96];F(t,e+0,16843009*ye(t[e-1-32],n,r)),F(t,e+32,
16843009*ye(n,r,i)),F(t,e+64,16843009*ye(r,i,a)),F(t,e+96,16843009*ye(i,a,a))}function Ae(t,e){var n,r=4;for(n=0;4>n;++n)r+=t[e+n-32]+t[e-1+32*n];for(r>>=3,n=0;4>n;++n)i(t,e+32*n,r,
4)}function Le(t,e){var n=t[e-1+0],r=t[e-1+32],i=t[e-1+64],a=t[e-1-32],s=t[e+0-32],o=t[e+1-32],h=t[e+2-32],l=t[e+3-32];t[e+0+96]=ye(r,i,t[e-1+96]),t[e+1+96]=t[e+0+64]=ye(n,r,i),t[e+2+96]=t[
e+1+64]=t[e+0+32]=ye(a,n,r),t[e+3+96]=t[e+2+64]=t[e+1+32]=t[e+0+0]=ye(s,a,n),t[e+3+64]=t[e+2+32]=t[e+1+0]=ye(o,s,a),t[e+3+32]=t[e+2+0]=ye(h,o,s),t[e+3+0]=ye(l,h,o)}function Ne(t,e){var n=t[
e+1-32],r=t[e+2-32],i=t[e+3-32],a=t[e+4-32],s=t[e+5-32],o=t[e+6-32],h=t[e+7-32];t[e+0+0]=ye(t[e+0-32],n,r),t[e+1+0]=t[e+0+32]=ye(n,r,i),t[e+2+0]=t[e+1+32]=t[e+0+64]=ye(r,i,a),t[e+3+0]=t[
e+2+32]=t[e+1+64]=t[e+0+96]=ye(i,a,s),t[e+3+32]=t[e+2+64]=t[e+1+96]=ye(a,s,o),t[e+2+32]=t[e+3+96]=ye(s,o,h),t[e+3+32]=ye(o,h,l)}function ke(t,e){var n=t[e+0-32],r=t[e+1-32],i=t[e+2-32],a=t[e+3-32],s=t[
e+4-32],o=t[e+5-32],h=t[e+6-32],l=t[e+7-32];t[e+0+0]=n+r+1>>1,t[e+1+0]=t[e+0+64]=r+i+1>>1,t[e+2+0]=t[e+1+64]=i+a+1>>1,t[e+3+0]=t[e+2+64]=a+s+1>>1,t[e+2+32]=t[e+3+96]=ye(s,o,h),t[e+3+32]=ye(o,h,l)}function Pe(t,e){var n=t[e-1+0],r=t[e-1+32],i=t[e-1+64],a=t[e-1+96];t[
e+0+0]=n+r+1>>1,t[e+2+0]=t[e+0+32]=r+i+1>>1,t[e+2+32]=t[e+0+64]=i+a+1>>1,t[e+1+0]=ye(n,r,i),t[e+3+0]=t[e+1+32]=ye(r,i,a),t[e+3+32]=t[e+1+64]=ye(i,a,a),t[e+3+64]=t[e+2+64]=t[e+0+96]=t[
e+1+96]=t[e+2+96]=t[e+3+96]=a}function Fe(t,e){var n=t[e-1+0],r=t[e-1+32],i=t[e-1+64],a=t[e-1+96],s=t[e-1-32],o=t[e+0-32],h=t[e+1-32],l=t[e+2-32];t[e+0+0]=t[e+2+32]=n+s+1>>1,t[e+0+32]=t[
e+2+64]=r+n+1>>1,t[e+0+64]=t[e+2+96]=i+r+1>>1,t[e+0+96]=a+i+1>>1,t[e+3+0]=ye(o,h,l),t[e+2+0]=ye(s,o,h),t[e+1+0]=t[e+3+32]=ye(n,s,o),t[e+1+32]=t[e+3+64]=ye(r,n,s),t[e+1+64]=t[e+3+96]=ye(i,r,
n),t[e+1+96]=ye(a,i,r)}function Ie(t,e){var n;for(n=0;8>n;++n)r(t,e+32*n,t,e-32,8)}function Ce(t,e){var n;for(n=0;8>n;++n)i(t,e,t[e-1],8),e+=32}function je(t,e,n){var r;for(r=8;8>r;++r)i(e,
n+32*r,t,8)}function Ee(t,e){var n,r=8;for(n=0;8>n;++n)r+=t[e+n-32]+t[e-1+32*n];je(r>>4,t,e)}function Oe(t,e){var n,r=4;for(n=0;8>n;++n)r+=t[e+n-32];je(r>>3,t,e)}function Be(t,e){var n,r=4;
for(n=0;8>n;++n)r+=t[e-1+32*n];je(r>>3,t,e)}function Me(t,e){je(128,t,e)}function Re(t,e,n){var r=t[e-n],i=t[e+0],a=3*(i-r)+Cr[1020+t[e-2*n]-t[e+n]],s=jr[112+(a+4>>3)]];t[e-n]=Er[255+r+jr[
112+(a+3>>3)]],t[e+0]=Er[255+i-s]}function Te(t,e,n,r){var i=t[e+0],a=t[e+n];return Or[255+t[e-2*n]-t[e-n]]>r||Or[255+a-i]>r}function De(t,e,n,r){return 4*Or[255+t[e-n]-t[e+0]]+Or[255+t[e-
2*n]-t[e+n]]<=r}function qe(t,e,n,r,i){var a=t[e-3*n],s=t[e-2*n],o=t[e-n],h=t[e+0],l=t[e+n],c=t[e+2*n],u=t[e+3*n];return 4*Or[255+o-h]+Or[255+s-l]>r?0:Or[255+t[e-4*n]-a]<=i&&Or[255+a-
s]<=i&&Or[255+s-o]<=i&&Or[255+u-c]<=i&&Or[255+c-l]<=i&&Or[255+l-h]<=i}function ze(t,e,n,r){var i=2*r+1;for(r=0;16>r;++r)De(t,e+r,n,i)&&Re(t,e+r,n)}function Ue(t,e,n,r){var i=2*r+1;for(r=0;
16>r;++r)De(t,e+r*n,1,i)&&Re(t,e+r*n,1)}function He(t,e,n,r){var i;for(i=3;0<i;--i)ze(t,e+=4*n,n,r)}function We(t,e,n,r){var i;for(i=3;0<i;--i)Ue(t,e+=4,n,r)}function Ve(t,e,n,r,i,a,s,o){
for(a=2*a+1;0<i--;){if(qe(t,e,n,a,s))if(Te(t,e,n,o))Re(t,e,n);else{var h=t,l=e,c=n,u=h[l-2*c],f=h[l-c],d=h[l+0],p=h[l+c],g=h[l+2*c],m=27*(v=Cr[1020+3*(d-f)+Cr[1020+u-p]])+63>>7,b=18*v+63>>7,
v=9*v+63>>7;h[l-3*c]=Er[255+h[l-3*c]+v],h[l-2*c]=Er[255+u+b],h[l-c]=Er[255+f+m],h[l+0]=Er[255+d-m],h[l+c]=Er[255+p-b],h[l+2*c]=Er[255+g-v]}e+=r}}function Ge(t,e,n,r,i,a,s,o){for(a=2*a+1;0<i-
-;){if(qe(t,e,n,a,s))if(Te(t,e,n,o))Re(t,e,n);else{var h=t,l=e,c=n,u=h[l-c],f=h[l+0],d=h[l+c],p=jr[112+(4+(g=3*(f-u))>>3)],g=jr[112+(g+3>>3)],m=p+1>>1;h[l-2*c]=Er[255+h[l-2*c]+m],h[l-c]=Er[
255+u+g],h[l+0]=Er[255+f-p],h[l+c]=Er[255+d-m]}e+=r}}function Ye(t,e,n,r,i,a){Ve(t,e,n,1,16,r,i,a)}function Ze(t,e,n,r,i,a){Ve(t,e,1,n,16,r,i,a)}function Je(t,e,n,r,i,a){var s;for(s=3;0<s;--
s)Ge(t,e+=4*n,n,1,16,r,i,a)}function Xe(t,e,n,r,i,a){var s;for(s=3;0<s;--s)Ge(t,e+=4,1,n,16,r,i,a)}function Ke(t,e,n,r,i,a,s,o){Ve(t,e,i,1,8,a,s,o),Ve(n,r,i,1,8,a,s,o)}function $e(t,e,n,r,i,
a,s,o){Ve(t,e,1,i,8,a,s,o),Ve(n,r,1,i,8,a,s,o)}function Qe(t,e,n,r,i,a,s,o){Ge(t,e+4*i,i,1,8,a,s,o),Ge(n,r+4*i,i,1,8,a,s,o)}function tn(t,e,n,r,i,a,s,o){Ge(t,e+4,1,1,8,a,s,o),Ge(n,r+4,1,1,8,
a,s,o)}function en(){this.ba=new st,this.ec=[],this.cc=[],this.Mc=[],this.Dc=this.Nc=this.dc=this.fc=0,this.Oa=new ht,this.memory=0,this.Ib="OutputFunc",this.Jb="OutputAlphaFunc",this.
Nd="OutputRowFunc"}function nn(){this.data=[],this.offset=this.kd=this.ha=this.w=0,this.na=[],this.xa=this.gb=this.Ja=this.Sa=this.P=0}function rn(){this.nc=this.Ea=this.b=this.hc=0,this.K=[
],this.w=0}function an(){this.ua=0,this.Wa=new B,this.vb=new B,this.md=this.xc=this.wc=0,this.vc=[],this.Wb=0,this.Ya=new d,this.yc=new u}function sn(){this.xb=this.a=0,this.l=new Gt,this.
ca=new st,this.V=[],this.Ba=0,this.Ta=[],this.Ua=0,this.m=new _,this.Pb=0,this.wd=new _,this.Ma=this.$=this.C=this.i=this.c=this.xd=0,this.s=new an,this.ab=0,this.gc=s(4,rn),this.
Oc=0}function on(){this.Lc=this.Z=this.$a=this.i=this.c=0,this.l=new Gt,this.ic=0,this.ca=[],this.tb=0,this.qd=null,this.rd=0}function hn(t,e,n,r,i,a,s){for(t=null==t?0:t[e+0],e=0;e<s;++e)i[
a+e]=t+n[r+e]&255,t=i[a+e]}function ln(t,e,n,r,i,a,s){var o;if(null==t)hn(null,null,n,r,i,a,s);else for(o=0;o<s;++o)i[a+o]=t[e+o]+n[r+o]&255}function cn(t,e,n,r,i,a,s){if(null==t)hn(null,
null,n,r,i,a,s);else{var o,h=t[e+0],l=h,c=h;for(o=0;o<s;++o)l=c+(h=t[e+o])-l,c=n[r+o]+(-256&l?0>1?0:255:l)&255,l=h,i[a+o]=c}}function un(t,n,i,s){var o=n.width,h=n.o;if(e(null!=t&&null!=n),
0>i||0>=s||i+s>h)return null;if(!t.Cc){if(null==t.ga){var l;if(t.ga=new on,(l=null==t.ga)||(l=n.width*n.o,e(0==t.Gb.length),t.Gb=a(l),t.Uc=0,null==t.Gb?l=0:(t.mb=t.Gb,t.nb=t.Uc,t.rc=null,
l=1),l=!l),!l){l=t.ga;var c=t.Fa,u=t.P,f=t.qc,d=t.mb,p=t.nb,g=u+1,m=f-1,v=l.l;if(e(null!=c&&null!=u&&null!=n),gi[0]=null,gi[1]=hn,gi[2]=ln,gi[3]=cn,l.ca=d,l.tb=p,l.c=n.width,l.i=n.height,e(
0<l.c&&0<l.i),1>=f)n=0;else if(l.$a=3&c[u+0],l.Z=c[u+0]>>2&3,l.Lc=c[u+0]>>4&3,u=c[u+0]>>6&3,0>l.$a||1<l.$a||4<=l.Z||1<l.Lc||u)n=0;else if(v.put=dt,v.ac=ft,v.bc=pt,v.ma=l,v.width=n.width,v.
height=n.height,v.Da=n.Da,v.v=n.v,v.va=n.va,v.j=n.j,v.o=n.o,l.$a)t:{e(1==l.$a),n=Pt();e:for(;;){if(null==n){n=0;break t}if(e(null!=l),l.mc=n,n.c=l.c,n.i=l.i,n.l=l.l,n.l.ma=l,n.l.width=l.c,n.
l.height=l.i,n.a=0,b(n.m,c,g,m),!Ft(l.c,l.i,1,n,null))break e;if(1==n.ab&&3==n.gc[0].hc&&At(n.s)?(l.ic=1,c=n.c*n.i,n.Ta=null,n.Ua=0,n.V=a(c),n.Ba=0,null==n.V?(n.a=1,n=0):n=1):(l.ic=0,n=It(n,
l.c)),!n)break e;n=1;break t}l.mc=null,n=0}else n=m>=l.c*l.i;l=!n}if(l)return null;1!=t.ga.Lc?t.Ga=0:s=h-i}e(null!=t.ga),e(i+s<=h);t:{if(n=(c=t.ga).c,h=c.l.o,0==c.$a){if(g=t.rc,m=t.Vc,v=t.
Fa,u=t.P+1+i*n,f=t.mb,d=t.nb+i*n,e(u<=t.P+t.qc),0!=c.Z)for(e(null!=gi[c.Z]),l=0;l<s;++l)gi[c.Z](g,m,v,u,f,d,n),g=f,m=d,d+=n,u+=n;else for(l=0;l<s;++l)r(f,d,v,u,n),g=f,m=d,d+=n,u+=n;t.rc=g,t.
Vc=m}else{if(e(null!=c.mc),n=i+s,e(null!=(l=c.mc)),e(n<=l.i),l.C>=n)n=1;else if(c.ic||mn(),c.ic){c=l.V,g=l.Ba,m=l.c;var w=l.i,y=(v=1,u=l.$/m,f=l.$%m,d=l.m,p=l.s,l.$),_=m*w,x=m*n,L=p.wc,
S=y<x?yt(p,f,u):null;e(y<=_),e(n<=w),e(At(p));e:for(;;){for(;!d.h&&y<x;){if(f&&!((S=yt(p,f,u)),e(null!=S),N(d),256>(w=vt(S.G[0],S.H[0],d))))c[g+y]=w,++y,++f>=m&&(f=0,++u<=n&&!(u%16)&&Nt(l,
u));else{if(!(280>w)){v=0;break e}w=mt(w-256,d);var k,P=vt(S.G[4],S.H[4],d);if(N(d),!(y>=(P=bt(m,P=mt(P,d)))&&_-y>=w)){v=0;break e}for(k=0;k<w;++k)c[g+y+k]=c[g+y+k-P];for(y+=w,f+=w;f>=m;)f-
=m,++u<=n&&!(u%16)&&Nt(l,u);y<x&&f&L&&(S=yt(p,f,u))}e(d.h==A(d))}Nt(l,u>n?n:u);break e}!v||d.h&&y<_?(v=0,l.a=d.h?5:3):l.$=y,n=v}else n=St(l,l.V,l.Ba,l.c,l.i,n,Ct);if(!n){s=0;break
t}}i+s>=h&&(t.Cc=1),s=1}if(!s)return null;if(t.Cc&&(null!=(s=t.ga)&&(s.mc=null),t.ga=null,0<t.Ga))return alert("todo:WebPDequantizeLevels"),null}return t.nb+i*o}function fn(t,e,n,r,i,a){for(
;0<i--;){var s,o=t,h=e+(n?1:0),l=t,c=e+(n?0:3);for(s=0;s<r;++s){var u=l[c+4*s];255!=u&&(u*=32897,o[h+4*s+0]=o[h+4*s+0]*u>>23,o[h+4*s+1]=o[h+4*s+1]*u>>23,o[h+4*s+2]=o[
h+4*s+2]*u>>23)}e+=a}}function dn(t,e,n,r,i){for(;0<--r;){var a;for(a=0;a<n;++a){var s=t[e+2*a+0],o=15&(l=t[e+2*a+1]),h=4369*o,l=(240&l|l>>4)*h>>16&2|(240&s|s>>4)*h>>16&240|(
15&s|s<<4)*h>>16>>4&15,t[e+2*a+1]=240&l|o}e+=i}}function pn(t,e,n,r,i,a,s,o){var h,l,c=255;for(l=0;l<i;++l){for(h=0;h<r;++h){var u=t[e+h];a[s+4*h]=u,c&=u}e+=n,s+=o}return 255!=c}function gn(
t,e,n,r,i){var a;for(a=0;a<i;++a)n[r+a]=t[e+a]>>8}function mn(){xr=fn,Ar=dn,Lr=pn,Nr=gn}function bn(n,r,i){t[n]=function(t,e,n,r,i,a,s,o,h,l,c,u,f,d,p,g,m,b,v,w){var y,_=w-1>>1,x=o[h+0]|l[
c+0]<<16,A=u[f+0]|d[p+0]<<16;e(null!=t);var L=3*x+A+131074>>2;for(r(t[n+0],255&L,L>>16,g,m),null!=a&&(L=3*A+x+131074>>2,r(a[s+0],255&L,L>>16,b,v)),y=1;y<=_;++y){var N=o[h+y]|l[c+y]<<16,S=u[
f+y]|d[p+y]<<16,k=x+N+A+S+524296,P=k+2*(N+A)>>3;L=P+x>>1,x=(k=k+2*(x+S)>>3)+N>>1,r(t[n+2*y-1],255&L,L>>16,g,m+(2*y-1)*i),r(t[n+2*y-0],255&x,x>>16,g,m+(2*y-0)*i),null!=a&&(L=k+A>>1,x=P+S>>1,
r(a[s+2*y-1],255&L,L>>16,b,v+(2*y-1)*i),r(a[s+2*y+0],255&x,x>>16,b,v+(2*y+0)*i)),x=N,A=S}1&w||(L=3*x+A+131074>>2,r(t[n+w-1],255&L,L>>16,g,m+(w-1)*i),null!=a&&(L=3*A+x+131074>>2,r(a[s+w-1],
255&L,L>>16,b,v+(w-1)*i)))}}function vn(){mi[Br]=bi,mi[Mr]=wi,mi[Rr]=vi,mi[Tr]=yi,mi[Dr]=_i,mi[qr]=xi,mi[zr]=Ai,mi[Ur]=wi,mi[Hr]=yi,mi[Wr]=_i,mi[Vr]=xi}function wn(t){return t&~Fi?0>t?0:255:
t>>Pi}function yn(t,e){return wn((19077*t>>8)+(26149*e>>8)-14234)}function _n(t,e,n){return wn((19077*t>>8)-(6419*e>>8)-(13320*n>>8)+8708)}function xn(t,e){return wn((19077*t>>8)+(
33050*e>>8)-17685)}function An(t,e,n,r,i){r[i+0]=yn(t,n),r[i+1]=_n(t,e,n),r[i+2]=xn(t,e)}function Ln(t,e,n,r,i){r[i+0]=xn(t,e),r[i+1]=_n(t,e,n),r[i+2]=yn(t,n)}function Nn(t,e,n,r,i){var
a=_n(t,e,n);e=a<<3&224|xn(t,e)>>3,r[i+0]=248&yn(t,n)|a>>5,r[i+1]=e}function Sn(t,e,n,r,i){var a=240&xn(t,e)|15;r[i+0]=240&yn(t,n)|_n(t,e,n)>>4,r[i+1]=a}function kn(t,e,n,r,i){r[i+0]=255,An(
t,e,n,r,i+1)}function Pn(t,e,n,r,i){Ln(t,e,n,r,i),r[i+3]=255}function Fn(t,e,n,r,i){An(t,e,n,r,i),r[i+3]=255}function In(e,n,r){t[e]=function(t,e,i,a,s,o,h,l,c){for(var u=l+(-2&c)*r;l!=u;)n(
t[e+0],i[a+0],s[o+0],h,l),n(t[e+1],i[a+0],s[o+0],h,l+r),e+=2,++a,++o,l+=2*r;1&c&&n(t[e+0],i[a+0],s[o+0],h,l)}}function Cn(t,e,n){return 0==n?0==t?0==e?6:5:0==e?4:0:n}function jn(t,e,n,r,i){
switch(t>>>30){case 3:ar(e,n,r,i,0);break;case 2:sr(e,n,r,i);break;case 1:hr(e,n,r,i)}}function En(t,e){var n,a,s=e.M,o=e.Nb,h=t.oc,l=t.pc+40,c=t.oc,u=t.pc+584,f=t.oc,d=t.pc+600;for(n=0;
16>n;++n)h[l+32*n-1]=129;for(n=0;8>n;++n)c[u+32*n-1]=129,f[d+32*n-1]=129;for(0<s?h[l-1-32]=c[u-1-32]=f[d-1-32]=129:(i(h,l-32-1,127,21),i(c,u-32-1,127,9),i(f,d-32-1,127,9)),a=0;a<t.za;++a){
var p=e.ya[e.aa+a];if(0<a){for(n=-1;16>n;++n)r(h,l+32*n-4,h,l+32*n+12,4);for(n=-1;8>n;++n)r(c,u+32*n-4,c,u+32*n+4,4),r(f,d+32*n+4,f,d+32*n+4,4)}var g=t.Gd,m=t.Hd+a,b=p.ad,v=p.Hc;if(0<s&&(r(
h,l-32,g[m].y,0,16),r(c,u-32,g[m].f,0,8),r(f,d-32,g[m].ea,0,8)),p.Za){var w=h,y=l-32+16;for(0<s&&(a>=t.za-1?i(w,y,g[m].y[15],4):r(w,y,g[m+1].y,0,4)),n=0;4>n;n++)w[y+128+n]=w[y+256+n]=w[
y+384+n]=w[y+0+n];for(n=0;16>n;++n,v<<=2)w=h,y=l+Ri[n],ui[p.Ob[n]](w,y),jn(v,b,16*+n,w,y)}else if(w=Cn(a,s,p.Ob[0]),ci[w](h,l),0!=v)for(n=0;16>n;++n,v<<=2)jn(v,b,16*+n,h,l+Ri[n]);for(n=p.Gc,
w=Cn(a,s,p.Dd),fi[w](c,u),fi[w](f,d),v=b,w=c,y=u,255&(p=0|n)&&(170&p?or(v,256,w,y):lr(v,256,w,y)),p=f,v=d,255&(n>>=8)&&(170&n?or(b,320,p,v):lr(b,320,p,v)),s<t.Ub-1&&(r(g[m].y,0,h,l+480,16),
r(g[m].f,0,c,u+224,8),r(g[m].ea,0,f,d+224,8)),n=8*o*t.B,g=t.sa,m=t.ta+16*a+16*o*t.R,b=t.qa,p=t.ra+8*a+n,v=t.Ha,w=t.Ia+8*a+n,n=0;16>n;++n)r(g,m+n*t.R,h,l+32*n,16);for(n=0;8>n;++n)r(b,p+n*t.B,
c,u+32*n,8),r(v,w+n*t.B,f,d+32*n,8)}}function On(t,r,i,a,s,o,h,l,c){var u=[0],f=[0],d=0,p=null!=c?c.kd:0,g=null!=c?c:new nn;if(null==t||12>i)return 7;g.data=t,g.w=r,g.ha=i,r=[r],i=[i],g.gb=[
g.gb];t:{var m=r,v=i,w=g.gb;if(e(null!=t),e(null!=v),e(null!=w),w[0]=0,12<=v[0]&&!n(t,m[0],"RIFF")){if(n(t,m[0]+8,"WEBP")){w=3;break t}var y=j(t,m[0]+4);if(12>y||4294967286<y){w=3;break
t}if(p&&y>v[0]-8){w=7;break t}w[0]=y,m[0]+=12,v[0]-=12}w=0}if(0!=w)return w;for(y=0<g.gb[0],i=i[0];;){t:{var x=t;v=r,w=i;var A=u,L=f,N=m=[0];if((P=d=[d])[0]=0,8>w[0])w=7;else{if(!(n(x,v[0],
"VP8X")){if(10!=j(x,v[0]+4)){w=3;break t}if(18>w[0]){w=7;break t}var S=j(x,v[0]+8),k=1+C(x,v[0]+12);if(2147483648<=k*(x=1+C(x,v[0]+15))){w=3;break t}null!=N&&(N[0]=S),null!=A&&(A[0]=k),
null!=L&&(L[0]=x),v[0]+=18,w[0]-=18,P[0]=1}w=0}if(d=d[0],m=m[0],0!=w)return w;if(v=!!(2&m),!y&&d)return 3;if(null!=o&&(o[0]=!!(16&m)),null!=h&&(h[0]=0),null!=l&&(l[0]=0),h=u[0],m=f[0],
d&&v&&null==c){w=0;break}if(4>i){w=7;break}if(y&&d||!y&&!d&&!n(t,r[0],"ALPH")){i=[i],g.na=[g.na],g.P=[g.P],g.Sa=[g.Sa];t:{S=t,w=r,y=i;var P=g.gb;A=g.na,L=g.P,N=g.Sa,k=22,e(null!=S),e(
null!=y),x=w[0];var F=y[0];for(e(null!=A),e(null!=N),A[0]=null,L[0]=null,N[0]=0;;){if(w[0]=x,y[0]=F,8>F){w=7;break t}var I=j(S,x+4);if(4294967286<I){w=3;break t}var E=8+I+1&-2;if(k+=E,
0<P&&k>P){w=3;break t}if(!n(S,x,"VP8 ")||!n(S,x,"VP8L")){w=0;break t}if(F[0]<E){w=7;break t}n(S,x,"ALPH")||(A[0]=S,L[0]=x+8,N[0]=I),x+=E,F-=E}}if(i=i[0],g.na=g.na[0],g.P=g.P[0],g.Ja=g.Sa[0],
0!=w)break}i=[i],g.Ja=[g.Ja],g.xa=[g.xa];t:if(P=t,w=r,y=i,A=g.gb[0],L=g.Ja,N=g.xa,S=w[0],x=!n(P,S,"VP8 "),k=!n(P,S,"VP8L"),e(null!=P),e(null!=y),e(null!=L),e(null!=N),8>y[0])w=7;else{if(
x||k){if(P=j(P,S+4),12<=A&&P>A-12){w=3;break t}if(p&&P>y[0]-8){w=7;break t}L[0]=P,w[0]+=8,N[0]=k}else N[0]=5<=y[0]&&47==P[S+0]&&!(P[S+4]>>5),L[0]=y[0];w=0}if(i=i[0],g.Ja=g.Ja[0],g.
xa=g.xa[0],r=[0],0!=w)break;if(4294967286<g.Ja)return 3;if(null==l||v||(l[0]=g.xa?2:1),h=[h],m=[m],g.xa){if(5>i){w=7;break}l=h,p=m,v=o,null==t||5>i?t=0:5<=i&&47==t[r+0]&&!(t[r+4]>>5)?(y=[
0],P=[0],A=[0],b(L=new _,t,r,i),gt(L,y,P,A)?(null!=l&&(l[0]=y[0]),null!=p&&(p[0]=P[0]),null!=v&&(v[0]=A[0]),t=1):t=0):t=0}else{if(10>i){w=7;break}l=m,null==t||10>i||!Jt(t,r+3,i-3)?t=0:(p=t[
r+0]|t[r+1]<<8|t[r+2]<<16,v=16383&(t[r+7]<<8|t[r+6]),t=16383&(t[r+9]<<8|t[r+8]),1&p||3<(p>>1&7)||!(p>>4&1)||p>>5>=g.Ja||!v||!t?t=0:(h&&(h[0]=v),l&&(l[0]=t),t=1))}if(!t)return 3;if(h=h[0],
m=m[0],d&&(u[0]!=h||f[0]!=m))return 3;null!=c&&(c[0]=g,c.offset=r-c.w,e(4294967286>r-c.w),e(c.offset==c.ha-i));break}return 0==w||7==w&&d&&null==c?(null!=o&&(o[0]=null!=g.na&&0<g.na.
length),null!=a&&(a[0]=h),null!=s&&(s[0]=m),0):w}function Bn(t,e,n){var r=e.width,i=e.height,a=0,s=0,o=r,h=i;if(e.Da=null!=t&&0<t.Da,e.Da&&(o=t.cd,h=t.bd,a=t.v,s=t.j,11>n||(a&=-2,s&=-2),
0>a||0>s||0>=o||0>=h||a+o>r||s+h>i))return 0;if(e.v=a,e.j=s,e.va=a+o,e.o=s+h,e.U=o,e.T=h,e.da=null!=t&&0<t.da,e.da){if(!M(o,h,n=[t.ib],a=[t.hb]))return 0;e.ib=n[0],e.hb=a[0]}return e.
ob=null!=t&&t.ob,e.Kb=null==t||!t.Sd,e.da&&(e.ob=e.ib<3*r/4&&e.hb<3*i/4,e.Kb=0),1}function Mn(t){if(null==t)return 2;if(11>t.S){var e=t.F.RGBA;e.fb+=(t.height-1)*e.A,e.A=-e.A}else e=t.f.kb,
t=t.height,e.O+=(t-1)*e.fa,e.fa=-e.fa,e.N+=(t-1>>1)*e.Ab,e.Ab=-e.Ab,e.W+=(t-1>>1)*e.Db,e.Db=-e.Db,null!=e.F&&(e.J+=(t-1)*e.lb,e.lb=-e.lb);return 0}function Rn(t,e,n,r){if(
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null==r||0>=t||0>=e)return 2;if(null!=n){if(n.Da){var i=n.cd,s=n.bd,o=-2&n.v,h=-2&n.j;if(0>o||0>h||0>=i||0>=s||o+i>t||h+s>e)return 2;t=i,e=s}if(n.da){if(!M(t,e,i=[n.ib],s=[n.hb]))return 2;
t=i[0],e=s[0]}}r.width=t,r.height=e;t:{var l=r.width,c=r.height;if(t=r.S,0>=l||0>=c||!(t>=Br&&13>t))t=2;else{if(0>=r.Rd&&null==r.sd){o=s=i=e=0;var u=(h=l*zi[t])*c;if(11>t||(s=(c+1)/2*(e=(
l+1)/2),12==t&&(o=(i=l)*c)),null==(c=a(u+2*s+o))){t=1;break t}r.sd=c,11>t?((l=r.f.RGBA).eb=c,l.fb=0,l.A=h,l.size=u):((l=r.f.kb).y=c,l.O=0,l.fa=h,l.Fd=u,l.f=c,l.N=0+u,l.Ab=e,l.Cd=s,l.ea=c,l.
W=0+u+s,l.Db=e,l.Ed=s,12==t&&(l.F=c,l.J=0+u+2*s),l.Tc=o,l.lb=i)}if(e=1,i=r.S,s=r.width,o=r.height,i>=Br&&13>i)if(11>i)t=r.f.RGBA,e&=(h=Math.abs(t.A))*(o-1)+s<=t.size,e&=h>=s*zi[i],
e&=null!=t.eb;else{t=r.f.kb,h=(s+1)/2,u=(o+1)/2,l=Math.abs(t.fa),c=Math.abs(t.Ab);var f=Math.abs(t.Db),d=Math.abs(t.lb),p=d*(o-1)+s;e&=l*(o-1)+s<=t.Fd,e&=c*(u-1)+h<=t.Cd,e=(e&=f*(u-1)+h<=t.
Ed)&l>=s&c>=h&f>=h,e&=null!=t.y,e&=null!=t.f,e&=null!=t.ea,12==i&&(e&=d>=s,e&=p<=t.Tc,e&=null!=t.F)}else e=0;t=e?0:2}}var 0!=t||null!=n&&n.fd&&(t=Mn(r)),t}var Tn=64,Dn=[0,1,3,7,15,31,63,
127,255,511,1023,2047,4095,8191,16383,32767,65535,131071,262143,524287,1048575,2097151,4194303,8388607,16777215],qn=24,zn=32,Un=8,Hn=[0,0,1,1,2,2,2,2,3,3,3,3,3,3,3,3,4,4,4,4,4,4,4,4,4,4,4,4,
4,4,4,4,5,5,5

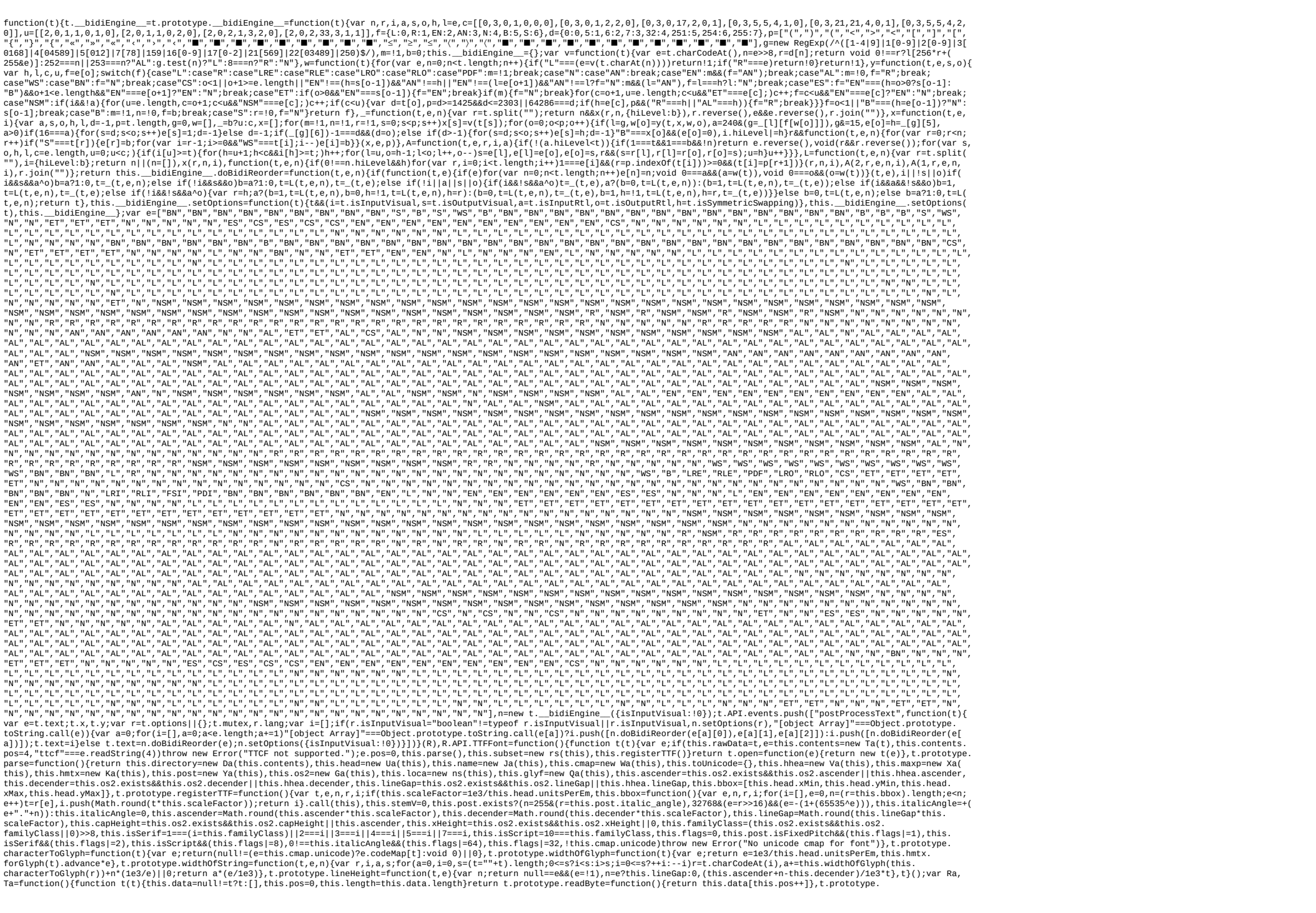
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255],[255,255,255,255,255,255,255,255,255,255,255,255],[255,255,255,255,255,255,255,255,255,255,255,255]],[[255,255,255,255,255,255,255,255,255,255,255,255],[255,255,255,255,255,255,255,255,255,255,255,255],[255,255,255,255,255,255,255,255,255,255,255,255]],[[[248,255,255,255,255,255,255,255,255,255,255,255],[250,254,252,254,255,255,255,255,255,255,255,255],[248,254,249,253,255,255,255,255,255,255,255,255]],[[255,253,253,255,255,255,255,255,255,255,255,255],[246,253,253,255,255,255,255,255,255,255,255,255],[252,254,251,254,254,255,255,255,255,255,255,255]],[[255,254,252,255,255,255,255,255,255,255,255,255],[248,254,253,255,255,255,255,255,255,255,255,255],[253,255,254,254,255,255,255,255,255,255,255,255]],[[255,251,254,255,255,255,255,255,255,255,255,255],[245,251,254,255,255,255,255,255,255,255,255,255],[253,253,254,255,255,255,255,255,255,255,255,255]],[[255,251,253,255,255,255,255,255,255,255,255,255],[252,253,254,255,255,255,255,255,255,255,255,255],[255,254,255,255,255,255,255,255,255,255,255,255]],[[255,252,255,255,255,255,255,255,255,255,255,255],[249,255,254,255,255,255,255,255,255,255,255,255],[255,255,254,255,255,255,255,255,255,255,255,255]],[[255,255,253,255,255,255,255,255,255,255,255,255],[250,255,255,255,255,255,255,255,255,255,255,255],[255,255,255,255,255,255,255,255,255,255,255,255]],[[255,255,255,255,255,255,255,255,255,255,255,255],[254,255,255,255,255,255,255,255,255,255,255,255],[255,255,255,255,255,255,255,255,255,255,255,255]]],li=[0,1,2,3,6,4,5,6,6,6,6,6,6,6,6,7,0],ci=[],ui=[],fi=[],di=1,pi=2,gi=[],mi=[];bn("UpsampleRgbLinePair",An,3),bn("UpsampleBgrLinePair",Ln,3),bn("UpsampleRgbaLinePair",Fn,4),bn("UpsampleArgbLinePair",Pn,4),bn("UpsampleArgbaLinePair",kn,4),bn("UpsampleRgba4444LinePair",Sn,2),bn("UpsampleRgb565LinePair",Nn,2);var bi=t.UpsampleRgbLinePair,vi=t.UpsampleBgrLinePair,wi=t.UpsampleRgbaLinePair,yi=t.UpsampleBgraLinePair,_i=t.UpsampleArgbLinePair,xi=t.UpsampleRgba4444LinePair,Ai=t.UpsampleRgb565LinePair,Li=16,Ni=1<<Li-1,Si=-227,ki=482,Pi=6,Fi=(256<<Pi)-1,Ii=0,Ci=a(256),ji=a(256),Ei=a(256),Oi=a(256),Bi=a(ki-Si),Mi=a(ki-Si);In("YuvToRgbRow",An,3),In("YuvToRgbaRow",Fn,4),In("YuvToBgraRow",Pn,4),In("YuvToArgbRow",kn,4),In("YuvToRgba4444Row",Sn,2),In("YuvToRgb565Row",Nn,2);var Ri=[0,4,8,12,128,132,136,140,256,260,264,268,384,388,392,396],Ti=[0,2,8],Di=[8,7,6,4,4,2,2,2,1,1,1,1],qi=1;this.WebPDecodeRGBA=function(t,n,o,h,l){var c=Mr,u=new en,f=new st;u.ba=f,f.S=c,f.width=[f.width],f.height=[f.height];var d=f.width,p=f.height,g=new ot;if(null==g||null==t)var m=2;else e(null!=g),m=On(t,n,o,g.width,g.height,g.Pd,g.Qd,g.format,null);if(0!=m?d=0:(null!=d&&(d[0]=g.width[0]),null!=p&&(p[0]=g.height[0]),d=1),d){f.width=f.width[0],f.height=f.height[0],null!=h&&(h[0]=f.width),null!=l&&(l[0]=f.height);t:{if(h=new Gt,(l=new nn).data=t,l.w=n,l.ha=o,l.kd=1,n=[0],e(null!=l),(0==(t=On(l.data,l.w,l.ha,null,null,null,n,null,l))||7==t)&&n[0]&&(t=4),0==(n=t)){if(e(null!=u),h.data=l.data,h.w=l.w+l.offset,h.ha=l.ha-l.offset,h.put=dt,h.ac=ft,h.bc=pt,h.ma=u,l.xa){if(null==(t=Pt())){u=1;break t}if(function(t,n){var r=[0],i=[0],a=[0];e:for(;;){if(null==t)return 0;if(null==n)return t.a=2,0;if(t.l=n,t.a=0,b(t.m,n.data,n.w,n.ha),!gt(t.m,r,i,a)){t.a=3;break e}if(t.xb=pi,n.width=r[0],n.height=i[0],!Ft(r[0],i[0],1,t,null))break e;return 1}return e(0!=t.a),0}(t,h)){if(h=0==(n=Rn(h.width,h.height,u.Oa,u.ba))){e:{h=t;n:for(;;){if(null==h){h=0;break e}if(e(null!=h.s.yc),e(null!=h.s.Ya),e(0<h.s.Wb),e(null!=(o=h.l)),e(null!=(l=o.ma)),0!=h.xb){if(h.ca=l.ba,h.tb=l.tb,e(null!=h.ca),!Bn(l.Oa,o,Tr)){h.a=2;break n}if(!It(h,o.width))break n;if(o.da)break n;if((o.da||rt(h.ca.S))&&mn(),11>h.ca.S||(alert("todo:WebPInitConvertARGBToYUV"),null!=h.ca.f.kb.F&&mn()),h.Pb&&0<h.s.ua&&null==h.s.vb.X&&!E(h.s.vb,h.s.Wa.Xa)){h.a=1;break n}h.xb=0}if(!St(h,h.V,h.Ba,h.c,h.i,o.o,xt))break n;l.Dc=h.Ma,h=1;break e}e(0!=h.a),h=0}h=!h}h&&(n=t.a)}else n=t.a}else{if(null==(t=new Yt)){u=1;break t}if(t.Fa=l.na,t.P=l.P,t.qc=l.Sa,Xt(t,h)){if(0==(n=Rn(h.width,h.height,u.Oa,u.ba))){if(t.Aa=0,o=u.Oa,e(null!=(l=t)),null!=o){if(0<(d=0>(d=o.Md)?0:100<d?255:255*d/100)){for(p=g=0;4>p;++p)12>(m=l.pb[p]).lc&&(m.ia=d*Di[0>m.lc?0:m.lc]>>3),g|=m.ia;g&&(alert("todo:VP8InitRandom"),l.ia=1)}l.Ga=o.Id,100<l.Ga?l.Ga=100:0>l.Ga&&(l.Ga=0)}(function(t,n){if(null==t)return 0;if(null==n)return Zt(t,2,"NULL VP8Io parameter in VP8Decode().");if(!t.cb&&!Xt(t,n))return 0;if(e(t.cb),null==n.ac||n.ac(n)){n.ob&&(t.L=0);var o=Ti[t.L];if(2==t.L?(t.yb=0,t.zb=0):(t.yb=n.v-o>>4,t.zb=n.j-o>>4,0>t.yb&&(t.yb=0),0>t.zb&&(t.zb=0)),t.Va=n.o+15+o>>4,t.Hb=n.va+15+o>>4,t.Hb>t.za&&(t.Hb=t.za),t.Va>t.Ub&&(t.Va=t.Ub),0<t.L){var h=t.ed;for(o=0;4>o;++o){var l;if(t.Qa.Cb){var c=t.Qa.Lb[o];t.Qa.Fb||(c+=h.Tb)}else c=h.Tb;for(l=0;1>=l;++l){var u=t.gd[o][l],f=c;if(h.Pc&&(f+=h.vd[0],l&&(f+=h.od[0])),0<(f=0>f?0:63<f?63:f)){var d=f;0<h.wb&&(d=4<h.wb?d>>2:d>>1)>9-h.wb&&(d=9-h.wb),1>d&&(d=1),u.dd=d,u.tc=2*f+d,u.ld=40<=f?2:15<=f?1:0}else u.tc=0;u.La=l}}}o=0}else Zt(t,6,"Frame setup failed"),o=t.a;if(o=0==o){if(o){t.$c=0,0<t.Aa||(t.Ic=qi);e:{o=t.Ic,h=4*(d=t.za);var p=32*d,g=d+1,m=0<t.L?d*(0<t.Aa?2:1):0,b=(2==t.Aa?2:1)*d;if((u=h+832+(l=3*(16*o+Ti[t.L])/2*p)+(c=null!=t.Fa&&0<t.Fa.length?t.Kc.c*t.Kc.i:0))!=u)o=0;else{if(u>t.Vb){if(t.Vb=0,t.Ec=a(u),t.Fc=0,null==t.Ec){o=Zt(t,1,"no memory during frame initialization.");break e}t.Vb=u}u=t.Ec,f=t.Fc,t.Ac=u,t.Bc=f,f+=h,t.Gd=s(p,Ht),t.Hd=0,t.rb=s(g+1,Dt),t.sb=1,t.wa=m?s(m,Tt):null,t.Y=0,t.D.Nb=0,t.D.wa=t.wa,t.D.Y=t.Y,0<t.Aa&&(t.D.Y+=d),e(!0),t.oc=u,t.pc=f,f+=832,t.ya=s(b,zt),t.aa=0,t.D.ya=t.ya,t.D.aa=t.aa,2==t.Aa&&(t.D.aa+=d),t.R=16*d,t.B=8*d,d=(p=Ti[t.L])*t.R,p=p/2*t.B,t.sa=u,t.ta=f+d,t.qa=t.sa,t.ra=t.ta+16*o*t.R+p,t.Ha=t.qa,t.Ia=t.ra+8*o*t.B+p,t.$c=0,f+=l,t.mb=c?u:null,t.nb=c?f:null,e(f+c<=t.Fc+t.Vb),$t(t),i(t.Ac,t.Bc,0,h),o=1}}if(o){if(n.ka=0,n.y=t.sa,n.O=t.ta,n.f=t.qa,n.N=t.ra,n.ea=t.Ha,n.Vd=t.Ia,n.fa=t.R,n.Rc=t.B,n.F=null,n.J=0,!Ir){for(o=-255;255>=o;++o)Sr[255+o]=0>o?-o:o;for(o=-1020;1020>=o;++o)kr[1020+o]=-128>o?-128:127<o?127:o;for(o=-112;112>=o;++o)Pr[112+o]=-16>o?-16:15<o?15:o;for(o=-255;510>=o;++o)Fr[255+o]=0>o?0:255<o?255:o;Ir=1}ir=he,ar=ie,or=ae,hr=se,lr=oe,sr=re,cr=Ye,ur=Ze,fr=Ke,dr=$e,pr=Je,gr=Xe,mr=Qe,br=tn,vr=ze,wr=Ue,yr=He,_r=We,ui[0]=Ae,ui[1]=ce,ui[2]=_e,ui[3]=xe,ui[4]=Le,ui[5]=Se,ui[6]=Ne,ui[7]=ke,ui[8]=Fe,ui[9]=Pe,ci[0]=me,ci[1]=fe,ci[2]=de,ci[3]=pe,ci[4]=be,ci[5]=ve,ci[6]=we,fi[0]=Ee,fi[1]=ue,fi[2]=Ie,fi[3]=Ce,fi[4]=Be,fi[5]=Oe,fi[6]=Me,o=1}else o=0}o&&(o=function(t,n){for(t.M=0;t.M<t.Va;++t.M){var s,o=t.Jc[t.M&t.Xb],h=t.m,l=t;for(s=0;s<l.za;++s){var c=h,u=l,f=u.Ac,d=u.Bc+4*s,p=u.zc,g=u.ya[u.aa+s];if(u.Qa.Bb?g.$b=P(c,u.Pa.jb[0])?2+P(c,u.Pa.jb[2]):P(c,u.Pa.jb[1]):g.$b=0,u.kc&&(g.Ad=P(c,u.Bd)),g.Za=!P(c,145)+0,g.Za){var m=g.Ob,b=0;for(u=0;4>u;++u){var v,w=p[0+u];for(v=0;4>v;++v){w=oi[f[d+v]][w];for(var y=ai[P(c,w[0])];0<y;)y=ai[2*y+P(c,w[y])];w=-y,f[d+v]=w}r(m,b,f,d,4),b+=4,p[0+u]=w}}else w=P(c,156)?P(c,128)?1:3:P(c,163)?2:0,g.Ob[0]=w,i(f,d,w,4),i(p,0,w,4);g.Dd=P(c,142)?P(c,114)?P(c,183)?1:3:2:0}if(l.m.Ka)return Zt(t,7,"Premature end-of-partition0 encountered.");for(;t.ja<t.za;++t.ja){if(l=o,c=(h=t).rb[h.sb-1],f=h.rb[h.sb+h.ja],s=h.ya[h.aa+h.ja],d=h.kc?s.Ad:0)c.la=f.la=0,s.Za||(c.Na=f.Na=0),s.Hc=0,s.Gc=0,s.ia=0;else{var _,x;if(c=f,f=l,d=h.Pa.Xc,p=h.ya[h.aa+h.ja],g=h.pb[p.$b],u=p.ad,m=0,b=h.rb[h.sb-1],w=v=0,i(u,m,0,384),p.Za)var A=0,L=d[3];else{y=a(16);var N=c.Na+b.Na;if(N=ni(f,d[1],N,g.Eb,0,y,0),c.Na=b.Na=(0<N)+0,1<N)ir(y,0,u,m);else{var S=y[0]+3>>3;for(y=0;256>y;y+=16)u[m+y]=S}A=1,L=d[0]}var k=15&c.la,F=15&b.la;for(y=0;4>y;++y){var I=1&F;for(S=x=0;4>S;++S)k=k>>1|(I=(N=ni(f,L,N=I+(1&k),g.Sc,A,u,m))>A)<<7,x=x<<2|(3<N?3:1<N?2:0!=u[m+0]),m+=16;k>>=4,F=F>>1|I<<7,v=(v<<8|x)>>>0}for(L=k,A=F>>4,_=0;4>_;_+=2){for(x=0,k=c.la>>4+_,F=b.la>>4+_,y=0;2>y;++y){for(I=1&F,S=0;2>S;++S)N=I+(1&k),k=k>>1|(I=0<(N=ni(f,d[2],N,g.Qc,0,u,m))<<3,x=x<<2|(3<N?3:1<N?2:0!=u[m+0]),m+=16;k>>=2,F=F>>1|I<<5}w|=x<<4*_,L|=k<<4<<_,A|=(240&F)<<_}c.la=L,b.la=A,p.Hc=v,p.Gc=w,p.ia=43690&w?0:g.ia,d=!(v|w)}if(0<h.L&&(h.wa[h.Y+h.ja]=h.gd[s.$b][s.Za],h.wa[h.Y+h.ja].La|=!d),l.Ka)return Zt(t,7,"Premature end-of-file encountered.")}if($t(t),h=n,l=1,s=(o=t).D,c=0<o.L&&o.M>=o.zb&&o.M<=o.Va,0==o.Aa)e:{if(s.M=o.M,s.uc=c,En(o,s),l=1,s=(x=o.D).Nb,c=(w=Ti[o.L])*o.R,f=w/2*o.B,y=16*s*o.R,S=8*s*o.B,d=o.sa,p=o.ta-c+y,g=o.qa,u=o.ra-f+S,m=o.Ha,b=o.Ia-f+S,F=0==(k=x.M),v=k>>o.Va-1,2==o.Aa&&En(o,x),x.uc)for(I=(N=o).D.M,e(N.D.uc),x=N.yb;x<N.Hb;++x){A=x,L=I;var C=(j=(z=N).D).Nb;_=z.R;var j=j.wa[j.Y+A],E=z.sa,O=z.ta+16*C*_+16*A,B=j.dd,M=j.tc;if(0!=M)if(e(3<=M),1==z.L)0<A&&wr(E,O,_,M+4),j.La&&_r(E,O,_,M),0<L&&vr(E,O,_,M+4),j.La&&yr(E,O,_,M);else{var R=z.B,T=z.qa,D=z.ra+8*C*R+8*A,q=z.Ha,z=z.Ia+8*C*R+8*A;C=j.ld,0<A&&(ur(E,O,_,M+4,B,C),dr(T,D,q,z,R,M+4,B,C)),j.La&&(gr(E,O,_,M,B,C),br(T,D,q,z,R,M,B,C)),0<L&&(cr(E,O,_,M+4,B,C),fr(T,D,q,z,R,M+4,B,C)),j.La&&(pr(E,O,_,M,B,C),mr(T,D,q,z,R,M,B,C))}}if(o.ia&&alert("todo:DitherRow"),null!=h.put){if(x=16*k,k=16*(k+1),F?(h.y=o.sa,h.O=o.ta+y,h.f=o.qa,h.N=o.ra+S,h.ea=o.Ha,h.W=o.Ia+S):(x-=w,h.y=d,h.O=p,h.f=g,h.N=u,h.ea=m,h.W=b),v||(k-=w),k>h.o&&(k=h.o),h.F=null,h.J=null,null!=o.Fa&&0<o.Fa.length&&x<k&&(h.J=un(o,h,x,k-x),h.F=o.mb,null==h.F&&0==h.F.length)){l=Zt(o,3,"Could not decode alpha data.");break e}x<h.j&&(w=h.j-x,x=h.j,e(!(1&w)),h.O+=o.R*w,h.N+=o.B*(w>>1),h.W+=o.B*(w>>1),null!=h.F&&(h.J+=h.width*w)),x<k&&(h.O+=h.v,h.N+=h.v>>1,h.W+=h.v>>1,null!=h.F&&(h.J+=h.v),h.ka=x-h.j,h.U=h.va-h.v,h.T=k-x,l=h.put(h))}s+1!=o.Ic||v||(r(o.sa,o.ta-c,d,p+16*o.R,c),r(o.qa,o.ra-f,g,u+8*o.B,f),r(o.Ha,o.Ia-f,m,b+8*o.B,f))}if(!l)return Zt(t,6,"Output aborted.")}return 1}(t,n)),null!=n.bc&&n.bc(n),o&=1}return o?(t.cb=0,o):0})(t,h)||(n=t.a)}}else n=t.a}0==n&&null!=u.Oa&&u.Oa.fd&&(n=Mn(u.ba))}u=n}c=0!=u?null:11>c?f.f.RGBA.eb:f.f.kb.y}else c=null;return c};var zi=[3,4,3,4,4,2,2,4,4,4,2,1,1];function l(t,e){for(var n="",r=0;r<4;r++)n+=String.fromCharCode(t[e++]);return n}function c(t,e){return t[e+0]|t[e+1]<<8}function u(t,e){return(t[e+0]|t[e+1]<<8|t[e+2]<<16)>>>0}function f(t,e){return(t[e+0]|t[e+1]<<8|t[e+2]<<16|t[e+3]<<24)>>>0}new h;var d=[0],p=[0],g=[],m=new h,b=t,v=function(t,e){var n={},r=0,i=!1,a=0,s=0;if(n.frames=[],!
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    https://webpjs.appspot.com
    WebPRiffParser dominikhlbg@gmail.com
    */
function(t,e){for(var n=0;n<4;n++)if(t[e+n]!="RIFF".charCodeAt(n))return!0;return!1}(t,e)){for(f(t,e+=4),e+=8;e<t.length;){var o=l(t,e),h=f(t,e+=4);e+=4;var d=h+(1&h);switch(o){case"VP8 ":case"VP8L":void 0===n.frames[r]&&(n.frames[r]={}),(m=n.frames[r]).src_off=i?s:e-8,m.src_size=a+h+8,r++,i&&(i=!1,a=0,s=0);break;case"VP8X":(m=n.header={}).feature_flags=t[e];var p=e+4;m.canvas_width=1+u(t,p),p+=3,m.canvas_height=1+u(t,p),p+=3;break;case"ALPH":i=!0,a=d+8,s=e-8;break;case"ANIM":(m=n.header).bgcolor=f(t,e),p=e+4,m.loop_count=c(t,p),p+=2;break;case"ANMF":var g,m;(m=n.frames[r]={}).offset_x=2*u(t,e),e+=3,m.offset_y=2*u(t,e),e+=3,m.width=1+u(t,e),e+=3,m.height=1+u(t,e),e+=3,m.duration=u(t,e),e+=3,g=t[e++],m.dispose=1&g,m.blend=g>>1&1}"ANMF"!=o&&(e+=d)}return n}}(b,0);v.response=b,v.rgbaoutput=!0,v.dataurl=!1;var w=v.header?v.header:null,y=v.frames?v.frames:null;if(w){w.loop_counter=w.loop_count,d=[w.canvas_height],p=[w.canvas_width];for(var _=0;_<y.length&&0!=y[_].blend;_++);var x=y[0],A=m.WebPDecodeRGBA(b,x.src_off,x.src_size,p,d);x.rgba=A,x.imgwidth=p[0],x.imgheight=d[0];for(var L=0;L<p[0]*d[0]*4;L++)g[L]=A[L];return this.width=p,this.height=d,this.data=g,this}
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   * ================================================================
   */
function Ba(){var t,e=this.internal.__metadata__.metadata,n=unescape(encodeURIComponent(e));t=this.internal.__metadata__.rawXml?n:'<x:xmpmeta xmlns:x="adobe:ns:meta/"><rdf:RDF xmlns:
rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"><rdf:Description rdf:about="" xmlns:jspdf="'+this.internal.__metadata__.namespaceUri+'"><jspdf:metadata>'+n.replace(/&/g,"&amp;").replace(/
</g,"&lt;").replace(/>/g,"&gt;").replace(/"/g,"&quot;").replace(/'/g,"&apos;")+"</jspdf:metadata></rdf:Description></rdf:RDF></x:xmpmeta>",this.internal.__metadata__.
metadataObjectNumber=this.internal.newObject(),this.internal.write("<< /Type /Metadata /Subtype /XML /Length "+t.length+" >>"),this.internal.write("stream"),this.internal.write(t),this.
internal.write("endstream"),this.internal.write("endobj")}function Ma(){this.internal.__metadata__.metadataObjectNumber&&this.internal.write("/Metadata "+this.internal.__metadata__.
metadataObjectNumber+" 0 R")}!function(t){t[t.UNKNOWN=0]="UNKNOWN",t[t.METRE=1]="METRE"}(da||(da={})),R.API.processPNG=function(t,e,r,i){if(this.__addimage__.isArrayBuffer(t)&&(t=new
Uint8Array(t)),this.__addimage__.isArrayBufferView(t)){var a,s=new fa(t,{checkCrc:!0}).decode(),o=s.width,h=s.height,l=s.channels,c=s.palette,u=s.depth;a=c&&1===l?function(t){for(var e=t.
width,r=t.height,i=t.data,a=t.palette,s=t.depth,o=!1,h=[],l=[],c=void 0,u=!1,f=0,d=0;d<a.length;d++){var p=n(a[d],4),g=p[0],m=p[1],b=p[2],v=p[3];h.push(g,m,b),null!=v&&(0===v?(f++,l.
length<1&&l.push(d)):v<255&&(u=!0))}if(u||f>1){o=!0,l=void 0;var w=e*r;c=new Uint8Array(w);for(var y=new DataView(i.buffer),_=0;_<w;_++){var x=n(a[ka(y,_,s)],4)[3];c[_]=x}}else 0===f&&(
l=void 0);return{colorSpace:"Indexed",colorsPerPixel:1,sMaskBitsPerComponent:o?8:void 0,colorBytes:i,alphaBytes:c,needSMask:o,palette:h,mask:l}}(s):2===l||4===l?function(t){for(var e=t.data,
n=t.width,r=t.height,i=t.channels,a=t.depth,s=2===i?"DeviceGray":"DeviceRGB",o=i-1,h=n*r,l=o,c=h*l,u=1*h,f=Math.ceil(c*a/8),d=Math.ceil(u*a/8),p=new Uint8Array(f),g=new Uint8Array(d),m=new
DataView(e.buffer),b=new DataView(p.buffer),v=new DataView(g.buffer),w=!1,y=0;y<h;y++){for(var _=y*i,x=0;x<l;x++)Pa(b,ka(m,_+x,a),y*l+x,a);var A=ka(m,_+l,a);A<(1<<a)-1&&(w=!0),Pa(v,A,1*y,
a)}return{colorSpace:s,colorsPerPixel:o,sMaskBitsPerComponent:w?a:void 0,colorBytes:p,alphaBytes:g,needSMask:w}}(s):function(t){var e,n=t.data,r=1===t.channels?"DeviceGray":"DeviceRGB",
i="DeviceGray"===r?1:3;return e=n instanceof Uint16Array?function(t){for(var e=t.length,n=new Uint8Array(2*e),r=new DataView(n.buffer,n.byteOffset,n.byteLength),i=0;i<e;i++)r.setUint16(2*i,
t[i],!1);return n}(n):n,{colorSpace:r,colorsPerPixel:i,colorBytes:e,needSMask:!1}}(s);var f,d,p,g=a,m=g.colorSpace,b=g.colorsPerPixel,v=g.sMaskBitsPerComponent,w=g.colorBytes,y=g.alphaBytes,
_=g.needSMask,x=g.palette,A=g.mask,L=null;return i!==R.API.image_compression.NONE?(L=function(t){var e;switch(t){case R.API.image_compression.FAST:e=11;break;case R.API.image_compression.
MEDIUM:e=13;break;case R.API.image_compression.SLOW:e=14;break;default:e=12}return e}(i),f=this.decode.FLATE_DECODE,d="/Predictor ".concat(L," /Colors ").concat(b," /BitsPerComponent ").
concat(u," /Columns ").concat(o),t=wa(w,Math.ceil(o*b*u/8),b,u,i),_&&(p=wa(y,Math.ceil(o*v/8),1,v,i))):(f=void 0,d=void 0,t=w,_&&(p=y)),(this.__addimage__.isArrayBuffer(t)||this.
__addimage__.isArrayBufferView(t))&&(t=this.__addimage__.arrayBufferToBinaryString(t)),(p&&this.__addimage__.isArrayBuffer(p)||this.__addimage__.isArrayBufferView(p))&&(p=this.__addimage__.
arrayBufferToBinaryString(p)),{alias:r,data:t,index:e,filter:f,decodeParameters:d,transparency:A,palette:x,sMask:p,predictor:L,width:o,height:h,bitsPerComponent:u,sMaskBitsPerComponent:v,
colorSpace:m}}},function(t){t.processGIF89A=function(e,n,r,i){var a=new Ia(e),s=a.width,o=a.height,h=[];a.decodeAndBlitFrameRGBA(0,h);var l={data:h,width:s,height:o},c=new ja(100).encode(l,
100);return t.processJPEG.call(this,c,n,r,i)},t.processGIF87A=t.processGIF89A}(R.API),Ea.prototype.parseHeader=function(){if(this.fileSize=this.datav.getUint32(this.pos,!0),this.pos+=4,this.
reserved=this.datav.getUint32(this.pos,!0),this.pos+=4,this.offset=this.datav.getUint32(this.pos,!0),this.pos+=4,this.headerSize=this.datav.getUint32(this.pos,!0),this.pos+=4,this.
width=this.datav.getUint32(this.pos,!0),this.pos+=4,this.height=this.datav.getInt32(this.pos,!0),this.pos+=4,this.planes=this.datav.getUint16(this.pos,!0),this.pos+=2,this.bitPP=this.datav.
getUint16(this.pos,!0),this.pos+=2,this.compress=this.datav.getUint32(this.pos,!0),this.pos+=4,this.rawSize=this.datav.getUint32(this.pos,!0),this.pos+=4,this.hr=this.datav.getUint32(this.
pos,!0),this.pos+=4,this.vr=this.datav.getUint32(this.pos,!0),this.pos+=4,this.colors=this.datav.getUint32(this.pos,!0),this.pos+=4,this.importantColors=this.datav.getUint32(this.pos,!0),
this.pos+=4,16===this.bitPP&&this.is_with_alpha&&(this.bitPP=15),this.bitPP<15){var t=0===this.colors?1<<this.bitPP:this.colors;this.palette=new Array(t);for(var e=0;e<t;e++){var n=this.
datav.getUint8(this.pos++,!0),r=this.datav.getUint8(this.pos++,!0),i=this.datav.getUint8(this.pos++,!0),a=this.datav.getUint8(this.pos++,!0);this.palette[e]={red:i,green:r,blue:n,quad:
a}}}this.height<0&&(this.height*=-1,this.bottom_up=!1)},Ea.prototype.parseBGR=function(){this.pos=this.offset;var t="bit"+this.bitPP,e=this.width*this.height*4;if(e>536870912)throw new
Error("Image dimensions exceed 512MB, which is too large.");this.data=new Uint8Array(e);try{this[t]()}catch(is){s.log("bit decode error:"+is)}},Ea.prototype.bit1=function(){var t,e=Math.
ceil(this.width/8),n=e%4;for(t=this.height-1;t>=0;t--){for(var r=this.bottom_up?t:this.height-1-t,i=0;i<e;i++)for(var a=this.datav.getUint8(this.pos++,!0),s=r*this.width*4+8*i*4,o=0;
o<8&&8*i+o<this.width;o++){var h=this.palette[a>>7-o&1];this.data[s+4*o]=h.blue,this.data[s+4*o+1]=h.green,this.data[s+4*o+2]=h.red,this.data[s+4*o+3]=255}0!==n&&(this.pos+=4-n)}},Ea.
prototype.bit4=function(){for(var t=Math.ceil(this.width/2),e=t%4,n=this.height-1;n>=0;n--){for(var r=this.bottom_up?n:this.height-1-n,i=0;i<t;i++){var a=this.datav.getUint8(this.pos++,!0),
s=r*this.width*4+2*i*4,o=a>>4,h=15&a,l=this.palette[o];if(this.data[s]=l.blue,this.data[s+1]=l.green,this.data[s+2]=l.red,this.data[s+3]=255,2*i+1>=this.width)break;l=this.palette[h],this.
data[s+4]=l.blue,this.data[s+4+1]=l.green,this.data[s+4+2]=l.red,this.data[s+4+3]=255}0!==e&&(this.pos+=4-e)}},Ea.prototype.bit8=function(){for(var t=this.width%4,e=this.height-1;e>=0;e--){
for(var n=this.bottom_up?e:this.height-1-e,r=0;r<this.width;r++){var i=this.datav.getUint8(this.pos++,!0),a=n*this.width*4+4*r;if(i<this.palette.length){var s=this.palette[i];this.data[a]=s.
red,this.data[a+1]=s.green,this.data[a+2]=s.blue,this.data[a+3]=255}else this.data[a]=255,this.data[a+1]=255,this.data[a+2]=255,this.data[a+3]=255}0!==t&&(this.pos+=4-t)}},Ea.prototype.
bit15=function(){for(var t=this.width%3,e=parseInt("11111",2),n=this.height-1;n>=0;n--){for(var r=this.bottom_up?n:this.height-1-n,i=0;i<this.width;i++){var a=this.datav.getUint16(this.pos,
!0);this.pos+=2;var s=(a&e)/e*255|0,o=(a>>5&e)/e*255|0,h=(a>>10&e)/e*255|0,l=a>>15?255:0,c=r*this.width*4+4*i;this.data[c]=h,this.data[c+1]=o,this.data[c+2]=s,this.data[c+3]=l}this.pos+=t}},
Ea.prototype.bit16=function(){for(var t=this.width%3,e=parseInt("11111",2),n=parseInt("111111",2),r=this.height-1;r>=0;r--){for(var i=this.bottom_up?r:this.height-1-r,a=0;a<this.width;a++){
var s=this.datav.getUint16(this.pos,!0);this.pos+=2;var o=(s&e)/e*255|0,h=(s>>5&n)/n*255|0,l=(s>>11)/e*255|0,c=i*this.width*4+4*a;this.data[c]=l,this.data[c+1]=h,this.data[c+2]=o,this.data[
c+3]=255}this.pos+=t}},Ea.prototype.bit24=function(){for(var t=this.height-1;t>=0;t--){for(var e=this.bottom_up?t:this.height-1-t,n=0;n<this.width;n++){var r=this.datav.getUint8(this.pos++,
!0),i=this.datav.getUint8(this.pos++,!0),a=this.datav.getUint8(this.pos++,!0),s=e*this.width*4+4*n;this.data[s]=a,this.data[s+1]=i,this.data[s+2]=r,this.data[s+3]=255}this.pos+=this.
width%4}},Ea.prototype.bit32=function(){for(var t=this.height-1;t>=0;t--)for(var e=this.bottom_up?t:this.height-1-t,n=0;n<this.width;n++){var r=this.datav.getUint8(this.pos++,!0),i=this.
datav.getUint8(this.pos++,!0),a=this.datav.getUint8(this.pos++,!0),s=this.datav.getUint8(this.pos++,!0),o=e*this.width*4+4*n;this.data[o]=a,this.data[o+1]=i,this.data[o+2]=r,this.data[
o+3]=s}},Ea.prototype.getData=function(){return this.data},
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   */
function(t){t.processBMP=function(e,n,r,i){var a=new Ea(e,!1),s=a.width,o=a.height,h={data:a.getData(),width:s,height:o},l=new ja(100).encode(h,100);return t.processJPEG.call(this,l,n,r,
i)}}(R.API),Oa.prototype.getData=function(){return this.data},
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   */
function(t){t.processWEBP=function(e,n,r,i){var a=new Oa(e),s=a.width,o=a.height,h={data:a.getData(),width:s,height:o},l=new ja(100).encode(h,100);return t.processJPEG.call(this,l,n,r,i)}}(
R.API),
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   * WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN THE SOFTWARE.
   * ================================================================
   */
function(t){t.processRGBA=function(t,e,n){for(var r=t.data,i=r.length,a=new Uint8Array(i/4*3),s=new Uint8Array(i/4),o=0,h=0,l=0;l<i;l+=4){var c=r[l],u=r[l+1],f=r[l+2],d=r[l+3];a[o++]=c,a[
o++]=u,a[o++]=f,s[h++]=d}var p=this.__addimage__.arrayBufferToBinaryString(a);return{alpha:this.__addimage__.arrayBufferToBinaryString(s),data:p,index:e,alias:n,colorSpace:"DeviceRGB",
bitsPerComponent:8,width:t.width,height:t.height}}}(R.API),
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   */
function(t){t.setLanguage=function(t){return void 0===this.internal.languageSettings&&(this.internal.languageSettings={},this.internal.languageSettings.isSubscribed=!1),void 0!=={af:
"Afrikaans",sq:"Albanian",ar:"Arabic (Standard)",ar-DZ":"Arabic (Algeria)","ar-BH":"Arabic (Bahrain)","ar-EG":"Arabic (Egypt)","ar-IQ":"Arabic (Iraq)","ar-JO":"Arabic (Jordan)","ar-KW":
"Arabic (Kuwait)","ar-LB":"Arabic (Lebanon)","ar-LY":"Arabic (Libya)","ar-MA":"Arabic (Morocco)","ar-OM":"Arabic (Oman)","ar-QA":"Arabic (Qatar)","ar-SA":"Arabic (Saudi Arabia)","ar-SY":
"Arabic (Syria)","ar-TN":"Arabic (Tunisia)","ar-AE":"Arabic (U.A.E.)","ar-YE":"Arabic (Yemen)",an:"Aragonese",hy:"Armenian",as:"Assamese",ast:"Asturian",az:"Azerbaijani",eu:"Basque",be:
"Belarusian",bn:"Bengali",bs:"Bosnian",br:"Breton",bg:"Bulgarian",my:"Burmese",ca:"Catalan",ch:"Chamorro",ce:"Chechen",zh:"Chinese","zh-HK":"Chinese (Hong Kong)","zh-CN":"Chinese (PRC)","zh-
SG":"Chinese (Singapore)","zh-TW":"Chinese (Taiwan)",cv:"Chuvash",co:"Corsican",cr:"Cree",hr:"Croatian",cs:"Czech",da:"Danish",nl:"Dutch (Standard)","nl-BE":"Dutch (Belgian)",en:"English",
"en-AU":"English (Australia)","en-BZ":"English (Belize)","en-CA":"English (Canada)","en-IE":"English (Ireland)","en-JM":"English (Jamaica)","en-NZ":"English (New Zealand)","en-PH":"English (
Philippines)","en-ZA":"English (South Africa)","en-TT":"English (Trinidad & Tobago)","en-GB":"English (United Kingdom)","en-US":"English (United States)","en-ZW":"English (Zimbabwe)",eo:
"Esperanto",et:"Estonian",fo:"Faeroese",fj:"Fijian",fi:"Finnish",fr:"French (Standard)","fr-BE":"French (Belgium)","fr-CA":"French (Canada)","fr-FR":"French (France)","fr-LU":"French (
Luxembourg)","fr-MC":"French (Monaco)","fr-CH":"French (Switzerland)",fy:"Frisian",fur:"Friulian",gd:"Gaelic (Scots)","gd-IE":"Gaelic (Irish)",gl:"Galacian",ka:"Georgian",de:"German (
Standard)","de-AT":"German (Austria)","de-DE":"German (Germany)","de-LI":"German (Liechtenstein)","de-LU":"German (Luxembourg)","de-CH":"German (Switzerland)",el:"Greek",gu:"Gujurati",ht:
"Haitian",he:"Hebrew",hi:"Hindi",hu:"Hungarian",is:"Icelandic",id:"Indonesian",iu:"Inuktitut",ga:"Irish",it:"Italian (Standard)","it-CH":"Italian (Switzerland)",ja:"Japanese",kn:"Kannada",
ks:"Kashmiri",kk:"Kazakh",km:"Khmer",ky:"Kirghiz",tlh:"klingon",ko:"Korean","ko-KP":"Korean (North Korea)","ko-KR":"Korean (South Korea)",la:"Latin",lv:"Latvian",lt:"Lithuanian",lb:
"Luxembourgish",mk:"North Macedonia",ms:"Malay",ml:"Malayalam",mt:"Maltese",mi:"Maori",mr:"Marathi",mo:"Moldavian",nv:"Navajo",ng:"Ndonga",ne:"Nepali",no:"Norwegian",nb:"Norwegian (Bokmal)",
nn:"Norwegian (Nynorsk)",oc:"Occitan",or:"Oriya",om:"Oromo",fa:"Persian","fa-IR":"Persian/Iran",pl:"Polish",pt:"Portuguese","pt-BR":"Portuguese (Brazil)",pa:"Punjabi","pa-IN":"Punjabi (
India)","pa-PK":"Punjabi (Pakistan)",qu:"Quechua",rm:"Rhaeto-Romanic",ro:"Romanian","ro-MO":"Romanian (Moldavia)",ru:"Russian","ru-MO":"Russian (Moldavia)",sz:"Sami (Lappish)",sg:"Sango",sa:
"Sanskrit",sc:"Sardinian",sd:"Sindhi",si:"Singhalese",sr:"Serbian",sk:"Slovak",sl:"Slovenian",so:"Somani",sb:"Sorbian",es:"Spanish","es-AR":"Spanish (Argentina)","es-BO":"Spanish (Bolivia)",
"es-CL":"Spanish (Chile)","es-CO":"Spanish (Colombia)","es-CR":"Spanish (Costa Rica)","es-DO":"Spanish (Dominican Republic)","es-EC":"Spanish (Ecuador)","es-SV":"Spanish (El Salvador)","es-
GT":"Spanish (Guatemala)","es-HN":"Spanish (Honduras)","es-MX":"Spanish (Mexico)","es-NI":"Spanish (Nicaragua)","es-PA":"Spanish (Panama)","es-PY":"Spanish (Paraguay)","es-PE":"Spanish (
Peru)","es-PR":"Spanish (Puerto Rico)","es-ES":"Spanish (Spain)","es-UY":"Spanish (Uruguay)","es-VE":"Spanish (Venezuela)",sx:"Sutu",sw:"Swahili",sv:"Swedish","sv-FI":"Swedish (Finland)",
"sv-SV":"Swedish (Sweden)",ta:"Tamil",tt:"Tatar",te:"Teluga",th:"Thai",tig:"Tigre",ts:"Tsonga",tn:"Tswana",tr:"Turkish",tk:"Turkmen",uk:"Ukrainian",hsb:"Upper Sorbian",ur:"Urdu",ve:"Venda",
vi:"Vietnamese",vo:"Volapuk",wa:"Walloon",cy:"Welsh",xh:"Xhosa",ji:"Yiddish",zu:"Zulu"}[t]&&(this.internal.languageSettings.languageCode=t,!1===this.internal.languageSettings.isSubscribed&&(
this.internal.events.subscribe("putCatalog",function(){this.internal.write("/Lang ("+this.internal.languageSettings.languageCode+")")}),this.internal.languageSettings.isSubscribed=!0)),
this}}(R.API),pa=R.API,ga=pa.getCharWidthsArray=function(t,e){var n,i,a=(e=e||{}).font||this.internal.getFont(),s=e.fontSize||this.internal.getFontSize(),o=e.charSpace||this.internal.
getCharSpace(),h=e.widths?e.widths:a.metadata.Unicode.widths,l=h.fof?h.fof:1,c=e.kerning?e.kerning:a.metadata.Unicode.kerning,u=c.fof?c.fof:1,f=!1!==e.doKerning,d=0,p=t.length,g=0,m=h[0]||l,
b=[];for(n=0;n<p;n++)i=t.charCodeAt(n),"function"==typeof a.metadata.widthOfString?b.push((a.metadata.widthOfGlyph(a.metadata.characterToGlyph(i))+o*(1e3/s)||0)/1e3):(d=f&&"object"===r(c[
i])&&!isNaN(parseInt(c[i][g],10))?c[i][g]/u:0,b.push((h[i]||m)/l+d)),g=i;return b},ma=pa.getStringUnitWidth=function(t,e){var n=(e=e||{}).fontSize||this.internal.getFontSize(),r=e.
font||this.internal.getFont(),i=e.charSpace||this.internal.getCharSpace();return pa.processArabic&&(t=pa.processArabic(t)),"function"==typeof r.metadata.widthOfString?r.metadata.
widthOfString(t,n,i)/n:ga.apply(this,arguments).reduce(function(t,e){return t+e},0)},ba=function(t,e,n,r){for(var i=[],a=0,s=t.length,o=0;a!==s&&o+e[a]<n;)o+=e[a],a++;i.push(t.slice(0,a));
var h=a;for(o=0;a!==s;)o+e[a]>r&&(i.push(t.slice(h,a)),o=0,h=a),o+=e[a],a++;return h!==a&&i.push(t.slice(h,a)),i},va=function(t,e,n){\n||(n={});var r,i,a,s,o,h,l,c=[],u=[c],f=n.textIndent||0,
d=0,p=0,g=t.split(" "),m=ga.apply(this,[" ",n])[0];if(h=-1===n.lineIndent?g[0].length+2:n.lineIndent||0){var b=Array(h).join(" "),v=[];g.map(function(t){(t=t.split(/\s*\n/)).length>1?v=v.
concat(t.map(function(t,e){return(e&&t.length?"\n":"")+t})):v.push(t[0])}),g=v,h=ma.apply(this,[b,n])}for(a=0,s=g.length;a<s;a++){var w=0;if(r=g[a],h&&"\n"==r[0]&&(r=r.substr(1),w=1),f+d+(
p=(i=ga.apply(this,[r,n])).reduce(function(t,e){return t+e},0))>e||w){if(p>e){for(o=ba.apply(this,[r,i,e-(f+d),e]),c.push(o.shift()),c=[o.pop()];o.length;)u.push([o.shift()]);p=i.slice(r.
length-(c[0]?c[0].length:0)).reduce(function(t,e){return t+e},0)}else c=[r];u.push(c),f=p+h,d=m}else c.push(r),f+=d+p,d=m}return l=h?function(t,e){return(e?b:"")+t.join(" ")}:function(t){
return t.join(" ")},u.map(l)},pa.splitTextToSize=function(t,e,n){var r,i=(n=n||{}).fontSize||this.internal.getFontSize(),a=function(t){if(t.widths&&t.kerning)return{widths:t.widths,kerning:
t.kerning};var e=this.internal.getFont(t.fontName,t.fontStyle),n="Unicode";return e.metadata[n]?{widths:e.metadata[n].widths||{0:1},kerning:e.metadata[n].kerning||{}}:{font:e.metadata,
fontSize:this.internal.getFontSize(),charSpace:this.internal.getCharSpace()}}.call(this,n);r=Array.isArray(t)?t:String(t).split(/\r?\n/);var s=1*this.internal.scaleFactor*e/i;a.textIndent=n.
textIndent?1*n.textIndent*this.internal.scaleFactor/i:0,a.lineIndent=n.lineIndent;var o,h,l=[];for(o=0,h=r.length;o<h;o++)l=l.concat(va.apply(this,[r[o],s,a]));return l},function(t){t.
__fontmetrics__=t.__fontmetrics__||{};for(var e="0123456789abcdef",n="klmnopqrstuvwxyz",i={},a={},s=0;s<16;s++)i[n[s]]=e[s],a[e[s]]=n[s];var o=function(t){return"0x"+parseInt(t,10).toString(
16)},h=t.__fontmetrics__.compress=function(t){var e,n,i,s,l="{";for(var c in t){if(e=t[c],isNaN(parseInt(c,10))?n="'"+c+"'":(c=parseInt(c,10),n=(n=o(c).slice(2)).slice(0,-1)+a[n.slice(-
1)]),"number"==typeof e)e<0?(i=o(e).slice(3),s="-"):(i=o(e).slice(2),s="");i=s+i.slice(0,-1)+a[i.slice(-1)];else{if("object"!=r(e))throw new Error("Don't know what to do with value type
"+r(e)+".");i=h(e)}l.push(n+i)}return l.push("}"),l.join("")},l=t.__fontmetrics__.uncompress=function(t){if("string"!=typeof t)throw new Error("Invalid argument passed to uncompress.");for(
var e,n,r,a,s={},o=1,h=s,l=[],c="",u="",f=t.length-1,d=1;d<f;d+=1)"'"==(a=t[d])?e?(r=e.join(""),e=void 0):e=[]:e?e.push(a):"{"==a?(l.push([h,r]),h={},r=void 0):"}"==a?((n=l.pop())[0][n[
1]]=h,r=void 0,h=n[0]):"-"==a?o=-1:void 0===r?i.hasOwnProperty(a)?(c+=i[a],r=parseInt(c,16)*o,o=1,c=""):c+=a:i.hasOwnProperty(a)?(u+=i[a],h[r]=parseInt(u,16)*o,o=1,r=void 0,u=""):u+=a;
return s},c={codePages:["WinAnsiEncoding"],WinAnsiEncoding:l("{
19m8n201n9q201o9r201s9l201t9m201u8m201w9n201x9o201y8o202k8q202l8r202m9p202q8p20aw8k203k8t203t8v203u9v2cq8s212m9t15m8w15n9w2dw9s16k8u16l9u17s9z17x8y17y9y}")},u={Unicode:{Courier:c,"Courier-
Bold":c,"Courier-BoldOblique":c,"Courier-Oblique":c,Helvetica:c,"Helvetica-Bold":c,"Helvetica-BoldOblique":c,"Helvetica-Oblique":c,"Times-Roman":c,"Times-Bold":c,"Times-BoldItalic":c,"Times-
Italic":c}},f={Unicode:{"Courier-Oblique":l("{'widths'{k3w'fof'6o}'kerning'{'fof'-6o}}"),"Times-BoldItalic":l("{'widths'{
k3o2q4ycx2r201n3m201o6o201s2l201t2l201u2l201w3m201x3m201y3m2k1t2l2r202m2n2n3m2o3m2p5n202q6o2r1w2s2l2t2l2u3m2v3t2w1t2x2l2y1t2z1w3k3m3l3m3m3m3n3o3m3p3m3q3m3r3m3s3m203t2l203u2l3v2l3w3t3x3t3y
t3z3m4k5n4l4m4m4m4n4m4o4s4p4m4q4m4r4s4s4y4t2r4u3m4v4m4w3x4x5t4y4s4z4s5k3x5l4s5m4m5n3r5o3x5p4s5q4m5r5t5s4m5t3x5u3x5v2l5w1w5x2l5y3t5z3m6k2l6l3m6m3m6n2w6o3m6p2w6q2l6r3m6s3r6t1w6u1w6v3m6w1w6x4y6
y3r6z3m7k3m7l3m7m2r7n2r7o1w7p3r7q2w7r4m7s3m7t2w7u2r7v2n7w1q7x2n7y3t202l3mcl4mal2ram3man3mao3map3mar3mas2lat4uau1uav3maw3way4uaz2lbk2sbl3t'fof'6obo2lbp3tbq3mbr1tbs2lbu1ybv3mbz3mck4m202k3mcm4m
cn4mco4mcp4mcq5ycr4mcs4mct4mcu4mcv4mcw2r2m3rcy2rcz2rdl4sdm4sdn4sdo4sdp4sdq4sds4sdt4sdu4sdv4sdw4sdz3mek3mel3mem3men3meo3mep3meq4ser2wes2wet2weu2wev2wew1wex1wey1wez1wfl3rfm3mfn3mfo3mfp3mfq3mfr
3tfs3mft3rfu3rfv3rfw3rfz2w203k6o212m6o2dw2l2cq2l3t3m3u2l17s3x19m3m}'kerning'{cl{4qu5kt5qt5rs17ss5ts}201s{201ss}201t{cks4lscmcsncscoscpscls2wu2yu201ts}201x{2wu2yu}2k{201ts}2w{
4qx5kx5ou5qx5rs17su5tu}2x{17su5tu5ou}2y{4qx5kx5ou5qx5rs17ss5ts}'fof'-6ofn{17sw5tw5ou5qw5rs}7t{cksclscmcsncscoscps4ls}3u{17su5tu5os5qs}3v{17su5tu5os5qs}7p{17su5tu}ck{4qu5kt5qt5rs17ss5ts}4l{
4qu5kt5qt5rs17ss5ts}cm{4qu5kt5qt5rs17ss5ts}cn{4qu5kt5qt5rs17ss5ts}co{4qu5kt5qt5rs17ss5ts}cp{4qu5kt5qt5rs17ss5ts}6l{4qu5ou5qw5rt17su5tu}5q{ckuclucmucnucoucpu4lu}5r{ckuclucmucnucoucpu4lu}7q{
cksclscmcsncscoscps4ls}6p{4qu5ou5qw5rt17sw5tw}ek{4qu5ou5qw5rt17su5tu}el{4qu5ou5qw5rt17su5tu}em{4qu5ou5qw5rt17su5tu}en{4qu5ou5qw5rt17su5tu}eo{4qu5ou5qw5rt17su5tu}ep{4qu5ou5qw5rt17su5tu}es{
17ss5ts5qs4qu}et{4qu5ou5qw5rt17sw5tw}eu{4qu5ou5qw5rt17ss5ts}ev{17ss5ts5qs4qu}6z{17sw5tw5ou5qw5rs}fm{17sw5tw5ou5qw5rs}7n{201ts}fo{17sw5tw5ou5qw5rs}fp{17sw5tw5ou5qw5rs}fq{17sw5tw5ou5qw5rs}7r{
cksclscmcsncscoscps4ls}fs{17sw5tw5ou5qw5rs}ft{17su5tu}fu{17su5tu}fv{17su5tu}fw{17su5tu}fz{cksclscmcsncscoscps4ls}}}"),"Helvetica-Bold":l("{'widths'{
k3s2q4scx1w201n3r201o6o201s1w201t1w201u1w201w3m201x3m201y3m2k1w2l2l202m2n2n3r2o3r2p5t202q6o2r1s2s2l2t2l2u2r2v3u2w1w2x2l2y1w2z1w3k3r3l3r3m3r3n3r3o3r3p3r3q3r3r3r3s3r203t2l203u2l3v2l3w3u3x3u3y3
u3z3x4k6l4l4s4m4s4n4s4o4s4p4m4q3x4r4y4s4s4t1w4u3r4v4s4w3x4x5n4y4s4z4y5k4m5l4y5m4s5n4m5o3x5p4s5q4m5r5y5s4m5t4m5u3x5v2l5w1w5x2l5y3u5z3r6k2l6l3r6m3r6n3r6o3x6p3r6q2l6r3x6s3x6t1w6u1w6v3r6w1w6x5t6
y3x6z3x7k3x7l3x7m2r7n3r7o2l7p3x7q3r7r4y7s3r7t3r7u3m7v2r7w1w7x2r7y3u202l3rcl4sal2lam3ran3rao3rap3rar3ras2lat4tau2pav3raw3uay4taz2lbk2sbl3u'fof'6obo2lbp3xbq3rbr1wbs2lbu2obv3rbz3xck4s202k3rcm4s
cn4sco4scp4scq6ocr4scs4mct4mcu4mcv4mcw1w2m2zcy1wdl4sdm4ydn4ydo4ydp4ydq4yds4ydt4sdu4sdv4sdw4sdz3xek3rel3rem3ren3reo3rep3req5ter3res3ret3reu3rev3rew1wex1wey1wez1wfl3xfm3xfn3xfo3xfp3xfq3xfr
3ufs3xft3xfu3xfv3xfw3xfz3r203k6o212m6o2dw2l2cq2l3t3r3u2l17s4m19m3r}'kerning'{cl{4qs5ku5ot5qs17sv5tv}201t{2ww4wy2yw}201w{2ks}201x{2ww4wy2yw}2k{201ts201xs}2w{
7qs4qu5kw5os5qw5rs17su5tu7tsfzs}2x{5ow5qs}2y{7qs4qu5kw5os5qw5rs17su5tu7tsfzs}'fof'-6o7p{17su5tu5ot}ck{4qs5ku5ot5qs17sv5tv}4l{4qs5ku5ot5qs17sv5tv}cm{4qs5ku5ot5qs17sv5tv}cn{
4qs5ku5ot5qs17sv5tv}co{4qs5ku5ot5qs17sv5tv}cp{4qs5ku5ot5qs17sv5tv}6l{17st5tt5os}17s{2kwclvcmcvncvcopcv4lv4wwckv}5o{2kucltcmtcntcotcpt4lt4wtckt}5q{2kuclvcmcvncvcopcv4lv4wwckv}5r{2ks4ws}5t{
2kwclvcmcvncvcopcv4lv4wwckv}eo{17st5tt5os}fu{17su5tu5ot}6p{17ss5ts}ek{17st5tt5os}el{17st5tt5os}em{17st5tt5os}en{17st5tt5os}6o{201ts}ep{17st5tt5os}es{17ss5ts}et{17ss5ts}eu{17ss5ts}ev{
```



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   * OF CONTRACT, TORT OR OTHERWISE, ARISING FROM, OUT OF OR IN CONNECTION
   * WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN THE SOFTWARE.
   * ====================================================================
   */
function(t){t.putTotalPages=function(t){var e,n=0;parseInt(this.internal.getFont().id.substr(1),10)<15?(e=new RegExp(t,"g"),n=this.internal.getNumberOfPages()):(e=new RegExp(this.
pdfEscape16(t,this.internal.getFont()),"g"),n=this.pdfEscape16(this.internal.getNumberOfPages()+"",this.internal.getFont()));for(var r=1;r<=this.internal.getNumberOfPages();r++)for(var i=0;
i<this.internal.pages[r].length;i++)this.internal.pages[r][i]=this.internal.pages[r][i].replace(e,n);return this}}(R.API),function(t){t.viewerPreferences=function(t,e){var n;t=t||{},e=e||!1;
var i,a,s,o={HideToolbar:{defaultValue:!1,value:!1,type:"boolean",explicitSet:!1,valueSet:[!0,!1],pdfVersion:1.3},HideMenubar:{defaultValue:!1,value:!1,type:"boolean",explicitSet:!1,
valueSet:[!0,!1],pdfVersion:1.3},HideWindowUI:{defaultValue:!1,value:!1,type:"boolean",explicitSet:!1,valueSet:[!0,!1],pdfVersion:1.3},FitWindow:{defaultValue:!1,value:!1,type:"boolean",
explicitSet:!1,valueSet:[!0,!1],pdfVersion:1.3},CenterWindow:{defaultValue:!1,value:!1,type:"boolean",explicitSet:!1,valueSet:[!0,!1],pdfVersion:1.3},DisplayDocTitle:{defaultValue:!1,value:
!1,type:"boolean",explicitSet:!1,valueSet:[!0,!1],pdfVersion:1.4},NonFullScreenPageMode:{defaultValue:"UseNone",value:"UseNone",type:"name",explicitSet:!1,valueSet:["UseNone","UseOutlines",
"UseThumbs","UseOC"],pdfVersion:1.3},Direction:{defaultValue:"L2R",value:"L2R",type:"name",explicitSet:!1,valueSet:["L2R","R2L"],pdfVersion:1.3},ViewArea:{defaultValue:"CropBox",value:
"CropBox",type:"name",explicitSet:!1,valueSet:["MediaBox","CropBox","TrimBox","BleedBox","ArtBox"],pdfVersion:1.4},ViewClip:{defaultValue:"CropBox",value:"CropBox",type:"name",explicitSet:
!1,valueSet:["MediaBox","CropBox","TrimBox","BleedBox","ArtBox"],pdfVersion:1.4},PrintArea:{defaultValue:"CropBox",value:"CropBox",type:"name",explicitSet:!1,valueSet:["MediaBox","CropBox",
"TrimBox","BleedBox","ArtBox"],pdfVersion:1.4},PrintClip:{defaultValue:"CropBox",value:"CropBox",type:"name",explicitSet:!1,valueSet:["MediaBox","CropBox","TrimBox","BleedBox","ArtBox"],
pdfVersion:1.4},PrintScaling:{defaultValue:"AppDefault",value:"AppDefault",type:"name",explicitSet:!1,valueSet:["AppDefault","None"],pdfVersion:1.6},Duplex:{defaultValue:"",value:"none",
type:"name",explicitSet:!1,valueSet:["Simplex","DuplexFlipShortEdge","DuplexFlipLongEdge","none"],pdfVersion:1.7},PickTrayByPDFSize:{defaultValue:!1,value:!1,type:"boolean",explicitSet:!1,
valueSet:[!0,!1],pdfVersion:1.7},PrintPageRange:{defaultValue:"",value:"",type:"array",explicitSet:!1,valueSet:null,pdfVersion:1.7},NumCopies:{defaultValue:1,value:1,type:"integer",
explicitSet:!1,valueSet:null,pdfVersion:1.7}},h=Object.keys(o),l=[],c=0,u=0,f=0;function d(t,e){var n,r=!1;for(n=0;n<t.length;n++=1)t[n]===e&&(r=!0);return r}if(void 0===this.internal.
viewerpreferences&&(this.internal.viewerpreferences={},this.internal.viewerpreferences.configuration=JSON.parse(JSON.stringify(o)),this.internal.viewerpreferences.isSubscribed=!1),n=this.
internal.viewerpreferences.configuration,"reset"===t||!0===e){var p=h.length;for(f=0;f<p;f+=1)n[h[f]].value=n[h[f]].defaultValue,n[h[f]].explicitSet=!1}if("object"===r(t))for(a in t)if(s=t[
a],d(h,a)&&void 0!==s){if("boolean"===n[a].type&&"boolean"==typeof s)n[a].value=s;else if("name"===n[a].type&&d(n[a].valueSet,s))n[a].value=s;else if("integer"===n[a].type&&Number.isInteger(
s))n[a].value=s;else if("array"===n[a].type){for(c=0;c<s.length;c+=1)if(i=!0,1===s[c].length&&"number"==typeof s[c][0])l.push(String(s[c]-1));else if(s[c].length>1){for(u=0;u<s[c].length;
u+=1)"number"!=typeof s[c][u]&&(i=!1);!0===i&&l.push([s[c][0]-1,s[c][1]-1].join(" "))}n[a].value="["+l.join(" ")+"]"}else n[a].value=n[a].defaultValue;n[a].explicitSet=!0}return!1===this.
internal.viewerpreferences.isSubscribed&&(this.internal.events.subscribe("putCatalog",function(){var t,e=[];for(t in n)!0===n[t].explicitSet&&("name"===n[t].type?e.push("/"+t+" /"+n[t].
value):e.push("/"+t+" "+n[t].value));0!==e.length&&this.internal.write("/ViewerPreferences\n<<\n"+e.join("\n")+"\n>>")}),this.internal.viewerpreferences.isSubscribed=!0),this.internal.
viewerpreferences.configuration=n,this}}(R.API),R.API.addMetadata=function(t,e){return void 0===this.internal.__metadata__&&(this.internal.__metadata__={metadata:t,namespaceUri:null!=e?e:
"http://jspdf.default.namespaceuri/",rawXml:"boolean"==typeof e&&e},this.internal.events.subscribe("putCatalog",Ma),this.internal.events.subscribe("postPutResources",Ba)),this},function(t){
var e=t.API,n=e.pdfEscape16=function(t,e){for(var n,r=e.metadata.Unicode.widths,i=["","0","00","000","0000"],a=[""],s=0,o=t.length;s<o;++s){if(n=e.metadata.characterToGlyph(t.charCodeAt(s)),
e.metadata.glyIdsUsed.push(n),e.metadata.toUnicode[n]=t.charCodeAt(s),-1==r.indexOf(n)&&(r.push(n),r.push([parseInt(e.metadata.widthOfGlyph(n),10)])),"0"==n)return a.join("");n=n.toString(
16),a.push(i[4-n.length],n)}return a.join("")},r=function(t){var e,n,r,i,a,s,o;for(a="/CIDInit /ProcSet findresource begin\n12 dict begin\nbegincmap\n/CIDSystemInfo <<\n  /Registry (Adobe)\
n  /Ordering (UCS)\n  /Supplement 0\n>> def\n/CMapName /Adobe-Identity-UCS def\n/CMapType 2 def\n1 begincodespacerange\n<0000><ffff>\nendcodespacerange",r=[],s=0,o=(n=Object.keys(t).sort(
function(t,e){return t-e})).length;s<o;s++)e=n[s],r.length>=100&&(a+="\n"+r.length+" beginbfchar\n"+r.join("\n")+"\nendbfchar",r=[]),void 0!==t[e]&&null!==t[e]&&"function"==typeof t[e].
toString&&(i="0000"+t[e].toString(16)).slice(-4),e=("0000"+(+e).toString(16)).slice(-4),r.push("<"+e+"><"+i+">"));return r.length&&(a+="\n"+r.length+" beginbfchar\n"+r.join("\n")+"\
nendbfchar\n"),a+"endcmap\nCMapName currentdict /CMap defineresource pop\nend\nend"};e.events.push(["putFont",function(e){!function(e){var n=e.font,i=e.out,a=e.newObject,s=e.putStream;if(n.
metadata instanceof t.API.TTFFont&&"Identity-H"===n.encoding){for(var o=n.metadata.Unicode.widths,h=n.metadata.subset.encode(n.metadata.glyIdsUsed,1),l="",c=0;c<h.length;c++)l+=String.
fromCharCode(h[c]);var u=a();s({data:l,addLength1:!0,objectId:u}),i("endobj");var f=a();s({data:r(n.metadata.toUnicode),addLength1:!0,objectId:f}),i("endobj");var d=a();i("<<"),i("/Type /
FontDescriptor"),i("/FontName /"+C(n.fontName)),i("/FontFile2 "+u+" 0 R"),i("/FontBBox "+t.API.PDFObject.convert(n.metadata.bbox)),i("/Flags "+n.metadata.flags),i("/StemV "+n.metadata.
stemV),i("/ItalicAngle "+n.metadata.italicAngle),i("/Ascent "+n.metadata.ascender),i("/Descent "+n.metadata.decender),i("/CapHeight "+n.metadata.capHeight),i(">>"),i("endobj");var p=a();i(
"<<"),i("/Type /Font"),i("/BaseFont /"+C(n.fontName)),i("/FontDescriptor "+d+" 0 R"),i("/W "+t.API.PDFObject.convert(o)),i("/CIDToGIDMap /Identity"),i("/DW 1000"),i("/Subtype /
CIDFontType2"),i("/CIDSystemInfo"),i("<<"),i("/Supplement 0"),i("/Registry (Adobe)"),i("/Ordering ("+n.encoding+")"),i(">>"),i(">>"),i("endobj"),n.objectNumber=a(),i("<<"),i("/Type /Font"),
i("/Subtype /Type0"),i("/ToUnicode "+f+" 0 R"),i("/BaseFont /"+C(n.fontName)),i("/Encoding /"+n.encoding),i("/DescendantFonts ["+p+" 0 R]"),i(">>"),i("endobj"),n.isAlreadyPutted=!0}}(e)}]),
e.events.push(["putFont",function(e){!function(e){var n=e.font,i=e.out,a=e.newObject,s=e.putStream;if(n.metadata instanceof t.API.TTFFont&&"WinAnsiEncoding"===n.encoding){for(var o=n.
metadata.rawData,h="",l=0;l<o.length;l++)h+=String.fromCharCode(o[l]);var c=a();s({data:h,addLength1:!0,objectId:c}),i("endobj");var u=a();s({data:r(n.metadata.toUnicode),addLength1:!0,
objectId:u}),i("endobj");var f=a();i("<<"),i("/Descent "+n.metadata.decender),i("/CapHeight "+n.metadata.capHeight),i("/StemV "+n.metadata.stemV),i("/Type /FontDescriptor"),i("/FontFile2
"+c+" 0 R"),i("/Flags 96"),i("/FontBBox "+t.API.PDFObject.convert(n.metadata.bbox)),i("/FontName /"+C(n.fontName)),i("/ItalicAngle "+n.metadata.italicAngle),i("/Ascent "+n.metadata.
ascender),i(">>"),i("endobj"),n.objectNumber=a();for(var d=0;d<n.metadata.hmtx.widths.length;d++)n.metadata.hmtx.widths[d]=parseInt(n.metadata.hmtx.widths[d]*(1e3/n.metadata.head.
unitsPerEm));i("<</Subtype/TrueType/Type/Font/ToUnicode "+u+" 0 R/BaseFont/"+C(n.fontName)+"/FontDescriptor "+f+" 0 R/Encoding/"+n.encoding+" /FirstChar 29 /LastChar 255 /Widths "+t.API.
PDFObject.convert(n.metadata.hmtx.widths)+">>"),i("endobj"),n.isAlreadyPutted=!0}}(e)}]);var i=function(t){var e,r=t.text||"",i=t.x,a=t.y,s=t.options||{},o=t.mutex||{},h=o.pdfEscape,l=o.
activeFontKey,c=o.fonts,u=l,f="",d=0,p="",g=c[u].encoding;if("Identity-H"!==c[u].encoding)return{text:r,x:i,y:a,options:s,mutex:o};for(p=r,u=l,Array.isArray(r)&&(p=r[0]),d=0;d<p.length;
d+=1)c[u].metadata.hasOwnProperty("cmap")&&(e=c[u].metadata.cmap.unicode.codeMap[p[d].charCodeAt(0)],e||p[d].charCodeAt(0)<256&&c[u].metadata.hasOwnProperty("Unicode")?f+=p[d]:f+="";var
m="";return parseInt(u.slice(1))<14||"WinAnsiEncoding"===g?m=h(f,u).split("").map(function(t){return t.charCodeAt(0).toString(16)}).join(""):"Identity-H"===g&&(m=n(f,c[u])),o.isHex=!0,{text:
m,x:i,y:a,options:s,mutex:o}};e.events.push(["postProcessText",function(t){var e=t.text||"",n=[],r={text:e,x:t.x,y:t.y,options:t.options,mutex:t.mutex};if(Array.isArray(e)){var a=0;for(a=0;
a<e.length;a+=1)Array.isArray(e[a])&&3===e[a].length?n.push([i(Object.assign({},r,{text:e[a][0]})).text,e[a][1],e[a][2]]):n.push(i(Object.assign({},r,{text:e[a]})).text);t.text=n}else t.
text=i(Object.assign({},r,{text:e})).text}])}(R),
/**
   * @license
   * jsPDF virtual FileSystem functionality
   *
   * Licensed under the MIT License.
   * http://opensource.org/licenses/mit-license
   */
function(t){var e=function(){return void 0===this.internal.vFS&&(this.internal.vFS={}),!0};t.existsFileInVFS=function(t){return e.call(this),void 0!==this.internal.vFS[t]},t.
addFileToVFS=function(t,n){return e.call(this),this.internal.vFS[t]=n,this},t.getFileFromVFS=function(t){return e.call(this),void 0!==this.internal.vFS[t]?this.internal.vFS[t]:null}}(R.API),
/**
   * @license
   * Unicode Bidi Engine based on the work of Alex Shensis (@asthensis)
   * MIT License
   */
```



```
writeByte=function(t){return this.data[this.pos++]=t},t.prototype.readUInt32=function(){return 16777216*this.readByte()+(this.readByte()<<16)+(this.readByte()<<8)+this.readByte()},t.
prototype.writeUInt32=function(t){return this.writeByte(t>>>24&255),this.writeByte(t>>>16&255),this.writeByte(t>>>8&255),this.writeByte(255&t)},t.prototype.readInt32=function(){var t;return(
t=this.readUInt32())>=21474836487t-4294967296:t},t.prototype.writeInt32=function(t){return t<0&&(t+=4294967296),this.writeUInt32(t)},t.prototype.readUInt16=function(){return this.readByte(
)<<8|this.readByte()},t.prototype.writeUInt16=function(t){return this.writeByte(t>>8&255),this.writeByte(255&t)},t.prototype.readInt16=function(){var t;return(t=this.readUInt16())>=32768?t-
65536:t},t.prototype.writeInt16=function(t){return t<0&&(t+=65536),this.writeUInt16(t)},t.prototype.readString=function(t){var e,n;for(n=[],e=0;0<=t?e<t:e>t;e=0<=t?++e:--e)n[e]=String.
fromCharCode(this.readByte());return n.join("")},t.prototype.writeString=function(t){var e,n,r;for(r=[],e=0,n=t.length;0<=n?e<n:e>n;e=0<=n?++e:--e)r.push(this.writeByte(t.charCodeAt(e)));
return r},t.prototype.readShort=function(){return this.readInt16()},t.prototype.writeShort=function(t){return this.writeInt16(t)},t.prototype.readLongLong=function(){var t,e,n,r,i,a,s,o;
return t=this.readByte(),e=this.readByte(),n=this.readByte(),r=this.readByte(),i=this.readByte(),a=this.readByte(),s=this.readByte(),o=this.readByte(),128&t?-1*(72057594037927940*(
255^t)+281474976710656*(255^e)+1099511627776*(255^n)+4294967296*(255^r)+16777216*(255^i)+65536*(255^a)+256*(255^s)+(255^o)+1):
72057594037927940*t+281474976710656*e+1099511627776*n+4294967296*r+16777216*i+65536*a+256*s+o},t.prototype.writeLongLong=function(t){var e,n;return e=Math.floor(t/4294967296),n=4294967295&t,
this.writeByte(e>>24&255),this.writeByte(e>>16&255),this.writeByte(e>>8&255),this.writeByte(255&e),this.writeByte(n>>24&255),this.writeByte(n>>16&255),this.writeByte(n>>8&255),this.
writeByte(255&n)},t.prototype.readInt=function(){return this.readInt32()},t.prototype.writeInt=function(t){return this.writeInt32(t)},t.prototype.read=function(t){var e,n;for(e=[],n=0;
0<=t?n<t:n>t;n=0<=t?++n:--n)e.push(this.readByte());return e},t.prototype.write=function(t){var e,n,r,i;for(i=[],n=0,r=t.length;n<r;n++)e=t[n],i.push(this.writeByte(e));return i},t(),
Da=function(){var t;function e(t){var e,n,r;for(this.scalarType=t.readInt(),this.tableCount=t.readShort(),this.searchRange=t.readShort(),this.entrySelector=t.readShort(),this.rangeShift=t.
readShort(),this.tables={},n=0,r=this.tableCount;0<=r?n<r:n>r;n=0<=r?++n:--n)e={tag:t.readString(4),checksum:t.readInt(),offset:t.readInt(),length:t.readInt()},this.tables[e.tag]=e}return e.
prototype.encode=function(e){var n,r,i,a,s,o,h,l,c,u,f,d,p;for(p in f=Object.keys(e).length,o=Math.log(2),c=16*Math.floor(Math.log(f)/o),a=Math.floor(c/o),l=16*f-c,(r=new Ta).writeInt(this.
scalarType),r.writeShort(f),r.writeShort(c),r.writeShort(a),r.writeShort(l),i=16*f,h=r.pos+i,s=null,d=[],e)for(u=e[p],r.writeString(p),r.writeInt(t(u)),r.writeInt(h),r.writeInt(u.length),
d=d.concat(u),"head"===p&&(s=h),h+=u.length;h%4;)d.push(0),h++;return r.write(d),n=2981146554-t(r.data),r.pos=s+8,r.writeUInt32(n),r.data},t=function(t){var e,n,r,i;for(t=$a.call(t);t.
length%4;)t.push(0);for(r=new Ta(t),n=0,e=0,i=t.length;e<i;e=e+=4)n+=r.readUInt32();return 4294967295&n},e}(),qa={}.hasOwnProperty,za=function(t,e){for(var n in e)qa.call(e,n)&&(t[n]=e[n]);
function r(){this.constructor=t}return r.prototype=e.prototype,t.prototype=new r,t.__super__=e.prototype,t},Ua=function(t){function e(){return e.__super__.constructor.apply(this,
arguments)}return za(e,t),e.prototype.tag="head",e.prototype.parse=function(t){return t.pos=this.offset,this.version=t.readInt(),this.revision=t.readInt(),this.checkSumAdjustment=t.readInt(
),this.magicNumber=t.readInt(),this.flags=t.readShort(),this.unitsPerEm=t.readShort(),this.created=t.readLongLong(),this.modified=t.readLongLong(),this.xMin=t.readShort(),this.yMin=t.
readShort(),this.xMax=t.readShort(),this.yMax=t.readShort(),this.macStyle=t.readShort(),this.lowestRecPPEM=t.readShort(),this.fontDirectionHint=t.readShort(),this.indexToLocFormat=t.
readShort(),this.glyphDataFormat=t.readShort()},e.prototype.encode=function(t){var e;return(e=new Ta).writeInt(this.version),e.writeInt(this.revision),e.writeInt(this.checkSumAdjustment),e.
writeInt(this.magicNumber),e.writeShort(this.flags),e.writeShort(this.unitsPerEm),e.writeLongLong(this.created),e.writeLongLong(this.modified),e.writeShort(this.xMin),e.writeShort(this.
yMin),e.writeShort(this.xMax),e.writeShort(this.yMax),e.writeShort(this.macStyle),e.writeShort(this.lowestRecPPEM),e.writeShort(this.fontDirectionHint),e.writeShort(t),e.writeShort(this.
glyphDataFormat),e.data},e}(Ra=function(){function t(t){var e;this.file=t,e=this.file.directory.tables[this.tag],this.exists=!!e,e&&(this.offset=e.offset,this.length=e.length,this.parse(
this.file.contents))}return t.prototype.parse=function(){},t.prototype.encode=function(){},t.prototype.raw=function(){return this.exists?(this.file.contents.pos=this.offset,this.file.
contents.read(this.length)):null},t}()),Ha=function(){function t(t,e){var n,r,i,a,s,o,h,l,c,u,f,d,p,g,m,b,v;switch(this.platformID=t.readUInt16(),this.encodingID=t.readShort(),this.
offset=e+t.readInt(),c=t.pos,t.pos=this.offset,this.format=t.readUInt16(),this.length=t.readUInt16(),this.language=t.readUInt16(),this.isUnicode=3===this.platformID&&1===this.
encodingID&&4===this.format||0===this.platformID&&4===this.format,this.codeMap={},this.format){case 0:for(o=0;o<256;++o)this.codeMap[o]=t.readByte();break;case 4:for(f=t.readUInt16(),u=f/2,
t.pos+=6,i=function(){var e,n;for(n=[],o=e=0;0<=u?e<u:e>u;o=0<=u?++e:--e)n.push(t.readUInt16());return n}(),t.pos+=2,p=function(){var e,n;for(n=[],o=e=0;0<=u?e<u:e>u;o=0<=u?++e:--e)n.push(t.
readUInt16());return n}(),h=function(){var e,n;for(n=[],o=e=0;0<=u?e<u:e>u;o=0<=u?++e:--e)n.push(t.readUInt16());return n}(),l=function(){var e,n;for(n=[],o=e=0;0<=u?e<u:e>u;o=0<=u?++e:--
e)n.push(t.readUInt16());return n}(),r=this.length-t.pos+this.offset)/2,s=function(){var e,n;for(n=[],o=e=0;0<=r?e<r:e>r;o=0<=r?++e:--e)n.push(t.readUInt16());return n}(),o=m=0,v=i.length;
m<v;o=++m)for(g=i[o],n=b=d=p[o];d<=g?b<=g:b>=g;n=d<=g?++b:--b)0===l[o]?a=n+h[o]:0!==(a=s[l[o]/2+(n-d)-(u-o)]||0)&&(a+=h[o]),this.codeMap[n]=65535&a}t.pos=c}return t.encode=function(t,e){var
n,r,i,a,s,o,h,l,c,u,f,d,p,g,m,b,v,w,y,_,x,A,L,N,S,k,P,F,I,C,j,E,O,B,M,R,T,D,q,z,U,H,W,V,G,Y;switch(F=new Ta,a=Object.keys(t).sort(function(t,e){return t-e}),e){case"macroman":for(p=0,
g=function(){var t=[];for(d=0;d<256;++d)t.push(0);return t}(),b={0:0},i={},I=0,O=a.length;I<O;I++)null===b[W=t[r=a[I]]]&&(b[W]=++p),i[r]={old:t[r],new:b[t[r]]},g[r]=b[t[r]];return F.
writeUInt16(1),F.writeUInt16(0),F.writeUInt32(12),F.writeUInt16(0),F.writeUInt16(262),F.writeUInt16(0),F.write(g),{charMap:i,subtable:F.data,maxGlyphID:p+1};case"unicode":for(k=[],c=[],v=0,
b={},n={},m=h=null,C=0,B=a.length;C<B;C++)null==b[y=t[r=a[C]]]&&(b[y]=++v),n[r]={old:y,new:b[y]},s=b[y]-r,null!=m&&s===h||(m&&c.push(m),k.push(r),h=s),m=r;for(m&&c.push(m),k.push(r),c.push(65535),k.
push(65535),N=2*(L=k.length),A=2*Math.pow(Math.log(L)/Math.LN2,2),u=Math.log(A/2)/Math.LN2,x=2*L-A,o=[],_=[],f=[],d=j=0,M=k.length;j<M;d=++j){if(S=k[d],l=c[d],65535===S){o.push(0),_.push(0);
break}if(S-(P=n[S].new)>=32768)for(o.push(0),_.push(2*(f.length+L-d)),r=E=S;S<=l?E<=l:E>=l;r=S<=l?++E:--E)f.push(n[r].new);else o.push(P-S),_.push(0)}for(F.writeUInt16(3),F.writeUInt16(1),F.
writeUInt32(12),F.writeUInt16(4),F.writeUInt16(16+8*L+2*f.length),F.writeUInt16(0),F.writeUInt16(N),F.writeUInt16(A),F.writeUInt16(u),F.writeUInt16(x),U=0,R=c.length;U<R;U++)r=c[U],F.
writeUInt16(r);for(F.writeUInt16(0),H=0,T=k.length;H<T;H++)r=k[H],F.writeUInt16(r);for(V=0,D=o.length;V<D;V++)s=o[V],F.writeUInt16(s);for(G=0,q=_.length;G<q;G++)w=_[G],F.writeUInt16(w);for(
Y=0,z=f.length;Y<z;Y++)p=f[Y],F.writeUInt16(p);return{charMap:n,subtable:F.data,maxGlyphID:v+1}}},t()),Wa=function(t){function e(){return e.__super__.constructor.apply(this,
arguments)}return za(e,t),e.prototype.tag="cmap",e.prototype.parse=function(t){var e,n,r;for(t.pos=this.offset,this.version=t.readUInt16(),r=t.readUInt16(),this.tables=[],this.unicode=null,
n=0;0<=r?n<r:n>r;n=0<=r?++n:--n)e=new Ha(t,this.offset),this.tables.push(e),e.isUnicode&&null==this.unicode&&(this.unicode=e);return!0},e.encode=function(t,e){var n,r;return null==e&&(
e="macroman"),n=Ha.encode(t,e),(r=new Ta).writeUInt16(0),r.writeUInt16(1),n.table=r.data.concat(n.subtable),n},e}(Ra),Va=function(t){function e(){return e.__super__.constructor.apply(this,
arguments)}return za(e,t),e.prototype.tag="hhea",e.prototype.parse=function(t){return t.pos=this.offset,this.version=t.readInt(),this.ascender=t.readShort(),this.decender=t.readShort(),this.
lineGap=t.readShort(),this.advanceWidthMax=t.readShort(),this.minLeftSideBearing=t.readShort(),this.minRightSideBearing=t.readShort(),this.xMaxExtent=t.readShort(),this.caretSlopeRise=t.
readShort(),this.caretSlopeRun=t.readShort(),this.caretOffset=t.readShort(),t.pos+=8,this.metricDataFormat=t.readShort(),this.numberOfMetrics=t.readUInt16()},e}(Ra),Ga=function(t){function
e(){return e.__super__.constructor.apply(this,arguments)}return za(e,t),e.prototype.tag="OS/2",e.prototype.parse=function(t){if(t.pos=this.offset,this.version=t.readUInt16(),this.
averageCharWidth=t.readShort(),this.weightClass=t.readUInt16(),this.widthClass=t.readUInt16(),this.type=t.readShort(),this.ySubscriptXSize=t.readShort(),this.ySubscriptYSize=t.readShort(),
this.ySubscriptXOffset=t.readShort(),this.ySubscriptYOffset=t.readShort(),this.ySuperscriptXSize=t.readShort(),this.ySuperscriptYSize=t.readShort(),this.ySuperscriptXOffset=t.readShort(),
this.ySuperscriptYOffset=t.readShort(),this.yStrikeoutSize=t.readShort(),this.yStrikeoutPosition=t.readShort(),this.familyClass=t.readShort(),this.panose=function(){var e,n;for(n=[],e=0;
e<10;++e)n.push(t.readByte());return n}(),this.charRange=function(){var e,n;for(n=[],e=0;e<4;++e)n.push(t.readInt());return n}(),this.vendorID=t.readString(4),this.selection=t.readShort(),
this.firstCharIndex=t.readShort(),this.lastCharIndex=t.readShort(),this.version>0&&(this.ascent=t.readShort(),this.descent=t.readShort(),this.lineGap=t.readShort(),this.winAscent=t.
readShort(),this.winDescent=t.readShort(),this.codePageRange=function(){var e,n;for(n=[],e=0;e<2;e=++e)n.push(t.readInt());return n}(),this.version>1))return this.xHeight=t.readShort(),this.
capHeight=t.readShort(),this.defaultChar=t.readShort(),this.breakChar=t.readShort(),this.maxContext=t.readShort()},e}(Ra),Ya=function(t){function e(){return e.__super__.constructor.apply(
this,arguments)}return za(e,t),e.prototype.tag="post",e.prototype.parse=function(t){var e,n,r;switch(t.pos=this.offset,this.format=t.readInt(),this.italicAngle=t.readInt(),this.
underlinePosition=t.readShort(),this.underlineThickness=t.readShort(),this.isFixedPitch=t.readInt(),this.minMemType42=t.readInt(),this.maxMemType42=t.readInt(),this.minMemType1=t.readInt(),
this.maxMemType1=t.readInt(),this.format){case 65536:case 196608:break;case 131072:var i;for(n=t.readUInt16(),this.glyphNameIndex=[],i=0;0<=n?i<n:i>n;i=0<=n?++i:--i)this.glyphNameIndex.push(
t.readUInt16());for(this.names=[],r=[];t.pos<this.offset+this.length;)e=t.readByte(),r.push(this.names.push(t.readString(e)));return r;case 151552:return n=t.readUInt16(),this.offsets=t.
read(n);case 262144:return this.map=function(){var e,n,r;for(r=[],i=e=0,n=this.file.maxp.numGlyphs;0<=n?e<n:e>n;i=0<=n?++e:--e)r.push(t.readUInt32());return r}.call(this)}},e}(Ra),
Za=function(t,e){this.raw=t,this.length=t.length,this.platformID=e.platformID,this.encodingID=e.encodingID,this.languageID=e.languageID},Ja=function(t){function e(){return e.__super__.
constructor.apply(this,arguments)}return za(e,t),e.prototype.tag="name",e.prototype.parse=function(t){var e,n,r,i,a,s,o,h,l,c,u;for(t.pos=this.offset,t.readShort(),e=t.readShort(),s=t.
readShort(),n=[],i=0;0<=e?i<e:i>e;i=0<=e?++i:--i)n.push({platformID:t.readShort(),encodingID:t.readShort(),languageID:t.readShort(),nameID:t.readShort(),length:t.readShort(),offset:this.
offset+s+t.readShort()});for(o={},i=l=0,c=n.length;l<c;i=++l)r=n[i],t.pos=r.offset,h=t.readString(r.length),a=new Za(h,r),null==o[u=r.nameID]&&(o[u]=[]),o[r.nameID].push(a);this.strings=o,
this.copyright=o[0],this.fontFamily=o[1],this.fontSubfamily=o[2],this.uniqueSubfamily=o[3],this.fontName=o[4],this.version=o[5];try{this.postscriptName=o[6][0].raw.replace(/[\x00-\x19\x80-\
xff]/g,"")}catch(is){this.postscriptName=o[4][0].raw.replace(/[\x00-\x19\x80-\xff]/g,"")}return this.trademark=o[7],this.manufacturer=o[8],this.designer=o[9],this.description=o[10],this.
vendorUrl=o[11],this.designerUrl=o[12],this.license=o[13],this.licenseUrl=o[14],this.preferredFamily=o[15],this.preferredSubfamily=o[17],this.compatibleFull=o[18],this.sampleText=o[19]},e}(
Ra),Xa=function(t){function e(){return e.__super__.constructor.apply(this,arguments)}return za(e,t),e.prototype.tag="maxp",e.prototype.parse=function(t){return t.pos=this.offset,this.
version=t.readInt(),this.numGlyphs=t.readUInt16(),this.maxPoints=t.readUInt16(),this.maxContours=t.readUInt16(),this.maxCompositePoints=t.readUInt16(),this.maxCompositeContours=t.readUInt16(
),this.maxZones=t.readUInt16(),this.maxTwilightPoints=t.readUInt16(),this.maxStorage=t.readUInt16(),this.maxFunctionDefs=t.readUInt16(),this.maxInstructionDefs=t.readUInt16(),this.
maxStackElements=t.readUInt16(),this.maxSizeOfInstructions=t.readUInt16(),this.maxComponentElements=t.readUInt16(),this.maxComponentDepth=t.readUInt16()},e}(Ra),Ka=function(t){function e(){
return e.__super__.constructor.apply(this,arguments)}return za(e,t),e.prototype.tag="hmtx",e.prototype.parse=function(t){var e,n,r,i,a,s,o;for(t.pos=this.offset,this.metrics=[],e=0,s=this.
file.hhea.numberOfMetrics;0<=s?e<s:e>s;e=0<=s?++e:--e)this.metrics.push({advance:t.readUInt16(),lsb:t.readInt16()});for(r=this.file.maxp.numGlyphs-this.file.hhea.numberOfMetrics,this.
leftSideBearings=function(){var n,i;for(i=[],e=n=0;0<=r?n<r:n>r;e=0<=r?++n:--n)i.push(t.readInt16());return i}(),this.widths=function(){var t,e,n,r;for(r=[],t=0,e=(n=this.metrics).length;
t<e;t++)i=n[t],r.push(i.advance);return r}.call(this),n=this.widths[this.widths.length-1],o=[],e=a=0;0<=r?a<r:a>r;e=0<=r?++a:--a)o.push(this.widths.push(n));return o},e.prototype.
forGlyph=function(t){return t in this.metrics?this.metrics[t]:{advance:this.metrics[this.metrics.length-1].advance,lsb:this.leftSideBearings[t-this.metrics.length]}},e}(Ra),$a=[].slice,
Qa=function(t){function e(){return e.__super__.constructor.apply(this,arguments)}return za(e,t),e.prototype.tag="glyf",e.prototype.parse=function(){return this.cache={}},e.prototype.
glyphFor=function(t){var e,n,r,i,a,s,o,h,l,c;return t in this.cache?this.cache[t]:(i=this.file.loca,e=this.file.contents,n=i.indexOf(t),0===(r=i.lengthOf(t))?this.cache[t]=null:(e.pos=this.
offset+n,a=(s=new Ta(e.read(r))).readShort(),h=s.readShort(),c=s.readShort(),o=s.readShort(),l=s.readShort(),this.cache[t]=-1===a?new es(s,h,c,o,l):new ts(s,a,h,c,o,l),this.cache[t]))},e.
prototype.encode=function(t,e,n){var r,i,a,s,o;for(a=[],i=[],s=0,o=e.length;s<o;s++)r=t[e[s]],i.push(a.length),r&&(a=a.concat(r.encode(n)));return i.push(a.length),{table:a,offsets:i}},e}(
Ra),ts=function(){function t(t,e,n,r,i,a){this.raw=t,this.numberOfContours=e,this.xMin=n,this.yMin=r,this.xMax=i,this.yMax=a,this.compound=!1}return t.prototype.encode=function(){return
this.raw.data},t}(),es=function(){function t(t,e,n,r,i){var a,s;for(this.raw=t,this.xMin=e,this.yMin=n,this.xMax=r,this.yMax=i,this.compound=!0,this.glyphIDs=[],this.glyphOffsets=[],a=this.
raw;s=a.readShort(),this.glyphOffsets.push(a.pos),this.glyphIDs.push(a.readUInt16()),32&s;)a.pos+=1&s?4:2,128&s?a.pos+=8:64&s?a.pos+=4:8&s&&(a.pos+=2)}return t.prototype.encode=function(){
var t,e,n;for(e=new Ta($a.call(this.raw.data)),t=0,n=this.glyphIDs.length;t<n;++t)e.pos=this.glyphOffsets[t];return e.data},t()},ns=function(t){function e(){return e.__super__.constructor.
```

```
apply(this,arguments)}return za(e,t),e.prototype.tag="loca",e.prototype.parse=function(t){var e,n;return t.pos=this.offset,e=this.file.head.indexToLocFormat,this.offsets=0===e?function(){
var e,r;for(r=[],n=0,e=this.length;n<e;n+=2)r.push(2*t.readUInt16());return r}.call(this):function(){var e,r;for(r=[],n=0,e=this.length;n<e;n+=4)r.push(t.readUInt32());return r}.call(this)},
e.prototype.indexOf=function(t){return this.offsets[t]},e.prototype.lengthOf=function(t){return this.offsets[t+1]-this.offsets[t]},e.prototype.encode=function(t,e){for(var n=new Uint32Array(
this.offsets.length),r=0,i=0,a=0;a<n.length;++a)if(n[a]=r,i<e.length&&e[i]==a){++i,n[a]=r;var s=this.offsets[a],o=this.offsets[a+1]-s;o>0&&(r+=o)}for(var h=new Array(4*n.length),l=0;l<n.
length;++l)h[4*l+3]=255&n[l],h[4*l+2]=(65280&n[l])>>8,h[4*l+1]=(16711680&n[l])>>16,h[4*l]=(4278190080&n[l])>>24;return h},e}(Ra),rs=function(){function t(t){this.font=t,this.subset={},this.
unicodes={},this.next=33}return t.prototype.generateCmap=function(){var t,e,n,r,i;for(e in r=this.font.cmap.tables[0].codeMap,t={},i=this.subset)n=i[e],t[e]=r[n];return t},t.prototype.
glyphsFor=function(t){var e,n,r,i,a,s,o;for(r={},a=0,s=t.length;a<s;a++)r[i=t[a]]=this.font.glyf.glyphFor(i);for(i in e=[],r)(null!=(n=r[i])?n.compound:void 0)&&e.push.apply(e,n.glyphIDs);
if(e.length>0)for(i in o=this.glyphsFor(e))n=o[i],r[i]=n;return r},t.prototype.encode=function(t,e){var n,r,i,a,s,o,h,l,c,u,f,d,p,g,m;for(r in n=Wa.encode(this.generateCmap(),"unicode"),
a=this.glyphsFor(t),f={0:0},m=n.charMap)f[(o=m[r]).old]=o.new;for(d in u=n.maxGlyphID,a)d in f||(f[d]=u++);return l=function(t){var e,n;for(e in n={},t)n[t[e]]=e;return n}(f),c=Object.keys(
l).sort(function(t,e){return t-e}),p=function(){var t,e,n;for(n=[],t=0,e=c.length;t<e;t++)s=c[t],n.push(l[s]);return n}(),i=this.font.glyf.encode(a,p,f),h=this.font.loca.encode(i.offsets,p),
g={cmap:this.font.cmap.raw(),glyf:i.table,loca:h,hmtx:this.font.hmtx.raw(),hhea:this.font.hhea.raw(),maxp:this.font.maxp.raw(),post:this.font.post.raw(),name:this.font.name.raw(),head:this.
font.head.encode(e)},this.font.os2.exists&&(g["OS/2"]=this.font.os2.raw()),this.font.directory.encode(g)},t}();R.API.PDFObject=function(){var t;function e(){}return t=function(t,e){return(
Array(e+1).join("0")+t).slice(-e)},e.convert=function(n){var r,i,a,s;if(Array.isArray(n))return"["+function(){var t,i,a;for(a=[],t=0,i=n.length;t<i;t++)r=n[t],a.push(e.convert(r));return a}(
).join(" ")+"]";if("string"==typeof n)return"/"+n;if(null!=n?n.isString:void 0)return"("+n+")";if(n instanceof Date)return"(D:"+t(n.getUTCFullYear(),4)+t(n.getUTCMonth(),2)+t(n.getUTCDate(),
2)+t(n.getUTCHours(),2)+t(n.getUTCMinutes(),2)+t(n.getUTCSeconds(),2)+"Z)";if("[object Object]"==={}.toString.call(n)){for(i in a=["<<"],n)s=n[i],a.push("/"+i+" "+e.convert(s));return a.
push(">>"),a.join("\n")}return""+n},e}(),t.AcroForm=St,t.AcroFormAppearance=Lt,t.AcroFormButton=bt,t.AcroFormCheckBox=_t,t.AcroFormChoiceField=dt,t.AcroFormComboBox=gt,t.AcroFormEditBox=mt,
t.AcroFormListBox=pt,t.AcroFormPasswordField=At,t.AcroFormPushButton=vt,t.AcroFormRadioButton=wt,t.AcroFormTextField=xt,t.GState=E,t.ShadingPattern=B,t.TilingPattern=M,t.default=R,t.jsPDF=R,
Object.defineProperty(t,"__esModule",{value:!0})});
//# sourceMappingURL=jspdf.umd.min.js.map
```

13. assets/vendor/pdf-fonts.js

```
/* Subsetted DejaVu Sans Condensed / Mono (Bitstream Vera licence, free to embed). Used by the PDF export
 * for real, selectable text. */
window.PDF_FONT_DATA = {
 "sans-400": "AAEAAAAPAIAAAwBwR0RFRkJ0R4UAAen4AAAAfEdQT1NEdkx1AAHqdAAAACBHU1VCJ6Q/
wwAB6pQAAACWTUFUSMQqKU0AAessAAAFAk9TLzINHTW8AAHl1AAAAFZjbWFw6eIH7QAB5iwAAAGkZ2FzcAAHAAcAAensAAAADGdseWYDZrXeAAAA/
AABvVBoZWFkJttMTwABy3QAAAA2aGhlYQ0BDP4AAeWwAAAAJGhtdHijZN91AAHLrAAAGgJsb2Nhl5wpLAABvmwAAA0GbWF4cAbCA8EAAb5MAAAAIG5hbWUr3EGfAAHn0AAAAfpwb3N0/9sAWgAB6cwAAAAgAAIAW/6WA/
YFpAADAAcAABMRIRElIREhXAOZ/M4CzP00/pYHDvjycgYpAAIBFgAAAc0F1QADAAkAACUzFSMRMxEDIwMBFre3txKSE/
7+BdX9cf6bAWUAAgCxA6oCnwXVAAMABwAAAREjESERIxEBSpkB7ZgF1f3VAiv91QIrAAIAjgAABXwFvgADAB8AAAEhAyElASETMwMhFSEDIRUhAyMTIQMjEyE1IRMhNSETA67++ksBBz1dAQZfkF0BGf7ESgEf/r5dkFz++V2RXv7kAT1M/
t4BRVwDhf6yA4f+YQGf/mGa/rKZ/mIBnv5iAZ6ZAU6aAZ8AAAMAmf7TA/wGFAAhACgALwAAASMDLgEnNR4BFxEuATU0Njc1MxUeARcVLgEnER4BFRQGBwMRDgEVFBYXET4BNTQmAm9aAV+9X1y8ZMe1xLhaU51LS51TzMHNwFppbWXLcnRv/
tMBLQItLbRAQQEByCSslqO8DuvoBB8bryouBP5VI7ScqcMPAwABmg1qWFZg1f5PEW5aWGgABQBl/
+MGcgXwAAsAFwAjACcAMwAAASIGFRQWMzI2NTQmJzIWFRQGIyImNTQ2ASIGFRQWMzI2NTQmJTMBIxMyFhUUBiMiJjU0NgU8TllZTk1ZWU2OqKmNkKeo/O5NWVhOT1laAt+Q/LeQHI+pqY+PpqcCkZSEgpWVgoOVf9y7u9vbu7zbAmGVgoSUlISBln/
58wYN27u92tu8utwAAgB0/+MFZQXwAkAkAMAAAAQ4BFRQWMzI2NwkBPgE3MwYCBxMjJw4BIyIANTQ2Ny4BNTQ2MzIWFxUuASMiBhUFBgHAUky/kFWWQf6iAcX1bAWoC15T++OBXc522P7seXgrLcilS5VMTo4+XnY1AyNRoViSwj9AAo/9+FnLcoT+/
n7+45NZVwET14DhYz99PKLFJCS2LzFvWDNnAAEAsQOqAUsF1QADAAABESMRAUqZBdX91QIrAAABAJ7+8gI8BhIADQAAAQYCFRQSFyMmAjU0EjcCPHl1dniQiIaGiAYS5v4+5+f+O+XrAcbg3wHE7AAAAQCT/
vICMQYSAA0AABMzFhIVFAIHIzYSNTQClJCHhoaHkHd2dgYS7P483+D+OuvlAcXn5wHCAAABADYCSgNjBfAAEQAAAQ0BByURIxEFJy0BNwURMxElA2P+vQFDNf7SZv7RNAFD/
r00AS9mAS4E38LDYsv+hwF5y2LDwmPLAXn+h8sAAQDDAAAFRgUEAAsAAAERIRUhESMRITUhEQNQAfX+C5f+CgH2BQT906r90wItqgItAAABAI7/EgGWAP4ABQAANzMVAyMT2L6UdEr+rP7AAUAAAAEAWgHfAkACgwADAAATIRUhWgHl/
hsCg6QAAQDFAAABgwD+AAMAADczFSPFvr7+/gAAAQAA/0ICbQXVAAMAAAEzASMB1Jn+LJkF1fltAAIAef/jBBsF8AALABcAAAEiAhEQEjMyEhEQAiMiAhEQEgJKjI6OjI2NjY3i7u7i4u7uBVD+zf7M/s3+zQEzATMBNAEzoP5z/ob+h/
5zAY0BeQF6AY0AAQDkAAAD6wXVAAoAADchEQU1JTMRIRUh5QEp/
rwBQrYBKfz6qgRzSLhI+tWqAAEAhwAAA90F8AAcAAAlIRUhNTYANz4BNTQmIyIGBzU+ATMyFhUUBgcGAgFiAnr8q2cBZi5XRZZ5Vb5sbr9P0fg+Uhfxqqqqdw GROm2XSXeWQkPMMTLowlylcB3+6wABAIz/
4wQCBfAAKAAAAR4BFRQEIyImJzUeATMyNjU0JisBNTMyNjU0JiMiBgc1PgEzMhYVFAYC7IKT/u/8VbNgTLRiq7OmlZyjho+TiUuqZ2e1UM/
ygAMlH8SQ3fIlJcMxMpaPhJWmd3BzeyQmtCAg0bJ8qwAAAgBaAAAELgXVAAIADQAACQEhAzMRMxUjESMRITUCuf41Acsw5b+/tf2hBSX84wPN/DOo/qABYMMAAAEAjv/jA/
QF1QAdAAATIRUhET4BMzIAFRQAIyImJzUeATMyNjU0JiMiBgfHAsr93ChPKOEBB/7y9lWvXlGtYJy2tpxJkUsF1ar+kg8P/u7q8f71ICDLMTC2nJy2JCYAAAIAgP/
jBCEF8AALACQAAAEiBhUUFjMyNjU0JhMVLgEjIgIDPgEzMhVIFAIjIgIREAAhMhYCYHqPj3p7j4/uRIxEtL4NNaBgy+v1zOn2AS4A/0SMAzu6oqG7u6GiugJ5uCQm/vL+71dd/u/r5v7qAY0BeQFiAaUeAAEAlwAAA/gF1QAGAAATIRUBIwEhlwNg/
hi9Acr9ewXVVvqBBSsAAwB9/
+MEFwXwAAsAIwAvAAABIgYVFBYzMjY1NCYlLgE1NDYzMhYVFAYHHgEVFAYjIiY1NDYTFBYzMjY1NCYjIgYCSoKUlIKClZX+yHWC5cjJ5IJ0g5Pv3t7vlEGCdnWEhHV2ggLFmoeHmpuGh5pWILKAs9DQs4CyICLGj9no6NmPxgFhdIKCdHSCggACAHT/
4wQUBfAAGAAkAAA3NR4BMzISEw4BIyICNTQSMzISERAAIyImATI2NTQmIyIGFRQWykWMRLS+DTSgYcrr9cvp9v7T/0WMAR57j497eo+PH7gkJgENARJWXAEP6+YBFv5z/ob+n/
5bHgKXuqKhu7uhoroAAgDYAAABlgQjAAMABwAANzMVIxEzFSPYvr6+vv7+BCP+AAIAjv8SAZYEIwADAAkAABMzFSMRMxUDIxPYvr6+lHRKBCP+/dms/sABQAAAAQDDAF4FRgSmAAYAAAkCFQE1AQVF/F8Doft+BIID8P6R/
pO2AdGmAdEAAgDDAWAFRgOiAAMABwAAEyEVIRUhFSHDBIL7fgSC+34Doqjwqq ABAMMAXgVGBKYABgAAEzUBFQE1AcMEgvt+A58D8Lb+L6b+L7YBbQAAAgCEAAADUgXwAAMAJAAAJTMVIxMjNTQ2PwE+ATU0JiMiBgc1PgEzMhYVFAYPAQ4BBw4BFQFgt7e
xrDNRUTMudmFHoVhVrlymyUFRTysjBwUG/v4BkZplglZZNV4xWW5GQ7w5OMKfTIlWVi81GRU8NAAAAgB5/
pwGswWiAAsATAAAARQWMzI2NTQmIyIGAQ4BIyImNTQ2MzIWFzUzET4BNTQmJyYkIyIGBwYCFRQSFx4BMzI2NxcGBCMiJCcmAjU0Ejc2JDMyBBceARUQAAcCroBvb3+Cbm2AAeo2i12bwcKaXYw1gYOVOTle/vOeb8xWjaBoYl75jXTgXlFx/
veJpv7Yc3N4enF0ATGqvgFHb0NH/uL8AhmPo6SOjKWk/khNSfnIyPpLTIP9IBbfsWu8UIOLQUBm/rXBn/7qamhtV1FvYWeDfX0BSb22AUp9f4euoGLme/75/tAGAAIADgAABN4F1QACAAoAAAEDIQEzASMDIQMjAnb3Ae/+oc8CAL17/aJ6wAUO/
RkDrvorAX/+gQAAAwC0AAAEbgXVAAgAEQAgAAABESEyNjU0JiMBESEyNjU0JiMlITIWFRQGBx4BFRQGIyEBawEjk42Nk/7dAQ2Fg4OF/j0B0NDhc3ChlPXi/h4Cyf3dh4uMhQJm/j5vcnFwpsCxiaIUIMuYyNoAAQBn/
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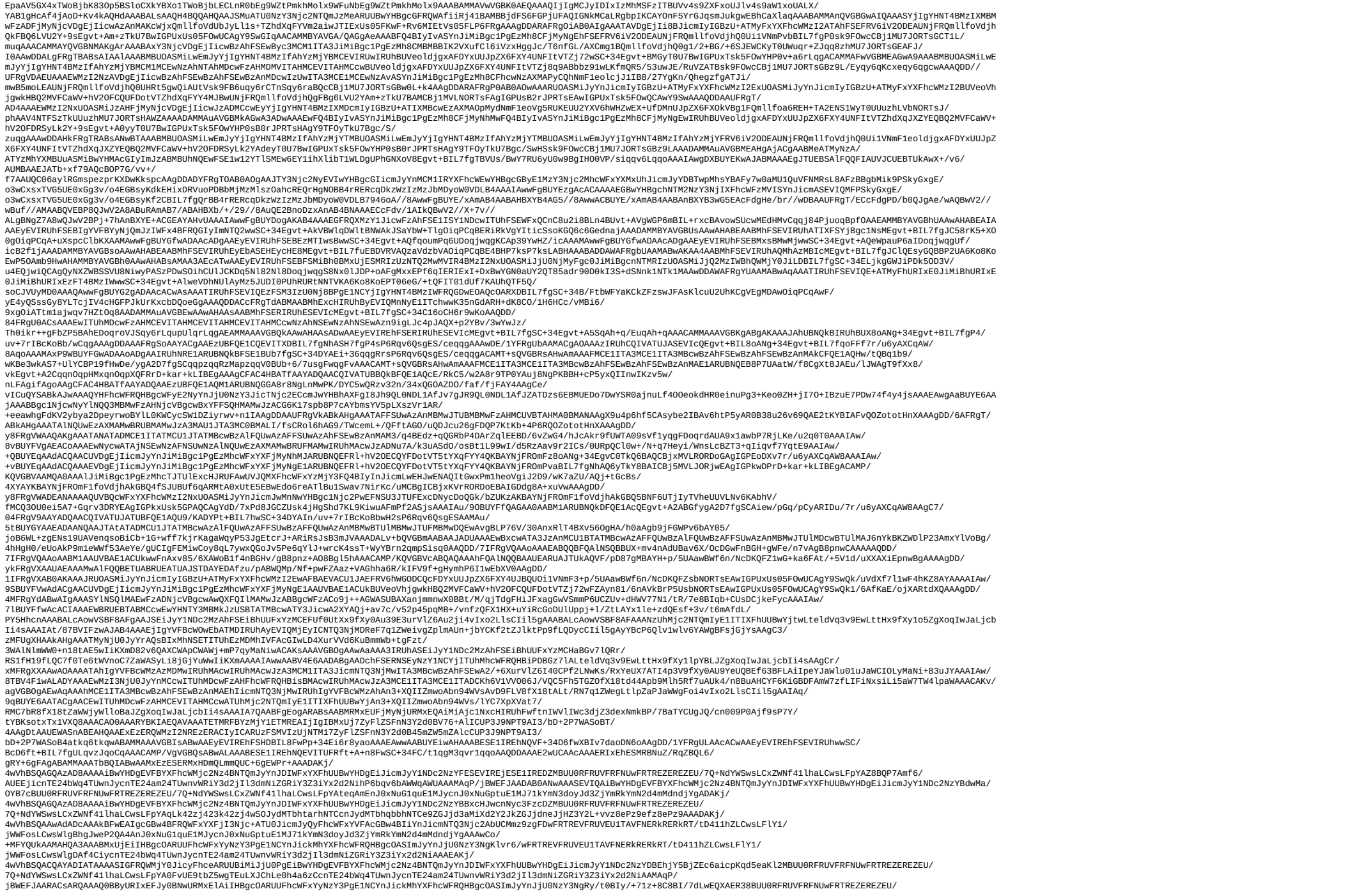
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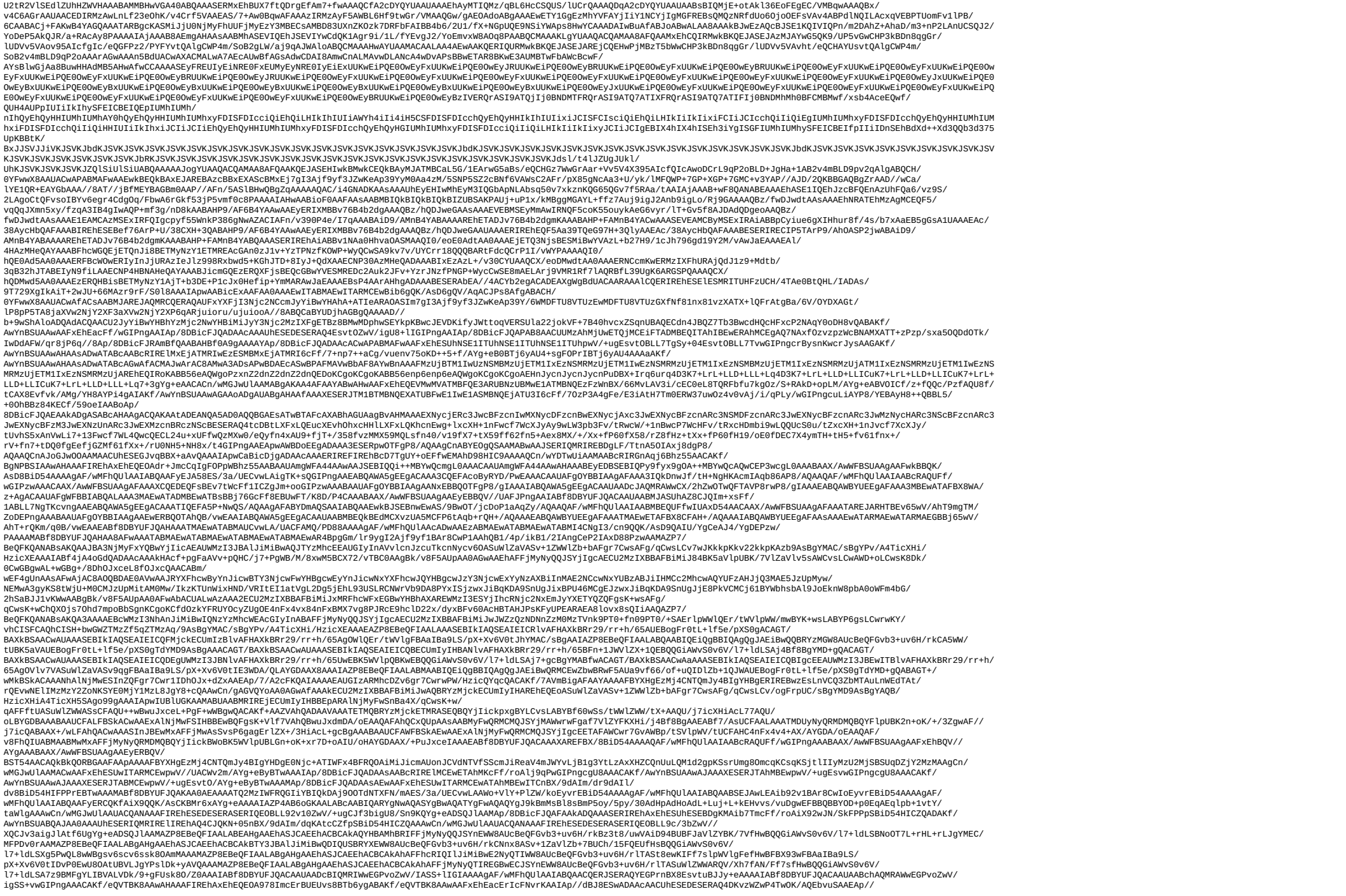

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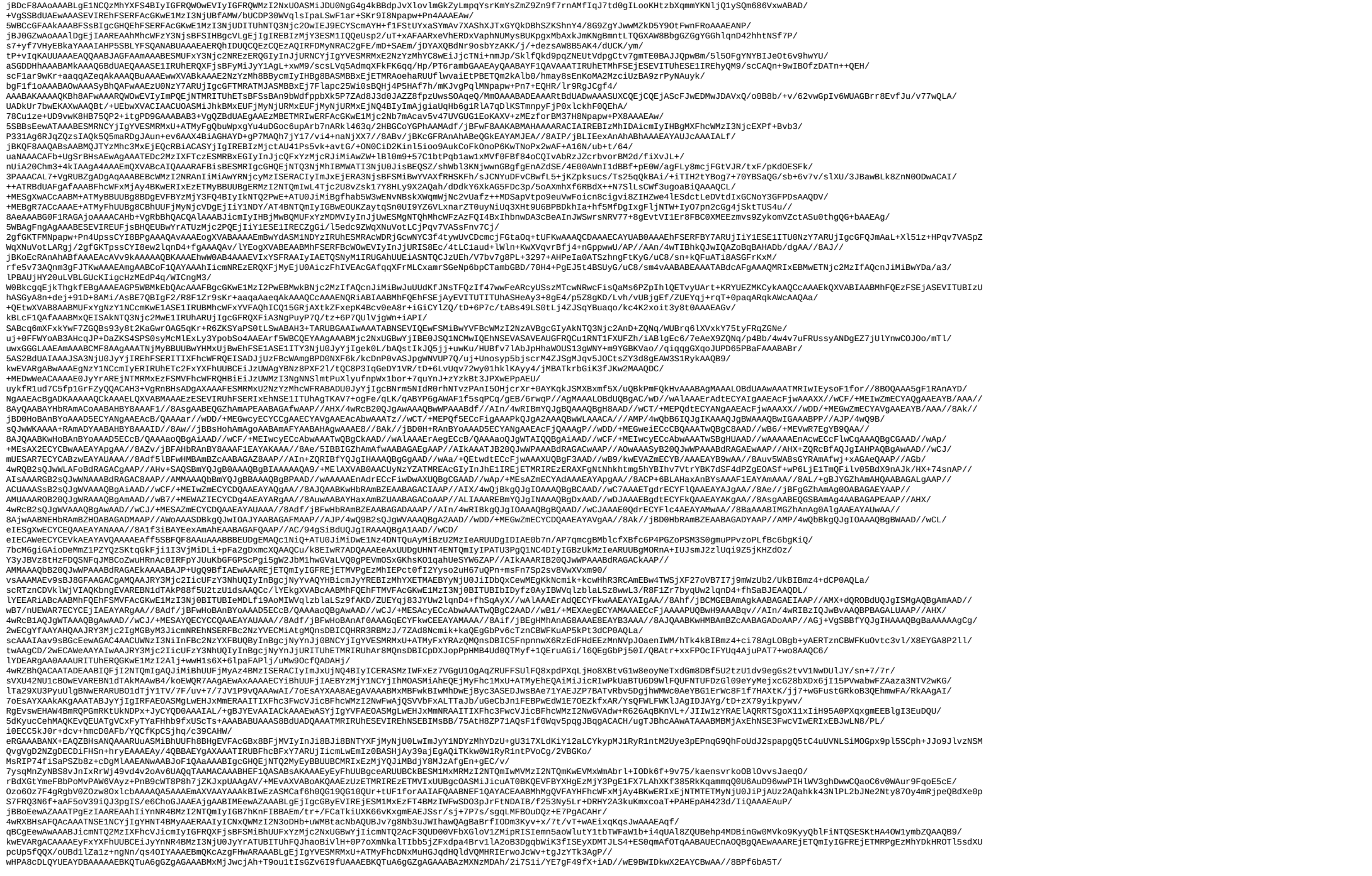
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Structure Analysis - Complete Source Code

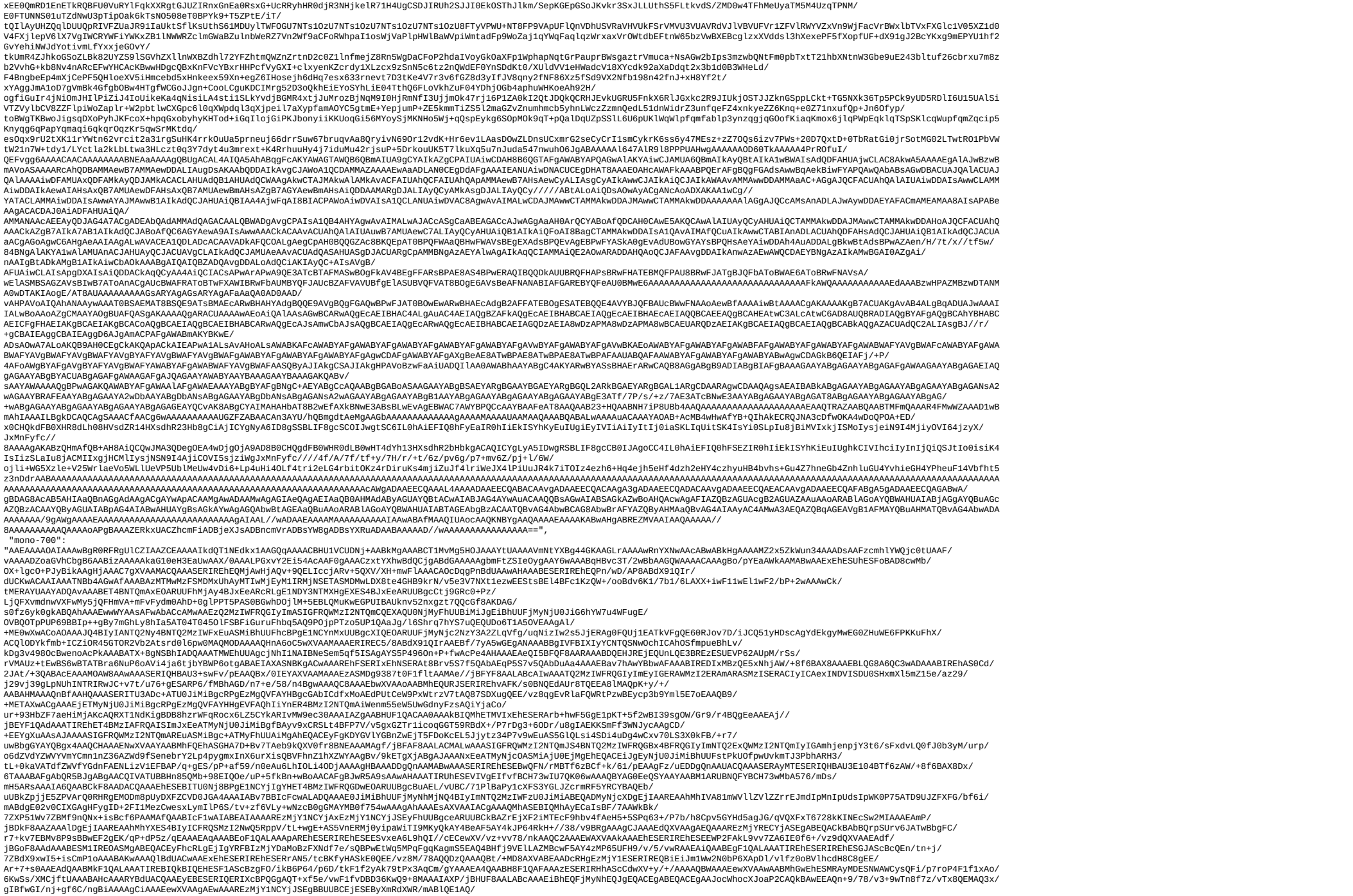
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NkF2whUAAEACAACAAAAAAAABNEAaAAAAAeMA5wACAKQAIQAlAecBfwE1AHkAQgFqAS0BwQBxAHsAvABzAH0AZgCPAIMAhwCBAG8BwQFzAFgAWABYAOkABgAhAH0AmACJAKgAtgB1AIkArABtAHUA4QBWAHcAXACiAFwAhQCBAFoAagA5AAAAGwAIAHMBpgB
vAS8AOQAAAMcAXgCWAKgAWgBcAK4AYgCsAI8AjwCuAFoAUgCsAGIAlgBaASMArABvAKAAUAAAADcAOwCiALAB9gC0AFgAAAHjAJMAdwC6AAgB9gCqAS0AAAEAAHcAWAEtAAABLQEbAFgBEgEfAdUArgBGAcEBbwE5ARIAwQAvAC8ALwCwACEAIQAhCEAI
QAhAAAAmACoAKgAqACoAKwArACsAKwAAAB3AFwAXABcAFwAXAB3//
oAagBqAGoAagAIAKIAfwBeAF4AXgBeAF4AXgAOAKgAXABcAFwAXACPAI8AjwCPAGIArABiAGIAYgBiAGIAQgAZAKAAoACgAKAAOwCWADsAIQBeACEAXgAhAF4AmACoAJgAqACYAKgAmACoAIkAWgAAAFoAqABcAKgAXACoAFwAqABcAKgAXAB1AGIAdQBi
AHUAYgB1AGIAif/DAAMADACsAI8ArACPAKwAjwCsAI8ArACP//gACQBtAI8AdQCuAK4A4QBaAOEAWgDhAFoA4QBa/9kAJQB3AKwAdwCsAHcArP+SAGoArABcAGIAXABiAFwAYgBEAA4AhQEjAIUA/
ACFASMAgQCsAIEArACBAKwAgQCsAFoAbwBaAG8AWgBvAGoAoABqAKAAagCgAGoAoABqAKAAagCgAAAAAAAIADsACABzAKIAcwCiAHMAogCuAAwAAACiAJYAKwAtAJgAmACoAAD/
5AB9AJYAYQCoAFwAfQBsAFsAJgAgAAwArACsABMArgBaAFAAUgAeAKwAXAAEAAkAHQASABQAlgCFAIEArABiADcAbwAVAG8AWgAFAAcAWgB1AAMAHQBzAJ0ACgAKAG4AgwBzAAoAbgCSAFEB4wD+AEkB4wAhAF4ArACPAFwAYgBqAKAAagCgAGoAoABqAK
AAagCgAFwAIQBeACEAXgAAAA4AdQBiAHUArgBcAGIAXABiAAoAbgCPAHUAYgAZAHcArAAAAA7/
+gAZACEAXgAhAF4AbwBcAKgAXABvAI4ArACPAFwAYgBcAGIAPQDyAIUBIwBqAI4AagCgAIEArABaAG8ANgBuAIkArABqAAgAcwCiACEAXgCoAFwAXABiAFwAYgBcAGIAXABiAAgAOwAxACUAbwCPABcAF//6//
oAGQAcAAwArACiAB0AAAAVADkAFQDAAKwAXACVAG4BSQFVAbAA8AwAS0BLQFMAb4BDAEnAUoBAQDHAdUBLQEZAd8BKQDwAI4BGQG/AaABdAFvAS0B9AG/AHcAmAIGAKwAqACsAXMAbQHVAS3/rwHB/gH+ef61/
1X+EP+CAQ4AIQB9ALYAIQCoAHMAiQBcAKwAdQAhAFYAdwCJAFwAiQCiAGIAWgAIAFwAGwBQAFoArAAIADYAlQCsAQ4ATAA2AIMAOABiAJUAjQCsAGEBDgCuAFAArgAvAI8AYgAOAJYAqABiAIgATABBAEgARQA+AQ4ATABiAEwAPgBnAGIAIv4kACIAOgA
eAC4AXABiAGAAWgC2AAwAYwByAAAALQAtAJYAqACPAFwAoACoAKIAlgCYAFYAVgAuAJgAmACYAKIAtgB3AIkAmACuADUAZgBpAA4AYgBhACkAKAArAPgA9AEiASwBRAFEASoBOgFQAU4BDAFEARoBMAEwAR4BMgEOARwBKADkASkBKQEwAPcBMAEwARwBH
AEkASQBLgEnASYBFwFNASIBTwEiASIBMAE5AU0A/
wEXARcBJgFVAU0BFwBiATOAagBaATABTwFKASUBJAFKAXQBMgFNASCBPwEnAScBHQEsASwBeAEXARcBDAEPATABIgDxAU8BHQE5ARwBGgEZARcBRwEBAS4BKQEiAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAEtAS0AAAAAAAAAAAEBAAABsAGwAWoBsACYAJY
AlgCWAJwAnAEAAQAAOQAAAAAAAAAGHAMsAEAGHAMkAEAFaAaQArwDpAAD/
vQGmAS8AJgAmAJ4AqAAAARYBJgD2ASUBHwEcARoBCwEOAQ4BHAG+Ab4BSQEwATkBEgEfAPYBJQEfARwBGgELAQ4BDgEcAb4BvgEpARwBIgEBARwBSQEmASCBFwFJATABSgE5AAAAWQBFAAAABgAnAAAABAAIAAAABAA8AAYAMgAEAAkAGQA8ACwAGAAAAC
4AWgBnAAoAZgCMAAYAFgA7ADsAMgAAAAAASgAAAAgAAAANAFoAdQAhAAD/7QBCAP4AQgD+AEIA/gCaAKQAowCaAEIAQgA7AEkAQgD+AEIA/gD+AEIAQgD+AEIA/gD+AEIAQgBCAEEAQgBCADkAtwCZALcAmQC6AD8AMgAyADIAQgA3ADcAQgBCAfgA/
gBCAEIB+AD+AEIADABCAEIADABCAAwAQgBCAEIAQgBCAEIA/gBCAP4AQgD+AH0AmwCbAH0AQgBCAEIAQgD+AP4AQgD+AEIA/gBCAEIABQDBAEEAwQDBAMEAwQDBAMEAwQBCACcALADBAEIADABCAEIAQgBCAAUAQgAFACEAFwB3AKMAowA///
r/+gBkAGQAYwBkAGQAYwD6AJgAmAB/AEIAWABxAHkBGwEAADEAMQAxALEADABYAFgB9gC2ALYAtgC2AJIAAP/
WAJMAlAHAAJMAQgBKAC8AWABYAFgAWABYAFgAWABYAFgAVwBYAFgAWABYAFgAWABRAFEAWABYAFgAWABYAFgAWABYAFcAWABYAFgAWABYAFgAWABYAFgAWABYAFgAVgBWAFgAWABYAFgAVgBWAFgAWABYAFgAWABYAFgAWgBaAFgAWABYAFgAPgA+ABoAGgAaABoAGgAaADIAMgAyADIAQgBCAEIAQgBYAFgAWABYABwAWgBaAAYBwQDhAFj//v/
+AFoAWgBYAFgAWABYAFgAWABYAFgAWABYADkAPwByAG4AkgCSAJIAkgGmAS8BpgEvAhYADQIWAA0AWABHAAYABgC4AJwAIwBYAQ0BHAENARwB5QBDAEAAABgBfADIABgA0AEIBAAAGAAYABgAGAAYABgAGAAYABg
```

14. assets/vendor/svg2pdf.umd.min.js

```
!function(t,e){"object"==typeof exports&&"undefined"!=typeof module?e(exports,require("jspdf")):"function"==typeof define&&define.amd?define(["exports","jspdf"],e):e((t="undefined"!=typeof
globalThis?globalThis:t||self).svg2pdf={},t.jspdf)}(this,(function(t,e){"use strict";var r=function(t,e){return r=Object.setPrototypeOf||{__proto__:[]}instanceof Array&&function(t,e){t.
__proto__=e}||function(t,e){for(var r in e)Object.prototype.hasOwnProperty.call(e,r)&&(t[r]=e[r])},r(t,e)};function i(t,e){if("function"!=typeof e&&null!==e)throw new TypeError("Class
extends value "+String(e)+" is not a constructor or null");function i(){this.constructor=t}r(t,e),t.prototype=null===e?Object.create(e):(i.prototype=e.prototype,new i)}var n=function(){
return n=Object.assign||function(t){for(var e,r=1,i=arguments.length;r<i;r++)for(var n in e=arguments[r])Object.prototype.hasOwnProperty.call(e,n)&&(t[n]=e[n]);return t},n.apply(this,
arguments)};function a(t,e,r,i){return new(r||(r=Promise))((function(n,a){function s(t){try{l(i.next(t))}catch(t){a(t)}}function o(t){try{l(i.throw(t))}catch(t){a(t)}}function l(t){var e;t.
done?n(t.value):(e=t.value,e instanceof r?e:new r((function(t){t(e)}))).then(s,o)}l((i=i.apply(t,e||[])).next())}))}function s(t,e){var r,i,n,a={label:0,sent:function(){if(1&n[0])throw n[1];
return n[1]},trys:[],ops:[]},s=Object.create(("function"==typeof Iterator?Iterator:Object).prototype);return s.next=o(0),s.throw=o(1),s.return=o(2),"function"==typeof Symbol&&(s[Symbol.
iterator]=function(){return this}),s;function o(o){return function(l){return function(o){if(r)throw new TypeError("Generator is already executing.");for(;s&&(s=0,o[0]&&(a=0)),a;)try{if(r=1,
i&&(n=2&o[0]?i.return:o[0]?i.throw||((n=i.return)&&n.call(i),0):i.next)&&!(n=n.call(i,o[1])).done)return n;switch(i=0,n&&(o=[2&o[0],n.value]),o[0]){case 0:case 1:n=o;break;case 4:return a.
label++,{value:o[1],done:!1};case 5:a.label++,i=o[1],o=[0];continue;case 7:o=a.ops.pop(),a.trys.pop();continue;default:if(!(n=a.trys,(n=n.length>0&&n[n.length-1])||6!==o[0]&&2!==o[0]))a=0;
continue}if(3===o[0]&&(!n||o[1]>n[0]&&o[1]<n[3])){a.label=o[1];break}if(6===o[0]&&a.label<n[1]){a.label=n[1],n=o;break}if(n&&a.label<n[2]){a.label=n[2],a.ops.push(o);break}n[2]&&a.ops.pop(),
a.trys.pop();continue}o=e.call(t,a)}catch(t){o=[6,t],i=0}finally{r=n=0}if(5&o[0])throw o[1];return{value:o[0]?o[1]:void 0,done:!0}}([o,l])}}}"function"==typeof
SuppressedError&&SuppressedError;var o,l,u=function(){function t(t){if(this.a=void 0,this.r=0,this.g=0,this.b=0,this.simpleColors={},this.colorDefs=[],this.ok=!1,t){for(var e in"#"==t.
charAt(0)&&(t=t.substr(1,6)),t=(t=t.replace(/ /g,"")).toLowerCase(),this.simpleColors={aliceblue:"f0f8ff",antiquewhite:"faebd7",aqua:"00ffff",aquamarine:"7fffd4",azure:"f0ffff",beige:
"f5f5dc",bisque:"ffe4c4",black:"000000",blanchedalmond:"ffebcd",blue:"0000ff",blueviolet:"8a2be2",brown:"a52a2a",burlywood:"deb887",cadetblue:"5f9ea0",chartreuse:"7fff00",chocolate:"d2691e",
coral:"ff7f50",cornflowerblue:"6495ed",cornsilk:"fff8dc",crimson:"dc143c",cyan:"00ffff",darkblue:"00008b",darkcyan:"008b8b",darkgoldenrod:"b8860b",darkgray:"a9a9a9",darkgrey:"a9a9a9",
darkgreen:"006400",darkkhaki:"bdb76b",darkmagenta:"8b008b",darkolivegreen:"556b2f",darkorange:"ff8c00",darkorchid:"9932cc",darkred:"8b0000",darksalmon:"e9967a",darkseagreen:"8fbc8f",
darkslateblue:"483d8b",darkslategray:"2f4f4f",darkslategrey:"2f4f4f",darkturquoise:"00ced1",darkviolet:"9400d3",deeppink:"ff1493",deepskyblue:"00bfff",dimgray:"696969",dimgrey:"696969",
dodgerblue:"1e90ff",feldspar:"d19275",firebrick:"b22222",floralwhite:"fffaf0",forestgreen:"228b22",fuchsia:"ff00ff",gainsboro:"dcdcdc",ghostwhite:"f8f8ff",gold:"ffd700",goldenrod:"daa520",
gray:"808080",grey:"808080",green:"008000",greenyellow:"adff2f",honeydew:"f0fff0",hotpink:"ff69b4",indianred:"cd5c5c",indigo:"4b0082",ivory:"fffff0",khaki:"f0e68c",lavender:"e6e6fa",
lavenderblush:"fff0f5",lawngreen:"7cfc00",lemonchiffon:"fffacd",lightblue:"add8e6",lightcoral:"f08080",lightcyan:"e0ffff",lightgoldenrodyellow:"fafad2",lightgray:"d3d3d3",lightgrey:"d3d3d3",
lightgreen:"90ee90",lightpink:"ffb6c1",lightsalmon:"ffa07a",lightseagreen:"20b2aa",lightskyblue:"87cefa",lightslateblue:"8470ff",lightslategray:"778899",lightslategrey:"778899",
lightsteelblue:"b0c4de",lightyellow:"ffffe0",lime:"00ff00",limegreen:"32cd32",linen:"faf0e6",magenta:"ff00ff",maroon:"800000",mediumaquamarine:"66cdaa",mediumblue:"0000cd",mediumorchid:
"ba55d3",mediumpurple:"9370d8",mediumseagreen:"3cb371",mediumslateblue:"7b68ee",mediumspringgreen:"00fa9a",mediumturquoise:"48d1cc",mediumvioletred:"c71585",midnightblue:"191970",mintcream:
"f5fffa",mistyrose:"ffe4e1",moccasin:"ffe4b5",navajowhite:"ffdead",navy:"000080",oldlace:"fdf5e6",olive:"808000",olivedrab:"6b8e23",orange:"ffa500",orangered:"ff4500",orchid:"da70d6",
palegoldenrod:"eee8aa",palegreen:"98fb98",paleturquoise:"afeeee",palevioletred:"d87093",papayawhip:"ffefd5",peachpuff:"ffdab9",peru:"cd853f",pink:"ffc0cb",plum:"dda0dd",powderblue:"b0e0e6",
purple:"800080",red:"ff0000",rosybrown:"bc8f8f",royalblue:"4169e1",saddlebrown:"8b4513",salmon:"fa8072",sandybrown:"f4a460",seagreen:"2e8b57",seashell:"fff5ee",sienna:"a0522d",silver:
"c0c0c0",skyblue:"87ceeb",slateblue:"6a5acd",slategray:"708090",slategrey:"708090",snow:"fffafa",springgreen:"00ff7f",steelblue:"4682b4",tan:"d2b48c",teal:"008080",thistle:"d8bfd8",tomato:
"ff6347",turquoise:"40e0d0",violet:"ee82ee",violetred:"d02090",wheat:"f5deb3",white:"ffffff",whitesmoke:"f5f5f5",yellow:"ffff00",yellowgreen:"9acd32"},this.simpleColors)t==e&&(t=this.
simpleColors[e]);this.colorDefs=[{re:/^rgb\((\d{1,3}),\s*(\d{1,3}),\s*(\d{1,3})\)$/,example:["rgb(123, 234, 45)","rgb(255,234,245)"],process:function(t){return[parseInt(t[1]),parseInt(t[2]),
parseInt(t[3])]}},{re:/^rgb\(([0-9.]+)%,\s*([0-9.]+)%,\s*([0-9.]+)%\)$/,example:["rgb(50.5%, 25.75%, 75.5%)","rgb(100%,0%,0%)"],process:function(t){return[Math.round(2.55*parseFloat(t[1])),
Math.round(2.55*parseFloat(t[2])),Math.round(2.55*parseFloat(t[3]))]}},{re:/^(\w{2})(\w{2})(\w{2})$/,example:["#00ff00","336699"],process:function(t){return[parseInt(t[1],16),parseInt(t[2],
16),parseInt(t[3],16)]}},{re:/^(\w{1})(\w{1})(\w{1})$/,example:["#fb0","f0f"],process:function(t){return[parseInt(t[1]+t[1],16),parseInt(t[2]+t[2],16),parseInt(t[3]+t[3],16)]}}];for(var r=0;
r<this.colorDefs.length;r++){var i=this.colorDefs[r].re,n=this.colorDefs[r].process,a=i.exec(t);if(a){var s=n(a);this.r=s[0],this.g=s[1],this.b=s[2],this.ok=!0}}this.r=this.r<0||isNaN(this.
r)?0:this.r>255?255:this.r,this.g=this.g<0||isNaN(this.g)?0:this.g>255?255:this.g,this.b=this.b<0||isNaN(this.b)?0:this.b>255?255:this.b}}return t.prototype.toRGB=function(){return"rgb(
"+this.r+", "+this.g+", "+this.b+")"},t.prototype.toRGBA=function(){return"rgba("+this.r+", "+this.g+", "+this.b+", "+(this.a||"1")+")"},t.prototype.toHex=function(){var t=this.r.toString(
16),e=this.g.toString(16),r=this.b.toString(16);return 1==t.length&&(t="0"+t),1==e.length&&(e="0"+e),1==r.length&&(r="0"+r),"#"+t+e+r},t.prototype.getHelpXML=function(){for(var e=[],r=0;
r<this.colorDefs.length;r++)for(var i=this.colorDefs[r].example,n=0;n<i.length;n++)e[e.length]=i[n];for(var a in this.simpleColors)e[e.length]=a;var s=document.createElement("ul");s.
setAttribute("id","rgbcolor-examples");for(r=0;r<e.length;r++)try{var o=document.createElement("li"),l=new t(e[r]),u=document.createElement("div");u.style.cssText="margin: 3px; border: 1px
solid black; background:"+l.toHex()+"; color:"+l.toHex(),u.appendChild(document.createTextNode("test"));var h=document.createTextNode(" "+e[r]+" -> "+l.toRGB()+" -> "+l.toHex());o.
appendChild(u),o.appendChild(h),s.appendChild(o)}catch(t){}return s},t}(),h=function(){function t(t){this.color=t}return t.prototype.getFillData=function(t,e){return a(this,void 0,void 0,(
function(){return s(this,(function(t){return[2,void 0]}))}))},t}(),c=function(){function t(){this.xmlSpace="",this.whiteSpace="",this.fill=null,this.fillOpacity=1,this.fontFamily="",this.
fontSize=16,this.fontStyle="",this.fontWeight="",this.opacity=1,this.stroke=null,this.strokeDasharray=null,this.strokeDashoffset=0,this.strokeLinecap="",this.strokeLinejoin="",this.
strokeMiterlimit=4,this.strokeOpacity=1,this.strokeWidth=1,this.alignmentBaseline="",this.textAnchor="",this.visibility="",this.color=null,this.contextFill=null,this.contextStroke=null,this.
fillRule=null}return t.prototype.clone=function(){var e=new t;return e.xmlSpace=this.xmlSpace,e.whiteSpace=this.whiteSpace,e.fill=this.fill,e.fillOpacity=this.fillOpacity,e.fontFamily=this.
fontFamily,e.fontSize=this.fontSize,e.fontStyle=this.fontStyle,e.fontWeight=this.fontWeight,e.opacity=this.opacity,e.stroke=this.stroke,e.strokeDasharray=this.strokeDasharray,e.
strokeDashoffset=this.strokeDashoffset,e.strokeLinecap=this.strokeLinecap,e.strokeLinejoin=this.strokeLinejoin,e.strokeMiterlimit=this.strokeMiterlimit,e.strokeOpacity=this.strokeOpacity,e.
strokeWidth=this.strokeWidth,e.textAnchor=this.textAnchor,e.alignmentBaseline=this.alignmentBaseline,e.visibility=this.visibility,e.color=this.color,e.fillRule=this.fillRule,e.
contextFill=this.contextFill,e.contextStroke=this.contextStroke,e},t.default=function(){var e=new t;return e.xmlSpace="default",e.whiteSpace="normal",e.fill=new h(new u("rgb(0, 0, 0)")),e.
fillOpacity=1,e.fontFamily="times",e.fontSize=16,e.fontStyle="normal",e.fontWeight="normal",e.opacity=1,e.stroke=null,e.strokeDasharray=null,e.strokeDashoffset=0,e.strokeLinecap="butt",e.
strokeLinejoin="miter",e.strokeMiterlimit=4,e.strokeOpacity=1,e.strokeWidth=1,e.alignmentBaseline="baseline",e.textAnchor="start",e.visibility="visible",e.color=new u("rgb(0, 0, 0)"),e.
fillRule="nonzero",e.contextFill=null,e.contextStroke=null,e},t.getContextColors=function(t,e){void 0===e&&(e=!1);var r={};return t.attributeState.contextFill&&(r.contextFill=t.
attributeState.contextFill),t.attributeState.contextStroke&&(r.contextStroke=t.attributeState.contextStroke),e&&t.attributeState.color&&(r.color=t.attributeState.color),r},t}(),f=function(){
function t(t,e){var r,i,n;this.pdf=t,this.svg2pdfParameters=e.svg2pdfParameters,this.attributeState=e.attributeState?e.attributeState.clone():c.default(),this.viewport=e.viewport,this.
refsHandler=e.refsHandler,this.styleSheets=e.styleSheets,this.textMeasure=e.textMeasure,this.transform=null!==(r=e.transform)&&void 0!==r?r:this.pdf.unitMatrix,this.withinClipPath=null!==(
i=e.withinClipPath)&&void 0!==i&&i,this.withinUse=null!==(n=e.withinUse)&&void 0!==n&&n}return t.prototype.clone=function(e){var r,i,n,a;return void 0===e&&(e={}),new t(this.pdf,{
svg2pdfParameters:this.svg2pdfParameters,attributeState:e.attributeState?e.attributeState.clone():this.attributeState.clone(),viewport:null!==(r=e.viewport)&&void 0!==r?r:this.viewport,
refsHandler:this.refsHandler,styleSheets:this.styleSheets,textMeasure:this.textMeasure,transform:null!==(i=e.transform)&&void 0!==i?i:this.transform,withinClipPath:null!==(n=e.
withinClipPath)&&void 0!==n?n:this.withinClipPath,withinUse:null!==(a=e.withinUse)&&void 0!==a?a:this.withinUse})},t}();function p(t){return t&&t.__esModule&&Object.prototype.hasOwnProperty.
call(t,"default")?t.default:t}
/*! https://mths.be/cssesc v3.0.0 by @mathias */var d=function(){if(l)return o;l=1;var t={}.hasOwnProperty,e=/[ -,\.\/:-@\[-\^`\{-~]/,r=/[ -,\.\/:-@\[\]\^`\{-~]/,i=/(^|\\+)?(\\[A-F0-9]{1,
6})\x20(?![a-fA-F0-9\x20])/g,n=function n(a,s){"single"!=(s=function(e,r){if(!e)return r;var i={};for(var n in r)i[n]=t.call(e,n)?e[n]:r[n];return i}(s,n.options)).quotes&&"double"!=s.
quotes&&(s.quotes="single");for(var o="double"==s.quotes?'"':"'",l=s.isIdentifier,u=a.charAt(0),h="",c=0,f=a.length;c<f;){var p=a.charAt(c++),d=p.charCodeAt(),g=void 0;if(d<32||d>126){if(
d>=55296&&d<=56319&&c<f){var m=a.charCodeAt(c++);56320==(64512&m)?d=((1023&d)<<10)+(1023&m)+65536:c--}g="\\"+d.toString(16).toUpperCase()+" "}else g=s.escapeEverything?e.test(p)?"\\"+p:"\\
"+d.toString(16).toUpperCase()+" ":/[\t\n\f\r\x0B]/.test(p)?"\\"+d.toString(16).toUpperCase()+" ":"\\"==p||!l&&('"'==p&&o==p||"'"==p&&o==p)||l&&r.test(p)?"\\"+p:p;h+=g}return l&&(/^-[-\d]/.
test(h)?h="\\-"+h.slice(1):/\d/.test(u)&&(h="\\3"+u+" "+h.slice(1))),h=h.replace(i,(function(t,e,r){return e&&e.length%2?t:(e||"")+r})),!l&&s.wrap?o+h+o:h};return n.options={
escapeEverything:!1,isIdentifier:!1,quotes:"single",wrap:!1},n.version="3.0.0",n}(),g=p(d),m=function(){function t(e){this.renderedElements={},this.idMap=e,this.idPrefix=String(t.
instanceCounter++)}return t.prototype.getRendered=function(t,e,r){return a(this,void 0,void 0,(function(){var i,n;return s(this,(function(a){switch(a.label){case 0:return i=this.generateKey(
t,e),this.renderedElements.hasOwnProperty(i)?[2,this.renderedElements[t]]:(n=this.get(t),this.renderedElements[i]=n,[4,r(n)]);case 1:return a.sent(),[2,n]}))}))},t.prototype.get=function(
t){return this.idMap[g(t,{isIdentifier:!0})]},t.prototype.generateKey=function(t,e){var r="";return e&&(r=["color","contextFill","contextStroke"].map((function(t){var r,i;return null!==(
i=null===(r=e[t])||void 0===r?void 0:r.toRGBA())&&void 0!==i?i:""})).join("|")),this.idPrefix+"|"+t+"|"+r},t.instanceCounter=0,t};function v(t,e){return Math.atan2(e[1]-t[1],e[0]-t[
0])}var y=2/3;function b(t,e){return[y*(e[0]-t[0])+t[0],y*(e[1]-t[1])+t[1]]}function x(t){var e=Math.sqrt(t[0]*t[0]+t[1]*t[1]);return[t[0]/e,t[1]/e]}function S(t,e){return x([e[0]-t[0],e[1]-
t[1]])}function w(t,e){return[t[0]+e[0],t[1]+e[1]]}function k(t,e){var r=t[0],i=t[1];return[e.a*r+e.c*i+e.e,e.b*r+e.d*i+e.f]}var M=function(){function t(){this.segments=[]}return t.
prototype.moveTo=function(t,e){return this.segments.push(new C(t,e)),this},t.prototype.lineTo=function(t,e){return this.segments.push(new F(t,e)),this},t.prototype.curveTo=function(t,e,r,i,
n,a){return this.segments.push(new A(t,e,r,i,n,a)),this},t.prototype.close=function(){return this.segments.push(new T),this},t.prototype.transform=function(t){this.segments.forEach((
function(e){if(e instanceof C||e instanceof F||e instanceof A){var r=k([e.x,e.y],t);e.x=r[0],e.y=r[1]}if(e instanceof A){var i=k([e.x1,e.y1],t),n=k([e.x2,e.y2],t);e.x1=i[0],e.y1=i[1],e.x2=n[
0],e.y2=n[1]}}))},t.prototype.draw=function(t){var e=t.pdf;this.segments.forEach((function(t){t instanceof C?e.moveTo(t.x,t.y):t instanceof F?e.lineTo(t.x,t.y):t instanceof A?e.curveTo(t.x1,
t.y1,t.x2,t.y2,t.x,t.y):e.close()}))},t}(),C=function(t,e){this.x=t,this.y=e},F=function(t,e){this.x=t,this.y=e},A=function(t,e,r,i,n,a){this.x1=t,this.y1=e,this.x2=r,this.y2=i,this.x=n,
this.y=a},T=function(){};function P(t,e){return e.split(",").indexOf((t.nodeName||t.tagName).toLowerCase())>=0}function B(t,e,r,i){var n;void 0===i&&(i=r);var a=null===(n=t.style)||void
0===n?void 0:n.getPropertyValue(i);if(a)return a;var s=e.getPropertyValue(t,i);return s||(t.hasAttribute(r)&&t.getAttribute(r)||void 0)}function N(t,e,r){if("none"===B(t.element,r.
styleSheets,"display"))return!1;var i=e,n=B(t.element,r.styleSheets,"visibility");return n&&(i="hidden"!==n),i}function E(t,e,r){var i=N(t,e,r);return 0!==t.element.childNodes.length&&(t.
children.forEach((function(t){t.isVisible(i,r)&&(i=!0)})),i)}var O,I,L=function(){function t(){this.markers=[]}return t.prototype.addMarker=function(t){this.markers.push(t)},t.prototype.
draw=function(t){return a(this,void 0,void 0,(function(){var e,r,i,n,a,o,l,u;return s(this,(function(s){switch(s.label){case 0:e=0,s.label=1;case 1:return e<this.markers.length?(r=this.
markers[e],i=void 0,n=r.angle,a=r.anchor,o=Math.cos(n),l=Math.sin(n),i=t.pdf.Matrix(o,l,-l,o,a[0],a[1]),i=t.pdf.matrixMult(t.pdf.Matrix(t.attributeState.strokeWidth,0,0,t.attributeState.
```



```
u&&(c["stroke-opacity"]=a),t.pdf.setGState(new e.GState(c))}if(t.attributeState.fill&&t.attributeState.fill!==r.attributeState.fill&&t.attributeState.fill instanceof h&&t.attributeState.
fill.color.ok&&!P(i,"text")&&t.pdf.setFillColor(t.attributeState.fill.color.r,t.attributeState.fill.color.g,t.attributeState.fill.color.b),t.attributeState.strokeWidth!==r.attributeState.
strokeWidth&&t.pdf.setLineWidth(t.attributeState.strokeWidth),t.attributeState.stroke!==r.attributeState.stroke&&t.attributeState.stroke instanceof h&&t.pdf.setDrawColor(t.attributeState.
stroke.color.r,t.attributeState.stroke.color.g,t.attributeState.stroke.color.b),t.attributeState.strokeLinecap!==r.attributeState.strokeLinecap&&t.pdf.setLineCap(t.attributeState.
strokeLinecap),t.attributeState.strokeLinejoin!==r.attributeState.strokeLinejoin&&t.pdf.setLineJoin(t.attributeState.strokeLinejoin),t.attributeState.strokeDasharray===r.attributeState.
strokeDasharray&&t.attributeState.strokeDashoffset===r.attributeState.strokeDashoffset||!t.attributeState.strokeDasharray||t.pdf.setLineDashPattern(t.attributeState.strokeDasharray,t.
attributeState.strokeDashoffset),t.attributeState.strokeMiterlimit!==r.attributeState.strokeMiterlimit&&t.pdf.setLineMiterLimit(t.attributeState.strokeMiterlimit),t.attributeState.
fontFamily!==r.attributeState.fontFamily&&(s=U.hasOwnProperty(t.attributeState.fontFamily)?U[t.attributeState.fontFamily]:t.attributeState.fontFamily),t.attributeState.fill&&t.
attributeState.fill!==r.attributeState.fill&&t.attributeState.fill instanceof h&&t.attributeState.fill.color.ok){var f=t.attributeState.fill.color;t.pdf.setTextColor(f.r,f.g,f.b)}t.
attributeState.fontWeight===r.attributeState.fontWeight&&t.attributeState.fontStyle===r.attributeState.fontStyle||(o=X(t.attributeState.fontStyle,t.attributeState.fontWeight)),void
0===s&&void 0===o||(void 0===s&&(s=U.hasOwnProperty(t.attributeState.fontFamily)?U[t.attributeState.fontFamily]:t.attributeState.fontFamily),t.pdf.setFont(s,o)),t.attributeState.
fontSize!==r.attributeState.fontSize&&t.pdf.setFontSize(t.attributeState.fontSize*t.pdf.internal.scaleFactor)}function ht(t,e,r){var i=R.exec(t);if(i){var n=i[1];return r.refsHandler.get(
n)||void 0}function ct(t,e,r){return a(this,void 0,void 0,(function(){var i,n;return s(this,(function(a){switch(a.label){case 0:return i=r.clone(),e.element.hasAttribute(
"clipPathUnits")&&"objectboundingbox"===e.element.getAttribute("clipPathUnits").toLowerCase()&&(n=t.getBoundingBox(r),i.transform=r.pdf.matrixMult(r.pdf.Matrix(n[2],0,0,n[3],n[0],n[1]),r.
transform)),[4,e.apply(i)];case 1:return a.sent(),[2]}}))}))}var ft=function(t){function e(){return null!==t&&t.apply(this,arguments)||this}return i(e,t),e.prototype.render=function(t){
return a(this,void 0,void 0,(function(){var e,r,i,n;return s(this,(function(a){switch(a.label){case 0:return this.isVisible("hidden"!==t.attributeState.visibility,t)?((e=t.clone()).
transform=e.pdf.matrixMult(this.computeNodeTransform(e),t.transform),lt(e,this),r=B(this.element,e.styleSheets,"clip-path"),(i=r&&"none"!==r)?(n=ht(r,0,e))?n.isVisible(!0,e)?(e.pdf.
saveGraphicsState(),[4,ct(this,n,e)]):[3,2]:[3,4]:[3,5]):[2];case 1:return a.sent(),[3,3];case 2:return[2];case 3:return[3,5];case 4:i=!1,a.label=5;case 5:return e.withClipPath||e.pdf.
saveGraphicsState(),ut(e,t,this.element),[4,this.renderCore(e)];case 6:return a.sent(),e.withClipPath||e.pdf.restoreGraphicsState(),i&&e.pdf.restoreGraphicsState(),[2]}}))}))},e)(J),
pt=function(t){function e(){return null!==t&&t.apply(this,arguments)||this}return i(e,t),e}(ft),dt=function(t){function e(e,r,i){var n=t.call(this,r,i)||this;return n.cachedPath=null,n.
hasMarkers=e,n}return i(e,t),e.prototype.renderCore=function(t){return a(this,void 0,void 0,(function(){var e;return s(this,(function(r){switch(r.label){case 0:return null===(e=this.
getCachedPath(t))||0===e.segments.length?[2]:(t.withinClipPath?e.transform(t.transform):t.pdf.setCurrentTransformationMatrix(t.transform),e.draw(t),[4,this.fillOrStroke(t)]);case 1:return r.
sent(),this.hasMarkers?[4,this.drawMarkers(t,e)]:[3,3];case 2:r.sent(),r.label=3;case 3:return[2]}}))}))},e.prototype.getCachedPath=function(t){return this.cachedPath||(this.cachedPath=this.
getPath(t))},e.prototype.drawMarkers=function(t,e){return a(this,void 0,void 0,(function(){return s(this,(function(r){switch(r.label){case 0:return[4,this.getMarkers(e,t).draw(t.clone({
transform:t.pdf.unitMatrix}))];case 1:return r.sent(),[2]}}))}))},e.prototype.fillOrStroke=function(t){return a(this,void 0,void 0,(function(){var e,r,i,n,a;return s(this,(function(s){
switch(s.label){case 0:return t.withinClipPath?[2]:(e=t.attributeState.fill,r=t.attributeState.stroke&&0!==t.attributeState.strokeWidth,e?[4,e.getFillData(this,t)]:[3,2]);case 1:return n=s.
sent(),[3,3];case 2:n=void 0,s.label=3;case 3:return i=n,a="evenodd"===t.attributeState.fillRule,e&&r||t.withinUse?a?t.pdf.fillStrokeEvenOdd(i):t.pdf.fillStroke(i):e?a?t.pdf.fillEvenOdd(i):
t.pdf.fill(i):r?t.pdf.stroke():t.pdf.discardPath(),[2]}}))}))},e.prototype.getBoundingBoxCore=function(t){var e=this.getCachedPath(t);if(!e||!e.segments.length)return[0,0,0,0];for(var
r=Number.POSITIVE_INFINITY,i=Number.POSITIVE_INFINITY,n=Number.NEGATIVE_INFINITY,a=Number.NEGATIVE_INFINITY,s=0,o=0,l=0;l<e.segments.length;l++){var u=e.segments[l];(u instanceof C||u
instanceof F||u instanceof A)&&(s=u.x,o=u.y),u instanceof A?(r=Math.min(r,s,u.x1,u.x2,u.x),n=Math.max(n,s,u.x1,u.x2,u.x),i=Math.min(i,o,u.y1,u.y2,u.y),a=Math.max(a,o,u.y1,u.y2,u.y)):(r=Math.
min(r,s),n=Math.max(n,s),i=Math.min(i,o),a=Math.max(a,o))}return[r,i,n-r,a-i]},e.prototype.getMarkers=function(t,e){var r=B(this.element,e.styleSheets,"marker-start"),i=B(this.element,e.
styleSheets,"marker-mid"),n=B(this.element,e.styleSheets,"marker-end"),a=new L;if(r||i||n){n&&(n=gt(n)),r&&(r=gt(r)),i&&(i=gt(i));for(var s=t.segments,o=[1,0],l=void 0,u=!1,h=[1,0],c=!1,
f=function(t){var e=s[t],f=r&&(1===t||!!(s[t]instanceof C)&&s[t-1]instanceof C);f&&s.forEach((function(e,r){if(!c&&e instanceof T&&r>t){var i=s[r-1];c=(i instanceof C||i instanceof F||i
instanceof A)&&i}}));var p=n&&(t===s.length-1||!(s[t]instanceof C)&&s[t+1]instanceof C),d=i&&t>0&&!(1===t&&s[t-1]instanceof C),g=s[t-1]||null;if(g instanceof C||g instanceof F||g instanceof
A){if(e instanceof A)f&&a.addMarker(new _(r,[g.x,g.y],v(c?[c.x,c.y]:[g.x,g.y],[e.x1,e.y1),!0)),p&&a.addMarker(new _(n,[e.x,e.y],v([e.x2,e.y2],[e.x,e.y])),d&&(l=S([g.x,g.y],[e.x1,e.y1)),
l=g instanceof C?l:x(w(o,l)),a.addMarker(new _(i,[g.x,g.y],Math.atan2(l[1],l[0])))),o=S([e.x2,e.y2],[e.x,e.y]);else if(e instanceof C||e instanceof F){if(l=S([g.x,g.y],[e.x,e.y]),f){var
m=c?S([c.x,c.y],[e.x,e.y]):l;a.addMarker(new _(r,[g.x,g.y],Math.atan2(m[1],m[0]),!0))}if(p&&a.addMarker(new _(n,[e.x,e.y],Math.atan2(l[1],l[0]))),d){m=e instanceof C?o:g instanceof C?l:x(w(
o,l));a.addMarker(new _(i,[g.x,g.y],Math.atan2(m[1],m[0])))}o=l}else if(e instanceof T){if(l=S([g.x,g.y],[u.x,u.y]),d){m=g instanceof C?l:x(w(o,l));a.addMarker(new _(i,[g.x,g.y],Math.atan2(
m[1],m[0])))}if(p){m=x(w(l,h));a.addMarker(new _(n,[u.x,u.y],Math.atan2(m[1],m[0])))}o=l}}else{u=e instanceof C&&e;var y=s[t+1];(y instanceof C||y instanceof F||y instanceof A)&&(h=S([u.x,u.
y],[y.x,y.y]))}},p=0;p<s.length;p++)f(p)}return a.markers.forEach((function(t){var r=e.refsHandler.get(t.id);if(r){var i=B(r.element,e.styleSheets,"orient");null!=i&&(t.isStartMarker&&"auto-
start-reverse"===i&&(t.angle+=Math.PI),isNaN(Number(i))||(t.angle=parseFloat(i)/180*Math.PI))}})),a},e)(pt);function gt(t){var e=R.exec(t);return e&&e[1]||void 0}var mt=function(t){function
e(e,r){return t.call(this,!0,e,r)||this}return i(e,t),e.prototype.getPath=function(t){if(t.withinClipPath||null===t.attributeState.stroke)return null;var e=parseFloat(this.element.
getAttribute("x1")||"0"),r=parseFloat(this.element.getAttribute("y1")||"0"),i=parseFloat(this.element.getAttribute("x2")||"0"),n=parseFloat(this.element.getAttribute("y2")||"0");return
e||i||r||n?(new M).moveTo(e,r).lineTo(i,n):null},e.prototype.computeNodeTransformCore=function(t){return t.pdf.unitMatrix},e.prototype.isVisible=function(t,e){return N(this,t,e)},e.
prototype.fillOrStroke=function(e){return a(this,void 0,void 0,(function(){return s(this,(function(r){switch(r.label){case 0:return e.attributeState.fill=null,[4,t.prototype.fillOrStroke.
call(this,e)];case 1:return r.sent(),[2]}}))}))},e}(dt),vt=function(t){function e(){return null!==t&&t.apply(this,arguments)||this}return i(e,t),e.prototype.apply=function(t){return a(this,
void 0,void 0,(function(){var e,r,i,n,a;return s(this,(function(s){switch(s.label){case 0:return this.isVisible("hidden"!==t.attributeState.visibility,t)?((e=t.clone()).transform=e.pdf.
unitMatrix,lt(e,this),r=B(this.element,e.styleSheets,"clip-path"),r&&"none"!==r&&(i=ht(r,0,e))?i.isVisible(!0,e)?[4,ct(this,i,e)]:[3,2]:[3,3]):[2];case 1:return s.sent(),[3,3];case 2:return[
2];case 3:ut(e,t,this.element),n=0,a=this.children,s.label=4;case 4:return n<a.length?[4,a[n].render(e)]:[3,7];case 5:s.sent(),s.label=6;case 6:return n++,[3,4];case 7:return[2]}}))}))},e.
prototype.getBoundingBoxCore=function(t){return Q(t,this)},e.prototype.isVisible=function(t,e){return E(this,t,e)},e.prototype.computeNodeTransformCore=function(t){var e=parseFloat(B(this.
element,t.styleSheets,"x")||"0"),r=parseFloat(B(this.element,t.styleSheets,"y")||"0"),i=this.element.getAttribute("viewBox");if(i){var n=V(i),a=parseFloat(B(this.element,t.styleSheets,
"width")||B(this.element.ownerSVGElement,t.styleSheets,"width")||i[2]),s=parseFloat(B(this.element,t.styleSheets,"height")||B(this.element.ownerSVGElement,t.styleSheets,"height")||i[3]);
return K(this.element,n,e,r,a,s,t)}return t.pdf.Matrix(1,0,0,1,e,r)},e}(tt),yt=function(t,e){this.width=t,this.height=e},bt=function(t){function e(){return null!==t&&t.apply(this,
arguments)||this}return i(e,t),e.prototype.renderCore=function(t){return a(this,void 0,void 0,(function(){var r,i,n,a,o,l,u,h,p,d,g,m,v;return s(this,(function(s){switch(s.label){case 0:
return r=parseFloat,(i=this.element.getAttribute("href")||this.element.getAttribute("xlink:href"))?(n=i.substring(1),a=t.refsHandler.get(n),o=P(a.element,"symbol,svg")&&a.element.
hasAttribute("viewBox"),l=r(B(this.element,t.styleSheets,"x")||"0"),u=r(B(this.element,t.styleSheets,"y")||"0"),h=void 0,p=void 0,o?(h=r(B(this.element,t.styleSheets,"width")||B(a.element,t.
styleSheets,"width")||"0"),p=r(B(this.element,t.styleSheets,"height")||B(a.element,t.styleSheets,"height")||"0"),l+=r(B(a.element,t.styleSheets,"x")||"0"),u+=r(B(a.element,t.styleSheets,
"y")||"0"),g=V(a.element.getAttribute("viewBox")),d=K(a.element,g,l,u,h,p,t)):d=t.pdf.Matrix(1,0,0,1,l,u),m=c.getContextColors(t,!0),v=new f(t.pdf,{refsHandler:t.refsHandler,styleSheets:t.
styleSheets,withinUse:!0,viewport:o?new yt(h,p):t.viewport,svg2pdfParameters:t.svg2pdfParameters,textMeasure:t.textMeasure,attributeState:Object.assign(c.default(),m)}),[4,t.refsHandler.
getRendered(n,m,(function(t){return e.renderReferencedNode(t,n,v)}))]):[2];case 1:return s.sent(),t.pdf.saveGraphicsState(),t.pdf.setCurrentTransformationMatrix(t.transform),
o&&"visible"!==B(a.element,t.styleSheets,"overflow")&&(t.pdf.rect(l,u,h,p),t.pdf.clip().discardPath()),t.pdf.doFormObject(t.refsHandler.generateKey(n,m),d),t.pdf.restoreGraphicsState(),[
2]}}))}))},e.renderReferencedNode=function(t,e,r){return a(this,void 0,void 0,(function(){var i;return s(this,(function(n){switch(n.label){case 0:return i=[(i=t.getBoundingBox(r))[0]-.5*i[
2],i[1]-.5*i[3],2*i[2],2*i[3]],r.pdf.beginFormObject(i[0],i[1],i[2],i[3],r.pdf.unitMatrix),t instanceof vt?[4,t.apply(r)]:[3,2];case 1:return n.sent(),[3,4];case 2:return[4,t.render(r)];
case 3:n.sent(),n.label=4;case 4:return r.pdf.endFormObject(r.refsHandler.generateKey(e,r.attributeState)),[2]}}))}))},e.prototype.getBoundingBoxCore=function(t){return $(this.element,t)},e.
prototype.isVisible=function(t,e){return N(this,t,e)},e.prototype.computeNodeTransformCore=function(t){return t.pdf.unitMatrix},e}(pt),xt=function(t){function e(e,r){return t.call(this,!1,e,
r)||this}return i(e,t),e.prototype.getPath=function(t){var e=parseFloat(B(this.element,t.styleSheets,"width")||"0"),r=parseFloat(B(this.element,t.styleSheets,"height")||"0");if(!isFinite(
e)||e<=0||!isFinite(r)||r<=0)return null;var i=B(this.element,t.styleSheets,"rx"),n=B(this.element,t.styleSheets,"ry"),a=Math.min(parseFloat(i||n||"0"),.5*e),s=Math.min(parseFloat(
n||i||"0"),.5*r),o=parseFloat(B(this.element,t.styleSheets,"x")||"0"),l=parseFloat(B(this.element,t.styleSheets,"y")||"0"),u=4/3*(Math.SQRT2-1);return 0===a&&0===s?(new M).moveTo(o,l).
lineTo(o+e,l).lineTo(o+e,l+r).lineTo(o,l+r).close():(new M).moveTo(o+=a,l).lineTo(o+=e-2*a,l).curveTo(o+a*u,l,o+a,l+(s-s*u),o+=a,l+=s).lineTo(o,l+=r-2*s).curveTo(o,l+s*u,o-a*u,l+s,o-=a,
l+=s).lineTo(o+=2*a-e,l).curveTo(o-a*u,l,o-a,l-s*u,o-=a,l-=s).lineTo(o,l+=2*s-r).curveTo(o,l-s*u,o+a*u,l-s,o+=a,l-=s).close()},e.prototype.computeNodeTransformCore=function(t){return t.pdf.
unitMatrix},e.prototype.isVisible=function(t,e){return N(this,t,e)},e}(dt),St=function(t){function e(e,r){return t.call(this,!1,e,r)||this}return i(e,t),e.prototype.getPath=function(t){var
e=this.getRx(t),r=this.getRy(t);if(!isFinite(e)||r<=0||!isFinite(r)||r<=0)return null;var i=parseFloat(B(this.element,t.styleSheets,"cx")||"0"),n=parseFloat(B(this.element,t.styleSheets,
"cy")||"0"),a=4/3*(Math.SQRT2-1)*e,s=4/3*(Math.SQRT2-1)*r;return(new M).moveTo(i+e,n).curveTo(i+e,n-s,i+a,n-r,i,n-r).curveTo(i-a,n-r,i-e,n-s,i-e,n).curveTo(i-e,n+s,i-a,n+r,i,n+r).curveTo(
i+a,n+r,i+e,n+s,i+e,n)},e.prototype.computeNodeTransformCore=function(t){return t.pdf.unitMatrix},e.prototype.isVisible=function(t,e){return N(this,t,e)},e}(dt),wt=function(t){function e(e,
r){return t.call(this,e,r)||this}return i(e,t),e.prototype.getRx=function(t){return parseFloat(B(this.element,t.styleSheets,"rx")||"0")},e.prototype.getRy=function(t){return parseFloat(B(
this.element,t.styleSheets,"ry")||"0")},e}(St);function kt(t){var e="invisible",r=t.stroke&&0!==t.strokeWidth,i=t.fill;return i&&r?e="fillThenStroke":i?e="fill":r&&(e="stroke"),e}function
Mt(t){return t.replace(/[\n\r]/g,"")}function Ct(t){return t.replace(/[\t]/g," ")}function Ft(t){return t.replace(/ +/g," ")}function At(t,e,r){switch(B(t,r.styleSheets,"text-transform")){
case"uppercase":return e.toUpperCase();case"lowercase":return e.toLowerCase();default:return e}}var Tt,Pt,Bt,Nt,Et,Ot,It,Lt,_t,Rt,Ht,Dt,jt,qt,Vt=function(){function t(t,e,r,i){this.
textNode=t,this.texts=[],this.textNodes=[],this.contexts=[],this.textAnchor=e,this.originX=r,this.originY=i,this.textMeasures=[]}return t.prototype.setX=function(t){this.originX=t},t.
prototype.setY=function(t){this.originY=t},t.prototype.add=function(t,e,r){this.texts.push(e),this.textNodes.push(t),this.contexts.push(r)},t.prototype.rightTrimText=function(){for(var
t=this.texts.length-1;t>=0;t--){if("preserve"===this.contexts[t].attributeState.xmlSpace||"pre"===this.contexts[t].attributeState.whiteSpace||(this.texts[t]=this.texts[t].replace(/\s+$/,
"")),this.texts[t].match(/[^\s]/))return!1}return!0},t.prototype.measureText=function(t){for(var e=0;e<this.texts.length;e++)this.textMeasures.push({width:t.textMeasure.measureTextWidth(
this.texts[e],this.contexts[e].attributeState),length:this.texts[e].length})},t.prototype.put=function(e,r){var i,n,a,s,o=[],l=[],u=[],h=this.originX,c=this.originY,f=h,p=h;for(i=0;i<this.
textNodes.length;i++){n=this.textNodes[i],a=this.contexts[i],s=this.textMeasures[i]||{width:e.textMeasure.measureTextWidth(this.texts[i],this.contexts[i].attributeState),length:this.texts[
i].length};var d=h,g=c;if("#text"!==n.nodeName&&!o.includes(n)){o.push(n);var m=t.resolveRelativePositionAttribute(n,"dx");null!==m&&(d+=j(m,a.attributeState.fontSize));var v=t.
resolveRelativePositionAttribute(n,"dy");null!==v&&(g+=j(v,a.attributeState.fontSize))}l[i]=d,u[i]=g,h=d+s.width+s.length*r,c=g,f=Math.min(f,d),p=Math.max(p,h)}var y=0;switch(this.
textAnchor){case"start":y=0;break;case"middle":y=(p-f)/2;break;case"end":y=p-f}for(i=0;i<this.textNodes.length;i++)if(n=this.textNodes[i],a=this.contexts[i],"#text"===n.
```

nodeName||"hidden"!==a.attributeState.visibility){e.pdf.saveGraphicsState(),ut(a,e,n);var b=a.attributeState.alignmentBaseline,x=kt(a.attributeState);e.pdf.text(this.texts[i],l[i]-y,u[i],{
baseline:q(b),angle:e.transform,renderingMode:"fill"===x?void 0:x,charSpace:0===r?void 0:r});e.pdf.restoreGraphicsState()}return[h,c]},t.resolveRelativePositionAttribute=function(t,e){for(
var r,i=t;i&&P(i,"tspan");){if(i.hasAttribute(e))return i.getAttribute(e);if((null===(r=t.parentElement)||void 0===r?void 0:r.firstChild)!==t)break;i=i.parentElement}return null},t}(),
Wt=function(t){function e(){var e=null!==t&&t.apply(this,arguments)||this;return e.boundingBox=[],e}return i(e,t),e.prototype.processTSpans=function(t,e,r,i,n,a){for(var s=r.pdf.getFontSize(
),o="preserve"===r.attributeState.xmlSpace||"pre"===r.attributeState.whiteSpace,l=!0,u=!1,h=0;h<e.childNodes.length;h++){var c=e.childNodes[h];if(c.textContent){var f=c.textContent;if(
"#text"===c.nodeName){var p=Mt(f);p=Ct(p),o||(p=Ft(p),l&&p.match(/^\s/)&&(u=!0),p.match(/[^\s]/)&&(l=!1),a.prevText.match(/\s$/)&&(p=p.replace(/^\s+/,"")));var d=At(e,p,r);n.add(e,d,r),a.
prevText=f,a.prevContext=r}else if(P(c,"title"));else if(P(c,"tspan")){var g=c,m=g.getAttribute("x");if(null!==m){var v=j(m,s);n=new Vt(this,B(g,r.styleSheets,"text-anchor")||r.
attributeState.textAnchor,v,0),i.push({type:"y",chunk:n})}var y=g.getAttribute("y");if(null!==y){var b=j(y,s);n=new Vt(this,B(g,r.styleSheets,"text-anchor")||r.attributeState.textAnchor,0,
b),i.push({type:"x",chunk:n})}var x=r.clone();lt(x,t,g),this.processTSpans(t,g,x,i,n,a)}}}return u},e.prototype.renderCore=function(t){return a(this,void 0,void 0,(function(){var e,r,i,n,a,
o,l,u,h,c,f,p,d,g,m,v,y,b,x,S,w,k,M,C;return s(this,(function(s){if(t.pdf.saveGraphicsState(),e=0,r=0,i=1,n=t.pdf.getFontSize(),a=j(this.element.getAttribute("x"),n),o=j(this.element.
getAttribute("y"),n),l=j(this.element.getAttribute("dx"),n),u=j(this.element.getAttribute("dy"),n),h=parseFloat(this.element.getAttribute("textLength"))||"0"),c=t.attributeState.visibility,
0===this.element.childElementCount)f=this.element.textContent||"",p=function(t,e){return t=Ct(t=Mt(t)),"preserve"===e.xmlSpace||"pre"===e.whiteSpace||(t=Ft(t=t.trim())),t}(f,t.
attributeState),d=At(this.element,p,t),e=t.textMeasure.getTextOffset(d,t.attributeState),h>0&&(g=t.textMeasure.measureTextWidth(d,t.attributeState),!("preserve"===t.attributeState.
xmlSpace||"pre"===t.attributeState.whiteSpace)&&f.match(/^\s/)&&(i=0),r=(h-g)/(d.length-i)||0),"visible"===c&&(m=t.attributeState.alignmentBaseline,v=kt(t.attributeState),t.pdf.text(d,a+l-e,
o+u,{baseline:q(m),angle:t.transform,renderingMode:"fill"===v?void 0:v,charSpace:0===r?void 0:r}),y=o+u+.1*n,this.boundingBox=[a+l-e,y-n,t.textMeasure.measureTextWidth(d,t.attributeState),
n]);else{for(b=[],x=new Vt(this,t.attributeState.textAnchor,a+l,o+u),b.push({type:"",chunk:x}),S=this.processTSpans(this,this.element,t,b,x,{prevText:" ",prevContext:t}),i=S?0:1,w=!0,k=b.
length-1;k>=0;k--)w&&(w=b[k].chunk.rightTrimText());h>0&&(M=0,C=0,b.forEach((function(e){var r=e.chunk;r.measureText(t),r.textMeasures.forEach((function(t){var e=t.width,r=t.length;M+=e,
C+=r})),r=(h-M)/(C-i)),b.reduce((function(e,i){var n=i.type,a=i.chunk;return"x"===n?a.setX(e[0]):"y"===n&&a.setY(e[1]),a.put(t,r)}),[0,0])}return t.pdf.restoreGraphicsState(),[2]})))})},
e.prototype.isVisible=function(t,e){return E(this,t,e)},e.prototype.getBoundingBoxCore=function(t){return this.boundingBox.length>0?this.boundingBox:$(this.element,t)},e.prototype.
computeNodeTransformCore=function(t){return t.pdf.unitMatrix},e}(pt);function Gt(){if(Nt)return Bt;function t(){if(!(this instanceof t))return new t;this.queue=[],this.cache=null}return
Nt=1,t.prototype.matrix=function(t){return 1===t[0]&&0===t[1]&&0===t[2]&&1===t[3]&&0===t[4]&&0===t[5]||(this.cache=null,this.queue.push(t)),this},t.prototype.translate=function(t,e){return
0===t&&0===e||(this.cache=null,this.queue.push([1,0,0,1,t,e])),this},t.prototype.scale=function(t,e){return 1===t&&1===e||(this.cache=null,this.queue.push([t,0,0,e,0,0])),this},t.prototype.
rotate=function(t,e,r){var i,n,a;return 0!==t&&(this.translate(e,r),i=t*Math.PI/180,n=Math.cos(i),a=Math.sin(i),this.queue.push([n,a,-a,n,0,0]),this.cache=null,this.translate(-e,-r)),this},
t.prototype.skewX=function(t){return 0!==t&&(this.cache=null,this.queue.push([1,0,Math.tan(t*Math.PI/180),1,0,0])),this},t.prototype.skewY=function(t){return 0!==t&&(this.cache=null,this.
queue.push([1,Math.tan(t*Math.PI/180),0,1,0,0])),this},t.prototype.toArray=function(){if(this.cache)return this.cache;if(!this.queue.length)return this.cache=[1,0,0,1,0,0],this.cache;if(
this.cache=this.queue[0],1===this.queue.length)return this.cache;for(var t=1;t<this.queue.length;t++)this.cache=(e=this.cache,r=this.queue[t],[e[0]*r[0]+e[2]*r[1],e[1]*r[0]+e[3]*r[1],e[0]*r[
2]+e[2]*r[3],e[1]*r[2]+e[3]*r[3],e[0]*r[4]+e[2]*r[5]+e[4],e[1]*r[4]+e[3]*r[5]+e[5]]);var e,r;return this.cache},t.prototype.calc=function(t,e,r){var i;return this.queue.length?(this.cache||(
this.cache=this.toArray()),[t*(i=this.cache)[0]+e*i[2]+(r?0:i[4]),t*i[1]+e*i[3]+(r?0:i[5])]):[t,e]},Bt=t}function Ut(){if(Lt)return It;Lt=1;var t=2*Math.PI;function e(t,e,r,i){var n=t*r+e*i;
return n>1&&(n=1),n<-1&&(n=-1),(t*i-e*r<0?-1:1)*Math.acos(n)}function r(t,e){var r=4/3*Math.tan(e/4),i=Math.cos(t),n=Math.sin(t),a=Math.cos(t+e),s=Math.sin(t+e);return[i,n,i-n*r,n+i*r,a+s*r,
s-a*r,a,s]}return It=function(i,n,a,s,o,l,u,h,c){var f=Math.sin(c*t/360),d=p*(i-a)/2+f*(n-s)/2,g=-f*(i-a)/2+p*(n-s)/2;if(0===d&&0===g)return[];if(0===u||0===h)return[];
u=Math.abs(u),h=Math.abs(h);var m=d*d/(u*u)+g*g/(h*h);m>1&&(u*=Math.sqrt(m),h*=Math.sqrt(m));var v=function(r,i,n,a,s,o,l,u,h,c){var f=c*(r-n)/2+h*(i-a)/2,p=-h*(r-n)/2+c*(i-a)/2,d=l*l,g=u*u,
m=f*f,v=p*p,y=d*g-d*v-g*m;y<0&&(y=0),y/=d*v+g*m;var b=(y=Math.sqrt(y)*(s===o?-1:1))*l/u*p,x=y*-u/l*f,S=c*b-h*x+(r+n)/2,w=h*b+c*x+(i+a)/2,k=(f-b)/l,M=(p-x)/u,C=(-f-b)/l,F=(-p-x)/u,A=e(1,0,k,
M),T=e(k,M,C,F);return 0===o&&T>0&&(T-=t),1===o&&T<0&&(T+=t),[S,w,A,T]}(i,n,a,s,o,l,u,h,f,p),y=[],b=v[2],x=v[3],S=Math.max(Math.ceil(Math.abs(x)/(t/4)),1);x/=S;for(var w=0;w<S;w++)y.push(r(
b,x)),b+=x;return y.map((function(t){for(var e=0;e<t.length;e+=2){var r=t[e+0],i=t[e+1],n=p*(r*u)-f*(i*=h),a=f*r+p*i;t[e+0]=n+v[0],t[e+1]=a+v[1]}return t}))}}function zt(){if(Rt)return _t;
Rt=1;var t=1e-10,e=Math.PI/180;function r(t,e,i){if(!(this instanceof r))return new r(t,e,i);this.rx=t,this.ry=e,this.ax=i}return r.prototype.transform=function(r){var i=Math.cos(this.ax*e),
n=Math.sin(this.ax*e),a=[this.rx*(r[0]*i+r[2]*n),this.rx*(r[1]*i+r[3]*n),this.ry*(-r[0]*n+r[2]*i),this.ry*(-r[1]*n+r[3]*i)],s=a[0]*a[0]+a[2]*a[2],o=a[1]*a[1]+a[3]*a[3],l=((a[0]-a[3])*(a[0]-
a[3])+(a[2]+a[1])*(a[2]+a[1]))*((a[0]+a[3])*(a[0]+a[3])+(a[2]-a[1])*(a[2]-a[1])),u=(s+o)/2;if(l<t*u)return this.rx=this.ry=Math.sqrt(u),this.ax=0,this;var h=a[0]*a[1]+a[2]*a[3],c=u+(l=Math.
sqrt(l))/2,f=u-l/2;return this.ax=Math.abs(h)<t&&Math.abs(c-o)<t?90:180*Math.atan(Math.abs(h)>Math.abs(c-o)?(c-s)/h:h/(c-o))/Math.PI,this.ax>=0?(this.rx=Math.sqrt(c),this.ry=Math.sqrt(f)):(
this.ax+=90,this.rx=Math.sqrt(f),this.ry=Math.sqrt(c)),this},r.prototype.isDegenerate=function(){return this.rx<t*this.ry||this.ry<t*this.rx},_t=r}function Yt(){if(Dt)return Ht;Dt=1;var
t=function(){if(Pt)return Tt;Pt=1;var t={a:7,c:6,h:1,l:2,m:2,r:4,q:4,s:4,t:2,v:1,z:0},e=[5760,6158,8192,8193,8194,8195,8196,8197,8198,8199,8200,8201,8202,8239,8287,12288,65279];function r(
t){return t>=48&&t<=57}function i(t){return t>=48&&t<=57||43===t||45===t||46===t}function n(t){this.index=0,this.path=t,this.max=t.length,this.result=[],this.param=0,this.err="",this.
segmentStart=0,this.data=[]}function a(t){for(;t.index<t.max&&(10===(r=t.path.charCodeAt(t.index))||13===r||8232===r||8233===r||32===r||9===r||11===r||12===r||160===r||r>=5760&&e.indexOf(
r)>=0);)t.index++;var r}function s(t){var e=t.path.charCodeAt(t.index);return 48===e?(t.param=0,void t.index++):49===e?(t.param=1,void t.index++):void(t.err="SvgPath: arc flag can be 0 or 1
only (at pos "+t.index+")")}function o(t){var e,i=t.index,n=i,a=t.max,s=!1,o=!1,l=!1,u=!1;if(n>=a)t.err="SvgPath: missed param (at pos "+n+")";else if(43!==(e=t.path.charCodeAt(
n))&&45!==e||(e=++n<a?t.path.charCodeAt(n):0),r(e)||46===e){if(46!==e){if(s=48===e,e=++n<a?t.path.charCodeAt(n):0,s&&n<a&&e&&r(e))return void(t.err="SvgPath: numbers started with `0` such
as `09` are illegal (at pos "+i+")");for(;n<a&&r(t.path.charCodeAt(n));)n++,o=!0;e=n<a?t.path.charCodeAt(n):0}if(46===e){for(u=!0,n++;r(t.path.charCodeAt(n));)n++,l=!0;e=n<a?t.path.
charCodeAt(n):0}if(101===e||69===e){if(u&&!o&&!l)return void(t.err="SvgPath: invalid float exponent (at pos "+n+")");if(43!==(e=++n<a?t.path.charCodeAt(n):0)&&45!==e||n++,!(n<a&&r(t.path.
charCodeAt(n))))return void(t.err="SvgPath: invalid float exponent (at pos "+n+")");for(;n<a&&r(t.path.charCodeAt(n));)n++}t.index=n,t.param=parseFloat(t.path.slice(i,n))+0}else t.
err="SvgPath: param should start with 0..9 or `.` (at pos "+n+")"}function l(e){var r,i;i=(r=e.path[e.segmentStart]).toLowerCase();var n=e.data;if("m"===i&&n.length>2&&(e.result.push([r,n[
0],n[1]]),n=n.slice(2),i="l",r="m"===r?"l":"L"),"r"===i)e.result.push([r].concat(n));else for(;n.length>=t[i]&&(e.result.push([r].concat(n.splice(0,t[i]))),t[i]););}function u(e){var r,n,u,
h,c,f=e.max;if(e.segmentStart=e.index,n=97==(32|(r=e.path.charCodeAt(e.index))),function(t){switch(32|t){case 109:case 122:case 108:case 104:case 118:case 99:case 115:case 113:case 116:case
97:case 114:return!0}return!1}(r))if(h=t[e.path[e.index].toLowerCase()],e.index++,a(e),e.data=[],h){for(u=!1;;){for(c=h;c>0;c--){if(!n||3!==c&&4!==c?o(e):s(e),e.err.length)return;e.data.
push(e.param),a(e),u=!1,e.index<f&&44===e.path.charCodeAt(e.index)&&(e.index++,a(e),u=!0)}if(!u){if(e.index>=e.max)break;if(!i(e.path.charCodeAt(e.index)))break}}l(e)}else l(e);else e.
err="SvgPath: bad command "+e.path[e.index]+" (at pos "+e.index+")"}return Tt=function(t){var e=new n(t),r=e.max;for(a(e);e.index<r&&!e.err.length;)u(e);return e.err.length?e.result=[]:e.
result.length&&("mM".indexOf(e.result[0][0])<0?(e.err="SvgPath: string should start with `M` or `m`",e.result=[]):e.result[0][0]="M"),{err:e.err,segments:e.result}}(),e=function(){if(
Ot)return Et;Ot=1;var t=Gt(),e={matrix:!0,scale:!0,rotate:!0,translate:!0,skewX:!0,skewY:!0},r=/\s*(matrix|translate|scale|rotate|skewX|skewY)\s*\(\s*(.+?)\s*\)[\s,]*/,i=/[\s,]+/;return
Et=function(n){var a,s,o=new t;return n.split(r).forEach((function(t){if(t.length)if(void 0===e[t])switch(s=t.split(i).map((function(t){return+t||0})),a){case"matrix":return void(6===s.
length&&o.matrix(s));case"scale":return void(1===s.length?o.scale(s[0],s[0]):2===s.length&&o.scale(s[0],s[1]));case"rotate":return void(1===s.length?o.rotate(s[0],0,0):3===s.length&&o.
rotate(s[0],s[1],s[2]));case"translate":return void(1===s.length?o.translate(s[0],0):2===s.length&&o.translate(s[0],s[1]));case"skewX":return void(1===s.length&&o.skewX(s[0]));case"skewY":
return void(1===s.length&&o.skewY(s[0]))}else a=t})),o},Et}(),r=Gt(),i=Ut(),n=zt();function a(e){if(!(this instanceof a))return new a(e);var r=t(e);this.segments=r.segments,this.err=r.err,
this.__stack=[]}return a.from=function(t){if("string"==typeof t)return new a(t);if(t instanceof a){var e=new a("");return e.err=t.err,e.segments=t.segments.map((function(t){return t.slice(
)})),e.__stack=t.__stack.map((function(t){return r().matrix(t.toArray())})),e}throw new Error("SvgPath.from: invalid param type "+t)},a.prototype.__matrix=function(t){var e,r=this;t.queue.
length&&this.iterate((function(i,a,s,o){var l,u,h,c;switch(i[0]){case"v":u=0===(l=t.calc(0,i[1],!0))[0]?["v",l[1]]:["l",l[0],l[1]];break;case"V":u=(l=t.calc(s,i[1],!1))[0]===t.calc(s,o,!1)[
0]?["V",l[1]]:["L",l[0],l[1]];break;case"h":u=0===(l=t.calc(i[1],0,!0))[1]?["h",l[0]]:["l",l[0],l[1]];break;case"H":u=(l=t.calc(i[1],o,!1))[1]===t.calc(s,o,!1)[1]?["H",l[0]]:["L",l[0],l[1]];
break;case"a":case"A":var f=t.toArray(),p=n(i[1],i[2],i[3]).transform(f);if(f[0]*f[3]-f[1]*f[2]<0&&(i[5]=i[5]?"0":"1"),t.calc(i[6],i[7],"a"===i[0]),"A"===i[0]&&i[6]===s&&i[7]===o||"a"===i[
0]&&0===i[6]&&0===i[7]){u=["a"===i[0]?"l":"L",i[0],i[1]];break}u=p.isDegenerate()?["a"===i[0]?"l":"L",i[0],i[1]]:[i[0],p.rx,p.ry,p.ax,i[4],i[5],i[0],i[1]];break;case"m":c=a>0,u=["m",(l=t.
calc(i[1],i[2],c))[0],l[1]];break;default:for(u=[h=i[0]],c=h.toLowerCase()===h,e=1;e<i.length;e+=2)l=t.calc(i[e],i[e+1],c),u.push(l[0],l[1])}r.segments[a]=u},!0)},a.prototype.
__evaluateStack=function(){var t,e;if(this.__stack.length){if(1===this.__stack.length)return this.__matrix(this.__stack[0]),void(this.__stack=[]);for(t=r(),e=this.__stack.length;--e>=0;)t.
matrix(this.__stack[e].toArray());this.__matrix(t),this.__stack=[]}},a.prototype.toString=function(){var t="",e="",r=!1;this.__evaluateStack();for(var i=0,n=this.segments.length;i<n;i++){
var a=this.segments[i],s=a[0];s!==e||"M"===s||"M"===s?("M"===s&&"z"===e&&(t+=" "),t+=s,r=!1):r=!0;for(var o=1;o<a.length;o++){var l=a[o];1===o?r&&l>=0&&(t+=" "):l>=0&&(t+=" "),
t+=l}e=s}return t},a.prototype.translate=function(t,e){return this.__stack.push(r().translate(t,e||0)),this},a.prototype.scale=function(t,e){return this.__stack.push(r().scale(t,e||0===e?e:
t)),this},a.prototype.rotate=function(t,e,i){return this.__stack.push(r().rotate(t,e||0,i||0)),this},a.prototype.skewX=function(t){return this.__stack.push(r().skewX(t)),this},a.prototype.
skewY=function(t){return this.__stack.push(r().skewY(t)),this},a.prototype.matrix=function(t){return this.__stack.push(r().matrix(t)),this},a.prototype.transform=function(t){return t.trim(
)?(this.__stack.push(e(t)),this):this},a.prototype.round=function(t){var e,r=0,i=0,n=0,a=0;return t=t||0,this.__evaluateStack(),this.segments.forEach((function(s){var o=s[0].toLowerCase(
)===s[0];switch(s[0]){case"H":case"h":return o&&(s[1]+=n),n=s[1]-s[1].toFixed(t),void(s[1]=+s[1].toFixed(t));case"V":case"v":return o&&(s[1]+=a),a=s[1]-s[1].toFixed(t),void(s[1]=+s[1].
toFixed(t));case"Z":case"z":return n=r,void(a=i);case"M":case"m":return o&&(s[1]+=n,s[2]+=a),n=s[1]-s[1].toFixed(t),a=s[2]-s[2].toFixed(t),r=n,i=a,s[1]=+s[1].toFixed(t),void(s[2]=+s[2].
toFixed(t));case"A":case"a":return o&&(s[6]+=n,s[7]+=a),n=s[6]-s[6].toFixed(t),a=s[7]-s[7].toFixed(t),s[1]=+s[1].toFixed(t),s[2]=+s[2].toFixed(t),s[3]=+s[3].toFixed(t+2),s[6]=+s[6].toFixed(
t),void(s[7]=+s[7].toFixed(t));default:return e=s.length,o&&(s[e-2]+=n,s[e-1]+=a),n=s[e-2]-s[e-2].toFixed(t),a=s[e-1]-s[e-1].toFixed(t),void s.forEach((function(e,r){r&&(s[r]=+s[r].toFixed(
t))}))}})),this},a.prototype.iterate=function(t,e){var r,i,n,a=this.segments,s={},o=!1,l=0,u=0,h=0,c=0;if(e||this.__evaluateStack(),a.forEach((function(e,r){var i=t(e,r,l,u);Array.isArray(
i)&&(s[r]=i,o=!0);var n=e[0]===e[0].toLowerCase();switch(e[0]){case"m":case"M":return l=e[1]+(n?l:0),u=e[2]+(n?u:0),h=l,void(c=u);case"h":case"H":return void(l=e[1]+(n?l:0));case"v":case"V":
return void(u=e[1]+(n?u:0));case"z":case"Z":return l=h,void(u=c);default:l=e[e.length-2]+(n?l:0),u=e[e.length-1]+(n?u:0)}})),!o)return this;for(n=[],r=0;r<a.length;r++)if(void 0!==s[r])for(
i=0;i<s[r].length;i++)n.push(s[r][i]);else n.push(a[r]);return this.segments=n,this},a.prototype.abs=function(){return this.iterate((function(t,e,r,i){var n,a=t[0],s=a.toUpperCase();if(
a!==s)switch(t[0]=s,a){case"v":return void(t[1]+=i);case"a":return t[6]+=r,void(t[7]+=i);default:for(n=1;n<t.length;n++)t[n]+=n%2?r:i})),!0),this},a.prototype.rel=function(){return this.
iterate((function(t,e,r,i){var n,a=t[0],s=a.toLowerCase();if(a!==s&&(0!==e||"M"!==a))switch(t[0]=s,a){case"V":return void(t[1]-=i);case"A":return t[6]-=r,void(t[7]-=i);default:for(n=1;n<t.
length;n++)t[n]-=n%2?r:i}}),!0),this},a.prototype.unarc=function(){return this.iterate((function(t,e,r,n){var a,s,o,l=[],u=t[0];return"A"!==u&&"a"!==u?null:("a"===u?(s=r+t[6],o=n+t[7]):(s=t[
6],o=t[7]),0===(a=i(r,n,s,o,t[4],t[5],t[1],t[2],t[3])).length?[["a"===t[0]?"l":"L",t[6],t[7]]]:(a.forEach((function(t){l.push(["C",t[2],t[3],t[4],t[5],t[6],t[7]])})),l))})),this},a.
prototype.unshort=function(){var t,e,r,i,n,a=this.segments;return this.iterate((function(s,o,l,u){var h,c=s[0],f=c.toUpperCase();o&&("T"===f?(h="t"===c,"Q"===(r=a[o-1])[0]?(t=r[1]-l,e=r[2]-

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u):"q"===r[0]?(t=r[1]-r[3],e=r[2]-r[4]):(t=0,e=0),i=-t,n=-e,h||(i+=l,n+=u),a[o]=[h?"q":"Q",i,n,s[1],s[2]]):"S"===f&&(h="s"===c,"C"===(r=a[o-1])[0]?(t=r[3]-l,e=r[4]-u):"c"===r[0]?(t=r[3]-r[
5],e=r[4]-r[6]):(t=0,e=0),i=-t,n=-e,h||(i+=l,n+=u),a[o]=[h?"c":"C",i,n,s[1],s[2],s[3],s[4]])))),this},Ht=a}var Xt=p(qt?jt:(qt=1,jt=Yt())),Qt=function(t){function e(e,r){return t.call(this,
!0,e,r)||this}return i(e,t),e.prototype.computeNodeTransformCore=function(t){return t.pdf.unitMatrix},e.prototype.isVisible=function(t,e){return N(this,t,e)},e.prototype.getPath=function(t){
var e,r,i=new Xt(B(this.element,t.styleSheets,"d"))||"").unshort().unarc().abs(),n=new M;return i.iterate((function(t){switch(t[0]){case"M":n.moveTo(t[1],t[2]);break;case"L":n.lineTo(t[1],t[
2]);break;case"H":n.lineTo(t[1],r);break;case"V":n.lineTo(e,t[1]);break;case"C":n.curveTo(t[1],t[2],t[3],t[4],t[5],t[6]);break;case"Q":var i=b([e,r],[t[1],t[2]]),a=b([t[3],t[4]],[t[1],t[
2]]);n.curveTo(i[0],i[1],a[0],a[1],t[3],t[4]);break;case"Z":n.close()}switch(t[0]){case"M":case"L":e=t[1],r=t[2];break;case"H":e=t[1];break;case"V":r=t[1];break;case"C":e=t[5],r=t[6];break;
case"Q":e=t[3],r=t[4]}})),n},e}(dt),$t=/^\s*data:(([^/,;]+\/[^/,;]+)(?:;([^,;=]+=[^,;=]+))?)?(?:;(base64))?,((?:.|\s)*)$/i,Kt=function(t){function e(r,i,n){var a=t.call(this,r,i)||this;
return a.imageLoadingPromise=null,a.imageUrl=a.element.getAttribute("xlink:href")||a.element.getAttribute("href"),a.imageUrl&&e.shouldLoadImage(a.imageUrl,n.svg2pdfParameters.loadImages)&&(
a.imageLoadingPromise=e.fetchImageData(a.imageUrl)),a}return i(e,t),e.shouldLoadImage=function(t,e){return!1!==e&&(!(e instanceof RegExp)||(e.lastIndex=0,e.test(t)))},e.prototype.
renderCore=function(t){return a(this,void 0,void 0,(function(){var r,i,n,a,o,l,u,h,c,p,d,g,v,y,b,x,S,w,k;return s(this,(function(s){switch(s.label){case 0:return this.imageLoadingPromise?(t.
pdf.setCurrentTransformationMatrix(t.transform),r=parseFloat(B(this.element,t.styleSheets,"width")||"0"),i=parseFloat(B(this.element,t.styleSheets,"height")||"0"),n=parseFloat(B(this.
element,t.styleSheets,"x")||"0"),a=parseFloat(B(this.element,t.styleSheets,"y")||"0"),!isFinite(r)||r<=0||!isFinite(i)||i<=0?[2]:[4,this.imageLoadingPromise]):[2];case 1:return o=s.sent(),
l=o.data,0!==(u=o.format).indexOf("svg")?[3,3]:(h=new DOMParser,c=h.parseFromString(l,"image/svg+xml").firstElementChild,(!(p=this.element.getAttribute("preserveAspectRatio"))||p.indexOf(
"defer")<0||!c.getAttribute("preserveAspectRatio"))&&c.setAttribute("preserveAspectRatio",p||""),c.setAttribute("x",String(n)),c.setAttribute("y",String(a)),c.setAttribute("width",String(
r)),c.setAttribute("height",String(i)),d={},g=new f(t.pdf,{refsHandler:new m(d),styleSheets:t.styleSheets,viewport:new yt(r,i),svg2pdfParameters:t.svg2pdfParameters,textMeasure:t.
textMeasure}),[4,ue(c,g,d).render(g)]);case 2:return s.sent(),[2];case 3:v="").concat(btoa(l)),s.label=4;case 4:return s.trys.push([4,6,,7]),[4,e.
getImageDimensions(v)];case 5:return y=s.sent(),b=y[0],x=y[1],S=[0,0,b,x],w=K(this.element,S,n,a,r,i,t),t.pdf.setCurrentTransformationMatrix(w),t.pdf.addImage(v,"",0,0,b,x),[3,7];case 6:
return k=s.sent(),"object"==typeof console&&console.warn&&console.warn("Could not load image ".concat(this.imageUrl,". \n").concat(k)),[3,7];case 7:return[2]}}))}))},e.prototype.
getBoundingBoxCore=function(t){return $(this.element,t)},e.prototype.computeNodeTransformCore=function(t){return t.pdf.unitMatrix},e.prototype.isVisible=function(t,e){return N(this,t,e)},e.
fetchImageData=function(t){return a(this,void 0,void 0,(function(){var r,i,n,a,o;return s(this,(function(s){switch(s.label){case 0:if(!(n=t.match($t)))return[3,1];if(a=n[2],"image"!==(o=a.
split("/"))[0])throw new Error("Unsupported image URL: ".concat(t));return i=o[1],r=n[5],"base64"===n[4]?(r=r.replace(/\s/g,""),r=atob(r)):r=decodeURIComponent(r),[3,3];case 1:return[4,e.
fetchImage(t)];case 2:r=s.sent(),i=t.substring(t.lastIndexOf(".")+1),s.label=3;case 3:return[2,{data:r,format:i}]}}))}))},e.fetchImage=function(t){return new Promise((function(e,r){var
i=new XMLHttpRequest;i.open("GET",t,!0),i.responseType="arraybuffer",i.onload=function(){if(200!==i.status)throw new Error("Error ".concat(i.status,": Failed to load image '").concat(t,
"'"));for(var r=new Uint8Array(i.response),n="",a=0;a<r.length;a++)n+=String.fromCharCode(r[a]);e(n)},i.onerror=r,i.onabort=r,i.send(null)}))},e.getMimeType=function(t){switch(t=t.
toLowerCase()){case"jpg":case"jpeg":return"image/jpeg";default:return"image/".concat(t)}},e.getImageDimensions=function(t){return new Promise((function(e,r){var i=new Image;i.
onload=function(){e([i.width,i.height])},i.onerror=r,i.src=t})},e}(pt),Zt=function(t){function e(e,r,i){var n=t.call(this,!0,r,i)||this;return n.closed=e,n}return i(e,t),e.prototype.
getPath=function(t){if(!this.element.hasAttribute("points")||""===this.element.getAttribute("points"))return null;var r=e.parsePointsString(this.element.getAttribute("points")),i=new M;if(r.
length<1)return i;i.moveTo(r[0][0],r[0][1]);for(var n=1;n<r.length;n++)i.lineTo(r[n][0],r[n][1]);return this.closed&&i.close(),i},e.prototype.isVisible=function(t,e){return N(this,t,e)},e.
prototype.computeNodeTransformCore=function(t){return t.pdf.unitMatrix},e.parsePointsString=function(t){for(var e=V(t),r=[],i=0;i<e.length-1;i+=2){var n=e[i],a=e[i+1];r.push([n,a])}return
r},e}(dt),Jt=function(t){function e(e,r){return t.call(this,!0,e,r)||this}return i(e,t),e}(Zt),te=function(t){function e(){return null!==t&&t.apply(this,arguments)||this}return i(e,t),e.
prototype.render=function(t){return Promise.resolve()},e.prototype.getBoundingBoxCore=function(t){return[0,0,0,0]},e.prototype.computeNodeTransformCore=function(t){return t.pdf.unitMatrix},
e.prototype.isVisible=function(t,e){return N(this,t,e)},e}(J),ee=function(t){function r(){return null!==t&&t.apply(this,arguments)||this}return i(r,t),r.prototype.apply=function(t){return a(
this,void 0,void 0,(function(){var r,i,n,a,o,l;return s(this,(function(s){switch(s.label){case 0:r=this.computeNodeTransform(t),i=this.getBoundingBox(t),t.pdf.beginFormObject(i[0],i[1],i[2],
i[3],r),n=c.getContextColors(t),function(t){var r=t.attributeState,i=t.pdf,n=1,a=1;n*=r.fillOpacity,n*=r.opacity,r.fill instanceof h&&void 0!==r.fill.color.a&&(n*=r.fill.color.a),a*=r.
strokeOpacity,a*=r.opacity,r.stroke instanceof h&&void 0!==r.stroke.color.a&&(a*=r.stroke.color.a);var s,o={};if(o.opacity=n,o["stroke-opacity"]=a,i.setGState(new e.GState(o)),r.fill&&r.
fill instanceof h&&r.fill.color.ok?i.setFillColor(r.fill.color.r,r.fill.color.g,r.fill.color.b):i.setFillColor(0,0,0),i.setLineWidth(r.strokeWidth),r.stroke instanceof h?i.setDrawColor(r.
stroke.color.r,r.stroke.color.g,r.stroke.color.b):i.setDrawColor(0,0,0),i.setLineCap(r.strokeLinecap),i.setLineJoin(r.strokeLinejoin),r.strokeDasharray?i.setLineDashPattern(r.
strokeDasharray,r.strokeDashoffset):i.setLineDashPattern([],0),i.setLineMiterLimit(r.strokeMiterlimit),s=U.hasOwnProperty(r.fontFamily)?U[r.fontFamily]:r.fontFamily,r.fill&&r.fill
instanceof h&&r.fill.color.ok){var l=r.fill.color;i.setTextColor(l.r,l.g,l.b)}else i.setTextColor(0,0,0);var u="";"bold"===r.fontWeight&&(u="bold"),"italic"===r.fontStyle&&(u+="italic"),
""===u&&(u="normal"),void 0!==s||void 0!==u?(void 0===s&&(s=U.hasOwnProperty(r.fontFamily)?U[r.fontFamily]:r.fontFamily),i.setFont(s,u)):i.setFont("helvetica",u),i.setFontSize(r.fontSize*i.
internal.scaleFactor)}(a=new f(t.pdf,{refsHandler:t.refsHandler,styleSheets:t.styleSheets,viewport:t.viewport,svg2pdfParameters:t.svg2pdfParameters,textMeasure:t.textMeasure,attributeState:
Object.assign(c.default(),n)})),o=0,l=this.children,s.label=1;case 1:return o<l.length?[4,l[o].render(a)]:[3,4];case 2:s.sent(),s.label=3;case 3:return o++,[3,1];case 4:return t.pdf.
endFormObject(a.refsHandler.generateKey(this.element.getAttribute("id"),n)),[2]}}))}))},r.prototype.getBoundingBoxCore=function(t){var e,r=this.element.getAttribute("viewBox");return r&&(
e=V(r)),[e&&e[0]||0,e&&e[1]||0,e&&e[2]||parseFloat(this.element.getAttribute("markerWidth")||"3"),e&&e[3]||parseFloat(this.element.getAttribute("markerHeight")||"3")]},r.prototype.
computeNodeTransformCore=function(t){var e,r=parseFloat(this.element.getAttribute("refX")||"0"),i=parseFloat(this.element.getAttribute("refY")||"0"),n=this.element.getAttribute("viewBox");
if(n){var a=V(n);e=K(this.element,a,0,0,parseFloat(this.element.getAttribute("markerWidth")||"3"),parseFloat(this.element.getAttribute("markerHeight")||"3"),t,!0),e=t.pdf.matrixMult(t.pdf.
Matrix(1,0,0,1,-r,-i),e)}else e=t.pdf.Matrix(1,0,0,1,-r,-i);return e},r.prototype.isVisible=function(t,e){return E(this,t,e)},r}(tt),re=function(t){function e(e,r){return t.call(this,e,
r)||this}return i(e,t),e.prototype.getR=function(t){var e;return null!==(e=this.r)&&void 0!==e?e:this.r=parseFloat(B(this.element,t.styleSheets,"r")||"0")},e.prototype.getRx=function(t){
return this.getR(t)},e.prototype.getRy=function(t){return this.getR(t)},e}(St),ie=function(t){function e(e,r){return t.call(this,!1,e,r)||this}return i(e,t),e}(Zt),ne=function(t){function e(
){return null!==t&&t.apply(this,arguments)||this}return i(e,t),e.prototype.renderCore=function(t){return a(this,void 0,void 0,(function(){var e,r;return s(this,(function(i){switch(i.label){
case 0:e=0,r=this.children,i.label=1;case 1:return e<r.length?[4,r[e].render(t)]:[3,4];case 2:i.sent(),i.label=3;case 3:return e++,[3,1];case 4:return[2]}}))},e.prototype.
getBoundingBoxCore=function(t){return Q(t,this)},e}(ft),ae=function(t){function e(){return null!==t&&t.apply(this,arguments)||this}return i(e,t),e.prototype.isVisible=function(t,e){return E(
this,t,e)},e.prototype.render=function(e){return a(this,void 0,void 0,(function(){var r,i,n,a,o;return s(this,(function(s){switch(s.label){case 0:return this.isVisible("hidden"!==e.
attributeState.visibility,e)?(r=this.getX(e),i=this.getY(e),n=this.getWidth(e),a=this.getHeight(e),e.pdf.saveGraphicsState(),o=e.transform,this.element.hasAttribute("transform")&&(o=e.pdf.
matrixMult(Z(this.element.getAttribute("transform"),e),o)),e.pdf.setCurrentTransformationMatrix(o),e.withinUse||"visible"===B(this.element,e.styleSheets,"overflow")||e.pdf.rect(r,i,n,a).
clip().discardPath(),[4,t.prototype.render.call(this,e.clone({transform:e.pdf.unitMatrix,viewport:e.withinUse?e.viewport:new yt(n,a)}))]):[2];case 1:return s.sent(),e.pdf.
restoreGraphicsState(),[2]}}))}))},e.prototype.computeNodeTransform=function(t){return this.computeNodeTransformCore(t)},e.prototype.computeNodeTransformCore=function(t){if(t.
withinUse)return t.pdf.unitMatrix;var e,r=this.getX(t),i=this.getY(t),n=this.getViewBox();if(n){var a=this.getWidth(t),s=this.getHeight(t);e=K(this.element,n,r,i,a,s,t)}else e=t.pdf.Matrix(
1,0,0,1,r,i);return e},e.prototype.getWidth=function(t){if(void 0!==this.width)return this.width;var e,r,i=t.svg2pdfParameters;if(this.isOutermostSvg(t))if(null!=i.width)e=i.width;else if(
r=B(this.element,t.styleSheets,"width"))e=parseFloat(r);else{var n=this.getViewBox();if(n&&(null!=i.height||B(this.element,t.styleSheets,"height"))){var a=n[2]/n[3];e=this.getHeight(
t)*a}else e=Math.min(300,t.viewport.width,2*t.viewport.height)}else e=(r=B(this.element,t.styleSheets,"width"))?parseFloat(r):t.viewport.width;return this.width=e},e.prototype.
getHeight=function(t){if(void 0!==this.height)return this.height;var e,r,i=t.svg2pdfParameters;if(this.isOutermostSvg(t))if(null!=i.height)e=i.height;else if(r=B(this.element,t.styleSheets,
"height"))e=parseFloat(r);else{var n=this.getViewBox();if(n){var a=n[2]/n[3];e=this.getWidth(t)/a}else e=Math.min(150,t.viewport.width/2,t.viewport.height)}else e=(r=B(this.element,t.
styleSheets,"height"))?parseFloat(r):t.viewport.height;return this.height=e},e.prototype.getX=function(t){if(void 0!==this.x)return this.x;if(this.isOutermostSvg(t))return this.x=0;var e=B(
this.element,t.styleSheets,"x");return this.x=e?parseFloat(e):0},e.prototype.getY=function(t){if(void 0!==this.y)return this.y;if(this.isOutermostSvg(t))return this.y=0;var e=B(this.element,
t.styleSheets,"y");return this.y=e?parseFloat(e):0},e.prototype.getViewBox=function(){if(void 0!==this.viewBox)return this.viewBox;var t=this.element.getAttribute("viewBox");return this.
viewBox=t?V(t):void 0},e.prototype.isOutermostSvg=function(t){return t.svg2pdfParameters.element===this.element},e}(ne),se=function(t){function e(){return null!==t&&t.apply(this,
arguments)||this}return i(e,t),e.prototype.isVisible=function(t,e){return E(this,t,e)},e.prototype.computeNodeTransformCore=function(t){return t.pdf.unitMatrix},e}(ne),oe=function(t){
function e(){return null!==t&&t.apply(this,arguments)||this}return i(e,t),e.prototype.renderCore=function(e){return a(this,void 0,void 0,(function(){var r,i,n,a;return s(this,(function(s){
switch(s.label){case 0:return[4,t.prototype.renderCore.call(this,e)];case 1:return s.sent(),(r=B(this.element,e.styleSheets,"href"))&&(i=this.getBoundingBox(e),n=e.pdf.internal.scaleFactor,
a=e.pdf.internal.pageSize.getHeight(),e.pdf.link(n*(i[0]*e.transform.sx+e.transform.tx),a-n*((i[1]+i[3])*e.transform.sy+e.transform.ty),n*i[2],n*i[3],{url:r})),[2]}}))}))},e}(se),
le=function(t){function e(){return null!==t&&t.apply(this,arguments)||this}return i(e,t),e.prototype.apply=function(t){return a(this,void 0,void 0,(function(){var e,r,i,n,a;return s(this,(
function(s){switch(s.label){case 0:if(!this.isVisible(!0,t))return[2];e=t.pdf.matrixMult(this.computeNodeTransform(t),t.transform),t.pdf.setCurrentTransformationMatrix(e),r=0,i=this.
children,s.label=1;case 1:return r<i.length?[4,i[r].render(new f(t.pdf,{refsHandler:t.refsHandler,styleSheets:t.styleSheets,viewport:t.viewport,withinClipPath:!0,svg2pdfParameters:t.
svg2pdfParameters,textMeasure:t.textMeasure}))]:[3,4];case 2:s.sent(),s.label=3;case 3:return r++,[3,1];case 4:return n=this.children.length>0&&!!B(this.children[0].element,t.styleSheets,
"clip-rule"),a=n?this.getClipRuleAttr(this.children[0].element,t.styleSheets):this.getClipRuleAttr(this.element,t.styleSheets),t.pdf.clip(a).discardPath(),t.pdf.
setCurrentTransformationMatrix(e.inversed()),[2]}}))}))},e.prototype.getBoundingBoxCore=function(t){return Q(t,this)},e.prototype.isVisible=function(t,e){return E(this,t,e)},e.prototype.
getClipRuleAttr=function(t,e){return"evenodd"===B(t,e,"clip-rule")?"evenodd":void 0},e}(tt);function ue(t,e,r){var i,n=[];switch(function(t,e){for(var r=[],i=0;i<t.childNodes.length;i++){
var n=t.childNodes[i];"#"!==n.nodeName.charAt(0)&&r.push(n)}for(i=0;i<r.length;i++)e(i,r[i])}(t,(function(t,i){return n.push(ue(i,e,r))})),t.tagName.toLowerCase()){case"a":i=new oe(t,n);
break;case"g":i=new se(t,n);break;case"circle":i=new re(t,n);break;case"clippath":i=new le(t,n);break;case"ellipse":i=new wt(t,n);break;case"lineargradient":i=new rt(t,n);break;case"image":
i=new Kt(t,n,e);break;case"line":i=new mt(t,n);break;case"marker":i=new ee(t,n);break;case"path":i=new Qt(t,n);break;case"pattern":i=new at(t,n);break;case"polygon":i=new Jt(t,n);break;
case"polyline":i=new ie(t,n);break;case"radialgradient":i=new it(t,n);break;case"rect":i=new xt(t,n);break;case"svg":i=new ae(t,n);break;case"symbol":i=new vt(t,n);break;case"text":i=new Wt(
t,n);break;case"use":i=new bt(t,n);break;default:i=new te(t,n)}if(null!=r&&i.element.hasAttribute("id")){var a=g(i.element.id,{isIdentifier:!0});r[a]=r[a]||i}return i.children.forEach((
function(t){return t.setParent(i)})),i}var he=function(t){var e,r,i=t,n={a:0,b:0,c:0},a=[];return e=function(e,r){var s,o,l,u,h,c;if(e.test(i))for(o=0,l=(s=i.match(e)).length;o<l;o+=1)n[
r]+=1,u=s[o],h=i.indexOf(u),c=u.length,a.push({selector:t.substr(h,c),type:r,index:h,length:c});i=i.replace(u,Array(c+1).join(" "))},(r=function(t){var e,r,n,a;if(t.test(i))for(r=0,n=(e=i.
match(t)).length;r<n;r+=1)a=e[r],i=i.replace(a,Array(a.length+1).join("A"))})(/\\[0-9A-Fa-f]{6}\s?/g),r(/\\[0-9A-Fa-f]{1,5}\s/g),r(/\\./g),function(){var t,e,r,n,a=/{[^]*/gm;if(a.test(
i))for(e=0,r=(t=i.match(a)).length;e<r;e+=1)n=t[e],i=i.replace(n,Array(n.length+1).join(" "))}(),e(/(\[[^\]]+\])/g,"b"),e(/(#[^\#\s\+>~\.\[:\)]+)/g,"a"),e(/(\.[^\s\+>~\.\[:\)]+)/g,"b"),e(/(:
```

```
:[^\s\+>~\.\[:]+|:first-line|:first-letter|:before|:after)/gi,"c"),e(/(:(?!not|global|local)[\w-]+\([^\)]*\))/gi,"b"),e(/(:(?!not|global|local)[^\s\+>~\.\[:]+)/g,"b"),i=(i=(i=(i=(i=(i=(i=i.
replace(/[\*\s\+>~]/g," ")).replace(/[#\.]/g," ")).replace(/:not/g,"    ")).replace(/:local/g,"      ")).replace(/:global/g,"       ")).replace(/[\(\)]/g," "),e(/([^\s\+>~\.\[:]+)/g,"c"),a.
sort((function(t,e){return t.index-e.index})),{selector:t,specificity:"0,"+n.a.toString()+","+n.b.toString()+","+n.c.toString(),specificityArray:[0,n.a,n.b,n.c],parts:a}},ce=function(){
function t(t,e){this.rootSvg=t,this.loadExternalSheets=e,this.styleSheets=[]}return t.prototype.load=function(){return a(this,void 0,void 0,(function(){var t;return s(this,(function(e){
switch(e.label){case 0:return[4,this.collectStyleSheetTexts()];case 1:return t=e.sent(),this.parseCssSheets(t),[2]}}))}))},t.prototype.collectStyleSheetTexts=function(){return a(this,void 0,
void 0,(function(){var e,r,i,n,a;return s(this,(function(s){switch(s.label){case 0:if(e=[],this.loadExternalSheets&&this.rootSvg.ownerDocument)for(n=0;n<this.rootSvg.ownerDocument.
childNodes.length;n++)"xml-stylesheet"===(r=this.rootSvg.ownerDocument.childNodes[n]).nodeName&&"string"==typeof r.data&&e.push(t.loadSheet(r.data.match(/href=["']*?["']/)[0].split("=")[1].
slice(1,-1)));for(i=this.rootSvg.querySelectorAll("style,link"),n=0;n<i.length;n++)P(a=i[n],"style")?e.push(a.textContent):this.loadExternalSheets&&P(a,"link")&&"stylesheet"===a.
getAttribute("rel")&&a.hasAttribute("href")&&e.push(t.loadSheet(a.getAttribute("href")));return[4,Promise.all(e)];case 1:return[2,s.sent().filter((function(t){return null!==t}))]}}))}))},t.
prototype.parseCssSheets=function(e){for(var r=document.implementation.createHTMLDocument(""),i=0,n=e;i<n.length;i++){var a=n[i],s=r.createElement("style");s.textContent=a,r.body.
appendChild(s);var o=s.sheet;if(o instanceof CSSStyleSheet){for(var l=o.cssRules.length-1;l>=0;l--){var u=o.cssRules[l];if(u instanceof CSSStyleRule){var h=u;if(h.selectorText.indexOf(",
")>=0){o.deleteRule(l);for(var c=h.cssText.substring(h.selectorText.length),f=t.splitSelectorAtCommas(h.selectorText),p=0;p<f.length;p++)o.insertRule(f[p]+c,l+p)}}else o.deleteRule(l)}this.
styleSheets.push(o)}}},t.splitSelectorAtCommas=function(t){if(0===t.trim().length)return[];for(var e,r=/,|["']/g,i=/\\[\s\S]|"/g,n=/\\[\s\S]|'/g,a=[],s="initial",o=-1,l=i,u=0;u<t.length;
)switch(s){case"initial":r.lastIndex=u,(e=r.exec(t))?(","===e[0]?(a.push(t.substring(o+1,r.lastIndex-1).trim()),o=r.lastIndex-1):(s="withinQuotes",l='"'===e[0]?i:n),u=r.lastIndex):(a.push(t.
substring(o+1).trim()),u=t.length);break;case"withinQuotes":l.lastIndex=u,(e=l.exec(t))?(u=l.lastIndex,1===e[0].length&&(s="initial")):u=t.length}return a},t.loadSheet=function(t){return
new Promise((function(e,r){var i=new XMLHttpRequest;i.open("GET",t,!0),i.responseType="text",i.onload=function(){200!==i.status&&r(new Error("Error ".concat(i.status,": Failed to load '").
concat(t,"'"))),e(i.responseText)},i.onerror=r,i.onabort=r,i.send(null)})).catch((function(){return null}))},t.prototype.getPropertyValue=function(t,e){for(var r=[],i=0,n=this.styleSheets;
i<n.length;i++)for(var a=n[i],s=0;s<a.cssRules.length;s++){var o=a.cssRules[s];o.style.getPropertyValue(e)&&t.matches(o.selectorText)&&r.push(o)}if(0!==r.length){var l=function(t,r){var i=t.
style.getPropertyPriority(e);return i!==r.style.getPropertyPriority(e)?"important"===i?1:-1:function(t,e){var r,i,n;if("string"==typeof t){if(-1!==t.indexOf(","))throw"Invalid CSS selector";
r=he(t).specificityArray}else{if(!Array.isArray(t))throw"Invalid CSS selector or specificity array";if(4!==t.filter((function(t){return"number"==typeof t})).length)throw"Invalid specificity
array";r=t}if("string"==typeof e){if(-1!==e.indexOf(","))throw"Invalid CSS selector";i=he(e).specificityArray}else{if(!Array.isArray(e))throw"Invalid CSS selector or specificity array";if(
4!==e.filter((function(t){return"number"==typeof t})).length)throw"Invalid specificity array";i=e}for(n=0;n<4;n+=1){if(r[n]<i[n])return-1;if(r[n]>i[n])return 1}return 0}(t.selectorText,r.
selectorText)};return r.reduce((function(t,e){return 1===l(t,e)?t:e})).style.getPropertyValue(e)||void 0}},t}(),fe=function(){function t(){this.measureMethods={}}return t.prototype.
getTextOffset=function(t,e){var r=e.textAnchor;if("start"===r)return 0;var i=this.measureTextWidth(t,e),n=0;switch(r){case"end":n=i;break;case"middle":n=i/2}return n},t.prototype.
measureTextWidth=function(t,e){if(0===t.length)return 0;var r=e.fontFamily;return this.getMeasureFunction(r).call(this,t,e.fontFamily,e.fontSize+"px",e.fontStyle,e.fontWeight)},t.prototype.
getMeasurementTextNode=function(){if(!this.textMeasuringTextElement){this.textMeasuringTextElement=document.createElementNS(D,"text");var t=document.createElementNS(D,"svg");t.appendChild(
this.textMeasuringTextElement),t.style.setProperty("position","absolute"),t.style.setProperty("visibility","hidden"),document.body.appendChild(t)}return this.textMeasuringTextElement},t.
prototype.canvasTextMeasure=function(t,e,r,i,n){var a=document.createElement("canvas").getContext("2d");return null!=a?(a.font=[i,n,r,e].join(" "),a.measureText(t).width):0},t.prototype.
svgTextMeasure=function(t,e,r,i,n,a){void 0===a&&(a=this.getMeasurementTextNode());var s=a;return s.setAttribute("font-family",e),s.setAttribute("font-size",r),s.setAttribute("font-style",
i),s.setAttribute("font-weight",n),s.setAttributeNS("http://www.w3.org/XML/1998/namespace","xml:space","preserve"),s.textContent=t,s.getBBox().width},t.prototype.getMeasureFunction=function(
e){var r=this.measureMethods[e];if(!r){var i="16px",n="normal",a="normal",s=this.canvasTextMeasure(t.testString,e,i,n,a),o=this.svgTextMeasure(t.testString,e,i,n,a);r=Math.abs(s-o)<t.
epsilon?this.canvasTextMeasure:this.svgTextMeasure,this.measureMethods[e]=r}return r},t.prototype.cleanupTextMeasuring=function(){if(this.textMeasuringTextElement){var t=this.
textMeasuringTextElement.parentNode;t&&document.body.removeChild(t),this.textMeasuringTextElement=void 0}},t.testString="abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ 0123456789!\
"$%&/()=?'\\+*-_.:,;^}][{#~|<>",t.epsilon=.1,t}();function pe(t,e){return a(this,arguments,void 0,(function(t,e,r){var i,a,o,l,u,h,c,p,d,g,v,y,b,x;return void 0===r&&(r={}),s(this,(function(
s){switch(s.label){case 0:return i=null!==(y=r.x)&&void 0!==y?y:0,a=null!==(b=r.y)&&void 0!==b?b:0,o=null!==(x=r.loadExternalStyleSheets)&&void 0!==x&&x,u=new m(l={}),[4,(h=new ce(t,o)).
load()];case 1:return s.sent(),c=new yt(e.internal.pageSize.getWidth(),e.internal.pageSize.getHeight()),p=n(n({},r),{element:t}),d=new fe,g=new f(e,{refsHandler:u,styleSheets:h,viewport:c,
svg2pdfParameters:p,textMeasure:d}),e.advancedAPI(),e.saveGraphicsState(),e.setCurrentTransformationMatrix(e.Matrix(1,0,0,1,i,a)),e.setLineWidth(g.attributeState.strokeWidth),v=g.
attributeState.fill.color,e.setFillColor(v.r,v.g,v.b),e.setFont(g.attributeState.fontFamily),e.setFontSize(g.attributeState.fontSize*e.internal.scaleFactor),[4,ue(t,g,l).render(g)];case 2:
return s.sent(),e.restoreGraphicsState(),e.compatAPI(),g.textMeasure.cleanupTextMeasuring(),[2,e]}}))}))}e.jsPDF.API.svg=function(t,e){return void 0===e&&(e={}),pe(t,this,e)},t.
svg2pdf=pe}));
//# sourceMappingURL=svg2pdf.umd.min.js.map
```

15. index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<meta name="theme-color" content="#111827">
<link rel="manifest" href="manifest.webmanifest">

<title>Structural Analysis — Input Model & Solver</title>

<!-- Global styles -->
<link rel="stylesheet" href="assets/css/main.css">

<!-- Page / module styles -->
<link rel="stylesheet" href="modules/home/home.css">
<link rel="stylesheet" href="modules/dashboard/dashboard.css">
<link rel="stylesheet" href="modules/results/results.css">
<link rel="stylesheet" href="modules/results/step-8/step-8.css">
<link rel="stylesheet" href="modules/results/diagrams/diagrams.css">

<!-- Onboarding tour: Driver.js base styles + tour theme -->
<link rel="stylesheet" href="assets/vendor/driver/driver.css">
<link rel="stylesheet" href="modules/tour/tour.css">
</head>

<body>

<!-- Vendor libraries (vector PDF export: jsPDF + svg2pdf + embedded fonts) -->
<script src="assets/vendor/jspdf.umd.min.js"></script>
<script src="assets/vendor/svg2pdf.umd.min.js"></script>
<script src="assets/vendor/pdf-fonts.js"></script>

<!-- Vendor library: Driver.js (onboarding tour) -->
<script src="assets/vendor/driver/driver.js.iife.js"></script>

<!-- Global JavaScript: shell markup, shared helpers, state, persistence, navigation, bootstrap -->
<script src="assets/js/main.js?v=20261012-temperature-loads-v1"></script>

<!-- Page modules -->
<script src="modules/home/home.js"></script>
<script src="modules/dashboard/dashboard.js"></script>
<script src="modules/nodes/nodes.js"></script>
<script src="modules/elements/elements.js?v=20261012-temperature-loads-v1"></script>
<script src="modules/supports/supports.js?v=20261012-prescribed-support-units-mm"></script>
<script src="modules/load-case/load-case.js"></script>
<script src="modules/load/load.js?v=20261012-temperature-loads-v2-edit-fields"></script>
<script src="modules/moment-case/moment-case.js"></script>
<script src="modules/moment/moment.js"></script>

<!-- Results module: frame + 8 calculation steps + diagrams -->
<script src="modules/results/results.js?v=20261011-grid-my-torsion-refresh"></script>
<script src="modules/results/step-1/step-1.js?v=20261012-temperature-loads-v1"></script>
<script src="modules/results/step-2/step-2.js?v=20261012-element-E-kn-m2"></script>
<script src="modules/results/step-3/step-3.js?v=20261012-thermal-AT-v1"></script>
<script src="modules/results/step-4/step-4.js?v=20261012-thermal-AT-v1"></script>
<script src="modules/results/step-5/step-5.js?v=20261012-prescribed-support-AD"></script>
<script src="modules/results/step-6/step-6.js?v=20261012-prescribed-support-AD"></script>
<script src="modules/results/step-7/step-7.js?v=20261012-prescribed-support-AD"></script>
<script src="modules/results/step-8/step-8.js"></script>
<script src="modules/results/diagrams/diagrams.js?v=20261012-element-E-kn-m2"></script>

<!-- Onboarding tour (must load after all page modules) -->
<script src="modules/tour/tour.js"></script>

<script src="assets/js/offline.js?v=20261013-offline-v1"></script>

</body>
</html>
```

16. manifest.webmanifest

```
{
  "id": "./",
  "name": "Structure Analysis",
  "short_name": "Structure Analysis",
  "description": "Offline-capable structural analysis for Beam, Truss, Frame and Grid models.",
  "start_url": "./",
  "scope": "./",
  "display": "standalone",
  "orientation": "any",
  "background_color": "#f4f7fb",
  "theme_color": "#111827",
  "icons": [
    {
      "src": "assets/icons/structure-analysis.svg",
      "sizes": "any",
      "type": "image/svg+xml",
      "purpose": "any"
    }
  ]
}
```

17. modules/dashboard/dashboard.css

```
/* ========================================================================
 * dashboard.css
 * Styles for the Dashboard page (statistics tiles and model note).
 * ======================================================================== */

/* DASHBOARD */
.stat-grid {
    display: grid;
    grid-template-columns: repeat(5, 1fr);
    gap: 14px;
    margin-bottom: 20px;
}

.stat {
    background: white;
    border: 1px solid var(--border);
    border-radius: 12px;
    padding: 17px;
}

.stat-label {
    color: var(--muted);
    font-size: 11px;
    font-weight: 700;
}

.stat-value {
    font-size: 26px;
    font-weight: 800;
    margin-top: 7px;
}

.model-note {
    background: var(--soft-blue);
    border: 1px solid #bfdbfe;
    padding: 15px;
    border-radius: 10px;
    font-size: 12px;
    line-height: 1.6;
    color: #1e3a8a;
}

@media (max-width: 1100px) {
    .stat-grid {
        grid-template-columns: repeat(3, 1fr);
    }
}

@media (max-width: 800px) {
    .stat-grid {
        grid-template-columns: repeat(2, 1fr);
    }
}

@media (max-width: 500px) {
    .stat-grid {
        grid-template-columns: 1fr;
    }
}
```

18. modules/dashboard/dashboard.js

```
/* ========================================================================
 * dashboard.js — Dashboard page module
 * Dashboard page — summary statistics of the current model.
 * Contains this page's markup (injected into .container) and its logic.
 * ======================================================================== */

/* ---- Markup ---- */
mountModuleHTML(".container", "beforeend", `
<!-- DASHBOARD -->
<section class="section" id="dashboard">
    <div class="stat-grid">
        <div class="stat"><div class="stat-label">NODES</div><div class="stat-value" id="statNodes">0</div></div>
        <div class="stat"><div class="stat-label">ELEMENTS</div><div class="stat-value" id="statElements">0</div></div>
        <div class="stat"><div class="stat-label">SUPPORTS</div><div class="stat-value" id="statSupports">0</div></div>
        <div class="stat"><div class="stat-label">LOADS</div><div class="stat-value" id="statLoads">0</div></div>
        <div class="stat"><div class="stat-label" id="statMomentsLabel">MOMENTS</div><div class="stat-value" id="statMoments">0</div></div>
    </div>

    <div class="card">
        <div class="card-header">
            <div>
                <h3>Input Model Workflow</h3>
                <p>Define the model in the following order.</p>
            </div>
        </div>
        <div class="card-body">
            <div class="model-note">
                <strong>Recommended input sequence:</strong><br><br>
                1. Create Nodes → define X, Y and Z coordinates.<br>
                2. Create Element Templates → <span id="dashboardElementTemplateInstruction">define I, E and A. Set each member's Self Weight (kN/m) in a Self Weight load case.</span><br>
                3. Assign Elements → connect start and end nodes.<br>
                4. Create Support Templates → define restraints and prescribed displacements.<br>
                5. Assign Supports → apply support definitions to nodes.<br>
                6. Create Load Cases → define Dead, Live or Self Weight cases.<br>
                7. Create Loads → apply nodal or element forces.<br>
                8. <span id="dashboardMomentCaseInstruction">Create Moment Cases → define Dead, Live or Self Weight cases.</span><br>
                9. <span id="dashboardMomentInstruction">Create Moments → apply nodal or element moments.</span><br><br>
                10. Go to the <b>Results</b> tab to run the Stiffness Method solver.<br><br>
                <strong>Note:</strong> <span id="dashboardModeNote">Z coordinates and Z moments are ignored in 2D Plane Frame Analysis. Length (L) is automatically calculated via node
coordinates.</span>
            </div>
        </div>
    </div>
</section>
`);

/* ---- Logic ---- */
function updateStats() {
    const isGrid = !!(model && model.analysisMode === "grid");
    const setText = (id, text) => {
        const el = document.getElementById(id);
        if (el) el.textContent = text;
    };

    // Dashboard wording follows the fixed solver type of the current model.
    setText("statMomentsLabel", isGrid ? "TORSIONS" : "MOMENTS");
    setText("dashboardElementTemplateInstruction", isGrid
        ? "define I, J, E and G."
        : "define I, E and A. Set each member's Self Weight (kN/m) in a Self Weight load case.");
    setText("dashboardMomentCaseInstruction", isGrid
        ? "Create Torsion Cases → define Dead, Live or Self Weight cases."
        : "Create Moment Cases → define Dead, Live or Self Weight cases.");
    setText("dashboardMomentInstruction", isGrid
        ? "Create Torsions → apply nodal or element torsions."
        : "Create Moments → apply nodal or element moments.");
    setText("dashboardModeNote", isGrid
        ? "Grid loads act along Z; torsion uses Mx (+ right / − left) and My (+ up / − down). Length (L) is automatically calculated from node coordinates."
        : "Z coordinates and Z moments are ignored in 2D Plane Frame Analysis. Length (L) is automatically calculated via node coordinates.");

    document.getElementById("statNodes").textContent = model.nodes.length;
    document.getElementById("statElements").textContent = model.elements.length;
    document.getElementById("statSupports").textContent = model.supportAssignments.length;
    document.getElementById("statLoads").textContent = model.loads.length;
    document.getElementById("statMoments").textContent = model.moments.length;
}
```

19. modules/elements/elements.js

```
/* ========================================================================
 * elements.js — Elements page module
 * Elements page — element templates (section / material) and element assignments.
 * Contains this page's markup (injected into .container) and its logic.
 * ======================================================================== */

/* ---- Markup ---- */
mountModuleHTML(".container", "beforeend", `
<!-- ELEMENTS -->
<section class="section" id="elements">
    <div class="card">
        <div class="card-header">
            <div><h3>Create Element Template</h3><p>Create a reusable element definition that can be assigned to multiple node pairs.</p></div>
        </div>
        <div class="card-body">
            <form id="elementTemplateForm">
                <div class="grid grid-3">
                    <div class="field"><label for="elementTemplateId">Element Template ID</label><input id="elementTemplateId" placeholder="E1" value="E1"></div>
                    <div class="field plane-only-property"><label for="elementMemberType">Member Type</label><select id="elementMemberType" onchange="syncElementTemplateTypeUI()"><option
value="frame" selected>Frame</option><option value="beam">Beam</option><option value="truss">Truss</option></select></div>
                    <div class="field"><label for="elementI">Moment of Inertia (I)</label><div class="input-wrap has-unit"><input id="elementI" type="text" inputmode="decimal"
placeholder="1" value="1"><span class="input-unit">m■</span></div><div class="length-chip-row scientific-shortcut-row"><button type="button" class="scientific-chip"
onclick="insertScientificExponent('elementI')">e</button></div></div>
                    <div class="field"><label for="elementE">Modulus of Elasticity (E)</label><div class="input-wrap has-unit"><input id="elementE" type="text" inputmode="decimal"
placeholder="1" value="1"><span class="input-unit">kN/m²</span></div><div class="length-chip-row scientific-shortcut-row"><button type="button" class="scientific-chip"
onclick="insertScientificExponent('elementE')">e</button></div></div>
                    <div class="field plane-only-property"><label for="elementA">Cross-sectional Area (A)</label><div class="input-wrap has-unit"><input id="elementA" type="text"
inputmode="decimal" placeholder="1" value="1"><span class="input-unit">m²</span></div><div class="length-chip-row scientific-shortcut-row"><button type="button" class="scientific-chip"
onclick="insertScientificExponent('elementA')">e</button></div></div>
                </div>
                <div class="grid grid-2 plane-only-property" id="planeThermalProperties" style="margin-top:14px;">
                    <div class="field"><label for="elementAlpha">Thermal Expansion Coefficient (α)</label><div class="input-wrap has-unit"><input id="elementAlpha" type="text"
inputmode="decimal" placeholder="e.g. 0.000012" value="0"><span class="input-unit">1/°C</span></div><div class="length-chip-row scientific-shortcut-row"><button type="button"
class="scientific-chip" onclick="insertScientificExponent('elementAlpha')">e</button></div></div>
                    <div class="field"><label for="elementThermalDepth">Section Depth for Thermal Gradient (d)</label><div class="input-wrap has-unit"><input id="elementThermalDepth"
type="text" inputmode="decimal" placeholder="e.g. 0.3" value="0"><span class="input-unit">m</span></div><div class="length-chip-row scientific-shortcut-row"><button type="button"
class="scientific-chip" onclick="insertScientificExponent('elementThermalDepth')">e</button></div></div>
                </div>
                <div id="gridElementProperties" class="grid grid-2" style="display:none; margin-top:14px;">
                    <div class="field"><label for="elementG">Shear Modulus (G)</label><div class="input-wrap has-unit"><input id="elementG" type="text" inputmode="decimal" placeholder="80"
value="80"><span class="input-unit">GPa</span></div><div class="length-chip-row scientific-shortcut-row"><button type="button" class="scientific-chip" onclick="insertScientificExponent(
'elementG')">e</button></div></div>
                    <div class="field"><label for="elementJ">Torsional Constant (J)</label><div class="input-wrap has-unit"><input id="elementJ" type="text" inputmode="decimal"
placeholder="1" value="1"><span class="input-unit">m■</span></div><div class="length-chip-row scientific-shortcut-row"><button type="button" class="scientific-chip"
onclick="insertScientificExponent('elementJ')">e</button></div></div>
                </div>
                <div class="form-footer">
                    <button type="button" class="btn btn-secondary" onclick="resetElementTemplateForm()">Reset</button>
                    <button id="elementTemplateSubmitBtn" type="submit" class="btn btn-primary">Create Element Template</button>
                </div>
            </form>
        </div>
    </div>

    <div class="card">
        <div class="card-header">
            <div><h3>Element Templates</h3><p>Reusable section/material definitions.</p></div>
            <input id="elementTemplateSearch" class="search-box" placeholder="Search..." oninput="renderElementTemplates()">
        </div>
        <div class="card-body">
            <div class="table-wrap">
                <table>
                    <thead><tr><th>Template ID</th><th class="plane-only-property">Type</th><th>I (m■)</th><th>E (kN/m²)</th><th class="grid-only-property" style="display:none;">G (GPa)</
th><th class="grid-only-property" style="display:none;">J (m■)</th><th class="plane-only-property">A (m²)</th><th class="plane-only-property">α (1/°C)</th><th class="plane-only-property">d (
m)</th><th>Actions</th></tr></thead>
                    <tbody id="elementTemplateTable"></tbody>
                </table>
            </div>
        </div>
    </div>

    <div class="card">
        <div class="card-header">
            <div><h3>Assign Element to Nodes</h3><p>Use an existing element template between two nodes.</p></div>
        </div>
        <div class="card-body">
            <form id="elementAssignmentForm">
                <div class="grid grid-4">
                    <div class="field"><label for="elementId">Element ID</label><input id="elementId" placeholder="E1-1" value="E1-1"></div>
                    <div class="field"><label for="elementTemplateSelect">Element Template</label><select id="elementTemplateSelect"><option value="">Select template</option></select></div>
                    <div class="field"><label for="elementStartNode">Start Node ID</label><select id="elementStartNode"><option value="">Select node</option></select></div>
                    <div class="field"><label for="elementEndNode">End Node ID</label><select id="elementEndNode"><option value="">Select node</option></select></div>
                </div>
                <div class="form-footer">
                    <button type="button" class="btn btn-secondary" onclick="resetElementAssignmentForm()">Reset</button>
```

```
                    <button id="elementSubmitBtn" type="submit" class="btn btn-primary">Assign Element</button>
                </div>
            </form>
        </div>
    </div>

    <div class="card">
        <div class="card-header">
            <div><h3>Element List</h3><p>Elements assigned to node pairs.</p></div>
        </div>
        <div class="card-body">
            <div class="table-wrap">
                <table>
                    <thead><tr><th>Element ID</th><th>Template</th><th>Start Node</th><th>End Node</th><th>Actions</th></tr></thead>
                    <tbody id="elementTable"></tbody>
                </table>
            </div>
        </div>
    </div>
</section>
`);

/* ---- Logic ---- */
// ---- Element Templates ----
let editingElementTemplateId = null;

function createElementTemplate(data) {
    model.elementTemplates.push(data);
    markAnalysisInvalidated();
}

function updateElementTemplate(existingId, data) {
    const template = model.elementTemplates.find((t) => t.id === existingId);
    if (!template) return false;
    const oldId = template.id;
    Object.assign(template, data);
    if (oldId !== data.id) model.elements.forEach((el) => { if (el.templateId === oldId) el.templateId = data.id; });
    markAnalysisInvalidated();
    return true;
}

document.getElementById("elementTemplateForm").addEventListener("submit", (e) => {
    e.preventDefault();
    const id = val("elementTemplateId");
    if (!id) return alert("Element Template ID is required.");
    if (hasId(model.elementTemplates, id, editingElementTemplateId)) return alert("Element Template ID already exists.");
    const gridMode = isGridAnalysisMode();
    const memberType = gridMode ? "" : val("elementMemberType");
    const I = memberType === "truss" ? 0 : num("elementI");
    const E_kN_m2 = num("elementE"), G_GPa = num("elementG"), J = num("elementJ"), A = num("elementA");
    const alpha = num("elementAlpha"), thermalDepth = num("elementThermalDepth");
    const validations = [
        [I, "Moment of Inertia (I)", gridMode || memberType !== "truss" ? { minExclusive: 0 } : { min: 0 }],
        [E_kN_m2, "Modulus of Elasticity (E)", { minExclusive: 0 }],
    ];
    if (gridMode) {
        validations.push([G_GPa, "Shear Modulus (G)", { minExclusive: 0 }], [J, "Torsional Constant (J)", { minExclusive: 0 }]);
    } else {
        validations.push(
            [A, "Cross-sectional Area (A)", { minExclusive: 0 }],
            [alpha, "Thermal Expansion Coefficient (α)", { min: 0 }],
            [thermalDepth, "Thermal Section Depth (d)", { min: 0 }]
        );
        if (!["beam", "truss", "frame"].includes(memberType)) return alert("Select Beam, Truss or Frame as the member type.");
    }
    for (const [value, label, options] of validations) {
        const error = requireFinite(value, label, options);
        if (error) return alert(error);
    }
    if (!gridMode && memberType === "beam" && I <= 0) return alert("A Beam requires both A > 0 and I > 0.");
    if (!gridMode && memberType === "frame" && I <= 0) return alert("A Frame requires both A > 0 and I > 0.");
    // Keep the two property sets separate. Values from the other solver mode
    // must not leak into the saved template when a template is edited.
    const data = {
        id,
        memberType: gridMode ? undefined : memberType,
        I,
        E_kN_m2,
        G_GPa: gridMode ? G_GPa : undefined,
        J: gridMode ? J : undefined,
        A: gridMode ? undefined : A,
        alpha: gridMode ? undefined : alpha,
        thermalDepth: gridMode ? undefined : thermalDepth
    };
    if (editingElementTemplateId !== null) {
        const updated = updateElementTemplate(editingElementTemplateId, data);
        if (!updated) { resetElementTemplateForm(); return; }
```

```
    } else {
        createElementTemplate(data);
    }
    resetElementTemplateForm();
    document.getElementById("elementTemplateId").value = getNextId(id);
    refreshAll();
});
function resetElementTemplateForm() {
    document.getElementById("elementTemplateForm").reset();
    syncElementTemplateTypeUI();
    editingElementTemplateId = null;
    document.getElementById("elementTemplateSubmitBtn").textContent = "Create Element Template";
}
function editElementTemplate(encodedId) {
    const id = decodeURIComponent(encodedId);
    const template = model.elementTemplates.find((t) => t.id === id);
    if (!template) return;
    editingElementTemplateId = template.id;
    document.getElementById("elementTemplateId").value = template.id;
    document.getElementById("elementI").value = template.I;
    document.getElementById("elementE").value = getElementEInternal(template);
    document.getElementById("elementG").value = template.G_GPa ?? "";
    document.getElementById("elementJ").value = template.J ?? "";
    document.getElementById("elementA").value = template.A ?? "1";
    document.getElementById("elementMemberType").value = getElementStructuralType({ templateId: template.id }) || "frame";
    document.getElementById("elementAlpha").value = template.alpha ?? 0;
    document.getElementById("elementThermalDepth").value = template.thermalDepth ?? 0;
    syncElementTemplateTypeUI();
    document.getElementById("elementTemplateSubmitBtn").textContent = "Update Element Template";
}
function syncElementTemplateTypeUI() {
    const typeInput = document.getElementById("elementMemberType");
    const inertiaInput = document.getElementById("elementI");
    if (!typeInput || !inertiaInput) return;
    if (isGridAnalysisMode()) {
        inertiaInput.disabled = false;
        return;
    }
    if (typeInput.value === "truss") {
        inertiaInput.value = "0";
        inertiaInput.disabled = true;
    } else {
        if (Number(inertiaInput.value) === 0 || inertiaInput.value === "") inertiaInput.value = "1";
        inertiaInput.disabled = false;
    }
}

function renderElementTemplates() {
    const tbody = document.getElementById("elementTemplateTable");
    const search = val("elementTemplateSearch").toLowerCase();
    const rows = model.elementTemplates.filter((e) => e.id.toLowerCase().includes(search));
    if (!rows.length) return showEmpty("elementTemplateTable", isGridAnalysisMode() ? 6 : 8);
    tbody.innerHTML = rows.map((e) => {
        const memberType = getElementStructuralType({ templateId: e.id }) || "—";
        return `<tr data-sort-collection="elementTemplates" data-sort-id="${escapeHtml(encodeInlineArgument(e.id))}"><td>${sortHandle(e.id)}<span class="badge badge-blue">${escapeHtml(e.
id)}</span></td><td class="plane-only-property">${escapeHtml(memberType)}</td><td>${formatNum(e.I)}</td><td>${formatNum(getElementEInternal(e))}</td><td class="grid-only-property"
style="display:none;">${e.G_GPa === undefined || e.G_GPa === "" ? "—" : formatNum(Number(e.G_GPa))}</td><td class="grid-only-property" style="display:none;">${e.J === undefined || e.J ===
"" ? "—" : formatNum(Number(e.J))}</td><td class="plane-only-property">${e.A === undefined || e.A === "" ? "—" : formatNum(Number(e.A))}</td><td class="plane-only-property">${e.alpha ===
undefined || e.alpha === "" ? "—" : formatNum(Number(e.alpha))}</td><td class="plane-only-property">${e.thermalDepth === undefined || e.thermalDepth === "" ? "—" : formatNum(Number(e.
thermalDepth))}</td><td><div class="table-actions"><button type="button" class="icon-btn" aria-label="Edit element template ${escapeHtml(e.id)}" title="Edit element template ${escapeHtml(e.
id)}" onclick="editElementTemplate('${encodeInlineArgument(e.id)}')">✎</button><button type="button" class="icon-btn" aria-label="Delete element template ${escapeHtml(e.id)}" title="Delete
element template ${escapeHtml(e.id)}" onclick="deleteItem('elementTemplates','${encodeInlineArgument(e.id)}','id')">×</button></div></td></tr>`;
    }).join("");
}

// ---- Element Assignments (connect nodes with a template) ----
let editingElementId = null;

function createElement(data) {
    model.elements.push(data);
    markAnalysisInvalidated();
}

function updateElement(existingId, data) {
    const element = model.elements.find((el) => el.id === existingId);
    if (!element) return false;
    const oldId = element.id;
    Object.assign(element, data);
    if (oldId !== data.id) {
        model.loads.forEach((load) => { if (load.elementId === oldId) load.elementId = data.id; });
        model.moments.forEach((moment) => { if (moment.elementId === oldId) moment.elementId = data.id; });
        model.loadCases.forEach((loadCase) => {
            const weights = loadCase.memberSelfWeights;
            if (!weights || typeof weights !== "object" || Array.isArray(weights)) return;
            if (Object.prototype.hasOwnProperty.call(weights, String(oldId))) {
                weights[String(data.id)] = weights[String(oldId)];
                delete weights[String(oldId)];
```

```
            }
        });
    }
    markAnalysisInvalidated();
    return true;
}

document.getElementById("elementAssignmentForm").addEventListener("submit", (e) => {
    e.preventDefault();
    const id = val("elementId");
    if (!id) return alert("Element ID is required.");
    if (hasId(model.elements, id, editingElementId)) return alert("Element ID already exists.");
    const templateId = val("elementTemplateSelect");
    const startNode = val("elementStartNode");
    const endNode = val("elementEndNode");
    let error = requireReference(model.elementTemplates, templateId, "Element Template");
    if (error) return alert(error);
    error = requireReference(model.nodes, startNode, "Start Node");
    if (error) return alert(error);
    error = requireReference(model.nodes, endNode, "End Node");
    if (error) return alert(error);
    if (startNode === endNode) return alert("Start Node and End Node must be different.");
    const n1 = model.nodes.find((n) => n.id === startNode);
    const n2 = model.nodes.find((n) => n.id === endNode);
    const L = Math.hypot(n2.x - n1.x, n2.y - n1.y);
    if (!Number.isFinite(L) || L <= 0) return alert("Element length must be greater than zero.");
    const data = { id, templateId, startNode, endNode };
    if (editingElementId !== null) {
        const updated = updateElement(editingElementId, data);
        if (!updated) { resetElementAssignmentForm(); return; }
    } else {
        createElement(data);
    }
    resetElementAssignmentForm();
    document.getElementById("elementId").value = getNextId(id);
    refreshAll();
});
function resetElementAssignmentForm() {
    document.getElementById("elementAssignmentForm").reset();
    editingElementId = null;
    document.getElementById("elementSubmitBtn").textContent = "Assign Element";
}
function editElement(encodedId) {
    const id = decodeURIComponent(encodedId);
    const element = model.elements.find((el) => el.id === id);
    if (!element) return;
    editingElementId = element.id;
    document.getElementById("elementId").value = element.id;
    document.getElementById("elementTemplateSelect").value = element.templateId;
    document.getElementById("elementStartNode").value = element.startNode;
    document.getElementById("elementEndNode").value = element.endNode;
    document.getElementById("elementSubmitBtn").textContent = "Update Element";
}
function renderElements() {
    const tbody = document.getElementById("elementTable");
    if (!model.elements.length) return showEmpty("elementTable", 5);
    tbody.innerHTML = model.elements.map((e) =>
        `<tr data-sort-collection="elements" data-sort-id="${escapeHtml(encodeInlineArgument(e.id))}"><td>${sortHandle(e.id)}<span class="badge badge-green">${escapeHtml(e.id)}</span></
td><td>${escapeHtml(e.templateId)}</td><td>${escapeHtml(e.startNode)}</td><td>${escapeHtml(e.endNode)}</td><td><div class="table-actions"><button type="button" class="icon-btn" aria-
label="Edit element ${escapeHtml(e.id)}" title="Edit element ${escapeHtml(e.id)}" onclick="editElement('${encodeInlineArgument(e.id)}')">✎</button><button type="button" class="icon-btn"
aria-label="Delete element ${escapeHtml(e.id)}" title="Delete element ${escapeHtml(e.id)}" onclick="deleteItem('elements','${encodeInlineArgument(e.id)}','id')">×</button></div></td></tr>`
    ).join("");
}
```

20. modules/home/home.css

```
/* ========================================================================
 * home.css
 * Styles for the Home page (standalone header, current-model strip, model library
 * cards, multi-select toolbar) and their responsive rules.
 * ======================================================================== */

/* STANDALONE HOME VIEW */
.app.home-mode {
    display: block;
    min-height: 100vh;
    background: #f4f7fb;
}

.app.home-mode .sidebar,
.app.home-mode .topbar,
.app.home-mode #liveModelDiagramCard {
    display: none !important;
}

.app.home-mode .main {
    width: 100%;
    min-height: 100vh;
}

.app.home-mode .container {
    max-width: 1420px;
    padding: 38px 34px 60px;
}

.home-standalone-head {
    display: flex;
    align-items: center;
    justify-content: space-between;
    gap: 20px;
    margin-bottom: 24px;
}

.home-brand {
    display: flex;
    align-items: center;
    gap: 13px;
}

.home-brand-mark {
    width: 46px;
    height: 46px;
    border-radius: 13px;
    display: flex;
    align-items: center;
    justify-content: center;
    background: #111827;
    color: #fff;
    font-size: 21px;
    box-shadow: 0 8px 20px rgba(15,23,42,.12);
}

.home-brand-text h1 {
    font-size: 20px;
    font-weight: 800;
    color: #182230;
    letter-spacing: -.3px;
}

.home-brand-text p {
    margin-top: 3px;
    font-size: 11px;
    color: var(--muted);
}

.home-standalone-actions {
    display: flex;
    gap: 8px;
    flex-wrap: wrap;
    justify-content: flex-end;
}

.home-standalone-title {
    margin-bottom: 7px;
    font-size: 30px;
    line-height: 1.15;
    letter-spacing: -.7px;
    color: #182230;
}

.home-standalone-subtitle {
```

```
    margin-bottom: 24px;
    color: var(--muted);
    font-size: 13px;
    line-height: 1.65;
    max-width: 720px;
}

.home-library-head {
    margin-top: 6px;
}

@media (max-width: 800px) {
    .app.home-mode .container {
        padding: 24px 16px 40px;
    }
    .home-standalone-head {
        align-items: flex-start;
        flex-direction: column;
    }
    .home-standalone-actions {
        justify-content: flex-start;
    }
    .home-standalone-title {
        font-size: 24px;
    }
}

.model-library-note {
    padding: 12px 14px;
    border: 1px solid #bfdbfe;
    border-radius: 9px;
    background: #eff6ff;
    color: #1e3a8a;
    font-size: 12px;
}

.model-library-table th,
.model-library-table td {
    vertical-align: middle;
}

.model-library-actions {
    display: flex;
    gap: 5px;
    flex-wrap: wrap;
}

.model-library-actions .icon-btn {
    white-space: nowrap;
    width: auto;
    padding: 0 9px;
    font-size: 11px;
}

/* HOME / MODEL LIBRARY */
.home-hero {
    border: 1px solid var(--border);
    border-radius: 16px;
    background: linear-gradient(135deg, #ffffff 0%, #f8fbff 100%);
    box-shadow: var(--shadow);
    overflow: hidden;
    margin-bottom: 20px;
}

.home-hero-inner {
    padding: 26px;
    display: flex;
    align-items: center;
    justify-content: space-between;
    gap: 20px;
}

.home-hero-copy h3 {
    font-size: 22px;
    margin-bottom: 7px;
    color: #182230;
}

.home-hero-copy p {
    color: var(--muted);
    font-size: 13px;
    line-height: 1.6;
    max-width: 680px;
}

.home-hero-actions {
    display: flex;
```

```
    gap: 8px;
    flex-wrap: wrap;
    justify-content: flex-end;
}

.home-current-strip {
    padding: 13px 26px;
    border-top: 1px solid var(--border);
    background: #f8fafc;
    display: flex;
    align-items: center;
    justify-content: space-between;
    gap: 12px;
    flex-wrap: wrap;
}

.home-current-info {
    display: flex;
    align-items: center;
    gap: 8px;
    min-width: 0;
    flex-wrap: wrap;
}

.home-current-label {
    font-size: 10px;
    color: var(--muted);
    font-weight: 800;
    text-transform: uppercase;
    letter-spacing: 0.5px;
}

.home-current-name {
    font-size: 13px;
    font-weight: 800;
    color: #1f2937;
    overflow: hidden;
    text-overflow: ellipsis;
    white-space: nowrap;
    max-width: min(420px, 70vw);
}

.home-library-head {
    display: flex;
    align-items: flex-end;
    justify-content: space-between;
    gap: 15px;
    margin-bottom: 12px;
}

.home-library-head h3 {
    font-size: 16px;
    margin-bottom: 4px;
}

.home-library-head p {
    color: var(--muted);
    font-size: 12px;
}

.home-model-count {
    font-size: 11px;
    font-weight: 800;
    color: #475569;
    background: #eef2f7;
    border: 1px solid #d6dee9;
    padding: 6px 10px;
    border-radius: 999px;
    white-space: nowrap;
}

.model-library-grid {
    display: grid;
    grid-template-columns: repeat(2, minmax(0, 1fr));
    gap: 18px;
}

.model-card {
    border: 1px solid var(--border);
    border-radius: 14px;
    background: white;
    overflow: hidden;
    box-shadow: 0 5px 18px rgba(15, 23, 42, 0.05);
    transition: transform 0.18s ease, box-shadow 0.18s ease, border-color 0.18s ease;
}

.model-card:hover {
```

```
    transform: translateY(-2px);
    box-shadow: 0 10px 24px rgba(15, 23, 42, 0.09);
    border-color: #cbd5e1;
}

.model-card.current {
    border-color: #93c5fd;
    box-shadow: 0 8px 22px rgba(37, 99, 235, 0.10);
}

.model-card-preview {
    height: 190px;
    background: #f8fafc;
    border-bottom: 1px solid var(--border);
    position: relative;
    overflow: hidden;
}

.model-card-preview svg {
    width: 100%;
    height: 100%;
    display: block;
}

.model-card-body {
    padding: 15px;
}

.model-card-title-row {
    display: flex;
    align-items: flex-start;
    justify-content: space-between;
    gap: 10px;
    margin-bottom: 12px;
}

.model-card-badges {
    display: flex;
    align-items: center;
    justify-content: flex-end;
    gap: 5px;
    flex: 0 0 auto;
    flex-wrap: wrap;
}

.model-card-name {
    flex: 1 1 auto;
    min-width: 0;
    font-size: 15px;
    font-weight: 800;
    color: #1f2937;
    line-height: 1.3;
    overflow-wrap: anywhere;
}

.model-card-meta {
    display: grid;
    grid-template-columns: repeat(4, minmax(0, 1fr));
    gap: 8px;
    margin-bottom: 13px;
}

.model-card-stat {
    border: 1px solid #e5eaf0;
    border-radius: 9px;
    background: #f8fafc;
    padding: 8px 9px;
    min-width: 0;
}

.model-card-stat-label {
    font-size: 9px;
    font-weight: 800;
    color: var(--muted);
    text-transform: uppercase;
    letter-spacing: 0.3px;
}

.model-card-stat-value {
    margin-top: 3px;
    font-size: 14px;
    font-weight: 800;
    color: #1f2937;
}

.model-card-foot {
    display: flex;
```

```
    align-items: center;
    justify-content: space-between;
    gap: 10px;
}

.model-card-saved {
    font-size: 10px;
    color: var(--muted);
}

.model-card-actions {
    display: flex;
    gap: 6px;
    flex-wrap: wrap;
    justify-content: flex-end;
}

.model-card-actions .icon-btn {
    width: auto;
    min-width: 58px;
    padding: 0 9px;
    font-size: 10px;
}

.home-empty {
    border: 1px dashed #cbd5e1;
    border-radius: 14px;
    background: #fafcff;
    padding: 46px 24px;
    text-align: center;
}

.home-empty-icon {
    width: 58px;
    height: 58px;
    margin: 0 auto 13px;
    border-radius: 15px;
    display: flex;
    align-items: center;
    justify-content: center;
    background: #eff6ff;
    border: 1px solid #bfdbfe;
    color: #2563eb;
    font-size: 27px;
}

.home-empty h4 {
    font-size: 16px;
    color: #1f2937;
    margin-bottom: 6px;
}

.home-empty p {
    color: var(--muted);
    font-size: 12px;
    line-height: 1.6;
    max-width: 500px;
    margin: 0 auto 15px;
}

.home-preview-label {
    position: absolute;
    top: 10px;
    left: 10px;
    z-index: 2;
    font-size: 9px;
    font-weight: 800;
    color: #475569;
    background: rgba(255,255,255,0.92);
    border: 1px solid #dce3ec;
    padding: 5px 8px;
    border-radius: 999px;
    text-transform: uppercase;
    letter-spacing: 0.4px;
}

@media (max-width: 900px) {
    .home-hero-inner {
        flex-direction: column;
        align-items: stretch;
    }
    .home-hero-actions {
        justify-content: flex-start;
    }
    .model-library-grid {
        grid-template-columns: 1fr;
    }
```

```
}

@media (max-width: 500px) {
    .home-hero-inner,
    .home-current-strip {
        padding: 18px;
    }
    .home-hero-copy h3 {
        font-size: 19px;
    }
    .home-hero-actions .btn {
        flex: 1 1 auto;
    }
    .model-card-meta {
        grid-template-columns: repeat(2, minmax(0, 1fr));
    }
    .model-card-foot {
        flex-direction: column;
        align-items: stretch;
    }
    .model-card-actions {
        justify-content: stretch;
    }
    .model-card-actions .icon-btn {
        flex: 1;
    }
}

@media (max-width: 800px) {
    .model-library-actions {
        flex-direction: column;
    }
    .model-library-actions .icon-btn {
        width: 100%;
    }
}

/* ===== Home: multi-select toolbar & card checkboxes ===== */
.model-select-bar {
    display: flex;
    align-items: center;
    flex-wrap: wrap;
    gap: 10px;
    margin-bottom: 14px;
    padding: 10px 12px;
    background: #fff;
    border: 1px solid var(--border);
    border-radius: 12px;
}

.model-selected-count {
    font-size: 12px;
    font-weight: 800;
    color: #475569;
    margin-right: auto;
}

.model-select-actions { display: flex; flex-wrap: wrap; gap: 8px; }

.model-select-bar .btn:disabled { opacity: 0.45; cursor: not-allowed; }

.model-select-delete { color: #b91c1c !important; }

.model-select-box {
    position: absolute;
    top: 10px;
    right: 10px;
    z-index: 3;
    width: 32px;
    height: 32px;
    display: flex;
    align-items: center;
    justify-content: center;
    background: rgba(255, 255, 255, 0.96);
    border: 1px solid #cbd5e1;
    border-radius: 8px;
    cursor: pointer;
    box-shadow: 0 2px 6px rgba(15, 23, 42, 0.12);
}

.model-select-box input { width: 18px; height: 18px; margin: 0; cursor: pointer; accent-color: var(--primary); }

.model-card.selected {
    border-color: var(--primary);
    box-shadow: 0 0 0 2px rgba(37, 99, 235, 0.28), 0 8px 22px rgba(37, 99, 235, 0.10);
}
@media (max-width: 600px) {
```

```
    .model-select-actions { width: 100%; }
    .model-select-actions .btn { flex: 1 1 0; }
}
```

21. modules/home/home.js

```
/* ========================================================================
 * home.js — Home page module
 * Home page — model library (saved models grid, current-model strip, multi-select export / delete).
 * Contains this page's markup (injected into .container) and its logic.
 * ======================================================================== */

/* ---- Markup ---- */
mountModuleHTML(".container", "beforeend", `
<!-- HOME / MODEL LIBRARY -->
<section class="section active" id="home">
    <div class="home-standalone-head">
        <div class="home-brand">
            <div class="home-brand-mark">■</div>
            <div class="home-brand-text">
                <h1>Structure Analysis</h1>
                <p>Structural Analysis Model Manager</p>
            </div>
        </div>
        <div class="home-standalone-actions">
            <button type="button" class="btn btn-secondary" onclick="document.getElementById('jsonImport').click()">Import Model</button>
            <button type="button" class="btn btn-secondary" onclick="exportAllModels()">Export All Models</button>
            <button type="button" class="btn btn-primary" onclick="newModel()">■ New Model</button>
        </div>
    </div>

    <h2 class="home-standalone-title">Your Structural Models</h2>
    <p class="home-standalone-subtitle">Create and manage separate structural-analysis projects from one place. Open a model to enter nodes, elements, supports, loads, moments and run the
existing analysis workflow.</p>

    <div class="home-hero">
        <div class="home-current-strip" style="border-top:none;">
            <div class="home-current-info">
                <span class="home-current-label">Current Model</span>
                <span id="homeCurrentModelName" class="home-current-name">Structural Model</span>
                <span id="homeCurrentModelType" class="badge badge-blue">Beam / Truss / Frame Model</span>
                <span id="homeCurrentModelStatus" class="badge badge-gray">Not saved</span>
            </div>
            <div class="home-standalone-actions">
                <button type="button" class="btn btn-secondary" onclick="saveModel()">Save Current</button>
                <button type="button" class="btn btn-secondary" onclick="exportModel()">Export Current</button>
                <button type="button" class="btn btn-secondary" onclick="newModel()">Start New Model</button>
            </div>
        </div>
    </div>

    <div class="home-library-head">
        <div>
            <h3>Saved Models</h3>
            <p>Open any saved model to continue working on its structure and analysis.</p>
        </div>
        <span id="homeModelCount" class="home-model-count">0 Models</span>
    </div>

    <div class="model-select-bar" id="modelSelectBar">
        <button type="button" class="btn btn-secondary" id="modelSelectAllBtn" onclick="toggleSelectAllModels()">Select All</button>
        <span class="model-selected-count" id="modelSelectedCount">0 selected</span>
        <div class="model-select-actions">
            <button type="button" class="btn btn-secondary" id="selExportJsonBtn" onclick="exportSelectedModelsJSON()" disabled>Export JSON</button>
            <button type="button" class="btn btn-secondary" id="selExportPdfBtn" onclick="exportSelectedModelsPDF()" disabled>Export PDF</button>
            <button type="button" class="btn btn-secondary model-select-delete" id="selDeleteBtn" onclick="deleteSelectedModels()" disabled>Delete</button>
        </div>
    </div>

    <div id="modelLibraryGrid" class="model-library-grid"></div>
</section>
`);

/* ---- Logic ---- */
function modelPreviewSVG(data) {
    const nodes = Array.isArray(data?.nodes) ? data.nodes : [];
    const elements = Array.isArray(data?.elements) ? data.elements : [];
    const supports = Array.isArray(data?.supportAssignments) ? data.supportAssignments : [];
    const loads = Array.isArray(data?.loads) ? data.loads : [];
    const moments = Array.isArray(data?.moments) ? data.moments : [];
    const isGridModel = data?.analysisMode === "grid";

    if (!nodes.length) {
        return `<svg viewBox="0 0 760 260" preserveAspectRatio="xMidYMid meet" aria-label="No structure defined">
            <rect x="0" y="0" width="760" height="260" fill="#f8fafc"/>
            <text x="380" y="126" text-anchor="middle" font-family="Inter,Arial,sans-serif" font-size="14" font-weight="700" fill="#64748b">No structure defined yet</text>
            <text x="380" y="148" text-anchor="middle" font-family="Inter,Arial,sans-serif" font-size="11" fill="#94a3b8">Add nodes and elements to preview the model</text>
        </svg>`;
    }

    const nodeById = Object.create(null);
```

```
    nodes.forEach(n => { nodeById[String(n.id)] = n; });

    const xs = nodes.map(n => Number(n.x) || 0);
    const ys = nodes.map(n => Number(n.y) || 0);
    const minX = Math.min(...xs), maxX = Math.max(...xs);
    const minY = Math.min(...ys), maxY = Math.max(...ys);
    const spanX = Math.max(maxX - minX, 1);
    const spanY = Math.max(maxY - minY, 1);
    const padX = Math.max(spanX * 0.14, 1);
    const padY = Math.max(spanY * 0.18, 1);
    const left = minX - padX, right = maxX + padX;
    const bottom = minY - padY, top = maxY + padY;
    const W = 760, H = 260;
    const sx = x => 55 + ((x - left) / Math.max(right - left, 1)) * 650;
    const sy = y => 210 - ((y - bottom) / Math.max(top - bottom, 1)) * 170;

    const templateById = Object.create(null);
    (Array.isArray(data.elementTemplates) ? data.elementTemplates : []).forEach(t => {
        templateById[String(t.id)] = t;
    });

    const elementLines = elements.map((el) => {
        const n1 = nodeById[String(el.startNode)], n2 = nodeById[String(el.endNode)];
        if (!n1 || !n2) return "";
        const x1=sx(Number(n1.x)||0), y1=sy(Number(n1.y)||0), x2=sx(Number(n2.x)||0), y2=sy(Number(n2.y)||0);
        const t = templateById[String(el.templateId)] || {};
        const A = Number(t.A) || 0, I = Number(t.I) || 0;
        const width = A === 0 ? 2 : I === 0 ? 1.8 : 2.7;
        return `<line x1="${x1.toFixed(2)}" y1="${y1.toFixed(2)}" x2="${x2.toFixed(2)}" y2="${y2.toFixed(2)}" stroke="#334155" stroke-width="${width}" stroke-linecap="round"/>`;
    }).join("");

    const supportIds = new Set(supports.map(s => String(s.nodeId)));
    const supportMarks = nodes.filter(n => supportIds.has(String(n.id))).map(n => {
        const x=sx(Number(n.x)||0), y=sy(Number(n.y)||0);
        return `<polygon points="${(x-9).toFixed(1)},${(y+14).toFixed(1)} ${(x+9).toFixed(1)},${(y+14).toFixed(1)} ${x.toFixed(1)},${(y+2).toFixed(1)}" fill="#94a3b8" opacity="0.9"/>`;
    }).join("");

    const getElementPoint = (element, distance) => {
        if (!element) return null;
        const a = nodeById[String(element.startNode)], b = nodeById[String(element.endNode)];
        if (!a || !b) return null;
        const x1 = Number(a.x) || 0, y1 = Number(a.y) || 0;
        const x2 = Number(b.x) || 0, y2 = Number(b.y) || 0;
        const L = Math.hypot(x2 - x1, y2 - y1);
        if (!(L > 0)) return null;
        const t = Math.max(0, Math.min(1, (Number(distance) || 0) / L));
        return { x: sx(x1 + (x2 - x1) * t), y: sy(y1 + (y2 - y1) * t), cx: (x2 - x1) / L, cy: (y2 - y1) / L, length: L };
    };
    const forceVector = (load, element, end = false) => {
        let fx = Number(end ? (load.FxEnd ?? load.Fx) : load.Fx) || 0;
        let fy = Number(end ? (load.FyEnd ?? load.Fy) : load.Fy) || 0;
        const axis = load.axisSystem || "Global";
        if (axis === "Local" && element) {
            const p = getElementPoint(element, 0);
            if (p) return { x: fx * p.cx - fy * p.cy, y: fx * p.cy + fy * p.cx };
        }
        if (axis === "Angled") {
            const angle = (Number(load.angle) || 0) * Math.PI / 180;
            [fx, fy] = [fx * Math.cos(angle) - fy * Math.sin(angle), fx * Math.sin(angle) + fy * Math.cos(angle)];
        }
        return { x: fx, y: fy };
    };
    const drawForceArrow = (x, y, fx, fy, color = "#dc2626") => {
        const magnitude = Math.hypot(fx, fy);
        if (!(magnitude > 1e-12)) return "";
        const ux = fx / magnitude, uy = -fy / magnitude;
        const tailX = x - ux * 23, tailY = y - uy * 23;
        const tipX = x - ux * 5, tipY = y - uy * 5;
        const px = -uy, py = ux;
        return `<g><line x1="${tailX.toFixed(1)}" y1="${tailY.toFixed(1)}" x2="${tipX.toFixed(1)}" y2="${tipY.toFixed(1)}" stroke="${color}" stroke-width="1.8"/><polygon points="${tipX.
toFixed(1)},${tipY.toFixed(1)} ${(tipX-ux*5+px*3).toFixed(1)},${(tipY-uy*5+py*3).toFixed(1)} ${(tipX-ux*5-px*3).toFixed(1)},${(tipY-uy*5-py*3).toFixed(1)}" fill="${color}"/></g>`;
    };
    const drawGridLoad = (x, y, fz) => {
        if (Math.abs(fz) <= 1e-12) return "";
        const r = Math.min(7.5, 4 + Math.sqrt(Math.abs(fz)) * 0.18);
        const color = fz < 0 ? "#dc2626" : "#16a34a";
        return `<circle cx="${x.toFixed(1)}" cy="${y.toFixed(1)}" r="${r.toFixed(1)}" fill="${fz < 0 ? color : "#ffffff"}" stroke="${color}" stroke-width="2"/>`;
    };
    const loadMarks = loads.map((load) => {
        if (!load) return "";
        const element = elements.find((el) => String(el.id) === String(load.elementId));
        if (load.type === "Nodal Load") {
            const node = nodeById[String(load.nodeId)];
            if (!node) return "";
            const x = sx(Number(node.x) || 0), y = sy(Number(node.y) || 0);
            return isGridModel ? drawGridLoad(x, y, Number(load.Fz) || 0) : ((() => { const f = forceVector(load, null); return drawForceArrow(x, y, f.x, f.y); })();
        }
```

```
        if (!element) return "";
        const start = Math.max(0, Number(load.distance) || 0);
        const end = load.type === "Element UDL" ? Math.max(start, Number(load.distanceEnd) || getElementPoint(element, 0)?.length || start) : start;
        const positions = load.type === "Element UDL" ? [0.2, 0.5, 0.8].map(frac => start + (end - start) * frac) : [start];
        return positions.map((distance) => {
            const point = getElementPoint(element, distance);
            if (!point) return "";
            if (isGridModel) {
                const q1 = Number(load.Fz) || 0, q2 = Number(load.FzEnd ?? load.Fz) || 0;
                const t = end > start ? (distance - start) / (end - start) : 0;
                return drawGridLoad(point.x, point.y, q1 + (q2 - q1) * t);
            }
            const f1 = forceVector(load, element), f2 = forceVector(load, element, true);
            const t = end > start ? (distance - start) / (end - start) : 0;
            return drawForceArrow(point.x, point.y, f1.x + (f2.x - f1.x) * t, f1.y + (f2.y - f1.y) * t);
        }).join("");
    }).join("");

    const momentMarks = moments.map((moment) => {
        if (!moment) return "";
        const nodal = moment.type === "Nodal Moment";
        const node = nodal ? nodeById[String(moment.nodeId)] : null;
        const element = nodal ? null : elements.find((el) => String(el.id) === String(moment.elementId));
        const point = node
            ? { x: sx(Number(node.x) || 0), y: sy(Number(node.y) || 0) }
            : getElementPoint(element, moment.distance);
        if (!point) return "";
        if (isGridModel) {
            const mx = Number(moment.Mx) || 0, my = Number(moment.My) || 0;
            const labels = [];
            if (Math.abs(mx) > 1e-12) labels.push(mx > 0 ? "→" : "←");
            if (Math.abs(my) > 1e-12) labels.push(my > 0 ? "↑" : "↓");
            return labels.length ? `<text x="${(point.x+9).toFixed(1)}" y="${(point.y-11).toFixed(1)}" font-size="13" font-weight="800" fill="#7c3aed">${labels.join("")}</text>` : "";
        }
        const mz = Number(moment.Mz) || 0;
        if (Math.abs(mz) <= 1e-12) return "";
        return `<text x="${(point.x+8).toFixed(1)}" y="${(point.y-9).toFixed(1)}" font-size="15" font-weight="800" fill="#7c3aed">${mz > 0 ? "■" : "■"}</text>`;
    }).join("");

    const nodeMarks = nodes.map(n => {
        const x=sx(Number(n.x)||0), y=sy(Number(n.y)||0);
        return `<g><circle cx="${x.toFixed(1)}" cy="${y.toFixed(1)}" r="5" fill="#2563eb" stroke="#ffffff" stroke-width="2"/><text x="${(x+8).toFixed(1)}" y="${(y-8).toFixed(1)}" font-
family="Inter,Arial,sans-serif" font-size="9" font-weight="700" fill="#475569">${escapeHtml(String(n.id))}</text></g>`;
    }).join("");

    return `<svg viewBox="0 0 ${W} ${H}" preserveAspectRatio="xMidYMid meet" aria-label="Structural model preview">
        <rect x="0" y="0" width="${W}" height="${H}" fill="#f8fafc"/>
        <line x1="48" y1="224" x2="712" y2="224" stroke="#e2e8f0" stroke-width="1"/>
        ${elementLines}
        ${supportMarks}
        ${loadMarks}
        ${momentMarks}
        ${nodeMarks}
    </svg>`;
}

// Home-screen multi-select state (ids of the selected saved models).
const selectedModelIds = new Set();

function renderModelLibrary() {
    const grid = document.getElementById("modelLibraryGrid");
    const currentNameEl = document.getElementById("homeCurrentModelName");
    const currentStatusEl = document.getElementById("homeCurrentModelStatus");
    const currentTypeEl = document.getElementById("homeCurrentModelType");
    const countEl = document.getElementById("homeModelCount");
    if (!grid) return;

    const currentName = getModelName();
    if (currentNameEl) currentNameEl.textContent = currentName;
    if (currentTypeEl) {
        const gridMode = model.analysisMode === "grid";
        currentTypeEl.textContent = gridMode ? "Grid Model" : "Beam / Truss / Frame Model";
        currentTypeEl.className = `badge ${gridMode ? "badge-green" : "badge-blue"}`;
    }

    const library = getModelLibrary();
    if (lastModelLibraryReadFailed) {
        if (currentStatusEl) {
            currentStatusEl.textContent = "Library unavailable";
            currentStatusEl.className = "badge badge-red";
        }
        if (countEl) countEl.textContent = "Unavailable";
        grid.innerHTML = `<div class="home-empty" style="grid-column:1 / -1;">
            <div class="home-empty-icon">■</div>
            <h4>Saved-model library could not be read</h4>
            <p>The stored library has not been overwritten. Export the current workspace as JSON and keep a backup before attempting browser-storage recovery.</p>
        </div>`;
```

```
        updateModelSelectionToolbar();
        return;
    }
    const activeSavedModel = currentModelLibraryId ? library[currentModelLibraryId] : null;
    const liveCurrentData = getPersistedInputModel();
    const currentModelHasUnsavedChanges = !!activeSavedModel && (
        getModelName() !== String(activeSavedModel.name || "Structural Model") ||
        modelDataFingerprint(liveCurrentData) !== modelDataFingerprint(activeSavedModel.data)
    );
    const models = Object.values(library).sort((a, b) => {
        const at = new Date(a.updatedAt || 0).getTime();
        const bt = new Date(b.updatedAt || 0).getTime();
        return bt - at;
    });
    Array.from(selectedModelIds).forEach((id) => { if (!library[id]) selectedModelIds.delete(id); });

    if (countEl) countEl.textContent = `${models.length} ${models.length === 1 ? "Model" : "Models"}`;

    if (currentStatusEl) {
        if (activeSavedModel) {
            currentStatusEl.textContent = currentModelHasUnsavedChanges ? "Unsaved changes" : "Saved";
            currentStatusEl.className = currentModelHasUnsavedChanges ? "badge badge-orange" : "badge badge-green";
        } else {
            currentStatusEl.textContent = "Not saved";
            currentStatusEl.className = "badge badge-gray";
        }
    }

    if (!models.length) {
        grid.innerHTML = `<div class="home-empty" style="grid-column:1 / -1;">
            <div class="home-empty-icon">■</div>
            <h4>No saved models yet</h4>
            <p>Create your first structural model, give it a name, and use <b>Save Model</b>. It will appear here with a visual preview of the structure.</p>
            <button type="button" class="btn btn-primary" onclick="newModel()">■ Create First Model</button>
        </div>`;
        updateModelSelectionToolbar();
        return;
    }

    grid.innerHTML = models.map((item) => {
        const savedData = item.data || {};
        const isCurrent = item.id === currentModelLibraryId;
        const data = isCurrent ? liveCurrentData : savedData;
        const displayName = isCurrent ? currentName : (item.name || "Structural Model");
        const nodes = Array.isArray(data.nodes) ? data.nodes.length : 0;
        const elements = Array.isArray(data.elements) ? data.elements.length : 0;
        const loads = Array.isArray(data.loads) ? data.loads.length : 0;
        const moments = Array.isArray(data.moments) ? data.moments.length : 0;
        const isGridModel = data.analysisMode === "grid";
        const modelTypeLabel = isGridModel ? "Grid Model" : "Beam / Truss / Frame Model";
        const modelTypeClass = isGridModel ? "badge-green" : "badge-blue";
        const safeId = encodeInlineArgument(String(item.id || ""));
        const isSelected = selectedModelIds.has(item.id);
        return `<article class="model-card${isCurrent ? " current" : ""}${isSelected ? " selected" : ""}">
            <div class="model-card-preview">
                <label class="model-select-box" title="Select this model"><input type="checkbox" aria-label="Select model ${escapeHtml(displayName)}" ${isSelected ? "checked" : ""}
onchange="toggleModelSelection('${safeId}', this.checked)"></label>
                <span class="home-preview-label">Structure Preview</span>
                ${modelPreviewSVG(data)}
            </div>
            <div class="model-card-body">
                <div class="model-card-title-row">
                    <div class="model-card-name">${escapeHtml(displayName)}</div>
                    <div class="model-card-badges">
                        <span class="badge ${modelTypeClass}">${modelTypeLabel}</span>
                        ${isCurrent ? '<span class="badge badge-gray">Current</span>' : ''}
                    </div>
                </div>
                <div class="model-card-meta">
                    <div class="model-card-stat"><div class="model-card-stat-label">Nodes</div><div class="model-card-stat-value">${nodes}</div></div>
                    <div class="model-card-stat"><div class="model-card-stat-label">Elements</div><div class="model-card-stat-value">${elements}</div></div>
                    <div class="model-card-stat"><div class="model-card-stat-label">Loads</div><div class="model-card-stat-value">${loads}</div></div>
                    <div class="model-card-stat"><div class="model-card-stat-label">${isGridModel ? "Torsions" : "Moments"}</div><div class="model-card-stat-value">${moments}</div></div>
                </div>
                <div class="model-card-foot">
                    <span class="model-card-saved">${isCurrent && currentModelHasUnsavedChanges ? "Unsaved changes" : `Saved ${escapeHtml(formatModelSavedTime(item.updatedAt))}`}</span>
                    <div class="model-card-actions">
                        <button type="button" class="icon-btn" onclick="openSavedModelFromInline('${safeId}')">Open</button>
                        <button type="button" class="icon-btn" onclick="exportSavedModelFromInline('${safeId}')">Export</button>
                        <button type="button" class="icon-btn" onclick="deleteSavedModelFromInline('${safeId}')">Delete</button>
                    </div>
                </div>
            </div>
        </article>`;
    }).join("");
    updateModelSelectionToolbar();
}
```

```
// ========================================================================
// 16c. HOME SCREEN: MULTI-SELECT (export JSON / export PDF / delete)
// ========================================================================
function getSortedLibraryModels() {
    return Object.values(getModelLibrary()).sort((a, b) => {
        const at = new Date(a.updatedAt || 0).getTime();
        const bt = new Date(b.updatedAt || 0).getTime();
        return bt - at;
    });
}

function toggleModelSelection(encodedId, checked) {
    let id;
    try {
        id = decodeURIComponent(encodedId);
    } catch (error) {
        console.error("Invalid encoded model ID in selection control.", error);
        return;
    }
    if (checked) selectedModelIds.add(id);
    else selectedModelIds.delete(id);
    const card = Array.from(document.querySelectorAll("#modelLibraryGrid .model-card input[type=checkbox]"))
        .find((cb) => (cb.getAttribute("onchange") || "").includes(`'${encodedId}'`));
    if (card) card.closest(".model-card").classList.toggle("selected", !!checked);
    updateModelSelectionToolbar();
}

function decodeSavedModelId(encodedId) {
    try {
        return decodeURIComponent(encodedId);
    } catch (error) {
        console.error("Invalid encoded saved-model ID.", error);
        return null;
    }
}

function openSavedModelFromInline(encodedId) {
    const id = decodeSavedModelId(encodedId);
    if (id !== null) openSavedModel(id);
}

function exportSavedModelFromInline(encodedId) {
    const id = decodeSavedModelId(encodedId);
    if (id !== null) exportSavedModel(id);
}

function deleteSavedModelFromInline(encodedId) {
    const id = decodeSavedModelId(encodedId);
    if (id !== null) deleteSavedModel(id);
}

function toggleSelectAllModels() {
    const models = getSortedLibraryModels();
    const allSelected = models.length > 0 && models.every((m) => selectedModelIds.has(m.id));
    selectedModelIds.clear();
    if (!allSelected) models.forEach((m) => selectedModelIds.add(m.id));
    renderModelLibrary();
}

function updateModelSelectionToolbar() {
    const total = Object.keys(getModelLibrary()).length;
    const n = selectedModelIds.size;
    const countEl = document.getElementById("modelSelectedCount");
    const allBtn = document.getElementById("modelSelectAllBtn");
    if (countEl) countEl.textContent = `${n} selected`;
    if (allBtn) {
        allBtn.textContent = total > 0 && n === total ? "Deselect All" : "Select All";
        allBtn.disabled = total === 0;
        allBtn.style.opacity = total === 0 ? "0.45" : "";
    }
    ["selExportJsonBtn", "selExportPdfBtn", "selDeleteBtn"].forEach((id) => {
        const b = document.getElementById(id);
        if (b) b.disabled = n === 0 || (pdfExportBusy && id !== "selExportJsonBtn");
    });
}

// Selected models in display order. The model that is open right now is taken
// from memory so unsaved edits are included (same rule as "Export All").
function getSelectedModelSnapshots() {
    return getSortedLibraryModels()
        .filter((m) => selectedModelIds.has(m.id))
        .map((m) => {
            if (m.id === currentModelLibraryId && hasModelInputData()) {
                return { id: m.id, name: getModelName(), data: getPersistedInputModel(), updatedAt: m.updatedAt };
            }
            return { id: m.id, name: m.name || "Structural Model", data: m.data || {}, updatedAt: m.updatedAt };
        });
```

```
}

function exportSelectedModelsJSON() {
    const items = getSelectedModelSnapshots();
    if (lastModelLibraryReadFailed) {
        alert(getModelLibraryReadFailureMessage());
        return;
    }
    if (!items.length) {
        alert("Select at least one model first.");
        return;
    }
    if (items.length === 1) {
        downloadJSONFile(`${sanitizeFileName(items[0].name)}.json`, items[0].data);
        return;
    }
    const bundle = {
        type: "StructuralAnalysisModelCollection",
        version: 1,
        exportedAt: new Date().toISOString(),
        models: items.map((item) => ({
            id: item.id,
            name: item.name,
            updatedAt: item.updatedAt || null,
            data: item.data,
        })),
    };
    downloadJSONFile(`Selected_Models (${items.length}).json`, bundle);
}

function deleteSelectedModels() {
    const library = getModelLibrary();
    if (lastModelLibraryReadFailed) {
        alert(getModelLibraryReadFailureMessage());
        return;
    }
    const ids = Array.from(selectedModelIds).filter((id) => library[id]);
    if (!ids.length) {
        alert("Select at least one model first.");
        return;
    }
    const names = ids.map((id) => `• ${library[id].name}`).slice(0, 12).join("\n");
    const more = ids.length > 12 ? `\n…and ${ids.length - 12} more` : "";
    if (!confirm(`Delete ${ids.length} selected model(s)? This cannot be undone.\n\n${names}${more}`)) return;

    const deletedCurrentModel = ids.includes(currentModelLibraryId);
    ids.forEach((id) => { delete library[id]; });
    const remainingLibrary = setModelLibrary(library);
    if (!lastModelLibraryWriteSucceeded) {
        alert("The selected models were not deleted because browser storage rejected the library update. No deletion is confirmed.");
        return;
    }
    selectedModelIds.clear();

    if (deletedCurrentModel || Object.keys(remainingLibrary).length === 0) {
        // Deleting every saved model must also remove the active unsaved
        // working copy and its localStorage snapshot.
        resetActiveModelAfterSavedModelDeletion();
    }
    renderModelLibrary();
}

// Choose a batch-PDF analysis case only when the saved model makes that
// choice unambiguous. Never silently export a different load case or omit an
// available self-weight case just because its case has no explicit load rows.
function chooseUnambiguousBatchCases(modelData) {
    try {
        modelData = normalizeLegacySelfWeightModel(modelData);
    } catch (error) {
        return { ok: false, message: error.message || "Legacy self-weight data could not be converted safely. Review the model before batch export." };
    }
    const loadCases = Array.isArray(modelData?.loadCases) ? modelData.loadCases : [];
    const loads = Array.isArray(modelData?.loads) ? modelData.loads : [];
    const elements = Array.isArray(modelData?.elements) ? modelData.elements : [];
    const momentCases = Array.isArray(modelData?.momentCases) ? modelData.momentCases : [];
    const moments = Array.isArray(modelData?.moments) ? modelData.moments : [];
    const selfWeightCases = loadCases.filter((loadCase) => loadCase?.type === "Self Weight");
    const elementIds = new Set(elements.map((element) => String(element?.id ?? "")));
    const hasSelfWeightData = selfWeightCases.some((loadCase) =>
        Object.entries(loadCase?.memberSelfWeights || {}).some(([elementId, weight]) => elementIds.has(String(elementId)) && Number(weight) > 0)
    );
    const incompleteSelfWeightCase = selfWeightCases.find((loadCase) =>
        getMissingBatchSelfWeightElement(loadCase, elements)
    );
    if (incompleteSelfWeightCase) {
        const missingElement = getMissingBatchSelfWeightElement(incompleteSelfWeightCase, elements);
        return { ok: false, message: `Self Weight case "${incompleteSelfWeightCase.id}" has no valid weight for element "${missingElement.id}". Edit the case and enter a weight in kN/m (
```

```
enter 0 only if zero is intentional).` };
    }
    if (hasSelfWeightData && selfWeightCases.length !== 1) {
        return { ok: false, message: "Batch PDF export cannot safely choose the self-weight analysis: this model needs exactly one Self Weight case. Open the model and export its Results
after choosing the intended case." };
    }

    const loadCaseIdsWithLoads = [...new Set(loads.map((load) => String(load?.loadCaseId ?? "")).filter(Boolean))];
    const candidateLoadCaseIds = new Set(loadCaseIdsWithLoads);
    if (hasSelfWeightData && selfWeightCases.length === 1) candidateLoadCaseIds.add(String(selfWeightCases[0].id));
    if (candidateLoadCaseIds.size > 1) {
        return { ok: false, message: "Batch PDF export cannot safely guess between multiple load cases. Open the model and export its Results after selecting the intended case." };
    }

    const momentCaseIdsWithMoments = [...new Set(moments.map((moment) => String(moment?.momentCaseId ?? "")).filter(Boolean))];
    if (momentCaseIdsWithMoments.length > 1) {
        return { ok: false, message: "Batch PDF export cannot safely guess between multiple moment/torsion cases. Open the model and export its Results after selecting the intended case." };
    }
    return {
        ok: true,
        loadCaseId: hasSelfWeightData ? String(selfWeightCases[0].id) : (loadCaseIdsWithLoads[0] || ""),
        momentCaseId: momentCaseIdsWithMoments[0] || "",
    };
}

function getMissingBatchSelfWeightElement(loadCase, elements) {
    const weights = loadCase?.memberSelfWeights && typeof loadCase.memberSelfWeights === "object"
        ? loadCase.memberSelfWeights : {};
    return elements.find((element) => {
        const id = String(element?.id ?? "");
        if (!Object.prototype.hasOwnProperty.call(weights, id)) return true;
        const weight = weights[id];
        return weight === undefined || weight === null || weight === "" ||
            !Number.isFinite(Number(weight)) || Number(weight) < 0;
    }) || null;
}

// A Home-screen batch export has no active case selector. For plane models,
// retain the established safety rule and refuse to guess between multiple
// cases. Grid models can export every defined, applicable load-case / torsion-
// case combination instead of failing just because the model has alternatives.
function chooseBatchCaseCombinations(modelData) {
    if (modelData?.analysisMode !== "grid") {
        const choice = chooseUnambiguousBatchCases(modelData);
        return choice.ok ? { ok: true, choices: [choice] } : choice;
    }

    let normalized;
    try {
        normalized = normalizeLegacySelfWeightModel(modelData);
    } catch (error) {
        return { ok: false, message: error.message || "Legacy self-weight data could not be converted safely. Review the model before batch export." };
    }

    const loadCases = Array.isArray(normalized?.loadCases) ? normalized.loadCases : [];
    const loads = Array.isArray(normalized?.loads) ? normalized.loads : [];
    const elements = Array.isArray(normalized?.elements) ? normalized.elements : [];
    const moments = Array.isArray(normalized?.moments) ? normalized.moments : [];
    const elementIds = new Set(elements.map((element) => String(element?.id ?? "")));

    // Every case containing explicit loads is a candidate. A Self Weight case
    // is also included when it has a positive weight for an existing element,
    // even if it has no explicit load rows. Set semantics avoid duplicate case
    // reports when a Self Weight case contains both types of input.
    const loadCaseIds = new Set(
        loads.map((load) => String(load?.loadCaseId ?? "")).filter(Boolean)
    );
    for (const loadCase of loadCases.filter((item) => item?.type === "Self Weight")) {
        const hasApplicableWeight = Object.entries(loadCase?.memberSelfWeights || {}).some(
            ([elementId, weight]) => elementIds.has(String(elementId)) && Number(weight) > 0
        );
        const missingElement = getMissingBatchSelfWeightElement(loadCase, elements);
        if (missingElement) {
            return { ok: false, message: `Self Weight case "${loadCase.id}" has no valid weight for element "${missingElement.id}". Edit the case and enter a weight in kN/m (enter 0 only if
zero is intentional).` };
        }
        if (hasApplicableWeight && loadCase.id !== undefined && loadCase.id !== null && String(loadCase.id)) {
            loadCaseIds.add(String(loadCase.id));
        }
    }

    const momentCaseIds = new Set(
        moments.map((moment) => String(moment?.momentCaseId ?? "")).filter(Boolean)
    );
    // Empty models retain the same default-case behavior as opening a model
    // and running its analysis from the Results page.
    if (!loadCaseIds.size) loadCaseIds.add("");
```

```
    if (!momentCaseIds.size) momentCaseIds.add("");

    const choices = [...loadCaseIds].flatMap((loadCaseId) =>
        [...momentCaseIds].map((momentCaseId) => ({
            ok: true,
            loadCaseId,
            momentCaseId,
        }))
    );
    return { ok: true, choices };
}

// runAnalysisCore uses a mode-specific success message. Recognize both the
// plane and Grid forms so a successful Grid solve is not misclassified as an
// error by the silent Home-screen exporter.
function isBatchAnalysisSuccessMessage(message) {
    return /^(?:Grid )?analysis completed successfully\b/i.test(String(message || ""));
}

// Loads a model's input data into the working model and solves it without any
// dialogs. Returns { ok, message, sections, caseLabel }.
function solveModelSilently(data, caseChoiceOverride = null) {
    const captured = [];
    const origAlert = window.alert;
    const origConfirm = window.confirm;
    window.alert = (m) => { captured.push(String(m)); };
    window.confirm = () => true;
    try {
        applyPersistedInputModel(JSON.parse(JSON.stringify(data || {})));
        migrateUnitSystem();
        ensureBuiltInSupportTemplates();

        const caseChoice = caseChoiceOverride || chooseUnambiguousBatchCases(model);
        if (!caseChoice.ok) return { ok: false, message: caseChoice.message };
        activeLoadCaseId = caseChoice.loadCaseId;
        activeMomentCaseId = caseChoice.momentCaseId;
        syncActiveAnalysisCaseSelectors();

        analysisState.status = "not-run";
        runAnalysisCore();

        const problems = captured.filter((m) => !isBatchAnalysisSuccessMessage(m));
        if (analysisState.status !== "valid" || problems.length) {
            return { ok: false, message: problems.join(" ") || "The analysis did not complete." };
        }
        const caseLabel = getActiveAnalysisCaseLabel();
        return { ok: true, sections: pdfCollectReportSections(), caseLabel };
    } catch (err) {
        console.error(err);
        return { ok: false, message: err && err.message ? err.message : "Unexpected error while solving the model." };
    } finally {
        window.alert = origAlert;
        window.confirm = origConfirm;
    }
}

async function exportSelectedModelsPDF() {
    if (pdfExportBusy) return;
    const items = getSelectedModelSnapshots();
    if (lastModelLibraryReadFailed) {
        alert(getModelLibraryReadFailureMessage());
        return;
    }
    if (!items.length) {
        alert("Select at least one model first.");
        return;
    }
    if (!pdfEngineReady()) {
        alert("The PDF engine could not be loaded. Please reload the page and try again.");
        return;
    }

    pdfExportBusy = true;
    updateModelSelectionToolbar();
    let frame = null;
    const solved = [];
    const failed = [];
    let hasExpandedCaseReports = false;

    // Keep the user's current work so it can be put back afterwards.
    const snap = {
        data: JSON.parse(JSON.stringify(getPersistedInputModel())),
        loadCase: activeLoadCaseId,
        momentCase: activeMomentCaseId,
        status: analysisState.status,
    };
    try {
```

```
    for (let i = 0; i < items.length; i++) {
        const item = items[i];
        const casePlan = chooseBatchCaseCombinations(item.data);
        if (!casePlan.ok) {
            failed.push({ name: item.name, message: casePlan.message });
            continue;
        }
        if (item.data?.analysisMode === "grid" && casePlan.choices.length > 1) {
            hasExpandedCaseReports = true;
        }

        for (let j = 0; j < casePlan.choices.length; j++) {
            const caseChoice = casePlan.choices[j];
            const caseDescription = `Load Case ${caseChoice.loadCaseId || "None"} / Torsion Case ${caseChoice.momentCaseId || "None"}`;
            const caseProgress = casePlan.choices.length > 1 ? ` — ${caseDescription}` : "";
            await pdfShowProgress(`Solving model ${i + 1} of ${items.length}: ${item.name}${caseProgress}…`);
            const r = solveModelSilently(item.data, caseChoice);
            if (r.ok) {
                solved.push({
                    modelKey: item.id || item.name,
                    name: item.name,
                    sections: r.sections,
                    caseLabel: r.caseLabel,
                });
            } else {
                failed.push({
                    name: item.name,
                    message: casePlan.choices.length > 1 ? `${caseDescription}: ${r.message}` : r.message,
                });
            }
        }
    }
} finally {
    // Restore the model that was open before the batch started.
    const origAlert = window.alert;
    const origConfirm = window.confirm;
    window.alert = () => {};
    window.confirm = () => true;
    try {
        applyPersistedInputModel(JSON.parse(JSON.stringify(snap.data)));
        migrateUnitSystem();
        ensureBuiltInSupportTemplates();
        activeLoadCaseId = snap.loadCase;
        activeMomentCaseId = snap.momentCase;
        refreshAll();
        if (snap.status === "valid") {
            runAnalysisCore();
        } else {
            analysisState.status = snap.status;
            const rc = document.getElementById("resultsContent");
            if (rc) rc.style.display = "none";
            updateAnalysisStatusUI();
        }
    } catch (err) {
        console.error(err);
    } finally {
        window.alert = origAlert;
        window.confirm = origConfirm;
    }
}

if (failed.length) {
    const list = failed.map((f) => `• ${f.name}: ${f.message}`).join("\n");
    if (!solved.length) {
        pdfHideProgress();
        pdfExportBusy = false;
        updateModelSelectionToolbar();
        alert(`No PDF was created. These models could not be solved:\n\n${list}`);
        return;
    }
    pdfHideProgress();
    const goOn = confirm(
        `These models could not be solved:\n\n${list}\n\nExport the PDF for the ${solved.length} model(s) that solved successfully?`,
    );
    if (!goOn) {
        pdfExportBusy = false;
        updateModelSelectionToolbar();
        return;
    }
}

try {
    await pdfShowProgress("Preparing report…");
    frame = pdfCreateFrame();
    await frame.stylesReady;
    const doc = frame.contentDocument;
    const win = frame.contentWindow;
```

```
        const root = doc.getElementById("pdfRoot");
        if (doc.fonts && doc.fonts.ready) await doc.fonts.ready;

        const jpegPages = [];
        const dateText = new Date().toLocaleDateString();
        for (let k = 0; k < solved.length; k++) {
            const multi = solved.length > 1;
            const reportPrefix = multi
                ? (hasExpandedCaseReports ? `Report ${k + 1} of ${solved.length}` : `Model ${k + 1} of ${solved.length}`)
                : "";
            const meta = {
                modelName: solved[k].name,
                caseLabel: solved[k].caseLabel,
                dateText,
                sectionPrefix: multi ? `${reportPrefix} · ` : "",
                progressPrefix: multi ? `${reportPrefix} — ` : "",
            };
            await pdfRenderSectionsToPages({ frame, doc, win, root }, solved[k].sections, meta, jpegPages);
        }

        await pdfShowProgress("Creating PDF file…");
        const title = solved.length === 1
            ? `${solved[0].name} - Analysis Report`
            : (hasExpandedCaseReports
                ? `Analysis Reports (${solved.length}) - Analysis Report`
                : `Selected Models (${solved.length}) - Analysis Report`);
        const blob = pdfBuildFile(jpegPages, title);
        pdfDownloadBlob(blob, `${sanitizeFileName(title)}.pdf`);
    } catch (err) {
        console.error(err);
        alert("PDF export failed. Check the browser console for details.");
    } finally {
        if (frame) frame.remove();
        pdfHideProgress();
        pdfExportBusy = false;
        updateModelSelectionToolbar();
    }
}
```

22. modules/load-case/load-case.js

```
/* ========================================================================
 * load-case.js — Load Case page module
 * Load Case page — load case definitions.
 * Contains this page's markup (injected into .container) and its logic.
 * ======================================================================== */

/* ---- Markup ---- */
mountModuleHTML(".container", "beforeend", `
<!-- LOAD CASES -->
<section class="section" id="loadCases">
    <div class="card">
        <div class="card-header">
            <div><h3>Create Load Case</h3><p>Create reusable load case definitions.</p></div>
        </div>
        <div class="card-body">
            <form id="loadCaseForm">
                <div class="grid grid-3">
                    <div class="field"><label for="loadCaseId">Load Case ID</label><input id="loadCaseId" placeholder="LC1" value="LC1"></div>
                    <div class="field"><label for="loadCaseType">Type</label><select id="loadCaseType" onchange="renderSelfWeightMemberInputs()"><option value="Dead">Dead</option><option
value="Live">Live</option><option value="Self Weight">Self Weight</option></select></div>
                    <div class="field"><label for="loadCaseDescription">Description</label><input id="loadCaseDescription" placeholder="Floor dead load" value="Dead Load"></div>
                </div>
                <div id="selfWeightCaseFields" class="restraint-box" style="display:none; margin-top:14px;">
                    <div class="restraint-title">Self Weight by Element</div>
                    <p style="color:#64748b; margin:0 0 12px; font-size:0.9em;">Enter each element's downward distributed self-weight individually (kN/m). Use 0 when that element has no
self-weight.</p>
                    <div id="selfWeightMemberInputs" class="grid grid-2"></div>
                </div>
                <div class="form-footer">
                    <button id="loadCaseSubmitBtn" type="submit" class="btn btn-primary">Create Load Case</button>
                </div>
            </form>
        </div>
    </div>

    <div class="card">
        <div class="card-header"><h3>Load Cases</h3></div>
        <div class="card-body">
            <div class="table-wrap">
                <table>
                    <thead><tr><th>Load Case ID</th><th>Type</th><th>Description</th><th>Self Weight by Element</th><th>Actions</th></tr></thead>
                    <tbody id="loadCaseTable"></tbody>
                </table>
            </div>
        </div>
    </div>
</section>
`);

/* ---- Logic ---- */
// ---- Load Cases ----
let editingLoadCaseId = null;

function createLoadCase(data) {
    model.loadCases.push(data);
    markAnalysisInvalidated();
}

function updateLoadCase(existingId, data) {
    const loadCase = model.loadCases.find((lc) => lc.id === existingId);
    if (!loadCase) return false;
    const oldId = loadCase.id;
    Object.assign(loadCase, data);
    if (data.type !== "Self Weight") delete loadCase.memberSelfWeights;
    if (oldId !== data.id) {
        model.loads.forEach((load) => { if (load.loadCaseId === oldId) load.loadCaseId = data.id; });
        if (activeLoadCaseId === oldId) activeLoadCaseId = data.id;
    }
    markAnalysisInvalidated();
    return true;
}

document.getElementById("loadCaseForm").addEventListener("submit", (e) => {
    e.preventDefault();
    const id = val("loadCaseId");
    if (!id) return alert("Load Case ID is required.");
    if (hasId(model.loadCases, id, editingLoadCaseId)) return alert("Load Case ID already exists.");
    const type = val("loadCaseType");
    const description = val("loadCaseDescription");
    if (!type) return alert("Load Case Type is required.");
    if (!description) return alert("Load Case Description is required.");
    const memberSelfWeights = Object.create(null);
    if (type === "Self Weight") {
        for (const element of model.elements) {
            const input = Array.from(document.querySelectorAll("#selfWeightMemberInputs [data-self-weight-element]"))
```

```
                .find((item) => String(item.dataset.selfWeightElement) === String(element.id));
            const raw = input ? input.value.trim() : "";
            const weight = raw === "" ? NaN : Number(raw);
            if (!Number.isFinite(weight) || weight < 0) {
                return alert(`Self-weight for element '${element.id}' must be a finite number greater than or equal to 0 kN/m.`);
            }
            memberSelfWeights[String(element.id)] = weight;
        }
    }
    const data = type === "Self Weight"
        ? { id, type, description, memberSelfWeights }
        : { id, type, description };
    if (editingLoadCaseId !== null) {
        const updated = updateLoadCase(editingLoadCaseId, data);
        if (!updated) { resetLoadCaseForm(); return; }
    } else {
        createLoadCase(data);
    }
    resetLoadCaseForm();
    document.getElementById("loadCaseId").value = getNextId(id);
    refreshAll();
});
function resetLoadCaseForm() {
    document.getElementById("loadCaseForm").reset();
    editingLoadCaseId = null;
    document.getElementById("loadCaseSubmitBtn").textContent = "Create Load Case";
    renderSelfWeightMemberInputs(false);
}
function editLoadCase(encodedId) {
    const id = decodeURIComponent(encodedId);
    const loadCase = model.loadCases.find((lc) => lc.id === id);
    if (!loadCase) return;
    editingLoadCaseId = loadCase.id;
    document.getElementById("loadCaseId").value = loadCase.id;
    document.getElementById("loadCaseType").value = loadCase.type;
    document.getElementById("loadCaseDescription").value = loadCase.description || "";
    document.getElementById("loadCaseSubmitBtn").textContent = "Update Load Case";
    renderSelfWeightMemberInputs(false);
}
function renderSelfWeightMemberInputs(preserveLiveValues = true) {
    const panel = document.getElementById("selfWeightCaseFields");
    const target = document.getElementById("selfWeightMemberInputs");
    const typeSelect = document.getElementById("loadCaseType");
    if (!panel || !target || !typeSelect) return;

    if (typeSelect.value !== "Self Weight") {
        panel.style.display = "none";
        return;
    }
    panel.style.display = "block";

    const liveValues = {};
    if (preserveLiveValues) {
        target.querySelectorAll("[data-self-weight-element]").forEach((input) => {
            liveValues[String(input.dataset.selfWeightElement)] = input.value;
        });
    }
    const editingCase = model.loadCases.find((item) => String(item.id) === String(editingLoadCaseId));
    const savedValues = editingCase?.memberSelfWeights && typeof editingCase.memberSelfWeights === "object"
        ? editingCase.memberSelfWeights : {};

    if (!model.elements.length) {
        target.innerHTML = '<p style="color:#b45309; margin:0;">Create an element first; its self-weight field will appear here.</p>';
        return;
    }

    target.innerHTML = model.elements.map((element) => {
        const id = String(element.id);
        const value = Object.prototype.hasOwnProperty.call(liveValues, id)
            ? liveValues[id]
            : (Object.prototype.hasOwnProperty.call(savedValues, id) ? savedValues[id] : "");
        return '<div class="field"><label>Element ' + escapeHtml(id) + '</label>' +
            '<div class="input-wrap has-unit"><input type="text" inputmode="decimal" data-self-weight-element="' + escapeHtml(id) +
            '" aria-label="Self-weight for element ' + escapeHtml(id) + '" value="' + escapeHtml(value) + '">' +
            '<span class="input-unit">kN/m</span></div></div>';
    }).join("");
}

function renderLoadCases() {
    const tbody = document.getElementById("loadCaseTable");
    if (!model.loadCases.length) {
        renderSelfWeightMemberInputs();
        return showEmpty("loadCaseTable", 5);
    }
    renderSelfWeightMemberInputs();
    tbody.innerHTML = model.loadCases.map((loadCase) => {
        const weights = loadCase.memberSelfWeights && typeof loadCase.memberSelfWeights === "object"
```

```
            ? loadCase.memberSelfWeights : {};
        const weightSummary = loadCase.type === "Self Weight"
            ? (model.elements.length
                ? model.elements.map((element) => {
                    const elementId = String(element.id);
                    const isSet = Object.prototype.hasOwnProperty.call(weights, elementId) &&
                        weights[elementId] !== undefined && weights[elementId] !== null && weights[elementId] !== "" &&
                        Number.isFinite(Number(weights[elementId])) && Number(weights[elementId]) >= 0;
                    return escapeHtml(elementId) + ": " + (isSet ? formatNum(Number(weights[elementId])) : '<strong style="color:#b45309;">Not set</strong>');
                }).join("; ") + " kN/m"
                : "No elements")
            : "—";
        return `<tr data-sort-collection="loadCases" data-sort-id="${escapeHtml(encodeInlineArgument(loadCase.id))}"><td>${sortHandle(loadCase.id)}${escapeHtml(loadCase.id)}</td><td><span
class="badge badge-blue">${escapeHtml(loadCase.type)}</span></td><td>${escapeHtml(loadCase.description)}</td><td>${weightSummary}</td><td><div class="table-actions"><button type="button"
class="icon-btn" aria-label="Edit load case ${escapeHtml(loadCase.id)}" title="Edit load case ${escapeHtml(loadCase.id)}" onclick="editLoadCase('${encodeInlineArgument(loadCase.id)}')">✎</
button><button type="button" class="icon-btn" aria-label="Delete load case ${escapeHtml(loadCase.id)}" title="Delete load case ${escapeHtml(loadCase.id)}" onclick="deleteItem('loadCases','${
encodeInlineArgument(loadCase.id)}','id')">×</button></div></td></tr>`;
    }).join("");
}
```

23. modules/load/load.js

```
/* ========================================================================
 * load.js — Load page module
 * Load page — nodal / member loads input and table.
 * Contains this page's markup (injected into .container) and its logic.
 * ======================================================================== */

/* ---- Markup ---- */
mountModuleHTML(".container", "beforeend", `
<!-- LOADS -->
<section class="section" id="loads">
    <div class="card">
        <div class="card-header">
            <div><h3>Create Load</h3><p>Create a reusable nodal or element force definition.</p></div>
        </div>
        <div class="card-body">
            <form id="loadForm">
                <div class="grid grid-3">
                    <div class="field"><label for="loadId">Load ID</label><input id="loadId" placeholder="L1" value="L1"></div>
                    <div class="field"><label for="loadCaseSelect">Load Case ID</label><select id="loadCaseSelect"><option value="">Select load case</option></select></div>
                    <div class="field"><label for="loadType">Load Type</label><select id="loadType" onchange="toggleLoadTarget()"><option value="Nodal Load">Nodal Load</option><option
value="Element Point Load">Element Point Load</option><option value="Element UDL">Element UDL</option><option class="plane-only-load-type" value="Uniform Temperature Change">Uniform
Temperature Change</option><option class="plane-only-load-type" value="Temperature Gradient">Temperature Gradient</option></select></div>
                </div>
                <div class="restraint-box" id="loadAxisBox" style="margin-top:15px;">
                    <div class="restraint-title">Load Axis & Direction</div>
                    <div class="grid grid-2">
                        <div class="field"><label for="loadAxisSystem">Axis System</label><select id="loadAxisSystem" onchange="toggleLoadAngle()"><option value="Global">Global</
option><option value="Local">Local (Elements only)</option><option value="Angled">Angled (from Global X)</option></select></div>
                        <div class="field" id="loadAngleField" style="display:none;"><label for="loadAngle">Angle (θ)</label><div class="input-wrap has-unit"><input id="loadAngle"
type="text" inputmode="decimal" placeholder="0" value="0"><span class="input-unit">deg</span></div><div class="length-chip-row scientific-shortcut-row"><button type="button"
class="scientific-chip" onclick="insertScientificExponent('loadAngle')">e</button></div></div>
                    </div>
                </div>
                <div id="nodalLoadTarget" class="restraint-box" style="margin-top:15px;">
                    <div class="restraint-title">Nodal Load Target</div>
                    <div class="grid grid-2">
                        <div class="field"><label for="loadNode">Node ID</label><select id="loadNode"><option value="">Select node</option></select></div>
                    </div>
                </div>
                <div id="elementLoadTarget" class="restraint-box" style="margin-top:15px; display:none;">
                    <div class="restraint-title">Element Load Target</div>
                    <div class="grid grid-3">
                        <div class="field"><label for="loadElement">Member / Element ID</label><select id="loadElement"><option value="">Select element</option></select></div>
                        <div class="field" id="loadDistanceField" style="display:none;"><label for="loadDistance" id="lblLoadDist">Start Distance (a)</label><div class="input-wrap has-
unit"><input id="loadDistance" type="text" placeholder="0 (or e.g. L/2)" value="0"><span class="input-unit">m</span></div><div class="length-chip-row"><button type="button" class="length-
chip" onclick="setLoadDistanceShortcut('L/4')">L/4</button><button type="button" class="length-chip" onclick="setLoadDistanceShortcut('L/3')">L/3</button><button type="button" class="length-
chip" onclick="setLoadDistanceShortcut('L/2')">L/2</button><button type="button" class="length-chip" onclick="setLoadDistanceShortcut('3L/4')">3L/4</button><button type="button"
class="length-chip" onclick="setLoadDistanceShortcut('L')">L</button><button type="button" class="scientific-chip" onclick="insertScientificExponent('loadDistance')">e</button></div></div>
                        <div class="field" id="loadEndDistanceField" style="display:none;"><label for="loadDistanceEnd">End Distance (b)</label><div class="input-wrap has-unit"><input
id="loadDistanceEnd" type="text" placeholder="L (or e.g. 3L/4)" value=""><span class="input-unit">m</span></div><div class="length-chip-row"><button type="button" class="length-chip"
onclick="setLoadDistanceEndShortcut('L/4')">L/4</button><button type="button" class="length-chip" onclick="setLoadDistanceEndShortcut('L/3')">L/3</button><button type="button" class="length-
chip" onclick="setLoadDistanceEndShortcut('L/2')">L/2</button><button type="button" class="length-chip" onclick="setLoadDistanceEndShortcut('3L/4')">3L/4</button><button type="button"
class="length-chip" onclick="setLoadDistanceEndShortcut('L')">L</button><button type="button" class="scientific-chip" onclick="insertScientificExponent('loadDistanceEnd')">e</button></div></
div>
                    </div>
                </div>
                <div class="restraint-box" style="margin-top:15px;">
                    <div class="restraint-title" id="loadComponentsTitle">Force Components</div>
                    <p id="gridLoadSignNote" style="display:none; color:#64748b; margin:0 0 12px; font-size:0.9em;">Grid Z load sign: + outside the page / upward; – inside the page /
downward.</p>
                    <div id="loadForceComponents" class="grid grid-3">
                        <div id="loadFxField" class="field"><label for="loadFx" id="lblLoadFx">F1 / Fx / Px</label><div class="input-wrap has-unit"><input id="loadFx" type="text"
inputmode="decimal" placeholder="0" value="0"><span id="unitLoadFx" class="input-unit">kN</span></div><div class="length-chip-row scientific-shortcut-row"><button type="button"
class="scientific-chip" onclick="insertScientificExponent('loadFx')">e</button></div></div>
                        <div id="loadFyField" class="field"><label for="loadFy" id="lblLoadFy">F2 / Fy / Py</label><div class="input-wrap has-unit"><input id="loadFy" type="text"
inputmode="decimal" placeholder="0" value="0"><span id="unitLoadFy" class="input-unit">kN</span></div><div class="length-chip-row scientific-shortcut-row"><button type="button"
class="scientific-chip" onclick="insertScientificExponent('loadFy')">e</button></div></div>
                        <div id="loadFzField" class="field"><label for="loadFz" id="lblLoadFz">Fz (Ignore)</label><div class="input-wrap has-unit"><input id="loadFz" type="text"
inputmode="decimal" placeholder="0" value="0"><span class="input-unit">kN</span></div><div class="length-chip-row scientific-shortcut-row"><button type="button" class="scientific-chip"
onclick="insertScientificExponent('loadFz')">e</button></div></div>
                    </div>
                    <div id="loadEndMagnitudes" class="grid grid-3" style="display:none; margin-top:14px;">
                        <div id="loadFxEndField" class="field"><label for="loadFxEnd" id="lblLoadFxEnd">End F1 / Fx / Px</label><div class="input-wrap has-unit"><input id="loadFxEnd"
type="text" inputmode="decimal" placeholder="0" value=""><span class="input-unit">kN/m</span></div><div class="length-chip-row scientific-shortcut-row"><button type="button"
class="scientific-chip" onclick="insertScientificExponent('loadFxEnd')">e</button></div></div>
                        <div id="loadFyEndField" class="field"><label for="loadFyEnd" id="lblLoadFyEnd">End F2 / Fy / Py</label><div class="input-wrap has-unit"><input id="loadFyEnd"
type="text" inputmode="decimal" placeholder="-10" value=""><span class="input-unit">kN/m</span></div><div class="length-chip-row scientific-shortcut-row"><button type="button"
class="scientific-chip" onclick="insertScientificExponent('loadFyEnd')">e</button></div></div>
                        <div id="loadFzEndField" class="field"><label for="loadFzEnd" id="lblLoadFzEnd">End Fz (Ignore)</label><div class="input-wrap has-unit"><input id="loadFzEnd"
type="text" inputmode="decimal" placeholder="0" value=""><span class="input-unit">kN/m</span></div><div class="length-chip-row scientific-shortcut-row"><button type="button"
class="scientific-chip" onclick="insertScientificExponent('loadFzEnd')">e</button></div></div>
                    </div>
                </div>
                <div class="restraint-box" id="loadTemperatureFields" style="display:none; margin-top:15px;">
                    <div class="restraint-title">Temperature Input</div>
```

```
                    <div id="temperatureDeltaField" class="field">
                        <label for="loadTemperatureDelta">Uniform Temperature Change (ΔT)</label>
                        <div class="input-wrap has-unit"><input id="loadTemperatureDelta" type="text" inputmode="decimal" placeholder="e.g. 30" value="0"><span class="input-unit">°C</span></
div>
                        <p style="color:#64748b; margin:8px 0 0; font-size:0.9em;">Free expansion: ΔL = LαΔT. If restrained, the axial thermal force magnitude is EAαΔT.</p>
                    </div>
                    <div id="temperatureGradientFields" class="grid grid-2" style="display:none;">
                        <div class="field"><label for="loadTemperatureT1">Face 1 Temperature Change (T1)</label><div class="input-wrap has-unit"><input id="loadTemperatureT1" type="text"
inputmode="decimal" placeholder="e.g. 30" value="0"><span class="input-unit">°C</span></div></div>
                        <div class="field"><label for="loadTemperatureT2">Face 2 Temperature Change (T2)</label><div class="input-wrap has-unit"><input id="loadTemperatureT2" type="text"
inputmode="decimal" placeholder="e.g. 10" value="0"><span class="input-unit">°C</span></div></div>
                    </div>
                    <p style="color:#64748b; margin:10px 0 0; font-size:0.9em;">The selected element template supplies α. Temperature gradients also use its section depth d. Face
temperatures are changes from the reference temperature.</p>
                </div>
                <div class="form-footer">
                    <button type="button" class="btn btn-secondary" onclick="resetLoadForm()">Reset</button>
                    <button id="loadSubmitBtn" type="submit" class="btn btn-primary">Create Load</button>
                </div>
            </form>
        </div>
    </div>

    <div class="card">
        <div class="card-header"><h3>Loads</h3></div>
        <div class="card-body">
            <div class="table-wrap">
                <table>
                    <thead id="loadTableHead"><tr><th>Load ID</th><th>Load Case</th><th>Type</th><th>Axis</th><th>Target</th><th>Distance (m)</th><th>F1 / Fx (kN or kN/m)</th><th>F2 / Fy (
kN or kN/m)</th><th>Fz (kN or kN/m)</th><th>Actions</th></tr></thead>
                    <tbody id="loadTable"></tbody>
                </table>
            </div>
        </div>
    </div>
</section>
`);

/* ---- Logic ---- */
// ---- Loads ----
function toggleLoadTarget() {
    const type = val("loadType");
    const isThermal = type === "Uniform Temperature Change" || type === "Temperature Gradient";
    const isElementPoint = type === "Element Point Load";
    const isUDL = type === "Element UDL";
    document.getElementById("nodalLoadTarget").style.display = type === "Nodal Load" ? "block" : "none";
    document.getElementById("elementLoadTarget").style.display = isElementPoint || isUDL || isThermal ? "block" : "none";
    document.getElementById("loadDistanceField").style.display = isElementPoint || isUDL ? "flex" : "none";
    document.getElementById("loadEndDistanceField").style.display = isUDL ? "flex" : "none";
    document.getElementById("loadEndMagnitudes").style.display = isUDL ? "grid" : "none";
    document.getElementById("lblLoadDist").innerText = isUDL ? "Start Distance (a)" : "Distance from Start Node (a)";
    const temperatureFields = document.getElementById("loadTemperatureFields");
    if (temperatureFields) temperatureFields.style.display = isThermal ? "block" : "none";
    const deltaField = document.getElementById("temperatureDeltaField");
    if (deltaField) deltaField.style.display = type === "Uniform Temperature Change" ? "block" : "none";
    const gradientFields = document.getElementById("temperatureGradientFields");
    if (gradientFields) gradientFields.style.display = type === "Temperature Gradient" ? "grid" : "none";
    syncLoadInputModeUI();
    toggleLoadAngle();
    if (typeof syncAnalysisModeUI === "function") syncAnalysisModeUI();
}

// Changes only the Grid load-input presentation. Stored BTF fields and the
// shared solver's internal load types remain compatible with existing models.
function syncLoadInputModeUI() {
    const gridMode = typeof isGridAnalysisMode === "function" && isGridAnalysisMode();
    const typeSelect = document.getElementById("loadType");
    const pointOption = typeSelect && Array.from(typeSelect.options).find((option) => option.value === "Element Point Load");
    if (pointOption) pointOption.textContent = gridMode ? "Point Load" : "Element Point Load";

    const setDisplay = (id, value) => {
        const el = document.getElementById(id);
        if (el) el.style.display = value;
    };
    const type = typeSelect ? typeSelect.value : "Nodal Load";
    const isThermal = type === "Uniform Temperature Change" || type === "Temperature Gradient";
    typeSelect?.querySelectorAll(".plane-only-load-type").forEach((option) => {
        option.hidden = gridMode;
        option.disabled = gridMode;
    });
    setDisplay("loadAxisBox", gridMode || isThermal ? "none" : "block");
    setDisplay("loadForceComponents", isThermal ? "none" : "grid");
    setDisplay("loadFxField", gridMode || isThermal ? "none" : "block");
    setDisplay("loadFyField", gridMode || isThermal ? "none" : "block");
    setDisplay("loadFxEndField", gridMode || isThermal ? "none" : "block");
    setDisplay("loadFyEndField", gridMode || isThermal ? "none" : "block");
    setDisplay("loadFzField", isThermal ? "none" : "block");
```

```
    setDisplay("loadFzEndField", isThermal ? "none" : "block");
    setDisplay("gridLoadSignNote", gridMode ? "block" : "none");

    const forceGrid = document.getElementById("loadForceComponents");
    if (forceGrid) forceGrid.style.gridTemplateColumns = gridMode ? "minmax(0, 1fr)" : "";
    const endGrid = document.getElementById("loadEndMagnitudes");
    if (endGrid) endGrid.style.gridTemplateColumns = gridMode ? "minmax(0, 1fr)" : "";

    const title = document.getElementById("loadComponentsTitle");
    if (title) title.textContent = gridMode ? "Grid Load Magnitude (Fz)" : "Force Components";
    const isUDL = type === "Element UDL";
    const fx = document.getElementById("lblLoadFx");
    const fy = document.getElementById("lblLoadFy");
    const fz = document.getElementById("lblLoadFz");
    const fxEnd = document.getElementById("lblLoadFxEnd");
    const fyEnd = document.getElementById("lblLoadFyEnd");
    const fzEnd = document.getElementById("lblLoadFzEnd");
    if (fx) fx.textContent = isUDL ? "Start F1" : "F1 / Fx / Px";
    if (fy) fy.textContent = isUDL ? "Start F2" : "F2 / Fy / Py";
    if (fz) fz.textContent = gridMode ? (isUDL ? "Start Fz" : "Fz") : (isUDL ? "Start F3" : "Fz (Ignore)");
    if (fxEnd) fxEnd.textContent = "End F1 / Fx / Px";
    if (fyEnd) fyEnd.textContent = "End F2 / Fy / Py";
    if (fzEnd) fzEnd.textContent = gridMode ? "End Fz" : "End Fz (Ignore)";
    const ux = document.getElementById("unitLoadFx");
    const uy = document.getElementById("unitLoadFy");
    if (ux) ux.textContent = isUDL ? "kN/m" : "kN";
    if (uy) uy.textContent = isUDL ? "kN/m" : "kN";
    const uz = document.getElementById("unitLoadFz");
    if (uz) uz.textContent = isUDL ? "kN/m" : "kN";
    const fzEndUnit = document.querySelector("#loadFzEndField .input-unit");
    if (fzEndUnit) fzEndUnit.textContent = "kN/m";

    const targetTitle = document.querySelector("#nodalLoadTarget .restraint-title");
    if (targetTitle) targetTitle.textContent = gridMode ? "Nodal Load Target" : "Nodal Load Target";
    const elementTitle = document.querySelector("#elementLoadTarget .restraint-title");
    if (elementTitle) elementTitle.textContent = gridMode ? "Grid Element Load Target" : "Element Load Target";
}

function toggleLoadAngle() {
    const axisSelect = document.getElementById("loadAxisSystem");
    const type = val("loadType");
    if (typeof isGridAnalysisMode === "function" && isGridAnalysisMode()) {
        document.getElementById("loadAngleField").style.display = "none";
        return;
    }
    const axis = val("loadAxisSystem");
    document.getElementById("loadAngleField").style.display = axis === "Angled" ? "flex" : "none";
    if (axis === "Local" && type === "Nodal Load") {
        alert("Local axis is only available for Element loads. Reverting to Global.");
        document.getElementById("loadAxisSystem").value = "Global";
    }
}

// Fills the Distance (a) field with a fractional-length formula (e.g. "L/2")
// so mobile users aren't forced to type "L" on a numeric-only keypad.
function setLoadDistanceShortcut(formula) {
    document.getElementById("loadDistance").value = formula;
}

// Same idea as setLoadDistanceShortcut, but for the UDL "End Distance (b)"
// field so it can also be filled with a fractional-length formula (e.g. "L/2").
function setLoadDistanceEndShortcut(formula) {
    document.getElementById("loadDistanceEnd").value = formula;
}

let editingLoadId = null;

function createLoad(data) {
    model.loads.push(data);
    markAnalysisInvalidated();
}

function updateLoad(existingId, data) {
    const load = model.loads.find((l) => l.id === existingId);
    if (!load) return false;
    // Remove any prior thermal-input fields when a load is edited to a
    // different type, so switching load types cannot preserve stale values.
    ["temperatureDelta", "temperatureT1", "temperatureT2"].forEach((key) => delete load[key]);
    Object.assign(load, data);
    markAnalysisInvalidated();
    return true;
}

document.getElementById("loadForm").addEventListener("submit", (e) => {
    e.preventDefault();
    const id = val("loadId");
    if (!id) return alert("Load ID is required.");
```

```
if (hasId(model.loads, id, editingLoadId)) return alert("Load ID already exists.");
const loadCaseId = val("loadCaseSelect");
const caseError = requireReference(model.loadCases, loadCaseId, "Load Case");
if (caseError) return alert(caseError);
const type = val("loadType");
const gridMode = typeof isGridAnalysisMode === "function" && isGridAnalysisMode();
const temperatureLoad = type === "Uniform Temperature Change" || type === "Temperature Gradient";
if (temperatureLoad) {
    if (gridMode) return alert("Temperature loads are supported only in Beam / Truss / Frame models.");
    const elementId = val("loadElement");
    let error = requireReference(model.elements, elementId, "Load Element");
    if (error) return alert(error);
    const element = model.elements.find((item) => item.id === elementId);
    const template = model.elementTemplates.find((item) => item.id === element.templateId);
    const alpha = Number(template?.alpha);
    if (!Number.isFinite(alpha) || alpha <= 0) return alert("Enter a positive thermal expansion coefficient α in the element template first.");
    let data;
    if (type === "Uniform Temperature Change") {
        const rawDelta = val("loadTemperatureDelta");
        if (rawDelta === "") return alert("Uniform temperature change ΔT is required.");
        const temperatureDelta = num("loadTemperatureDelta");
        error = requireFinite(temperatureDelta, "Uniform Temperature Change ΔT");
        if (error) return alert(error);
        if (!Number.isFinite(Number(template?.A)) || Number(template.A) <= 0) {
            return alert("Uniform axial thermal effects require a positive cross-sectional area A on the element template.");
        }
        data = { id, loadCaseId, type, elementId, temperatureDelta, axisSystem: "Global", angle: 0 };
    } else {
        const rawT1 = val("loadTemperatureT1"), rawT2 = val("loadTemperatureT2");
        if (rawT1 === "" || rawT2 === "") return alert("Both face temperature changes T1 and T2 are required.");
        const temperatureT1 = num("loadTemperatureT1"), temperatureT2 = num("loadTemperatureT2");
        error = requireFinite(temperatureT1, "Face 1 Temperature Change T1") || requireFinite(temperatureT2, "Face 2 Temperature Change T2");
        if (error) return alert(error);
        if (!(Number(template?.I) > 0)) return alert("Temperature gradients require a member with I > 0; a truss has I = 0 in this solver.");
        if (!(Number(template?.thermalDepth) > 0)) return alert("Enter a positive section depth d in the element template before applying a temperature gradient.");
        data = { id, loadCaseId, type, elementId, temperatureT1, temperatureT2, axisSystem: "Global", angle: 0 };
    }
    if (editingLoadId !== null) {
        const updated = updateLoad(editingLoadId, data);
        if (!updated) { resetLoadForm(); return; }
    } else {
        createLoad(data);
    }
    resetLoadForm();
    document.getElementById("loadId").value = getNextId(id);
    refreshAll();
    return;
}
if (!["Nodal Load", "Element Point Load", "Element UDL"].includes(type)) return alert("Invalid Load Type.");
const axisSystem = gridMode ? "Global" : val("loadAxisSystem");
// An angle is meaningful only for an Angled load. Do not turn invalid
// text into zero through `|| 0`; validate the raw parsed value whenever
// the selected axis system actually uses it.
const angle = gridMode || axisSystem !== "Angled" ? 0 : num("loadAngle");
let error = !gridMode && axisSystem === "Angled" ? requireFinite(angle, "Load Angle") : null;
if (error) return alert(error);
const Fx = gridMode ? 0 : num("loadFx"), Fy = gridMode ? 0 : num("loadFy"), Fz = gridMode ? num("loadFz") : 0;
const componentsToValidate = gridMode ? [[Fz, "Fz"]] : [[Fx, "Fx / F1"], [Fy, "Fy / F2"]];
for (const [value, label] of componentsToValidate) { error = requireFinite(value, label); if (error) return alert(error); }
let FzEnd = gridMode ? Fz : 0;
let nodeId = "", elementId = "", distance = 0, distanceEnd = "";
if (type === "Nodal Load") {
    nodeId = val("loadNode");
    error = requireReference(model.nodes, nodeId, "Load Node");
    if (error) return alert(error);
    if (axisSystem === "Local") return alert("Local axis is only available for Element loads.");
} else {
    elementId = val("loadElement");
    error = requireReference(model.elements, elementId, "Load Element");
    if (error) return alert(error);
    const distResult = parseLengthInputStrict(val("loadDistance"), elementId, "Start Distance (a)");
    if (!distResult.ok) return alert(distResult.message);
    distance = distResult.value;
    if (distance < 0 || distance > distResult.length) return alert("Start Distance (a) must satisfy 0 ≤ a ≤ L.");
    if (type === "Element UDL") {
        const rawEnd = val("loadDistanceEnd");
        const endResult = rawEnd === "" ? { ok: true, value: distResult.length, length: distResult.length } : parseLengthInputStrict(rawEnd, elementId, "End Distance (b)");
        if (!endResult.ok) return alert(endResult.message);
        distanceEnd = endResult.value;
        if (distanceEnd < 0 || distanceEnd > distResult.length) return alert("End Distance (b) must satisfy 0 ≤ b ≤ L.");
        if (!(distance < distanceEnd)) return alert("UDL interval must satisfy 0 ≤ a < b ≤ L.");
        const FxEnd = gridMode ? 0 : (val("loadFxEnd") === "" ? Fx : num("loadFxEnd"));
        const FyEnd = gridMode ? 0 : (val("loadFyEnd") === "" ? Fy : num("loadFyEnd"));
        FzEnd = gridMode ? (val("loadFzEnd") === "" ? Fz : num("loadFzEnd")) : 0;
        const endComponentsToValidate = gridMode ? [[FzEnd, "End Fz"]] : [[FxEnd, "End Fx / F1"], [FyEnd, "End Fy / F2"]];
        for (const [value, label] of endComponentsToValidate) { error = requireFinite(value, label); if (error) return alert(error); }
    }
```

```
    }
    const data = { id, loadCaseId, type, nodeId, elementId, axisSystem, angle, distance, distanceEnd, Fx, Fy, Fz,
        FxEnd: type === "Element UDL" ? (gridMode ? 0 : (val("loadFxEnd") === "" ? Fx : num("loadFxEnd"))) : "",
        FyEnd: type === "Element UDL" ? (gridMode ? 0 : (val("loadFyEnd") === "" ? Fy : num("loadFyEnd"))) : "",
        FzEnd: type === "Element UDL" ? FzEnd : "" };
    if (editingLoadId !== null) {
        const updated = updateLoad(editingLoadId, data);
        if (!updated) { resetLoadForm(); return; }
    } else {
        createLoad(data);
    }
    resetLoadForm();
    document.getElementById("loadId").value = getNextId(id);
    refreshAll();
});
function resetLoadForm() {
    document.getElementById("loadForm").reset();
    editingLoadId = null;
    document.getElementById("loadSubmitBtn").textContent = "Create Load";
    toggleLoadTarget();
}
function editLoad(encodedId) {
    const id = decodeURIComponent(encodedId);
    const load = model.loads.find((l) => l.id === id);
    if (!load) return;
    editingLoadId = load.id;
    document.getElementById("loadId").value = load.id;
    document.getElementById("loadCaseSelect").value = load.loadCaseId || "";
    document.getElementById("loadType").value = load.type;
    document.getElementById("loadAxisSystem").value = load.axisSystem || "Global";
    document.getElementById("loadAngle").value = load.angle ?? 0;
    document.getElementById("loadNode").value = load.nodeId || "";
    document.getElementById("loadElement").value = load.elementId || "";
    document.getElementById("loadDistance").value = load.distance ?? 0;
    document.getElementById("loadDistanceEnd").value = load.distanceEnd ?? "";
    document.getElementById("loadFx").value = load.Fx ?? 0;
    document.getElementById("loadFy").value = load.Fy ?? 0;
    document.getElementById("loadFz").value = load.Fz ?? 0;
    document.getElementById("loadFxEnd").value = load.FxEnd ?? "";
    document.getElementById("loadFyEnd").value = load.FyEnd ?? "";
    document.getElementById("loadFzEnd").value = load.FzEnd ?? "";
    document.getElementById("loadTemperatureDelta").value = load.temperatureDelta ?? 0;
    document.getElementById("loadTemperatureT1").value = load.temperatureT1 ?? 0;
    document.getElementById("loadTemperatureT2").value = load.temperatureT2 ?? 0;
    document.getElementById("loadSubmitBtn").textContent = "Update Load";
    // Re-display the controls associated with the restored type. This is
    // required for thermal edits so the T1/T2 fields are visible again.
    toggleLoadTarget();
}
function renderLoads() {
    const tbody = document.getElementById("loadTable");
    const gridMode = typeof isGridAnalysisMode === "function" && isGridAnalysisMode();
    const thead = document.getElementById("loadTableHead");
    if (thead) {
        thead.innerHTML = gridMode
            ? `<tr><th>Load ID</th><th>Load Case</th><th>Type</th><th>Target</th><th>Distance (m)</th><th>Fz (kN or kN/m)</th><th>Actions</th></tr>`
            : `<tr><th>Load ID</th><th>Load Case</th><th>Type</th><th>Axis</th><th>Target</th><th>Distance (m)</th><th>F1 / Fx (kN or kN/m)</th><th>F2 / Fy (kN or kN/m)</th><th>Fz (kN or kN/
m)</th><th>Actions</th></tr>`;
    }
    if (!model.loads.length) return showEmpty("loadTable", gridMode ? 7 : 10);
    tbody.innerHTML = model.loads.map((l) => {
        const target = l.type === "Nodal Load" ? `Node ${escapeHtml(l.nodeId)}` : `Element ${escapeHtml(l.elementId)}`;
        let dist = "-";
        if (l.type === "Element Point Load") dist = `${escapeHtml(l.distance)} m`;
        else if (l.type === "Element UDL") dist = `${escapeHtml(l.distance)}m to ${l.distanceEnd !== undefined && l.distanceEnd !== "" ? escapeHtml(l.distanceEnd) : "L"}m`;
        const typeLabel = gridMode && l.type === "Element Point Load" ? "Point Load" : l.type;
        if (gridMode) {
            const magnitude = l.type === "Element UDL"
                ? `${escapeHtml(l.Fz)} to ${l.FzEnd !== undefined && l.FzEnd !== "" ? escapeHtml(l.FzEnd) : escapeHtml(l.Fz)}`
                : escapeHtml(l.Fz);
            return `<tr data-sort-collection="loads" data-sort-id="${escapeHtml(encodeInlineArgument(l.id))}"><td>${sortHandle(l.id)}${escapeHtml(l.id)}</td><td>${escapeHtml(l.loadCaseId)}</
td><td><span class="badge badge-blue">${escapeHtml(typeLabel)}</span></td><td>${target}</td><td>${dist}</td><td>${magnitude}</td><td><div class="table-actions"><button type="button"
class="icon-btn" aria-label="Edit load ${escapeHtml(l.id)}" title="Edit load ${escapeHtml(l.id)}" onclick="editLoad('${encodeInlineArgument(l.id)}')">✎</button><button type="button"
class="icon-btn" aria-label="Delete load ${escapeHtml(l.id)}" title="Delete load ${escapeHtml(l.id)}" onclick="deleteItem('loads','${encodeInlineArgument(l.id)}','id')">×</button></div></
td></tr>`;
        }
        if (l.type === "Uniform Temperature Change" || l.type === "Temperature Gradient") {
            const thermalText = l.type === "Uniform Temperature Change"
                ? `ΔT = ${escapeHtml(l.temperatureDelta)} °C`
                : `T1 = ${escapeHtml(l.temperatureT1)} °C; T2 = ${escapeHtml(l.temperatureT2)} °C`;
            return `<tr data-sort-collection="loads" data-sort-id="${escapeHtml(encodeInlineArgument(l.id))}"><td>${sortHandle(l.id)}${escapeHtml(l.id)}</td><td>${escapeHtml(l.loadCaseId)}</
td><td><span class="badge badge-blue">${escapeHtml(l.type)}</span></td><td>—</td><td>${target}</td><td>—</td><td colspan="3">${thermalText}</td><td><div class="table-actions"><button
type="button" class="icon-btn" aria-label="Edit load ${escapeHtml(l.id)}" title="Edit load ${escapeHtml(l.id)}" onclick="editLoad('${encodeInlineArgument(l.id)}')">✎</button><button
type="button" class="icon-btn" aria-label="Delete load ${escapeHtml(l.id)}" title="Delete load ${escapeHtml(l.id)}" onclick="deleteItem('loads','${encodeInlineArgument(l.id)}','id')">×</
button></div></td></tr>`;
        }
        const fxStr = l.type === "Element UDL" ? `${escapeHtml(l.Fx)} to ${l.FxEnd !== undefined && l.FxEnd !== "" ? escapeHtml(l.FxEnd) : escapeHtml(l.Fx)}` : escapeHtml(l.Fx);
```

```
        const fyStr = l.type === "Element UDL" ? `${escapeHtml(l.Fy)} to ${l.FyEnd !== undefined && l.FyEnd !== "" ? escapeHtml(l.FyEnd) : escapeHtml(l.Fy)}` : escapeHtml(l.Fy);
        const fzStr = l.type === "Element UDL" ? `${escapeHtml(l.Fz)} to ${l.FzEnd !== undefined && l.FzEnd !== "" ? escapeHtml(l.FzEnd) : escapeHtml(l.Fz)}` : escapeHtml(l.Fz);
        const axisStr = l.axisSystem === "Angled" ? `Angled (${l.angle}°)` : escapeHtml(l.axisSystem || "Global");
        return `<tr data-sort-collection="loads" data-sort-id="${escapeHtml(encodeInlineArgument(l.id))}"><td>${sortHandle(l.id)}${escapeHtml(l.id)}</td><td>${escapeHtml(l.loadCaseId)}</
td><td><span class="badge badge-blue">${escapeHtml(l.type)}</span></td><td><span class="badge badge-gray">${axisStr}</span></td><td>${target}</td><td>${dist}</td><td>${fxStr}</td><td>${
fyStr}</td><td>${fzStr}</td><td><div class="table-actions"><button type="button" class="icon-btn" aria-label="Edit load ${escapeHtml(l.id)}" title="Edit load ${escapeHtml(l.id)}"
onclick="editLoad('${encodeInlineArgument(l.id)}')">✎</button><button type="button" class="icon-btn" aria-label="Delete load ${escapeHtml(l.id)}" title="Delete load ${escapeHtml(l.id)}"
onclick="deleteItem('loads','${encodeInlineArgument(l.id)}','id')">×</button></div></td></tr>`;
    }).join("");
}
```

24. modules/moment-case/moment-case.js

```
/* ========================================================================
 * moment-case.js — Moment Case page module
 * Moment Case page — moment case definitions.
 * Contains this page's markup (injected into .container) and its logic.
 * ======================================================================== */

/* ---- Markup ---- */
mountModuleHTML(".container", "beforeend", `
<!-- MOMENT CASES -->
<section class="section" id="momentCases">
    <div class="card">
        <div class="card-header">
            <div><h3 id="momentCaseFormTitle">Create Moment Case</h3><p id="momentCaseFormDescription">Create reusable moment case definitions.</p></div>
        </div>
        <div class="card-body">
            <form id="momentCaseForm">
                <div class="grid grid-3">
                    <div class="field"><label for="momentCaseId" id="momentCaseIdLabel">Moment Case ID</label><input id="momentCaseId" placeholder="MC1" value="MC1"></div>
                    <div class="field"><label for="momentCaseType">Type</label><select id="momentCaseType"><option value="Dead">Dead</option><option value="Live">Live</option><option
value="Self Weight">Self Weight</option></select></div>
                    <div class="field"><label for="momentCaseDescription">Description</label><input id="momentCaseDescription" placeholder="Beam live moment" value="Dead Moment"></div>
                </div>
                <div class="form-footer"><button id="momentCaseSubmitBtn" type="submit" class="btn btn-primary">Create Moment Case</button></div>
            </form>
        </div>
    </div>

    <div class="card">
        <div class="card-header"><h3 id="momentCaseTableTitle">Moment Cases</h3></div>
        <div class="card-body">
            <div class="table-wrap">
                <table>
                    <thead id="momentCaseTableHead"><tr><th>Moment Case ID</th><th>Type</th><th>Description</th><th>Actions</th></tr></thead>
                    <tbody id="momentCaseTable"></tbody>
                </table>
            </div>
        </div>
    </div>
</section>
`);


// Keep Grid torsion case wording isolated from the BTF moment terminology.
function syncMomentCaseInputModeUI() {
    const gridMode = typeof isGridAnalysisMode === "function" && isGridAnalysisMode();
    const setText = (id, text) => { const el = document.getElementById(id); if (el) el.textContent = text; };
    setText("momentCaseFormTitle", gridMode ? "Create Torsion Case" : "Create Moment Case");
    setText("momentCaseFormDescription", gridMode ? "Create reusable torsion case definitions." : "Create reusable moment case definitions.");
    setText("momentCaseIdLabel", gridMode ? "Torsion Case ID" : "Moment Case ID");
    const description = document.getElementById("momentCaseDescription");
    if (description) {
        description.placeholder = gridMode ? "Grid torsion case" : "Beam live moment";
        if (description.value === "Dead Moment" || description.value === "Dead Torsion") {
            description.value = gridMode ? "Dead Torsion" : "Dead Moment";
        }
    }
    setText("momentCaseSubmitBtn", editingMomentCaseId !== null ? (gridMode ? "Update Torsion Case" : "Update Moment Case") : (gridMode ? "Create Torsion Case" : "Create Moment Case"));
    setText("momentCaseTableTitle", gridMode ? "Torsion Cases" : "Moment Cases");
    const head = document.getElementById("momentCaseTableHead");
    if (head) head.innerHTML = gridMode
        ? `<tr><th>Torsion Case ID</th><th>Type</th><th>Description</th><th>Actions</th></tr>`
        : `<tr><th>Moment Case ID</th><th>Type</th><th>Description</th><th>Actions</th></tr>`;
}

/* ---- Logic ---- */
// ---- Moment Cases ----
let editingMomentCaseId = null;

function createMomentCase(data) {
    model.momentCases.push(data);
    markAnalysisInvalidated();
}

function updateMomentCase(existingId, data) {
    const momentCase = model.momentCases.find((mc) => mc.id === existingId);
    if (!momentCase) return false;
    const oldId = momentCase.id;
    Object.assign(momentCase, data);
    if (oldId !== data.id) {
        model.moments.forEach((moment) => { if (moment.momentCaseId === oldId) moment.momentCaseId = data.id; });
        if (activeMomentCaseId === oldId) activeMomentCaseId = data.id;
    }
    markAnalysisInvalidated();
    return true;
}
```

```
document.getElementById("momentCaseForm").addEventListener("submit", (e) => {
    e.preventDefault();
    const gridMode = typeof isGridAnalysisMode === "function" && isGridAnalysisMode();
    const id = val("momentCaseId");
    if (!id) return alert(gridMode ? "Torsion Case ID is required." : "Moment Case ID is required.");
    if (hasId(model.momentCases, id, editingMomentCaseId)) return alert(gridMode ? "Torsion Case ID already exists." : "Moment Case ID already exists.");
    const type = val("momentCaseType");
    const description = val("momentCaseDescription");
    if (!type) return alert(gridMode ? "Torsion Case Type is required." : "Moment Case Type is required.");
    if (!description) return alert(gridMode ? "Torsion Case Description is required." : "Moment Case Description is required.");
    const data = { id, type, description };
    if (editingMomentCaseId !== null) {
        const updated = updateMomentCase(editingMomentCaseId, data);
        if (!updated) { resetMomentCaseForm(); return; }
    } else {
        createMomentCase(data);
    }
    resetMomentCaseForm();
    document.getElementById("momentCaseId").value = getNextId(id);
    refreshAll();
});
function resetMomentCaseForm() {
    document.getElementById("momentCaseForm").reset();
    editingMomentCaseId = null;
    document.getElementById("momentCaseSubmitBtn").textContent = "Create Moment Case";
    syncMomentCaseInputModeUI();
}
function editMomentCase(encodedId) {
    const id = decodeURIComponent(encodedId);
    const momentCase = model.momentCases.find((mc) => mc.id === id);
    if (!momentCase) return;
    editingMomentCaseId = momentCase.id;
    document.getElementById("momentCaseId").value = momentCase.id;
    document.getElementById("momentCaseType").value = momentCase.type;
    document.getElementById("momentCaseDescription").value = momentCase.description || "";
    document.getElementById("momentCaseSubmitBtn").textContent = typeof isGridAnalysisMode === "function" && isGridAnalysisMode() ? "Update Torsion Case" : "Update Moment Case";
    syncMomentCaseInputModeUI();
}
function renderMomentCases() {
    syncMomentCaseInputModeUI();
    const tbody = document.getElementById("momentCaseTable");
    if (!model.momentCases.length) return showEmpty("momentCaseTable", 4);
    tbody.innerHTML = model.momentCases.map((m) =>
        `<tr data-sort-collection="momentCases" data-sort-id="${escapeHtml(encodeInlineArgument(m.id))}"><td>${sortHandle(m.id)}${escapeHtml(m.id)}</td><td><span class="badge badge-blue">${
escapeHtml(m.type)}</span></td><td>${escapeHtml(m.description)}</td><td><div class="table-actions"><button type="button" class="icon-btn" aria-label="Edit moment case ${escapeHtml(m.id)}"
title="Edit moment case ${escapeHtml(m.id)}" onclick="editMomentCase('${encodeInlineArgument(m.id)}')">\</button><button type="button" class="icon-btn" aria-label="Delete moment case ${
escapeHtml(m.id)}" title="Delete moment case ${escapeHtml(m.id)}" onclick="deleteItem('momentCases','${encodeInlineArgument(m.id)}','id')">×</button></div></td></tr>`
    ).join("");
}
```

25. modules/moment/moment.js

```
/* ========================================================================
 * moment.js — Moment page module
 * Moment page — nodal / member moments input and table.
 * Contains this page's markup (injected into .container) and its logic.
 * ======================================================================== */

/* ---- Markup ---- */
mountModuleHTML(".container", "beforeend", `
<!-- MOMENTS -->
<section class="section" id="moments">
    <div class="card">
        <div class="card-header">
            <div><h3 id="momentFormTitle">Create Moment</h3><p id="momentFormDescription">Create a reusable nodal or element moment definition.</p></div>
        </div>
        <div class="card-body">
            <form id="momentForm">
                <div class="grid grid-3">
                    <div class="field"><label for="momentId" id="momentIdLabel">Moment ID</label><input id="momentId" placeholder="M1" value="M1"></div>
                    <div class="field"><label for="momentCaseSelect" id="momentCaseSelectLabel">Moment Case ID</label><select id="momentCaseSelect"><option value="">Select moment case</
option></select></div>
                    <div class="field"><label for="momentType" id="momentTypeLabel">Moment Type</label><select id="momentType" onchange="toggleMomentTarget()"><option value="Nodal
Moment">Nodal Moment</option><option value="Element Moment">Element Moment</option></select></div>
                </div>
                <div id="nodalMomentTarget" class="restraint-box" style="margin-top:15px;">
                    <div class="restraint-title" id="nodalMomentTargetTitle">Nodal Moment Target</div>
                    <div class="grid grid-2">
                        <div class="field"><label for="momentNode">Node ID</label><select id="momentNode"><option value="">Select node</option></select></div>
                    </div>
                </div>
                <div id="elementMomentTarget" class="restraint-box" style="margin-top:15px; display:none;">
                    <div class="restraint-title" id="elementMomentTargetTitle">Element Moment Target</div>
                    <div class="grid grid-2">
                        <div class="field"><label for="momentElement">Member / Element ID</label><select id="momentElement"><option value="">Select element</option></select></div>
                        <div class="field" id="momentDistanceField" style="display:none;"><label for="momentDistance">Distance from Start Node (a)</label><div class="input-wrap has-
unit"><input id="momentDistance" type="text" placeholder="0 (or e.g. L/2)" value="0"><span class="input-unit">m</span></div><div class="length-chip-row"><button type="button" class="length-
chip" onclick="setMomentDistanceShortcut('L/4')">L/4</button><button type="button" class="length-chip" onclick="setMomentDistanceShortcut('L/3')">L/3</button><button type="button"
class="length-chip" onclick="setMomentDistanceShortcut('L/2')">L/2</button><button type="button" class="length-chip" onclick="setMomentDistanceShortcut('3L/4')">3L/4</button><button
type="button" class="length-chip" onclick="setMomentDistanceShortcut('L')">L</button><button type="button" class="scientific-chip" onclick="insertScientificExponent('momentDistance')">e</
button></div></div>
                    </div>
                </div>
                <div class="restraint-box" style="margin-top:15px;">
                    <div class="restraint-title" id="momentComponentsTitle">Moment Components</div>
                    <p id="gridTorsionSignNote" style="display:none; color:#64748b; margin:0 0 12px; font-size:0.9em;">Grid torsion signs: My is + up / − down; Mx is + right / − left.</p>
                    <div id="momentComponentsGrid" class="grid grid-3">
                        <div id="momentXField" class="field"><label for="momentX" id="lblMomentX">Mx (Ignore)</label><div class="input-wrap has-unit"><input id="momentX" type="text"
inputmode="decimal" placeholder="0" value="0"><span class="input-unit">kN·m</span></div><div class="length-chip-row scientific-shortcut-row"><button type="button" class="scientific-chip"
onclick="insertScientificExponent('momentX')">e</button></div></div>
                        <div id="momentYField" class="field"><label for="momentY" id="lblMomentY">My (Ignore)</label><div class="input-wrap has-unit"><input id="momentY" type="text"
inputmode="decimal" placeholder="0" value="0"><span class="input-unit">kN·m</span></div><div class="length-chip-row scientific-shortcut-row"><button type="button" class="scientific-chip"
onclick="insertScientificExponent('momentY')">e</button></div></div>
                        <div id="momentZField" class="field"><label for="momentZ" id="lblMomentZ">Mz (Rotation)</label><div class="input-wrap has-unit"><input id="momentZ" type="text"
inputmode="decimal" placeholder="0" value="0"><span class="input-unit">kN·m</span></div><div class="length-chip-row scientific-shortcut-row"><button type="button" class="scientific-chip"
onclick="insertScientificExponent('momentZ')">e</button></div></div>
                    </div>
                </div>
                <div class="form-footer">
                    <button type="button" class="btn btn-secondary" onclick="resetMomentForm()">Reset</button>
                    <button id="momentSubmitBtn" type="submit" class="btn btn-primary">Create Moment</button>
                </div>
            </form>
        </div>
    </div>

    <div class="card">
        <div class="card-header"><h3 id="momentTableTitle">Moments</h3></div>
        <div class="card-body">
            <div class="table-wrap">
                <table>
                    <thead id="momentTableHead"><tr><th>Moment ID</th><th>Moment Case</th><th>Type</th><th>Target</th><th>Mx (kN·m)</th><th>My (kN·m)</th><th>Mz (kN·m)</th><th>Actions</th></
tr></thead>
                    <tbody id="momentTable"></tbody>
                </table>
            </div>
        </div>
    </div>
</section>
`);

/* ---- Logic ---- */
// ---- Moments ----
// Same idea as setLoadDistanceShortcut, but for the Element Moment
// "Distance from Start Node (a)" field.
function setMomentDistanceShortcut(formula) {
    document.getElementById("momentDistance").value = formula;
```

```
}

function toggleMomentTarget() {
    const type = val("momentType");
    document.getElementById("nodalMomentTarget").style.display =
        type === "Nodal Moment" ? "block" : "none";
    document.getElementById("elementMomentTarget").style.display =
        type === "Element Moment" ? "block" : "none";
    document.getElementById("momentDistanceField").style.display =
        type === "Element Moment" ? "flex" : "none";
    syncMomentInputModeUI();
    if (typeof syncAnalysisModeUI === "function") syncAnalysisModeUI();
}

// Grid torsions share the solver's existing Mx/My data fields and internal
// nodal/element moment types, while presenting only the two Grid DOF directions.
function syncMomentInputModeUI() {
    const gridMode = typeof isGridAnalysisMode === "function" && isGridAnalysisMode();
    const typeSelect = document.getElementById("momentType");
    if (typeSelect) {
        const nodalOption = Array.from(typeSelect.options).find((option) => option.value === "Nodal Moment");
        const elementOption = Array.from(typeSelect.options).find((option) => option.value === "Element Moment");
        if (nodalOption) nodalOption.textContent = gridMode ? "Nodal Torsion" : "Nodal Moment";
        if (elementOption) elementOption.textContent = gridMode ? "Element Torsion" : "Element Moment";
    }
    const setText = (id, text) => { const el = document.getElementById(id); if (el) el.textContent = text; };
    const setDisplay = (id, display) => { const el = document.getElementById(id); if (el) el.style.display = display; };
    setText("momentFormTitle", gridMode ? "Create Torsion" : "Create Moment");
    setText("momentFormDescription", gridMode ? "Create a reusable nodal or element torsion definition." : "Create a reusable nodal or element moment definition.");
    setText("momentIdLabel", gridMode ? "Torsion ID" : "Moment ID");
    setText("momentCaseSelectLabel", gridMode ? "Torsion Case ID" : "Moment Case ID");
    const casePlaceholder = document.querySelector("#momentCaseSelect option[value='']");
    if (casePlaceholder) casePlaceholder.textContent = gridMode ? "Select torsion case" : "Select moment case";
    setText("momentTypeLabel", gridMode ? "Torsion Type" : "Moment Type");
    setText("nodalMomentTargetTitle", gridMode ? "Nodal Torsion Target" : "Nodal Moment Target");
    setText("elementMomentTargetTitle", gridMode ? "Element Torsion Target" : "Element Moment Target");
    setText("momentComponentsTitle", gridMode ? "Torsion Magnitudes" : "Moment Components");
    setText("momentSubmitBtn", editingMomentId !== null ? (gridMode ? "Update Torsion" : "Update Moment") : (gridMode ? "Create Torsion" : "Create Moment"));
    setText("gridTorsionSignNote", "Grid torsion signs: My is + up / − down; Mx is + right / − left.");
    setDisplay("gridTorsionSignNote", gridMode ? "block" : "none");
    setDisplay("momentZField", gridMode ? "none" : "block");
    const componentGrid = document.getElementById("momentComponentsGrid");
    if (componentGrid) componentGrid.style.gridTemplateColumns = gridMode ? "repeat(2, minmax(0, 1fr))" : "";
    setText("lblMomentX", gridMode ? "Mx — Right / Left (+ right, − left)" : "Mx (Ignore)");
    setText("lblMomentY", gridMode ? "My — Up / Down (+ up, − down)" : "My (Ignore)");
    setText("lblMomentZ", "Mz (Rotation)");
    setText("momentTableTitle", gridMode ? "Torsions" : "Moments");
    const tableHead = document.getElementById("momentTableHead");
    if (tableHead) tableHead.innerHTML = gridMode
        ? `<tr><th>Torsion ID</th><th>Torsion Case</th><th>Type</th><th>Target</th><th>Distance (m)</th><th>Mx — Right/Left (kN·m)</th><th>My — Up/Down (kN·m)</th><th>Actions</th></tr>`
        : `<tr><th>Moment ID</th><th>Moment Case</th><th>Type</th><th>Target</th><th>Mx (kN·m)</th><th>My (kN·m)</th><th>Mz (kN·m)</th><th>Actions</th></tr>`;
}

let editingMomentId = null;

function createMoment(data) {
    model.moments.push(data);
    markAnalysisInvalidated();
}

function updateMoment(existingId, data) {
    const moment = model.moments.find((m) => m.id === existingId);
    if (!moment) return false;
    Object.assign(moment, data);
    markAnalysisInvalidated();
    return true;
}

document.getElementById("momentForm").addEventListener("submit", (e) => {
    e.preventDefault();
    const gridMode = typeof isGridAnalysisMode === "function" && isGridAnalysisMode();
    const id = val("momentId");
    if (!id) return alert(gridMode ? "Torsion ID is required." : "Moment ID is required.");
    if (hasId(model.moments, id, editingMomentId)) return alert(gridMode ? "Torsion ID already exists." : "Moment ID already exists.");
    const momentCaseId = val("momentCaseSelect");
    const caseError = requireReference(model.momentCases, momentCaseId, gridMode ? "Torsion Case" : "Moment Case");
    if (caseError) return alert(caseError);
    const type = val("momentType");
    if (!["Nodal Moment", "Element Moment"].includes(type)) return alert(gridMode ? "Invalid torsion type." : "Invalid Moment Type.");
    const Mx = gridMode ? num("momentX") : 0, My = gridMode ? num("momentY") : 0, Mz = gridMode ? 0 : num("momentZ");
    const componentsToValidate = gridMode ? [[Mx, "Mx (Right/Left torsion)"], [My, "My (Up/Down torsion)"]] : [[Mz, "Mz"]];
    for (const [value, label] of componentsToValidate) { const error = requireFinite(value, label); if (error) return alert(error); }
    let nodeId = "", elementId = "", distance = 0;
    if (type === "Nodal Moment") {
        nodeId = val("momentNode");
        const error = requireReference(model.nodes, nodeId, gridMode ? "Torsion Node" : "Moment Node");
        if (error) return alert(error);
```

```
    } else {
        elementId = val("momentElement");
        const error = requireReference(model.elements, elementId, gridMode ? "Torsion Element" : "Moment Element");
        if (error) return alert(error);
        const distResult = parseLengthInputStrict(val("momentDistance"), elementId, gridMode ? "Torsion Distance (a)" : "Moment Distance (a)");
        if (!distResult.ok) return alert(distResult.message);
        distance = distResult.value;
        if (distance < 0 || distance > distResult.length) return alert(`${gridMode ? "Torsion" : "Moment"} Distance (a) must satisfy 0 ≤ a ≤ L.`);
    }
    const data = { id, momentCaseId, type, nodeId, elementId, distance, Mx, My, Mz };
    if (editingMomentId !== null) {
        const updated = updateMoment(editingMomentId, data);
        if (!updated) { resetMomentForm(); return; }
    } else {
        createMoment(data);
    }
    resetMomentForm();
    document.getElementById("momentId").value = getNextId(id);
    refreshAll();
});
function resetMomentForm() {
    document.getElementById("momentForm").reset();
    editingMomentId = null;
    document.getElementById("momentSubmitBtn").textContent = "Create Moment";
    toggleMomentTarget();
}
function editMoment(encodedId) {
    const id = decodeURIComponent(encodedId);
    const moment = model.moments.find((m) => m.id === id);
    if (!moment) return;
    editingMomentId = moment.id;
    document.getElementById("momentId").value = moment.id;
    document.getElementById("momentCaseSelect").value = moment.momentCaseId || "";
    document.getElementById("momentType").value = moment.type;
    toggleMomentTarget();
    document.getElementById("momentNode").value = moment.nodeId || "";
    document.getElementById("momentElement").value = moment.elementId || "";
    document.getElementById("momentDistance").value = moment.distance ?? 0;
    document.getElementById("momentX").value = moment.Mx ?? 0;
    document.getElementById("momentY").value = moment.My ?? 0;
    document.getElementById("momentZ").value = moment.Mz ?? 0;
    document.getElementById("momentSubmitBtn").textContent = typeof isGridAnalysisMode === "function" && isGridAnalysisMode() ? "Update Torsion" : "Update Moment";
    syncMomentInputModeUI();
}
function renderMoments() {
    const tbody = document.getElementById("momentTable");
    const gridMode = typeof isGridAnalysisMode === "function" && isGridAnalysisMode();
    syncMomentInputModeUI();
    if (!model.moments.length) return showEmpty("momentTable", 8);
    tbody.innerHTML = model.moments.map((m) => {
        const targetName = m.type === "Nodal Moment" ? `Node ${escapeHtml(m.nodeId)}` : `Element ${escapeHtml(m.elementId)}`;
        const dist = m.type === "Element Moment" ? `${escapeHtml(m.distance)} m` : "-";
        if (gridMode) {
            const typeLabel = m.type === "Nodal Moment" ? "Nodal Torsion" : "Element Torsion";
            return `<tr data-sort-collection="moments" data-sort-id="${escapeHtml(encodeInlineArgument(m.id))}"><td>${sortHandle(m.id)}${escapeHtml(m.id)}</td><td>${escapeHtml(m.
momentCaseId)}</td><td><span class="badge badge-orange">${typeLabel}</span></td><td>${targetName}</td><td>${dist}</td><td>${escapeHtml(m.Mx ?? 0)}</td><td>${escapeHtml(m.My ?? 0)}</
td><td><div class="table-actions"><button type="button" class="icon-btn" aria-label="Edit moment ${escapeHtml(m.id)}" title="Edit moment ${escapeHtml(m.id)}" onclick="editMoment('${
encodeInlineArgument(m.id)}')">✎</button><button type="button" class="icon-btn" aria-label="Delete moment ${escapeHtml(m.id)}" title="Delete moment ${escapeHtml(m.id)}" onclick="deleteItem(
'moments','${encodeInlineArgument(m.id)}','id')">×</button></div></td></tr>`;
        }
        const target = `${targetName} / dist: ${dist}`;
        return `<tr data-sort-collection="moments" data-sort-id="${escapeHtml(encodeInlineArgument(m.id))}"><td>${sortHandle(m.id)}${escapeHtml(m.id)}</td><td>${escapeHtml(m.momentCaseId)}</
td><td><span class="badge badge-orange">${escapeHtml(m.type)}</span></td><td>${target}</td><td>${escapeHtml(m.Mx)}</td><td>${escapeHtml(m.My)}</td><td>${escapeHtml(m.Mz)}</td><td><div
class="table-actions"><button type="button" class="icon-btn" aria-label="Edit moment ${escapeHtml(m.id)}" title="Edit moment ${escapeHtml(m.id)}" onclick="editMoment('${encodeInlineArgument(
m.id)}')">✎</button><button type="button" class="icon-btn" aria-label="Delete moment ${escapeHtml(m.id)}" title="Delete moment ${escapeHtml(m.id)}" onclick="deleteItem('moments','${
encodeInlineArgument(m.id)}','id')">×</button></div></td></tr>`;
    }).join("");
}
```

26. modules/nodes/nodes.js

```
/* ========================================================================
 * nodes.js — Nodes page module
 * Nodes page — node input form, node table, search.
 * Contains this page's markup (injected into .container) and its logic.
 * ======================================================================== */

/* ---- Markup ---- */
mountModuleHTML(".container", "beforeend", `
<!-- NODES -->
<section class="section" id="nodes">
    <div class="card">
        <div class="card-header">
            <div><h3>Create Node</h3><p>Define the global coordinates of a node.</p></div>
        </div>
        <div class="card-body">
            <form id="nodeForm">
                <div class="grid grid-4">
                    <div class="field"><label for="nodeId">Node ID</label><input id="nodeId" placeholder="N1" value="N1"></div>
                    <div class="field"><label for="nodeX">X Coordinate</label><div class="input-wrap has-unit"><input id="nodeX" type="text" inputmode="decimal" placeholder="0"
value="0"><span class="input-unit">m</span></div><div class="length-chip-row scientific-shortcut-row"><button type="button" class="scientific-chip" onclick="insertScientificExponent(
'nodeX')">e</button></div></div>
                    <div class="field"><label for="nodeY">Y Coordinate</label><div class="input-wrap has-unit"><input id="nodeY" type="text" inputmode="decimal" placeholder="0"
value="0"><span class="input-unit">m</span></div><div class="length-chip-row scientific-shortcut-row"><button type="button" class="scientific-chip" onclick="insertScientificExponent(
'nodeY')">e</button></div></div>
                    <div class="field plane-only-property"><label for="nodeZ">Z Coordinate (unused in 2D analysis)</label><div class="input-wrap has-unit"><input id="nodeZ" type="text"
inputmode="decimal" placeholder="0" value="0"><span class="input-unit">m</span></div><div class="length-chip-row scientific-shortcut-row"><button type="button" class="scientific-chip"
onclick="insertScientificExponent('nodeZ')">e</button></div></div>
                </div>
                <div class="form-footer">
                    <button type="button" class="btn btn-secondary" onclick="resetNodeForm()">Reset</button>
                    <button type="submit" class="btn btn-primary" id="nodeSubmitBtn">Create Node</button>
                </div>
            </form>
        </div>
    </div>

    <div class="card">
        <div class="card-header">
            <div><h3>Node List</h3><p>All nodes currently present in the model.</p></div>
            <input id="nodeSearch" class="search-box" placeholder="Search Node..." oninput="renderNodes()">
        </div>
        <div class="card-body">
            <div class="table-wrap">
                <table>
                    <thead><tr><th>Node ID</th><th>X (m)</th><th>Y (m)</th><th class="plane-only-property">Z (m)</th><th>Actions</th></tr></thead>
                    <tbody id="nodeTable"></tbody>
                </table>
            </div>
        </div>
    </div>
</section>
`);

/* ---- Logic ---- */
// ---- Nodes ----
let editingNodeId = null;

function createNode(id, x, y, z) {
    model.nodes.push({ id, x, y, z });
    markAnalysisInvalidated();
}

function findCollapsedConnectedElement(nodeId, x, y) {
    return model.elements.find((element) => {
        let otherNodeId = null;
        if (String(element.startNode) === String(nodeId)) otherNodeId = element.endNode;
        else if (String(element.endNode) === String(nodeId)) otherNodeId = element.startNode;
        if (otherNodeId === null) return false;
        const otherNode = model.nodes.find((candidate) => String(candidate.id) === String(otherNodeId));
        return !!otherNode && Number(otherNode.x) === Number(x) && Number(otherNode.y) === Number(y);
    }) || null;
}

function findNodeMoveDistanceViolation(nodeId, x, y) {
    for (const element of model.elements) {
        const touchesStart = String(element.startNode) === String(nodeId);
        const touchesEnd = String(element.endNode) === String(nodeId);
        if (!touchesStart && !touchesEnd) continue;

        const startNode = model.nodes.find((node) => String(node.id) === String(element.startNode));
        const endNode = model.nodes.find((node) => String(node.id) === String(element.endNode));
        if (!startNode || !endNode) continue;
        const startX = touchesStart ? Number(x) : Number(startNode.x);
        const startY = touchesStart ? Number(y) : Number(startNode.y);
        const endX = touchesEnd ? Number(x) : Number(endNode.x);
```

```
        const endY = touchesEnd ? Number(y) : Number(endNode.y);
        const length = Math.hypot(endX - startX, endY - startY);
        if (!Number.isFinite(length) || length <= 0) continue; // zero length has its own validation message

        for (const load of model.loads) {
            if (String(load.elementId) !== String(element.id) ||
                !["Element Point Load", "Element Load", "Element UDL"].includes(load.type)) continue;
            const a = Number(load.distance);
            if (!Number.isFinite(a) || a < 0 || a > length) {
                return `Moving this node would make load "${load.id}" start distance invalid: it must satisfy 0 ≤ a ≤ the new element length (${length} m).`;
            }
            if (load.type === "Element UDL") {
                const endDistance = load.distanceEnd === undefined || load.distanceEnd === null || load.distanceEnd === ""
                    ? length : Number(load.distanceEnd);
                if (!Number.isFinite(endDistance) || endDistance < 0 || endDistance > length || !(a < endDistance)) {
                    return `Moving this node would make UDL "${load.id}" invalid: its interval must satisfy 0 ≤ a < b ≤ the new element length (${length} m).`;
                }
            }
        }
        for (const moment of model.moments) {
            if (String(moment.elementId) !== String(element.id) || moment.type !== "Element Moment") continue;
            const distance = Number(moment.distance);
            if (!Number.isFinite(distance) || distance < 0 || distance > length) {
                return `Moving this node would make moment/torsion "${moment.id}" distance invalid: it must satisfy 0 ≤ a ≤ the new element length (${length} m).`;
            }
        }
    }
    return null;
}

function updateNode(existingId, id, x, y, z) {
    const node = model.nodes.find((n) => n.id === existingId);
    if (!node) return false;
    const oldId = node.id;
    node.id = id;
    node.x = x;
    node.y = y;
    node.z = z;
    if (oldId !== id) {
        model.elements.forEach((el) => {
            if (el.startNode === oldId) el.startNode = id;
            if (el.endNode === oldId) el.endNode = id;
        });
        model.supportAssignments.forEach((sa) => { if (sa.nodeId === oldId) sa.nodeId = id; });
        model.loads.forEach((load) => { if (load.nodeId === oldId) load.nodeId = id; });
        model.moments.forEach((moment) => { if (moment.nodeId === oldId) moment.nodeId = id; });
    }
    markAnalysisInvalidated();
    return true;
}

document.getElementById("nodeForm").addEventListener("submit", (e) => {
    e.preventDefault();
    const id = val("nodeId");
    const x = num("nodeX"), y = num("nodeY"), z = isGridAnalysisMode() ? 0 : num("nodeZ");
    const validationError = validateNodeInput(id, x, y, z, editingNodeId);
    if (validationError) return alert(validationError);

    if (editingNodeId !== null) {
        const collapsedElement = findCollapsedConnectedElement(editingNodeId, x, y);
        if (collapsedElement) {
            return alert(`Node edit would make element "${collapsedElement.id}" zero-length. Move the node to a different X/Y coordinate before saving.`);
        }
        const distanceViolation = findNodeMoveDistanceViolation(editingNodeId, x, y);
        if (distanceViolation) return alert(distanceViolation);
        const updated = updateNode(editingNodeId, id, x, y, z);
        if (!updated) {
            editingNodeId = null;
            resetNodeForm();
            return;
        }
        const nextId = getNextId(id);
        resetNodeForm();
        document.getElementById("nodeId").value = nextId;
    } else {
        createNode(id, x, y, z);
        resetNodeForm();
        document.getElementById("nodeId").value = getNextId(id);
    }

    refreshAll();
});

function resetNodeForm() {
    document.getElementById("nodeForm").reset();
    editingNodeId = null;
    document.getElementById("nodeSubmitBtn").textContent = "Create Node";
```

```
}
function renderNodes() {
    const tbody = document.getElementById("nodeTable");
    const search = val("nodeSearch").toLowerCase();
    const rows = model.nodes.filter((n) => n.id.toLowerCase().includes(search));
    if (!rows.length) return showEmpty("nodeTable", 5);
    tbody.innerHTML = rows
        .map(
            (n) =>
                `<tr data-sort-collection="nodes" data-sort-id="${escapeHtml(encodeInlineArgument(n.id))}"><td>${sortHandle(n.id)}<span class="badge badge-blue">${escapeHtml(n.id)}</span></
td><td>${escapeHtml(n.x)}</td><td>${escapeHtml(n.y)}</td><td class="plane-only-property" style="display:${isGridAnalysisMode() ? "none" : "table-cell"}">${escapeHtml(n.z ?? 0)}</td><td><div
class="table-actions"><button type="button" class="icon-btn" aria-label="Edit node ${escapeHtml(n.id)}" title="Edit node ${escapeHtml(n.id)}" onclick="editNode('${encodeInlineArgument(n.
id)}')">✎</button><button type="button" class="icon-btn" aria-label="Delete node ${escapeHtml(n.id)}" title="Delete node ${escapeHtml(n.id)}" onclick="deleteItem('nodes','${
encodeInlineArgument(n.id)}','id')">×</button></div></td></tr>`,
        )
        .join("");
}
function editNode(encodedId) {
    const id = decodeURIComponent(encodedId);
    const node = model.nodes.find((n) => n.id === id);
    if (!node) return;

    editingNodeId = node.id;
    document.getElementById("nodeId").value = node.id;
    document.getElementById("nodeX").value = node.x;
    document.getElementById("nodeY").value = node.y;
    document.getElementById("nodeZ").value = node.z;
    document.getElementById("nodeSubmitBtn").textContent = "Update Node";
}
```

27. modules/results/diagrams/diagrams.css

```
/* ============================================================================
 * diagrams.css
 * Styles for all diagrams: canvas containers, member focus, the full-screen
 * diagram modal and its navigation toolbar (+ responsive rules).
 * ============================================================================ */

/* CANVAS UI */
.canvas-container {
    text-align: center;
    overflow: hidden;
    padding: 10px;
    background: #f8fafc;
    border-radius: 8px;
    border: 1px solid #e2e8f0;
}

/* Inline diagrams are display-only. Interaction is enabled only inside
   the dedicated fullscreen diagram viewer. */
.canvas-container > svg {
    width: 100% !important;
    height: 100% !important;
    display: block;
}

/* FULL SCREEN DIAGRAM MODAL
   Built as a flex column (header row + diagram body) rather than free-floating
   absolutely-positioned elements, so the close button and mode toolbar can
   never be covered by the diagram and never overlap each other, on any
   screen size. */
.fs-modal {
    display: none;
    position: fixed;
    top: 0;
    right: 0;
    bottom: 0;
    left: 0;
    width: 100vw;
    height: 100vh;
    height: 100dvh; /* dynamic viewport height where supported, so mobile
                       browser toolbars can't hide the bottom of the modal */
    background: #f4f7fb;
    z-index: 10000;
    overflow: hidden;
}

.fs-modal.active {
    display: flex;
    flex-direction: column;
}

.fs-modal-header {
    flex: 0 0 auto;
    display: flex;
    align-items: center;
    justify-content: space-between;
    gap: 10px;
    padding: 16px 20px;
}

.fs-toolbar {
    display: flex;
    align-items: center;
    gap: 10px;
    min-width: 0;
    flex: 1 1 auto;
    max-width: calc(100% - 52px); /* always leaves room for the close button */
    background: white;
    padding: 8px 16px;
    border-radius: 8px;
    border: 1px solid var(--border);
    box-shadow: 0 4px 10px rgba(0, 0, 0, 0.1);
}

.fs-toolbar select {
    flex: 1 1 auto;
    min-width: 0;
    max-width: 100%;
    height: 32px;
    font-size: 12px;
    font-weight: bold;
    padding: 0 8px;
    border-radius: 6px;
    border: 1px solid #cbd5e1;
    outline: none;
    background: white;
```

```
    color: var(--text);
    cursor: pointer;
    overflow: hidden;
    text-overflow: ellipsis;
    white-space: nowrap;
}

.fs-close {
    flex: 0 0 auto;
    background: white;
    border: 1px solid var(--border);
    color: var(--text);
    font-size: 22px;
    width: 40px;
    height: 40px;
    border-radius: 8px;
    cursor: pointer;
    box-shadow: 0 4px 10px rgba(0, 0, 0, 0.1);
    display: flex;
    align-items: center;
    justify-content: center;
}

.fs-close:hover {
    background: #eef2f7;
}

.fs-body {
    flex: 1 1 auto;
    min-height: 0;
    position: relative;
    overflow: hidden;
    padding: 0 20px;
    padding-bottom: max(20px, env(safe-area-inset-bottom));
}

#fsSVG {
    width: 100%;
    height: 100%;
    background: white;
}

/* Keep the diagram legend fully inside the visible fullscreen area */
#fsModal .result-diagram-fixed-legend {
    max-width: calc(100% - 20px) !important;
}

@media (max-width: 800px) {

    /* FULL SCREEN MODAL — MOBILE */
    .fs-modal-header {
        padding: 12px;
        gap: 8px;
    }
    .fs-toolbar {
        padding: 7px 10px;
    }
    .fs-toolbar select {
        font-size: 12px;
    }
    .fs-close {
        width: 36px;
        height: 36px;
        font-size: 20px;
    }
    .fs-body {
        padding: 0 12px;
        padding-bottom: max(12px, env(safe-area-inset-bottom));
    }
}

@media (max-width: 500px) {

    /* FULL SCREEN MODAL — SMALL PHONES */
    .fs-modal-header {
        padding: 10px;
        gap: 6px;
    }
    .fs-toolbar {
        padding: 6px 8px;
    }
    .fs-toolbar select {
        font-size: 11px;
    }
    .fs-close {
        width: 32px;
        height: 32px;
```

```
        font-size: 18px;
    }
    .fs-body {
        padding: 0 8px;
        padding-bottom: max(8px, env(safe-area-inset-bottom));
    }
    #fsModal .result-diagram-fixed-legend {
        font-size: 9px !important;
        padding: 7px 8px !important;
        line-height: 1.5 !important;
        max-width: calc(100% - 12px) !important;
    }
}

/* ===== Click-to-front + fade-others for AFD / SFD / BMD members (no color/weight change) ===== */
.sf-member, .sf-diagram, .sf-label { cursor: pointer; }

.sf-member, .sf-diagram, .sf-label, .sf-node { transition: opacity 0.15s; }

svg.sf-focus-on .sf-dim { opacity: 0.15; }

/* ===== Fullscreen canvas navigation toolbar (Zoom / Pan / Member Highlight) ===== */
.fs-nav-tools {
    position: absolute;
    top: 10px;
    right: 28px;
    z-index: 30;
    display: flex;
    gap: 6px;
    padding: 5px;
    background: rgba(255, 255, 255, 0.97);
    border: 1px solid var(--border);
    border-radius: 8px;
    box-shadow: 0 4px 10px rgba(0, 0, 0, 0.12);
}

.fs-nav-btn {
    width: 38px;
    height: 38px;
    padding: 0;
    display: flex;
    align-items: center;
    justify-content: center;
    background: white;
    color: #475569;
    border: 1px solid #cbd5e1;
    border-radius: 6px;
    cursor: pointer;
    -webkit-tap-highlight-color: transparent;
    touch-action: manipulation;
    user-select: none;
}

.fs-nav-btn svg { width: 20px; height: 20px; pointer-events: none; }

.fs-nav-btn:hover { background: #eef2f7; }

.fs-nav-btn.active {
    background: #2563eb;
    border-color: #1d4ed8;
    color: #ffffff;
}

.fs-nav-btn.active:hover { background: #1d4ed8; }

/* Navigation cursors / touch handling only while the matching tool is active */
#fsModal.fs-nav-zoom #fsSVG svg { cursor: zoom-in; }

#fsModal.fs-nav-pan #fsSVG svg { cursor: grab; }

#fsModal.fs-nav-pan.fs-panning #fsSVG svg { cursor: grabbing; }

#fsModal.fs-nav-zoom #fsSVG svg,
#fsModal.fs-nav-pan #fsSVG svg { touch-action: none; }

@media (max-width: 768px) {
    .fs-nav-tools { right: 16px; top: 8px; gap: 5px; padding: 4px; }
    .fs-nav-btn { width: 40px; height: 40px; }
}

@media (max-width: 500px) {
    .fs-nav-tools { right: 12px; top: 6px; }
}

/* Grid load/torsion symbols in the structure model diagram are independent,
   accessible SVG controls. Hover/focus/selection draws a visible blue halo. */
.grid-diagram-symbol { cursor: pointer; outline: none; }
```

```
.grid-diagram-symbol .grid-symbol-halo { transition: stroke 0.12s, stroke-width 0.12s; }
.grid-diagram-symbol:hover .grid-symbol-halo,
.grid-diagram-symbol:focus .grid-symbol-halo,
.grid-diagram-symbol.grid-symbol-selected .grid-symbol-halo { stroke: #2563eb; stroke-width: 2.4; }
.grid-diagram-symbol.grid-symbol-selected .grid-symbol-core { stroke: #2563eb; stroke-width: 2.2; }
.grid-torsion-symbol:hover path,
.grid-torsion-symbol:focus path,
.grid-torsion-symbol.grid-symbol-selected path { stroke-width: 3.4; }
```

28. modules/results/diagrams/diagrams.js

```
/* ================================================================================
 * diagrams.js — Results part 9: diagrams
 * Everything related to drawing: live model diagram card, Reaction + Combined
 * (AFD / SFD / BMD / Displacement) diagram cards, the shared SVG diagram engine
 * (getStructureSVG + utilities), DoF diagram, member focus, pinch-zoom / pan and the
 * full-screen diagram modal.
 * ================================================================================ */

/* ---- Markup ---- */
/* Live model diagram card: first element inside .container (visible on every page) */
mountModuleHTML(".container", "afterbegin", `
<!-- LIVE DIAGRAM SVG UI (Always visible at top) -->
<div class="card" id="liveModelDiagramCard">
    <div class="card-header">
        <h3>Live Structural Model Diagram</h3>
        <p>Updates instantly as you add or edit components.</p>
    </div>
    <div class="card-body canvas-container">
        <div id="structureSVGSmall" style="width: 100%; height: 350px; background: white; border: 1px solid #cbd5e1; border-radius: 6px; overflow: hidden; position: relative;"></div>
    </div>
    <button class="btn btn-secondary" style="margin-top: 15px; width: 100%;" onclick="openFullScreen('model')">■ Open Full Structure Diagram</button>
</div>
`);

/* Reaction + Combined result diagram cards: inside #resultsContent, right after the equilibrium card */
mountModuleHTML("#resultsContent .equilibrium-card", "afterend", `
<!-- DIAGRAM SVG UI : Reaction Diagram -->
<div class="card">
    <div class="card-header">
        <div>
            <h3>Reaction Diagram</h3>
            <p>Structure showing calculated support reactions.</p>
        </div>
    </div>
    <div class="card-body canvas-container">
        <div id="reactionSVGSmall" style="width: 100%; height: 350px; background: white; border: 1px solid #cbd5e1; border-radius: 6px; overflow: hidden; position: relative;"></div>
    </div>
    <button class="btn btn-secondary" style="margin-top: 15px; width: 100%;" onclick="openFullScreen('reactions')">■ Open Full Screen Diagram</button>
</div>

<!-- DIAGRAM SVG UI : Combined Results Diagram -->
<div class="card">
    <div class="card-header">
        <div>
            <h3>Analysis Diagrams</h3>
            <p>Derived using member end actions [AM] and intermediate loads.</p>
        </div>
        <select id="resultsDiagramMode" onchange="updateResultsDiagram()" style="width: auto; height: 32px; font-size: 12px; padding: 0 10px; border-radius: 6px; border: 1px solid #cbd5e1;
outline: none; background: #f8fafc;">
            <option value="afd">AFD</option>
            <option value="sfd" selected>SFD</option>
            <option value="bmd">BMD</option>
            <option value="displacement">Displacement</option>
        </select>
    </div>
    <div class="card-body canvas-container">
        <div id="resultsSVGSmall" style="width: 100%; height: 350px; background: white; border: 1px solid #cbd5e1; border-radius: 6px; overflow: hidden; position: relative;"></div>
    </div>
    <button class="btn btn-secondary" style="margin-top: 15px; width: 100%;" onclick="openFullScreen('results')">■ Open Full Screen Diagram</button>
</div>
`);

/* Full-screen diagram modal: sibling of .app */
mountModuleHTML(".app", "afterend", `
<!-- FULL SCREEN DIAGRAM MODAL -->
<div id="fsModal" class="fs-modal">
    <div class="fs-modal-header">
        <div class="fs-toolbar">
            <select id="fsModeSelect" onchange="changeFSMode(this.value)">
                <option value="model">Structure Model</option>
                <option value="dof">DoF & Numbering Diagram</option>
                <option value="reactions">Reaction Diagram</option>
                <option value="afd">AFD</option>
                <option value="sfd">SFD</option>
                <option value="bmd">BMD</option>
                <option value="displacement">Displacement</option>
            </select>
        </div>
        <button type="button" class="fs-close" aria-label="Close full-screen diagram" title="Close full-screen diagram" onclick="closeFullScreen()">×</button>
    </div>
    <div class="fs-body">
        <div class="fs-nav-tools" id="fsNavTools" role="toolbar" aria-label="Canvas navigation controls">
            <button type="button" class="fs-nav-btn" id="fsNavZoomBtn" data-tool="zoom" aria-pressed="false" title="Zoom In / Out (Ctrl + mouse wheel, or pinch)" aria-label="Zoom In / Out
toggle">
                <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2" stroke-linecap="round" stroke-linejoin="round"><circle cx="10" cy="10" r="6"/><line x1="21"
```

```
y1="21" x2="14.5" y2="14.5"/><line x1="7.5" y1="10" x2="12.5" y2="10"/><line x1="10" y1="7.5" x2="10" y2="12.5"/></svg>
            </button>
            <button type="button" class="fs-nav-btn" id="fsNavPanBtn" data-tool="pan" aria-pressed="false" title="Pan Mode (right-click or middle-click + drag, or one-finger drag)" aria-
label="Pan Mode toggle">
                <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2" stroke-linecap="round" stroke-linejoin="round"><polyline points="5 9 2 12 5 15"/><polyline
points="9 5 12 2 15 5"/><polyline points="15 19 12 22 9 19"/><polyline points="19 9 22 12 19 15"/><line x1="2" y1="12" x2="22" y2="12"/><line x1="12" y1="2" x2="12" y2="22"/></svg>
            </button>
            <button type="button" class="fs-nav-btn" id="fsNavHighlightBtn" data-tool="highlight" aria-pressed="false" title="Member Highlight (click / tap a member)" aria-label="Member
Highlight toggle">
                <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2" stroke-linecap="round" stroke-linejoin="round"><line x1="4" y1="20" x2="20" y2="4"/><circle
cx="4" cy="20" r="2" fill="currentColor"/><circle cx="20" cy="4" r="2" fill="currentColor"/><line x1="4" y1="4" x2="9" y2="4" opacity="0.5"/><line x1="15" y1="20" x2="20" y2="20" opacity="0.
5"/></svg>
            </button>
        </div>
        <div id="fsSVG"></div>
    </div>
</div>
`);

/* ---- Logic ---- */
let inlineModelDiagramRedrawTimer = null;

function scheduleInlineModelDiagramRedraw() {
    if (inlineModelDiagramRedrawTimer !== null) {
        clearTimeout(inlineModelDiagramRedrawTimer);
    }
    inlineModelDiagramRedrawTimer = setTimeout(() => {
        inlineModelDiagramRedrawTimer = null;
        renderInlineModelDiagram();
    }, 80);
}

// ========================================================================

function showHeavyDiagramPlaceholder(containerId, title = "Large model — inline diagram paused") {
    const container = document.getElementById(containerId);
    if (!container) return;

    container.innerHTML = `
        <div style="
            height:100%;
            display:flex;
            align-items:center;
            justify-content:center;
            flex-direction:column;
            gap:8px;
            padding:20px;
            text-align:center;
            color:#64748b;
            font-family:Arial,sans-serif;
        ">
            <div style="font-size:26px;">■</div>
            <div style="font-size:14px;font-weight:800;color:#334155;">
                ${title}
            </div>
            <div style="font-size:11px;line-height:1.5;max-width:420px;">
                ${model.nodes.length} nodes detected. This inline diagram is hidden
                to keep the page responsive. Use <b>Open Full Screen Diagram</b>
                to render the complete diagram.
            </div>
        </div>`;
}

function renderInlineModelDiagram() {
    const container = document.getElementById("structureSVGSmall");
    if (!container) return;

    const nodeCount = model.nodes.length;
    const INLINE_DIAGRAM_NODE_LIMIT = 10;

    if (nodeCount > INLINE_DIAGRAM_NODE_LIMIT) {
        container.innerHTML = `
            <div style="
                height:100%;
                display:flex;
                align-items:center;
                justify-content:center;
                flex-direction:column;
                gap:8px;
                padding:20px;
                text-align:center;
                color:#64748b;
                font-family:Arial,sans-serif;
            ">
                <div style="font-size:26px;">■</div>
                <div style="font-size:14px;font-weight:800;color:#334155;">
                    Large model — inline diagram paused
```

```
                </div>
                <div style="font-size:11px;line-height:1.5;max-width:420px;">
                    ${nodeCount} nodes detected. The inline diagram is hidden
                    to keep the page responsive. Open <b>Full Structure Diagram</b>
                    to render the complete model.
                </div>
            </div>`;
        return;
    }

    drawDiagramSVG("structureSVGSmall", "model");
}

// 9. STRUCTURE DIAGRAM (SVG) RENDERING
// Draws the model / AFD / SFD / BMD / reactions diagrams as inline SVG.
// ============================================================================
function getBoundingBox() {
    let minX = Infinity,
        maxX = -Infinity,
        minY = Infinity,
        maxY = -Infinity;
    model.nodes.forEach((n) => {
        if (n.x < minX) minX = n.x;
        if (n.x > maxX) maxX = n.x;
        if (n.y < minY) minY = n.y;
        if (n.y > maxY) maxY = n.y;
    });
    return { minX, maxX, minY, maxY };
}

// Dynamic diagram baseline scaling:
// The shortest real structural member is treated as the visual baseline.
// That minimum member is forced to occupy 200 px in every proportional
// diagram that uses the shared structure renderer. Every longer member and
// every annotation drawn in the same SVG coordinate system then scales by
// exactly the same factor, keeping the complete visual system proportional.
//
// Structure Model already uses this method directly in drawDiagramSVG().
// Reactions / AFD / SFD / BMD / Displacement use it through the shared
// calculateAutoFit() path. DoF/Numbering is intentionally NOT affected:
// it has its own dedicated schematic renderer.
function getMinimumElementLength() {
    let minL = Infinity;

    model.elements.forEach((el) => {
        if (!el) return;

        const n1 = model.nodes.find((n) => n.id === el.startNode);
        const n2 = model.nodes.find((n) => n.id === el.endNode);
        if (!n1 || !n2) return;

        const dx = Number(n2.x) - Number(n1.x);
        const dy = Number(n2.y) - Number(n1.y);
        const L = Math.hypot(dx, dy);

        if (Number.isFinite(L) && L > 1e-9) {
            minL = Math.min(minL, L);
        }
    });

    return Number.isFinite(minL) ? minL : null;
}

function calculateAutoFit(width, height, mode = "model") {
    if (model.nodes.length === 0) return { scale: 100, offsetX: width / 2, offsetY: height / 2 };

    let bb = getBoundingBox();
    let dX = bb.maxX - bb.minX;
    if (dX === 0) dX = 1;
    let dY = bb.maxY - bb.minY;
    if (dY === 0) dY = 1;

    // The reaction diagram places labelled cards outside the structure's own
    // bounding box, so it needs more breathing room than the other diagram
    // modes to keep those cards from being clipped at the SVG edge.
    const paddingX = mode === "reactions" ? 130 : 80;
    const paddingY = mode === "reactions" ? 130 : 80;
    let scaleX = (width - 2 * paddingX) / dX;
    let scaleY = (height - 2 * paddingY) / dY;

    // First calculate the original fit scale. This is the reference scale
    // that previously made a 7 m member readable.
    let fitScale = Math.min(scaleX, scaleY);
    if (!Number.isFinite(fitScale) || fitScale <= 0) fitScale = 100;

    // Shared proportional baseline for result/reaction diagrams. It is a
    // target, not an absolute guarantee: when the shortest member would force
```

```
    // the full model outside the viewport, prefer the fit scale to avoid
    // clipping long members or node labels.
    const minElementLength = getMinimumElementLength();
    const baselineScale =
        minElementLength && Number.isFinite(minElementLength) && minElementLength > 0
            ? AppConfig.sharedDiagramBaselinePx / minElementLength
            : fitScale;
    let scale = Math.min(baselineScale, fitScale);

    // Keep a finite upper safety bound for malformed/extreme coordinate data.
    const MAX_DIAGRAM_SCALE = 1500;
    if (scale > MAX_DIAGRAM_SCALE) scale = MAX_DIAGRAM_SCALE;

    let midX = (bb.minX + bb.maxX) / 2;
    let midY = (bb.minY + bb.maxY) / 2;

    let offsetX = width / 2 - midX * scale;
    let offsetY = height / 2 + midY * scale;

    return { scale, offsetX, offsetY };
}

// ============================================================================
// UNIVERSAL DIAGRAM LABEL ENGINE
// Every diagram (Live/model view, AFD, SFD, BMD, Displacement, Reactions,
// DoF/Numbering) funnels ALL of its text labels through this single engine
// instead of writing <text> straight into the SVG string. Each caller just
// "queues" a label describing what it wants to show and, optionally, the
// point on the drawing it refers to (its anchor). Once a diagram has queued
// every label it needs, resolve() runs once and:
//   1. Works out each label's real on-screen footprint from its text length,
//      font size and rotation (so short and long labels are treated fairly).
//   2. Nudges any labels whose footprints overlap apart from one another
//      using an iterative separation pass, in every direction, until nothing
//      overlaps (or the iteration budget is used up on very dense models).
//   3. Clamps every label back inside the visible canvas so nothing ends up
//      drawn off-screen (a common cause of "invisible" text).
//   4. Draws every label LAST, after all lines/fills/paths, so a label can
//      never be hidden underneath a diagram curve or fill that was drawn
//      after it — and adds a soft white halo behind the text so it stays
//      legible over member lines, curves, and fills of any color.
//   5. If a label had to move noticeably away from the point it describes,
//      a thin dashed leader line is drawn back to that point so it is never
//      ambiguous which value/label belongs to which node, load, or curve.
// This is intentionally generic (addText for simple labels, addBlock for
// pre-built multi-line boxes like the reaction cards) so it works the same
// way for every diagram type without special-casing any one of them.
// ============================================================================
function createLabelLayer() {
    const items = [];

    function estimateTextWidth(text, fontSize, fontWeight) {
        const str = String(text);
        const bold = fontWeight === "bold" || fontWeight === "700" || fontWeight === "800" || fontWeight === "600";
        const factor = bold ? 0.62 : 0.56;
        return Math.max(str.length * fontSize * factor, fontSize * 0.6);
    }

    function rotatedHalfExtents(w, h, angleDeg) {
        if (!angleDeg) return { hw: w / 2, hh: h / 2 };
        const rad = (angleDeg * Math.PI) / 180;
        const cos = Math.abs(Math.cos(rad));
        const sin = Math.abs(Math.sin(rad));
        return { hw: (w * cos + h * sin) / 2, hh: (w * sin + h * cos) / 2 };
    }

    // Queue a single-line text label. (x, y) is the SVG baseline position it
    // would use in isolation; (anchorX, anchorY) is the point in the drawing
    // it describes (defaults to x, y) used only for the optional leader line.
    function addText(opts) {
        const {
            x, y, text, fontSize = 11, fontWeight = "bold", color = "#1e293b",
            textAnchor = "start", angle = 0, anchorX, anchorY, halo = true,
            leader = true, leaderColor, padding = 3, lockPosition = false, memberKey,
        } = opts;
        const w = estimateTextWidth(text, fontSize, fontWeight) + padding * 2;
        const h = fontSize * 1.3 + padding * 2;
        const ext = rotatedHalfExtents(w, h, angle);

        let cx;
        if (textAnchor === "middle") cx = x;
        else if (textAnchor === "end") cx = x - w / 2 + padding;
        else cx = x + w / 2 - padding;
        let cy = y - fontSize * 0.32;

        items.push({
            type: "text", text, fontSize, fontWeight, color, textAnchor, angle,
```

```
        halfW: ext.hw, halfH: ext.hh, boxW: w, boxH: h, cx, cy, padding,
        anchorX: anchorX !== undefined ? anchorX : x,
        anchorY: anchorY !== undefined ? anchorY : y,
        halo, leader, leaderColor: leaderColor || color, lockPosition, memberKey,
    });
}

// Queue an arbitrary pre-sized block (e.g. a multi-line reaction card)
// that knows how to render itself at a given center via render(cx, cy).
// It still takes part in overlap avoidance with every other label.
function addBlock(opts) {
    const { x, y, w, h, anchorX, anchorY, render, leader = true, leaderColor = "#94a3b8" } = opts;
    items.push({
        type: "block", boxW: w, boxH: h, halfW: w / 2, halfH: h / 2,
        cx: x, cy: y, render,
        anchorX: anchorX !== undefined ? anchorX : x,
        anchorY: anchorY !== undefined ? anchorY : y,
        leader, leaderColor,
    });
}

// Fixed, non-text obstacles (e.g. node markers) that labels must stay
// clear of. Obstacles never move; only labels are pushed away from them.
const obstacles = [];
function addObstacle(cx, cy, halfW, halfH) {
    obstacles.push({ cx, cy, halfW, halfH });
}

function resolve(bounds, options = {}) {
    const iterations = options.iterations || 300;
    const gap = options.gap ?? 3;

    for (let iter = 0; iter < iterations; iter++) {
        let moved = false;
        for (let i = 0; i < items.length && obstacles.length; i++) {
            const a = items[i];
            if (a.lockPosition) continue;
            for (let k = 0; k < obstacles.length; k++) {
                const o = obstacles[k];
                const dx = a.cx - o.cx, dy = a.cy - o.cy;
                const overlapX = a.halfW + o.halfW + gap - Math.abs(dx);
                const overlapY = a.halfH + o.halfH + gap - Math.abs(dy);
                if (overlapX > 0 && overlapY > 0) {
                    moved = true;
                    if (overlapX < overlapY) a.cx += (dx === 0 ? 1 : Math.sign(dx)) * (overlapX + 0.4);
                    else a.cy += (dy === 0 ? -1 : Math.sign(dy)) * (overlapY + 0.4);
                }
            }
        }
        for (let i = 0; i < items.length; i++) {
            for (let j = i + 1; j < items.length; j++) {
                const a = items[i], b = items[j];
                if (a.lockPosition || b.lockPosition) continue;
                const dx = b.cx - a.cx, dy = b.cy - a.cy;
                const overlapX = a.halfW + b.halfW + gap - Math.abs(dx);
                const overlapY = a.halfH + b.halfH + gap - Math.abs(dy);
                if (overlapX > 0 && overlapY > 0) {
                    moved = true;
                    if (overlapX < overlapY) {
                        const push = overlapX / 2 + 0.4;
                        const dir = dx === 0 ? (i < j ? -1 : 1) : Math.sign(dx);
                        a.cx -= dir * push;
                        b.cx += dir * push;
                    } else {
                        const push = overlapY / 2 + 0.4;
                        const dir = dy === 0 ? (i < j ? -1 : 1) : Math.sign(dy);
                        a.cy -= dir * push;
                        b.cy += dir * push;
                    }
                }
            }
        }
        if (!moved) break;
    }

    if (bounds) {
        items.forEach((it) => {
            const minX = bounds.minX + it.halfW + 2, maxX = bounds.maxX - it.halfW - 2;
            const minY = bounds.minY + it.halfH + 2, maxY = bounds.maxY - it.halfH - 2;
            if (minX <= maxX) it.cx = Math.min(Math.max(it.cx, minX), maxX);
            if (minY <= maxY) it.cy = Math.min(Math.max(it.cy, minY), maxY);
        });
    }

    let out = "";
    items.forEach((it) => {
        const dist = Math.hypot(it.cx - it.anchorX, it.cy - it.anchorY);
```

```
            const movedFar = dist > (it.type === "text" ? 9 : 6);
            // Label-to-diagram connector lines are intentionally omitted.
            if (it.type === "block") {
                out += it.render(it.cx, it.cy);
            } else {
                const baseX = it.textAnchor === "middle" ? it.cx : it.textAnchor === "end" ? it.cx + it.boxW / 2 - it.padding : it.cx - it.boxW / 2 + it.padding;
                const baseY = it.cy + it.fontSize * 0.32;
                const anchorAttr = it.textAnchor && it.textAnchor !== "start" ? ` text-anchor="${it.textAnchor}"` : "";
                const rotateAttr = it.angle ? ` transform="rotate(${it.angle} ${baseX.toFixed(2)} ${baseY.toFixed(2)})"` : "";
                const haloAttr = it.halo ? ` paint-order="stroke" stroke="white" stroke-width="${it.memberKey !== undefined ? 4 : 3}" stroke-linejoin="round"` : "";
                const memberAttr = it.memberKey !== undefined ? ` class="sf-label" data-member="${escapeHtml(it.memberKey)}"` : "";
                out += `<text${memberAttr} x="${baseX.toFixed(2)}" y="${baseY.toFixed(2)}" fill="${it.color}" font-family="Arial" font-size="${it.fontSize}" font-weight="${it.fontWeight}"${
it.memberKey !== undefined ? ' text-rendering="geometricPrecision"' : ""}${anchorAttr}${rotateAttr}${haloAttr}>${escapeHtml(it.text)}</text>`;
            }
        });
        return out;
    }

    return { addText, addBlock, addObstacle, resolve };
}

// ============================================================================
// 10. SVG RENDERING UTILITIES
// Low-level, purely-geometric SVG fragment builders used by the diagram
// engine below. None of these know anything about the stiffness solver —
// they only turn already-computed numbers into markup.
// ============================================================================
function getGlobalLoadComponentsForDraw(ld, el) {
    let Fx = ld.Fx || 0,
        Fy = ld.Fy || 0;
    let Fx2 = ld.FxEnd !== undefined && ld.FxEnd !== "" ? Number(ld.FxEnd) : Fx;
    let Fy2 = ld.FyEnd !== undefined && ld.FyEnd !== "" ? Number(ld.FyEnd) : Fy;
    let g1 = { x: Fx, y: Fy },
        g2 = { x: Fx2, y: Fy2 };

    if (ld.axisSystem === "Local" && el) {
        // el.cx/el.cy (direction cosines) are only populated once "Run
        // Analysis" has executed. For the live preview diagram they may
        // still be undefined, which produced NaN projections. Recompute
        // them here directly from the element's node coordinates instead,
        // guarding against a zero-length element to avoid a division by 0.
        let n1 = model.nodes.find((n) => n.id === el.startNode);
        let n2 = model.nodes.find((n) => n.id === el.endNode);
        let cx = 1,
            cy = 0;
        if (n1 && n2) {
            let dx = n2.x - n1.x,
                dy = n2.y - n1.y;
            let L = Math.hypot(dx, dy);
            if (L > 0) {
                cx = dx / L;
                cy = dy / L;
            }
        }
        g1.x = Fx * cx - Fy * cy;
        g1.y = Fx * cy + Fy * cx;
        g2.x = Fx2 * cx - Fy2 * cy;
        g2.y = Fx2 * cy + Fy2 * cx;
    } else if (ld.axisSystem === "Angled") {
        let rad = ((ld.angle || 0) * Math.PI) / 180;
        g1.x = Fx * Math.cos(rad) - Fy * Math.sin(rad);
        g1.y = Fx * Math.sin(rad) + Fy * Math.cos(rad);
        g2.x = Fx2 * Math.cos(rad) - Fy2 * Math.sin(rad);
        g2.y = Fx2 * Math.sin(rad) + Fy2 * Math.cos(rad);
    }
    return { g1, g2 };
}

function drawSVGArrow(fromx, fromy, tox, toy, color) {
    const headlen = 10;
    const dx = tox - fromx;
    const dy = toy - fromy;
    const angle = Math.atan2(dy, dx);
    const p1x = tox - headlen * Math.cos(angle - Math.PI / 6);
    const p1y = toy - headlen * Math.sin(angle - Math.PI / 6);
    const p2x = tox - headlen * Math.cos(angle + Math.PI / 6);
    const p2y = toy - headlen * Math.sin(angle + Math.PI / 6);
    return `<line x1="${fromx}" y1="${fromy}" x2="${tox}" y2="${toy}" stroke="${color}" stroke-width="2"/>
                <polyline points="${p1x},${p1y} ${tox},${toy} ${p2x},${p2y}" fill="none" stroke="${color}" stroke-width="2"/>`;
}

// Draw a clear circular moment arrow with the arrowhead tangent to the arc.
// Positive Mz is shown anticlockwise on the screen; negative Mz is shown clockwise.
function drawMomentArrow(cx, cy, radius, anticlockwise, color) {
    const startDeg = anticlockwise ? 135 : -135;
    const endDeg = anticlockwise ? -135 : 135;
    const steps = 32;
```

```
    const points = [];

    for (let i = 0; i <= steps; i++) {
        const deg = startDeg + ((endDeg - startDeg) * i) / steps;
        const rad = (deg * Math.PI) / 180;
        points.push({
            x: cx + radius * Math.cos(rad),
            y: cy + radius * Math.sin(rad),
        });
    }

    let d = `M ${points[0].x.toFixed(2)} ${points[0].y.toFixed(2)}`;
    for (let i = 1; i < points.length; i++) {
        d += ` L ${points[i].x.toFixed(2)} ${points[i].y.toFixed(2)}`;
    }

    const lineTip = points[points.length - 1];
    const prev = points[points.length - 2];
    const vx = lineTip.x - prev.x;
    const vy = lineTip.y - prev.y;
    const vLen = Math.hypot(vx, vy) || 1;
    const ux = vx / vLen;
    const uy = vy / vLen;
    const tip = { x: lineTip.x + ux * 2, y: lineTip.y + uy * 2 };
    const headLen = Math.min(12, radius * 0.42);
    const headWidth = Math.min(7, radius * 0.30);
    const bx = tip.x - ux * headLen;
    const by = tip.y - uy * headLen;
    const px = -uy;
    const py = ux;

    const p1 = `${tip.x.toFixed(2)},${tip.y.toFixed(2)}`;
    const p2 = `${(bx + px * headWidth).toFixed(2)},${(by + py * headWidth).toFixed(2)}`;
    const p3 = `${(bx - px * headWidth).toFixed(2)},${(by - py * headWidth).toFixed(2)}`;

    return `<path d="${d}" fill="none" stroke="${color}" stroke-width="2.5" stroke-linecap="round" stroke-linejoin="round"/>` +
        `<polygon points="${p1} ${p2} ${p3}" fill="${color}"/>`;
}

function getSupportVisualType(sa) {
    const template = model.supportTemplates.find((s) => s.id === sa.supportId);
    if (!template) return "custom";
    if (template.type === "fixed" || template.builtIn && template.id === "FIXED") return "fixed";
    if (template.type === "roller" || template.builtIn && template.id === "ROLLER") return "roller";
    if (template.type === "hinged" || template.builtIn && template.id === "HINGED") return "hinged";
    return "custom";
}

function drawSupportSymbol(x, y, type) {
    // Compact support markers: intentionally small so they identify the
    // restraint type without obscuring nodes, loads, dimensions, or labels.
    const triangleTop = y + 1;
    if (type === "fixed" || type === "custom") {
        const fill = type === "fixed" ? "#111827" : "#8b5a2b";
        const stroke = type === "fixed" ? "#030712" : "#6f451f";
        const baseY = y + 14;
        return `
                <rect x="${x - 7}" y="${baseY}" width="14" height="11" rx="1.5" fill="${fill}" stroke="${stroke}" stroke-width="1"/>
            `;
    }

    const color = type === "roller" ? "#16a34a" : "#2563eb";
    let out = `<polygon points="${x},${triangleTop} ${x - 8},${y + 12} ${x + 8},${y + 12}" fill="${color}" stroke="${color}" stroke-width="1"/>`;
    if (type === "roller") {
        const wheelY = y + 15;
        out += `<circle cx="${x - 4}" cy="${wheelY}" r="2" fill="white" stroke="${color}" stroke-width="1"/>`;
        out += `<circle cx="${x + 4}" cy="${wheelY}" r="2" fill="white" stroke="${color}" stroke-width="1"/>`;
        out += `<line x1="${x - 10}" y1="${y + 18}" x2="${x + 10}" y2="${y + 18}" stroke="${color}" stroke-width="1.2"/>`;
        for (let gx = x - 9; gx <= x + 9; gx += 6) {
            out += `<line x1="${gx}" y1="${y + 18}" x2="${gx - 3}" y2="${y + 21}" stroke="${color}" stroke-width="0.8"/>`;
        }
    } else {
        out += `<line x1="${x - 10}" y1="${y + 15}" x2="${x + 10}" y2="${y + 15}" stroke="${color}" stroke-width="1.2"/>`;
        for (let gx = x - 9; gx <= x + 9; gx += 6) {
            out += `<line x1="${gx}" y1="${y + 15}" x2="${gx - 3}" y2="${y + 18}" stroke="${color}" stroke-width="0.8"/>`;
        }
    }
    return out;
}

// ============================================================================
// 11. DIAGRAM ENGINE
// getStructureSVG() is the shared rendering engine behind every structural
// diagram (Structure Model, Reactions, AFD, SFD, BMD, Displacement) — the
// six modes share one continuous coordinate system, label-collision pass,
// and node/support drawing, so they are driven by a single `mode` switch
// rather than six independently-scrolled copies of that shared machinery.
```

```
// Inside getStructureSVG, each mode's own drawing logic is a nested,
// individually-named function (drawModelModeLoadsAndMoments,
// drawDisplacementDiagram, drawReactionsDiagram, drawInternalActionDiagram)
// declared as closures so they can share the coordinate transform (tx/ty),
// the SVG accumulator, and the label-collision layer without re-deriving or
// re-passing them — the top-level body of getStructureSVG is then just the
// ordered sequence of "which phase runs for this mode".
// The named, independent entry point for each diagram (as used everywhere
// else in this file, and matching the diagram naming used elsewhere in this
// codebase) is the render*Diagram() family defined right after it — callers
// outside this section should use those, not getStructureSVG directly.
// ========================================================================
// Axial force, shear force, and bending moment at a distance `x` along a
// member's local axis, given its already-solved member-end plot data `pd`
// (see solveStep8 / el.plotData). Pure function of (pd, x) — used by both
// the AFD/SFD/BMD max-magnitude scan and the actual diagram point sampling
// inside getStructureSVG below, so the internal-action formula exists in
// exactly one place instead of two identical copies.
function calculateInternalActionsAtPosition(pd, x) {
    if (pd.grid || isGridAnalysisMode()) {
        // Grid end-action order is [Mx, My, Fz] at each end. These reusable
        // diagrams project the three member quantities onto AFD/SFD/BMD:
        // AFD -> torsion Mx, SFD -> vertical shear Fz, BMD -> bending My.
        let torsion = Number(pd.fx1) || 0;
        let shear = Number(pd.fy1) || 0;
        let bending = -(Number(pd.mz1) || 0) + shear * x;

        (pd.gridUDLs || []).forEach((udl) => {
            if (!(x > udl.a)) return;
            const span = udl.b - udl.a;
            if (!(span > 0)) return;
            const len = Math.min(x, udl.b) - udl.a;
            if (!(len > 0)) return;
            const q1 = Number(udl.q1) || 0;
            const qEnd = q1 + ((Number(udl.q2) || 0) - q1) * (len / span);
            shear += ((q1 + qEnd) / 2) * len;
            const vRect = q1 * len;
            const mRect = vRect * (x - (udl.a + len / 2));
            const vTri = 0.5 * (qEnd - q1) * len;
            const mTri = vTri * (x - (udl.a + (2 * len) / 3));
            bending += mRect + mTri;
        });

        (pd.gridPointLoads || []).forEach((load) => {
            if (x > load.a) {
                const fz = Number(load.Fz) || 0;
                shear += fz;
                bending += fz * (x - load.a);
            }
        });
        (pd.gridPointTorsions || []).forEach((moment) => {
            if (x > moment.a) torsion -= Number(moment.Mx) || 0;
        });
        (pd.gridPointMoments || []).forEach((moment) => {
            if (x > moment.a) bending -= Number(moment.My) || 0;
        });
        return { A: torsion, V: shear, M: bending };
    }

    let A_ax = pd.fx1;
    let V = pd.fy1;
    let M = -pd.mz1 + pd.fy1 * x;

    // Add overall self-weight
    A_ax += pd.wx * x;
    V += pd.wy * x;
    M += (pd.wy * x * x) / 2;

    // Complex Custom Partial UDLs
    pd.localUDLs.forEach((udl) => {
        let wx1 = udl.Fx1 * pd.cx + udl.Fy1 * pd.cy;
        let wy1 = -udl.Fx1 * pd.cy + udl.Fy1 * pd.cx;
        let wx2 = udl.Fx2 * pd.cx + udl.Fy2 * pd.cy;
        let wy2 = -udl.Fx2 * pd.cy + udl.Fy2 * pd.cx;

        if (x > udl.a) {
            let L_udl = udl.b - udl.a;
            let len = Math.min(x, udl.b) - udl.a;

            let w1_x = wx1;
            let w2_x = wx1 + ((wx2 - wx1) * len) / L_udl;
            A_ax += ((w1_x + w2_x) / 2) * len;

            let w1_y = wy1;
            let w2_y = wy1 + ((wy2 - wy1) * len) / L_udl;
            V += ((w1_y + w2_y) / 2) * len;
            let V_rect = w1_y * len;
```

```
            let M_rect = V_rect * (x - (udl.a + len / 2));
            let V_tri = 0.5 * (w2_y - w1_y) * len;
            let M_tri = V_tri * (x - (udl.a + (2 * len) / 3));

            M += M_rect + M_tri;
        }
    });

    pd.pointLoads.forEach((pl) => {
        if (x > pl.a) {
            A_ax += pl.Px;
            V += pl.Py;
            M += pl.Py * (x - pl.a);
        }
    });

    pd.localPointMoments.forEach((pm) => {
        if (x > pm.a) {
            M -= pm.Mz;
        }
    });

    return { A: A_ax, V, M };
}

// Scans all nodal/element loads for their largest global magnitude, and all
// nodal moments for the largest |Mz|. Used only by the "model" diagram mode
// to auto-scale load-arrow lengths and moment-arc radii so magnitude stays
// visually comparable across a load/moment case. Pure function of `model` —
// no DOM access, no SVG string building.
function computeMaxLoadAndMomentMagnitudes() {
    let maxLoad = 0;
    let maxMoment = 0;
    model.loads.forEach((ld) => {
        let el = model.elements.find((e) => e.id === ld.elementId);
        let comps = getGlobalLoadComponentsForDraw(ld, el);
        let mag1 = Math.hypot(comps.g1.x, comps.g1.y);
        let mag2 = Math.hypot(comps.g2.x, comps.g2.y);
        if (mag1 > maxLoad) maxLoad = mag1;
        if (mag2 > maxLoad) maxLoad = mag2;
    });
    model.moments.forEach((m) => {
        let mag = Math.abs(m.Mz || 0);
        if (mag > maxMoment) maxMoment = mag;
    });
    return { maxLoad, maxMoment };
}

// Scans every solved element's plot data (see solveStep8 / el.plotData) at a
// dense set of candidate positions — plus just before/after every point
// load, point moment, and UDL boundary, where axial/shear/moment can jump —
// to find the largest |axial|, |shear|, and |moment| anywhere in the
// structure. calculateInternalActionsAtPosition() above remains the single
// source of truth for the internal-action formula itself; this function only
// decides WHERE to sample it for the purpose of auto-scaling the AFD/SFD/BMD
// diagrams. Pure function of `model` — no DOM access, no SVG string building.
function computeInternalActionExtremes() {
    let maxA = 0,
        maxV = 0,
        maxM = 0;
    model.elements.forEach((el) => {
        if (!el.plotData) return;
        let pd = el.plotData;
        let xs = [];
        for (let i = 0; i <= 50; i++) xs.push((i / 50) * pd.L);
        pd.pointLoads.forEach((pl) => {
            xs.push(pl.a - 1e-6);
            xs.push(pl.a + 1e-6);
        });
        pd.localPointMoments.forEach((pm) => {
            xs.push(pm.a - 1e-6);
            xs.push(pm.a + 1e-6);
        });
        pd.localUDLs.forEach((udl) => {
            xs.push(udl.a);
            xs.push(udl.b);
        });
        xs.sort((a, b) => a - b);

        xs.forEach((x) => {
            const { A: A_ax, V, M } = calculateInternalActionsAtPosition(pd, x);
            if (Math.abs(A_ax) > maxA) maxA = Math.abs(A_ax);
            if (Math.abs(V) > maxV) maxV = Math.abs(V);
            if (Math.abs(M) > maxM) maxM = Math.abs(M);
        });
    });
    return { maxA, maxV, maxM };
```

```
}

function getStructureSVG(width, height, camera, mode = "model") {
    if (model.nodes.length === 0) {
        return `<svg width="100%" height="100%" viewBox="0 0 ${width} ${height}"><text x="20" y="30" fill="#94a3b8" font-family="Arial" font-size="14">No nodes to draw.</text></svg>`;
    }

    let hasAnalysisData = analysisState.status === "valid" && model.elements.length > 0 && model.elements.some((el) => el && el.plotData);
    if (
        (mode === "afd" || mode === "sfd" || mode === "bmd" || mode === "displacement" || mode === "reactions") &&
        !hasAnalysisData
    ) {
        const message = analysisState.status === "invalidated"
            ? "Model changed since last analysis — Run Analysis again."
            : "Run Analysis first to view this diagram.";
        return `<svg width="100%" height="100%" viewBox="0 0 ${width} ${height}"><text x="20" y="30" fill="#94a3b8" font-family="Arial" font-size="14">${message}</text></svg>`;
    }

    const svgIntrinsicSize = camera && camera.intrinsicCanvas;
    let svg = svgIntrinsicSize
        ? `<svg width="${Math.ceil(width)}" height="${Math.ceil(height)}" viewBox="0 0 ${width} ${height}" xmlns="http://www.w3.org/2000/svg" style="background: white; display:block;">`
        : `<svg width="100%" height="100%" viewBox="0 0 ${width} ${height}" xmlns="http://www.w3.org/2000/svg" style="background: white;">`;

    // Every text label this diagram draws (node IDs, member lengths, load/
    // moment values, displacement values, diagram peak values, reaction
    // cards, ...) is queued here and placed without overlap once, at the
    // very end of this function. See createLabelLayer() above.
    const labelLayer = createLabelLayer();

    function tx(x) {
        return camera.offsetX + x * camera.scale;
    }
    function ty(y) {
        return camera.offsetY - y * camera.scale;
    }

    let bb = getBoundingBox();
    let maxDim = Math.max(bb.maxX - bb.minX || 1, bb.maxY - bb.minY || 1);

    let maxLoad = 0,
        maxMoment = 0;
    if (mode === "model") {
        ({ maxLoad, maxMoment } = computeMaxLoadAndMomentMagnitudes());
    }

    let maxA = 0,
        maxV = 0,
        maxM = 0;
    if (mode === "afd" || mode === "sfd" || mode === "bmd") {
        ({ maxA, maxV, maxM } = computeInternalActionExtremes());
    }

    let scaleA = maxA > 0 ? (0.15 * maxDim) / maxA : 0;
    let scaleV = maxV > 0 ? (0.15 * maxDim) / maxV : 0;
    let scaleM = maxM > 0 ? (0.15 * maxDim) / maxM : 0;

    let elStroke = mode === "model" || mode === "reactions" ? "#4f8df7" : "#cbd5e1";
    let textStroke = mode === "model" || mode === "reactions" ? "#64748b" : "#94a3b8";
    if (mode === "displacement") {
        elStroke = "#cbd5e1";
        textStroke = "#94a3b8";
    }

    const isInternalActionMode = mode === "afd" || mode === "sfd" || mode === "bmd";
    // Compact node marker used only by the AFD / SFD / BMD views.
    const actionNodeRadius = 7;

    // Leader lines + anchor dots for AFD/SFD/BMD values. Kept apart from the
    // main markup so they can be drawn ABOVE the node markers (see below).
    let actionOverlaySvg = "";

    model.elements.forEach((el, ei) => {
        let n1 = model.nodes.find((n) => n.id === el.startNode);
        let n2 = model.nodes.find((n) => n.id === el.endNode);
        if (n1 && n2) {
            let x1 = tx(n1.x),
                y1 = ty(n1.y),
                x2 = tx(n2.x),
                y2 = ty(n2.y);
            const elementStrokeWidth = mode === "model" ? 9 : 3;
            if (isInternalActionMode) {
                // Clickable member: a wide invisible hit-line makes thin members easy to tap.
                svg += `<g class="sf-member" data-member="${ei}" data-start="${escapeHtml(String(el.startNode))}" data-end="${escapeHtml(String(el.endNode))}"><line x1="${x1}" y1="${y1}"
x2="${x2}" y2="${y2}" stroke="rgba(0,0,0,0)" stroke-width="16" pointer-events="stroke"/><line class="sf-member-line" x1="${x1}" y1="${y1}" x2="${x2}" y2="${y2}" stroke="${elStroke}" stroke-
width="${elementStrokeWidth}"/></g>`;
            } else {
                svg += `<line x1="${x1}" y1="${y1}" x2="${x2}" y2="${y2}" stroke="${elStroke}" stroke-width="${elementStrokeWidth}"/>`;
```

```
            }
            if (mode === "model" && !isGridAnalysisMode()) {
                const elementLength = Math.hypot(Number(n2.x) - Number(n1.x), Number(n2.y) - Number(n1.y));
                if (Number.isFinite(elementLength) && elementLength > 0) {
                    const midX = (x1 + x2) / 2;
                    const midY = (y1 + y2) / 2;
                    let labelAngle = Math.atan2(y2 - y1, x2 - x1) * 180 / Math.PI;
                    if (labelAngle > 90) labelAngle -= 180;
                    if (labelAngle < -90) labelAngle += 180;

                    svg += `<g transform="translate(${midX.toFixed(2)} ${midY.toFixed(2)}) rotate(${labelAngle.toFixed(2)})">
                        <text x="0" y="0.3" text-anchor="middle" dominant-baseline="middle"
                              fill="#000000" font-family="Arial" font-size="8" font-weight="bold">
                            ${escapeHtml(String(el.id))}
                        </text>
                    </g>`;
                }
            }
        }
    });

    // Support symbols are shown consistently in every structural/result
    // diagram so the restrained boundary condition remains identifiable.
    model.supportAssignments.forEach((sa) => {
        let n = model.nodes.find((node) => node.id === sa.nodeId);
        if (n) {
            const supportType = getSupportVisualType(sa);
            const supportY =
                (supportType === "roller" || supportType === "hinged") ? ty(n.y) + 8 : ty(n.y) - 5;
            svg += drawSupportSymbol(tx(n.x), supportY, supportType);
        }
    });

    // Renders every input load and moment for the plain "model" diagram:
    // nodal loads, element point loads, element UDLs (including linearly-
    // varying and angled/local-axis variants), and nodal/element moments.
    // Kept as one function (mirroring the original inline block) because
    // every branch shares the same `svg`/`labelLayer` accumulators and the
    // same `maxLoad`/`maxMoment` auto-scale factors computed just above in
    // getStructureSVG — splitting further would mean threading half a dozen
    // shared values through extra parameters for no real gain in clarity.
    function drawModelModeLoadsAndMoments() {
        // Grid loads act normal to the XY plane (Fz), and Grid torsions use
        // Mx / My. They must not pass through the BTF-only Fx / Fy / Mz
        // renderer below, which would silently omit them from the model view.
        function drawGridLoadsAndTorsions() {
            const gridLoads = Array.isArray(model.loads) ? model.loads : [];
            const gridTorsions = Array.isArray(model.moments) ? model.moments : [];
            let maxFz = 0;
            let maxGridTorsion = 0;
            gridLoads.forEach((ld) => {
                const fz = Math.abs(Number(ld.Fz) || 0);
                maxFz = Math.max(maxFz, fz, Math.abs(Number(ld.FzEnd) || 0));
            });
            gridTorsions.forEach((m) => {
                maxGridTorsion = Math.max(maxGridTorsion, Math.abs(Number(m.Mx) || 0), Math.abs(Number(m.My) || 0));
            });

            const ratioOf = (value, maxValue) => maxValue > 0
                ? Math.max(0, Math.min(1, Math.abs(Number(value) || 0) / maxValue))
                : 0;
            const loadRadius = (value) => 2.8 + 2.6 * Math.sqrt(ratioOf(value, maxFz));
            const amount = (value) => formatNum(Math.abs(Number(value) || 0));
            // Negative loads are red; positive loads are light green. The
            // filled/open circle convention remains: filled = into the page,
            // open = out of the page.
            const gridLoadColor = (value) => Number(value) < 0 ? '#ef4444' : '#4ade80';

            // Filled red circle means Fz into the page; open light-green circle
            // means Fz out of the page. An optional radius lets a nodal load be
            // centered on the joint while leaving a colored ring around the
            // node marker, whose ID/coordinates remain on the top drawing layer.
            function drawGridLoadSymbol(x, y, fz, load, detail, radiusOverride = null, colorOverride = null) {
                const value = Number(fz) || 0;
                if (Math.abs(value) <= 1e-12) return 0;
                const intoPage = value < 0;
                const radius = radiusOverride !== null && radiusOverride !== undefined && Number.isFinite(Number(radiusOverride)) ? Number(radiusOverride) : loadRadius(value);
                const haloRadius = radius + 1.35;
                const color = colorOverride || gridLoadColor(value);
                const id = String(load.id ?? '');
                const label = `${id ? `Load ${id}: ` : ''}${detail}; ${amount(value)} kN${intoPage ? ', into the page' : ', out of the page'}`;
                svg += `<g class="grid-diagram-symbol grid-load-symbol ${intoPage ? 'grid-load-into-page' : 'grid-load-out-of-page'}" data-grid-load-id="${escapeHtml(id)}" role="button"
tabindex="0" aria-pressed="false" aria-label="${escapeHtml(label)}"><title>${escapeHtml(label)}</title>
                    <circle class="grid-symbol-hit" cx="${x.toFixed(2)}" cy="${y.toFixed(2)}" r="${(haloRadius + 5).toFixed(2)}" fill="transparent" stroke="transparent"/>
                    <circle class="grid-symbol-halo" cx="${x.toFixed(2)}" cy="${y.toFixed(2)}" r="${haloRadius.toFixed(2)}" fill="#ffffff" stroke="#ffffff" stroke-width="1.5"/>
                    <circle class="grid-symbol-core" cx="${x.toFixed(2)}" cy="${y.toFixed(2)}" r="${radius.toFixed(2)}" fill="${intoPage ? color : '#ffffff'}" stroke="${color}" stroke-
width="${Math.max(1.1, radius * 0.28).toFixed(2)}"/>
```

```
            </g>`;
            return radius;
        }

        // Point along an element in screen coordinates. Grid load circles
        // deliberately use normalOffset=0 so they sit ON the blue member.
        function gridElementPoint(el, distance, normalOffset = 0) {
            const n1 = model.nodes.find((n) => String(n.id) === String(el.startNode));
            const n2 = model.nodes.find((n) => String(n.id) === String(el.endNode));
            if (!n1 || !n2) return null;
            const dx = Number(n2.x) - Number(n1.x);
            const dy = Number(n2.y) - Number(n1.y);
            const L = Math.hypot(dx, dy);
            if (L <= 1e-12) return null;
            const t = Math.max(0, Math.min(1, (Number(distance) || 0) / L));
            const x1 = tx(n1.x), y1 = ty(n1.y);
            const x2 = tx(n2.x), y2 = ty(n2.y);
            const sx = x1 + t * (x2 - x1);
            const sy = y1 + t * (y2 - y1);
            const screenL = Math.hypot(x2 - x1, y2 - y1);
            const nx = screenL > 1e-12 ? (y2 - y1) / screenL : 0;
            const ny = screenL > 1e-12 ? -(x2 - x1) / screenL : -1;
            return { x: sx + nx * normalOffset, y: sy + ny * normalOffset, baseX: sx, baseY: sy, n1, n2, L, t, nx, ny, x1, y1, x2, y2 };
        }

        // Choose a visually clear side of a joint for the nodal-load
        // magnitude label. No arrow is drawn for Grid nodal loads.
        function nodeLoadDirection(node) {
            const nx = tx(node.x), ny = ty(node.y);
            let vx = 0, vy = 0;
            model.elements.forEach((el) => {
                const a = model.nodes.find((n) => String(n.id) === String(el.startNode));
                const b = model.nodes.find((n) => String(n.id) === String(el.endNode));
                if (!a || !b) return;
                let other = null;
                if (String(a.id) === String(node.id)) other = b;
                else if (String(b.id) === String(node.id)) other = a;
                if (!other) return;
                const ox = tx(other.x) - nx, oy = ty(other.y) - ny;
                const d = Math.hypot(ox, oy);
                if (d > 1e-9) { vx += ox / d; vy += oy / d; }
            });
            // Point the label away from attached members.
            vx = -vx; vy = -vy;
            let d = Math.hypot(vx, vy);
            if (d < 0.2) { vx = -0.7071; vy = 0.7071; d = Math.hypot(vx, vy); }
            return { x: vx / d, y: vy / d };
        }

        // Single shaft, two stacked chevron heads pointing in the signed
        // torsion direction. Shaft length and thickness scale by magnitude.
        function drawGridDoubleArrow(x, y, ux, uy, value, id, targetLabel, color, labelSide) {
            if (Math.abs(value) <= 1e-12) return;
            const r = ratioOf(value, maxGridTorsion);
            const length = 14 + 17 * Math.sqrt(r);
            const width = 1.8 + r * 0.9;
            const headLength = 5 + r * 1.8;
            const halfWidth = 3.1 + r * 1.5;
            const gap = 6 + r * 1.5;
            const ex = x + ux * length;
            const ey = y + uy * length;
            const px = -uy, py = ux;
            const head = (hx, hy) => {
                const bx = hx - ux * headLength;
                const by = hy - uy * headLength;
                return `M ${(bx + px * halfWidth).toFixed(2)} ${(by + py * halfWidth).toFixed(2)} L ${hx.toFixed(2)} ${hy.toFixed(2)} L ${(bx - px * halfWidth).toFixed(2)} ${(by - py *
halfWidth).toFixed(2)}`;
            };
            const hx2 = ex - ux * gap, hy2 = ey - uy * gap;
            const idText = String(id ?? '');
            const description = `${idText ? `Torsion ${idText}: ` : ''}${amount(value)} kN·m, ${targetLabel}`;
            const hitRadius = length / 2 + halfWidth + 8;
            svg += `<g class="grid-diagram-symbol grid-torsion-symbol" data-grid-torsion-id="${escapeHtml(idText)}" role="button" tabindex="0" aria-pressed="false" aria-label="${
escapeHtml(description)}"><title>${escapeHtml(description)}</title><circle class="grid-symbol-hit" cx="${((x + ex) / 2).toFixed(2)}" cy="${((y + ey) / 2).toFixed(2)}" r="${hitRadius.toFixed(
2)}" fill="transparent" stroke="transparent"/><path d="M ${x.toFixed(2)} ${y.toFixed(2)} L ${ex.toFixed(2)} ${ey.toFixed(2)}" fill="none" stroke="${color}" stroke-width="${width.toFixed(
2)}" stroke-linecap="round"/><path d="${head(ex, ey)} ${head(hx2, hy2)}" fill="none" stroke="${color}" stroke-width="${width.toFixed(2)}" stroke-linecap="round" stroke-linejoin="round"/></
g>`;
            // Keep the torsion value tied to the arrow end. The shared
            // collision-avoidance label layer can otherwise move nodal
            // torsion text far away when a load is present at the same joint.
            let labelX, labelY, textAnchor;
            if (labelSide === 'left' || labelSide === 'right') {
                labelX = ex + ux * 5;
                labelY = ey - 7;
                textAnchor = labelSide === 'right' ? 'start' : 'end';
            } else {
                labelX = ex + 8;
```

```
                    labelY = ey + (uy < 0 ? 0 : 10);
                    textAnchor = 'start';
                }
                svg += `<text class="grid-torsion-value" x="${labelX.toFixed(2)}" y="${labelY.toFixed(2)}" text-anchor="${textAnchor}" dominant-baseline="middle" font-family="Arial" font-
size="10" font-weight="700" fill="${color}" stroke="#ffffff" stroke-width="2.6" stroke-linejoin="round" paint-order="stroke fill" pointer-events="none">${amount(value)} kN·m</text>`;
            }

            gridLoads.forEach((ld) => {
                const fz = Number(ld.Fz) || 0;
                const type = String(ld.type || '');
                if (Math.abs(fz) <= 1e-12 && type !== 'Element UDL') return;

                if (type === 'Nodal Load') {
                    const node = model.nodes.find((n) => String(n.id) === String(ld.nodeId));
                    if (!node || Math.abs(fz) <= 1e-12) return;
                    const nodeX = tx(node.x), nodeY = ty(node.y);
                    const nodeRadius = 9.3333333333;
                    // Use the exact same magnitude-scaled circle radius as an
                    // element point load. Keep it concentric with the joint so
                    // the orange node marker naturally covers its center; the
                    // red/light-green magnitude label identifies the nodal load.
                    drawGridLoadSymbol(nodeX, nodeY, fz, ld,
                        `Nodal load at node ${ld.nodeId}`);

                    // Put only the magnitude beside the joint. If a torsion is
                    // applied at the same node, use the opposite side from its
                    // arrow direction to reduce label overlap. No load arrow is
                    // drawn for Grid nodal loads.
                    let direction = nodeLoadDirection(node);
                    const nodalTorsion = gridTorsions.filter((m) =>
                        String(m.nodeId) === String(node.id) &&
                        (m.type === 'Nodal Moment' || m.type === 'Nodal Torsion' || !m.elementId)
                    );
                    let torsionX = 0, torsionY = 0;
                    nodalTorsion.forEach((m) => {
                        torsionX += Math.sign(Number(m.Mx) || 0);
                        torsionY -= Math.sign(Number(m.My) || 0); // screen Y is inverted
                    });
                    const torsionLength = Math.hypot(torsionX, torsionY);
                    if (torsionLength > 1e-9) {
                        torsionX /= torsionLength; torsionY /= torsionLength;
                        if (direction.x * torsionX + direction.y * torsionY > 0.1) {
                            direction = { x: -direction.x, y: -direction.y };
                        }
                    }
                    const labelDistance = nodeRadius + 9;
                    const labelX = nodeX + direction.x * labelDistance;
                    const labelY = nodeY + direction.y * labelDistance + 2;
                    const textAnchor = direction.x > 0.25 ? 'start' : direction.x < -0.25 ? 'end' : 'middle';
                    labelLayer.addText({
                        x: labelX, y: labelY, text: `${amount(fz)} kN`,
                        fontSize: 10, fontWeight: 'bold', color: gridLoadColor(fz),
                        textAnchor, anchorX: nodeX, anchorY: nodeY,
                        leader: false, halo: true,
                    });
                    return;
                }

                if (type === 'Element Point Load' || type === 'Element Load' || type === 'Point Load') {
                    const el = model.elements.find((e) => String(e.id) === String(ld.elementId));
                    if (!el || Math.abs(fz) <= 1e-12) return;
                    const point = gridElementPoint(el, ld.distance, 0);
                    if (!point) return;
                    // Element point loads are purple (matching the existing
                    // BTF moment color), distinct from red/green nodal loads.
                    const pointLoadColor = '#8b5cf6';
                    const radius = drawGridLoadSymbol(point.x, point.y, fz, ld,
                        `Point load on element ${ld.elementId} at ${formatNum(Number(ld.distance) || 0)} m`,
                        null, pointLoadColor);
                    labelLayer.addText({
                        x: point.x + point.nx * (radius + 10),
                        y: point.y + point.ny * (radius + 8) - 2,
                        text: `${amount(fz)} kN`, fontSize: 10, fontWeight: 'bold',
                        color: pointLoadColor,
                        textAnchor: point.nx > 0.25 ? 'start' : point.nx < -0.25 ? 'end' : 'middle',
                        anchorX: point.x, anchorY: point.y, leader: false, halo: true,
                    });
                    return;
                }

                if (type === 'Element UDL') {
                    const el = model.elements.find((e) => String(e.id) === String(ld.elementId));
                    if (!el) return;
                    const n1 = model.nodes.find((n) => String(n.id) === String(el.startNode));
                    const n2 = model.nodes.find((n) => String(n.id) === String(el.endNode));
                    if (!n1 || !n2) return;
                    const L = Math.hypot(Number(n2.x) - Number(n1.x), Number(n2.y) - Number(n1.y));
```

```
            if (L <= 1e-12) return;
            const a = Math.max(0, Math.min(L, Number(ld.distance) || 0));
            const rawEnd = ld.distanceEnd === '' || ld.distanceEnd == null ? L : Number(ld.distanceEnd);
            const b = Math.max(0, Math.min(L, Number.isFinite(rawEnd) ? rawEnd : L));
            if (!(b > a)) return;
            const q1 = Number(ld.Fz) || 0;
            const q2 = ld.FzEnd !== '' && ld.FzEnd != null && Number.isFinite(Number(ld.FzEnd)) ? Number(ld.FzEnd) : q1;
            const count = Math.max(5, Math.min(9, Math.round((b - a) / L * 8) + 3));
            let middlePoint = null;
            for (let i = 0; i <= count; i++) {
                const ratio = i / count;
                const dist = a + (b - a) * ratio;
                const value = q1 + (q2 - q1) * ratio;
                const point = gridElementPoint(el, dist, 0);
                if (!point) continue;
                drawGridLoadSymbol(point.x, point.y, value, ld,
                    `Distributed load on element ${ld.elementId}, ${formatNum(a)}–${formatNum(b)} m`, null, '#111827');
                if (i === Math.floor(count / 2)) middlePoint = point;
            }
            if (middlePoint && (Math.abs(q1) > 1e-12 || Math.abs(q2) > 1e-12)) {
                const intensity = Math.abs(q1 - q2) <= 1e-12
                    ? `${amount(q1)} kN/m`
                    : `${amount(q1)} → ${amount(q2)} kN/m`;
                labelLayer.addText({
                    x: middlePoint.x + middlePoint.nx * 20,
                    y: middlePoint.y + middlePoint.ny * 20 - 3,
                    text: intensity, fontSize: 10, fontWeight: 'bold', color: '#111827',
                    textAnchor: middlePoint.nx > 0.25 ? 'start' : middlePoint.nx < -0.25 ? 'end' : 'middle',
                    anchorX: middlePoint.x, anchorY: middlePoint.y, leader: false, halo: true,
                });
            }
        }
    });

    // Grid torsions use signed directional arrows: positive Mx points
    // right, negative Mx left; positive My points up, negative My down.
    // Each directional arrow has one shaft and a double-chevron head.
    gridTorsions.forEach((m) => {
        const mx = Number(m.Mx) || 0;
        const my = Number(m.My) || 0;
        const isNodal = m.type === 'Nodal Moment' || m.type === 'Nodal Torsion' || (!!m.nodeId && !m.elementId);
        const isElement = m.type === 'Element Moment' || m.type === 'Element Torsion' || (!!m.elementId && !m.nodeId);
        if ((!isNodal && !isElement) || (Math.abs(mx) <= 1e-12 && Math.abs(my) <= 1e-12)) return;
        let point = null;
        if (isNodal) {
            const node = model.nodes.find((n) => String(n.id) === String(m.nodeId));
            if (!node) return;
            point = { x: tx(node.x), y: ty(node.y) };
        } else {
            const el = model.elements.find((e) => String(e.id) === String(m.elementId));
            if (!el) return;
            point = gridElementPoint(el, m.distance, 0);
            if (!point) return;
        }
        const id = String(m.id ?? '');
        const targetLabel = isNodal ? `at node ${m.nodeId}` : `on element ${m.elementId} at ${formatNum(Number(m.distance) || 0)} m`;
        if (Math.abs(mx) > 1e-12) {
            const dir = mx > 0 ? 1 : -1;
            // Start at the actual node/element point so the horizontal
            // arrow visibly grows from the joint instead of floating.
            drawGridDoubleArrow(point.x, point.y, dir, 0, mx, id, `${targetLabel}, horizontal direction`, '#7c3aed', dir > 0 ? 'right' : 'left');
        }
        if (Math.abs(my) > 1e-12) {
            const dir = my > 0 ? -1 : 1; // SVG y-axis is inverted.
            // For nodal torsion this begins at the node centre; the
            // node marker is painted afterwards and stays on top.
            drawGridDoubleArrow(point.x, point.y, 0, dir, my, id, `${targetLabel}, vertical direction`, '#0f766e', dir < 0 ? 'up' : 'down');
        }
    });

    // Element identifiers are redrawn after all load/torsion markers so
    // they stay readable over UDL circles. Node circles and coordinates
    // are still rendered later by the shared topmost node-marker layer.
    if (mode === 'model') {
        model.elements.forEach((el) => {
            const n1 = model.nodes.find((n) => String(n.id) === String(el.startNode));
            const n2 = model.nodes.find((n) => String(n.id) === String(el.endNode));
            if (!n1 || !n2) return;
            const x1 = tx(n1.x), y1 = ty(n1.y), x2 = tx(n2.x), y2 = ty(n2.y);
            const len = Math.hypot(x2 - x1, y2 - y1);
            if (len <= 1e-9) return;
            const midX = (x1 + x2) / 2, midY = (y1 + y2) / 2;
            let angle = Math.atan2(y2 - y1, x2 - x1) * 180 / Math.PI;
            if (angle > 90) angle -= 180;
            if (angle < -90) angle += 180;
            svg += `<g class="grid-member-id" pointer-events="none" transform="translate(${midX.toFixed(2)} ${midY.toFixed(2)}) rotate(${angle.toFixed(2)})"><text x="0" y="0.3" text-
anchor="middle" dominant-baseline="middle" fill="#111827" stroke="#ffffff" stroke-width="2.7" stroke-linejoin="round" paint-order="stroke fill" font-family="Arial" font-size="8" font-
```

```
weight="800">${escapeHtml(String(el.id))}</text></g>`;
                });
            }
        }

        if (isGridAnalysisMode()) {
            drawGridLoadsAndTorsions();
            return;
        }

        // Plane-mode Beam / Truss / Frame drawing logic below is intentionally
        // left unchanged; it continues to use Fx / Fy and Mz.
        model.loads.forEach((ld) => {
                let isUDL = ld.type === "Element UDL";
                let color = isUDL ? "#f97316" : "#ef4444";

                if (ld.type === "Nodal Load") {
                    let comps = getGlobalLoadComponentsForDraw(ld, null);
                    let fx = comps.g1.x,
                        fy = comps.g1.y;
                    let mag = Math.hypot(fx, fy);
                    if (mag == 0) return;
                    let displayMag = Number(mag.toFixed(2));
                    let arrowLen = maxLoad > 0 ? 25 + (mag / maxLoad) * 30 : 40;
                    let dirX = fx / mag,
                        dirY = -fy / mag;

                    let n = model.nodes.find((node) => node.id === ld.nodeId);
                    if (n) {
                        let x = tx(n.x),
                            y = ty(n.y);

                        // Keep nodal-load graphics just above the joint so the
                        // arrowhead never enters or sits on top of the node circle.
                        // The same screen-space offset is used for vertical,
                        // horizontal and angled nodal loads.
                        const NODE_LOAD_OFFSET_Y = 11.3333333333;
                        const loadX = x;
                        const loadY = y - NODE_LOAD_OFFSET_Y;
                        let tx_tail = loadX - dirX * arrowLen,
                            ty_tail = loadY - dirY * arrowLen;
                        svg += drawSVGArrow(tx_tail, ty_tail, loadX, loadY, color);
                        labelLayer.addText({
                            x: tx_tail + 2, y: ty_tail + 8, text: `${displayMag} kN`,
                            fontSize: 11, fontWeight: "bold", color,
                            anchorX: tx_tail, anchorY: ty_tail, leader: false, halo: false,
                        });
                    }
                } else if (ld.type === "Element Point Load" || ld.type === "Element Load") {
                    let el = model.elements.find((e) => e.id === ld.elementId);
                    if (el) {
                        let comps = getGlobalLoadComponentsForDraw(ld, el);
                        let fx = comps.g1.x,
                            fy = comps.g1.y;
                        let mag = Math.hypot(fx, fy);
                        if (mag == 0) return;
                        let displayMag = Number(mag.toFixed(2));
                        let arrowLen = maxLoad > 0 ? 25 + (mag / maxLoad) * 45 : 40;
                        let dirX = fx / mag,
                            dirY = -fy / mag;

                        let n1 = model.nodes.find((n) => n.id === el.startNode);
                        let n2 = model.nodes.find((n) => n.id === el.endNode);
                        if (n1 && n2) {
                            let L = Math.hypot(n2.x - n1.x, n2.y - n1.y);
                            let a = ld.distance || 0,
                                t = L > 0 ? a / L : 0.5;
                            if (t < 0) t = 0;
                            if (t > 1) t = 1;
                            let sx = tx(n1.x) + t * (tx(n2.x) - tx(n1.x)),
                                sy = ty(n1.y) + t * (ty(n2.y) - ty(n1.y)) - 6.5;
                            let tx_tail = sx - dirX * arrowLen,
                                ty_tail = sy - dirY * arrowLen;
                            svg += drawSVGArrow(tx_tail, ty_tail, sx, sy, color);
                            labelLayer.addText({
                                x: tx_tail, y: ty_tail - 5, text: `${displayMag} kN`,
                                fontSize: 11, fontWeight: "bold", color,
                                anchorX: tx_tail, anchorY: ty_tail, leaderColor: color,
                            });
                        }
                    }
                } else if (ld.type === "Element UDL") {
                    let el = model.elements.find((e) => e.id === ld.elementId);
                    if (el) {
                        let n1 = model.nodes.find((n) => n.id === el.startNode);
                        let n2 = model.nodes.find((n) => n.id === el.endNode);
                        if (n1 && n2) {
```

```
let L = Math.hypot(n2.x - n1.x, n2.y - n1.y);
let a = ld.distance || 0;
let b =
    ld.distanceEnd !== undefined && ld.distanceEnd !== ""
        ? Number(ld.distanceEnd)
        : L;

// Draw the UDL using its actual GLOBAL direction.
// For Angled loads, the entered angle is measured counter-clockwise
// from global +X; therefore every arrow must use that rotated
// direction in the diagram, independent of the member angle.
const g1 = getGlobalLoadComponentsForDraw(ld, el).g1;
const g2 = getGlobalLoadComponentsForDraw(ld, el).g2;
let Fx1 = g1.x,
    Fy1 = g1.y;
let Fx2 = g2.x,
    Fy2 = g2.y;

let t_start = L > 0 ? a / L : 0;
let t_end = L > 0 ? b / L : 1;
if (t_start < 0) t_start = 0;
if (t_end > 1) t_end = 1;
if (t_start >= t_end) return;

let numArrows = 6,
    tailX = [],
    tailY = [],
    pathD = "";

// The UDL top line is defined by arrow heights, not by the
// fixed minimum arrow length used for ordinary point loads.
// Take the first arrow height as the reference, then linearly
// interpolate that height by distance so every arrow lies on
// the same UDL slope and a zero-magnitude end reaches zero.
const startMag = Math.hypot(Fx1, Fy1);
const endMag = Math.hypot(Fx2, Fy2);
const firstArrowLen = startMag > 0
    ? (maxLoad > 0 ? 25 + (startMag / maxLoad) * 45 : 40)
    : 0;
const lastArrowLen = endMag > 0
    ? (maxLoad > 0 ? 25 + (endMag / maxLoad) * 45 : 40)
    : 0;

for (let i = 0; i <= numArrows; i++) {
    let t_local = i / numArrows;
    let t = t_start + t_local * (t_end - t_start);
    let sx = tx(n1.x) + t * (tx(n2.x) - tx(n1.x));
    let sy = ty(n1.y) + t * (ty(n2.y) - ty(n1.y)) - 6.5;

    let curFx = Fx1 + t_local * (Fx2 - Fx1);
    let curFy = Fy1 + t_local * (Fy2 - Fy1);
    const mag = Math.hypot(curFx, curFy);

    if (mag > 0) {
        // SVG uses +Y downward, while structural global +Y is upward.
        // Convert the real global load vector to screen direction only here.
        const dirX = curFx / mag;
        const dirY = -curFy / mag;
        const arrowLen = firstArrowLen + (lastArrowLen - firstArrowLen) * t_local;
        let tx_tail = sx - dirX * arrowLen,
            ty_tail = sy - dirY * arrowLen;
        tailX.push(tx_tail);
        tailY.push(ty_tail);
        svg += drawSVGArrow(tx_tail, ty_tail, sx, sy, color);
        if (i === 0) pathD += `M ${tx_tail} ${ty_tail} `;
        else pathD += `L ${tx_tail} ${ty_tail} `;
    } else if (i === numArrows) {
        // A zero-intensity endpoint is part of the same straight
        // UDL taper, so its arrow height is exactly zero and the
        // slope line terminates at the actual zero-magnitude point.
        pathD += `L ${sx} ${sy} `;
    }
}
if (tailX.length > 0) {
    svg += `<path d="${pathD}" fill="none" stroke="${color}" stroke-width="2"/>`;

    // For a varying UDL, place each end magnitude at its
    // corresponding end of the load diagram rather than
    // combining both values into one centered label.
    // The label for the larger magnitude is above the
    // visually highest end; the smaller magnitude is below
    // the visually lowest end. No connecting arrow is used.
    if (Math.abs(startMag - endMag) > 1e-12) {
        const startT = t_start;
        const endT = t_end;
        const startSx = tx(n1.x) + startT * (tx(n2.x) - tx(n1.x));
        const startSy = ty(n1.y) + startT * (ty(n2.y) - ty(n1.y)) - 6.5;
```

```
                        const endSx = tx(n1.x) + endT * (tx(n2.x) - tx(n1.x));
                        const endSy = ty(n1.y) + endT * (ty(n2.y) - ty(n1.y)) - 6.5;

                        const startArrowLen = firstArrowLen;
                        const endArrowLen = lastArrowLen;

                        const startDirX = startMag > 0 ? Fx1 / startMag : 0;
                        const startDirY = startMag > 0 ? -Fy1 / startMag : 0;
                        const endDirX = endMag > 0 ? Fx2 / endMag : 0;
                        const endDirY = endMag > 0 ? -Fy2 / endMag : 0;

                        const startPoint = {
                            x: startSx - startDirX * startArrowLen,
                            y: startSy - startDirY * startArrowLen
                        };
                        const endPoint = {
                            x: endSx - endDirX * endArrowLen,
                            y: endSy - endDirY * endArrowLen
                        };

                        const highMag = Math.max(startMag, endMag);
                        const lowMag = Math.min(startMag, endMag);
                        const highPoint = startMag >= endMag ? startPoint : endPoint;
                        const lowPoint = startMag >= endMag ? endPoint : startPoint;

                        // Keep the UDL magnitude labels aligned with the actual
                        // slope of the UDL load diagram. Normalize the angle so
                        // the text stays readable rather than appearing upside down.
                        let udlLabelAngle = Math.atan2(
                            endPoint.y - startPoint.y,
                            endPoint.x - startPoint.x,
                        ) * 180 / Math.PI;
                        if (udlLabelAngle > 90) udlLabelAngle -= 180;
                        if (udlLabelAngle <= -90) udlLabelAngle += 180;

                        labelLayer.addText({
                            x: highPoint.x,
                            y: highPoint.y - 8,
                            text: `${Number(highMag.toFixed(2))} kN/m`,
                            fontSize: 11, fontWeight: "bold", color, angle: udlLabelAngle,
                            anchorX: highPoint.x, anchorY: highPoint.y,
                            leader: false, leaderColor: color,
                        });

                        labelLayer.addText({
                            x: lowPoint.x,
                            y: lowPoint.y - 11,
                            text: `${Number(lowMag.toFixed(2))} kN/m`,
                            fontSize: 11, fontWeight: "bold", color, angle: udlLabelAngle,
                            anchorX: lowPoint.x, anchorY: lowPoint.y,
                            leader: false, leaderColor: color,
                        });
                    } else {
                        const mx_tail = (tailX[0] + tailX[tailX.length - 1]) / 2;
                        const my_tail = (tailY[0] + tailY[tailY.length - 1]) / 2;
                        // The constant UDL follows the same straight load-diagram
                        // line, so align its single magnitude label to that slope.
                        let udlLabelAngle = Math.atan2(
                            tailY[tailY.length - 1] - tailY[0],
                            tailX[tailX.length - 1] - tailX[0],
                        ) * 180 / Math.PI;
                        if (udlLabelAngle > 90) udlLabelAngle -= 180;
                        if (udlLabelAngle <= -90) udlLabelAngle += 180;
                        labelLayer.addText({
                            x: mx_tail - 20, y: my_tail - 8, text: `${Number(startMag.toFixed(2))} kN/m`,
                            fontSize: 11, fontWeight: "bold", color, angle: udlLabelAngle,
                            anchorX: mx_tail, anchorY: my_tail, leader: false, leaderColor: color,
                        });
                    }
                }
            }
        }
    });

model.moments.forEach((m) => {
    const mz = Number(m.Mz) || 0;
    if (mz === 0) return;
    const r = maxMoment > 0 ? 9.3333333333 + (Math.abs(mz) / maxMoment) * 4.4444444445 : 12;
    const color = "#8b5cf6";
    const anticlockwise = mz > 0;

    if (m.type === "Nodal Moment") {
        const n = model.nodes.find((node) => node.id === m.nodeId);
        if (n) {
            const x = tx(n.x);
            const y = ty(n.y);
```

```
                    svg += drawMomentArrow(x, y, r, anticlockwise, color);
                    labelLayer.addText({
                        x: x + r - 1, y: y + r - 20.6666666667, text: `${Number(Math.abs(mz).toFixed(2))} kN·m`,
                        fontSize: 11, fontWeight: "bold", color,
                        anchorX: x, anchorY: y, leader: false, leaderColor: color,
                    });
                }
            } else if (m.type === "Element Moment") {
                const el = model.elements.find((e) => e.id === m.elementId);
                if (el) {
                    const n1 = model.nodes.find((n) => n.id === el.startNode);
                    const n2 = model.nodes.find((n) => n.id === el.endNode);
                    if (n1 && n2) {
                        const L = Math.hypot(n2.x - n1.x, n2.y - n1.y);
                        let a = Number(m.distance) || 0;
                        const t = L > 0 ? Math.max(0, Math.min(1, a / L)) : 0.5;
                        const x = tx(n1.x) + t * (tx(n2.x) - tx(n1.x));
                        const y = ty(n1.y) + t * (ty(n2.y) - ty(n1.y));
                        svg += drawMomentArrow(x, y, r, anticlockwise, color);
                        labelLayer.addText({
                            x: x + r + 2, y: y + r - 30, text: `${Number(Math.abs(mz).toFixed(2))} kN·m`,
                            fontSize: 11, fontWeight: "bold", color,
                            anchorX: x, anchorY: y, leader: false, leaderColor: color,
                        });
                    }
                }
            }
        });
    }

    if (mode === "model") {
        drawModelModeLoadsAndMoments();
    }

    if (mode === "displacement") {
        drawDisplacementDiagram();
    }

    // Draws the "displacement" (deformed shape) diagram: scales every
    // solved nodal translation by a fixed visual envelope (28% of the
    // structure's largest dimension) so the deformation stays readable
    // regardless of actual magnitude, then interpolates a smooth deformed
    // shape along each element using its local translations and end
    // rotations (cubic Hermite shape functions), and overlays the
    // undeformed structure as a dashed reference outline.
    function drawDisplacementDiagram() {
        const disp = model.nodeDisplacements || {};
        let maxDisp = 0;
        Object.values(disp).forEach((d) => {
            if (!d) return;
            const mag = isGridAnalysisMode()
                ? Math.abs(Number(d.deltaZ) || 0)
                : Math.hypot(Number(d.ux) || 0, Number(d.uy) || 0);
            if (mag > maxDisp) maxDisp = mag;
        });

        if (maxDisp <= 1e-14) {
            svg += `<text x="${width / 2}" y="${height / 2}" fill="#64748b" font-family="Arial" font-size="14" font-weight="600" text-anchor="middle">All calculated nodal translations are
zero — no visible deformation.</text>`;
        } else {
            // Keep deformation visually readable while preserving the actual
            // relative deformation pattern. The scale is visualization-only.
            const deformationScale = (0.28 * maxDim) / maxDisp;

            const screenPoint = (x, y) => ({ x: tx(x), y: ty(y) });
            const interpolateDeformedPoint = (n1, n2, t, L, cx, cy, d1, d2) => {
                // Grid's solved ΔZ is normal to the XY layout. The shared 2D
                // diagram projects it onto the displayed Y direction and uses
                // the member-local θY component to preserve the bending shape.
                if (isGridAnalysisMode()) {
                    const w1 = Number(d1.deltaZ) || 0;
                    const w2 = Number(d2.deltaZ) || 0;
                    const slope1 = -(Number(d1.thetaX) || 0) * cy + (Number(d1.thetaY) || 0) * cx;
                    const slope2 = -(Number(d2.thetaX) || 0) * cy + (Number(d2.thetaY) || 0) * cx;
                    const H1 = 1 - 3 * t * t + 2 * t * t * t;
                    const H2 = L * (t - 2 * t * t + t * t * t);
                    const H3 = 3 * t * t - 2 * t * t * t;
                    const H4 = L * (-t * t + t * t * t);
                    return {
                        x: n1.x + (n2.x - n1.x) * t,
                        y: n1.y + (n2.y - n1.y) * t + (H1 * w1 + H2 * slope1 + H3 * w2 + H4 * slope2) * deformationScale,
                    };
                }
                const u1 = (d1.ux || 0) * cx + (d1.uy || 0) * cy;
                const v1 = -(d1.ux || 0) * cy + (d1.uy || 0) * cx;
                const u2 = (d2.ux || 0) * cx + (d2.uy || 0) * cy;
                const v2 = -(d2.ux || 0) * cy + (d2.uy || 0) * cx;
```

```
    const th1 = d1.rz || 0;
    const th2 = d2.rz || 0;

    // Linear axial interpolation + cubic Hermite bending shape
    // functions for transverse displacement.
    const N1 = 1 - t;
    const N2 = t;
    const H1 = 1 - 3 * t * t + 2 * t * t * t;
    const H2 = L * (t - 2 * t * t + t * t * t);
    const H3 = 3 * t * t - 2 * t * t * t;
    const H4 = L * (-t * t + t * t * t);

    const u = (N1 * u1 + N2 * u2) * deformationScale;
    const v = (H1 * v1 + H2 * th1 + H3 * v2 + H4 * th2) * deformationScale;

    const baseX = n1.x + (n2.x - n1.x) * t;
    const baseY = n1.y + (n2.y - n1.y) * t;
    return {
        x: baseX + u * cx - v * cy,
        y: baseY + u * cy + v * cx,
    };
};

// Original geometry as a quiet reference.
model.elements.forEach((el) => {
    const n1 = model.nodes.find((n) => n.id === el.startNode);
    const n2 = model.nodes.find((n) => n.id === el.endNode);
    if (!n1 || !n2) return;
    const p1 = screenPoint(n1.x, n1.y);
    const p2 = screenPoint(n2.x, n2.y);
    svg += `<line x1="${p1.x}" y1="${p1.y}" x2="${p2.x}" y2="${p2.y}" stroke="#cbd5e1" stroke-width="2" stroke-dasharray="6 4"/>`;
});

// Deformed shape, sampled finely enough to make frame bending
// smooth instead of a simple straight-node displacement plot.
model.elements.forEach((el) => {
    const n1 = model.nodes.find((n) => n.id === el.startNode);
    const n2 = model.nodes.find((n) => n.id === el.endNode);
    const d1 = disp[n1?.id];
    const d2 = disp[n2?.id];
    if (!n1 || !n2 || !d1 || !d2) return;

    const dx = n2.x - n1.x;
    const dy = n2.y - n1.y;
    const L = Math.hypot(dx, dy);
    if (L <= 1e-12) return;
    const cx = dx / L;
    const cy = dy / L;

    let path = "";
    const samples = 48;
    for (let i = 0; i <= samples; i++) {
        const t = i / samples;
        const p = interpolateDeformedPoint(n1, n2, t, L, cx, cy, d1, d2);
        const sp = screenPoint(p.x, p.y);
        path += `${i === 0 ? "M" : "L"} ${sp.x} ${sp.y} `;
    }
    svg += `<path d="${path}" fill="none" stroke="#2563eb" stroke-width="3" stroke-linecap="round"/>`;

    // A few deformation vectors make the scale and direction obvious.
    for (let i = 0; i <= 4; i++) {
        const t = i / 4;
        const base = { x: n1.x + dx * t, y: n1.y + dy * t };
        const p = interpolateDeformedPoint(n1, n2, t, L, cx, cy, d1, d2);
        const deltaX = p.x - base.x;
        const deltaY = p.y - base.y;
        const mag = Math.hypot(deltaX, deltaY);
        if (mag <= 1e-10) continue;
        const b = screenPoint(base.x, base.y);
        const q = screenPoint(p.x, p.y);
        svg += drawSVGArrow(b.x, b.y, q.x, q.y, "#60a5fa").replace(/stroke-width="2"/g, 'stroke-width="1.2"');
    }
});

// Deformed joint locations and displacement labels.
model.nodes.forEach((n) => {
    const d = disp[n.id];
    if (!d) return;
    const projectedOffset = isGridAnalysisMode()
        ? (Number(d.deltaZ) || 0) * deformationScale
        : (Number(d.uy) || 0) * deformationScale;
    const p = screenPoint(n.x + (isGridAnalysisMode() ? 0 : (d.ux || 0) * deformationScale), n.y + projectedOffset);
    const original = screenPoint(n.x, n.y);
    svg += `<circle cx="${p.x}" cy="${p.y}" r="5" fill="#2563eb" stroke="#1e3a8a" stroke-width="1"/>`;
    svg += `<line x1="${original.x}" y1="${original.y}" x2="${p.x}" y2="${p.y}" stroke="#93c5fd" stroke-width="1" stroke-dasharray="3 3"/>`;
    const mag = isGridAnalysisMode() ? Math.abs(Number(d.deltaZ) || 0) : Math.hypot(d.ux || 0, d.uy || 0);
    labelLayer.addText({
```

```
                x: p.x + 8, y: p.y - 8, text: `${n.id}: ${formatNum(mag)}`,
                fontSize: 10, fontWeight: "700", color: "#1d4ed8",
                anchorX: p.x, anchorY: p.y, leaderColor: "#1d4ed8",
            });
        });

    }
}

if (mode === "reactions") {
    drawReactionsDiagram();
}

// Draws the "reactions" diagram: for every support, a leader line to a
// label card listing GLOBAL Rx/Ry/Mz and LOCAL (member-axis) Fx'/Fy'/Mz',
// laid out via the shared label-collision engine (labelLayer) so cards
// never overlap the structure or each other, at any zoom level.
function drawReactionsDiagram() {
    // Reactions are solved in GLOBAL coordinates. The diagram reports
    // both GLOBAL and LOCAL reaction components so the plotted values
    // can be matched directly to the two coordinate systems shown in
    // the final reaction table. In 2D, Mz is invariant under in-plane
    // axis rotation, so Global Mz = Local Mz.
    //
    // Pure label-box layout (STAAD Pro / Bentley style): no arrows or
    // arcs are drawn at the joint at all. Each support only gets a
    // dashed leader line to a value card listing Rx/Ry/Mz. Card
    // positions are pushed out into open space, then run through a
    // simple separation pass so no two cards can ever land on top of
    // one another - this holds for any number of supports at any angle.
    const frameNodes = model.nodes.filter(Boolean);
    const frameCenter = frameNodes.length
        ? {
            x: frameNodes.reduce((sum, node) => sum + node.x, 0) / frameNodes.length,
            y: frameNodes.reduce((sum, node) => sum + node.y, 0) / frameNodes.length,
          }
        : { x: 0, y: 0 };

    const cards = [];

    model.supportAssignments.forEach((sa) => {
        const n = model.nodes.find((node) => node.id === sa.nodeId);
        if (!n) return;

        const local = getReactionLocalComponents(sa);
        const globalFx = Number(sa.rx) || 0;
        const globalFy = Number(sa.ry) || 0;
        const globalMz = Number(sa.rz) || 0;
        const nodeX = tx(n.x);
        const nodeY = ty(n.y);

        const lines = [];
        if (isGridAnalysisMode()) {
            if (Math.abs(globalFx) > 1e-5) lines.push({ text: `Mx = ${Number(globalFx.toFixed(2))} kN·m`, color: "#7c3aed" });
            if (Math.abs(globalFy) > 1e-5) lines.push({ text: `My = ${Number(globalFy.toFixed(2))} kN·m`, color: "#7c3aed" });
            if (Math.abs(globalMz) > 1e-5) lines.push({ text: `Fz = ${Number(globalMz.toFixed(2))} kN`, color: "#2563eb" });
        } else {
            if (Math.abs(globalFx) > 1e-5 || Math.abs(globalFy) > 1e-5) {
                lines.push({ text: `Global Fx = ${Number(globalFx.toFixed(2))} kN`, color: "#2563eb" });
                lines.push({ text: `Global Fy = ${Number(globalFy.toFixed(2))} kN`, color: "#2563eb" });
            }
            if (Math.abs(globalMz) > 1e-5) {
                lines.push({ text: `Global Mz = ${Number(globalMz.toFixed(2))} kN·m`, color: "#7c3aed" });
            }
        }
        if (lines.length === 0) return;

        // --- Outward direction, used to push the label card into open
        // space (away from the structure's own centroid on screen) -----
        const worldDx = n.x - frameCenter.x;
        const worldDy = n.y - frameCenter.y;
        let ox = worldDx, oy = -worldDy;
        let omag = Math.hypot(ox, oy);
        if (omag < 1e-6) { ox = 0; oy = 1; omag = 1; }
        ox /= omag; oy /= omag;

        cards.push({ nodeX, nodeY, ox, oy, lines });
    });

    // --- Size each card to fit exactly the text it needs to show -------
    const lineHeight = 14, padX = 8, padY = 6, charWidth = 5.4;
    cards.forEach((c) => {
        const maxChars = Math.max(...c.lines.map((l) => l.text.length));
        c.width = maxChars * charWidth + padX * 2;
        c.height = c.lines.length * lineHeight + padY * 2;
    });
    // --- Place each card out past the joint. Final separation from
```

```
    // every other label on the diagram (other cards, node IDs, member
    // length labels, ...) is handled once, uniformly, by the shared
    // label-collision engine (labelLayer.resolve, called once at the
    // end of this function) instead of a separate bespoke pass. -------
    const baseGap = 20;
    cards.forEach((c) => {
        const dist = baseGap + Math.hypot(c.width, c.height) / 2;
        c.cx = c.nodeX + c.ox * dist;
        c.cy = c.nodeY + c.oy * dist;
    });

    cards.forEach((c) => {
        labelLayer.addBlock({
            x: c.cx, y: c.cy, w: c.width + 12, h: c.height + 10,
            anchorX: c.nodeX, anchorY: c.nodeY, leader: false,
            render: (cx, cy) => {
                const left = cx - c.width / 2;
                const top = cy - c.height / 2;
                const dx = c.nodeX - cx, dy = c.nodeY - cy;
                let edgeX = cx, edgeY = cy;
                if (dx !== 0 || dy !== 0) {
                    const scaleX = dx !== 0 ? (c.width / 2) / Math.abs(dx) : Infinity;
                    const scaleY = dy !== 0 ? (c.height / 2) / Math.abs(dy) : Infinity;
                    const s = Math.min(scaleX, scaleY);
                    edgeX = cx + dx * s;
                    edgeY = cy + dy * s;
                }
                let out = "";
                out += `<rect x="${left}" y="${top}" width="${c.width}" height="${c.height}" rx="5" fill="white" fill-opacity="0.97" stroke="#cbd5e1" stroke-width="1.1"/>`;
                c.lines.forEach((line, idx) => {
                    const ly = top + padY + idx * lineHeight + lineHeight * 0.74;
                    out += `<text x="${cx}" y="${ly}" fill="${line.color}" font-family="Arial" font-size="10" font-weight="700" text-anchor="middle">${line.text}</text>`;
                });
                return out;
            },
        });
    });
}

if (mode === "afd" || mode === "sfd" || mode === "bmd") {
    drawInternalActionDiagram();
}

// Draws the AFD/SFD/BMD internal-action diagram curve for every solved
// element: samples calculateInternalActionsAtPosition() at each of the
// candidate x-positions, offsets each point perpendicular to the member
// by the appropriate pre-computed scale (scaleA/scaleV/scaleM from
// computeInternalActionExtremes(), see above), fills contiguous
// positive/negative sign regions, hatches them, and — for BMD only —
// labels every local peak/valley.
function drawInternalActionDiagram() {
    // Labels already queued for BMD, shared across members so the same
    // value at a shared joint is printed once instead of once per member.
    const placedBmdLabels = [];
    // Per-member diagram outlines / axes and the value labels waiting to be
    // placed. Labels are placed AFTER every member is drawn so each one can
    // avoid the other members' diagrams (see placeActionLabels below).
    const actionRegions = [];
    const pendingActionLabels = [];
    model.elements.forEach((el, ei) => {
        if (!el.plotData) return;
        let n1 = model.nodes.find((n) => n.id === el.startNode);
        let pd = el.plotData;

        let xs = [];
        for (let i = 0; i <= 50; i++) xs.push((i / 50) * pd.L);
        pd.pointLoads.forEach((pl) => {
            xs.push(pl.a - 1e-6);
            xs.push(pl.a + 1e-6);
        });
        pd.localPointMoments.forEach((pm) => {
            xs.push(pm.a - 1e-6);
            xs.push(pm.a + 1e-6);
        });
        pd.localUDLs.forEach((udl) => {
            xs.push(udl.a);
            xs.push(udl.b);
        });
        xs.sort((a, b) => a - b);

        // Builds one plotted point of this member's diagram at position x.
        const makePt = (x) => {
            const { A: A_ax, V, M } = calculateInternalActionsAtPosition(pd, x);

            let offset = 0;
            if (mode === "afd") offset = A_ax * scaleA;
            if (mode === "sfd") offset = V * scaleV;
```

```
            if (mode === "bmd") offset = M * scaleM;

            let gx = n1.x + x * pd.cx;
            let gy = n1.y + x * pd.cy;

            let ox = gx - offset * pd.cy;
            let oy = gy + offset * pd.cx;

            return {
                tx: tx(ox),
                ty: ty(oy),
                baseTx: tx(gx),
                baseTy: ty(gy),
                A: A_ax,
                V: V,
                M: M,
                x: x,
            };
        };
        let pts = xs.map(makePt);

        if (pts.length === 0) return;

        // Everything drawn for this member's diagram (fills + hatching) is
        // grouped so it can be focused / brought to front as one unit.
        svg += `<g class="sf-diagram" data-member="${ei}">`;

        // Preserve the original diagram geometry, opacity, hatching, line widths,
        // and hatch spacing. Only the COLOR changes according to the sign of the
        // plotted quantity. Positive/negative regions are split only for coloring.
        const diagramColors = {
            afd: { positive: "#16a34a", negative: "#db2777" },
            sfd: { positive: "#2563eb", negative: "#f59e0b" },
            bmd: { positive: "#dc2626", negative: "#7c3aed" },
        };
        const valueForMode = (p) =>
            mode === "afd" ? p.A : mode === "sfd" ? p.V : p.M;
        const colorForValue = (v) =>
            v < -1e-12 ? diagramColors[mode].negative : diagramColors[mode].positive;

        // Draw the filled diagram in contiguous sign regions. All original SVG
        // styling values are kept exactly: fill opacity 0.2, stroke-width 1.5,
        // hatch stroke-width 0.5 and hatch opacity 0.5.
        function addSignRegion(start, end, color) {
            if (end - start < 1) return;
            let rd = `M ${pts[start].baseTx} ${pts[start].baseTy} `;
            for (let j = start; j <= end; j++) rd += `L ${pts[j].tx} ${pts[j].ty} `;
            for (let j = end; j >= start; j--) rd += `L ${pts[j].baseTx} ${pts[j].baseTy} `;
            svg += `<path d="${rd}" fill="${mode === "afd" ? (color === diagramColors[mode].positive ? "rgba(34, 197, 94, 0.2)" : "rgba(219, 39, 119, 0.2)") : mode === "sfd" ? (color
=== diagramColors[mode].positive ? "rgba(59, 130, 246, 0.2)" : "rgba(245, 158, 11, 0.2)") : (color === diagramColors[mode].positive ? "rgba(239, 68, 68, 0.2)" : "rgba(124, 58, 237, 0.2)")}"
stroke="${color}" stroke-width="1.5"/>`;
        }

        // The original base fill colors remain the fill colors; only the visible
        // stroke is sign-coded. This avoids changing opacity/shading/patterns.
        let regionStart = 0;
        let regionColor = colorForValue(valueForMode(pts[0]));
        for (let i = 1; i < pts.length; i++) {
            const nextColor = colorForValue(valueForMode(pts[i]));
            if (nextColor !== regionColor) {
                addSignRegion(regionStart, i - 1, regionColor);
                regionStart = i - 1;
                regionColor = nextColor;
                if (regionStart > 0) {
                    // Re-draw the boundary point as part of the new region so the
                    // original continuous diagram edge remains visually identical.
                    continue;
                }
            }
        }
        addSignRegion(regionStart, pts.length - 1, regionColor);

        // Keep the original hatch pattern exactly: same every-2-point spacing,
        // direction, width and opacity. Only hatch color follows the local sign.
        for (let i = 0; i < pts.length; i += 2) {
            const hatchColor = colorForValue(valueForMode(pts[i]));
            svg += `<line x1="${pts[i].baseTx}" y1="${pts[i].baseTy}" x2="${pts[i].tx}" y2="${pts[i].ty}" stroke="${hatchColor}" stroke-width="0.5" opacity="0.5"/>`;
        }

        svg += `</g>`;

        {
            const poly = pts.map((q) => [q.tx, q.ty]).concat(pts.slice().reverse().map((q) => [q.baseTx, q.baseTy]));
            actionRegions.push({
                key: ei, poly,
                axis: [pts[0].baseTx, pts[0].baseTy, pts[pts.length - 1].baseTx, pts[pts.length - 1].baseTy],
            });
```

```
        }

        {
            // Value labels for AFD / SFD / BMD. Every important value of the
            // member is labelled:
            //   1. both end values,
            //   2. the largest value of every positive / negative part
            //      (for the BMD also the exact peak where the shear is zero),
            //   3. both sides of every jump of the diagram (a point load / moment
            //      makes the diagram jump, e.g. from + to - or - to +).
            const tol = 1e-9;
            const unit = mode === "bmd" || (isGridAnalysisMode() && mode === "afd") ? "kN·m" : "kN";
            const valueOf = (q) => valueForMode(q);
            const shown = (q) => Math.abs(Number(valueOf(q)) || 0).toFixed(2);

            // Candidate points: the plotted samples (+ the exact BMD peaks).
            const cand = pts.slice();
            if (mode === "bmd") {
                for (let i = 0; i < pts.length - 1; i++) {
                    const v0 = pts[i].V, v1 = pts[i + 1].V;
                    if (pts[i + 1].x - pts[i].x < 1e-4) continue; // a jump, not a smooth span
                    if (v0 * v1 < 0) {
                        const xr = pts[i].x + ((pts[i + 1].x - pts[i].x) * v0) / (v0 - v1);
                        cand.push(makePt(xr));
                    }
                }
                cand.sort((p, q) => p.x - q.x);
            }

            const chosen = [];
            const choose = (q) => { if (q && !chosen.includes(q)) chosen.push(q); };

            // 1. both ends
            choose(pts[0]);
            choose(pts[pts.length - 1]);

            // 2. maximum of every sign region
            const signOf = (q) => {
                const v = Number(valueOf(q)) || 0;
                return Math.abs(v) <= tol ? 0 : v > 0 ? 1 : -1;
            };
            let regBest = null, regSign = 0;
            cand.forEach((q) => {
                const sg = signOf(q);
                if (sg === 0) return;
                if (regBest === null || sg !== regSign) {
                    if (regBest) choose(regBest);
                    regBest = q; regSign = sg;
                } else if (Math.abs(valueOf(q)) > Math.abs(valueOf(regBest))) {
                    regBest = q;
                }
            });
            if (regBest) choose(regBest);

            // 3. both sides of every jump (samples taken either side of a load)
            for (let i = 0; i < pts.length; ) {
                let j = i;
                while (j + 1 < pts.length && pts[j + 1].x - pts[j].x < 1e-4) j++;
                if (j > i) {
                    const first = pts[i], last = pts[j];
                    if (Math.abs(valueOf(first) - valueOf(last)) > 1e-6) {
                        choose(first);
                        choose(last);
                    }
                }
                i = j + 1;
            }

            chosen.forEach((p) => {
                const txt = shown(p);
                if (Math.abs(Number(valueOf(p)) || 0) <= tol || txt === "0.00") return;
                // Same value already labelled at (almost) the same spot, e.g. at a
                // joint shared by two members: print it once.
                const signKey = (Number(valueOf(p)) < 0 ? '-' : '+') + txt;
                if (placedBmdLabels.some((q) => q.key === signKey && Math.hypot(q.x - p.tx, q.y - p.ty) < 6)) return;
                // Same value right beside a label of this member: skip the repeat.
                if (pendingActionLabels.some((q) => q.key === ei && q.text === `${txt} ${unit}` && q.color === colorForValue(valueOf(p)) && Math.hypot(q.p.tx - p.tx, q.p.ty - p.ty) <
40)) return;
                placedBmdLabels.push({ key: signKey, x: p.tx, y: p.ty });
                pendingActionLabels.push({
                    p, text: `${txt} ${unit}`, color: colorForValue(valueOf(p)), key: ei,
                });
            });
        }
    });
    placeActionLabels(actionRegions, pendingActionLabels);
}
```

```
// Places every AFD/SFD/BMD value label so it is unambiguous which diagram
// it belongs to. For each label many candidate spots around its own point
// are scored; a spot is rejected if it would sit on another member's
// diagram, a member line, a node, or another label. The nearest clean spot
// wins, and whenever a label could not stay right at its point a thin
// leader line (same colour) with a dot marks the exact value it describes.
function placeActionLabels(regions, labels) {
    const fontSize = 12, pad = 3;
    const nodeBoxes = model.nodes.map((n) => ({ x: tx(n.x), y: ty(n.y), r: actionNodeRadius }));
    const placed = [];

    // Bounding box per region so far-away members are skipped immediately
    // (keeps placement fast on models with many members).
    regions.forEach((rg) => {
        let x0 = Infinity, y0 = Infinity, x1 = -Infinity, y1 = -Infinity;
        rg.poly.forEach((q) => {
            if (q[0] < x0) x0 = q[0];
            if (q[0] > x1) x1 = q[0];
            if (q[1] < y0) y0 = q[1];
            if (q[1] > y1) y1 = q[1];
        });
        rg.bb = [x0, y0, x1, y1];
    });

    const inPoly = (poly, x, y) => {
        let inside = false;
        for (let i = 0, j = poly.length - 1; i < poly.length; j = i++) {
            const xi = poly[i][0], yi = poly[i][1], xj = poly[j][0], yj = poly[j][1];
            if ((yi > y) !== (yj > y) && x < ((xj - xi) * (y - yi)) / (yj - yi || 1e-12) + xi) inside = !inside;
        }
        return inside;
    };
    const distSeg = (x, y, x1, y1, x2, y2) => {
        const dx = x2 - x1, dy = y2 - y1;
        const l2 = dx * dx + dy * dy;
        let t = l2 ? ((x - x1) * dx + (y - y1) * dy) / l2 : 0;
        t = t < 0 ? 0 : t > 1 ? 1 : t;
        return Math.hypot(x - (x1 + t * dx), y - (y1 + t * dy));
    };

    // Candidate spots, tried nearest-first. Because every penalty only adds
    // to the base cost, the search can stop as soon as the next candidate's
    // base cost is no better than the best clean spot already found.
    const cands = [];
    [4, 10, 18, 28, 40, 54, 70, 90].forEach((d) => {
        [0, 25, -25, 50, -50, 80, -80, 110, -110, 150, -150, 180].forEach((t) => {
            cands.push({ d, t, base: d + Math.abs(t) * 0.15, rad: (t * Math.PI) / 180 });
        });
    });
    cands.sort((u, v) => u.base - v.base);

    labels.forEach((lb) => {
        const w = String(lb.text).length * fontSize * 0.62 + pad * 2;
        const h = fontSize * 1.3 + pad * 2;
        const hw = w / 2, hh = h / 2;
        const ax = lb.p.tx, ay = lb.p.ty;

        // Outward direction (axis -> plotted point); fall back to "up".
        let ox = lb.p.tx - lb.p.baseTx, oy = lb.p.ty - lb.p.baseTy;
        const ol = Math.hypot(ox, oy);
        if (ol < 1e-6) { ox = 0; oy = -1; } else { ox /= ol; oy /= ol; }
        const baseAng = Math.atan2(oy, ox);

        // Only things within reach of this label's search area matter.
        const R = 100 + 2 * Math.max(hw, hh);
        const nearRegions = regions.filter((rg) => rg.bb[0] < ax + R && rg.bb[2] > ax - R && rg.bb[1] < ay + R && rg.bb[3] > ay - R);
        const nearNodes = nodeBoxes.filter((nb) => Math.abs(nb.x - ax) < R && Math.abs(nb.y - ay) < R);
        const nearPlaced = placed.filter((q) => Math.abs(q.cx - ax) < R + q.hw && Math.abs(q.cy - ay) < R + q.hh);
        const nearLabels = labels.filter((o) => o !== lb && Math.abs(o.p.tx - ax) < R && Math.abs(o.p.ty - ay) < R);

        // Sample grid inside the label box used for overlap tests.
        const samples = [];
        for (let ix = -2; ix <= 2; ix++) for (let iy = -1; iy <= 1; iy++) samples.push([ix * hw * 0.5, iy * hh * 0.8]);

        let best = null;
        for (let ci = 0; ci < cands.length; ci++) {
            const c = cands[ci];
            if (best && c.base >= best.score) break;
            const ang = baseAng + c.rad;
            const dx = Math.cos(ang), dy = Math.sin(ang);
            const cx = ax + dx * c.d + dx * hw * Math.abs(dx);
            const cy = ay + dy * c.d + dy * hh * Math.abs(dy);
            let score = c.base;

            // Canvas bounds.
            if (cx - hw < 2 || cx + hw > width - 2 || cy - hh < 2 || cy + hh > height - 2) score += 20000;
            // Other labels.
```

```
                for (let k = 0; k < nearPlaced.length; k++) {
                    const q = nearPlaced[k];
                    if (Math.abs(cx - q.cx) < hw + q.hw + 3 && Math.abs(cy - q.cy) < hh + q.hh + 2) score += 9000;
                }
                // The leader (anchor -> nearest box point) must not run through
                // another label's anchor dot, a node circle, or a placed label.
                const lx = Math.max(cx - hw, Math.min(ax, cx + hw));
                const ly = Math.max(cy - hh, Math.min(ay, cy + hh));
                if (Math.hypot(lx - ax, ly - ay) > 6) {
                    for (let k = 0; k < nearLabels.length; k++) {
                        if (distSeg(nearLabels[k].p.tx, nearLabels[k].p.ty, ax, ay, lx, ly) < 6) score += 7000;
                    }
                    for (let k = 0; k < nearNodes.length; k++) {
                        const nb = nearNodes[k];
                        if (distSeg(nb.x, nb.y, ax, ay, lx, ly) < nb.r + 3 && Math.hypot(nb.x - ax, nb.y - ay) > nb.r + 1) score += 7000;
                    }
                    for (let k = 0; k < nearPlaced.length; k++) {
                        const q = nearPlaced[k];
                        for (let m = 1; m < 8; m++) {
                            const px = ax + ((lx - ax) * m) / 8, py = ay + ((ly - ay) * m) / 8;
                            if (Math.abs(px - q.cx) < q.hw && Math.abs(py - q.cy) < q.hh) { score += 7000; break; }
                        }
                    }
                }
                // Node markers.
                for (let k = 0; k < nearNodes.length; k++) {
                    const nb = nearNodes[k];
                    if (Math.abs(cx - nb.x) < hw + nb.r + 2 && Math.abs(cy - nb.y) < hh + nb.r + 2) score += 6000;
                }
                // Member lines and diagrams.
                for (let k = 0; k < nearRegions.length; k++) {
                    const rg = nearRegions[k];
                    if (rg.bb[0] > cx + hw + 4 || rg.bb[2] < cx - hw - 4 || rg.bb[1] > cy + hh + 4 || rg.bb[3] < cy - hh - 4) continue;
                    const own = rg.key === lb.key;
                    for (let m = 0; m < samples.length; m++) {
                        const x = cx + samples[m][0], y = cy + samples[m][1];
                        if (distSeg(x, y, rg.axis[0], rg.axis[1], rg.axis[2], rg.axis[3]) < 4) score += own ? 150 : 700;
                        if (x >= rg.bb[0] && x <= rg.bb[2] && y >= rg.bb[1] && y <= rg.bb[3] && inPoly(rg.poly, x, y)) score += own ? 40 : 1200;
                    }
                }

                if (!best || score < best.score) best = { score, cx, cy, d: c.d };
            }

            placed.push({ cx: best.cx, cy: best.cy, hw, hh });

            // Every value gets a dot on the exact point it describes; a leader
            // line joins dot and label whenever the label is not right beside it.
            const nx = Math.max(best.cx - hw, Math.min(ax, best.cx + hw));
            const ny = Math.max(best.cy - hh, Math.min(ay, best.cy + hh));
            const gap = Math.hypot(nx - ax, ny - ay);
            const leaderLine = gap > 3
                ? `<line x1="${ax.toFixed(2)}" y1="${ay.toFixed(2)}" x2="${nx.toFixed(2)}" y2="${ny.toFixed(2)}" stroke="${lb.color}" stroke-width="1.4" stroke-linecap="round"/>`
                : "";
            actionOverlaySvg += `<g class="sf-label" data-member="${escapeHtml(String(lb.key))}">${leaderLine}<circle cx="${ax.toFixed(2)}" cy="${ay.toFixed(2)}" r="3" fill="${lb.color}"
stroke="white" stroke-width="1.2"/></g>`;

            labelLayer.addText({
                x: best.cx, y: best.cy + fontSize * 0.32, text: lb.text,
                fontSize, fontWeight: "bold", color: lb.color, textAnchor: "middle",
                anchorX: ax, anchorY: ay, leaderColor: lb.color,
                lockPosition: true, memberKey: lb.key,
            });
        });
    }

    // Resolve all ordinary labels first. The node markers are intentionally
    // rendered afterwards so the node circle and node number stay at the top
    // SVG drawing level and can never be hidden by loads, moments, diagrams,
    // support symbols, deformation graphics, or other result annotations.
    const resolvedLabelsSvg = labelLayer.resolve({ minX: 0, minY: 0, maxX: width, maxY: height });
    if (!isInternalActionMode) svg += resolvedLabelsSvg;

    // Top-level node markers for all normal diagrams. The dedicated DoF
    // renderer is separate and never reaches this function.
    if (isInternalActionMode) svg += `<g class="sf-nodes">`;
    model.nodes.forEach((n) => {
        const x = tx(n.x);
        const y = ty(n.y);
        const nFill = mode === "model" || mode === "reactions" ? "#f59e0b" : "#e2e8f0";
        const nodeRadius = isInternalActionMode ? actionNodeRadius : 9.3333333333;
        const nodeStrokeWidth = mode === "model" ? 1 : 1;
        if (isInternalActionMode) svg += `<g class="sf-node" data-node="${escapeHtml(String(n.id))}">`;
        svg += `<circle cx="${x}" cy="${y}" r="${nodeRadius}" fill="${nFill}" stroke="#1e293b" stroke-width="${nodeStrokeWidth}"/>`;
        if (mode === "model") {
            const nodeIdSize = Math.max(5.0666666667, Math.min(5.8666666667, 18 / String(n.id).length));
            const coordXText = `x=${Number(n.x).toFixed(2)}`;
```

```
            const coordYText = `y=${Number(n.y).toFixed(2)}`;
            svg += `<text x="${x}" y="${y - 1.2}" text-anchor="middle"
                font-family="Arial" font-size="${nodeIdSize}" font-weight="700" fill="#000000">${escapeHtml(String(n.id))}</text>`;
            svg += `<text x="${x}" y="${y + 2.1}" text-anchor="middle"
                font-family="Arial" font-size="3.6666666667" font-weight="700" fill="#000000">${coordXText}</text>`;
            svg += `<text x="${x}" y="${y + 5.6}" text-anchor="middle"
                font-family="Arial" font-size="3.6666666667" font-weight="700" fill="#000000">${coordYText}</text>`;
        } else if (isInternalActionMode) {
            // Compact marker: node ID only (N1, N2, ...), no coordinates.
            const rawId = String(n.id);
            const idText = /^n/i.test(rawId) ? rawId : `N${rawId}`;
            const idSize = Math.max(3.8, Math.min(6.4, (2 * actionNodeRadius * 0.82) / (idText.length * 0.62)));
            svg += `<text x="${x}" y="${y + idSize * 0.35}" text-anchor="middle"
                font-family="Arial" font-size="${idSize.toFixed(2)}" font-weight="700" fill="#000000">${escapeHtml(idText)}</text>`;
        } else {
            const nodeLabelSize = Math.max(5.8, Math.min(6.5, 21 / String(n.id).length));
            const coordXText = `x=${Number(n.x).toFixed(2)}`;
            const coordYText = `y=${Number(n.y).toFixed(2)}`;
            svg += `<text x="${x}" y="${y - 1.2}" text-anchor="middle"
                font-family="Arial" font-size="${nodeLabelSize}" font-weight="700" fill="#000000">${escapeHtml(String(n.id))}</text>`;
            svg += `<text x="${x}" y="${y + 2.1}" text-anchor="middle"
                font-family="Arial" font-size="3.6666666667" font-weight="700" fill="#000000">${coordXText}</text>`;
            svg += `<text x="${x}" y="${y + 5.6}" text-anchor="middle"
                font-family="Arial" font-size="3.6666666667" font-weight="700" fill="#000000">${coordYText}</text>`;
        }
        if (isInternalActionMode) svg += `</g>`;
    });
    if (isInternalActionMode) {
        svg += `</g>`;
        // Value dots, leader lines and value text sit on top of the node
        // markers, so a node circle can never come between a value and the
        // diagram point it describes.
        svg += actionOverlaySvg + resolvedLabelsSvg;
    }

    svg += `</svg>`;
    return svg;
}

// ---------------------------------------------------------------------------
// Named, independent entry points — one per diagram — for everything outside
// this file's diagram engine (drawDiagramSVG, the fullscreen controller,
// the 8-step analysis report, etc.) to use. Each one simply fixes the
// `mode` argument of the shared renderer above; callers should never pass a
// raw mode string to getStructureSVG directly.
// ---------------------------------------------------------------------------
function renderStructureModelDiagram(width, height, camera) {
    return getStructureSVG(width, height, camera, "model");
}
function renderReactionDiagram(width, height, camera) {
    return getStructureSVG(width, height, camera, "reactions");
}
function renderAFDDiagram(width, height, camera) {
    return getStructureSVG(width, height, camera, "afd");
}
function renderSFDDiagram(width, height, camera) {
    return getStructureSVG(width, height, camera, "sfd");
}
function renderBMDDiagram(width, height, camera) {
    return getStructureSVG(width, height, camera, "bmd");
}
function renderDisplacementDiagram(width, height, camera) {
    return getStructureSVG(width, height, camera, "displacement");
}

function getReactionLocalComponents(sa) {
    const node = model.nodes.find((n) => n.id === sa.nodeId);
    let cx = 1;
    let cy = 0;

    // A support reaction is a global nodal quantity. For the optional local
    // display, resolve it into the lowest-ID connected member's local x-y axes.
    // If no element is connected, fall back to the global axes.
    // Select a deterministic reference member. Array order can change when a
    // user reorders elements; sorting by ID keeps these display-only local
    // components stable and makes the reference axis explicit in the heading.
    const connectedElement = model.elements
        .filter((el) => String(el.startNode) === String(sa.nodeId) || String(el.endNode) === String(sa.nodeId))
        .sort((a, b) => String(a.id).localeCompare(String(b.id)))[0];

    if (connectedElement) {
        const n1 = model.nodes.find((n) => n.id === connectedElement.startNode);
        const n2 = model.nodes.find((n) => n.id === connectedElement.endNode);
        if (n1 && n2) {
            const dx = n2.x - n1.x;
            const dy = n2.y - n1.y;
            const L = Math.hypot(dx, dy);
```

```
            if (L > 1e-12) {
                cx = dx / L;
                cy = dy / L;
            }
        }
    }

    const gx = sa.rx || 0;
    const gy = sa.ry || 0;

    return {
        x: gx * cx + gy * cy,
        y: -gx * cy + gy * cx,
        axisX: { x: cx, y: cy },
        axisY: { x: -cy, y: cx },
    };
}

function getCurrentDeformationScale() {
    const disp = model.nodeDisplacements || {};
    let maxDisp = 0;
    Object.values(disp).forEach((d) => {
        if (!d) return;
        const magnitude = isGridAnalysisMode()
            ? Math.abs(Number(d.deltaZ) || 0)
            : Math.hypot(Number(d.ux) || 0, Number(d.uy) || 0);
        maxDisp = Math.max(maxDisp, magnitude);
    });
    if (maxDisp <= 1e-12) return 1;
    const bb = getBoundingBox();
    const maxDim = Math.max(bb.maxX - bb.minX, bb.maxY - bb.minY, 1);
    return (0.28 * maxDim) / maxDisp;
}

// Legend of the structure (model) diagram: one row per element with its ID and
// section / material properties (same values and units as the Step 2 member table).
function buildElementPropertiesLegendRows() {
    const grid = isGridAnalysisMode();
    return (model.elements || []).map((el) => {
        const n1 = model.nodes.find((n) => n.id === el.startNode);
        const n2 = model.nodes.find((n) => n.id === el.endNode);
        const tmpl = model.elementTemplates.find((t) => t.id === el.templateId);
        const L = n1 && n2 ? Math.hypot(Number(n2.x) - Number(n1.x), Number(n2.y) - Number(n1.y)) : NaN;
        const num = (v) => (Number.isFinite(v) ? formatNum(v) : '—');
        const base = {
            id: String(el.id),
            L: num(L),
            E: tmpl ? num(getElementEInternal(tmpl)) : '—',
            I: tmpl ? num(Number(tmpl.I)) : '—',
        };
        if (grid) {
            const G = !tmpl ? NaN
                : tmpl.G_GPa !== undefined && tmpl.G_GPa !== '' && Number.isFinite(Number(tmpl.G_GPa))
                    ? Number(tmpl.G_GPa)
                    : tmpl.G !== undefined && tmpl.G !== '' && Number.isFinite(Number(tmpl.G))
                        ? Number(tmpl.G) / 1e9
                        : NaN;
            return { ...base, J: tmpl ? num(Number(tmpl.J)) : '—', G: num(G) };
        }
        return { ...base, A: tmpl ? num(Number(tmpl.A)) : '—' };
    });
}

function buildResultDiagramLegend(mode, deformationScale = 1) {
    if (!['model', 'reactions', 'sfd', 'bmd', 'afd', 'displacement'].includes(mode)) return '';

    const legendStyle = [
        'position:absolute',
        'left:12px',
        'bottom:12px',
        'z-index:20',
        'background:rgba(255,255,255,0.97)',
        'border:1px solid #cbd5e1',
        'border-radius:8px',
        'box-shadow:0 3px 10px rgba(15,23,42,0.12)',
        'padding:9px 11px',
        'font-family:Arial,sans-serif',
        'font-size:10px',
        'line-height:1.65',
        'color:#334155',
        'pointer-events:none',
        'white-space:nowrap'
    ].join(';');

    const legendSwatch = (color, label) =>
        `<span style="display:inline-block;width:24px;height:4px;background:${color};border-radius:2px;vertical-align:middle;margin-right:6px"></span>${label}`;
    if (mode === 'model') {
```

```
        const rows = buildElementPropertiesLegendRows();
        if (!rows.length) return '';
        const grid = isGridAnalysisMode();
        // Inline resets: the page's global table styles (uppercase headers, row lines,
        // header background) must not leak into this legend.
        const cell = 'border:none;background:transparent;text-transform:none;letter-spacing:0;font-size:10px;line-height:1.55;color:#334155;white-space:nowrap';
        const th = cell + ';font-weight:800;text-align:right;padding:0 0 2px 14px';
        const td = cell + ';font-weight:400;text-align:right;padding:0 0 0 14px';
        const idTd = cell + ';font-weight:700;text-align:left;padding:0';
        const columns = grid
            ? [{ key: 'L', label: 'L (m)' }, { key: 'I', label: 'I (m■)' }, { key: 'J', label: 'J (m■)' }, { key: 'E', label: 'E (kN/m²)' }, { key: 'G', label: 'G (GPa)' }]
            : [{ key: 'L', label: 'L (m)' }, { key: 'A', label: 'A (m²)' }, { key: 'E', label: 'E (kN/m²)' }, { key: 'I', label: 'I (m■)' }];
        // The list may be long: it scrolls inside the legend instead of covering the diagram.
        const scrollStyle = legendStyle.replace('pointer-events:none', 'pointer-events:auto') + ';max-height:45%;overflow:auto';
        return `<div class="result-diagram-fixed-legend" style="${scrollStyle}">
            <div style="font-weight:800;margin-bottom:2px;text-align:left">Element properties</div>
            <table style="border-collapse:collapse;border:none;font-size:10px;line-height:1.55;background:transparent;min-width:0;width:auto;margin:0;box-shadow:none">
                <thead style="background:transparent"><tr style="background:transparent">
                    <th style="${cell};font-weight:800;text-align:left;padding:0 0 2px 0">Element</th>
                    ${columns.map((column) => `<th style="${th}">${column.label}</th>`).join('')}
                </tr></thead>
                <tbody>${rows.map((r) => `<tr style="background:transparent">
                    <td style="${idTd}">${escapeHtml(r.id)}</td>
                    ${columns.map((column) => `<td style="${td}">${r[column.key]}</td>`).join('')}
                </tr>`).join('')}</tbody>
            </table>
        </div>`;
    }
    if (mode === 'reactions') {
        return `<div class="result-diagram-fixed-legend" style="${legendStyle}">
            <div style="font-weight:800;margin-bottom:2px">${isGridAnalysisMode() ? 'Grid support reactions' : 'Reaction directions'}</div>
            ${isGridAnalysisMode()
                ? '<div>Mx / My — reaction torsions (kN·m)</div><div>Fz — vertical reaction (kN)</div>'
                : '<div>↑ Positive Fy — global +Y</div><div>→ Positive Fx — global +X</div><div>■ Positive Mz — anticlockwise</div>'}
        </div>`;
    }
    if (mode === 'sfd') {
        return `<div class="result-diagram-fixed-legend" style="${legendStyle}">
            <div style="font-weight:800;margin-bottom:2px">${isGridAnalysisMode() ? 'Grid Fz diagram' : 'SFD convention'}</div>
            <div>${legendSwatch('#2563eb', isGridAnalysisMode() ? 'Positive Fz shear' : 'Positive shear (+V)')}</div>
            <div>${legendSwatch('#f59e0b', isGridAnalysisMode() ? 'Negative Fz shear' : 'Negative shear (−V)')}</div>
            <div style="margin-top:2px">${isGridAnalysisMode() ? 'Out-of-plane load direction' : 'Local +Y convention'}</div>
        </div>`;
    }
    if (mode === 'bmd') {
        return `<div class="result-diagram-fixed-legend" style="${legendStyle}">
            <div style="font-weight:800;margin-bottom:2px">${isGridAnalysisMode() ? 'Grid My torsion diagram' : 'BMD convention'}</div>
            <div>${legendSwatch('#dc2626', isGridAnalysisMode() ? 'Positive My torsion' : 'Positive moment (+M)')}</div>
            <div>${legendSwatch('#7c3aed', isGridAnalysisMode() ? 'Negative My torsion' : 'Negative moment (−M)')}</div>
            <div style="margin-top:2px">${isGridAnalysisMode() ? 'Grid torsion about Y' : 'Local +Mz / positive BMD'}</div>
        </div>`;
    }
    if (mode === 'afd') {
        return `<div class="result-diagram-fixed-legend" style="${legendStyle}">
            <div style="font-weight:800;margin-bottom:2px">${isGridAnalysisMode() ? 'Grid Mx / torsion diagram' : 'AFD convention'}</div>
            <div>${legendSwatch('#16a34a', isGridAnalysisMode() ? 'Positive Mx torsion' : 'Positive axial force (+A)')}</div>
            <div>${legendSwatch('#db2777', isGridAnalysisMode() ? 'Negative Mx torsion' : 'Negative axial force (−A)')}</div>
            <div style="margin-top:2px">${isGridAnalysisMode() ? 'Member torsion action' : 'Local +X (tension)'}</div>
        </div>`;
    }
    return `<div class="result-diagram-fixed-legend" style="${legendStyle}">
        <div style="font-weight:800;margin-bottom:2px">Displacement diagram</div>
        <div style="border-top:2px dashed #cbd5e1;padding-top:1px">Dashed = undeformed</div>
        <div><span style="display:inline-block;width:22px;border-top:3px solid #2563eb;vertical-align:middle;margin-right:4px"></span>Solid = deformed</div>
        <div>Scale factor = ×${deformationScale.toExponential(2)}</div>
    </div>`;
}

function drawDiagramSVG(containerId, mode) {
    const container = document.getElementById(containerId);
    if (!container) return;
    const rect = container.getBoundingClientRect();
    const isFullscreenDiagram = containerId === "fsSVG";

    container.style.position = 'relative';
    if (!isFullscreenDiagram) {
        container.style.overflow = "hidden";
    }
    if (mode === 'dof') {
        // The DoF & numbering diagram uses its own dedicated renderer
        // (fixed logical canvas + viewBox scaling) rather than the shared
        // getStructureSVG pipeline used by the other diagram modes.
        container.innerHTML = renderDOFDiagram(rect.width || 800, rect.height || 400);
        if (isFullscreenDiagram) {
            enableSVGZoomPan(container.querySelector("svg"));
        }
        return;
```

```
}
let camera;
let renderWidth = rect.width || 800;
let renderHeight = rect.height || 400;

// Inline diagrams are only an overview of the model. Fit the complete
// diagram inside the fixed viewport instead of using the larger intrinsic
// drawing canvas reserved for fullscreen, so no part of the structure is
// clipped on the normal page. Fullscreen keeps the existing readable
// intrinsic sizing and zoom/pan behavior below.
if (!isFullscreenDiagram && model.nodes.length > 0 && mode !== "dof") {
    const bb = getBoundingBox();
    const dX = Math.max(bb.maxX - bb.minX, 1e-9);
    const dY = Math.max(bb.maxY - bb.minY, 1e-9);
    const fitPadding = 35;
    const availableW = Math.max((rect.width || 800) - 2 * fitPadding, 1);
    const availableH = Math.max((rect.height || 400) - 2 * fitPadding, 1);
    const fitScale = Math.min(availableW / dX, availableH / dY);
    const safeFitScale = Number.isFinite(fitScale) && fitScale > 0 ? fitScale : 1;
    const midX = (bb.minX + bb.maxX) / 2;
    const midY = (bb.minY + bb.maxY) / 2;

    camera = {
        scale: safeFitScale,
        offsetX: (rect.width || 800) / 2 - midX * safeFitScale,
        offsetY: (rect.height || 400) / 2 + midY * safeFitScale
    };
    renderWidth = rect.width || 800;
    renderHeight = rect.height || 400;
} else if (model.nodes.length > 0 && mode !== "dof") {
    // With nodes but no structural elements there is no real member
    // length to use as the proportional baseline. Fall back to the
    // stable viewport-fit camera so every node remains visible immediately
    // after node creation. Once elements exist, keep the new proportional
    // scaling used by the full structural model diagram.
    if (mode === "model" && model.elements.length === 0) {
        camera = calculateAutoFit(rect.width || 800, rect.height || 400, mode);
    } else {
        const proportionalModes = new Set([
            "model",
            "reactions",
            "afd",
            "sfd",
            "bmd",
            "displacement"
        ]);

        if (proportionalModes.has(mode)) {
        const bb = getBoundingBox();
        const dX = Math.max(bb.maxX - bb.minX, 1e-9);
        const dY = Math.max(bb.maxY - bb.minY, 1e-9);
        const minL = getMinimumElementLength();
        const baselinePx = mode === "model" ? AppConfig.modelDiagramBaselinePx : AppConfig.sharedDiagramBaselinePx;
        const diagramScale =
            minL && Number.isFinite(minL) && minL > 0
                ? baselinePx / minL
                : baselinePx;

        // Keep the same readable intrinsic-canvas approach as the
        // Structure Model diagram. Reactions receive a little extra
        // breathing room because their labelled cards sit outside nodes.
        const padding = mode === "reactions" ? 180 : 140;

        renderWidth = Math.max(rect.width || 800, dX * diagramScale + 2 * padding);
        renderHeight = Math.max(rect.height || 400, dY * diagramScale + 2 * padding);

        camera = {
            scale: diagramScale,
            offsetX: padding - bb.minX * diagramScale,
            offsetY: padding + bb.maxY * diagramScale,
            intrinsicCanvas: true
        };
        } else {
            camera = calculateAutoFit(rect.width, rect.height, mode);
        }
    }
} else {
    camera = calculateAutoFit(rect.width, rect.height, mode);
}

const diagramRenderers = {
    model: renderStructureModelDiagram,
    reactions: renderReactionDiagram,
    afd: renderAFDDiagram,
    sfd: renderSFDDiagram,
    bmd: renderBMDDiagram,
    displacement: renderDisplacementDiagram,
```

```
    };
    const renderDiagram = diagramRenderers[mode];
    container.innerHTML = renderDiagram
        ? renderDiagram(renderWidth, renderHeight, camera)
        : getStructureSVG(renderWidth, renderHeight, camera, mode);
    const legendScale = mode === 'displacement' ? getCurrentDeformationScale() : 1;
    container.insertAdjacentHTML('beforeend', buildResultDiagramLegend(mode, legendScale));
    if (isFullscreenDiagram) {
        enableSVGZoomPan(container.querySelector("svg"));
    }
    applyMemberFocus(container);
}

// Grid model-diagram symbols are selectable: selecting one highlights every
// glyph belonging to the same load/torsion record. This does not mutate the
// structural model or affect solver behavior.
function toggleGridDiagramSymbolSelection(target) {
    const symbol = target && target.closest
        ? target.closest('.grid-diagram-symbol[data-grid-load-id], .grid-diagram-symbol[data-grid-torsion-id]')
        : null;
    if (!symbol) return false;
    const svgRoot = symbol.ownerSVGElement;
    if (!svgRoot) return false;
    const isLoad = symbol.dataset.gridLoadId !== undefined;
    const id = isLoad ? symbol.dataset.gridLoadId : symbol.dataset.gridTorsionId;
    const wasSelected = symbol.classList.contains('grid-symbol-selected');
    svgRoot.querySelectorAll('.grid-diagram-symbol').forEach((item) => {
        const sameType = isLoad ? item.dataset.gridLoadId !== undefined : item.dataset.gridTorsionId !== undefined;
        const sameId = isLoad ? item.dataset.gridLoadId === id : item.dataset.gridTorsionId === id;
        const selected = !wasSelected && sameType && sameId;
        item.classList.toggle('grid-symbol-selected', selected);
        item.setAttribute('aria-pressed', selected ? 'true' : 'false');
    });
    return true;
}

document.addEventListener('click', (event) => {
    if (toggleGridDiagramSymbolSelection(event.target)) event.stopPropagation();
});
document.addEventListener('keydown', (event) => {
    if (event.key !== 'Enter' && event.key !== ' ') return;
    if (toggleGridDiagramSymbolSelection(event.target)) {
        event.preventDefault();
        event.stopPropagation();
    }
});

// ========================================================================
// MEMBER CLICK-TO-FOCUS (AFD / SFD / BMD)
// Click a member (or its diagram / value labels) to bring that member's
// diagram and labels to the front and fade all other members (no color/weight change). Click the same
// member again, or the empty canvas, to reset. The state is applied by
// editing the existing SVG in place, so zoom/pan in fullscreen is preserved.
// ========================================================================
let activeMemberFocus = null;

// Fullscreen canvas navigation toggles (Zoom / Pan / Member Highlight).
// Every tool is inactive until its toolbar button is switched on.
const fsNavState = { zoom: false, pan: false, highlight: false };
let lastMemberTouchTap = 0;

function applyMemberFocus(root) {
    const svgs = root
        ? [root.querySelector("svg")]
        : Array.from(document.querySelectorAll("#resultsSVGSmall svg, #fsSVG svg"));
    svgs.forEach((svg) => {
        if (!svg || !svg.querySelector(".sf-member")) return;

        // Remember the original drawing order once so it can be restored on reset.
        Array.from(svg.children).forEach((child, i) => {
            if (child.dataset.sfOrder === undefined) child.dataset.sfOrder = i;
        });
        if (svg.dataset.sfReordered === "1") {
            Array.from(svg.children)
                .sort((a, b) => a.dataset.sfOrder - b.dataset.sfOrder)
                .forEach((child) => svg.appendChild(child));
            svg.dataset.sfReordered = "0";
        }

        // In the fullscreen viewer, member highlight only applies while its toolbar toggle is active.
        const fsHighlightOff = !!svg.closest("#fsSVG") && !fsNavState.highlight;
        const key = fsHighlightOff ? null : activeMemberFocus;
        const active = key !== null && !!svg.querySelector(`.sf-member[data-member="${key}"]`);
        svg.classList.toggle("sf-focus-on", active);
        svg.querySelectorAll("[data-member]").forEach((node) => {
            const isActive = active && node.dataset.member === key;
            node.classList.toggle("sf-active", isActive);
```

```
            node.classList.toggle("sf-dim", active && !isActive);
        });
        // Nodes: only the two end nodes of the focused member stay visible,
        // so it is clear which joints the focused diagram spans.
        const focusedMember = active ? svg.querySelector(`.sf-member[data-member="${key}"]`) : null;
        svg.querySelectorAll(".sf-node").forEach((nodeEl) => {
            const spans = !!focusedMember &&
                (nodeEl.dataset.node === focusedMember.dataset.start || nodeEl.dataset.node === focusedMember.dataset.end);
            nodeEl.classList.toggle("sf-active", spans);
            nodeEl.classList.toggle("sf-dim", active && !spans);
        });
        if (active) {
            const own = (cls) => Array.from(svg.children).filter(
                (c) => c.classList.contains(cls) && c.dataset.member === key);
            // Order: node markers sent to the very bottom, then the focused
            // member + diagram, then its value labels on top.
            const nodes = svg.querySelector(":scope > .sf-nodes");
            if (nodes) svg.insertBefore(nodes, svg.firstChild);
            own("sf-member").concat(own("sf-diagram")).forEach((c) => svg.appendChild(c));
            own("sf-label").forEach((c) => svg.appendChild(c));
            svg.dataset.sfReordered = "1";
        }
    });
}

function handleMemberFocusTap(target) {
    if (!target || !target.closest) return;
    const host = target.closest("#resultsSVGSmall, #fsSVG");
    if (!host) return;
    if (host.id === "fsSVG" && !fsNavState.highlight) return; // highlight toggle is off
    const svg = host.querySelector("svg");
    if (!svg || !svg.querySelector(".sf-member")) return;

    const hit = target.closest("[data-member]");
    if (hit && svg.contains(hit)) {
        const key = hit.dataset.member;
        activeMemberFocus = activeMemberFocus === key ? null : key;
    } else if (target === host || target.closest("svg") === svg) {
        activeMemberFocus = null; // empty canvas
    } else {
        return; // e.g. legend
    }
    applyMemberFocus();
}

document.addEventListener("click", (e) => {
    if (Date.now() - lastMemberTouchTap < 700) return;
    handleMemberFocusTap(e.target);
});

// The fullscreen zoom/pan handler calls preventDefault() on touchstart, which
// suppresses the browser's synthetic click, so detect taps there explicitly.
let memberTouchStart = null;
document.addEventListener("touchstart", (e) => {
    memberTouchStart = e.touches.length === 1
        ? { x: e.touches[0].clientX, y: e.touches[0].clientY, t: Date.now() }
        : null;
}, { passive: true });
document.addEventListener("touchend", (e) => {
    const start = memberTouchStart;
    memberTouchStart = null;
    if (!start || e.changedTouches.length !== 1) return;
    const t = e.changedTouches[0];
    if (Math.hypot(t.clientX - start.x, t.clientY - start.y) > 10 || Date.now() - start.t > 500) return;
    if (!(e.target.closest && e.target.closest("#fsSVG"))) return;
    lastMemberTouchTap = Date.now();
    handleMemberFocusTap(e.target);
}, { passive: true });

// ============================================================================
// 12. PINCH-TO-ZOOM & PAN
// Touch (mobile) and mouse-wheel (desktop) viewBox navigation for the SVGs.
// ============================================================================
function enableSVGZoomPan(svg) {
    if (!svg) return;

    let viewBox = null;
    let vbAttr = svg.getAttribute("viewBox");
    if (vbAttr) {
        let vb = vbAttr.split(" ").map(Number);
        viewBox = { x: vb[0], y: vb[1], w: vb[2], h: vb[3] };
    } else {
        return;
    }

    const baseW = viewBox.w;
    const baseH = viewBox.h;
```

```
const maxW = baseW * 15;
const minW = baseW / 15;

let initialDistance = null;
let initialViewBox = null;
let initialPinchCenter = null;
let isPanning = false;
let lastPanPos = null;

svg.addEventListener(
    "touchstart",
    function (e) {
        if (e.touches.length === 2) {
            if (!fsNavState.zoom) { initialDistance = null; isPanning = false; return; }
            e.preventDefault();
            isPanning = false;
            let dx = e.touches[0].clientX - e.touches[1].clientX;
            let dy = e.touches[0].clientY - e.touches[1].clientY;
            initialDistance = Math.hypot(dx, dy);
            initialViewBox = Object.assign({}, viewBox);

            let rect = svg.getBoundingClientRect();
            let rW = rect.width || 1;
            let rH = rect.height || 1;

            let cx = (e.touches[0].clientX + e.touches[1].clientX) / 2 - rect.left;
            let cy = (e.touches[0].clientY + e.touches[1].clientY) / 2 - rect.top;

            initialPinchCenter = {
                x: initialViewBox.x + (cx / rW) * initialViewBox.w,
                y: initialViewBox.y + (cy / rH) * initialViewBox.h,
            };
        } else if (e.touches.length === 1) {
            if (!fsNavState.pan) { isPanning = false; lastPanPos = null; return; }
            e.preventDefault();
            isPanning = true;
            lastPanPos = { x: e.touches[0].clientX, y: e.touches[0].clientY };
        }
    },
    { passive: false },
);

svg.addEventListener(
    "touchmove",
    function (e) {
        if (e.touches.length === 2 && initialDistance && fsNavState.zoom) {
            e.preventDefault();
            let dx = e.touches[0].clientX - e.touches[1].clientX;
            let dy = e.touches[0].clientY - e.touches[1].clientY;
            let currentDistance = Math.hypot(dx, dy);

            let scale = initialDistance / currentDistance;
            let newW = initialViewBox.w * scale;
            let newH = initialViewBox.h * scale;

            if (newW > maxW) {
                scale = maxW / initialViewBox.w;
                newW = maxW;
                newH = initialViewBox.h * scale;
            } else if (newW < minW) {
                scale = minW / initialViewBox.w;
                newW = minW;
                newH = initialViewBox.h * scale;
            }

            let rect = svg.getBoundingClientRect();
            let rW = rect.width || 1;
            let rH = rect.height || 1;

            let currentCx = (e.touches[0].clientX + e.touches[1].clientX) / 2 - rect.left;
            let currentCy = (e.touches[0].clientY + e.touches[1].clientY) / 2 - rect.top;

            viewBox.x = initialPinchCenter.x - (currentCx / rW) * newW;
            viewBox.y = initialPinchCenter.y - (currentCy / rH) * newH;
            viewBox.w = newW;
            viewBox.h = newH;

            svg.setAttribute("viewBox", `${viewBox.x} ${viewBox.y} ${viewBox.w} ${viewBox.h}`);
        } else if (e.touches.length === 1 && isPanning && lastPanPos && fsNavState.pan) {
            e.preventDefault();
            let dx = e.touches[0].clientX - lastPanPos.x;
            let dy = e.touches[0].clientY - lastPanPos.y;

            let rect = svg.getBoundingClientRect();
            let rW = rect.width || 1;
            let rH = rect.height || 1;
            viewBox.x -= (dx / rW) * viewBox.w;
```

```
            viewBox.y -= (dy / rH) * viewBox.h;

            svg.setAttribute("viewBox", `${viewBox.x} ${viewBox.y} ${viewBox.w} ${viewBox.h}`);
            lastPanPos = { x: e.touches[0].clientX, y: e.touches[0].clientY };
        }
    },
    { passive: false },
);

svg.addEventListener(
    "touchend",
    function (e) {
        if (e.touches.length < 2) {
            initialDistance = null;
        }
        if (e.touches.length === 0) {
            isPanning = false;
            lastPanPos = null;
        } else if (e.touches.length === 1) {
            isPanning = fsNavState.pan;
            lastPanPos = { x: e.touches[0].clientX, y: e.touches[0].clientY };
        }
    },
    { passive: false },
);

svg.addEventListener(
    "wheel",
    function (e) {
        // PC zoom: only while the Zoom toggle is ON and Ctrl is held.
        // Plain scrolling (no Ctrl) is ignored.
        if (!fsNavState.zoom || !e.ctrlKey) return;
        e.preventDefault();
        let zoomOut = e.deltaY > 0;
        let scale = zoomOut ? 1.1 : 0.9;

        let newW = viewBox.w * scale;
        let newH = viewBox.h * scale;

        if (newW > maxW) {
            scale = maxW / viewBox.w;
            newW = maxW;
            newH = viewBox.h * scale;
        } else if (newW < minW) {
            scale = minW / viewBox.w;
            newW = minW;
            newH = viewBox.h * scale;
        }

        let rect = svg.getBoundingClientRect();
        let rW = rect.width || 1;
        let rH = rect.height || 1;

        let cx = e.clientX - rect.left;
        let cy = e.clientY - rect.top;

        let pinchX = viewBox.x + (cx / rW) * viewBox.w;
        let pinchY = viewBox.y + (cy / rH) * viewBox.h;

        viewBox.x = pinchX - (cx / rW) * newW;
        viewBox.y = pinchY - (cy / rH) * newH;
        viewBox.w = newW;
        viewBox.h = newH;

        svg.setAttribute("viewBox", `${viewBox.x} ${viewBox.y} ${viewBox.w} ${viewBox.h}`);
    },
    { passive: false },
);

// PC: right-button drag OR middle-button (wheel click) drag pans the canvas,
// only while Pan Mode is active. Left-click is left untouched.
let mouseDrag = null;
svg.addEventListener("contextmenu", function (e) {
    if (fsNavState.pan) e.preventDefault(); // no context menu while Pan Mode is ON
});
svg.addEventListener("mousedown", function (e) {
    if ((e.button !== 1 && e.button !== 2) || !fsNavState.pan) return;
    e.preventDefault(); // also stops middle-click autoscroll
    mouseDrag = { x: e.clientX, y: e.clientY, button: e.button };
    const modal = document.getElementById("fsModal");
    if (modal) modal.classList.add("fs-panning");

    const onMove = function (ev) {
        if (!mouseDrag) return;
        if (!fsNavState.pan) { onUp(); return; }
        const dx = ev.clientX - mouseDrag.x;
        const dy = ev.clientY - mouseDrag.y;
```

```
            const rect = svg.getBoundingClientRect();
            viewBox.x -= (dx / (rect.width || 1)) * viewBox.w;
            viewBox.y -= (dy / (rect.height || 1)) * viewBox.h;
            svg.setAttribute("viewBox", `${viewBox.x} ${viewBox.y} ${viewBox.w} ${viewBox.h}`);
            mouseDrag.x = ev.clientX;
            mouseDrag.y = ev.clientY;
        };
        const onUp = function () {
            window.removeEventListener("mousemove", onMove);
            window.removeEventListener("mouseup", onUp);
            if (modal) modal.classList.remove("fs-panning");
            mouseDrag = null;
        };
        window.addEventListener("mousemove", onMove);
        window.addEventListener("mouseup", onUp);
    });
}

// Toolbar wiring for the fullscreen viewer (Zoom / Pan / Member Highlight toggles).
function setFsNavTool(tool, on) {
    fsNavState[tool] = !!on;
    const ids = { zoom: "fsNavZoomBtn", pan: "fsNavPanBtn", highlight: "fsNavHighlightBtn" };
    const btn = document.getElementById(ids[tool]);
    if (btn) {
        btn.classList.toggle("active", fsNavState[tool]);
        btn.setAttribute("aria-pressed", fsNavState[tool] ? "true" : "false");
    }
    const modal = document.getElementById("fsModal");
    if (modal) {
        modal.classList.toggle("fs-nav-zoom", fsNavState.zoom);
        modal.classList.toggle("fs-nav-pan", fsNavState.pan);
        if (!fsNavState.pan) modal.classList.remove("fs-panning");
    }
    if (tool === "highlight") {
        if (!fsNavState.highlight) activeMemberFocus = null;
        applyMemberFocus();
    }
}

function resetFsNavTools() {
    ["zoom", "pan", "highlight"].forEach((t) => setFsNavTool(t, false));
}

document.querySelectorAll("#fsNavTools .fs-nav-btn").forEach((btn) => {
    btn.addEventListener("click", function (e) {
        e.stopPropagation();
        const tool = btn.dataset.tool;
        setFsNavTool(tool, !fsNavState[tool]);
    });
});

// ========================================================================
// 13. FULLSCREEN DIAGRAM MODAL & ANALYSIS DIAGRAM RESIZE HANDLING
// ========================================================================
function updateResultsDiagram() {
    let mode = document.getElementById("resultsDiagramMode").value;
    if (model.nodes.length > 10) {
        showHeavyDiagramPlaceholder("resultsSVGSmall");
        return;
    }
    if (model.nodes.length > 10) {
            showHeavyDiagramPlaceholder("resultsSVGSmall");
        } else {
            drawDiagramSVG("resultsSVGSmall", mode);
        }
}

function openFullScreen(source) {
    if (source === "results") {
        currentFSMode = document.getElementById("resultsDiagramMode").value;
    } else if (source === "reactions") {
        currentFSMode = "reactions";
    } else if (source === "dof") {
        currentFSMode = "dof";
    } else {
        currentFSMode = "model";
    }
    // Reapply solver-specific text immediately before the picker becomes visible.
    // The shared internal values (afd/sfd/bmd) remain unchanged.
    if (typeof syncDiagramModeLabels === "function") syncDiagramModeLabels();
    document.getElementById("fsModeSelect").value = currentFSMode;
    const modal = document.getElementById("fsModal");
    modal.classList.add("active");
    // Lock the background page so it can't be scrolled/peeked at while the
    // fullscreen diagram is open (prevents underlying content from showing
    // through on mobile browsers).
    document.body.style.overflow = "hidden";
```

```
    setTimeout(() => {
        renderFullScreen();
    }, 10);
}

function changeFSMode(mode) {
    currentFSMode = mode;
    renderFullScreen();
}

function closeFullScreen() {
    document.getElementById("fsModal").classList.remove("active");
    document.body.style.overflow = "";
    resetFsNavTools();
}

function renderFullScreen() {
    const fsContainer = document.getElementById("fsSVG");
    if (fsContainer) {
        fsContainer.style.overflow = currentFSMode === "model" ? "auto" : "hidden";
    }
    drawDiagramSVG("fsSVG", currentFSMode);
}

// Redraw statically scaled diagrams universally on window resize
window.addEventListener("resize", () => {
    if (document.getElementById("fsModal").classList.contains("active")) renderFullScreen();
    renderInlineModelDiagram();
    if (document.getElementById("resultsContent").style.display !== "none") {
        if (model.nodes.length > 10) {
            showHeavyDiagramPlaceholder("reactionSVGSmall");
        } else {
            drawDiagramSVG("reactionSVGSmall", "reactions");
        }
        let mode = document.getElementById("resultsDiagramMode").value;
        if (model.nodes.length > 10) {
            showHeavyDiagramPlaceholder("resultsSVGSmall");
        } else {
            drawDiagramSVG("resultsSVGSmall", mode);
        }
    }
});

// ============================================================================
// 14. DEGREES-OF-FREEDOM (DoF) DIAGRAM
// Step 1 visual used inside the analysis report.
// ============================================================================
// Named entry point for the DoF/numbering diagram, matching the naming of
// the render*Diagram() family above. It uses its own dedicated renderer
// (fixed logical canvas + viewBox scaling) rather than the shared
// getStructureSVG pipeline used by the other six diagram modes.
function renderDOFDiagram(width, height) {
    return getDoFDiagramSVG(width, height);
}

function getNodeGlobalDofsForDiagram(nodeId) {
    const numbering = buildStructuralDofNumbering();
    return numbering.dofMap[nodeId] || { x: null, y: null, z: null };
}

function getDoFDiagramSVG(width, height) {
    if (model.nodes.length === 0) return `<svg width="${width}" height="${height}"></svg>`;

    /*
     * Dedicated schematic renderer for the Step-1 DoF/numbering diagram.
     *
     * IMPORTANT: this function is intentionally isolated from the solver and
     * from the normal model/reaction/result renderers.  It changes only the
     * presentation of the DoF diagram so that members, joints and DoF labels
     * have fixed, readable spacing instead of fighting a generic collision
     * engine.
     *
     * The schematic keeps the model's topology and numbering exactly as the
     * solver uses them:
     *   - node number = index in model.nodes + 1
     *   - member number = index in model.elements + 1
     *   - DoFs are shown using the global compressed numbering configuration
     */

    const W = Math.max(Number(width) || 800, ((new Set(model.nodes.map(n => Number(n.x) || 0)).size - 1) * 160 + 170), 640);
    const H = Math.max(Number(height) || 400, 300);

    // ------------------------------------------------------------------
    // 1) Build a clean, orthogonalised schematic coordinate system.
    //    Unique X/Y coordinates are kept in their original order, but
    //    equally-ranked positions are given deliberate breathing room.
    // ------------------------------------------------------------------
```

```
const EPS = 1e-9;
const xs = [...new Set(model.nodes.map(n => Number(n.x) || 0).sort((a, b) => a - b))];
const ys = [...new Set(model.nodes.map(n => Number(n.y) || 0).sort((a, b) => a - b))];

const leftPad = 85;
const rightPad = 85;
const topPad = 72;
const bottomPad = 72;

const usableW = Math.max(W - leftPad - rightPad, 1);
const usableH = Math.max(H - topPad - bottomPad, 1);

// Horizontal spacing: preserve the number/order of coordinate columns.
// Fixed visual spacing for the schematic. The model's actual member
// lengths must not resize the symbols or squeeze their annotation slots.
const FIXED_X_GAP = 160;
const FIXED_Y_GAP = 150;

// Keep the logical node-to-node spacing constant regardless of how many
// elements/coordinate columns exist. This prevents the element bubble
// from moving relative to the fixed +X annotation slot on different
// structures.
const xGap = xs.length <= 1 ? 0 : FIXED_X_GAP;
const xSpan = xGap * Math.max(xs.length - 1, 0);
const xStart = (W - xSpan) / 2;

// Vertical spacing: put the highest model row at the top and the lowest
// row at the bottom.  This is deliberately schematic rather than a
// perspective/scale drawing.
const yGap = ys.length <= 1 ? 0 : FIXED_Y_GAP;
const ySpan = yGap * Math.max(ys.length - 1, 0);
const yStart = (H - ySpan) / 2;

const xRank = new Map(xs.map((v, i) => [String(v), i]));
const yRank = new Map(ys.map((v, i) => [String(v), i]));
const schematic = new Map();

model.nodes.forEach((n, i) => {
    const nx = Number(n.x) || 0;
    const ny = Number(n.y) || 0;
    const xi = xRank.get(String(nx));
    const yi = yRank.get(String(ny));
    const sx = xStart + (Number.isFinite(xi) ? xi : 0) * xGap;
    // SVG Y grows downward, while structural Y grows upward.
    const sy = yStart + (ys.length - 1 - (Number.isFinite(yi) ? yi : 0)) * yGap;
    schematic.set(n.id, { x: sx, y: sy, index: i });
});

// ---------------------------------------------------------------------
// 1A) Universal fixed-member-length schematic layout.
//      The DoF diagram is a schematic, so actual member length must not
//      change how long an angled member appears.  Every structural member
//      is therefore constrained to the same logical visual length as the
//      existing horizontal member.  The original member direction is kept
//      so horizontal, vertical and angled members remain symmetric.
//      A small deterministic relaxation handles shared joints and frames.
// ---------------------------------------------------------------------
const FIXED_MEMBER_LENGTH = 160;
const fixedLengthEdges = model.elements.map((el) => {
    const n1 = model.nodes.find(n => n.id === el.startNode);
    const n2 = model.nodes.find(n => n.id === el.endNode);
    const dx = (Number(n2?.x) || 0) - (Number(n1?.x) || 0);
    const dy = (Number(n2?.y) || 0) - (Number(n1?.y) || 0);
    const L = Math.hypot(dx, dy) || 1;
    return {
        start: el.startNode,
        end: el.endNode,
        ux: dx / L,
        uy: -dy / L, // convert structural +Y to SVG +Y-down
    };
});

if (fixedLengthEdges.length) {
    const anchorCentroid = () => {
        let sx = 0, sy = 0, count = 0;
        schematic.forEach(pt => { sx += pt.x; sy += pt.y; count++; });
        return count ? { x: sx / count, y: sy / count } : { x: W / 2, y: H / 2 };
    };

    const anchor = anchorCentroid();
    for (let iter = 0; iter < 300; iter++) {
        for (const edge of fixedLengthEdges) {
            const a = schematic.get(edge.start);
            const b = schematic.get(edge.end);
            if (!a || !b) continue;

            const desiredDx = edge.ux * FIXED_MEMBER_LENGTH;
```

```
            const desiredDy = edge.uy * FIXED_MEMBER_LENGTH;
            const actualDx = b.x - a.x;
            const actualDy = b.y - a.y;
            const ex = actualDx - desiredDx;
            const ey = actualDy - desiredDy;

            // Equal/opposite correction keeps the joint at the midpoint.
            a.x += ex * 0.5;
            a.y += ey * 0.5;
            b.x -= ex * 0.5;
            b.y -= ey * 0.5;
        }

        // Prevent the relaxation from drifting the complete model.
        const c = anchorCentroid();
        const tx = anchor.x - c.x;
        const ty = anchor.y - c.y;
        schematic.forEach(pt => { pt.x += tx; pt.y += ty; });
    }

    // Recenter the relaxed schematic and translate it into the visible
    // logical canvas without applying a scale (scaling would destroy the
    // fixed member length relationship with the symbols).
    const bounds = [...schematic.values()];
    const minX = Math.min(...bounds.map(p => p.x));
    const maxX = Math.max(...bounds.map(p => p.x));
    const minY = Math.min(...bounds.map(p => p.y));
    const maxY = Math.max(...bounds.map(p => p.y));
    const bbCx = (minX + maxX) / 2;
    const bbCy = (minY + maxY) / 2;
    const moveX = W / 2 - bbCx;
    const moveY = H / 2 - bbCy;
    schematic.forEach(pt => { pt.x += moveX; pt.y += moveY; });
}

function pointForNode(nodeId) {
    return schematic.get(nodeId);
}

function esc(value) {
    return String(value)
        .replace(/&/g, '&amp;')
        .replace(/</g, '&lt;')
        .replace(/>/g, '&gt;')
        .replace(/"/g, '&quot;')
        .replace(/'/g, '&#39;');
}

const green = '#10b981';
const red = '#ef4444';
const yellow = '#facc15';
const blue = '#3b82f6';
const memberGray = '#cbd5e1';
const labelBlack = '#111827';

// Fixed symbol dimensions: these remain independent of member length or
// node spacing so the DoF/numbering graphics cannot resize dynamically.
const MEMBER_STROKE = 7;
// Keep element bubbles compact so they cannot collide with the nearby +X
// annotation slot. The number font scales down for long element IDs so
// values such as 1000 still stay inside the fixed-size circle.
const MEMBER_BUBBLE_R = 10;
const MEMBER_BUBBLE_STROKE = 1.5;
const MEMBER_NUMBER_SIZE = 10;
const NODE_HALF_W = 14;
const NODE_TOP = 14;
const NODE_BOTTOM = 10;
const NODE_STROKE = 1.5;
const NODE_NUMBER_SIZE = 11;
const AXIS_STROKE = 1.4;
const AXIS_ARROW_LEN = 30;
const AXIS_ARROW_GAP = 18;
const AXIS_NUMBER_SIZE = 10.5;
const MOMENT_STROKE = 1.4;
const MOMENT_NUMBER_SIZE = 10;

let svg = `<svg width="100%" height="100%" viewBox="0 0 ${W} ${H}" xmlns="http://www.w3.org/2000/svg" preserveAspectRatio="xMidYMid meet">`;

svg += `<defs>
    <marker id="dof-arrow-red" markerWidth="8" markerHeight="8" refX="7" refY="4" orient="auto" markerUnits="userSpaceOnUse">
        <path d="M0,0 L8,4 L0,8 z" fill="${red}"/>
    </marker>
    <marker id="dof-arrow-moment" markerWidth="8" markerHeight="8" refX="7" refY="4" orient="auto" markerUnits="userSpaceOnUse">
        <path d="M0,0 L8,4 L0,8 Z" fill="#000000"/>
    </marker>
</defs>`;
// ---------------------------------------------------------------------
```

```
// 2) Members first.  Member-number bubbles use a fixed-size,
//    collision-safe anchor on each member.  The horizontal case stays
//    centered exactly as before; for angled members the anchor is moved
//    along the same member only when a fixed annotation corridor around
//    either end node would intersect it.  This keeps the visual grammar
//    symmetric without changing member topology or solver data.
// ---------------------------------------------------------------
const placedMemberBubbles = [];

function rectsOverlapCircle(cx, cy, r, rect, pad = 0) {
    const qx = Math.max(rect.x, Math.min(cx, rect.x + rect.w));
    const qy = Math.max(rect.y, Math.min(cy, rect.y + rect.h));
    const dx = cx - qx;
    const dy = cy - qy;
    return (dx * dx + dy * dy) <= Math.pow(r + pad, 2);
}

function getNodeAnnotationObstacles(point) {
    if (!point) return [];
    // Fixed corridors reserved for the +X arrow/number and +Y arrow/number.
    return [
        // +X corridor (left of the node).
        { x: point.x - 58, y: point.y - 9, w: 50, h: 18 },
        // +Y corridor (above the node).
        { x: point.x - 12, y: point.y - 58, w: 24, h: 50 },
        // Node triangle itself.
        { x: point.x - 16, y: point.y - 15, w: 32, h: 30 },
    ];
}

function chooseMemberBubbleAnchor(a, b) {
    const dx = b.x - a.x;
    const dy = b.y - a.y;
    const len = Math.hypot(dx, dy) || 1;
    const ux = dx / len;
    const uy = dy / len;

    // Keep the exact horizontal appearance at t=0.5 unless that would
    // actually intersect a fixed annotation corridor.
    const candidates = [0.50, 0.42, 0.58, 0.35, 0.65, 0.28, 0.72];
    const obstacles = [...getNodeAnnotationObstacles(a), ...getNodeAnnotationObstacles(b)];

    let best = null;
    let bestScore = -Infinity;

    for (const t of candidates) {
        const cx = a.x + dx * t;
        const cy = a.y + dy * t;
        let score = 100 - Math.abs(t - 0.5) * 120;

        let blocked = false;
        obstacles.forEach(rect => {
            if (rectsOverlapCircle(cx, cy, MEMBER_BUBBLE_R, rect, 5)) blocked = true;
        });

        placedMemberBubbles.forEach(prev => {
            const d = Math.hypot(cx - prev.x, cy - prev.y);
            if (d < MEMBER_BUBBLE_R * 2 + 8) blocked = true;
            else score += Math.min(d, 40) * 0.05;
        });

        // Prefer a point that is visually central on angled members while
        // keeping a generous fixed safety corridor from node annotations.
        const endpointClearance = Math.min(t, 1 - t) * len;
        score += Math.min(endpointClearance, 80) * 0.08;

        if (!blocked && score > bestScore) {
            bestScore = score;
            best = { x: cx, y: cy };
        }
    }

    // Long fixed schematic members should always provide at least one safe
    // candidate.  Fall back to the exact midpoint only in the unlikely
    // case that every candidate is occupied by another annotation.
    return best || { x: a.x + dx * 0.5, y: a.y + dy * 0.5 };
}

model.elements.forEach((el, index) => {
    const a = pointForNode(el.startNode);
    const b = pointForNode(el.endNode);
    if (!a || !b) return;

    svg += `<line x1="${a.x}" y1="${a.y}" x2="${b.x}" y2="${b.y}"
        stroke="${memberGray}" stroke-width="${MEMBER_STROKE}" stroke-linecap="butt"/>`;

    const bubble = chooseMemberBubbleAnchor(a, b);
```

```
        const mx = bubble.x;
        const my = bubble.y;
        placedMemberBubbles.push({ x: mx, y: my });

        svg += `<circle cx="${mx}" cy="${my}" r="${MEMBER_BUBBLE_R}"
            fill="white" stroke="${blue}" stroke-width="${MEMBER_BUBBLE_STROKE}"/>`;
        const memberLabel = String(index + 1);
        const memberFontSize = memberLabel.length >= 4 ? 7 : memberLabel.length === 3 ? 8.5 : MEMBER_NUMBER_SIZE;
        svg += `<text x="${mx}" y="${my + memberFontSize * 0.36}" text-anchor="middle"
            font-family="Inter, Arial, Helvetica, sans-serif" font-size="${memberFontSize}"
            font-weight="700" fill="${blue}">${memberLabel}</text>`;
    });

    // ------------------------------------------------------------------
    // 3) Joint glyphs.
    //    Normal joints use an upward triangle.  A leaf node that sits below
    //    another row is turned sideways so the vertical member and +Y DoF do
    //    not stack on top of the joint number — matching the Canva design.
    // ------------------------------------------------------------------
    const neighbours = new Map(model.nodes.map(n => [n.id, []]));
    model.elements.forEach(el => {
        if (neighbours.has(el.startNode) && neighbours.has(el.endNode)) {
            neighbours.get(el.startNode).push(el.endNode);
            neighbours.get(el.endNode).push(el.startNode);
        }
    });

    function isLowerLeaf(n) {
        const p = pointForNode(n.id);
        const ns = neighbours.get(n.id) || [];
        if (!p || ns.length !== 1) return false;
        const q = pointForNode(ns[0]);
        return !!q && q.y < p.y - EPS;
    }

    // Grid torsion DoFs are rendered as the bottom visual layer. Their shafts
    // start at the joint center; the joint triangle/green number is rendered
    // above them, and the open Delta-Z symbol is added as the top layer below.
    // This ordering keeps the three Grid DoF indicators centered on each joint
    // without allowing red torsion graphics to obscure the green node number.
    if (isGridAnalysisMode()) {
        const gridDofRed = '#ef4444';
        const drawGridDofArrow = (sx, sy, ux, uy, number, direction) => {
            if (number === null || number === undefined) return;
            const length = 30;
            const stroke = 2.4;
            const headLength = 6.1;
            const halfWidth = 4.0;
            const secondHeadGap = 7.0;
            const ex = sx + ux * length;
            const ey = sy + uy * length;
            const px = -uy;
            const py = ux;
            const head = (hx, hy) => {
                const bx = hx - ux * headLength;
                const by = hy - uy * headLength;
                return `M ${(bx + px * halfWidth).toFixed(2)} ${(by + py * halfWidth).toFixed(2)} L ${hx.toFixed(2)} ${hy.toFixed(2)} L ${(bx - px * halfWidth).toFixed(2)} ${(by - py *
halfWidth).toFixed(2)}`;
            };
            const secondX = ex - ux * secondHeadGap;
            const secondY = ey - uy * secondHeadGap;
            svg += `<g aria-label="Grid ${direction} torsion degree of freedom ${number}">`;
            svg += `<path d="M ${sx.toFixed(2)} ${sy.toFixed(2)} L ${ex.toFixed(2)} ${ey.toFixed(2)}" fill="none" stroke="${gridDofRed}" stroke-width="${stroke}" stroke-linecap="round"/>`;
            svg += `<path d="${head(ex, ey)} ${head(secondX, secondY)}" fill="none" stroke="${gridDofRed}" stroke-width="${stroke}" stroke-linecap="round" stroke-linejoin="round"/>`;
            if (direction === 'right') {
                svg += `<text x="${(ex + 4).toFixed(2)}" y="${(ey + 3.5).toFixed(2)}" text-anchor="start" font-family="Inter, Arial, Helvetica, sans-serif" font-size="${AXIS_NUMBER_SIZE}"
font-weight="700" fill="${gridDofRed}">${number}</text>`;
            } else {
                svg += `<text x="${(ex + 1).toFixed(2)}" y="${(ey - 7).toFixed(2)}" text-anchor="middle" font-family="Inter, Arial, Helvetica, sans-serif" font-size="${AXIS_NUMBER_SIZE}"
font-weight="700" fill="${gridDofRed}">${number}</text>`;
            }
            svg += `</g>`;
        };

        model.nodes.forEach((n) => {
            const p = pointForNode(n.id);
            if (!p) return;
            const d = getNodeGlobalDofsForDiagram(n.id);
            const dofX = d.x === null ? null : d.x + 1;
            const dofY = d.y === null ? null : d.y + 1;
            // Both torsion indicators are anchored at the joint center.
            drawGridDofArrow(p.x, p.y, 1, 0, dofX, 'right');
            drawGridDofArrow(p.x, p.y, 0, -1, dofY, 'up');
        });
    }

    model.nodes.forEach((n, i) => {
```

```
        const p = pointForNode(n.id);
        if (!p) return;

        // All joint triangles use one fixed orientation: point upward.
        // Do not rotate the triangle for vertical/lower-leaf members.
        svg += `<polygon points="${p.x},${p.y - NODE_TOP} ${p.x - NODE_HALF_W},${p.y + NODE_BOTTOM} ${p.x + NODE_HALF_W},${p.y + NODE_BOTTOM}"
            fill="white" stroke="${green}" stroke-width="${NODE_STROKE}"/>`;

        svg += `<text x="${p.x}" y="${p.y + 4}" text-anchor="middle"
            font-family="Inter, Arial, Helvetica, sans-serif" font-size="${NODE_NUMBER_SIZE}"
            font-weight="700" fill="${green}">${i + 1}</text>`;
    });

    // ------------------------------------------------------------------
    // 4) Fixed DoF annotation slots.
    //    +X is always a red right-pointing arrow.  At a right boundary node
    //    it is placed outside the joint; otherwise it is placed on the left.
    //    +Y is upward.  +Mz is anticlockwise and yellow, matching the reference.
    // ------------------------------------------------------------------
    model.nodes.forEach((n, i) => {
        const p = pointForNode(n.id);
        if (!p) return;

        const d = getNodeGlobalDofsForDiagram(n.id);
        const dofX = d.x === null ? null : d.x + 1;
        const dofY = d.y === null ? null : d.y + 1;
        const dofM = d.z === null ? null : d.z + 1;

        // Grid DoF annotations use Grid-specific glyphs only: an open black
        // circle for the out-of-page translation (Delta Z), and matching
        // single-shaft/double-chevron torsion arrows for thetaX (right) and
        // thetaY (up). The Plane/BTF annotations below remain unchanged.
        if (isGridAnalysisMode()) {
            const gridDofBlack = '#111111';

            // Delta Z: open black circle is centered exactly on the joint and
            // rendered after the green node number, making it the top Grid DoF
            // symbol. A transparent fill keeps the node number readable inside.
            const outX = p.x;
            const outY = p.y;
            svg += `<g aria-label="Grid out-of-page translation degree of freedom ${dofM}">`;
            svg += `<circle cx="${outX.toFixed(2)}" cy="${outY.toFixed(2)}" r="7.8" fill="none" stroke="${gridDofBlack}" stroke-width="1.8"/>`;
            if (dofM !== null) {
                svg += `<text x="${outX.toFixed(2)}" y="${(outY + 16).toFixed(2)}" text-anchor="middle" font-family="Inter, Arial, Helvetica, sans-serif" font-size="${AXIS_NUMBER_SIZE}"
font-weight="700" fill="${gridDofBlack}">${dofM}</text>`;
            }
            svg += `</g>`;
            return;
        }

        const ns = neighbours.get(n.id) || [];
        const hasRightNeighbour = ns.some(id => {
            const q = pointForNode(id);
            return q && q.x > p.x + EPS;
        });
        const hasAnyRightColumn = xs.some(v => {
            const nx = Number(n.x) || 0;
            return v > nx + EPS;
        });

        // +X arrow/label is omitted when the global axial switch is active.
        if (dofX !== null) {
            const ax1 = p.x - 40;
            const ax2 = ax1 + AXIS_ARROW_LEN;
            svg += `<line x1="${ax1}" y1="${p.y}" x2="${ax2}" y2="${p.y}"
                stroke="${red}" stroke-width="${AXIS_STROKE}" marker-end="url(#dof-arrow-red)"/>`;
            svg += `<text x="${ax1 - 7}" y="${p.y + 4}" text-anchor="end"
                font-family="Inter, Arial, Helvetica, sans-serif" font-size="${AXIS_NUMBER_SIZE}"
                font-weight="700" fill="${red}">${dofX}</text>`;
        }

        // +Y arrow. Keep the arrow and label vertically aligned with the
        // centerline of the green node triangle for every node.
        const yShift = 0;
        const ay1 = p.y - AXIS_ARROW_GAP;
        const ay2 = ay1 - AXIS_ARROW_LEN;
        svg += `<line x1="${p.x + yShift}" y1="${ay1}" x2="${p.x + yShift}" y2="${ay2}"
            stroke="${red}" stroke-width="${AXIS_STROKE}" marker-end="url(#dof-arrow-red)"/>`;
        svg += `<text x="${p.x + yShift}" y="${ay2 - 7}" text-anchor="middle"
            font-family="Inter, Arial, Helvetica, sans-serif" font-size="${AXIS_NUMBER_SIZE}"
            font-weight="700" fill="${red}">${dofY}</text>`;

        // Moment symbol — preserve the reference yellow symbol's exact
        // logical position/scale while using the new black curved-tail design.
        if (dofM !== null) {
            const cx = p.x - 38;
            const cy = p.y - 34;
```

```
            svg += `<path d="M ${cx - 11} ${cy - 15}
                L ${cx - 11} ${cy - 5}
                C ${cx - 11} ${cy + 7}, ${cx - 2} ${cy + 13}, ${cx + 9} ${cy + 13}
                L ${cx + 14} ${cy + 13}"
                fill="none" stroke="#000000"
                stroke-linecap="round" stroke-linejoin="round"
                stroke-width="${MOMENT_STROKE}"
                marker-end="url(#dof-arrow-moment)"/>`;
            svg += `<text x="${cx - 2}" y="${cy - 15}" text-anchor="middle"
                font-family="Inter, Arial, Helvetica, sans-serif" font-size="${MOMENT_NUMBER_SIZE}"
                font-weight="700" fill="#000000">${dofM}</text>`;
        }
    });

    svg += `</svg>`;
    return svg;
}
```

29. modules/results/results.css

```
/* ========================================================================
 * results.css
 * Styles shared by the Results module: calculation-step typography, equations,
 * matrices, analysis summary, equilibrium check and the step menu / panels.
 * ======================================================================== */

/* CALC STEPS STYLING */
.calc-step {
    margin-bottom: 30px;
    padding-bottom: 20px;
    border-bottom: 1px dashed #cbd5e1;
}

.calc-step h4 {
    color: #1e3a8a;
    margin-bottom: 10px;
    font-size: 16px;
}

.calc-step h5 {
    color: #2563eb;
    margin-top: 25px;
    margin-bottom: 12px;
    padding-bottom: 6px;
    font-size: 14px;
    border-bottom: 1px solid #bfdbfe;
}

.calc-step p {
    font-size: 13px;
    color: #334155;
    margin-bottom: 12px;
    line-height: 1.6;
}

.eq-wrap {
    display: flex;
    flex-direction: column;
    align-items: stretch;
    justify-content: flex-start;
    gap: 18px;
    overflow-x: auto;
    padding: 20px;
    margin-bottom: 20px;
    background: #f8fafc;
    border-radius: 12px;
    border: 1px solid #e2e8f0;
    box-shadow: inset 0 2px 4px rgba(0, 0, 0, 0.02);
}

.eq-row {
    display: flex;
    align-items: center;
    justify-content: center;
    flex-wrap: nowrap;
    min-width: max-content;
}

.math-operator {
    font-weight: 800;
    font-size: 20px;
    color: #64748b;
    margin: 0 16px;
    display: flex;
    align-items: center;
    justify-content: center;
}

/* MATRICES */
.matrix-wrapper {
    display: inline-grid;
    grid-template-rows: 24px 14px auto;
    align-items: center;
    justify-items: center;
    margin: 0 4px;
    gap: 4px;
}

.matrix-title {
    font-weight: 700;
    font-size: 11px;
    color: #475569;
    background: #e2e8f0;
    padding: 4px 12px;
    border-radius: 12px;
```

```
    letter-spacing: 0.5px;
    margin: 0;
    align-self: end;
}

.matrix-table {
    font-family: "SFMono-Regular", Consolas, "Liberation Mono", Menlo, monospace;
    font-size: 13px;
    border-collapse: collapse;
    background: transparent;
    min-width: unset;
    width: auto;
    margin: 0;
    padding: 0;
}

.matrix-table td {
    padding: 6px 14px;
    text-align: right;
    color: #0f172a;
    white-space: nowrap;
    border: none;
    border-bottom: none;
}

.matrix-table tr:hover td {
    background: transparent;
}

/* ANALYSIS SUMMARY */
.analysis-summary-card {
    border: 1px solid var(--border);
    border-radius: 12px;
    background: white;
    margin-bottom: 20px;
    overflow: hidden;
}

.analysis-summary-grid {
    display: grid;
    grid-template-columns: repeat(3, minmax(0, 1fr));
    gap: 10px;
    padding: 16px;
}

.analysis-summary-metric {
    border: 1px solid #e5eaf0;
    border-radius: 9px;
    padding: 12px;
    background: #f8fafc;
}

.analysis-summary-label {
    color: var(--muted);
    font-size: 10px;
    font-weight: 800;
    text-transform: uppercase;
}

.analysis-summary-value {
    margin-top: 5px;
    font-size: 15px;
    font-weight: 800;
    color: #1f2937;
}

@media (max-width: 900px) {
    .analysis-summary-grid {
        grid-template-columns: repeat(2, minmax(0, 1fr));
    }
}

/* EQUILIBRIUM CHECK */
.equilibrium-card {
    border: 1px solid var(--border);
    border-radius: 12px;
    background: white;
    margin-bottom: 20px;
    overflow: hidden;
}

.equilibrium-status {
    display: flex;
    align-items: center;
    gap: 10px;
    padding: 14px 16px;
    font-weight: 800;
```

```
    font-size: 13px;
}

.equilibrium-status.ok {
    background: #f0fdf4;
    color: #166534;
}

.equilibrium-status.warn {
    background: #fff7ed;
    color: #9a3412;
}

.equilibrium-grid {
    display: grid;
    grid-template-columns: repeat(4, minmax(0, 1fr));
    gap: 10px;
    padding: 0 16px 16px;
}

.equilibrium-metric {
    border: 1px solid #e5eaf0;
    border-radius: 9px;
    padding: 11px 12px;
    background: #f8fafc;
}

.equilibrium-metric-label {
    color: var(--muted);
    font-size: 10px;
    font-weight: 800;
    text-transform: uppercase;
}

.equilibrium-metric-value {
    margin-top: 5px;
    font-size: 13px;
    font-weight: 800;
    color: #1f2937;
}

.equilibrium-note {
    padding: 0 16px 16px;
    color: var(--muted);
    font-size: 11px;
}

@media (max-width: 900px) {
    .equilibrium-grid {
        grid-template-columns: repeat(2, minmax(0, 1fr));
    }
}

/* RESULTS DETAIL MENU */
.results-details-card {
    overflow: visible;
}

.results-detail-menu {
    display: grid;
    grid-template-columns: repeat(4, minmax(0, 1fr));
    gap: 8px;
    padding: 14px 20px;
    border-bottom: 1px solid var(--border);
    background: #f8fafc;
}

.results-detail-btn {
    border: 1px solid #d6dee9;
    background: white;
    color: #334155;
    min-height: 42px;
    padding: 8px 10px;
    border-radius: 9px;
    font-size: 11px;
    font-weight: 800;
    line-height: 1.25;
    transition: 0.2s;
}

.results-detail-btn:hover {
    background: #eef2f7;
    border-color: #cbd5e1;
}

.results-detail-btn.active {
    background: var(--primary);
```

```
    border-color: var(--primary);
    color: white;
}

.results-detail-body {
    padding: 20px;
}

.result-detail-panel {
    display: none;
    animation: fade 0.2s ease;
}

.result-detail-panel.active {
    display: block;
}

.result-detail-panel > .calc-step {
    margin-bottom: 0;
    padding-bottom: 0;
    border-bottom: none;
}

.calc-steps-source {
    display: none;
}

@media (max-width: 900px) {
    .results-detail-menu {
        grid-template-columns: repeat(2, minmax(0, 1fr));
    }
}
```

30. modules/results/results.js

```
/* ========================================================================
 * results.js — Results module (container + orchestration)
 * Results page frame: analysis execution card, active load/moment case selectors,
 * analysis summary, equilibrium check, step menu container, analysis orchestration
 * (runAnalysis / runAnalysisCore), load-transformation helpers and the PDF export.
 * The nine result parts live in sub-folders and plug into this frame:
 *   step-1 ... step-8  -> menu button + panel of each calculation step / final results
 *   diagrams           -> reaction + combined result diagram cards, live model diagram,
 *                         diagram engine and full-screen modal
 * ======================================================================== */

/* ---- Markup ---- */
mountModuleHTML(".container", "beforeend", `
<!-- RESULTS -->
<section class="section" id="results">
    <div class="card">
        <div class="card-header">
            <div>
                <h3>Analysis Execution</h3>
                <p><span id="analysisModeNote">Plane analysis classifies each element as Beam, Truss or Frame from its A and I values.</span> Units: kN, m, kN/m, kN·m. <span
id="activeAnalysisCaseStatus" style="font-weight:700; color:#1d4ed8;"></span></p>
            </div>
            <button class="btn btn-primary" onclick="runAnalysis()">Run Analysis</button>
        </div>
        <div class="card-body" style="border-top: 1px solid var(--border); padding-top: 16px;">
            <div class="restraint-box">
                <div class="restraint-title">Active Analysis Cases</div>
                <div class="grid grid-2">
                    <div class="field">
                        <label for="activeLoadCaseSelect">Active Load Case</label>
                        <select id="activeLoadCaseSelect" onchange="setActiveAnalysisCases()"><option value="">No load case selected</option></select>
                    </div>
                    <div class="field">
                        <label for="activeMomentCaseSelect">Active Moment Case</label>
                        <select id="activeMomentCaseSelect" onchange="setActiveAnalysisCases()"><option value="">No moment case selected</option></select>
                    </div>
                </div>
                <small style="display:block; margin-top:10px; color:#64748b; line-height:1.5;">Only loads and moments belonging to the selected cases are included in the next analysis. Load
cases and moment cases are selected independently.</small>
            </div>

            <div class="restraint-box" id="planeDofNumberingBox" style="margin-top:14px;">
                <div class="restraint-title">Global DoF Numbering</div>
                <div class="grid grid-2">
                    <label class="check-item" style="cursor:pointer;">
                        <input type="checkbox" id="skipAxialDofGlobal" onchange="setGlobalDofNumberingOption('skipAxial', this.checked)">
                        <span>Neglect X DOF in numbering</span>
                    </label>
                    <label class="check-item" style="cursor:pointer;">
                        <input type="checkbox" id="skipMomentDofGlobal" onchange="setGlobalDofNumberingOption('skipMoment', this.checked)">
                        <span>Neglect Moment M DOF in numbering</span>
                    </label>
                </div>
                <small id="globalDofNumberingNote" style="display:block; margin-top:10px; color:#64748b; line-height:1.5;">Select Neglect X and/or Neglect Moment M to omit only the
corresponding DOFs that are unnecessary at each node according to connected element A/I values. Leave them unselected to keep all X,Y,M DOFs numbered.</small>
            </div>
        </div>
        <div id="analysisStatusBanner" style="margin-top:12px; padding:12px 14px; border-radius:9px; border:1px solid #bfdbfe; background:#eff6ff; color:#1e3a8a; font-size:12px; font-weight:
700; line-height:1.5;">Analysis not run yet — Run Analysis to view results.</div>
    </div>

    <div id="resultsContent" style="display:none;">

        <!-- ANALYSIS SUMMARY -->
        <div class="analysis-summary-card">
            <div class="card-header">
                <div>
                    <h3>Analysis Summary</h3>
                    <p>Key results from the current analysis.</p>
                </div>
            </div>
            <div class="analysis-summary-grid">
                <div class="analysis-summary-metric"><div class="analysis-summary-label">Maximum displacement</div><div class="analysis-summary-value" id="summaryMaxDisplacement">—</div></
div>
                <div class="analysis-summary-metric"><div class="analysis-summary-label">Maximum reaction</div><div class="analysis-summary-value" id="summaryMaxReaction">—</div></div>
                <div class="analysis-summary-metric"><div class="analysis-summary-label" id="summaryAxialLabel">Maximum axial force</div><div class="analysis-summary-value"
id="summaryMaxAxial">—</div></div>
                <div class="analysis-summary-metric"><div class="analysis-summary-label" id="summaryShearLabel">Maximum shear</div><div class="analysis-summary-value"
id="summaryMaxShear">—</div></div>
                <div class="analysis-summary-metric"><div class="analysis-summary-label" id="summaryMomentLabel">Maximum bending moment</div><div class="analysis-summary-value"
id="summaryMaxMoment">—</div></div>
                <div class="analysis-summary-metric"><div class="analysis-summary-label">Equilibrium status</div><div class="analysis-summary-value" id="summaryEquilibriumStatus">—</div></
div>
            </div>
        </div>
```

```
        <!-- EQUILIBRIUM CHECK -->
        <div class="equilibrium-card">
            <div class="card-header">
                <div>
                    <h3>Global Equilibrium Check</h3>
                    <p>External loads and support reactions must balance in global coordinates.</p>
                </div>
            </div>
            <div id="equilibriumStatus" class="equilibrium-status ok">✓ Equilibrium satisfied</div>
            <div class="equilibrium-grid">
                <div class="equilibrium-metric"><div class="equilibrium-metric-label" id="eqFxLabel">ΣFx Residual</div><div class="equilibrium-metric-value" id="eqFxResidual">0.000 kN</
div></div>
                <div class="equilibrium-metric"><div class="equilibrium-metric-label" id="eqFyLabel">ΣFy Residual</div><div class="equilibrium-metric-value" id="eqFyResidual">0.000 kN</
div></div>
                <div class="equilibrium-metric"><div class="equilibrium-metric-label" id="eqMzLabel">ΣMz Residual</div><div class="equilibrium-metric-value" id="eqMzResidual">0.000 kN·m</
div></div>
                <div class="equilibrium-metric"><div class="equilibrium-metric-label" id="eqMaxResidualLabel">Force / Moment Residuals</div><div class="equilibrium-metric-value"
id="eqMaxResidual">0.000</div></div>
            </div>
            <div class="equilibrium-note" id="equilibriumNote">External load vector + support reaction vector = global equilibrium residual.</div>
        </div>

        <!-- CALCULATION STEPS UI -->
                        <!-- CALCULATION STEPS + FINAL TABLES MENU -->
        <div class="card results-details-card">
            <div class="card-header" style="flex-wrap: wrap;">
                <div>
                    <h3>Analysis Calculation & Final Results</h3>
                    <p>Select one option to view only that calculation step or the final result tables.</p>
                </div>
                <button type="button" class="btn btn-secondary" id="exportPdfBtn" style="flex-shrink: 0; white-space: nowrap;" onclick="exportAnalysisPDF()" title="Export the full 8-step
solution as a PDF">■ Export PDF</button>
            </div>
            <div class="results-detail-menu" role="tablist" aria-label="Analysis calculation steps">
            </div>

            <div class="results-detail-body">
            </div>

            <!-- Hidden source container used by the existing solver; its generated step content is moved into the menu panels after each analysis. -->
            <div id="calcStepsContainer" class="calc-steps-source" aria-hidden="true"></div>
        </div>
    </div>
</section>
`);

/* ---- Logic ---- */
// ---- Cross-entity helpers: dropdown syncing, deletion, stats, full refresh ----
function updateAnalysisStatusUI() {
    const banner = document.getElementById("analysisStatusBanner");
    const statusEl = document.getElementById("activeAnalysisCaseStatus");
    if (!banner) return;

    if (analysisState.status === "valid") {
        banner.textContent = "Analysis results are current for the selected analysis cases.";
        banner.style.borderColor = "#bbf7d0";
        banner.style.background = "#f0fdf4";
        banner.style.color = "#166534";
    } else if (analysisState.status === "failed") {
        banner.textContent = "Equilibrium check failed — review the results before relying on them.";
        banner.style.borderColor = "#fecaca";
        banner.style.background = "#fef2f2";
        banner.style.color = "#991b1b";
    } else if (analysisState.status === "invalidated") {
        banner.textContent = "Model changed since last analysis — Run Analysis again.";
        banner.style.borderColor = "#fecaca";
        banner.style.background = "#fef2f2";
        banner.style.color = "#991b1b";
    } else {
        banner.textContent = "Analysis not run yet — Run Analysis to view results.";
        banner.style.borderColor = "#bfdbfe";
        banner.style.background = "#eff6ff";
        banner.style.color = "#1e3a8a";
    }

    if (statusEl) {
        if (analysisState.status === "valid") {
            statusEl.textContent = "";
        } else if (analysisState.status === "failed") {
            statusEl.textContent = " • Equilibrium check failed";
            statusEl.style.color = "#991b1b";
        } else {
            statusEl.textContent = analysisState.status === "invalidated" ? " • Results outdated" : " • Not run";
            statusEl.style.color = analysisState.status === "invalidated" ? "#b91c1c" : "#64748b";
        }
    }
}
```

```
function markAnalysisInvalidated() {
    updateGlobalDofNumberingUI();
    analysisState.status = "invalidated";
    model.nodeDisplacements = undefined;
    model.elements.forEach((el) => {
        if (el) delete el.plotData;
    });
    const resultsContent = document.getElementById("resultsContent");
    if (resultsContent) resultsContent.style.display = "none";
    updateAnalysisStatusUI();
}

function markAnalysisValid() {
    analysisState.status = "valid";
    updateAnalysisStatusUI();
}

function markAnalysisFailed() {
    analysisState.status = "failed";
    updateAnalysisStatusUI();
}

function setActiveAnalysisCases() {
    const previousLoad = activeLoadCaseId;
    const previousMoment = activeMomentCaseId;
    const loadSelect = document.getElementById("activeLoadCaseSelect");
    const momentSelect = document.getElementById("activeMomentCaseSelect");
    if (loadSelect) activeLoadCaseId = loadSelect.value;
    if (momentSelect) activeMomentCaseId = momentSelect.value;
    if (activeLoadCaseId !== previousLoad || activeMomentCaseId !== previousMoment) {
        markAnalysisInvalidated();
    }
    updateAnalysisStatusUI();
}

function updateGlobalDofNumberingUI() {
    const axialEl = document.getElementById("skipAxialDofGlobal");
    const momentEl = document.getElementById("skipMomentDofGlobal");
    const noteEl = document.getElementById("globalDofNumberingNote");
    const config = getGlobalDofNumberingConfig();

    // The two controls are the optional manual switches for the Excel-style
    // per-node numbering. They do not remove a DOF globally from the model.
    if (axialEl) { axialEl.checked = config.skipAxial; axialEl.disabled = false; }
    if (momentEl) { momentEl.checked = config.skipMoment; momentEl.disabled = false; }
    if (noteEl) {
        noteEl.textContent =
            "Neglect X: " + (config.skipAxial ? "ON" : "OFF") +
            " · Neglect Moment M: " + (config.skipMoment ? "ON" : "OFF") +
            ". When ON, only the DOF made unnecessary by the connected element A/I properties is omitted at that node. Shared nodes keep the union of required DOFs and numbering remains
compressed.";
    }
}

function syncActiveAnalysisCaseSelectors() {
    const loadSelect = document.getElementById("activeLoadCaseSelect");
    const momentSelect = document.getElementById("activeMomentCaseSelect");
    if (!loadSelect || !momentSelect) return;

    const previousLoad = activeLoadCaseId;
    const previousMoment = activeMomentCaseId;

    loadSelect.innerHTML = model.loadCases.length
        ? model.loadCases.map((lc) => `<option value="${escapeHtml(lc.id)}">${escapeHtml(lc.id)} — ${escapeHtml(lc.type)}</option>`).join("")
        : `<option value="">No load cases available</option>`;

    momentSelect.innerHTML = model.momentCases.length
        ? model.momentCases.map((mc) => `<option value="${escapeHtml(mc.id)}">${escapeHtml(mc.id)} — ${escapeHtml(mc.type)}</option>`).join("")
        : `<option value="">No moment cases available</option>`;

    if (model.loadCases.some((lc) => lc.id === previousLoad)) {
        activeLoadCaseId = previousLoad;
    } else {
        activeLoadCaseId = model.loadCases.length ? model.loadCases[0].id : "";
    }

    if (model.momentCases.some((mc) => mc.id === previousMoment)) {
        activeMomentCaseId = previousMoment;
    } else {
        activeMomentCaseId = model.momentCases.length ? model.momentCases[0].id : "";
    }

    loadSelect.value = activeLoadCaseId;
    momentSelect.value = activeMomentCaseId;
}

function isActiveLoadCase(load) {
```

```
    return !!activeLoadCaseId && String(load.loadCaseId || "") === String(activeLoadCaseId);
}

function isActiveMomentCase(moment) {
    return !!activeMomentCaseId && String(moment.momentCaseId || "") === String(activeMomentCaseId);
}

function getActiveAnalysisCaseLabel() {
    const load = model.loadCases.find((lc) => lc.id === activeLoadCaseId);
    const moment = model.momentCases.find((mc) => mc.id === activeMomentCaseId);
    return `Load Case: ${load ? load.id : "None"} | Moment Case: ${moment ? moment.id : "None"}`;
}

// ============================================================================
// 15. LOAD TRANSFORMATION HELPERS
// Plane-frame convention from the lab manual: [AMS] = [RT]^T [AML]
//
// For an inclined member, any load entered in GLOBAL/ANGLED axes is first
// resolved into the member axes (xm, ym). AML is then formed entirely in
// member axes. Only after AML is complete do we transform it to structure
// axes through RT^T.
// ============================================================================
function rotateForceToGlobal(Fx, Fy, axisSystem, angleDeg) {
    let x = Number(Fx) || 0;
    let y = Number(Fy) || 0;

    if (axisSystem === "Angled") {
        const rad = ((Number(angleDeg) || 0) * Math.PI) / 180;
        return {
            x: x * Math.cos(rad) - y * Math.sin(rad),
            y: x * Math.sin(rad) + y * Math.cos(rad),
        };
    }

    return { x, y };
}

function globalForceToLocal(gx, gy, cx, cy) {
    // [Fxm]   [ Cx   Cy ] [Fx_global]
    // [Fym] = [-Cy   Cx ] [Fy_global]
    return {
        x: gx * cx + gy * cy,
        y: -gx * cy + gy * cx,
    };
}

function inputForceToLocal(Fx, Fy, axisSystem, angleDeg, cx, cy) {
    if (axisSystem === "Local") {
        return { x: Number(Fx) || 0, y: Number(Fy) || 0 };
    }

    const g = rotateForceToGlobal(Fx, Fy, axisSystem, angleDeg);
    return globalForceToLocal(g.x, g.y, cx, cy);
}

function localForceToGlobal(Fx, Fy, cx, cy) {
    return {
        x: Fx * cx - Fy * cy,
        y: Fx * cy + Fy * cx,
    };
}

function addLocalPointLoadToAML(AML, Px, Py, a, L) {
    AML[0][0] += (-Px * (L - a)) / L;
    AML[1][0] += (-Py * Math.pow(L - a, 2) * (L + 2 * a)) / Math.pow(L, 3);
    AML[2][0] += (-Py * a * Math.pow(L - a, 2)) / Math.pow(L, 2);
    AML[3][0] += (-Px * a) / L;
    AML[4][0] += (-Py * Math.pow(a, 2) * (3 * L - 2 * a)) / Math.pow(L, 3);
    AML[5][0] += (Py * Math.pow(a, 2) * (L - a)) / Math.pow(L, 2);
}

function integrateLinearLocalUDLToAML(AML, wx1, wy1, wx2, wy2, a, b, L, nSlices = 100) {
    if (b <= a) return;

    const h = (b - a) / nSlices;

    // Simpson integration of the exact point-load AML influence
    // functions. The load intensity is interpolated in LOCAL axes.
    for (let i = 0; i <= nSlices; i++) {
        const x = a + i * h;
        const t = (x - a) / (b - a);
        const wx = wx1 + t * (wx2 - wx1);
        const wy = wy1 + t * (wy2 - wy1);

        const mult = i === 0 || i === nSlices ? 1 : i % 2 === 0 ? 2 : 4;
        const weight = (h / 3) * mult;
        AML[0][0] += -wx * ((L - x) / L) * weight;
```

```
        AML[1][0] += -wy * ((Math.pow(L - x, 2) * (L + 2 * x)) / Math.pow(L, 3)) * weight;
        AML[2][0] += -wy * ((x * Math.pow(L - x, 2)) / Math.pow(L, 2)) * weight;
        AML[3][0] += -wx * (x / L) * weight;
        AML[4][0] += -wy * ((Math.pow(x, 2) * (3 * L - 2 * x)) / Math.pow(L, 3)) * weight;
        AML[5][0] += wy * ((Math.pow(x, 2) * (L - x)) / Math.pow(L, 2)) * weight;
    }
}

// ============================================================================
// 16. STIFFNESS METHOD SOLVER
// Runs the full LPW-3 stiffness method and renders the step-by-step report.
// ============================================================================
function showResultDetail(panelId, button) {
    document.querySelectorAll(".result-detail-panel").forEach((panel) => {
        panel.classList.toggle("active", panel.id === panelId);
    });

    document.querySelectorAll(".results-detail-btn").forEach((btn) => {
        btn.classList.toggle("active", btn === button);
    });
}

function setupResultDetailPanels() {
    const source = document.getElementById("calcStepsContainer");
    if (!source) return;

    const steps = Array.from(source.querySelectorAll(":scope > .calc-step"));
    steps.forEach((step, index) => {
        const stepNumber = index + 1;
        const slot = document.querySelector(`.calc-step-slot[data-step-number="${stepNumber}"]`);
        if (slot) {
            slot.innerHTML = "";
            slot.appendChild(step);
        }
    });

    const firstButton = document.querySelector('.results-detail-btn[data-result-panel="result-step-1"]');
    if (firstButton) showResultDetail("result-step-1", firstButton);
}

// Reconstructs global static equilibrium (ΣFx, ΣFy, ΣMz) directly from the
// input loads/moments/self-weight and the solved reactions, independent of
// the assembled [AC] load vector (see note below). Pure calculation — no
// DOM access. Rendering the result happens separately in
// renderEquilibriumCheck().
function calculateEquilibriumResiduals(AC, dofMap, modelNodes, restIndices, AR) {
    // IMPORTANT: Do not use the assembled [AC] vector for global statics.
    // [AC] contains finite-element equivalent nodal actions from member loads.
    // This check independently reconstructs the PHYSICAL external loads and
    // moments so that distributed loads are not mistaken for an unbalanced
    // nodal moment (e.g. a 10 kN/m UDL over 5 m gives 125 kN·m about its end).
    let sumFx = 0;
    let sumFy = 0;
    let sumMz = 0;

    const refNode = modelNodes[0] || { x: 0, y: 0 };
    const refX = Number(refNode.x) || 0;
    const refY = Number(refNode.y) || 0;

    const addForce = (x, y, fx, fy) => {
        const rx = x - refX;
        const ry = y - refY;
        sumFx += Number(fx) || 0;
        sumFy += Number(fy) || 0;
        sumMz += rx * (Number(fy) || 0) - ry * (Number(fx) || 0);
    };

    const addPureMoment = (mzStructuralSign) => {
        sumMz += Number(mzStructuralSign) || 0;
    };

    // ------------------------------------------------------------------
    // 1. Nodal forces and nodal moments from the active cases.
    // ------------------------------------------------------------------
    model.loads.forEach((ld) => {
        if (!isActiveLoadCase(ld) || ld.type !== "Nodal Load") return;
        const node = modelNodes.find((n) => n.id === ld.nodeId);
        if (!node) return;

        const g = rotateForceToGlobal(ld.Fx, ld.Fy, ld.axisSystem, ld.angle);
        addForce(Number(node.x) || 0, Number(node.y) || 0, g.x, g.y);
    });

    model.moments.forEach((m) => {
        if (!isActiveMomentCase(m) || m.type !== "Nodal Moment") return;
        // UI convention: +Mz is anticlockwise, matching the global structural
        // equilibrium convention.
```

```
    addPureMoment(Number(m.Mz) || 0);
});

// ---------------------------------------------------------------------
// 2. Element point loads, partial/linearly varying UDLs, and self-weight.
//    Reconstruct their actual global resultants directly from the input.
// ---------------------------------------------------------------------
const addElementUDLResultant = (el, a, b, qx1, qy1, qx2, qy2) => {
    const n1 = modelNodes.find((n) => n.id === el.startNode);
    const n2 = modelNodes.find((n) => n.id === el.endNode);
    if (!n1 || !n2) return;

    const x1 = Number(n1.x) || 0;
    const y1 = Number(n1.y) || 0;
    const Lseg = b - a;
    if (!(Lseg > 0)) return;

    const dx = (Number(n2.x) || 0) - x1;
    const dy = (Number(n2.y) || 0) - y1;
    const L = Math.hypot(dx, dy);
    if (!(L > 0)) return;

    const cx = dx / L;
    const cy = dy / L;

    const ix = ((qx1 + qx2) * 0.5) * Lseg;
    const iy = ((qy1 + qy2) * 0.5) * Lseg;

    // First moment of the line-load intensities with respect to the
    // element start node: ∫ s*q(s) ds over s in [a,b].
    const kx = (qx2 - qx1) / Lseg;
    const ky = (qy2 - qy1) / Lseg;
    // Equivalent expression using u=s-a is clearer and numerically safer.
    const sxSafe = a * qx1 * Lseg + a * kx * Lseg * Lseg / 2 + qx1 * Lseg * Lseg / 2 + kx * Math.pow(Lseg, 3) / 3;
    const sySafe = a * qy1 * Lseg + a * ky * Lseg * Lseg / 2 + qy1 * Lseg * Lseg / 2 + ky * Math.pow(Lseg, 3) / 3;

    sumFx += ix;
    sumFy += iy;
    sumMz += (x1 - refX) * iy - (y1 - refY) * ix + cx * sySafe - cy * sxSafe;
};

model.elements.forEach((el) => {
    const n1 = modelNodes.find((n) => n.id === el.startNode);
    const n2 = modelNodes.find((n) => n.id === el.endNode);
    const tmpl = model.elementTemplates.find((t) => t.id === el.templateId);
    if (!n1 || !n2 || !tmpl) return;

    const x1 = Number(n1.x) || 0;
    const y1 = Number(n1.y) || 0;
    const x2 = Number(n2.x) || 0;
    const y2 = Number(n2.y) || 0;
    const dx = x2 - x1;
    const dy = y2 - y1;
    const L = Math.hypot(dx, dy);
    if (!(L > 0)) return;
    const cx = dx / L;
    const cy = dy / L;

    model.loads.forEach((ld) => {
        if (!isActiveLoadCase(ld) || ld.elementId !== el.id) return;

        if (ld.type === "Element Point Load" || ld.type === "Element Load") {
            const a = Math.max(0, Math.min(Number(ld.distance) || 0, L));
            const local = inputForceToLocal(ld.Fx, ld.Fy, ld.axisSystem, ld.angle, cx, cy);
            const global = localForceToGlobal(local.x, local.y, cx, cy);
            addForce(x1 + cx * a, y1 + cy * a, global.x, global.y);
            return;
        }

        if (ld.type === "Element UDL") {
            const a = Math.max(0, Math.min(Number(ld.distance) || 0, L));
            const bRaw = ld.distanceEnd !== undefined && ld.distanceEnd !== "" ? Number(ld.distanceEnd) : L;
            const b = Math.max(0, Math.min(Number.isFinite(bRaw) ? bRaw : L, L));
            if (!(a < b)) return;

            const Fx1 = Number(ld.Fx) || 0;
            const Fy1 = Number(ld.Fy) || 0;
            const Fx2 = ld.FxEnd !== undefined && ld.FxEnd !== "" ? Number(ld.FxEnd) : Fx1;
            const Fy2 = ld.FyEnd !== undefined && ld.FyEnd !== "" ? Number(ld.FyEnd) : Fy1;

            const local1 = inputForceToLocal(Fx1, Fy1, ld.axisSystem, ld.angle, cx, cy);
            const local2 = inputForceToLocal(Fx2, Fy2, ld.axisSystem, ld.angle, cx, cy);
            const global1 = localForceToGlobal(local1.x, local1.y, cx, cy);
            const global2 = localForceToGlobal(local2.x, local2.y, cx, cy);

            addElementUDLResultant(el, a, b, global1.x, global1.y, global2.x, global2.y);
        }
```

```
        });

        model.moments.forEach((m) => {
            if (!isActiveMomentCase(m) || m.type !== "Element Moment" || m.elementId !== el.id) return;
            // Applied element point moment is independent of its location.
            // +Mz in the UI is anticlockwise and therefore positive in the
            // global structural equilibrium convention.
            addPureMoment(Number(m.Mz) || 0);
        });

        // Per-element self-weight is applied only from the active Self Weight case.
        const activeLoadCase = model.loadCases.find((lc) => String(lc.id) === String(activeLoadCaseId));
        if (activeLoadCase?.type === "Self Weight") {
            const selfWeight = Number(activeLoadCase.memberSelfWeights?.[String(el.id)]) || 0;
            if (selfWeight > 0) {
                const weightPerLength = -selfWeight;
                addElementUDLResultant(el, 0, L, 0, weightPerLength, 0, weightPerLength);
            }
        }
    });

    // ---------------------------------------------------------------------
    // 3. Add solved support reactions in the same global equilibrium sign
    //    convention. AR is already expressed in structural/global DOFs.
    // ---------------------------------------------------------------------
    restIndices.forEach((ri, i) => {
        const reaction = Number(AR[i][0]) || 0;
        let reactionNode = null;
        let reactionDof = null;

        for (const node of modelNodes) {
            const d = dofMap[node.id];
            if (d.x === ri) { reactionNode = node; reactionDof = "UX"; break; }
            if (d.y === ri) { reactionNode = node; reactionDof = "UY"; break; }
            if (d.z === ri) { reactionNode = node; reactionDof = "RZ"; break; }
        }

        if (!reactionNode) return;

        if (reactionDof === "UX") {
            addForce(Number(reactionNode.x) || 0, Number(reactionNode.y) || 0, reaction, 0);
        } else if (reactionDof === "UY") {
            addForce(Number(reactionNode.x) || 0, Number(reactionNode.y) || 0, 0, reaction);
        } else if (reactionDof === "RZ") {
            sumMz += reaction;
        }
    });

    const absFx = Math.abs(sumFx);
    const absFy = Math.abs(sumFy);
    const absMz = Math.abs(sumMz);
    const maxResidual = Math.max(absFx, absFy, absMz);

    let loadScale = Math.max(Math.abs(sumFx), Math.abs(sumFy));
    let momentScale = Math.abs(sumMz);
    model.loads.forEach((ld) => {
        if (!isActiveLoadCase(ld)) return;
        loadScale = Math.max(loadScale, Math.abs(Number(ld.Fx) || 0), Math.abs(Number(ld.Fy) || 0));
    });
    model.moments.forEach((m) => {
        if (!isActiveMomentCase(m)) return;
        momentScale = Math.max(momentScale, Math.abs(Number(m.Mz) || 0));
    });
    restIndices.forEach((ri, i) => {
        loadScale = Math.max(loadScale, Math.abs(Number(AR[i][0]) || 0));
    });

    const structureSpan = Math.max(1, ...modelNodes.map((n) => Math.hypot((Number(n.x) || 0) - refX, (Number(n.y) || 0) - refY)));
    const tolerance = Math.max(1e-8, loadScale * 1e-8);
    const momentTolerance = Math.max(1e-8, Math.max(momentScale, loadScale * structureSpan) * 1e-8);
    const forceResidualOk = absFx <= tolerance && absFy <= tolerance;
    const momentResidualOk = absMz <= momentTolerance;
    const balanced = forceResidualOk && momentResidualOk;

    return {
        sumFx, sumFy, sumMz, maxResidual,
        maxForceResidual: Math.max(absFx, absFy),
        maxMomentResidual: absMz,
        balanced, tolerance, momentTolerance,
    };
}

// Grid equilibrium is audited from physical global load resultants and solved
// support reactions, including the positional moments caused by vertical Fz loads.
function calculateGridEquilibriumResiduals(AC, dofMap, modelNodes, restIndices, AR) {
    // The Grid equilibrium audit must check physical global statics, not the
    // unweighted sum of generalized nodal DOFs. A vertical Fz at (x,y) also
```

```
// contributes Mx = (y-y0)Fz and My = -(x-x0)Fz about the common reference.
// AC remains an assembled solver vector and is intentionally not used here.
const refNode = modelNodes[0] || { x: 0, y: 0 };
const refX = Number(refNode.x) || 0;
const refY = Number(refNode.y) || 0;
let sumMx = 0;
let sumMy = 0;
let sumFz = 0;
let forceScale = 0;
let momentScale = 0;

const addVerticalForce = (x, y, fz) => {
    const fx = Number(x) || 0;
    const fy = Number(y) || 0;
    const force = Number(fz) || 0;
    const mx = (fy - refY) * force;
    const my = -(fx - refX) * force;
    sumFz += force;
    sumMx += mx;
    sumMy += my;
    forceScale = Math.max(forceScale, Math.abs(force));
    momentScale = Math.max(momentScale, Math.abs(mx), Math.abs(my));
};

const addAppliedTorsion = (mx, my) => {
    const tx = Number(mx) || 0;
    const ty = Number(my) || 0;
    sumMx += tx;
    sumMy += ty;
    momentScale = Math.max(momentScale, Math.abs(tx), Math.abs(ty));
};

// Resultant of a linearly varying vertical grid load over [a,b].
// firstMoment is integral(s*q(s) ds), measured from the element start.
const addGridUDLResultant = (el, a, b, q1, q2) => {
    const n1 = modelNodes.find((node) => String(node.id) === String(el.startNode));
    const n2 = modelNodes.find((node) => String(node.id) === String(el.endNode));
    if (!n1 || !n2) return;
    const x1 = Number(n1.x) || 0;
    const y1 = Number(n1.y) || 0;
    const dx = (Number(n2.x) || 0) - x1;
    const dy = (Number(n2.y) || 0) - y1;
    const L = Math.hypot(dx, dy);
    if (!(L > 0)) return;
    const lo = Math.max(0, Math.min(Number(a) || 0, L));
    const hi = Math.max(0, Math.min(Number(b), L));
    const span = hi - lo;
    if (!(span > 0)) return;
    const startLoad = Number(q1) || 0;
    const endLoad = Number(q2) || 0;
    const resultant = (startLoad + endLoad) * span / 2;
    const firstMoment = lo * resultant + span * span * (startLoad + 2 * endLoad) / 6;
    const cx = dx / L;
    const cy = dy / L;
    sumFz += resultant;
    const mx = (y1 - refY) * resultant + cy * firstMoment;
    const my = -((x1 - refX) * resultant + cx * firstMoment);
    sumMx += mx;
    sumMy += my;
    forceScale = Math.max(forceScale, Math.abs(startLoad * span), Math.abs(endLoad * span), Math.abs(resultant));
    momentScale = Math.max(momentScale, Math.abs(mx), Math.abs(my), Math.abs(firstMoment));
};

// 1. Applied nodal loads.
model.loads.forEach((load) => {
    if (!isActiveLoadCase(load)) return;
    if (load.type === "Nodal Load") {
        const node = modelNodes.find((item) => String(item.id) === String(load.nodeId));
        if (node) addVerticalForce(node.x, node.y, load.Fz);
        return;
    }

    const elementLoad = load.type === "Element Point Load" || load.type === "Element Load" || load.type === "Element UDL";
    if (!elementLoad) return;
    const el = model.elements.find((item) => String(item.id) === String(load.elementId));
    if (!el) return;
    const n1 = modelNodes.find((node) => String(node.id) === String(el.startNode));
    const n2 = modelNodes.find((node) => String(node.id) === String(el.endNode));
    if (!n1 || !n2) return;
    const x1 = Number(n1.x) || 0;
    const y1 = Number(n1.y) || 0;
    const dx = (Number(n2.x) || 0) - x1;
    const dy = (Number(n2.y) || 0) - y1;
    const L = Math.hypot(dx, dy);
    if (!(L > 0)) return;
    const cx = dx / L;
    const cy = dy / L;
```

```
    if (load.type === "Element Point Load" || load.type === "Element Load") {
        const a = Math.max(0, Math.min(Number(load.distance) || 0, L));
        addVerticalForce(x1 + cx * a, y1 + cy * a, load.Fz);
    } else {
        const a = Math.max(0, Math.min(Number(load.distance) || 0, L));
        const rawEnd = load.distanceEnd === "" || load.distanceEnd == null ? L : Number(load.distanceEnd);
        const b = Math.max(0, Math.min(Number.isFinite(rawEnd) ? rawEnd : L, L));
        if (!(b > a)) return;
        const q1 = Number(load.Fz) || 0;
        const q2 = load.FzEnd !== undefined && load.FzEnd !== "" ? (Number(load.FzEnd) || 0) : q1;
        addGridUDLResultant(el, a, b, q1, q2);
    }
});

// 2. Self-weight, only for an active Self Weight load case, matching Grid
//    Step 3's existing rule and property conversion.
const activeCase = model.loadCases.find((item) => String(item.id) === String(activeLoadCaseId));
if (activeCase?.type === "Self Weight") {
    model.elements.forEach((el) => {
        const n1 = modelNodes.find((node) => String(node.id) === String(el.startNode));
        const n2 = modelNodes.find((node) => String(node.id) === String(el.endNode));
        const template = model.elementTemplates.find((item) => String(item.id) === String(el.templateId));
        const selfWeight = Number(activeCase.memberSelfWeights?.[String(el.id)]) || 0;
        if (!n1 || !n2 || !template || !(selfWeight > 0)) return;
        const dx = (Number(n2.x) || 0) - (Number(n1.x) || 0);
        const dy = (Number(n2.y) || 0) - (Number(n1.y) || 0);
        const L = Math.hypot(dx, dy);
        if (!(L > 0)) return;
        // Self-weight is stored directly as a line load (kN/m).
        const q = -selfWeight;
        if (q !== 0) addGridUDLResultant(el, 0, L, q, q);
    });
}

// 3. Applied Grid torsions. Mx and My are already global moment components.
model.moments.forEach((moment) => {
    if (!isActiveMomentCase(moment)) return;
    if (moment.type === "Nodal Moment") {
        addAppliedTorsion(moment.Mx, moment.My);
    } else if (moment.type === "Element Moment") {
        addAppliedTorsion(moment.Mx, moment.My);
        // Legacy Mz is not shown in Grid mode, but if an older model still
        // contains it, Step 3 treats it as torsion about the member axis.
        const mz = Number(moment.Mz) || 0;
        if (mz !== 0) {
            const el = model.elements.find((item) => String(item.id) === String(moment.elementId));
            const n1 = el && modelNodes.find((node) => String(node.id) === String(el.startNode));
            const n2 = el && modelNodes.find((node) => String(node.id) === String(el.endNode));
            if (n1 && n2) {
                const dx = (Number(n2.x) || 0) - (Number(n1.x) || 0);
                const dy = (Number(n2.y) || 0) - (Number(n1.y) || 0);
                const L = Math.hypot(dx, dy);
                if (L > 0) addAppliedTorsion(mz * dx / L, mz * dy / L);
            }
        }
    }
});

// 4. Solved support reactions: θX reaction = Mx, θY = My and ΔZ = Fz.
//    A support's Fz reaction also contributes its coordinate-dependent Mx/My.
restIndices.forEach((idx, i) => {
    const reaction = Number(AR[i]?.[0]) || 0;
    for (const node of modelNodes) {
        const d = dofMap[node.id];
        if (!d) continue;
        if (d.x === idx) {
            addAppliedTorsion(reaction, 0);
            break;
        }
        if (d.y === idx) {
            addAppliedTorsion(0, reaction);
            break;
        }
        if (d.z === idx) {
            addVerticalForce(node.x, node.y, reaction);
            break;
        }
    }
});

const absMx = Math.abs(sumMx);
const absMy = Math.abs(sumMy);
const absFz = Math.abs(sumFz);
const maxResidual = Math.max(absMx, absMy, absFz);
const structureSpan = Math.max(1, ...modelNodes.map((node) => Math.hypot((Number(node.x) || 0) - refX, (Number(node.y) || 0) - refY)));
const forceTolerance = Math.max(1e-8, forceScale * 1e-8);
const momentTolerance = Math.max(1e-8, momentScale * 1e-8, forceScale * structureSpan * 1e-8);
```

```
    return {
        // Existing result property names are retained for the shared UI: in
        // Grid mode these are ΣMx, ΣMy and ΣFz, as the labels already indicate.
        sumFx: sumMx,
        sumFy: sumMy,
        sumMz: sumFz,
        maxResidual,
        maxForceResidual: absFz,
        maxMomentResidual: Math.max(absMx, absMy),
        balanced: absMx <= momentTolerance && absMy <= momentTolerance && absFz <= forceTolerance,
        tolerance: forceTolerance,
        forceTolerance,
        momentTolerance,
        grid: true,
    };
}

// Writes an equilibrium-check result (from calculateEquilibriumResiduals())
// into the Results panel. Pure DOM update — no calculation.
function renderEquilibriumCheck(result) {
    const { sumFx, sumFy, sumMz, maxResidual, balanced, tolerance, momentTolerance } = result;

    const statusEl = document.getElementById("equilibriumStatus");
    const fxEl = document.getElementById("eqFxResidual");
    const fyEl = document.getElementById("eqFyResidual");
    const mzEl = document.getElementById("eqMzResidual");
    const maxEl = document.getElementById("eqMaxResidual");
    const noteEl = document.getElementById("equilibriumNote");

    if (result.grid) {
        if (fxEl) fxEl.textContent = `${formatNum(sumFx)} kN·m`;
        if (fyEl) fyEl.textContent = `${formatNum(sumFy)} kN·m`;
        if (mzEl) mzEl.textContent = `${formatNum(sumMz)} kN`;
    } else {
        if (fxEl) fxEl.textContent = `${formatNum(sumFx)} kN`;
        if (fyEl) fyEl.textContent = `${formatNum(sumFy)} kN`;
        if (mzEl) mzEl.textContent = `${formatNum(sumMz)} kN·m`;
    }
    const forceResidual = Number.isFinite(result.maxForceResidual)
        ? result.maxForceResidual
        : (result.grid ? Math.abs(sumMz) : Math.max(Math.abs(sumFx), Math.abs(sumFy)));
    const momentResidual = Number.isFinite(result.maxMomentResidual)
        ? result.maxMomentResidual
        : (result.grid ? Math.max(Math.abs(sumFx), Math.abs(sumFy)) : Math.abs(sumMz));
    const residualSummary = `Force: ${formatNum(forceResidual)} kN · Moment: ${formatNum(momentResidual)} kN·m`;
    if (maxEl) maxEl.textContent = residualSummary;

    if (statusEl) {
        statusEl.classList.toggle("ok", balanced);
        statusEl.classList.toggle("warn", !balanced);
        statusEl.textContent = balanced
            ? `✓ Equilibrium satisfied — ${residualSummary}`
            : `■ Equilibrium check failed — ${residualSummary}`;
    }
    if (noteEl) {
        noteEl.textContent = result.grid
            ? (balanced
                ? `ΣMx = ${formatNum(sumFx)} kN·m, ΣMy = ${formatNum(sumFy)} kN·m, ΣFz = ${formatNum(sumMz)} kN. Global Grid equilibrium residuals are within tolerance.`
                : `Grid global equilibrium residuals exceed tolerance. Force tolerance: ${formatNum(result.forceTolerance ?? tolerance)} kN; moment tolerance: ${formatNum(momentTolerance)}
kN·m. Check member loads, nodal loads, torsion signs and support conditions.`)
            : (balanced
                ? `ΣFx = ${formatNum(sumFx)} kN, ΣFy = ${formatNum(sumFy)} kN, ΣMz = ${formatNum(sumMz)} kN·m. Residuals are within tolerance.`
                : `Check load directions, moment signs, support reactions, and member load equilibrium. Force tolerance: ${formatNum(tolerance)} kN; moment tolerance: ${formatNum(
momentTolerance)} kN·m.`);
    }
}

// Compatibility wrapper preserving the original combined signature/behavior
// used by solveStep7 (calculate, then immediately render).
function updateEquilibriumCheck(AC, dofMap, modelNodes, restIndices, AR) {
    const result = isGridAnalysisMode()
        ? calculateGridEquilibriumResiduals(AC, dofMap, modelNodes, restIndices, AR)
        : calculateEquilibriumResiduals(AC, dofMap, modelNodes, restIndices, AR);
    renderEquilibriumCheck(result);
    return result;
}



// Maximum-magnitude displacement, reaction, axial force, shear, and moment
// across the whole solved structure, for the compact Results summary strip.
// Pure calculation — no DOM access.
function calculateAnalysisSummary(dofMap, D, AR, restIndices) {
    if (isGridAnalysisMode()) {
        let maxDisp = 0;
        model.nodes.forEach((node) => {
            const d = dofMap[node.id];
```

```
        if (d && d.z !== null) maxDisp = Math.max(maxDisp, Math.abs(Number(D[d.z]?.[0]) || 0));
    });
    let maxReaction = 0;
    model.supportAssignments.forEach((sa) => { maxReaction = Math.max(maxReaction, Math.abs(Number(sa.rz) || 0)); });
    let maxA = 0, maxV = 0, maxM = 0;
    model.elements.forEach((el) => {
        const AM = el.gridEndActions;
        if (!Array.isArray(AM) || AM.length < 6) return;
        maxA = Math.max(maxA, Math.abs(Number(AM[0]?.[0]) || 0), Math.abs(Number(AM[3]?.[0]) || 0));
        maxV = Math.max(maxV, Math.abs(Number(AM[1]?.[0]) || 0), Math.abs(Number(AM[4]?.[0]) || 0));
        maxM = Math.max(maxM, Math.abs(Number(AM[2]?.[0]) || 0), Math.abs(Number(AM[5]?.[0]) || 0));
    });
    return { maxDisp, maxReaction, maxA, maxV, maxM, grid: true };
}
let maxDisp = 0;
model.nodes.forEach((n) => {
    const d = dofMap[n.id];
    if (!d) return;
    const ux = Number(D[d.x]?.[0]) || 0;
    const uy = Number(D[d.y]?.[0]) || 0;
    const mag = Math.hypot(ux, uy);
    if (mag > maxDisp) maxDisp = mag;
});

let maxReaction = 0;
model.supportAssignments.forEach((sa) => {
    // Reactions are reported as a force resultant in kN; moment reaction
    // is intentionally excluded because it has different units.
    const mag = Math.hypot(Number(sa.rx) || 0, Number(sa.ry) || 0);
    if (mag > maxReaction) maxReaction = mag;
});

let maxA = 0, maxV = 0, maxM = 0;
model.elements.forEach((el) => {
    if (!el.plotData) return;
    const pd = el.plotData;
    const xs = [];
    for (let i = 0; i <= 50; i++) xs.push((i / 50) * pd.L);
    pd.pointLoads.forEach((pl) => { xs.push(pl.a - 1e-6, pl.a + 1e-6); });
    pd.localPointMoments.forEach((pm) => { xs.push(pm.a - 1e-6, pm.a + 1e-6); });
    pd.localUDLs.forEach((udl) => { xs.push(udl.a, udl.b); });
    xs.sort((a, b) => a - b);

    xs.forEach((x) => {
        let A_ax = Number(pd.fx1) || 0;
        let V = Number(pd.fy1) || 0;
        let M = -(Number(pd.mz1) || 0) + (Number(pd.fy1) || 0) * x;

        A_ax += (Number(pd.wx) || 0) * x;
        V += (Number(pd.wy) || 0) * x;
        M += ((Number(pd.wy) || 0) * x * x) / 2;

        pd.localUDLs.forEach((udl) => {
            const wx1 = (Number(udl.Fx1) || 0) * pd.cx + (Number(udl.Fy1) || 0) * pd.cy;
            const wy1 = -(Number(udl.Fx1) || 0) * pd.cy + (Number(udl.Fy1) || 0) * pd.cx;
            const wx2 = (Number(udl.Fx2) || 0) * pd.cx + (Number(udl.Fy2) || 0) * pd.cy;
            const wy2 = -(Number(udl.Fx2) || 0) * pd.cy + (Number(udl.Fy2) || 0) * pd.cx;
            if (x > udl.a) {
                const L_udl = udl.b - udl.a;
                const len = Math.min(x, udl.b) - udl.a;
                const w2x = wx1 + ((wx2 - wx1) * len) / L_udl;
                A_ax += ((wx1 + w2x) / 2) * len;
                const w2y = wy1 + ((wy2 - wy1) * len) / L_udl;
                V += ((wy1 + w2y) / 2) * len;
                const V_rect = wy1 * len;
                const M_rect = V_rect * (x - (udl.a + len / 2));
                const V_tri = 0.5 * (w2y - wy1) * len;
                const M_tri = V_tri * (x - (udl.a + (2 * len) / 3));
                M += M_rect + M_tri;
            }
        });

        pd.pointLoads.forEach((pl) => {
            if (x > pl.a) {
                A_ax += Number(pl.Px) || 0;
                V += Number(pl.Py) || 0;
                M += (Number(pl.Py) || 0) * (x - pl.a);
            }
        });
        pd.localPointMoments.forEach((pm) => {
            if (x > pm.a) M -= Number(pm.Mz) || 0;
        });

        maxA = Math.max(maxA, Math.abs(A_ax));
        maxV = Math.max(maxV, Math.abs(V));
        maxM = Math.max(maxM, Math.abs(M));
    });
```

```
    });

    return { maxDisp, maxReaction, maxA, maxV, maxM };
}

// Writes an analysis summary (from calculateAnalysisSummary()) into the
// Results summary strip. Pure DOM update — no calculation. The equilibrium
// status label is intentionally read back from the equilibriumStatus element
// rather than recomputed, matching the original behavior of reflecting
// whatever renderEquilibriumCheck() most recently displayed there.
function renderAnalysisSummary(summary) {
    const { maxDisp, maxReaction, maxA, maxV, maxM } = summary;

    const dispEl = document.getElementById("summaryMaxDisplacement");
    const reactEl = document.getElementById("summaryMaxReaction");
    const axialEl = document.getElementById("summaryMaxAxial");
    const shearEl = document.getElementById("summaryMaxShear");
    const momentEl = document.getElementById("summaryMaxMoment");
    const eqEl = document.getElementById("summaryEquilibriumStatus");

    if (dispEl) dispEl.textContent = `${formatNum(maxDisp * 1000)} mm`;
    if (reactEl) reactEl.textContent = `${formatNum(maxReaction)} kN`;
    if (axialEl) axialEl.textContent = summary.grid ? `${formatNum(maxA)} kN·m` : `${formatNum(maxA)} kN`;
    if (shearEl) shearEl.textContent = summary.grid ? `${formatNum(maxV)} kN·m` : `${formatNum(maxV)} kN`;
    if (momentEl) momentEl.textContent = summary.grid ? `${formatNum(maxM)} kN` : `${formatNum(maxM)} kN·m`;
    if (eqEl) {
        const eq = document.getElementById("equilibriumStatus");
        eqEl.textContent = eq && eq.textContent.includes("satisfied") ? "Satisfied" : "Check failed";
    }
}

// Compatibility wrapper preserving the original combined signature/behavior
// used by runAnalysisCore (calculate, then immediately render).
function updateAnalysisSummary(dofMap, D, AR, restIndices) {
    const summary = calculateAnalysisSummary(dofMap, D, AR, restIndices);
    renderAnalysisSummary(summary);
    return summary;
}

function showAnalysisLoading() {
    const overlay = document.getElementById("analysisLoadingOverlay");
    const text = document.getElementById("analysisLoadingText");
    if (!overlay) return;
    if (text) {
        text.textContent =
            `Loading ${model.nodes.length} nodes, ${model.elements.length} elements, ` +
            `${model.loads.length} loads and ${model.moments.length} moments…`;
    }
    overlay.classList.add("active");
    overlay.setAttribute("aria-hidden", "false");
    document.body.style.overflow = "hidden";
}

function hideAnalysisLoading() {
    const overlay = document.getElementById("analysisLoadingOverlay");
    if (!overlay) return;
    overlay.classList.remove("active");
    overlay.setAttribute("aria-hidden", "true");
    document.body.style.overflow = "";
}

function runAnalysis() {
    showAnalysisLoading();

    // Give the browser one full paint cycle so the loading screen is actually
    // rendered before the synchronous stiffness calculation begins. This is
    // especially important for large imported models.
    requestAnimationFrame(() => {
        setTimeout(() => {
            try {
                runAnalysisCore();
            } catch (error) {
                console.error(error);
                alert("Analysis failed. Check the browser console for details.");
            } finally {
                hideAnalysisLoading();
            }
        }, 30);
    });
}

// ========================================================================
// 16b. ANALYSIS ORCHESTRATOR
// The only top-level analysis controller. Validates input, establishes the
// active load/moment cases, then runs the eight solver steps in order and
// hands their results to the matching render/UI-update calls. No structural
// mathematics lives directly in this function.
```

```
// ========================================================================
function updateSolverModeResultHeaders() {
    const setText = (id, value) => { const el = document.getElementById(id); if (el) el.textContent = value; };
    if (isGridAnalysisMode()) {
        setText("resDispHdr1", "θX (rad)"); setText("resDispHdr2", "θY (rad)"); setText("resDispHdr3", "ΔZ (m)");
        setText("resReactionGlobalHeading", "Grid DOF Reactions"); setText("resReactionLocalHeading", "Local Components");
        setText("resReactionHdr1", "Mx (kN·m)"); setText("resReactionHdr2", "My (kN·m)"); setText("resReactionHdr3", "Fz (kN)");
        setText("resReactionHdr4", "—"); setText("resReactionHdr5", "—"); setText("resReactionHdr6", "—");
        setText("resForceHdr1", "Mx (kN·m)"); setText("resForceHdr2", "My (kN·m)"); setText("resForceHdr3", "Fz (kN)");
        setText("eqFxLabel", "ΣMx Residual"); setText("eqFyLabel", "ΣMy Residual"); setText("eqMzLabel", "ΣFz Residual");
        setText("eqMaxResidualLabel", "Force / Moment Residuals");
        setText("summaryAxialLabel", "Maximum Mx action"); setText("summaryShearLabel", "Maximum My action"); setText("summaryMomentLabel", "Maximum Fz action");
    } else {
        setText("resDispHdr1", "UX (m)"); setText("resDispHdr2", "UY (m)"); setText("resDispHdr3", "RZ (rad)");
        setText("resReactionGlobalHeading", "Global Axes"); setText("resReactionLocalHeading", "Local Axes (lowest-ID connected member)");
        setText("resReactionHdr1", "Fx (kN)"); setText("resReactionHdr2", "Fy (kN)"); setText("resReactionHdr3", "Mz (kN·m)");
        setText("resReactionHdr4", "Fx′ (kN)"); setText("resReactionHdr5", "Fy′ (kN)"); setText("resReactionHdr6", "Mz′ (kN·m)");
        setText("resForceHdr1", "Axial (kN)"); setText("resForceHdr2", "Shear (kN)"); setText("resForceHdr3", "Moment (kN·m)");
        setText("eqFxLabel", "ΣFx Residual"); setText("eqFyLabel", "ΣFy Residual"); setText("eqMzLabel", "ΣMz Residual");
        setText("eqMaxResidualLabel", "Force / Moment Residuals");
        setText("summaryAxialLabel", "Maximum axial force"); setText("summaryShearLabel", "Maximum shear"); setText("summaryMomentLabel", "Maximum bending moment");
    }
}

function runAnalysisCore() {
    setActiveAnalysisCases();

    if (model.nodes.length === 0 || model.elements.length === 0) {
        return alert("Model is empty. Please define nodes and elements.");
    }

    if (model.loads.length > 0 && !activeLoadCaseId) {
        return alert("Select an Active Load Case before running analysis.");
    }
    if (model.moments.length > 0 && !activeMomentCaseId) {
        return alert("Select an Active Moment Case before running analysis.");
    }

    const activeLoadCase = model.loadCases.find((lc) => String(lc.id) === String(activeLoadCaseId));
    const activeLoadCount = model.loads.filter(isActiveLoadCase).length;
    const activeMomentCount = model.moments.filter(isActiveMomentCase).length;
    if (model.loads.length > 0 && activeLoadCount === 0 && activeLoadCase?.type !== "Self Weight") {
        return alert(`No loads belong to Active Load Case "${activeLoadCaseId}". Choose a case containing loads or select a Self Weight case with per-element weights.`);
    }
    if (model.moments.length > 0 && activeMomentCount === 0) {
        return alert(`No moments belong to Active Moment Case "${activeMomentCaseId}". Choose a case containing moments.`);
    }

    const missingSelfWeightElement = getMissingSelfWeightElement(activeLoadCase);
    if (missingSelfWeightElement) {
        return alert(`Self Weight Load Case "${activeLoadCaseId}" has no valid weight for element "${missingSelfWeightElement.id}". Edit the load case and enter its self-weight in kN/m (
enter 0 only if zero is intentional).`);
    }

    // Grid's supplied calculation engine is a 3-DOF plate-grid formulation:
    // θX, θY and ΔZ. Reject components that its model does not represent
    // rather than silently dropping them during assembly.
    if (isGridAnalysisMode()) {
        const unsupportedLoad = model.loads.find((load) => {
            if (!isActiveLoadCase(load)) return false;
            const fx = Number(load.Fx) || 0;
            const fy = Number(load.Fy) || 0;
            const fxEnd = Number(load.FxEnd) || 0;
            const fyEnd = Number(load.FyEnd) || 0;
            return Math.abs(fx) > 1e-12 || Math.abs(fy) > 1e-12 ||
                (load.type === "Element UDL" && (Math.abs(fxEnd) > 1e-12 || Math.abs(fyEnd) > 1e-12));
        });
        if (unsupportedLoad) {
            return alert(`Grid load "${unsupportedLoad.id}" contains Fx/Fy components. The Grid solver only supports Fz for nodal and member forces; set unsupported components to zero.`);
        }
        const unsupportedNodalMoment = model.moments.find((moment) =>
            isActiveMomentCase(moment) && moment.type === "Nodal Moment" && Math.abs(Number(moment.Mz) || 0) > 1e-12
        );
        if (unsupportedNodalMoment) {
            return alert(`Nodal moment "${unsupportedNodalMoment.id}" contains Mz, but Grid joints only have θX, θY and ΔZ DOFs. Use nodal Mx/My; Mz is supported only as a member moment
input.`);
        }
    }

    const activeCaseLabel = getActiveAnalysisCaseLabel();
    const hasSelfWeightData = model.loadCases.some((loadCase) => loadCase.type === "Self Weight" && Object.values(loadCase.memberSelfWeights || {}).some((weight) => Number(weight) > 0));
    if (hasSelfWeightData && activeLoadCase?.type !== "Self Weight") {
        const useSelfWeight = confirm(
            `The selected Load Case "${activeLoadCaseId}" is not a Self Weight case. Self-weight from the other case will be excluded from this analysis.\n\nContinue?`,
        );
        if (!useSelfWeight) return;
    }
```

```
    // Matrix algebra helper (zeros/add/sub/mul/transpose/invert) — see the
    // "MATRIX ALGEBRA UTILITIES" section near the top of this file.
    const heavyModel = model.nodes.length > 10;
    const ctx = { activeCaseLabel, activeLoadCase, heavyModel };

    const step1 = solveStep1(ctx);
    if (!step1) return;

    const step2 = solveStep2(ctx, step1);
    const step3 = solveStep3(ctx, step1);

    const step4 = solveStep4(ctx, step1, step3);
    if (!step4) return;

    const step5 = solveStep5(ctx, step1, step4);
    const step6 = solveStep6(ctx, step1, step5);
    if (!step6) return;

    const step7 = solveStep7(ctx, step1, step4, step5, step6);
    const step8 = solveStep8(ctx, step7);

    let stepsHTML = "";
    stepsHTML += renderStep1(step1);
    stepsHTML += renderStep2(step2);
    stepsHTML += renderStep3(step3);
    stepsHTML += renderStep4(step4);
    stepsHTML += renderStep5(step5);
    stepsHTML += renderStep6(step6);
    stepsHTML += renderStep7(step7);
    stepsHTML += renderStep8(step8);

    updateSolverModeResultHeaders();
    document.getElementById("resDispTable").innerHTML = step7.resDispHTML;
    document.getElementById("resReactionsTable").innerHTML = step7.resReactHTML;
    document.getElementById("resForcesTable").innerHTML = step7.resForceHTML;
    updateAnalysisSummary(step1.dofMap, step6.D, step7.AR, step5.restIndices);

    document.getElementById("calcStepsContainer").innerHTML = stepsHTML;
    setupResultDetailPanels();
    const equilibriumPassed = step7.equilibriumResult?.balanced !== false;
    if (equilibriumPassed) markAnalysisValid();
    else markAnalysisFailed();
    const statusEl = document.getElementById("activeAnalysisCaseStatus");
    if (statusEl) {
        statusEl.textContent = equilibriumPassed
            ? ` Active: ${activeCaseLabel}`
            : ` Active: ${activeCaseLabel} • Equilibrium check failed`;
        if (!equilibriumPassed) statusEl.style.color = "#991b1b";
    }
    document.getElementById("resultsContent").style.display = "block";
    if (model.nodes.length > 10) {
        showHeavyDiagramPlaceholder("reactionSVGSmall");
    } else {
        drawDiagramSVG("reactionSVGSmall", "reactions");
    }
    updateResultsDiagram();
    if (equilibriumPassed) {
        alert(isGridAnalysisMode()
            ? "Grid analysis completed successfully using the Grid Stiffness Method."
            : "Analysis completed successfully using the LPW-3 Stiffness Method approach.");
    } else {
        alert("Analysis results were computed, but the global equilibrium check failed. Review the residuals, load cases, load directions and support conditions before relying on these
results. PDF export is disabled until equilibrium passes.");
    }
}

// ============================================================================
// 16b. PDF EXPORT (full 8-step solution report, generated in the page)
// Builds the report pages in a hidden frame from the SAME step content the
// page shows (Steps 1-7 are cloned from the results panels), draws the
// AFD / SFD / BMD diagrams as Step 8 (instead of the final tables), turns
// every report page is written as REAL vector PDF content (selectable text,
// vector diagrams - no images) and the PDF file is downloaded directly. No print dialog or device print feature is used.
// ============================================================================
const PdfLayout = {
    pageW: 1160, pageH: 820, // page size in CSS px (A4 landscape ratio)
    marginX: 30, headerH: 46, footerH: 38,
    contentW: 1100,
    ptW: 841.89, ptH: 595.28, // A4 landscape in PDF points
};
PdfLayout.contentH = PdfLayout.pageH - PdfLayout.headerH - PdfLayout.footerH;
let pdfExportBusy = false;

const PDF_FRAME_CSS = `
html, body { margin: 0; padding: 0; background: #fff !important; overflow: visible !important; max-width: none !important; min-height: 0 !important; height: auto !important; display: block
!important; }
#pdfRoot { width: ${PdfLayout.contentW}px; background: #fff; font-family: Inter, Arial, Helvetica, sans-serif; color: #182230; }
```

```
#pdfRoot .calc-step { margin: 0 !important; padding: 0 !important; border: 0 !important; }
#pdfRoot .eq-wrap, #pdfRoot .table-wrap, #pdfRoot .canvas-container { overflow: visible !important; }
#pdfRoot * { animation: none !important; transition: none !important; }
#pdfRoot .pdf-diagram h5 { margin-top: 0; }
#pdfRoot .pdf-diagram-box { border: 1px solid #cbd5e1; border-radius: 6px; background: #fff; padding: 6px; }
#pdfRoot .pdf-diagram-box svg { width: 100% !important; height: auto !important; display: block; }
#pdfRoot .result-diagram-fixed-legend { position: static !important; z-index: auto !important; opacity: 1 !important; background: #fff !important; display: block; width: max-content; margin-
top: 8px; box-shadow: none !important; max-height: none !important; overflow: visible !important; }
`;

function buildPdfDiagramBlock(mode) {
    const bb = getBoundingBox();
    const dX = Math.max(bb.maxX - bb.minX, 1e-9);
    const dY = Math.max(bb.maxY - bb.minY, 1e-9);
    const minL = getMinimumElementLength();
    const baselinePx = AppConfig.sharedDiagramBaselinePx;
    const diagramScale = minL && Number.isFinite(minL) && minL > 0 ? baselinePx / minL : baselinePx;
    const padding = 140;
    const renderWidth = Math.max(800, dX * diagramScale + 2 * padding);
    const renderHeight = Math.max(400, dY * diagramScale + 2 * padding);
    const camera = {
        scale: diagramScale,
        offsetX: padding - bb.minX * diagramScale,
        offsetY: padding + bb.maxY * diagramScale,
        intrinsicCanvas: true,
    };
    const renderers = { model: renderStructureModelDiagram, afd: renderAFDDiagram, sfd: renderSFDDiagram, bmd: renderBMDDiagram };
    const svg = renderers[mode](renderWidth, renderHeight, camera);
    const legend = buildResultDiagramLegend(mode, 1);
    return `<div class="pdf-diagram-box">${svg}</div>${legend}`;
}

function pdfShowProgress(text) {
    let ov = document.getElementById("pdfExportOverlay");
    if (!ov) {
        ov = document.createElement("div");
        ov.id = "pdfExportOverlay";
        ov.setAttribute("role", "status");
        ov.style.cssText = "position:fixed;top:0;left:0;right:0;bottom:0;z-index:2147483000;background:rgba(15,23,42,0.72);display:flex;align-items:center;justify-content:center;font-family:
Arial,sans-serif;";
        ov.innerHTML = '<div style="background:#fff;border-radius:12px;padding:22px 26px;max-width:86vw;text-align:center;box-shadow:0 12px 40px rgba(0,0,0,.35)"><div style="font-size:30px;
margin-bottom:8px">■</div><div style="font-weight:800;color:#1e3a8a;font-size:16px;margin-bottom:6px">Generating PDF…</div><div id="pdfExportOverlayText" style="font-size:13px;color:
#475569"></div></div>';
        document.body.appendChild(ov);
    }
    const t = document.getElementById("pdfExportOverlayText");
    if (t) t.textContent = text;
    return new Promise((resolve) => setTimeout(resolve, 20)); // let the browser repaint
}

function pdfHideProgress() {
    const ov = document.getElementById("pdfExportOverlay");
    if (ov) ov.remove();
}

function pdfCreateFrame() {
    // The application stylesheets live in external files (assets/css, modules/**),
    // so the off-screen layout frame links the very same files instead of copying
    // <style> text. Any remaining inline <style> blocks are still copied as before.
    const styleLinks = Array.from(document.querySelectorAll('link[rel="stylesheet"]'))
        .map((lk) => `<link rel="stylesheet" href="${lk.href}">`).join("");
    const pageStyles = Array.from(document.querySelectorAll("style")).map((st) => st.textContent).join("\n");
    const frame = document.createElement("iframe");
    frame.id = "pdfRenderFrame";
    frame.setAttribute("aria-hidden", "true");
    frame.style.cssText = "position:fixed;left:-30000px;top:0;width:1500px;height:1000px;border:0;pointer-events:none;";
    document.body.appendChild(frame);
    const doc = frame.contentDocument;
    doc.open();
    doc.write(`<!DOCTYPE html><html><head><meta charset="utf-8">${styleLinks}<style>${pageStyles}</style><style>${pdfFrameFontCss()}</style><style>${PDF_FRAME_CSS}</style></head><body><div
id="pdfRoot"></div></body></html>`);
    doc.close();
    frame.stylesReady = pdfWaitForFrameStyles(doc).then(() => pdfLoadFrameFonts(doc));
    return frame;
}

// Resolves once every <link rel="stylesheet"> inside the render frame has loaded
// (or failed / timed out), so layout measurements use the final styles.
function pdfWaitForFrameStyles(doc) {
    const links = Array.from(doc.querySelectorAll('link[rel="stylesheet"]'));
    const waits = links.map((lk) => new Promise((resolve) => {
        if (lk.sheet) return resolve();
        lk.addEventListener("load", () => resolve());
        lk.addEventListener("error", () => resolve());
    }));
    const timeout = new Promise((resolve) => setTimeout(resolve, 8000));
    return Promise.race([Promise.all(waits), timeout]);
```

```
}

// Replaces every inline <svg> by an <img> of the same size so the diagrams
// are drawn by the browser's own SVG renderer (sharp, exact colours/text).
async function pdfRasterizeSvgs(doc, root) {
    // (name kept for compatibility) The diagrams stay REAL <svg> elements - they
    // are written to the PDF as vector graphics. Here they only receive a fixed
    // size so the layout (and the page planning) is stable.
    Array.from(root.querySelectorAll("svg")).forEach((svg) => {
        const r = svg.getBoundingClientRect();
        let w = Math.round(r.width);
        let h = Math.round(r.height);
        if (!w || !h) {
            const vb = (svg.getAttribute("viewBox") || "").split(/[\s,]+/).map(Number);
            w = vb[2] || 300;
            h = vb[3] || 150;
        }
        const box = svg.closest(".pdf-diagram-box");
        if (box) {
            const maxH = 500;
            const boxW = box.clientWidth - 14;
            const k = Math.min(1, maxH / h, boxW / w);
            w = Math.round(w * k);
            h = Math.round(h * k);
            svg.style.setProperty("margin", "0 auto", "important");
        }
        svg.setAttribute("width", w);
        svg.setAttribute("height", h);
        svg.style.setProperty("width", w + "px", "important");
        svg.style.setProperty("height", h + "px", "important");
        svg.style.setProperty("display", "block", "important");
        if (!svg.getAttribute("xmlns")) svg.setAttribute("xmlns", "http://www.w3.org/2000/svg");
    });
}

// Splits a simple table that is wider than the page into several tables
// by column groups (first column repeated) so its text stays readable.
function pdfSplitWideTables(root) {
    const doc = root.ownerDocument;
    const maxW = PdfLayout.contentW;
    Array.from(root.querySelectorAll("table")).forEach((table) => {
        if (table.parentElement && table.parentElement.closest("table")) return;
        if (table.classList.contains("matrix-table")) return;
        const r = table.getBoundingClientRect();
        if (r.width <= maxW * 1.12) return;
        const head = table.tHead && table.tHead.rows.length === 1 ? table.tHead.rows[0] : null;
        const bodyRows = Array.from(table.tBodies).flatMap((b) => Array.from(b.rows));
        if (!head || !bodyRows.length) return;
        const nCols = head.cells.length;
        const simple = [head, ...bodyRows].every(
            (row) => row.cells.length === nCols && Array.from(row.cells).every((c) => c.colSpan === 1 && c.rowSpan === 1),
        );
        if (!simple || nCols < 3) return;
        const colW = Array.from(head.cells).map((c) => c.getBoundingClientRect().width);
        const groups = [];
        let cur = [];
        let curW = colW[0];
        for (let c = 1; c < nCols; c++) {
            if (cur.length && curW + colW[c] > maxW) {
                groups.push(cur);
                cur = [];
                curW = colW[0];
            }
            cur.push(c);
            curW += colW[c];
        }
        if (cur.length) groups.push(cur);
        if (groups.length < 2) return;
        groups.forEach((cols, gi) => {
            const keep = new Set([0, ...cols]);
            const t = table.cloneNode(false);
            t.style.width = "100%";
            t.style.minWidth = "0";
            if (gi > 0) t.style.marginTop = "14px";
            const prune = (row) => {
                const rc = row.cloneNode(true);
                Array.from(rc.cells).forEach((cell, idx) => {
                    if (!keep.has(idx)) cell.remove();
                });
                return rc;
            };
            const th = doc.createElement("thead");
            th.appendChild(prune(head));
            t.appendChild(th);
            const tb = doc.createElement("tbody");
            bodyRows.forEach((row) => tb.appendChild(prune(row)));
            t.appendChild(tb);
```

```
            table.parentNode.insertBefore(t, table);
        });
        table.remove();
    });
}

// Equations (matrix products) much wider than the page are allowed to wrap
// onto several lines instead of being shrunk to an unreadable size.
// DOF numbering of a matrix: the column numbers are centred over the text of
// their column and the row numbers get exactly the height of their matrix row,
// so both line up precisely with the numbers they belong to.
function pdfAlignMatrixLabels(root) {
    root.querySelectorAll(".matrix-wrapper").forEach((wrap) => {
        const table = wrap.querySelector("table.matrix-table");
        if (!table || !table.rows.length) return;
        const kids = Array.from(wrap.children);
        const mainRow = kids.find((k) => k.contains(table));
        const colRow = kids.find((k) => k !== mainRow && !k.classList.contains("matrix-title") && k.children.length > 0);
        const rowLabels = mainRow && mainRow.lastElementChild && mainRow.lastElementChild !== table &&
            !mainRow.lastElementChild.contains(table) && mainRow.lastElementChild.children.length
            ? mainRow.lastElementChild : null;
        const doc = wrap.ownerDocument;
        const wr = wrap.getBoundingClientRect();
        const tr0 = table.getBoundingClientRect();
        const rows = Array.from(table.rows);

        // ---- column numbers
        if (colRow && colRow !== rowLabels) {
            const labels = Array.from(colRow.children);
            const cells = Array.from(rows[0].cells);
            if (labels.length === cells.length) {
                const range = doc.createRange();
                const plan = cells.map((cell, j) => {
                    const cr = cell.getBoundingClientRect();
                    let l = Infinity, r = -Infinity;
                    rows.forEach((row) => {
                        const c = row.cells[j];
                        if (!c || !c.firstChild) return;
                        range.selectNodeContents(c);
                        Array.from(range.getClientRects()).forEach((q) => {
                            if (q.width > 0) { l = Math.min(l, q.left); r = Math.max(r, q.right); }
                        });
                    });
                    const textCenter = isFinite(l) ? (l + r) / 2 : cr.left + cr.width / 2;
                    return { width: cr.width, shift: textCenter - (cr.left + cr.width / 2) };
                });
                colRow.style.width = tr0.width + "px";
                colRow.style.padding = "0";
                colRow.style.justifySelf = "start";
                colRow.style.marginLeft = tr0.left - wr.left + "px";
                labels.forEach((lab, j) => {
                    lab.style.flex = "none";
                    lab.style.width = plan[j].width + "px";
                    lab.style.textAlign = "center";
                    lab.style.position = "relative";
                    lab.style.left = plan[j].shift + "px";
                });
            }
        }

        // ---- row numbers
        if (rowLabels && rowLabels.children.length === rows.length) {
            const rr = rows.map((row) => row.getBoundingClientRect());
            const cr = rowLabels.getBoundingClientRect();
            const labels = Array.from(rowLabels.children);
            rowLabels.style.padding = "0";
            rowLabels.style.paddingTop = Math.max(0, rr[0].top - cr.top) + "px";
            rowLabels.style.justifyContent = "flex-start";
            labels.forEach((lab, i) => {
                const h = i < rr.length - 1 ? rr[i + 1].top - rr[i].top : rr[i].height;
                lab.style.height = h + "px";
                lab.style.display = "flex";
                lab.style.alignItems = "center";
                lab.style.lineHeight = "1";
            });
        }
    });
}

// Packs the matrices / equations of a step side by side: every equation (".eq-row")
// is kept as ONE unbroken unit and as many units as fit are placed on one line.
// Each finished line is followed by exactly one blank line.
function pdfPackEquationRows(root) {
    const contentW = PdfLayout.contentW;
    const GAP = 40; // horizontal space between two units of a line
    const win = root.ownerDocument.defaultView;
    const doc = root.ownerDocument;
```

```
    const parents = new Set();
    root.querySelectorAll(".eq-wrap").forEach((w) => parents.add(w.parentElement));
    parents.forEach((parent) => {
        // one blank line = the line height of the surrounding text
        const probe = parent.querySelector("p") || parent;
        const pcs = win.getComputedStyle(probe);
        const fsz = parseFloat(pcs.fontSize) || 14;
        const lineGap = Math.round(parseFloat(pcs.lineHeight) || fsz * 1.5);

        const kids = Array.from(parent.children);
        let i = 0;
        while (i < kids.length) {
            if (!kids[i].classList.contains("eq-wrap")) { i++; continue; }
            const run = [];
            while (i < kids.length && kids[i].classList.contains("eq-wrap")) run.push(kids[i++]);
            const rows = [];
            run.forEach((w) => Array.from(w.children).forEach((c) => c.classList.contains("eq-row") && rows.push(c)));
            if (!rows.length) continue;

            const wcs = win.getComputedStyle(run[0]);
            const chrome = parseFloat(wcs.paddingLeft) + parseFloat(wcs.paddingRight) +
                parseFloat(wcs.borderLeftWidth) + parseFloat(wcs.borderRightWidth);
            const inner = contentW - chrome;
            rows.forEach((r) => {
                r.style.flex = "none";
                r.style.width = "max-content";
                r.style.minWidth = "0";
                r.style.margin = "0";
            });
            const widths = rows.map((r) => r.getBoundingClientRect().width);

            const lines = [];
            let cur = null;
            let curW = 0;
            rows.forEach((r, k) => {
                if (cur && curW + GAP + widths[k] <= inner) {
                    cur.push(r);
                    curW += GAP + widths[k];
                } else {
                    cur = [r];
                    curW = widths[k];
                    lines.push(cur);
                }
            });

            const anchor = run[0];
            lines.forEach((items) => {
                const line = doc.createElement("div");
                line.className = "eq-wrap pdf-eq-line";
                line.style.flexDirection = "row";
                line.style.flexWrap = "nowrap";
                // a unit wider than the line starts at the left edge: its overflow then
                // grows to the right, where the page-fit logic widens the card around it
                const lineW = items.reduce((a, r) => a + r.getBoundingClientRect().width, 0) + GAP * (items.length - 1);
                line.style.justifyContent = lineW > inner ? "flex-start" : "center";
                line.style.alignItems = "flex-start";
                line.style.gap = GAP + "px";
                line.style.marginBottom = lineGap + "px";
                items.forEach((r) => line.appendChild(r));
                parent.insertBefore(line, anchor);
            });
            run.forEach((w) => w.remove());
        }
    });
}

function pdfWrapWideEquations(root) {
    // An equation stays on ONE line (the page is scaled to fit it). Only an
    // extremely wide one - which would become unreadably small - is wrapped.
    root.querySelectorAll(".eq-row").forEach((row) => {
        if (row.getBoundingClientRect().width <= PdfLayout.contentW * 2.5) return;
        row.style.flexWrap = "wrap";
        row.style.minWidth = "0";
        row.style.width = "100%";
        row.style.rowGap = "18px";
        row.style.justifyContent = "center";
    });
}

// Splits any table taller than a page into several tables (same header,
// same column widths) so long tables can flow across pages.
function pdfSplitTallTables(root, availH) {
    const doc = root.ownerDocument;
    Array.from(root.querySelectorAll("table")).forEach((table) => {
        if (table.parentElement && table.parentElement.closest("table")) return;
        const r = table.getBoundingClientRect();
        if (r.height <= availH) return;
```

```
        const thead = table.tHead;
        const rows = Array.from(table.tBodies).flatMap((b) => Array.from(b.rows));
        if (rows.length < 2) return;
        const headH = thead ? thead.getBoundingClientRect().height : 0;
        const widths = Array.from(rows[0].cells).map((c) => c.getBoundingClientRect().width);
        const groups = [];
        let cur = [];
        let curH = headH;
        rows.forEach((row) => {
            const h = row.getBoundingClientRect().height;
            if (cur.length && curH + h > availH - 4) {
                groups.push(cur);
                cur = [];
                curH = headH;
            }
            cur.push(row);
            curH += h;
        });
        if (cur.length) groups.push(cur);
        if (groups.length < 2) return;
        groups.forEach((g) => {
            const t = table.cloneNode(false);
            t.style.width = r.width + "px";
            t.style.tableLayout = "fixed";
            const cg = doc.createElement("colgroup");
            widths.forEach((wd) => {
                const col = doc.createElement("col");
                col.style.width = wd + "px";
                cg.appendChild(col);
            });
            t.appendChild(cg);
            if (thead) t.appendChild(thead.cloneNode(true));
            const tb = doc.createElement("tbody");
            g.forEach((row) => tb.appendChild(row.cloneNode(true)));
            t.appendChild(tb);
            table.parentNode.insertBefore(t, table);
        });
        table.remove();
    });
}

// Breaks the section into "blocks" (heading, paragraph, matrix row, table…)
// that are never cut through; blocks taller than a page are split further.
function pdfCollectBlocks(root, availH) {
    const win = root.ownerDocument.defaultView;
    const out = [];
    const walk = (parent) => {
        Array.from(parent.children).forEach((c) => {
            const cs = win.getComputedStyle(c);
            if (cs.display === "none") return;
            const r = c.getBoundingClientRect();
            if (r.height < 1 && r.width < 1) return;
            const onlyElements = Array.from(c.childNodes).every(
                (n) => n.nodeType === 1 || (n.nodeType === 3 && !n.textContent.trim()),
            );
            const stacks = cs.display === "block" || (cs.display === "flex" && cs.flexDirection === "column");
            if (
                r.height > availH * 0.45 && c.children.length && onlyElements && stacks &&
                c.tagName !== "TABLE" && !c.classList.contains("pdf-diagram")
            ) {
                walk(c);
            } else {
                out.push(c);
            }
        });
    };
    walk(root);
    return out;
}

function pdfPlanPages(root, blocks, availH) {
    const contentW = PdfLayout.contentW;
    const win = root.ownerDocument.defaultView;
    const px = (v) => parseFloat(v) || 0;
    const rootTop = root.getBoundingClientRect().top;

    // 1. Measure the true content width of every block, pin every block to
    //    that width, then widen the containers of the wide ones so panel
    //    backgrounds / borders cover the whole content. Pinning first keeps
    //    the other blocks that share a container at their normal size.
    const measured = blocks.map((el) => {
        // True content width: the block itself or any descendant sticking out
        // to the right (scrollWidth alone misses overflowing tables).
        const br = el.getBoundingClientRect();
        let w = Math.max(el.offsetWidth || 0, br.width);
        el.querySelectorAll("*").forEach((d) => {
            const dr = d.getBoundingClientRect();
```

```
        if (dr.width > 0 && dr.right - br.left > w) w = dr.right - br.left;
    });
    return w;
});
blocks.forEach((el, i) => {
    el.style.width = Math.ceil(measured[i]) + "px";
    el.style.maxWidth = "none";
});
const widths = blocks.map((el, i) => {
    const w = measured[i];
    if (w <= contentW + 1) return w;
    let acc = 0;
    let a = el.parentElement;
    while (a && a !== root) {
        const cs = win.getComputedStyle(a);
        acc += px(cs.paddingLeft) + px(cs.paddingRight) + px(cs.borderLeftWidth) + px(cs.borderRightWidth);
        const need = Math.ceil(w + acc);
        if ((a.offsetWidth || 0) < need) {
            a.style.width = need + "px";
            a.style.maxWidth = "none";
        }
        a = a.parentElement;
    }
    return w + acc;
});

// 2. Measure every block AFTER widening. When a block is the first / last
//    child of a container, the container's padding and border are included.
const items = blocks.map((el, i) => {
    const r = el.getBoundingClientRect();
    let top = r.top;
    let bottom = r.bottom;
    let cur = el;
    while (cur.parentElement && cur.parentElement !== root && cur.parentElement.firstElementChild === cur) {
        cur = cur.parentElement;
        top = Math.min(top, cur.getBoundingClientRect().top);
    }
    cur = el;
    while (cur.parentElement && cur.parentElement !== root && cur.parentElement.lastElementChild === cur) {
        cur = cur.parentElement;
        bottom = Math.max(bottom, cur.getBoundingClientRect().bottom);
    }
    return {
        el,
        w: widths[i],
        top: top - rootTop,
        bottom: bottom - rootTop,
        br: el.classList.contains("pdf-break-before"),
    };
});

// 3. Keep a heading together with the block that follows it.
const units = [];
for (let i = 0; i < items.length; i++) {
    const it = items[i];
    const isHeading = /^H[1-6]$/.test(it.el.tagName);
    const nx = items[i + 1];
    if (isHeading && nx && !nx.br && nx.bottom - it.top <= availH) {
        units.push({ top: it.top, bottom: nx.bottom, w: Math.max(it.w, nx.w), br: it.br });
        i++;
    } else {
        units.push({ top: it.top, bottom: it.bottom, w: it.w, br: it.br });
    }
}

// 4. Fill pages. Wide content (which has to be shrunk) is kept on its
//    own pages so normal content keeps its full size.
const pages = [];
let cur = null;
units.forEach((u) => {
    if (cur) {
        const wide = u.w > contentW + 1;
        const curWide = cur.w > contentW + 1;
        const w = Math.max(cur.w, u.w);
        const s = Math.min(1, contentW / w);
        const mixed = wide !== curWide && cur.bottom - cur.top > 150;
        if (u.br || mixed || (u.bottom - cur.top) * s > availH) {
            pages.push(cur);
            cur = null;
        }
    }
    if (!cur) cur = { top: u.top, bottom: u.bottom, w: u.w };
    else {
        cur.bottom = Math.max(cur.bottom, u.bottom);
        cur.w = Math.max(cur.w, u.w);
    }
});
```

```
    if (cur) pages.push(cur);
    return pages;
}




// ---------------------------------------------------------------------------
// Vector PDF engine: the report is written with jsPDF as REAL text (embedded
// fonts, selectable / searchable) and REAL vector graphics (borders, fills,
// SVG diagrams through svg2pdf). No page is ever turned into an image.
// ---------------------------------------------------------------------------
const PDF_FONT_FILES = [
    ["sans-400", "PdfSans", "normal"],
    ["sans-700", "PdfSans", "bold"],
    ["sans-400i", "PdfSans", "italic"],
    ["sans-700i", "PdfSans", "bolditalic"],
    ["mono-400", "PdfMono", "normal"],
    ["mono-700", "PdfMono", "bold"],
];
const PDF_FONT_METRICS = { ascent: 0.92822, descent: 0.23584 }; // DejaVu Sans (hhea)
let pdfFontBlobUrls = null;

function pdfSvg2PdfFn() {
    const g = window.svg2pdf;
    if (!g) return null;
    return typeof g === "function" ? g : g.svg2pdf || null;
}

function pdfEngineReady() {
    return !!(window.jspdf && window.jspdf.jsPDF && window.PDF_FONT_DATA && pdfSvg2PdfFn());
}

// @font-face rules for the hidden layout frame. The embedded PDF fonts are
// registered under the family names the page uses, so the frame is laid out
// with exactly the glyph widths that end up in the PDF.
function pdfFrameFontCss() {
    if (!pdfFontBlobUrls) {
        pdfFontBlobUrls = {};
        PDF_FONT_FILES.forEach(([key]) => {
            const bin = atob(window.PDF_FONT_DATA[key]);
            const bytes = new Uint8Array(bin.length);
            for (let i = 0; i < bin.length; i++) bytes[i] = bin.charCodeAt(i);
            pdfFontBlobUrls[key] = URL.createObjectURL(new Blob([bytes], { type: "font/ttf" }));
        });
    }
    const face = (family, weight, style, key) =>
        `@font-face{font-family:"${family}";font-weight:${weight};font-style:${style};src:url(${pdfFontBlobUrls[key]}) format("truetype");}`;
    let css = "";
    ["Inter", "Arial", "Helvetica", "PdfSans"].forEach((f) => {
        css += face(f, 400, "normal", "sans-400") + face(f, 700, "normal", "sans-700");
        css += face(f, 400, "italic", "sans-400i") + face(f, 700, "italic", "sans-700i");
    });
    ["SFMono-Regular", "Consolas", "Liberation Mono", "Menlo", "PdfMono"].forEach((f) => {
        css += face(f, 400, "normal", "mono-400") + face(f, 700, "normal", "mono-700");
        css += face(f, 400, "italic", "mono-400") + face(f, 700, "italic", "mono-700");
    });
    return css;
}

function pdfLoadFrameFonts(doc) {
    const specs = [];
    ["Arial", "PdfSans"].forEach((f) => {
        ["normal", "italic"].forEach((st) => ["400", "700"].forEach((w) => specs.push(`${st} ${w} 16px "${f}"`)));
    });
    ["Menlo", "PdfMono"].forEach((f) => ["400", "700"].forEach((w) => specs.push(`${w} 16px "${f}"`)));
    return Promise.all(specs.map((s) => doc.fonts.load(s, "Aa1Δ²").catch(() => null))).then(() => (doc.fonts.ready || null));
}

function pdfCreateDocument() {
    const L = PdfLayout;
    const pdf = new window.jspdf.jsPDF({ orientation: "landscape", unit: "pt", format: [L.ptW, L.ptH], compress: true });
    PDF_FONT_FILES.forEach(([key, name, style]) => {
        const file = `${key}.ttf`;
        pdf.addFileToVFS(file, window.PDF_FONT_DATA[key]);
        pdf.addFont(file, name, style, undefined, "Identity-H");
    });
    pdf.setLineJoin("miter");
    return pdf;
}

// The page list passed around by the callers doubles as the holder of the
// PDF document being written (one document per export).
function pdfDocumentFor(pages) {
    if (!pages.pdf) pages.pdf = pdfCreateDocument();
```

```
    return pages.pdf;
}

function pdfParseColor(str) {
    if (!str || str === "transparent") return null;
    const m = String(str).match(/rgba?\(([^)]+)\)/);
    if (!m) return null;
    const p = m[1].split(/[\s,\/]+/).filter(Boolean).map(parseFloat);
    if (p.length < 3) return null;
    const a = p.length > 3 ? p[3] : 1;
    return { r: Math.round(p[0]), g: Math.round(p[1]), b: Math.round(p[2]), a: isNaN(a) ? 1 : a };
}

function pdfFontFor(cs) {
    const fam = String(cs.fontFamily || "").toLowerCase();
    const mono = /mono|consolas|menlo|courier/.test(fam);
    const w = cs.fontWeight;
    const bold = w === "bold" || w === "bolder" || parseInt(w, 10) >= 600;
    const italic = /italic|oblique/.test(cs.fontStyle || "");
    if (mono) return { name: "PdfMono", style: bold ? "bold" : "normal" };
    return { name: "PdfSans", style: bold && italic ? "bolditalic" : bold ? "bold" : italic ? "italic" : "normal" };
}

function pdfFitText(pdf, text, maxW) {
    let t = String(text);
    if (pdf.getTextWidth(t) <= maxW) return t;
    while (t.length > 1 && pdf.getTextWidth(t + "…") > maxW) t = t.slice(0, -1);
    return t + "…";
}

// Fills / strokes with an optional alpha.
function pdfWithAlpha(pdf, alpha, fn) {
    if (alpha >= 0.999) return fn();
    pdf.saveGraphicsState();
    pdf.setGState(new pdf.GState({ opacity: Math.max(0, alpha), "stroke-opacity": Math.max(0, alpha) }));
    try {
        return fn();
    } finally {
        pdf.restoreGraphicsState();
    }
}

function pdfBoxRadius(cs, w, h) {
    const one = (v) => {
        const s = String(v || "0").split(" ")[0];
        const n = parseFloat(s) || 0;
        return s.endsWith("%") ? (n / 100) * Math.min(w, h) : n;
    };
    const r = Math.max(one(cs.borderTopLeftRadius), one(cs.borderTopRightRadius), one(cs.borderBottomRightRadius), one(cs.borderBottomLeftRadius));
    return Math.min(r, Math.min(w, h) / 2);
}

// Background + borders of one element as real vector shapes.
function pdfDrawBox(c, cs, rel) {
    const pdf = c.pdf;
    const s = c.s;
    const x = c.x(rel.left), y = c.y(rel.top), w = rel.width * s, h = rel.height * s;
    if (w <= 0 || h <= 0) return;
    const rad = pdfBoxRadius(cs, rel.width, rel.height) * s;

    const bg = pdfParseColor(cs.backgroundColor);
    if (bg && bg.a > 0) {
        pdfWithAlpha(pdf, bg.a, () => {
            pdf.setFillColor(bg.r, bg.g, bg.b);
            if (rad > 0.2) pdf.roundedRect(x, y, w, h, rad, rad, "F");
            else pdf.rect(x, y, w, h, "F");
        });
    }

    const sides = ["Top", "Right", "Bottom", "Left"].map((n) => ({
        n,
        w: parseFloat(cs["border" + n + "Width"]) || 0,
        st: cs["border" + n + "Style"],
        col: pdfParseColor(cs["border" + n + "Color"]),
    }));
    const active = sides.filter((sd) => sd.w > 0 && sd.st !== "none" && sd.st !== "hidden" && sd.col && sd.col.a > 0);
    if (!active.length) return;

    const same = active.length === 4 && active.every((sd) => sd.w === active[0].w && sd.st === active[0].st &&
        sd.col.r === active[0].col.r && sd.col.g === active[0].col.g && sd.col.b === active[0].col.b && sd.col.a === active[0].col.a);
    const dash = (sd) => (sd.st === "dashed" ? [sd.w * 3 * s, sd.w * 2 * s] : sd.st === "dotted" ? [sd.w * s, sd.w * s] : null);
    if (same && rad > 0.2) {
        const sd = active[0];
        const lw = sd.w * s;
        pdfWithAlpha(pdf, sd.col.a, () => {
            pdf.setDrawColor(sd.col.r, sd.col.g, sd.col.b);
            pdf.setLineWidth(lw);
```

```
            const d = dash(sd);
            if (d) pdf.setLineDashPattern(d, 0);
            pdf.roundedRect(x + lw / 2, y + lw / 2, w - lw, h - lw, Math.max(0, rad - lw / 2), Math.max(0, rad - lw / 2), "S");
            if (d) pdf.setLineDashPattern([], 0);
        });
        return;
    }
    active.forEach((sd) => {
        const lw = sd.w * s;
        pdfWithAlpha(pdf, sd.col.a, () => {
            const d = dash(sd);
            if (d) {
                pdf.setDrawColor(sd.col.r, sd.col.g, sd.col.b);
                pdf.setLineWidth(lw);
                pdf.setLineDashPattern(d, 0);
                if (sd.n === "Top") pdf.line(x, y + lw / 2, x + w, y + lw / 2);
                if (sd.n === "Bottom") pdf.line(x, y + h - lw / 2, x + w, y + h - lw / 2);
                if (sd.n === "Left") pdf.line(x + lw / 2, y, x + lw / 2, y + h);
                if (sd.n === "Right") pdf.line(x + w - lw / 2, y, x + w - lw / 2, y + h);
                pdf.setLineDashPattern([], 0);
                return;
            }
            pdf.setFillColor(sd.col.r, sd.col.g, sd.col.b);
            if (sd.n === "Top") pdf.rect(x, y, w, lw, "F");
            if (sd.n === "Bottom") pdf.rect(x, y + h - lw, w, lw, "F");
            if (sd.n === "Left") pdf.rect(x, y, lw, h, "F");
            if (sd.n === "Right") pdf.rect(x + w - lw, y, lw, h, "F");
        });
    });
}

// Draws every word of a text node as real PDF text at the position the
// browser laid it out.
function pdfDrawText(c, node) {
    const parent = node.parentElement;
    if (!parent) return;
    const text = node.nodeValue;
    if (!text || !text.trim()) return;
    const cs = c.win.getComputedStyle(parent);
    if (cs.visibility === "hidden" || cs.display === "none") return;
    const col = pdfParseColor(cs.color) || { r: 0, g: 0, b: 0, a: 1 };
    const fs = parseFloat(cs.fontSize) || 12;
    const font = pdfFontFor(cs);
    const pdf = c.pdf;
    const s = c.s;
    const ls = parseFloat(cs.letterSpacing) || 0;
    const tt = cs.textTransform;
    const underline = /underline/.test(cs.textDecorationLine || "");
    const strike = /line-through/.test(cs.textDecorationLine || "");
    const fsPt = fs * s;
    const lineH = (PDF_FONT_METRICS.ascent + PDF_FONT_METRICS.descent) * fs;
    const range = c.doc.createRange();

    pdf.setFont(font.name, font.style);
    pdf.setFontSize(fsPt);
    pdf.setTextColor(col.r, col.g, col.b);

    const place = (str, rect, perChar) => {
        const mid = rect.top - c.rootTop + rect.height / 2;
        if (mid < c.plan.top || mid > c.plan.bottom) return;
        const baseCss = rect.top - c.rootTop + (rect.height - lineH) / 2 + PDF_FONT_METRICS.ascent * fs;
        const px = c.x(rect.left - c.rootLeft);
        const py = c.y(baseCss);
        let charSpace = ls * s;
        if (!perChar && str.length > 0) {
            const natural = pdf.getTextWidth(str) + charSpace * str.length;
            const target = rect.width * s;
            const diff = target - natural;
            if (Math.abs(diff) > 0.15 && Math.abs(diff) < natural * 0.3) charSpace += diff / str.length;
        }
        pdfWithAlpha(pdf, col.a, () => {
            if (Math.abs(charSpace) > 0.001) pdf.text(str, px, py, { charSpace });
            else pdf.text(str, px, py);
            if (underline || strike) {
                pdf.setDrawColor(col.r, col.g, col.b);
                pdf.setLineWidth(Math.max(0.3, fsPt * 0.06));
                const wpt = rect.width * s;
                if (underline) pdf.line(px, py + fsPt * 0.12, px + wpt, py + fsPt * 0.12);
                if (strike) pdf.line(px, py - fsPt * 0.28, px + wpt, py - fsPt * 0.28);
            }
        });
    };

    const re = /\S+/g;
    let m;
    while ((m = re.exec(text))) {
        range.setStart(node, m.index);
```

```
        range.setEnd(node, m.index + m[0].length);
        const rects = Array.from(range.getClientRects()).filter((r) => r.width > 0.01 && r.height > 0.01);
        if (!rects.length) continue;
        let word = m[0];
        if (tt === "uppercase") word = word.toUpperCase();
        else if (tt === "lowercase") word = word.toLowerCase();
        else if (tt === "capitalize") word = word.charAt(0).toUpperCase() + word.slice(1);
        if (rects.length === 1) {
            place(word, rects[0], false);
        } else {
            // word broken over several lines: position each character on its own
            for (let i = 0; i < m[0].length; i++) {
                range.setStart(node, m.index + i);
                range.setEnd(node, m.index + i + 1);
                const rr = Array.from(range.getClientRects()).filter((r) => r.width > 0.01 && r.height > 0.01)[0];
                if (rr) place(word.charAt(i) || m[0].charAt(i), rr, true);
            }
        }
    }
    range.detach && range.detach();
}

// SVG diagrams become real vector graphics (paths, lines, text) in the PDF.
async function pdfDrawSvg(c, svg, rel) {
    const svg2pdf = pdfSvg2PdfFn();
    const holder = document.createElement("div");
    holder.style.cssText = "position:fixed;left:-30000px;top:0;pointer-events:none;opacity:0;";
    const clone = svg.cloneNode(true);
    clone.setAttribute("width", rel.width);
    clone.setAttribute("height", rel.height);
    clone.style.width = rel.width + "px";
    clone.style.height = rel.height + "px";
    if (!clone.getAttribute("xmlns")) clone.setAttribute("xmlns", "http://www.w3.org/2000/svg");
    clone.style.fontFamily = "PdfSans";
    clone.querySelectorAll("text, tspan").forEach((t) => {
        t.style.fontFamily = "PdfSans";
    });
    // svg2pdf ignores "dominant-baseline", so labels centred on a line with
    // dominant-baseline="middle" (e.g. the member length written over the member)
    // would sit too high. Apply the same vertical offset the browser uses
    // (half the x-height for "middle", half of ascent - descent for "central").
    Array.from(clone.querySelectorAll("text[dominant-baseline], tspan[dominant-baseline]")).forEach((t) => {
        const mode = t.getAttribute("dominant-baseline");
        const k = mode === "middle" ? 0.26 : mode === "central" || mode === "mathematical" ? 0.3465 : mode === "hanging" ? 0.72 : 0;
        t.removeAttribute("dominant-baseline");
        if (!k || t.tagName.toLowerCase() === "tspan") return;
        const fs = parseFloat(t.getAttribute("font-size") || window.getComputedStyle(t).fontSize) || 12;
        t.setAttribute("y", (parseFloat(t.getAttribute("y")) || 0) + k * fs);
    });
    // svg2pdf has no "paint-order": labels that use a white stroke as a halo behind the
    // text would be painted over their own fill. Split such labels into a halo copy
    // (drawn first) and the plain filled text on top - the same look as the page.
    Array.from(clone.querySelectorAll("text[paint-order]")).forEach((t) => {
        const stroke = t.getAttribute("stroke");
        if (!stroke || stroke === "none" || !/stroke/.test(t.getAttribute("paint-order") || "")) return;
        const halo = t.cloneNode(true);
        halo.setAttribute("fill", stroke);
        halo.removeAttribute("paint-order");
        t.parentNode.insertBefore(halo, t);
        t.setAttribute("stroke", "none");
        t.removeAttribute("stroke-width");
        t.removeAttribute("paint-order");
    });
    holder.appendChild(clone);
    document.body.appendChild(holder);
    try {
        await svg2pdf(clone, c.pdf, {
            x: c.x(rel.left),
            y: c.y(rel.top),
            width: rel.width * c.s,
            height: rel.height * c.s,
        });
    } finally {
        holder.remove();
    }
}

async function pdfDrawNode(c, node) {
    if (node.nodeType === 3) {
        pdfDrawText(c, node);
        return;
    }
    if (node.nodeType !== 1) return;
    const tag = node.tagName.toLowerCase();
    if (tag === "script" || tag === "style" || tag === "noscript" || tag === "template") return;
    const cs = c.win.getComputedStyle(node);
    if (cs.display === "none" || cs.visibility === "hidden") return;
```

```
    if (cs.display === "contents") {
        for (const ch of Array.from(node.childNodes)) await pdfDrawNode(c, ch);
        return;
    }
    const r = node.getBoundingClientRect();
    const rel = { left: r.left - c.rootLeft, top: r.top - c.rootTop, width: r.width, height: r.height };
    if (r.height > 0 && (rel.top > c.plan.bottom + 2 || rel.top + rel.height < c.plan.top - 2)) return;
    const op = parseFloat(cs.opacity);
    if (op === 0) return;
    const layer = !isNaN(op) && op < 1;
    if (layer) {
        c.pdf.saveGraphicsState();
        c.pdf.setGState(new c.pdf.GState({ opacity: op, "stroke-opacity": op }));
    }
    try {
        if (tag === "svg") {
            if (rel.width > 0 && rel.height > 0) await pdfDrawSvg(c, node, rel);
            return;
        }
        pdfDrawBox(c, cs, rel);
        for (const ch of Array.from(node.childNodes)) await pdfDrawNode(c, ch);
    } finally {
        if (layer) c.pdf.restoreGraphicsState();
    }
}

// Running header / footer of a report page.
function pdfDrawPageChrome(pdf, meta) {
    const L = PdfLayout;
    const S = L.ptW / L.pageW; // pt per page px
    pdf.setFont("PdfSans", "bold");
    pdf.setFontSize(13 * S);
    pdf.setTextColor(30, 58, 138);
    pdf.text(pdfFitText(pdf, `Structural Analysis Report — ${meta.modelName}`, 760 * S), L.marginX * S, 29 * S);
    pdf.setFontSize(12 * S);
    pdf.setTextColor(71, 85, 105);
    pdf.text(String(meta.sectionLabel), (L.pageW - L.marginX) * S, 29 * S, { align: "right" });
    pdf.setDrawColor(30, 58, 138);
    pdf.setLineWidth(1.5 * S);
    pdf.line(L.marginX * S, 38 * S, (L.pageW - L.marginX) * S, 38 * S);

    pdf.setDrawColor(203, 213, 225);
    pdf.setLineWidth(1 * S);
    pdf.line(L.marginX * S, (L.pageH - L.footerH + 6) * S, (L.pageW - L.marginX) * S, (L.pageH - L.footerH + 6) * S);
    pdf.setFont("PdfSans", "normal");
    pdf.setFontSize(11 * S);
    pdf.setTextColor(100, 116, 139);
    pdf.text(pdfFitText(pdf, `Load case: ${meta.caseLabel}  |  ${meta.dateText}`, 800 * S), L.marginX * S, (L.pageH - 14) * S);
    pdf.text(`Page ${meta.pageNo}`, (L.pageW - L.marginX) * S, (L.pageH - 14) * S, { align: "right" });
}

// Draws one planned page: chrome + the report region as vector content.
async function pdfDrawReportPage(pdf, root, plan, fc, meta) {
    const L = PdfLayout;
    const S = L.ptW / L.pageW;
    const w = Math.ceil(plan.w);
    const h = Math.ceil(plan.bottom - plan.top) + 1;
    const fit = Math.min(1, L.contentW / w, L.contentH / h);
    plan.fit = fit;

    pdfDrawPageChrome(pdf, meta);

    const rr = root.getBoundingClientRect();
    const s = fit * S; // pt per CSS px
    const c = {
        pdf, doc: fc.doc, win: fc.win, plan, s,
        rootLeft: rr.left, rootTop: rr.top,
        x: (cssX) => (L.marginX + cssX * fit) * S,
        y: (cssY) => (L.headerH + (cssY - plan.top) * fit) * S,
    };
    pdf.saveGraphicsState();
    pdf.rect(0, L.headerH * S, L.pageW * S, h * fit * S, null);
    pdf.clip();
    pdf.discardPath();
    try {
        for (const ch of Array.from(root.childNodes)) await pdfDrawNode(c, ch);
    } finally {
        pdf.restoreGraphicsState();
    }
}

function pdfBuildFile(pages, title) {
    const pdf = pages.pdf;
    pdf.setProperties({ title: String(title), creator: "Structural Analysis App" });
    return pdf.output("blob");
}
function pdfDownloadBlob(blob, fileName) {
```

```
    const url = URL.createObjectURL(blob);
    const a = document.createElement("a");
    a.href = url;
    a.download = fileName;
    a.style.display = "none";
    document.body.appendChild(a);
    a.click();
    setTimeout(() => {
        a.remove();
        URL.revokeObjectURL(url);
    }, 60000);
}

// Collects the report sections (structure model diagram first, then Steps 1-7 from the results panels, Step 8 =
// AFD / SFD / BMD diagrams). Throws an Error carrying a user-facing message.
function pdfCollectReportSections() {
    const sections = [];

    // First page: the structure (model) diagram, followed by the solution steps.
    try {
        const nNodes = model.nodes.length, nEls = model.elements.length;
        const nSup = model.supportAssignments.length;
        sections.push({
            label: "Structure Model",
            html: `<div class="calc-step"><h4>Structure Model Diagram</h4><p>Structure analysed in this report: ${nNodes} node(s), ${nEls} element(s), ${nSup} support(s). The step-by-step
solution follows.</p><div class="pdf-diagram">${buildPdfDiagramBlock("model")}</div></div>`,
        });
    } catch (err) {
        console.error(err);
        throw new Error("Could not draw the structure model diagram for the PDF. Check the browser console for details.");
    }
    for (let n = 1; n <= 7; n++) {
        const slot = document.querySelector(`.calc-step-slot[data-step-number="${n}"]`);
        const step = slot && slot.querySelector(":scope > .calc-step");
        if (!step) throw new Error("Step results are not available. Please run the analysis again.");
        const clone = step.cloneNode(true);
        clone.querySelectorAll("button, .btn").forEach((b) => b.remove());
        clone.querySelectorAll("[onclick]").forEach((el) => el.removeAttribute("onclick"));
        sections.push({ label: `Step ${n} of 8`, html: clone.outerHTML });
    }

    try {
        const titles = isGridAnalysisMode()
            ? {
                afd: "Torsion Diagram (Mx)",
                sfd: "Vertical Shear Diagram (Fz)",
                bmd: "Torsion Diagram (My)",
            }
            : {
                afd: "Axial Force Diagram (AFD)",
                sfd: "Shear Force Diagram (SFD)",
                bmd: "Bending Moment Diagram (BMD)",
            };
        const blocks = ["afd", "sfd", "bmd"].map(
            (mode, i) => `<div class="pdf-diagram${i > 0 ? " pdf-break-before" : ""}"><h5>${titles[mode]}</h5>${buildPdfDiagramBlock(mode)}</div>`,
        );
        sections.push({
            label: "Step 8 of 8",
            html: `<div class="calc-step"><h4>${isGridAnalysisMode() ? "Step 8: Grid Torsion (Mx, My) and Vertical Shear (Fz) Diagrams" : "Step 8: Axial Force, Shear Force and Bending
Moment Diagrams (AFD, SFD, BMD)"}</h4><p>Diagrams derived from the member end actions [AM] recovered in Step 7 and the intermediate member loads.</p>${blocks.join("")}</div>`,
        });
    } catch (err) {
        console.error(err);
        throw new Error("Could not draw the Step 8 diagrams for the PDF. Check the browser console for details.");
    }
    return sections;
}

// Lays out the sections in the hidden frame and draws every page as vector PDF content
// (the "pages" array only counts pages and carries the jsPDF document).
async function pdfRenderSectionsToPages(fc, sections, meta, jpegPages) {
    const { frame, doc, win, root } = fc;
    const progressPrefix = meta.progressPrefix || "";
    for (let s = 0; s < sections.length; s++) {
        const sec = sections[s];
        await pdfShowProgress(`${progressPrefix}${sec.label}: laying out…`);
        root.innerHTML = sec.html;
        await pdfRasterizeSvgs(doc, root);
        pdfAlignMatrixLabels(root);
        pdfPackEquationRows(root);
        pdfSplitWideTables(root);
        pdfWrapWideEquations(root);
        pdfSplitTallTables(root, PdfLayout.contentH);
        const blocks = pdfCollectBlocks(root, PdfLayout.contentH);
        const plans = pdfPlanPages(root, blocks, PdfLayout.contentH);

        // Widen the frame if some block is wider than the page content.
```

```
        const maxW = Math.max(PdfLayout.contentW, ...plans.map((p) => Math.ceil(p.w)));
        const frameW = Math.max(1500, maxW + 80);
        const frameH = Math.max(1000, Math.ceil(root.getBoundingClientRect().height) + 40);
        frame.style.width = frameW + "px";
        frame.style.height = frameH + "px";
        await new Promise((r) => setTimeout(r, 30));

        for (let p = 0; p < plans.length; p++) {
            await pdfShowProgress(`${progressPrefix}${sec.label}: page ${p + 1} of ${plans.length}…`);
            const plan = plans[p];
            const pdf = pdfDocumentFor(jpegPages);
            if (jpegPages.length > 0) pdf.addPage();
            await pdfDrawReportPage(pdf, root, plan, { doc, win }, {
                ...meta,
                sectionLabel: (meta.sectionPrefix || "") + sec.label,
                pageNo: jpegPages.length + 1,
            });
            jpegPages.push({ vector: true }); // one entry per PDF page
        }
    }
}

async function exportAnalysisPDF() {
    if (pdfExportBusy) return;
    if (analysisState.status !== "valid") {
        const message = analysisState.status === "invalidated"
            ? "The model changed since the last analysis. Please run the analysis again before exporting the PDF."
            : analysisState.status === "failed"
                ? "The last analysis failed its equilibrium check. Review the model until equilibrium passes before exporting a report."
                : "Please run the analysis first, then export the PDF.";
        alert(message);
        return;
    }
    if (!pdfEngineReady()) {
        alert("The PDF engine could not be loaded. Please reload the page and try again.");
        return;
    }

    let sections;
    try {
        sections = pdfCollectReportSections();
    } catch (err) {
        alert(err.message);
        return;
    }

    pdfExportBusy = true;
    const btn = document.getElementById("exportPdfBtn");
    if (btn) btn.disabled = true;
    let frame = null;
    try {
        await pdfShowProgress("Preparing report…");
        frame = pdfCreateFrame();
        await frame.stylesReady;
        const doc = frame.contentDocument;
        const win = frame.contentWindow;
        const root = doc.getElementById("pdfRoot");
        if (doc.fonts && doc.fonts.ready) await doc.fonts.ready;

        const meta = {
            modelName: getModelName(),
            caseLabel: getActiveAnalysisCaseLabel(),
            dateText: new Date().toLocaleDateString(),
        };
        const jpegPages = [];
        await pdfRenderSectionsToPages({ frame, doc, win, root }, sections, meta, jpegPages);

        await pdfShowProgress("Creating PDF file…");
        const blob = pdfBuildFile(jpegPages, `${meta.modelName} - Analysis Report`);
        pdfDownloadBlob(blob, `${sanitizeFileName(meta.modelName)} - Analysis Report.pdf`);
    } catch (err) {
        console.error(err);
        alert("PDF export failed. Check the browser console for details.");
    } finally {
        if (frame) frame.remove();
        pdfHideProgress();
        pdfExportBusy = false;
        if (btn) btn.disabled = false;
    }
}
```

31. modules/results/step-1/step-1.js

```
/* ================================================================================
 * step-1.js — Results part: Step 1
 * Step 1 — Degrees of freedom & numbering (solver + DoF numbering helpers)
 * Adds its menu button + panel to the Results frame, then defines its solver / render code.
 * ================================================================================ */

/* ---- Markup ---- */
mountModuleHTML(".results-detail-menu", "beforeend", `<button type="button" class="results-detail-btn active" data-result-panel="result-step-1" onclick="showResultDetail('result-step-1',
this)">1. DoF &amp;amp; Numbering</button>
`);
mountModuleHTML(".results-detail-body", "beforeend", `
<div class="result-detail-panel active" id="result-step-1">
    <div class="calc-step-slot" data-step-number="1"></div>
</div>
`);

/* ---- Logic ---- */
function buildStructuralDofNumbering() {
    if (isGridAnalysisMode()) {
        const dofMap = {};
        const dofInfo = [];
        let dofCounter = 0;
        model.nodes.forEach((node, index) => {
            const d = { x: dofCounter++, y: dofCounter++, z: dofCounter++, num: index };
            dofInfo[d.x] = { nodeId: node.id, dofLabel: "0X" };
            dofInfo[d.y] = { nodeId: node.id, dofLabel: "0Y" };
            dofInfo[d.z] = { nodeId: node.id, dofLabel: "ΔZ" };
            dofMap[node.id] = d;
        });
        return { dofMap, dofInfo, totalDof: dofCounter };
    }
    const dofMap = {};
    const dofInfo = [];
    let dofCounter = 0;
    const dofConfig = getGlobalDofNumberingConfig();

    // The Neglect X / Neglect Moment M switches are optional controls.
    // New Beam templates use their actual area when X is included; when Ignore
    // X is checked, beam-only joints do not get axial DOFs. Truss and Frame
    // retain their existing axial behavior. Shared nodes use the union of DOFs.
    // The numbering is compressed, so an omitted DOF never reserves a number.
    model.nodes.forEach((node, index) => {
        const relevance = getNodeStructuralDofRelevance(node.id);
        const d = { x: null, y: null, z: null, num: index };

        if (!dofConfig.skipAxial || relevance.ux) {
            d.x = dofCounter++;
            dofInfo[d.x] = { nodeId: node.id, dofLabel: "UX" };
        }

        if (relevance.uy) {
            d.y = dofCounter++;
            dofInfo[d.y] = { nodeId: node.id, dofLabel: "UY" };
        }

        if (!dofConfig.skipMoment || relevance.rz) {
            d.z = dofCounter++;
            dofInfo[d.z] = { nodeId: node.id, dofLabel: "RZ" };
        }

        dofMap[node.id] = d;
    });

    return { dofMap, dofInfo, totalDof: dofCounter };
}

function getElementLocalDofMask(element) {
    if (isGridAnalysisMode()) return [true, true, true, true, true, true];
    const type = getElementStructuralType(element);
    if (type === "beam") {
        // Beam axial stiffness uses the section area only when Ignore X is OFF.
        const template = model.elementTemplates.find((item) => item.id === element.templateId);
        const axialActive = Number(template?.A) > 0 && !getGlobalDofNumberingConfig().skipAxial;
        return [axialActive, true, true, axialActive, true, true];
    }
    if (type === "truss") {
        // Truss: X, Y at both ends. M/rotation is released.
        return [true, true, false, true, true, false];
    }
    if (type === "frame") {
        // Frame: X, Y, M at both ends.
        return [true, true, true, true, true, true];
    }
    return [false, false, false, false, false, false];
}
```

```
function getElementGlobalDofIndices(element, n1, n2, dofMap) {
    if (!element || !n1 || !n2 || !dofMap) {
        return { indices: [null, null, null, null, null, null], mask: [false, false, false, false, false, false] };
    }

    const mask = getElementLocalDofMask(element);
    const nodeDofs = dofMap;
    const globalCandidates = [
        nodeDofs[n1.id]?.x ?? null,
        nodeDofs[n1.id]?.y ?? null,
        nodeDofs[n1.id]?.z ?? null,
        nodeDofs[n2.id]?.x ?? null,
        nodeDofs[n2.id]?.y ?? null,
        nodeDofs[n2.id]?.z ?? null,
    ];

    // A member uses only the DOFs allowed by its own type. This matters at a
    // mixed node: a truss can create a global X DOF at the node, but a beam
    // connected to the same node must still ignore X in that beam's member
    // matrix. Likewise, a beam/frame can create a global M DOF that a truss
    // must not use.
    const indices = globalCandidates.map((idx, i) => mask[i] ? idx : null);
    return { indices, mask };
}

// ============================================================================
// 16a. SOLVER ENGINE
// The LPW-3 plane-frame stiffness method solver, split into the same eight
// steps the analysis report displays. Each solveStepN() function performs
// ONLY that step's calculation and returns a structured result (or null to
// signal that a fatal condition was already reported to the user via
// alert()/confirm() and the analysis must stop). HTML report generation for
// each step lives in the matching renderStepN() function further below —
// calculation and rendering are kept separate, but each step function does
// still assemble the small amount of step-local HTML text that its own
// numbers feed into (e.g. table rows), exactly as the original single
// analysis routine did, so the displayed report is pixel-for-pixel
// unchanged.
// ============================================================================

function solveGridStep1(ctx) {
    let gridResult;
    try {
        gridResult = GridSolver.calculateSteps1To3(model, {
            loadCaseId: activeLoadCaseId,
            momentCaseId: activeMomentCaseId,
        });
    } catch (error) {
        alert(`Grid Steps 1–3 failed: ${error && error.message ? error.message : error}`);
        return null;
    }
    ctx.gridSteps13 = gridResult;
    const source = gridResult.steps.step1;
    const dofMap = source.dofMap;
    const totalDof = source.totalDof;
    const isRestrained = new Array(totalDof).fill(false);
    const prescribedDisp = new Array(totalDof).fill(0);
    const dofInfo = new Array(totalDof);
    source.nodes.forEach((item) => {
        const map = dofMap[item.nodeId];
        const items = [
            [map.x, item.dof.thetaX, "θX"],
            [map.y, item.dof.thetaY, "θY"],
            [map.z, item.dof.deltaZ, "ΔZ"],
        ];
        items.forEach(([idx, details, label]) => {
            isRestrained[idx] = !!details.restrained;
            prescribedDisp[idx] = Number(details.prescribed) || 0;
            dofInfo[idx] = { nodeId: item.nodeId, dofLabel: label };
        });
    });

    const heavyModel = ctx.heavyModel;
    const dofSvg = model.nodes.length > 10
        ? `<div style="height:380px;display:flex;align-items:center;justify-content:center;flex-direction:column;gap:8px;text-align:center;color:#64748b;font-family:Arial,sans-serif;border:
1px dashed #cbd5e1;border-radius:8px;background:#f8fafc;"><div style="font-size:26px;">■</div><div style="font-size:14px;font-weight:800;color:#334155;">Large model — DoF diagram paused</
div><div style="font-size:11px;line-height:1.5;max-width:420px;">Open the full-screen DoF diagram to inspect the Grid numbering.</div></div>`
        : renderDOFDiagram(800, 400);
    const html = `<div class="calc-step">
        <h4>Step 1: Grid Joint DoF Numbering and Restraints</h4>
        <p>Grid uses three degrees of freedom per joint: θX, θY and ΔZ. The supplied Grid calculation engine numbers these in node order.</p>
        <div class="canvas-container" style="background:white;border:1px solid #cbd5e1;border-radius:6px;padding:10px;margin-bottom:15px;width:100%;max-width:100%;overflow-x:auto;height:
400px;">${dofSvg}</div>
        <button class="btn btn-secondary" style="width:100%;" onclick="openFullScreen('dof')">■ Open Full Screen Diagram</button>
    </div>`;

    return {
```

```
        dofMap, dofInfo, numNodes: model.nodes.length, totalDof, isRestrained, prescribedDisp,
        SJ: Matrix.zeros(totalDof, totalDof), AE: Matrix.zeros(totalDof, 1), AJ: Matrix.zeros(totalDof, 1),
        heavyModel, html, gridSteps13: gridResult,
    };
}

// Step 1: Divide the structure into members, assign member/joint/DoF numbers.
function solveStep1(ctx) {
    if (isGridAnalysisMode()) return solveGridStep1(ctx);
    const numbering = buildStructuralDofNumbering();
    const { dofMap, dofInfo, totalDof } = numbering;
    let numNodes = model.nodes.length;

    // Validate structural element definitions and identify zero-stiffness modes.
    for (const el of model.elements) {
        const tmpl = model.elementTemplates.find((t) => t.id === el.templateId);
        if (!tmpl) {
            alert(`Element ${el.id} references a missing element template.`);
            return null;
        }
        const A = Number(tmpl.A);
        const I = Number(tmpl.I);
        if (!Number.isFinite(A) || !Number.isFinite(I) || A < 0 || I < 0 || (A === 0 && I === 0)) {
            alert(`Element ${el.id} must have A > 0 or I > 0. Use A = 0 for a beam or I = 0 for a truss.`);
            return null;
        }
    }

    // Calculate restrained DoFs early for Step 2 UI output
    let isRestrained = new Array(totalDof).fill(false);
    let prescribedDisp = new Array(totalDof).fill(0);

    model.supportAssignments.forEach((sa) => {
        let d = dofMap[sa.nodeId];
        let t = model.supportTemplates.find((s) => s.id === sa.supportId);
        if (d && t) {
            const relevant = getNodeStructuralDofRelevance(sa.nodeId);
            const movement = getSupportAssignmentPrescribedDisplacements(sa, t);
            // A = 0 -> pure beam/flexural behavior: axial RFx is irrelevant.
            // I = 0 -> pure truss behavior: rotational RMz is irrelevant.
            if (t.restraints.RFx && relevant.ux && d.x !== null) {
                isRestrained[d.x] = true;
                prescribedDisp[d.x] = Number(movement.dx) || 0;
            }
            if (t.restraints.RFy && relevant.uy && d.y !== null) {
                isRestrained[d.y] = true;
                prescribedDisp[d.y] = Number(movement.dy) || 0;
            }
            if (t.restraints.RMz && relevant.rz && d.z !== null) {
                isRestrained[d.z] = true;
                prescribedDisp[d.z] = Number(movement.tz) || 0;
            }
        }
    });

    let SJ = Matrix.zeros(totalDof, totalDof);
    let AE = Matrix.zeros(totalDof, 1);
    let AJ = Matrix.zeros(totalDof, 1);
    const heavyModel = ctx.heavyModel;

    // Step 1: SVG DIAGRAM INSERTION
    let step1SVG = model.nodes.length > 10
        ? `<div style="height:380px;display:flex;align-items:center;justify-content:center;flex-direction:column;gap:8px;text-align:center;color:#64748b;font-family:Arial,sans-serif;border:
1px dashed #cbd5e1;border-radius:8px;background:#f8fafc;">
                <div style="font-size:26px;">■</div>
                <div style="font-size:14px;font-weight:800;color:#334155;">Large model — DoF diagram paused</div>
                <div style="font-size:11px;line-height:1.5;max-width:420px;">The DoF / numbering diagram is rendered only in Full Screen for models above 10 nodes.</div>
           </div>`
        : renderDOFDiagram(800, 400);

    let html = `<div class="calc-step">
        <h4>Step 1: Divide the structure into number of members and assign member number, joint number and degrees of freedom (DoF) number</h4>
        <p>Structure is divided into ${model.elements.length} members and ${model.nodes.length} joints. DoF numbering is compressed only for the neglect options selected above.</p>
        <p><strong>Numbering rule:</strong> Beam → Y,M (and X when area is positive and Ignore X is OFF)  |  Truss → X,Y  |  Frame → X,Y,M. Ignore X also suppresses
axial thermal actions. Shared nodes receive the union of the applicable member DOFs.</p>
        <div class="canvas-container" style="background: white; border: 1px solid #cbd5e1; border-radius: 6px; padding: 10px; margin-bottom:15px; width: 100%; max-width: 100%; overflow-x:
auto; height: 400px;">
            ${step1SVG}
        </div>
        <button class="btn btn-secondary" style="width: 100%;" onclick="openFullScreen('dof')">■ Open Full Screen Diagram</button>
    </div>`;

    return { dofMap, dofInfo, numNodes, totalDof, isRestrained, prescribedDisp, SJ, AE, AJ, heavyModel, html };
}

function renderStep1(step1) {
    return step1.html;
```

```
}
```

32. modules/results/step-2/step-2.js

```
/* ========================================================================
 * step-2.js — Results part: Step 2
 * Step 2 — Joint & member properties
 * Adds its menu button + panel to the Results frame, then defines its solver / render code.
 * ======================================================================== */

/* ---- Markup ---- */
mountModuleHTML(".results-detail-menu", "beforeend", `<button type="button" class="results-detail-btn" data-result-panel="result-step-2" onclick="showResultDetail('result-step-2', this)">2.
Joint &amp;amp; Member Properties</button>
`);
mountModuleHTML(".results-detail-body", "beforeend", `
<div class="result-detail-panel" id="result-step-2">
    <div class="calc-step-slot" data-step-number="2"></div>
</div>
`);

/* ---- Logic ---- */
// Step 2: Joint and Member Properties Tables (display-only; does not feed
// later steps — the actual stiffness terms are recomputed in Step 3).
function solveStep2(ctx, step1) {
    if (isGridAnalysisMode()) {
        const grid = ctx.gridSteps13;
        const joints = grid.steps.step2.joints;
        const members = grid.steps.step2.members;
        const jointRows = joints.map((n) => {
            const d = step1.dofMap[n.nodeId];
            const label = (idx) => step1.isRestrained[idx] ? `R (${formatNum(step1.prescribedDisp[idx])})` : `D${idx + 1}`;
            return `<tr><td>${escapeHtml(n.nodeId)}</td><td>${formatNum(n.x)}</td><td>${formatNum(n.y)}</td><td>${label(d.x)}</td><td>${label(d.y)}</td><td>${label(d.z)}</td></tr>`;
        }).join("");
        // Display the same Grid member stiffness coefficients used by the Grid
        // formulation. E is already in kN/m²; convert G from GPa to kN/m².
        const memberRows = members.map((m) => {
            const L = Number(m.length);
            const E = Number(m.E_kN_m2);
            const G = Number(m.G_GPa) * 1e6;
            const I = Number(m.I);
            const J = Number(m.J);
            const valid = Number.isFinite(L) && L > 0 &&
                Number.isFinite(E) && E >= 0 &&
                Number.isFinite(G) && G >= 0 &&
                Number.isFinite(I) && I >= 0 &&
                Number.isFinite(J) && J >= 0;
            const mTerm = valid ? (G * J) / L : NaN;
            const aTerm = valid ? (12 * E * I) / Math.pow(L, 3) : NaN;
            const bTerm = valid ? (6 * E * I) / Math.pow(L, 2) : NaN;
            const cTerm = valid ? (4 * E * I) / L : NaN;
            return `<tr><td>${escapeHtml(m.elementId)}</td><td>${escapeHtml(m.startNode)}</td><td>${escapeHtml(m.endNode)}</td><td>${formatNum(L)}</td><td>${formatNum(m.E_kN_m2)}</td><td>${
formatNum(m.G_GPa)}</td><td>${formatNum(I)}</td><td>${formatNum(J)}</td><td>${formatNum(mTerm)}</td><td>${formatNum(aTerm)}</td><td>${formatNum(bTerm)}</td><td>${formatNum(cTerm)}</td></
tr>`;
        }).join("");
        const html = `<div class="calc-step"><h4>Step 2: Grid Joint and Member Properties</h4>
            <h5>Joint Properties</h5><div class="table-wrap" style="margin-bottom:20px;border:1px solid #e2e8f0;border-radius:8px;"><table style="min-width:100%;background:white;
"><thead><tr><th>Joint</th><th>X (m)</th><th>Y (m)</th><th>0X</th><th>0Y</th><th>ΔZ</th></tr></thead><tbody>${jointRows}</tbody></table></div>
            <h5>Member Properties and Grid Stiffness Terms</h5><div class="table-wrap" style="border:1px solid #e2e8f0;border-radius:8px;overflow-x:auto;"><table style="min-width:1150px;
background:white;"><thead><tr><th>Member</th><th>Start Node</th><th>End Node</th><th>L (m)</th><th>E (kN/m²)</th><th>G (GPa)</th><th>I (m■)</th><th>J (m■)</th><th>m = GJ/L (kN·m)</th><th>a
= 12EI/L³ (kN/m)</th><th>b = 6EI/L² (kN)</th><th>c = 4EI/L (kN·m)</th></tr></thead><tbody>${memberRows}</tbody></table></div>
            <p>Grid self-weight, when needed, must be entered directly as a line weight in kN/m.</p></div>`;
        return { html };
    }
    const { dofMap, isRestrained, prescribedDisp, heavyModel } = step1;

    let jointTableRows = "";
    model.nodes.forEach((n, i) => {
        let d = dofMap[n.id];
        let dxStr = d.x === null
            ? "—"
            : isRestrained[d.x]
                ? prescribedDisp[d.x] === 0
                    ? "0"
                    : prescribedDisp[d.x]
                : "D" + (d.x + 1);
        let dyStr = d.y === null
            ? "—"
            : isRestrained[d.y]
                ? prescribedDisp[d.y] === 0
                    ? "0"
                    : prescribedDisp[d.y]
                : "D" + (d.y + 1);
        let dzStr = d.z === null
            ? "—"
            : isRestrained[d.z]
                ? prescribedDisp[d.z] === 0
                    ? "0"
                    : prescribedDisp[d.z]
                : "D" + (d.z + 1);
```

```
        jointTableRows += `<tr><td>${i + 1} (${escapeHtml(n.id)})</td><td>${formatNum(n.x)}</td><td>${formatNum(n.y)}</td><td>${formatNum(n.z)}</td><td>${escapeHtml(dxStr)}</td><td>${
escapeHtml(dyStr)}</td><td>${escapeHtml(dzStr)}</td></tr>`;
    });

    let memberTableRows = "";
    model.elements.forEach((el, index) => {
        let n1 = model.nodes.find((n) => n.id === el.startNode);
        let n2 = model.nodes.find((n) => n.id === el.endNode);
        let tmpl = model.elementTemplates.find((t) => t.id === el.templateId);
        if (!n1 || !n2 || !tmpl) return;

        let dx = n2.x - n1.x,
            dy = n2.y - n1.y;
        let L = Math.hypot(dx, dy);
        let cx = L > 0 ? dx / L : 0,
            cy = L > 0 ? dy / L : 0;
        let A = tmpl.A,
            E = getElementEInternal(tmpl),
            I = tmpl.I;
        const elementType = getElementStructuralType(el);

        let d_val = L > 0 ? (A * E) / L : 0;
        let dCx2 = d_val * cx * cx;
        let dCy2 = d_val * cy * cy;
        let dCxCy = d_val * cx * cy;

        let a_val = L > 0 ? (12 * E * I) / Math.pow(L, 3) : 0;
        let b_val = L > 0 ? (6 * E * I) / Math.pow(L, 2) : 0;
        let c_val = L > 0 ? (4 * E * I) / L : 0;

        let e_val = dCx2 + a_val * cy * cy;
        let f_val = (d_val - a_val) * cx * cy;
        let g_val = -b_val * cy;
        let h_val = dCy2 + a_val * cx * cx;
        let j_val = b_val * cx;

        memberTableRows += `<tr>
            <td>${index + 1}</td><td>${escapeHtml(el.startNode)}</td><td>${escapeHtml(el.endNode)}</td>
            <td>${formatNum(A)}</td><td>${formatNum(E)}</td><td>${formatNum(I)}</td><td>${formatNum(L)}</td>
            <td>${formatNum(d_val)}</td><td>${formatNum(cx)}</td><td>${formatNum(cy)}</td>
            <td>${formatNum(dCx2)}</td><td>${formatNum(dCy2)}</td><td>${formatNum(dCxCy)}</td>
            <td>${formatNum(a_val)}</td><td>${formatNum(b_val)}</td><td>${formatNum(c_val)}</td>
            <td>${formatNum(e_val)}</td><td>${formatNum(f_val)}</td><td>${formatNum(g_val)}</td>
            <td>${formatNum(h_val)}</td><td>${formatNum(j_val)}</td>
        </tr>`;
    });

    let html = "";
    if (!heavyModel) {
        html += `<div class="calc-step">
        <h4>Step 2: Joint and Member Properties Tables</h4>
        <h5>Joint table</h5>
        <div class="table-wrap" style="margin-bottom: 20px; border: 1px solid #e2e8f0; border-radius: 8px;">
            <table style="min-width: 100%; background: white;">
                <thead><tr><th>Joint number</th><th>x coordinate</th><th>y coordinate</th><th>z coordinate</th><th>displacement-x</th><th>displacement-y</th><th>displacement-z</th></tr></
thead>
                <tbody>${jointTableRows}</tbody>
            </table>
        </div>
        <h5>Members table</h5>
        <div class="table-wrap" style="border: 1px solid #e2e8f0; border-radius: 8px;">
            <table style="min-width: 100%; background: white;">
                <thead><tr>
                    <th style="text-transform: none;">member number</th><th style="text-transform: none;">node id start</th><th style="text-transform: none;">node id end</th>
                    <th style="text-transform: none;">A (m²)</th><th style="text-transform: none;">E (kN/m²)</th><th style="text-transform: none;">I (m■)</th><th style="text-transform: none;
">L (m)</th>
                    <th style="text-transform: none;">d=AE/L (kN/m)</th><th style="text-transform: none;">Cx=(xk-xj)/L</th><th style="text-transform: none;">Cy=(yk-yj)/L</th>
                    <th style="text-transform: none;">d(Cx²)</th><th style="text-transform: none;">d(Cy²)</th><th style="text-transform: none;">d(Cx*Cy)</th>
                    <th style="text-transform: none;">a=12EI/(L³) (kN/m)</th><th style="text-transform: none;">b=6EI/(L²) (kN)</th><th style="text-transform: none;">c=4EI/L (kN·m)</th>
                    <th style="text-transform: none;">e (kN/m)</th><th style="text-transform: none;">f (kN/m)</th><th style="text-transform: none;">g (kN)</th>
                    <th style="text-transform: none;">h (kN/m)</th><th style="text-transform: none;">j (kN)</th>
                </tr></thead>
                <tbody>${memberTableRows}</tbody>
            </table>
        </div>

        <h5>General Matrix Formulas</h5>
        <div style="display:flex; flex-direction:column; align-items:center; gap: 20px; margin-top: 15px; margin-bottom: 25px;">
            <!-- SM -->
            <div class="matrix-wrapper">
                <div class="matrix-title">[SM]■</div>
                <div></div>
                <div style="display: flex; align-items: stretch; gap: 2px;">
                    <div style="width: 6px; border-left: 2px solid #1e293b; border-top: 2px solid #1e293b; border-bottom: 2px solid #1e293b;"></div>
                    <table class="matrix-table" style="text-align:center;">
                        <tr><td>d</td><td>0</td><td>0</td><td>-d</td><td>0</td><td>0</td></tr>
                        <tr><td>0</td><td>a</td><td>b</td><td>0</td><td>-a</td><td>b</td></tr>
```

```
                    <tr><td>0</td><td>b</td><td>c</td><td>0</td><td>-b</td><td>c/2</td></tr>
                    <tr><td>-d</td><td>0</td><td>0</td><td>0</td><td>d</td><td>0</td><td>0</td></tr>
                    <tr><td>0</td><td>-a</td><td>-b</td><td>0</td><td>a</td><td>-b</td></tr>
                    <tr><td>0</td><td>b</td><td>c/2</td><td>0</td><td>-b</td><td>c</td></tr>
                </table>
                <div style="width: 6px; border-right: 2px solid #1e293b; border-top: 2px solid #1e293b; border-bottom: 2px solid #1e293b;"></div>
            </div>
        </div>

        <!-- RT -->
        <div class="matrix-wrapper">
            <div class="matrix-title">[Rt]■</div>
            <div></div>
            <div style="display: flex; align-items: stretch; gap: 2px;">
                <div style="width: 6px; border-left: 2px solid #1e293b; border-top: 2px solid #1e293b; border-bottom: 2px solid #1e293b;"></div>
                <table class="matrix-table" style="text-align:center;">
                    <tr><td>Cx</td><td>Cy</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
                    <tr><td>-Cy</td><td>Cx</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
                    <tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td></tr>
                    <tr><td>0</td><td>0</td><td>0</td><td>Cx</td><td>Cy</td><td>0</td></tr>
                    <tr><td>0</td><td>0</td><td>0</td><td>-Cy</td><td>Cx</td><td>0</td></tr>
                    <tr><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td></tr>
                </table>
                <div style="width: 6px; border-right: 2px solid #1e293b; border-top: 2px solid #1e293b; border-bottom: 2px solid #1e293b;"></div>
            </div>
        </div>

        <!-- SMS -->
        <div class="matrix-wrapper">
            <div class="matrix-title">[Sms]■</div>
            <div></div>
            <div style="display: flex; align-items: stretch; gap: 2px;">
                <div style="width: 6px; border-left: 2px solid #1e293b; border-top: 2px solid #1e293b; border-bottom: 2px solid #1e293b;"></div>
                <table class="matrix-table" style="text-align:center;">
                    <tr><td>e</td><td>f</td><td>g</td><td>-e</td><td>-f</td><td>g</td></tr>
                    <tr><td>f</td><td>h</td><td>j</td><td>-f</td><td>-h</td><td>j</td></tr>
                    <tr><td>g</td><td>j</td><td>c</td><td>-g</td><td>-j</td><td>c/2</td></tr>
                    <tr><td>-e</td><td>-f</td><td>-g</td><td>e</td><td>f</td><td>-g</td></tr>
                    <tr><td>-f</td><td>-h</td><td>-j</td><td>f</td><td>h</td><td>-j</td></tr>
                    <tr><td>g</td><td>j</td><td>c/2</td><td>-g</td><td>-j</td><td>c</td></tr>
                </table>
                <div style="width: 6px; border-right: 2px solid #1e293b; border-top: 2px solid #1e293b; border-bottom: 2px solid #1e293b;"></div>
            </div>
        </div>
    </div>
</div>`;
}

    return { html };
}

function renderStep2(step2) {
    return step2.html;
}
```

33. modules/results/step-3/step-3.js

```
/* ========================================================================
 * step-3.js — Results part: Step 3
 * Step 3 — Member stiffness & AML (actions at member ends)
 * Adds its menu button + panel to the Results frame, then defines its solver / render code.
 * ======================================================================== */

/* ---- Markup ---- */
mountModuleHTML(".results-detail-menu", "beforeend", `<button type="button" class="results-detail-btn" data-result-panel="result-step-3" onclick="showResultDetail('result-step-3', this)">3.
Member Stiffness &amp;amp; AML</button>
`);
mountModuleHTML(".results-detail-body", "beforeend", `
<div class="result-detail-panel" id="result-step-3">
    <div class="calc-step-slot" data-step-number="3"></div>
</div>
`);

/* ---- Logic ---- */
// Step 3: Member stiffness matrices [SMS] and member end actions [AML],
// assembled directly into the global [SJ]/[AE] as each element is processed
// (matching the original single-pass implementation exactly), followed by
// assembly of nodal loads/moments into [AJ].
function solveGridStep3(ctx, step1) {
    const { dofMap, SJ, AE, AJ, heavyModel } = step1;
    const gridMembers = ctx.gridSteps13.steps.step3.members;
    let html = `<div class="calc-step"><h4>Step 3: Grid Member Stiffness Matrices and Fixed-End Actions</h4>`;
    model.elements.forEach((el) => {
        const member = gridMembers.find((m) => String(m.elementId) === String(el.id));
        const n1 = model.nodes.find((n) => n.id === el.startNode);
        const n2 = model.nodes.find((n) => n.id === el.endNode);
        if (!member || !n1 || !n2) throw new Error(`Missing Grid calculation result for element ${el.id}.`);
        el.L = member.length;
        el.cx = member.directionCosines.cx;
        el.cy = member.directionCosines.cy;
        el.SM = member.SM;
        el.RT = member.RT;
        el.RTT = member.RTT;
        el.SMS = member.SMS;
        el.AML = member.AML;
        el.AMS = member.AMS;
        el.localwx = 0;
        el.localwy = 0;
        el.localPointLoads = (member.localPointLoads || []).map((load) => ({ a: load.distance, Px: 0, Py: load.Fz, Pz: load.Fz, inputAxis: "Grid Z" }));
        el.localUDLs = (member.localUDLs || []).map((load) => ({ a: load.startDistance, b: load.endDistance, Fx1: 0, Fx2: 0, Fy1: load.q1, Fy2: load.q2, localFx1: 0, localFx2: 0, localFy1:
load.q1, localFy2: load.q2, inputAxis: "Grid Z" }));
        el.localPointMoments = (member.localPointMoments || []).map((moment) => ({ a: moment.distance, Mz: moment.Mx, Mx: moment.Mx, My: moment.My }));
        const mapped = getElementGlobalDofIndices(el, n1, n2, dofMap);
        const indices = mapped.indices;
        const labels = indices.map((idx) => idx === null ? "—" : idx + 1);
        for (let r = 0; r < 6; r++) {
            if (indices[r] === null) continue;
            for (let c = 0; c < 6; c++) {
                if (indices[c] === null) continue;
                SJ[indices[r]][indices[c]] += member.SMS[r][c];
            }
            AE[indices[r]][0] -= member.AMS[r][0];
        }
        if (!heavyModel) {
            html += `<h5>Element ${escapeHtml(el.id)}</h5>`;
            html += matrixEqHTML([
                { m: el.SM, t: "[SM] (Grid Local)", top: labels, right: labels },
                { m: el.RT, t: "[RT] (Grid Transformation)", top: labels, right: labels },
            ], [" "]);
            html += matrixEqHTML([
                { m: el.SMS, t: "[SMS] = [RT]■[SM][RT]", top: labels, right: labels },
                { m: el.AML, t: "[AML] (Local Fixed-End Actions)", right: labels },
                { m: el.RTT, t: "[RT]■", right: labels },
                { m: el.AMS, t: "[AMS] = [RT]■[AML]", right: labels },
            ], [" ", "×", "="]);
        }
    });

    // Grid nodal loads act in the ΔZ direction; nodal Mx/My act in 0X/0Y.
    model.loads.forEach((load) => {
        if (!isActiveLoadCase(load) || load.type !== "Nodal Load") return;
        const dof = dofMap[load.nodeId];
        if (dof && dof.z !== null) AJ[dof.z][0] += Number(load.Fz) || 0;
    });
    model.moments.forEach((moment) => {
        if (!isActiveMomentCase(moment) || moment.type !== "Nodal Moment") return;
        const dof = dofMap[moment.nodeId];
        if (!dof) return;
        if (dof.x !== null) AJ[dof.x][0] += Number(moment.Mx) || 0;
        if (dof.y !== null) AJ[dof.y][0] += Number(moment.My) || 0;
    });
    html += `</div>`;
```

```
    // Grid models do not currently accept temperature loads, so [AT] is zero.
    return { SJ, AE, AJ, AT: Matrix.zeros(AE.length, 1), html };
}

function solveStep3(ctx, step1) {
    if (isGridAnalysisMode()) return solveGridStep3(ctx, step1);
    const { dofMap, SJ, AE, AJ, heavyModel } = step1;
    // [AT] is kept separate from [AE] so Step 4 can assemble thermal actions
    // explicitly as [AC] = [AE] + [AJ] + [AD] + [AT].
    const AT = Matrix.zeros(AE.length, 1);
    let html = "";

    if (!heavyModel) {
        html += `<div class="calc-step"><h4>Step 3: Member Stiffness Matrices [SMS] and Member End Actions [AML]</h4>`;
    }

    model.elements.forEach((el) => {
        let n1 = model.nodes.find((n) => n.id === el.startNode);
        let n2 = model.nodes.find((n) => n.id === el.endNode);
        let tmpl = model.elementTemplates.find((t) => t.id === el.templateId);
        if (!n1 || !n2 || !tmpl) return;

        let dx = n2.x - n1.x,
            dy = n2.y - n1.y;
        let L = Math.hypot(dx, dy);
        if (L === 0) return;
        let cx = dx / L,
            cy = dy / L;

        let E = getElementEInternal(tmpl),
            A = tmpl.A,
            I = tmpl.I;
        let SM = Matrix.zeros(6, 6);
        SM[0][0] = (E * A) / L;
        SM[0][3] = (-E * A) / L;
        SM[1][1] = (12 * E * I) / Math.pow(L, 3);
        SM[1][2] = (6 * E * I) / Math.pow(L, 2);
        SM[1][4] = (-12 * E * I) / Math.pow(L, 3);
        SM[1][5] = (6 * E * I) / Math.pow(L, 2);
        SM[2][1] = (6 * E * I) / Math.pow(L, 2);
        SM[2][2] = (4 * E * I) / L;
        SM[2][4] = (-6 * E * I) / Math.pow(L, 2);
        SM[2][5] = (2 * E * I) / L;
        SM[3][0] = (-E * A) / L;
        SM[3][3] = (E * A) / L;
        SM[4][1] = (-12 * E * I) / Math.pow(L, 3);
        SM[4][2] = (-6 * E * I) / Math.pow(L, 2);
        SM[4][4] = (12 * E * I) / Math.pow(L, 3);
        SM[4][5] = (-6 * E * I) / Math.pow(L, 2);
        SM[5][1] = (6 * E * I) / Math.pow(L, 2);
        SM[5][2] = (2 * E * I) / L;
        SM[5][4] = (-6 * E * I) / Math.pow(L, 2);
        SM[5][5] = (4 * E * I) / L;

        let RT = Matrix.zeros(6, 6);
        RT[0][0] = cx;
        RT[0][1] = cy;
        RT[1][0] = -cy;
        RT[1][1] = cx;
        RT[2][2] = 1;
        RT[3][3] = cx;
        RT[3][4] = cy;
        RT[4][3] = -cy;
        RT[4][4] = cx;
        RT[5][5] = 1;

        let RTT = Matrix.transpose(RT);
        let SMS = Matrix.mul(Matrix.mul(RTT, SM), RT);

        el.L = L;
        el.cx = cx;
        el.cy = cy;
        el.SM = SM;
        el.RT = RT;
        el.RTT = RTT;
        el.SMS = SMS;

        // Map the member's six local positions to the global system.
        // The node numbering is global, but X/M may still be intentionally
        // absent for this particular member type.
        const elementDofMap = getElementGlobalDofIndices(el, n1, n2, dofMap);
        let idxs = elementDofMap.indices;
        let elLabels = idxs.map((idx) => idx === null ? "—" : idx + 1);

        // Keep the existing 6×6 local element formulation, but assemble only
        // the DOFs that belong to this member type.
        for (let r = 0; r < 6; r++) {
```

```
    if (idxs[r] === null) continue;
    for (let c = 0; c < 6; c++) {
        if (idxs[c] === null) continue;
        SJ[idxs[r]][idxs[c]] += SMS[r][c];
    }
}

el.AML = Matrix.zeros(6, 1);
el.localwx = 0;
el.localwy = 0;
el.localPointLoads = [];
el.localUDLs = [];
el.localPointMoments = [];

const activeLoadCase = model.loadCases.find((lc) => String(lc.id) === String(activeLoadCaseId));
const selfWeight = Number(activeLoadCase?.memberSelfWeights?.[String(el.id)]) || 0;
const includeSelfWeight = activeLoadCase?.type === "Self Weight";
if (selfWeight > 0 && includeSelfWeight) {
    const W_Y = -selfWeight;
    let wx = W_Y * cy,
        wy = W_Y * cx;
    el.localwx += wx;
    el.localwy += wy;

    el.AML[0][0] += (-wx * L) / 2;
    el.AML[1][0] += (-wy * L) / 2;
    el.AML[2][0] += (-wy * Math.pow(L, 2)) / 12;
    el.AML[3][0] += (-wx * L) / 2;
    el.AML[4][0] += (-wy * L) / 2;
    el.AML[5][0] += (wy * Math.pow(L, 2)) / 12;
}

// ------------------------------------------------------------
// MEMBER LOADS: GLOBAL/ANGLED -> LOCAL -> AML -> GLOBAL AMS
// ------------------------------------------------------------
model.loads.forEach((ld) => {
    if (!isActiveLoadCase(ld)) return;
    if (ld.elementId !== el.id) return;

    // ELEMENT POINT LOAD
    if (ld.type === "Element Point Load" || ld.type === "Element Load") {
        let a = Number(ld.distance) || 0;
        a = Math.max(0, Math.min(a, el.L));

        // IMPORTANT: convert the applied load to member axes FIRST.
        // AML is defined only in member-oriented axes by the lab manual.
        const local = inputForceToLocal(
            ld.Fx,
            ld.Fy,
            ld.axisSystem,
            ld.angle,
            el.cx,
            el.cy,
        );

        el.localPointLoads.push({
            a: a,
            Px: local.x,
            Py: local.y,
            inputAxis: ld.axisSystem || "Global",
        });

        addLocalPointLoadToAML(el.AML, local.x, local.y, a, el.L);
        return;
    }

    // ELEMENT UDL / LINEARLY VARYING DISTRIBUTED LOAD
    if (ld.type === "Element UDL") {
        let a = Number(ld.distance) || 0;
        let b =
            ld.distanceEnd !== undefined && ld.distanceEnd !== ""
                ? Number(ld.distanceEnd)
                : el.L;

        a = Math.max(0, Math.min(a, el.L));
        b = Math.max(0, Math.min(b, el.L));
        if (a >= b) return;

        const Fx1 = Number(ld.Fx) || 0;
        const Fy1 = Number(ld.Fy) || 0;
        const Fx2 = ld.FxEnd !== undefined && ld.FxEnd !== "" ? Number(ld.FxEnd) : Fx1;
        const Fy2 = ld.FyEnd !== undefined && ld.FyEnd !== "" ? Number(ld.FyEnd) : Fy1;

        // IMPORTANT: transform both UDL endpoint intensities to LOCAL axes
        // before calculating AML. For a member with fixed orientation,
        // a linearly varying global/angled UDL remains linearly varying
        // after this constant rotation.
```

```
        const local1 = inputForceToLocal(Fx1, Fy1, ld.axisSystem, ld.angle, el.cx, el.cy);
        const local2 = inputForceToLocal(Fx2, Fy2, ld.axisSystem, ld.angle, el.cx, el.cy);

        // Store GLOBAL components only for plotting / visualization.
        const global1 =
            ld.axisSystem === "Local"
                ? localForceToGlobal(local1.x, local1.y, el.cx, el.cy)
                : rotateForceToGlobal(Fx1, Fy1, ld.axisSystem, ld.angle);
        const global2 =
            ld.axisSystem === "Local"
                ? localForceToGlobal(local2.x, local2.y, el.cx, el.cy)
                : rotateForceToGlobal(Fx2, Fy2, ld.axisSystem, ld.angle);

        el.localUDLs.push({
            a: a,
            b: b,
            Fx1: global1.x,
            Fx2: global2.x,
            Fy1: global1.y,
            Fy2: global2.y,
            localFx1: local1.x,
            localFx2: local2.x,
            localFy1: local1.y,
            localFy2: local2.y,
            inputAxis: ld.axisSystem || "Global",
        });

        integrateLinearLocalUDLToAML(
            el.AML,
            local1.x,
            local1.y,
            local2.x,
            local2.y,
            a,
            b,
            el.L,
        );
    }
});

model.moments.forEach((m) => {
    if (!isActiveMomentCase(m)) return;
    // UI convention: +Mz is anticlockwise, matching the
    // structural stiffness convention.
    let Mz = Number(m.Mz) || 0;
    if (m.type === "Element Moment" && m.elementId === el.id) {
        let a = m.distance || 0;
        if (a < 0) a = 0;
        if (a > el.L) a = el.L;
        let b = el.L - a;
        el.AML[1][0] += (-6 * Mz * a * b) / Math.pow(el.L, 3);
        el.AML[2][0] += (Mz * b * (2 * a - b)) / Math.pow(el.L, 2);
        el.AML[4][0] += (6 * Mz * a * b) / Math.pow(el.L, 3);
        el.AML[5][0] += (Mz * a * (2 * b - a)) / Math.pow(el.L, 2);

        el.localPointMoments.push({ a: a, Mz: Mz });
    }
});

// Assemble ordinary member loads into [AE] exactly as before.
// Thermal loads get their own member-end vector and global vector [AT].
const mechanicalAMS = Matrix.mul(el.RTT, el.AML);
for (let r = 0; r < 6; r++) {
    if (idxs[r] === null) continue;
    AE[idxs[r]][0] -= mechanicalAMS[r][0];
}

const thermalAML = Matrix.zeros(6, 1);
const ignoreAxialThermal = getGlobalDofNumberingConfig().skipAxial;
model.loads.forEach((thermalLoad) => {
    if (!isActiveLoadCase(thermalLoad) || thermalLoad.elementId !== el.id) return;
    if (thermalLoad.type !== "Uniform Temperature Change" && thermalLoad.type !== "Temperature Gradient") return;
    const thermalActions = calculatePlaneThermalFixedEndActions(tmpl, thermalLoad, ignoreAxialThermal);
    thermalActions.forEach((value, index) => { thermalAML[index][0] += value; });
});

const thermalAMS = assembleThermalEquivalentNodalActions(AT, el.RTT, thermalAML, idxs);
el.thermalAML = thermalAML;
el.thermalAMS = thermalAMS;

// Keep the combined local end-action vector for Step 7 member-force
// recovery, while [AE] remains strictly mechanical and [AT] is thermal.
for (let r = 0; r < 6; r++) el.AML[r][0] += thermalAML[r][0];
const AMS = Matrix.mul(el.RTT, el.AML);
el.AMS = AMS;

if (!heavyModel) {
```

```
            html += `<h5>Element ${escapeHtml(el.id)}</h5>`;
        }
        if (!heavyModel) {
            html += matrixEqHTML(
                [
                    { m: el.SM, t: "[SM] (Local)", top: elLabels, right: elLabels },
                    { m: el.RT, t: "[Rt] (Rotational)", top: elLabels, right: elLabels },
                ],
                [" "],
            );
        }
        if (!heavyModel) {
            html += matrixEqHTML(
                [
                    { m: el.SMS, t: "[Sms] = [Rt]■[SM][Rt]", top: elLabels, right: elLabels },
                    { m: el.AML, t: "[AML] (Local Actions)", right: elLabels },
                    { m: el.RTT, t: "[RT]■", right: elLabels },
                    { m: el.AMS, t: "[AMS] = [RT]■[AML]", right: elLabels },
                ],
                [" ", "x", "="],
            );
        }
    });

    model.loads.forEach((ld) => {
        if (!isActiveLoadCase(ld)) return;
        let Fx = ld.Fx || 0,
            Fy = ld.Fy || 0;
        if (ld.type === "Nodal Load") {
            let d = dofMap[ld.nodeId];
            if (d) {
                // Use the single authoritative rotation helper (also used for
                // member loads) instead of a duplicated inline formula.
                const g = rotateForceToGlobal(Fx, Fy, ld.axisSystem, ld.angle);
                if (d.x !== null) AJ[d.x][0] += g.x;
                if (d.y !== null) AJ[d.y][0] += g.y;
            }
        }
    });

    model.moments.forEach((m) => {
        if (!isActiveMomentCase(m)) return;
        // UI convention: +Mz is anticlockwise, matching the global
        // stiffness formulation.
        let Mz = Number(m.Mz) || 0;
        if (m.type === "Nodal Moment") {
            let d = dofMap[m.nodeId];
            if (d && d.z !== null) AJ[d.z][0] += Mz;
        }
    });

    if (!heavyModel) {
        html += `</div>`;
    }

    return { SJ, AE, AJ, AT, html };
}

// Assemble the negative transformed thermal fixed-end actions into [AT].
// This mirrors the sign used for equivalent member-load actions in [AE] but
// stores thermal contributions separately to avoid double counting.
function assembleThermalEquivalentNodalActions(AT, RTT, thermalAML, indices) {
    const thermalAMS = Matrix.mul(RTT, thermalAML);
    for (let r = 0; r < indices.length; r++) {
        if (indices[r] === null) continue;
        AT[indices[r]][0] -= thermalAMS[r][0];
    }
    return thermalAMS;
}

// Return the six local fixed-end actions produced by a thermal input. The
// established solver convention is AML + SM·RT·DS for recovered member ends,
// and the global equivalent nodal vector is assembled as -RT■·AML.
function calculatePlaneThermalFixedEndActions(template, load, ignoreAxial = false) {
    const actions = [0, 0, 0, 0, 0, 0];
    if (!template || !load) return actions;
    const E = getElementEInternal(template);
    const A = Number(template.A) || 0;
    const I = Number(template.I) || 0;
    const alpha = Number(template.alpha) || 0;
    if (!(E > 0) || !(alpha > 0)) return actions;

    if (load.type === "Uniform Temperature Change") {
        // Fully restrained axial thermal action magnitude: EAαΔT.
        // Heating (ΔT > 0) produces compression; the end-action signs below
        // match the element's local axial DOF convention.
        const deltaT = Number(load.temperatureDelta);
```

```
        if (!ignoreAxial && A > 0 && Number.isFinite(deltaT)) {
            const axialAction = E * A * alpha * deltaT;
            actions[0] = axialAction;
            actions[3] = -axialAction;
        }
    } else if (load.type === "Temperature Gradient") {
        // Face 1 is the local –Y face and Face 2 the local +Y face, so the
        // thermal curvature is α(T1–T2)/d. Section depth is measured in metres.
        const T1 = Number(load.temperatureT1);
        const T2 = Number(load.temperatureT2);
        const depth = Number(template.thermalDepth);
        if (I > 0 && depth > 0 && Number.isFinite(T1) && Number.isFinite(T2)) {
            const thermalMoment = E * I * alpha * (T1 - T2) / depth;
            actions[2] = thermalMoment;
            actions[5] = -thermalMoment;
        }
    }
    return actions;
}

function renderStep3(step3) {
    return step3.html;
}
```

34. modules/results/step-4/step-4.js

```
/* ================================================================================
 * step-4.js — Results part: Step 4
 * Step 4 — Global stiffness assembly
 * Adds its menu button + panel to the Results frame, then defines its solver / render code.
 * ================================================================================ */

/* ---- Markup ---- */
mountModuleHTML(".results-detail-menu", "beforeend", `<button type="button" class="results-detail-btn" data-result-panel="result-step-4" onclick="showResultDetail('result-step-4', this)">4.
Global Assembly</button>
`);
mountModuleHTML(".results-detail-body", "beforeend", `
<div class="result-detail-panel" id="result-step-4">
    <div class="calc-step-slot" data-step-number="4"></div>
</div>
`);

/* ---- Logic ---- */
// Prescribed movements are converted to equivalent nodal actions using the
// already-assembled global stiffness matrix. [AD] = -[SJ][Dp], where [Dp]
// contains imposed support movements at restrained DOFs and zero elsewhere.
// The solution then uses zero incremental movement at those supports; the
// actual displacement vector is reconstructed as D = Dp + D_increment.
function buildPrescribedMovementActionVector(SJ, prescribedDisp) {
    const totalDof = SJ.length;
    const Dp = Matrix.zeros(totalDof, 1);
    for (let i = 0; i < totalDof; i++) {
        const raw = prescribedDisp?.[i];
        const value = raw === undefined || raw === null || raw === "" ? 0 : Number(raw);
        Dp[i][0] = Number.isFinite(value) ? value : 0;
    }

    const stiffnessTimesMovement = Matrix.mul(SJ, Dp);
    const AD = Matrix.zeros(totalDof, 1);
    for (let i = 0; i < totalDof; i++) {
        AD[i][0] = -(Number(stiffnessTimesMovement[i]?.[0]) || 0);
    }
    return { Dp, AD };
}

// Step 4: Assemble [AC] = [AE] + [AJ] + [AD] + [AT], display the assembled
// system, then guard against zero-stiffness (unstable) DoFs before partitioning.
function assembleGlobalLoadVector(AE, AJ, AD, AT) {
    const physicalAC = Matrix.add(AE, AJ);
    let AC = Matrix.add(physicalAC, AD);
    AC = Matrix.add(AC, AT);
    return { AC, physicalAC };
}

function solveStep4(ctx, step1, step3) {
    const { dofMap, dofInfo, totalDof, isRestrained, prescribedDisp, heavyModel } = step1;
    const { SJ, AE, AJ } = step3;
    const AT = step3.AT || Matrix.zeros(AE.length, 1);
    const activeCaseLabel = ctx.activeCaseLabel;

    let globalLabels = Array.from({ length: totalDof }, (_, i) => i + 1);
    const { Dp, AD } = buildPrescribedMovementActionVector(SJ, prescribedDisp);
    const { AC, physicalAC } = assembleGlobalLoadVector(AE, AJ, AD, AT);

    let html = "";
    if (!heavyModel) {
        html += `<div class="calc-step"><h4>Step 4: Assembled Stiffness Matrix [SJ] & Load Vector [AC]</h4><p><strong>Active analysis:</strong> ${escapeHtml(activeCaseLabel)}</p>`;
    }
    if (!heavyModel) {
        html += matrixEqHTML(
            [{ m: SJ, t: "[SJ] (Global Assembly)", top: globalLabels, right: globalLabels }],
            [],
        );
    }
    if (!heavyModel) {
        html += `<p>[AD] = −[SJ][Dp]. Prescribed translations are entered in mm and converted to m before solving; rotations remain in rad. Lengths are m, I is m■, E is kN/m², and α is 1/°C.
 Therefore [AD] entries are kN at translational DOFs and kN·m at rotational DOFs. [AT] contains the equivalent global nodal actions from member temperature effects; its translation entries
are kN and rotation entries are kN·m. The assembled load vector is [AC] = [AE] + [AJ] + [AD] + [AT].</p>`;
        html += matrixEqHTML(
            [
                { m: AE, t: "[AE]", right: globalLabels },
                { m: AJ, t: "[AJ]", right: globalLabels },
                { m: AD, t: "[AD]", right: globalLabels },
                { m: AT, t: "[AT] (Thermal Actions)", right: globalLabels },
                { m: AC, t: "[AC]", right: globalLabels },
            ],
            ["+", "+", "+", "="],
        );
    }
    if (!heavyModel) {
        html += `</div>`;
```

```
    }

    // --- ZERO-STIFFNESS DOF WARNING ---
    // Never silently change the mathematical model by auto-restraining a DOF.
    // First determine which joint DOFs are structurally relevant. Intentional
    // dummy DOFs are not part of the stiffness system: UX for a pure beam node
    // and RZ for a pure truss node are therefore excluded, not restrained.
    const structurallyRelevantDof = new Array(totalDof).fill(false);
    model.nodes.forEach((node) => {
        const d = dofMap[node.id];
        const relevance = getNodeStructuralDofRelevance(node.id);
        if (d.x !== null) structurallyRelevantDof[d.x] = !!relevance.ux;
        if (d.y !== null) structurallyRelevantDof[d.y] = !!relevance.uy;
        if (d.z !== null) structurallyRelevantDof[d.z] = !!relevance.rz;
    });

    const zeroStiffnessDofs = [];
    for (let i = 0; i < totalDof; i++) {
        // Intentional beam/truss dummy DOFs are excluded from this warning.
        if (!structurallyRelevantDof[i]) continue;
        if (!isNumericallyZeroStiffnessDof(SJ, i) || isRestrained[i]) continue;

        const info = dofInfo?.[i] || { nodeId: `DOF ${i + 1}`, dofLabel: "DOF" };

        zeroStiffnessDofs.push({
            nodeId: info.nodeId,
            dofLabel: info.dofLabel,
            hasLoad: Math.abs(AC[i][0]) >= 1e-10,
        });
    }

    if (zeroStiffnessDofs.length > 0) {
        const detail = zeroStiffnessDofs
            .map((d) => `${d.nodeId} ${d.dofLabel}${d.hasLoad ? " (load applied)" : ""}`)
            .join(", ");
        alert(
            `Warning: Zero-stiffness DOF detected at ${detail}.\n\n` +
            `The solver will NOT automatically restrain these DOFs.\n` +
            `Check the model connectivity, element properties, and support conditions before running the analysis again.`,
        );
        return null;
    }

    return { AC, physicalAC, AD, Dp, globalLabels, structurallyRelevantDof, html };
}

function renderStep4(step4) {
    return step4.html;
}
```

35. modules/results/step-5/step-5.js

```
/* ================================================================================
 * step-5.js — Results part: Step 5
 * Step 5 — Boundary conditions
 * Adds its menu button + panel to the Results frame, then defines its solver / render code.
 * ================================================================================ */

/* ---- Markup ---- */
mountModuleHTML(".results-detail-menu", "beforeend", `<button type="button" class="results-detail-btn" data-result-panel="result-step-5" onclick="showResultDetail('result-step-5', this)">5.
Boundary Conditions</button>
`);
mountModuleHTML(".results-detail-body", "beforeend", `
<div class="result-detail-panel" id="result-step-5">
    <div class="calc-step-slot" data-step-number="5"></div>
</div>
`);

/* ---- Logic ---- */
// Step 5: Incorporate boundary conditions exactly as prescribed in the LPW-3 manual.
// Free DoFs are placed first (ascending order), followed by restrained DoFs
// (ascending order). The assembled stiffness matrix [SJ] and load vector [AC]
// are then partitioned into the four sub-matrices/vectors required by Step 6.
function solveStep5(ctx, step1, step4) {
    const { totalDof, isRestrained, prescribedDisp, SJ, heavyModel } = step1;
    const { AC, structurallyRelevantDof } = step4;

    // The manual numbers only the actual structural joint displacements.
    // Keep the existing model's beam/truss dummy-DoF handling intact by using
    // the structural-relevance mask created in Step 4.
    const activeIndices = [];
    for (let i = 0; i < totalDof; i++) {
        if (structurallyRelevantDof[i]) activeIndices.push(i);
    }

    const freeIndices = activeIndices.filter((i) => !isRestrained[i]);
    const restIndices = activeIndices.filter((i) => isRestrained[i]);

    const numF = freeIndices.length;
    const numR = restIndices.length;

    // Rearranged numbering required by the manual: free DoFs first, then
    // restrained DoFs, with each group remaining in ascending global order.
    const reorderedIndices = [...freeIndices, ...restIndices];
    const reorderedLabels = reorderedIndices.map((i) => i + 1);
    const freeLabels = freeIndices.map((i) => i + 1);
    const restLabels = restIndices.map((i) => i + 1);

    // Build [SJ]R and [AC]R from the reordered global system:
    // [SJ]R = [ SFF  SFR ]
    //         [ SRF  SRR ]
    // [AC]R = [ AFC ]
    //         [ ARC ]
    const SJR = Matrix.zeros(reorderedIndices.length, reorderedIndices.length);
    const ACR = Matrix.zeros(reorderedIndices.length, 1);

    reorderedIndices.forEach((ri, r) => {
        ACR[r][0] = AC[ri][0];
        reorderedIndices.forEach((ci, c) => {
            SJR[r][c] = SJ[ri][ci];
        });
    });

    const SFF = Matrix.zeros(numF, numF);
    const SFR = Matrix.zeros(numF, numR);
    const SRF = Matrix.zeros(numR, numF);
    const SRR = Matrix.zeros(numR, numR);
    const AFC = Matrix.zeros(numF, 1);
    const ARC = Matrix.zeros(numR, 1);
    // DpR stores the actual imposed support values for reporting. DR is the
    // incremental restrained displacement vector used in the transformed
    // system; it is zero because its effect is already in AD from Step 4.
    const prescribedDR = Matrix.zeros(numR, 1);
    const DR = Matrix.zeros(numR, 1);

    for (let i = 0; i < numF; i++) {
        AFC[i][0] = ACR[i][0];
        for (let j = 0; j < numF; j++) SFF[i][j] = SJR[i][j];
        for (let j = 0; j < numR; j++) SFR[i][j] = SJR[i][numF + j];
    }

    for (let i = 0; i < numR; i++) {
        ARC[i][0] = ACR[numF + i][0];
        prescribedDR[i][0] = Number.isFinite(Number(prescribedDisp[restIndices[i]]))
            ? Number(prescribedDisp[restIndices[i]])
            : 0;
        DR[i][0] = 0;
```

```
        for (let j = 0; j < numF; j++) SRF[i][j] = SJR[numF + i][j];
        for (let j = 0; j < numR; j++) SRR[i][j] = SJR[numF + i][numF + j];
    }

    let html = `<div class="calc-step"><h4>Step 5: Incorporate Boundary Conditions</h4>`;
    html += `<p>[SJ] and [AC] are rearranged so that free degrees of freedom are listed first, followed by restrained degrees of freedom. Prescribed movements have already been converted to
equivalent actions [AD] in Step 4. The solve therefore uses zero incremental restrained displacement [DR] = 0; the actual prescribed movement vector [Dp] is restored in Step 6.</p>`;

    if (reorderedIndices.length > 0) {
        html += `<h5>Rearranged System [SJR] and [AC]R</h5>`;
        if (!heavyModel) {
            html += matrixEqHTML(
                [
                    { m: SJR, t: "[SJ]R", top: reorderedLabels, right: reorderedLabels },
                    { m: ACR, t: "[AC]R", right: reorderedLabels },
                ],
                [" "]
            );
        } else {
            html += `<p><strong>[SJ]R:</strong> ${SJR.length} × ${SJR.length}    <strong>[AC]R:</strong> ${ACR.length} × 1</p>`;
        }
    }

    html += `<h5>Free / Restrained Partition</h5>`;
    if (numF > 0) {
        if (!heavyModel) {
            html += matrixEqHTML(
                [
                    { m: SFF, t: "[SFF]", top: freeLabels, right: freeLabels },
                    { m: SFR, t: "[SFR]", top: restLabels, right: freeLabels },
                    { m: AFC, t: "[AFC]", right: freeLabels },
                ],
                [" ", " "]
            );
        } else {
            html += `<p><strong>[SFF]:</strong> ${numF} × ${numF}    <strong>[SFR]:</strong> ${numF} × ${numR}    <strong>[AFC]:</strong> ${numF} × 1</p>`;
        }
    } else {
        html += `<p>No free structural degrees of freedom are present.</p>`;
    }

    if (numR > 0) {
        html += `<p><strong>Restrained DoFs:</strong> ${restLabels.join(", ")}    <strong>[Dp]R:</strong> prescribed support movements; <strong>[DR]:</strong> zero incremental
values.</p>`;
        if (!heavyModel) {
            html += matrixEqHTML(
                [
                    { m: SRF, t: "[SRF]", top: freeLabels, right: restLabels },
                    { m: SRR, t: "[SRR]", top: restLabels, right: restLabels },
                    { m: ARC, t: "[ARC]", right: restLabels },
                    { m: prescribedDR, t: "[Dp]R (actual prescribed)", right: restLabels },
                    { m: DR, t: "[DR] (incremental = 0)", right: restLabels },
                ],
                [" ", " ", " ", " ", " "]
            );
        }
    } else {
        html += `<p><strong>Restrained DoFs:</strong> None.</p>`;
    }

    html += `</div>`;

    return {
        freeIndices,
        restIndices,
        reorderedIndices,
        numF,
        numR,
        SJR,
        ACR,
        SFF,
        SFR,
        SRF,
        SRR,
        AFC,
        ARC,
        prescribedDR,
        DR,
        freeLabels,
        restLabels,
        html,
    };
}

function renderStep5(step5) {
    return step5.html;
}
```

36. modules/results/step-6/step-6.js

```
/* ========================================================================
 * step-6.js — Results part: Step 6
 * Step 6 — Displacements
 * Adds its menu button + panel to the Results frame, then defines its solver / render code.
 * ======================================================================== */

/* ---- Markup ---- */
mountModuleHTML(".results-detail-menu", "beforeend", `<button type="button" class="results-detail-btn" data-result-panel="result-step-6" onclick="showResultDetail('result-step-6', this)">6.
Displacements</button>
`);
mountModuleHTML(".results-detail-body", "beforeend", `
<div class="result-detail-panel" id="result-step-6">
    <div class="calc-step-slot" data-step-number="6"></div>
</div>
`);

/* ---- Logic ---- */
// Reconstruct actual absolute nodal displacements from solved increments and
// prescribed support movements. For the equivalent-action formulation the
// restrained incremental displacements are zero, and [Dp] is added back here.
function reconstructActualDisplacementVector(totalDof, prescribedDisp, freeIndices, DF) {
    const D = Matrix.zeros(totalDof, 1);
    for (let i = 0; i < totalDof; i++) {
        const raw = prescribedDisp?.[i];
        const value = raw === undefined || raw === null || raw === "" ? 0 : Number(raw);
        D[i][0] = Number.isFinite(value) ? value : 0;
    }
    freeIndices.forEach((freeIndex, i) => {
        D[freeIndex][0] += Number(DF[i]?.[0]) || 0;
    });
    return D;
}

// Step 6: The prescribed-movement effect is already contained in [AD] and [AC].
// Solve the incremental free displacements as [DF] = [SFF]^-1 [AFC], then
// restore the imposed movements to recover the actual displacement vector [D].
function solveStep6(ctx, step1, step5) {
    const { totalDof, dofMap, prescribedDisp, heavyModel } = step1;
    const { numF, numR, SFF, AFC, DR, freeIndices, restIndices, freeLabels, restLabels } = step5;

    let html = `<div class="calc-step"><h4>Step 6: Calculate Displacements</h4>`;
    let DF = Matrix.zeros(numF, 1);
    let RHS = AFC.map((row) => [Number(row[0]) || 0]);
    let SFF_inv = null;
    const SFR_DR = Matrix.zeros(numF, 1); // [DR] is identically zero in the incremental system.

    if (numF === 0) {
        html += `<p>No free degrees of freedom exist. The actual nodal movements are the prescribed values [Dp].</p>`;
    } else {
        // Because [AD] = -[SJ][Dp] has already been assembled, the incremental
        // solution uses [AFC] directly; subtracting [SFR][Dp] again would double-count settlement.
        SFF_inv = Matrix.invert(SFF);
        if (!SFF_inv) {
            alert("Step 6 failed: [SFF] is singular. The structure is unstable or the boundary conditions are insufficient.");
            return null;
        }

        DF = Matrix.mul(SFF_inv, RHS);

        html += `<p>[DF] = [SFF]■⁻¹[AFC]. The equivalent movement actions are already included in [AC]; actual displacements are reconstructed as [D] = [Dp] + [D_increment].</p>`;
        if (!heavyModel) {
            html += matrixEqHTML(
                [
                    { m: SFF_inv, t: "[SFF]■⁻¹", top: freeLabels, right: freeLabels },
                    { m: RHS, t: "[AFC]", right: freeLabels },
                    { m: DF, t: "[DF]", right: freeLabels },
                ],
                ["×", "="]
            );
        } else {
            html += `<p><strong>[DF]:</strong> ${numF} × 1 incremental displacement vector calculated successfully.</p>`;
        }
    }

    // [Dp] is nonzero only at prescribed/ restrained DOFs. [DF] is added to
    // the free DOFs; the resulting [D] is the actual displacement used in [AM].
    const D = reconstructActualDisplacementVector(totalDof, prescribedDisp, freeIndices, DF);

    // Persist solved nodal displacements for the Results displacement diagram.
    model.nodeDisplacements = {};
    model.nodes.forEach((node) => {
        const d = dofMap[node.id];
        model.nodeDisplacements[node.id] = isGridAnalysisMode()
            ? {
                // Keep Grid's three generalized joint DOFs distinct from the
```

```
                // plane solver's UX/UY/RZ fields used by the shared diagrams.
                ux: 0, uy: 0, rz: 0,
                thetaX: d.x === null ? 0 : D[d.x][0],
                thetaY: d.y === null ? 0 : D[d.y][0],
                deltaZ: d.z === null ? 0 : D[d.z][0],
            }
            : {
                ux: d.x === null ? 0 : D[d.x][0],
                uy: d.y === null ? 0 : D[d.y][0],
                rz: d.z === null ? 0 : D[d.z][0],
            };
    });

    if (numR > 0) {
        html += `<p><strong>Restrained displacements:</strong> ${restLabels.join(", ")} use zero incremental [DR]; the actual prescribed movement values are restored from [Dp] in [D].</p>`;
    }
    html += `</div>`;

    return { DF, D, SFF_inv, RHS, SFR_DR, html };
}

function renderStep6(step6) {
    return step6.html;
}
```

37. modules/results/step-7/step-7.js

```
/* ========================================================================
 * step-7.js — Results part: Step 7
 * Step 7 — Reactions & member end forces
 * Adds its menu button + panel to the Results frame, then defines its solver / render code.
 * ======================================================================== */

/* ---- Markup ---- */
mountModuleHTML(".results-detail-menu", "beforeend", `<button type="button" class="results-detail-btn" data-result-panel="result-step-7" onclick="showResultDetail('result-step-7', this)">7.
Reactions &amp;amp; Member Actions</button>
`);
mountModuleHTML(".results-detail-body", "beforeend", `
<div class="result-detail-panel" id="result-step-7">
    <div class="calc-step-slot" data-step-number="7"></div>
</div>
`);

/* ---- Logic ---- */
// Step 7: Recover support reactions from the incremental system.
// Since prescribed-movement actions [AD] are included in [AC] and restrained
// incremental displacements [DR] are zero, [AR] = -[ARC] + [SRF][DF].
// Member end actions always use actual displacements restored in [D].
function solveStep7(ctx, step1, step4, step5, step6) {
    const { dofMap, heavyModel } = step1;
    const { AC } = step4;
    const { numF, numR, SRF, ARC, restIndices, freeLabels, restLabels } = step5;
    const { DF, D } = step6;

    let SRF_DF = Matrix.zeros(numR, 1);
    if (numF > 0 && numR > 0) SRF_DF = Matrix.mul(SRF, DF);

    let negARC = Matrix.zeros(numR, 1);
    for (let i = 0; i < numR; i++) negARC[i][0] = -ARC[i][0];

    let AR = Matrix.zeros(numR, 1);
    for (let i = 0; i < numR; i++) {
        AR[i][0] = -ARC[i][0] + SRF_DF[i][0];
    }

    let html = "";
    if (!heavyModel) {
        html += `<div class="calc-step"><h4>Step 7: Support Reactions [AR] and Member End Actions [AM]</h4>`;
    }
    if (!heavyModel) {
        html += `<h5>Support Reactions [AR]</h5>`;
    }
    if (numR > 0) {
        if (!heavyModel) {
            html += numF > 0
            ? `<p>[AR] = -[ARC] + [SRF][DF], because incremental [DR] = 0.</p>`
            : `<p>[AR] = -[ARC], because no free degrees of freedom exist.</p>`;
        }

        if (numF > 0) {
            if (!heavyModel) {
                html += matrixEqHTML(
                    [
                        { m: negARC, t: "-[ARC]", right: restLabels },
                        { m: SRF, t: "[SRF]", top: freeLabels, right: restLabels },
                        { m: DF, t: "[DF]", right: freeLabels },
                        { m: AR, t: "[AR]", right: restLabels },
                    ],
                    ["+", "×", "="],
                );
            }
        } else {
            if (!heavyModel) {
                html += matrixEqHTML(
                    [
                        { m: negARC, t: "-[ARC]", right: restLabels },
                        { m: AR, t: "[AR]", right: restLabels },
                    ],
                    ["="],
                );
            }
        }
    } else {
        if (!heavyModel) {
            html += `<p>No restrained degrees of freedom.</p>`;
        }
    }

    if (!heavyModel) {
        html += `<h5>Member End Actions [AM]</h5>`;
    }
    if (!heavyModel) {
```

```
        html += `<p>[AM]■ = [AML]■ + [SM]■[RT]■[DS]■</p>`;
    }

    // Validate global static equilibrium before the orchestrator marks results as valid.
    const equilibriumResult = updateEquilibriumCheck(AC, dofMap, model.nodes, restIndices, AR);

    // Update UI Tables
    let resDispHTML = "";
    model.nodes.forEach((n) => {
        let d = dofMap[n.id];
        resDispHTML += `<tr><td>${escapeHtml(n.id)}</td><td>${d.x === null ? "—" : formatNum(D[d.x][0])}</td><td>${d.y === null ? "—" : formatNum(D[d.y][0])}</td><td>${d.z === null ? "—" :
formatNum(D[d.z][0])}</td></tr>`;
    });

    let resReactHTML = "";
    model.supportAssignments.forEach((sa) => {
        let d = dofMap[sa.nodeId];
        let rx = 0,
            ry = 0,
            rz = 0;
        restIndices.forEach((ri, i) => {
            if (d.x !== null && ri === d.x) rx = AR[i][0];
            if (d.y !== null && ri === d.y) ry = AR[i][0];
            if (d.z !== null && ri === d.z) rz = AR[i][0];
        });
        sa.rx = rx;
        sa.ry = ry;
        sa.rz = rz; // Attach reactions for SVG diagram
        const localReaction = isGridAnalysisMode() ? null : getReactionLocalComponents(sa);
        const localX = localReaction ? formatNum(localReaction.x) : "—";
        const localY = localReaction ? formatNum(localReaction.y) : "—";
        const localZ = localReaction ? formatNum(rz) : "—";
        resReactHTML += `<tr><td>${escapeHtml(sa.nodeId)}</td><td>${formatNum(rx)}</td><td>${formatNum(ry)}</td><td>${formatNum(rz)}</td><td>${localX}</td><td>${localY}</td><td>${localZ}</
td></tr>`;
    });

    let resForceHTML = "";
    const elementResults = [];
    model.elements.forEach((el) => {
        let n1 = model.nodes.find((n) => n.id === el.startNode);
        let n2 = model.nodes.find((n) => n.id === el.endNode);
        const elementDofMap = getElementGlobalDofIndices(el, n1, n2, dofMap);
        let idxs = elementDofMap.indices;
        let elLabels = idxs.map((idx) => idx === null ? "—" : idx + 1);

        let DS = Matrix.zeros(6, 1);
        for (let r = 0; r < 6; r++) DS[r][0] = idxs[r] === null ? 0 : D[idxs[r]][0];

        let RT_DS = Matrix.mul(el.RT, DS);
        let SM_RT_DS = Matrix.mul(el.SM, RT_DS);
        let AM = Matrix.add(el.AML, SM_RT_DS);
        if (isGridAnalysisMode()) el.gridEndActions = AM;

        // Step 7 [AM] direct formula injection
        if (!heavyModel) {
            html += `<h6 style="margin-top:10px; color:#475569;">Element ${escapeHtml(el.id)}</h6>`;
        }
        if (!heavyModel) {
            html += matrixEqHTML(
                [
                    { m: el.AML, t: `[AML]_${el.id}`, right: elLabels },
                    { m: el.SM, t: `[SM]_${el.id}`, top: elLabels, right: elLabels },
                    { m: el.RT, t: `[RT]_${el.id}`, top: elLabels, right: elLabels },
                    { m: DS, t: `[DS]_${el.id}`, right: elLabels },
                    { m: AM, t: `[AM]_${el.id}`, right: elLabels },
                ],
                ["+", "x", "x", "="],
            );
        }

        if (isGridAnalysisMode()) {
            resForceHTML += `<tr><td rowspan="2">${escapeHtml(el.id)}</td><td>${escapeHtml(el.startNode)} (Start)</td><td>${formatNum(AM[0][0])}</td><td>${formatNum(AM[1][0])}</td><td>${
formatNum(AM[2][0])}</td></tr>
                            <tr><td>${escapeHtml(el.endNode)} (End)</td><td>${formatNum(AM[3][0])}</td><td>${formatNum(AM[4][0])}</td><td>${formatNum(AM[5][0])}</td></tr>`;
        } else {
            resForceHTML += `<tr><td rowspan="2">${escapeHtml(el.id)}</td><td>${escapeHtml(el.startNode)} (Start)</td><td>${formatNum(AM[0][0])}</td><td>${formatNum(AM[1][0])}</td><td>${
formatNum(AM[2][0])}</td></tr>
                            <tr><td>${escapeHtml(el.endNode)} (End)</td><td>${formatNum(AM[3][0])}</td><td>${formatNum(AM[4][0])}</td><td>${formatNum(AM[5][0])}</td></tr>`;
        }

        elementResults.push({ el, AM });
    });
    if (!heavyModel) {
        html += `</div>`;
    } // Close step 7 container here

    return { AR, resDispHTML, resReactHTML, resForceHTML, elementResults, equilibriumResult, html };
```

```
}

function renderStep7(step7) {
    return step7.html;
}
```

38. modules/results/step-8/step-8.css

```
/* ========================================================================
 * step-8.css
 * Styles specific to Results > Step 8 (final result tables).
 * ======================================================================== */

.final-results-grid {
    display: grid;
    gap: 18px;
}

.final-result-block {
    border: 1px solid var(--border);
    border-radius: 12px;
    background: white;
    overflow: hidden;
}

.final-result-title {
    padding: 14px 16px;
    border-bottom: 1px solid var(--border);
    background: #f8fafc;
    font-size: 14px;
    font-weight: 800;
    color: #1f2937;
}

.final-result-block .table-wrap {
    padding: 0;
}
```

39. modules/results/step-8/step-8.js

```
/* ============================================================================
 * step-8.js — Results part: Step 8
 * Step 8 — Final results (tables of displacements, reactions, member end forces)
 * Adds its menu button + panel to the Results frame, then defines its solver / render code.
 * ============================================================================ */

/* ---- Markup ---- */
mountModuleHTML(".results-detail-menu", "beforeend", `<button type="button" class="results-detail-btn" data-result-panel="result-final" onclick="showResultDetail('result-final', this)">8.
Final Results</button>
`);
mountModuleHTML(".results-detail-body", "beforeend", `
<div class="result-detail-panel" id="result-final">
    <div class="final-results-grid">
        <div class="final-result-block">
            <div class="final-result-title">Nodal Displacements</div>
            <div class="table-wrap">
                <table>
                    <thead><tr><th>Node</th><th id="resDispHdr1">UX (m)</th><th id="resDispHdr2">UY (m)</th><th id="resDispHdr3">RZ (rad)</th></tr></thead>
                    <tbody id="resDispTable"></tbody>
                </table>
            </div>
        </div>

        <div class="final-result-block">
            <div class="final-result-title">Support Reactions</div>
            <div class="table-wrap">
                <table>
                    <thead><tr><th rowspan="2">Node</th><th colspan="3" id="resReactionGlobalHeading">Global Axes</th><th colspan="3" id="resReactionLocalHeading">Local Axes</th></
tr><tr><th id="resReactionHdr1">Fx (kN)</th><th id="resReactionHdr2">Fy (kN)</th><th id="resReactionHdr3">Mz (kN·m)</th><th id="resReactionHdr4">Fx′ (kN)</th><th id="resReactionHdr5">Fy′ (
kN)</th><th id="resReactionHdr6">Mz′ (kN·m)</th></tr></thead>
                    <tbody id="resReactionsTable"></tbody>
                </table>
            </div>
        </div>

        <div class="final-result-block">
            <div class="final-result-title">Member End Forces (Local Axes)</div>
            <div class="table-wrap">
                <table>
                    <thead><tr><th>Element</th><th>Node</th><th id="resForceHdr1">Axial (kN)</th><th id="resForceHdr2">Shear (kN)</th><th id="resForceHdr3">Moment (kN·m)</th></tr></thead>
                    <tbody id="resForcesTable"></tbody>
                </table>
            </div>
        </div>
    </div>
</div>
`);

/* ---- Logic ---- */
// Step 8: Prepare per-element plot data (used by the AFD/SFD/BMD/Displacement
// diagrams) from the member end actions [AM] recovered in Step 7. This step
// produces no visible calc-report section, matching the original analysis
// routine, which never rendered a distinct "Step 8" block.
function solveStep8(ctx, step7) {
    const grid = isGridAnalysisMode();
    step7.elementResults.forEach(({ el, AM }) => {
        const gridPointMoments = (el.localPointMoments || []).map((moment) => ({ a: moment.a, Mx: Number(moment.Mx) || 0, My: Number(moment.My) || 0 }));
        el.plotData = {
            grid,
            L: el.L,
            cx: el.cx,
            cy: el.cy,
            wx: grid ? 0 : el.localwx,
            wy: grid ? 0 : el.localwy,
            pointLoads: el.localPointLoads || [],
            localUDLs: el.localUDLs || [],
            localPointMoments: el.localPointMoments || [],
            gridUDLs: grid ? (el.localUDLs || []).map((udl) => ({ a: udl.a, b: udl.b, q1: Number(udl.localFy1) || 0, q2: Number(udl.localFy2) || 0 })) : [],
            gridPointLoads: grid ? (el.localPointLoads || []).map((load) => ({ a: load.a, Fz: Number(load.Pz) || Number(load.Py) || 0 })) : [],
            gridPointTorsions: grid ? gridPointMoments.map((moment) => ({ a: moment.a, Mx: moment.Mx })) : [],
            gridPointMoments: grid ? gridPointMoments : [],
            fx1: -AM[0][0],
            fy1: grid ? AM[2][0] : AM[1][0],
            mz1: grid ? AM[1][0] : AM[2][0],
        };
    });
    return { count: step7.elementResults.length };
}

function renderStep8() {
    return "";
}
```

40. modules/supports/supports.js

```
/* ========================================================================
 * supports.js — Supports page module
 * Supports page — support templates (fixed / roller / hinged / custom) and support assignments.
 * Contains this page's markup (injected into .container) and its logic.
 * ======================================================================== */

/* ---- Markup ---- */
mountModuleHTML(".container", "beforeend", `
<!-- SUPPORTS -->
<section class="section" id="supports">
    <div class="card">
        <div class="card-header">
            <div><h3>Built-in Support Templates</h3><p>These three standard support templates are already available. No creation is required.</p></div>
        </div>
        <div class="card-body">
            <div class="grid grid-3">
                <div class="restraint-box">
                    <div class="restraint-title">FIXED SUPPORT</div>
                    <p id="supportFixedDescription" style="font-size:12px; color:#475569; line-height:1.5; margin-bottom:10px;">RFx + RFy + RFz + RMx + RMy + RMz restrained</p>
                    <button type="button" class="btn btn-primary" onclick="selectBuiltInSupport('FIXED')">Use Fixed</button>
                </div>
                <div class="restraint-box">
                    <div class="restraint-title">ROLLER SUPPORT</div>
                    <p id="supportRollerDescription" style="font-size:12px; color:#475569; line-height:1.5; margin-bottom:10px;">RFy restrained only<br><span style="color:#64748b;">All
other 2D restraints free</span></p>
                    <button type="button" class="btn btn-primary" onclick="selectBuiltInSupport('ROLLER')">Use Roller</button>
                </div>
                <div class="restraint-box">
                    <div class="restraint-title">HINGED SUPPORT</div>
                    <p id="supportHingedDescription" style="font-size:12px; color:#475569; line-height:1.5; margin-bottom:10px;">RFx + RFy restrained<br><span style="color:#64748b;
">Rotation RZ remains free</span></p>
                    <button type="button" class="btn btn-primary" onclick="selectBuiltInSupport('HINGED')">Use Hinged</button>
                </div>
            </div>
        </div>
    </div>

    <div class="card">
        <div class="card-header">
            <div><h3>Create Custom Support Template</h3><p>Use this form only when you need a support condition other than the three built-in templates.</p></div>
        </div>
        <div class="card-body">
            <form id="supportTemplateForm">
                <div class="grid grid-3">
                    <div class="field"><label for="supportId">Support ID</label><input id="supportId" placeholder="S1" value="S1"></div>
                    <div class="field"><label for="supportDescription">Description</label><input id="supportDescription" placeholder="Pinned Support" value="Pinned Support"></div>
                </div>
                <div class="restraint-box" style="margin-top:18px;">
                    <div class="restraint-title">Restraints</div>
                    <div class="check-grid">
                        <label class="check-item"><input type="checkbox" id="RFx"><span id="supportRFxLabel">RFx</span></label>
                        <label class="check-item"><input type="checkbox" id="RFy"><span id="supportRFyLabel">RFy</span></label>
                        <label class="check-item"><input type="checkbox" id="RFz"><span id="supportRFzLabel">RFz (Ignore)</span></label>
                        <label class="check-item"><input type="checkbox" id="RMx"><span id="supportRMxLabel">RMx (Ignore)</span></label>
                        <label class="check-item"><input type="checkbox" id="RMy"><span id="supportRMyLabel">RMy (Ignore)</span></label>
                        <label class="check-item"><input type="checkbox" id="RMz"><span id="supportRMzLabel">RMz</span></label>
                    </div>
                </div>
                <div class="form-footer">
                    <button type="button" class="btn btn-secondary" onclick="resetSupportForm()">Reset</button>
                    <button id="supportTemplateSubmitBtn" type="submit" class="btn btn-primary">Create Custom Support</button>
                </div>
            </form>
        </div>
    </div>

    <div class="card">
        <div class="card-header">
            <div><h3>Support Templates</h3><p>Reusable support definitions.</p></div>
        </div>
        <div class="card-body">
            <div class="table-wrap">
                <table>
                    <thead><tr><th>Support ID</th><th>Description</th><th>Restraints</th><th>Actions</th></tr></thead>
                    <tbody id="supportTemplateTable"></tbody>
                </table>
            </div>
        </div>
    </div>

    <div class="card">
        <div class="card-header">
            <div><h3>Assign Support to Node</h3><p>Apply a reusable support template to any node.</p></div>
        </div>
        <div class="card-body">
```

```
            <form id="supportAssignmentForm">
                <div class="grid grid-3">
                    <div class="field"><label for="supportAssignmentId">Support Assignment ID</label><input id="supportAssignmentId" placeholder="SA1" value="SA1"></div>
                    <div class="field"><label for="supportTemplateSelect">Support Template</label><select id="supportTemplateSelect"><option value="">Select support</option></select></div>
                    <div class="field"><label for="supportNodeSelect">Node ID</label><select id="supportNodeSelect"><option value="">Select node</option></select></div>
                </div>
                <div class="restraint-box" style="margin-top:14px;">
                    <div class="restraint-title">Prescribed Movement at This Node (optional)</div>
                    <p style="font-size:12px;color:#64748b;line-height:1.5;margin:0 0 12px;">These values apply only to the selected node. A nonzero value requires the selected support
template to restrain that degree of freedom.</p>
                    <div class="grid grid-3">
                        <div class="field plane-support-movement"><label for="assignmentDispX">Prescribed X displacement (Δx)</label><div class="input-wrap has-unit"><input
id="assignmentDispX" type="text" inputmode="decimal" placeholder="0" value="0"><span class="input-unit">mm</span></div><div class="length-chip-row scientific-shortcut-row"><button
type="button" class="scientific-chip" onclick="insertScientificExponent('assignmentDispX')">e</button></div></div>
                        <div class="field plane-support-movement"><label for="assignmentDispY">Prescribed Y displacement (Δy)</label><div class="input-wrap has-unit"><input
id="assignmentDispY" type="text" inputmode="decimal" placeholder="0" value="0"><span class="input-unit">mm</span></div><div class="length-chip-row scientific-shortcut-row"><button
type="button" class="scientific-chip" onclick="insertScientificExponent('assignmentDispY')">e</button></div></div>
                        <div class="field plane-support-movement"><label for="assignmentRotZ">Prescribed rotation (θz)</label><div class="input-wrap has-unit"><input id="assignmentRotZ"
type="text" inputmode="decimal" placeholder="0" value="0"><span class="input-unit">rad</span></div><div class="length-chip-row scientific-shortcut-row"><button type="button"
class="scientific-chip" onclick="insertScientificExponent('assignmentRotZ')">e</button></div></div>
                        <div class="field grid-support-movement" style="display:none"><label for="assignmentDispZ">Prescribed Grid settlement (ΔZ)</label><div class="input-wrap has-
unit"><input id="assignmentDispZ" type="text" inputmode="decimal" placeholder="0" value="0"><span class="input-unit">mm</span></div><div class="length-chip-row scientific-shortcut-
row"><button type="button" class="scientific-chip" onclick="insertScientificExponent('assignmentDispZ')">e</button></div></div>
                        <div class="field grid-support-movement" style="display:none"><label for="assignmentRotX">Prescribed rotation (θX)</label><div class="input-wrap has-unit"><input
id="assignmentRotX" type="text" inputmode="decimal" placeholder="0" value="0"><span class="input-unit">rad</span></div><div class="length-chip-row scientific-shortcut-row"><button
type="button" class="scientific-chip" onclick="insertScientificExponent('assignmentRotX')">e</button></div></div>
                        <div class="field grid-support-movement" style="display:none"><label for="assignmentRotY">Prescribed rotation (θY)</label><div class="input-wrap has-unit"><input
id="assignmentRotY" type="text" inputmode="decimal" placeholder="0" value="0"><span class="input-unit">rad</span></div><div class="length-chip-row scientific-shortcut-row"><button
type="button" class="scientific-chip" onclick="insertScientificExponent('assignmentRotY')">e</button></div></div>
                    </div>
                </div>
                <div class="form-footer">
                    <button id="supportAssignmentSubmitBtn" type="submit" class="btn btn-primary">Assign Support</button>
                </div>
            </form>
        </div>
    </div>

    <div class="card">
        <div class="card-header">
            <div><h3>Assigned Supports</h3></div>
        </div>
        <div class="card-body">
            <div class="table-wrap">
                <table>
                    <thead><tr><th>Assignment ID</th><th>Support</th><th>Node</th><th>Prescribed Movement</th><th>Actions</th></tr></thead>
                    <tbody id="supportAssignmentTable"></tbody>
                </table>
            </div>
        </div>
    </div>
</section>
`);

/* ---- Logic ---- */
// ---- Support Templates ----
let editingSupportTemplateId = null;

function createSupportTemplate(data) {
    model.supportTemplates.push(data);
    markAnalysisInvalidated();
}

function updateSupportTemplate(existingId, data) {
    const support = model.supportTemplates.find((s) => s.id === existingId && !s.builtIn);
    if (!support) return false;
    const oldId = support.id;
    Object.assign(support, data);
    if (oldId !== data.id) model.supportAssignments.forEach((sa) => { if (sa.supportId === oldId) sa.supportId = data.id; });
    markAnalysisInvalidated();
    return true;
}

document.getElementById("supportTemplateForm").addEventListener("submit", (e) => {
    e.preventDefault();
    const id = val("supportId");
    if (!id) return alert("Support ID is required.");
    if (hasId(model.supportTemplates, id, editingSupportTemplateId)) return alert("Support ID already exists.");
    const data = {
        id, description: val("supportDescription"), type: "custom", builtIn: false,
        restraints: { RFx: document.getElementById("RFx").checked, RFy: document.getElementById("RFy").checked, RFz: document.getElementById("RFz").checked, RMx: document.getElementById(
"RMx").checked, RMy: document.getElementById("RMy").checked, RMz: document.getElementById("RMz").checked },
        displacement: { ...ZERO_SUPPORT_MOVEMENTS },
    };
    if (editingSupportTemplateId !== null) {
        const gridMode = isGridAnalysisMode();
        const activeFields = gridMode ? ["tx", "ty", "dz"] : ["dx", "dy", "tz"];
```

```
        for (const assignment of model.supportAssignments.filter((item) => String(item.supportId) === String(editingSupportTemplateId))) {
            const movements = getSupportAssignmentPrescribedDisplacements(assignment, model.supportTemplates.find((item) => String(item.id) === String(editingSupportTemplateId)));
            const incompatible = activeFields.find((field) =>
                Number(movements[field]) !== 0 && !supportTemplateRestrainsMovement(data, field, gridMode)
            );
            if (incompatible) {
                return alert(`Cannot remove the restraint for ${incompatible}: support assignment "${assignment.id}" has a nonzero prescribed movement on that degree of freedom. Set that
assignment's movement to zero first.`);
            }
        }
        const updated = updateSupportTemplate(editingSupportTemplateId, data);
        if (!updated) { resetSupportForm(); return; }
    } else {
        createSupportTemplate(data);
    }
    resetSupportForm();
    document.getElementById("supportId").value = getNextId(id);
    refreshAll();
});
function resetSupportForm() {
    document.getElementById("supportTemplateForm").reset();
    editingSupportTemplateId = null;
    document.getElementById("supportTemplateSubmitBtn").textContent = "Create Custom Support";
}
function editSupportTemplate(encodedId) {
    const id = decodeURIComponent(encodedId);
    const support = model.supportTemplates.find((s) => s.id === id);
    if (!support || support.builtIn) return;
    editingSupportTemplateId = support.id;
    document.getElementById("supportId").value = support.id;
    document.getElementById("supportDescription").value = support.description || "";
    ["RFx","RFy","RFz","RMx","RMy","RMz"].forEach((key) => { document.getElementById(key).checked = !!support.restraints?.[key]; });
    document.getElementById("supportTemplateSubmitBtn").textContent = "Update Custom Support";
}
function selectBuiltInSupport(id) {
    const select = document.getElementById("supportTemplateSelect");
    if (!select) return;
    select.value = id;
    const assignmentCard = document.getElementById("supportAssignmentForm");
    if (assignmentCard) assignmentCard.scrollIntoView({ behavior: "smooth", block: "center" });
}
function renderSupportTemplates() {
    const tbody = document.getElementById("supportTemplateTable");
    if (!model.supportTemplates.length) return showEmpty("supportTemplateTable", 4);
    const gridMode = typeof isGridAnalysisMode === "function" && isGridAnalysisMode();
    const gridRestraintLabels = { RFx: "0X", RFy: "0Y", RMz: "ΔZ" };
    const allRestraintLabels = { RFx: "RFx", RFy: "RFy", RFz: "RFz", RMx: "RMx", RMy: "RMy", RMz: "RMz" };
    tbody.innerHTML = model.supportTemplates.map((s) => {
        const restraints = s.restraints || {};
        const active = Object.entries(restraints).filter(([key, value]) => !!value).map(([key]) => key);
        const effective = active.filter((key) => gridMode ? Object.prototype.hasOwnProperty.call(gridRestraintLabels, key) : Object.prototype.hasOwnProperty.call(allRestraintLabels, key));
        const ignored = gridMode ? active.filter((key) => !Object.prototype.hasOwnProperty.call(gridRestraintLabels, key)) : [];
        const labels = gridMode ? gridRestraintLabels : allRestraintLabels;
        const r = effective.map((key) => labels[key]).join(", ") || "None";
        const rText = `${escapeHtml(r)}${ignored.length ? `<br><span class="muted">Ignored in Grid: ${escapeHtml(ignored.map((key) => allRestraintLabels[key] || key).join(", "))}</span>` :
""}`;
        const kind = s.builtIn ? `<span class="badge badge-blue">Built-in</span>` : `<span class="badge badge-gray">Custom</span>`;
        const action = s.builtIn ? `<span class="badge badge-gray">Protected</span>` : `<div class="table-actions"><button type="button" class="icon-btn" aria-label="Edit support template ${
escapeHtml(s.id)}" title="Edit support template ${escapeHtml(s.id)}" onclick="editSupportTemplate('${encodeInlineArgument(s.id)}')">✎</button><button type="button" class="icon-btn" aria-
label="Delete support template ${escapeHtml(s.id)}" title="Delete support template ${escapeHtml(s.id)}" onclick="deleteItem('supportTemplates','${encodeInlineArgument(s.id)}','id')">×</
button></div>`;
        return `<tr data-sort-collection="supportTemplates" data-sort-id="${escapeHtml(encodeInlineArgument(s.id))}"><td>${sortHandle(s.id)}${escapeHtml(s.id)}</td><td>${escapeHtml(s.
description)}<br>${kind}</td><td>${rText}</td><td>${action}</td></tr>`;
    }).join("");
}

// ---- Prescribed movement unit conversion ----
// Model/solver translations use metres. The support-assignment UI accepts mm.
// Rotations remain radians and are never scaled.
function supportMovementInputToModelUnits(field, value) {
    return ["dx", "dy", "dz"].includes(field) ? value / 1000 : value;
}
function supportMovementModelToInputUnits(field, value) {
    return ["dx", "dy", "dz"].includes(field) ? value * 1000 : value;
}

// ---- Support Assignments (apply a support template to a node) ----
let editingSupportAssignmentId = null;

function createSupportAssignment(data) {
    model.supportAssignments.push(data);
    markAnalysisInvalidated();
}

function updateSupportAssignment(existingId, data) {
    const assignment = model.supportAssignments.find((sa) => sa.id === existingId);
    if (!assignment) return false;
```

```
    Object.assign(assignment, data);
    markAnalysisInvalidated();
    return true;
}

document.getElementById("supportAssignmentForm").addEventListener("submit", (e) => {
    e.preventDefault();
    const id = val("supportAssignmentId");
    if (!id) return alert("Support Assignment ID is required.");
    if (hasId(model.supportAssignments, id, editingSupportAssignmentId)) return alert("Support Assignment ID already exists.");
    const supportId = val("supportTemplateSelect");
    const nodeId = val("supportNodeSelect");
    let error = requireReference(model.supportTemplates, supportId, "Support Template");
    if (error) return alert(error);
    error = requireReference(model.nodes, nodeId, "Support Node");
    if (error) return alert(error);
    const otherAssignmentOnNode = model.supportAssignments.find((assignment) =>
        String(assignment.nodeId) === String(nodeId) &&
        (editingSupportAssignmentId === null || String(assignment.id) !== String(editingSupportAssignmentId))
    );
    if (otherAssignmentOnNode) {
        return alert(`Node "${nodeId}" already has support assignment "${otherAssignmentOnNode.id}". Create one custom support template with all required restraints instead of assigning
multiple supports to the same node.`);
    }
    const supportTemplate = model.supportTemplates.find((item) => String(item.id) === String(supportId)) ||
        AppConfig.builtInSupportTemplates.find((item) => String(item.id) === String(supportId));
    const gridMode = isGridAnalysisMode();
    const prescribedDisplacements = { ...ZERO_SUPPORT_MOVEMENTS };
    const activeMovementInputs = gridMode
        ? [
            { inputId: "assignmentDispZ", field: "dz", label: "ΔZ" },
            { inputId: "assignmentRotX", field: "tx", label: "θX" },
            { inputId: "assignmentRotY", field: "ty", label: "θY" },
        ]
        : [
            { inputId: "assignmentDispX", field: "dx", label: "Δx" },
            { inputId: "assignmentDispY", field: "dy", label: "Δy" },
            { inputId: "assignmentRotZ", field: "tz", label: "θz" },
        ];
    for (const item of activeMovementInputs) {
        const raw = val(item.inputId);
        const value = raw === "" ? 0 : Number(raw);
        if (!Number.isFinite(value)) return alert(`Prescribed movement ${item.label} must be a finite number.`);
        prescribedDisplacements[item.field] = supportMovementInputToModelUnits(item.field, value);
        if (value !== 0 && !supportTemplateRestrainsMovement(supportTemplate, item.field, gridMode)) {
            return alert(`Cannot prescribe ${item.label} at node "${nodeId}" because support "${supportId}" does not restrain that degree of freedom. Select a support template that
restrains it, or set the movement to zero.`);
        }
    }
    const data = { id, supportId, nodeId, prescribedDisplacements };
    if (editingSupportAssignmentId !== null) {
        const updated = updateSupportAssignment(editingSupportAssignmentId, data);
        if (!updated) { resetSupportAssignmentForm(); return; }
    } else {
        createSupportAssignment(data);
    }
    resetSupportAssignmentForm();
    document.getElementById("supportAssignmentId").value = getNextId(id);
    refreshAll();
});
function resetSupportAssignmentForm() {
    document.getElementById("supportAssignmentForm").reset();
    editingSupportAssignmentId = null;
    document.getElementById("supportAssignmentSubmitBtn").textContent = "Assign Support";
}
function editSupportAssignment(encodedId) {
    const id = decodeURIComponent(encodedId);
    const assignment = model.supportAssignments.find((sa) => sa.id === id);
    if (!assignment) return;
    editingSupportAssignmentId = assignment.id;
    document.getElementById("supportAssignmentId").value = assignment.id;
    document.getElementById("supportTemplateSelect").value = assignment.supportId;
    document.getElementById("supportNodeSelect").value = assignment.nodeId;
    const template = model.supportTemplates.find((item) => String(item.id) === String(assignment.supportId));
    const movement = getSupportAssignmentPrescribedDisplacements(assignment, template);
    document.getElementById("assignmentDispX").value = supportMovementModelToInputUnits("dx", Number(movement.dx) || 0);
    document.getElementById("assignmentDispY").value = supportMovementModelToInputUnits("dy", Number(movement.dy) || 0);
    document.getElementById("assignmentDispZ").value = supportMovementModelToInputUnits("dz", Number(movement.dz) || 0);
    document.getElementById("assignmentRotX").value = movement.tx ?? 0;
    document.getElementById("assignmentRotY").value = movement.ty ?? 0;
    document.getElementById("assignmentRotZ").value = movement.tz ?? 0;
    document.getElementById("supportAssignmentSubmitBtn").textContent = "Update Support";
}
function renderSupportAssignments() {
    const tbody = document.getElementById("supportAssignmentTable");
    if (!model.supportAssignments.length) return showEmpty("supportAssignmentTable", 5);
    const gridMode = isGridAnalysisMode();
```

```
    tbody.innerHTML = model.supportAssignments.map((s) => {
        const template = model.supportTemplates.find((item) => String(item.id) === String(s.supportId));
        const movement = getSupportAssignmentPrescribedDisplacements(s, template);
        const movementText = gridMode
            ? `ΔZ=${escapeHtml(supportMovementModelToInputUnits("dz", Number(movement.dz) || 0))} mm<br>0X=${escapeHtml(movement.tx ?? 0)} rad, 0Y=${escapeHtml(movement.ty ?? 0)} rad`
            : `Δx=${escapeHtml(supportMovementModelToInputUnits("dx", Number(movement.dx) || 0))} mm, Δy=${escapeHtml(supportMovementModelToInputUnits("dy", Number(movement.dy) || 0))}
mm<br>0z=${escapeHtml(movement.tz ?? 0)} rad`;
        return `<tr data-sort-collection="supportAssignments" data-sort-id="${escapeHtml(encodeInlineArgument(s.id))}"><td>${sortHandle(s.id)}${escapeHtml(s.id)}</td><td><span class="badge
badge-orange">${escapeHtml(s.supportId)}</span></td><td>${escapeHtml(s.nodeId)}</td><td>${movementText}</td><td><div class="table-actions"><button type="button" class="icon-btn" aria-
label="Edit support assignment ${escapeHtml(s.id)}" title="Edit support assignment ${escapeHtml(s.id)}" onclick="editSupportAssignment('${encodeInlineArgument(s.id)}')">✎</button><button
type="button" class="icon-btn" aria-label="Delete support assignment ${escapeHtml(s.id)}" title="Delete support assignment ${escapeHtml(s.id)}" onclick="deleteItem('supportAssignments','${
encodeInlineArgument(s.id)}','id')">×</button></div></td></tr>`;
    }).join("");
}
```

41. modules/tour/tour.css

```
/* ========================================================================
 * tour.css
 * Look & feel of the onboarding tour: the step popover (a themed Driver.js
 * popover, scoped under .sa-tour so Driver's defaults are overridden) and the
 * permanent "?" replay button. Uses the colour tokens from main.css.
 * ======================================================================== */

/* ---- Step popover ---- */
.driver-popover.sa-tour {
    min-width: 280px;
    max-width: min(360px, calc(100vw - 24px));
    padding: 18px 18px 14px;
    border-radius: 14px;
    background: #ffffff;
    color: var(--text, #182230);
    box-shadow: 0 18px 50px rgba(15, 23, 42, 0.35), 0 2px 8px rgba(15, 23, 42, 0.18);
}

.driver-popover.sa-tour,
.driver-popover.sa-tour * {
    font-family: Inter, Arial, Helvetica, sans-serif;
}

.driver-popover.sa-tour .driver-popover-arrow {
    border-width: 6px;
}

.driver-popover.sa-tour .driver-popover-title {
    font-size: 17px;
    font-weight: 800;
    letter-spacing: -0.2px;
    line-height: 1.35;
    padding-right: 26px; /* room for the × button */
    color: var(--text, #182230);
}

.driver-popover.sa-tour .driver-popover-description {
    font-size: 13.5px;
    line-height: 1.6;
    color: #334155;
}

.driver-popover.sa-tour .driver-popover-title[style*="block"] + .driver-popover-description {
    margin-top: 8px;
}

.driver-popover.sa-tour .driver-popover-description b {
    color: var(--text, #182230);
    font-weight: 700;
}

/* × close button */
.driver-popover.sa-tour .driver-popover-close-btn {
    top: 10px;
    right: 10px;
    width: 28px;
    height: 28px;
    line-height: 28px;
    border-radius: 8px;
    font-size: 20px;
    color: #64748b;
}

.driver-popover.sa-tour .driver-popover-close-btn:hover,
.driver-popover.sa-tour .driver-popover-close-btn:focus-visible {
    background: #f1f5f9;
    color: var(--text, #182230);
}

/* Footer: [progress ........ Back Next]
 *         [      Skip tutorial       ]   */
.driver-popover.sa-tour .driver-popover-footer {
    margin-top: 16px;
    flex-wrap: wrap;
    row-gap: 12px;
}

.driver-popover.sa-tour .driver-popover-progress-text {
    font-size: 12px;
    font-weight: 600;
    color: #64748b;
}

.driver-popover.sa-tour .driver-popover-footer button {
    padding: 8px 14px;
```

```
    border-radius: 9px;
    border: 1px solid var(--border, #dce3ec);
    background: #ffffff;
    color: var(--text, #182230);
    font-size: 13px;
    font-weight: 700;
    line-height: 1.2;
    text-shadow: none;
    transition: background 0.15s, border-color 0.15s, color 0.15s;
}

.driver-popover.sa-tour .driver-popover-footer button:hover {
    background: #f1f5f9;
}

.driver-popover.sa-tour .driver-popover-footer button:focus-visible,
.driver-popover.sa-tour .driver-popover-close-btn:focus-visible {
    outline: 3px solid rgba(37, 99, 235, 0.45);
    outline-offset: 2px;
}

.driver-popover.sa-tour .driver-popover-footer .driver-popover-next-btn {
    background: var(--primary, #2563eb);
    border-color: var(--primary, #2563eb);
    color: #ffffff;
}

.driver-popover.sa-tour .driver-popover-footer .driver-popover-next-btn:hover {
    background: var(--primary-dark, #1d4ed8);
    border-color: var(--primary-dark, #1d4ed8);
}

/* Nothing to go back to on the first step: hide the disabled Back button.
   !important is needed because Driver sets an inline display on its buttons. */
.driver-popover.sa-tour .driver-popover-prev-btn.driver-popover-btn-disabled {
    display: none !important;
}

.driver-popover.sa-tour .driver-popover-navigation-btns button + button {
    margin-left: 8px;
}

/* "Skip tutorial": always present, own row, visible but secondary */
.driver-popover.sa-tour .driver-popover-footer .sa-tour-skip {
    order: 3;
    flex: 0 0 100%;
    padding: 9px 0 0;
    border: 0;
    border-top: 1px solid #e2e8f0;
    border-radius: 0;
    background: transparent;
    color: #475569;
    font-size: 12.5px;
    font-weight: 600;
    text-align: center;
}

.driver-popover.sa-tour .driver-popover-footer .sa-tour-skip:hover {
    background: transparent;
    color: var(--text, #182230);
    text-decoration: underline;
}

/* ---- Permanent "replay tour" button ----
   z-index sits below Driver's overlay (10000) and the app's modals, so it is
   dimmed during the tour and never covers a dialog. */
.tour-help-btn {
    position: fixed;
    right: 18px;
    bottom: 18px;
    z-index: 9000;
    display: inline-flex;
    align-items: center;
    gap: 8px;
    min-height: 44px;
    padding: 6px 16px 6px 8px;
    border: 1px solid var(--border, #dce3ec);
    border-radius: 999px;
    background: #ffffff;
    color: var(--primary-dark, #1d4ed8);
    font-size: 13px;
    font-weight: 700;
    box-shadow: 0 8px 25px rgba(15, 23, 42, 0.14);
    transition: background 0.15s, box-shadow 0.15s, transform 0.15s;
}

.tour-help-btn:hover {
```

```
    background: var(--soft-blue, #eff6ff);
    box-shadow: 0 10px 28px rgba(15, 23, 42, 0.2);
}

.tour-help-btn:active {
    transform: translateY(1px);
}

.tour-help-btn:focus-visible {
    outline: 3px solid rgba(37, 99, 235, 0.45);
    outline-offset: 2px;
}

.tour-help-icon {
    display: inline-flex;
    align-items: center;
    justify-content: center;
    width: 30px;
    height: 30px;
    border-radius: 50%;
    background: var(--primary, #2563eb);
    color: #ffffff;
    font-size: 16px;
    font-weight: 800;
    line-height: 1;
}

/* Small screens: icon-only, so it does not cover page content */
@media (max-width: 640px) {
    .tour-help-btn {
        right: 12px;
        bottom: 12px;
        padding: 6px;
    }
    .tour-help-label {
        position: absolute;
        width: 1px;
        height: 1px;
        overflow: hidden;
        clip: rect(0 0 0 0);
        white-space: nowrap;
    }
}

@media (prefers-reduced-motion: reduce) {
    .tour-help-btn,
    .driver-popover.sa-tour .driver-popover-footer button {
        transition: none;
    }
}

@media print {
    .tour-help-btn {
        display: none;
    }
}
```

42. modules/tour/tour.js

```
/* ============================================================================
 * tour.js — Onboarding tour module
 * Guided product tour (built on Driver.js, vendored in assets/vendor/driver/).
 *
 *  - Starts automatically ONLY the first time the app is opened in a browser.
 *    "Seen" is remembered in localStorage under STORAGE_KEY.
 *  - Every step has a "Skip tutorial" button (plus the × button and the Esc key).
 *  - A permanent floating "?" button (bottom-right) replays the tour any time.
 *  - The tour walks through Home and the workspace tabs by calling the app's own
 *    goHome() / switchSection(); when it ends (finished or skipped) the user is
 *    returned to the section and scroll position they started from.
 *  - It never reads or writes model data.
 *
 * Public API (window.StructuralTour):
 *   start()      replay the tour now
 *   isActive()   true while the tour is on screen
 *   hasSeen()    true once the tour has been started (auto or manual) before
 *   resetSeen()  forget that the tour was seen (it will auto-start on next load)
 *
 * Load order: after every page module (see index.html) so all markup exists.
 * ============================================================================ */
(function () {
    "use strict";

    if (window.StructuralTour) return; // guard against the script being loaded twice

    /* ------------------------------------------------------------------------
     * Configuration
     * ---------------------------------------------------------------------- */
    const STORAGE_KEY = "structuralOnboardingTour";
    const AUTO_START_DELAY_MS = 500; // let the app finish its first render

    /* Each step:
     *   section : app section that must be visible ("home", "nodes", ...)
     *   element : CSS selector to highlight (omit for a centred welcome/finish card)
     *   popover : title / description (HTML allowed) / side / align / button labels
     * Steps whose element is missing or hidden at run time are skipped automatically. */
    const STEPS = [
        {
            section: "home",
            popover: {
                title: "Welcome to Structure Analysis",
                description:
                    "Take a quick tour (about two minutes) of how to build a structural model, " +
                    "run the stiffness-method analysis and read the results." +
                    "<br><br>You can skip at any time and replay the tour later from the " +
                    "<b>?</b> button at the bottom-right of the screen.",
                nextBtnText: "Start tour",
            },
        },
        {
            section: "home",
            element: "#home .home-standalone-head .btn-primary",
            popover: {
                title: "Create a model",
                description:
                    "This highlighted button starts a new project after you finish or skip the tour. " +
                    "You will name the model and choose its structure type — " +
                    "<b>Beam / Truss / Frame</b> (2D) or <b>Grid</b>. The type is fixed once the model is created.",
                side: "bottom",
                align: "end",
            },
        },
        {
            section: "home",
            element: "#home .home-standalone-head .home-standalone-actions",
            popover: {
                title: "Import &amp;amp; export",
                description:
                    "<b>Import Model</b> loads a JSON file you exported earlier. " +
                    "<b>Export All Models</b> downloads every saved model in one file — handy as a backup or for sharing. " +
                    "After the tour closes, use the floating <b>?</b> button to replay the tutorial.",
                side: "bottom",
                align: "end",
            },
        },
        {
            section: "home",
            element: "#home .home-current-strip",
            popover: {
                title: "Current model",
                description:
                    "This strip shows the model you are working on, its type and whether it has been saved. " +
                    "After the tour, use <b>Save Current</b>, <b>Export Current</b> or <b>Start New Model</b> from here.",
                side: "bottom",
```

```
        align: "center",
    },
},
{
    section: "home",
    element: "#modelLibraryGrid",
    popover: {
        title: "Your saved models",
        description:
            "Saved models appear here as cards with a preview of the structure. After the tour, open one to keep working on it, " +
            "or select several to export or delete them together.",
        side: "top",
        align: "center",
    },
},
{
    section: "dashboard",
    element: ".sidebar .nav-buttons",
    popover: {
        title: "Model input tabs",
        description:
            "Once a model is open, each part of the input lives in its own tab. " +
            "Work from top to bottom — <b>Nodes → Elements → Supports → Load Cases → Loads → " +
            "Moment Cases → Moments</b> — then open <b>Results</b>.",
        side: "right",
        align: "start",
    },
},
{
    section: "dashboard",
    element: "#dashboard .stat-grid",
    popover: {
        title: "Model at a glance",
        description:
            "Live counts of the nodes, elements, supports, loads and moments (torsions for Grid models) " +
            "in the current model — a quick way to check that nothing is missing.",
        side: "bottom",
        align: "center",
    },
},
{
    section: "dashboard",
    element: "#dashboard .model-note",
    popover: {
        title: "Recommended input order",
        description:
            "Follow this sequence when defining a model. Later items depend on earlier ones — " +
            "for example, elements need nodes, and loads need a load case.",
        side: "top",
        align: "center",
    },
},
{
    section: "dashboard",
    element: "#liveModelDiagramCard",
    popover: {
        title: "Live model diagram",
        description:
            "Your structure is drawn here and updates instantly as you add or edit nodes, elements, " +
            "supports and loads. Use it to spot mistakes early.",
        side: "bottom",
        align: "center",
    },
},
{
    section: "nodes",
    element: "#nodes > .card:first-child > .card-header",
    popover: {
        title: "1 · Nodes",
        description:
            "Use the form below after the tour to define each joint with an ID and its X, Y and Z coordinates. " +
            "Z is ignored in both 2D plane and Grid geometry; Grid members lie in the XY plane. " +
            "Element lengths are calculated automatically from X and Y coordinates.",
        side: "bottom",
        align: "start",
    },
},
{
    section: "elements",
    element: "#elements > .card:first-child > .card-header",
    popover: {
        title: "2 · Element templates and elements",
        description:
            "Create a reusable template with the section and material properties in the form below, then use " +
            "the assignment card further down to connect a start node and an end node. " +
            "One template can be used for many members.",
        side: "bottom",
```

```
            align: "start",
        },
    },
    {
        section: "supports",
        element: "#supports > .card:first-child",
        popover: {
            title: "3 · Supports",
            description:
                "Pick a built-in <b>Fixed</b>, <b>Roller</b> or <b>Hinged</b> template — or define a custom one " +
                "further down — then assign it to nodes to restrain the structure.",
            side: "bottom",
            align: "center",
        },
    },
    {
        section: "loads",
        element: "#loads > .card:first-child > .card-header",
        popover: {
            title: "4 · Loads and moments",
            description:
                "Loads are grouped into <b>load cases</b> (Dead, Live, Self Weight): create a case in " +
                "<b>Load Cases</b> first, then add nodal or element loads with the form below. For Grid models, " +
                "self-weight must be entered directly as a line weight in kN/m. Moments " +
                "(torsions for Grid models) work the same way in <b>Moment Cases</b> and <b>Moments</b>.",
            side: "bottom",
            align: "start",
        },
    },
    {
        section: "results",
        element: "#results > .card:first-child > .card-header .btn-primary",
        popover: {
            title: "5 · Run the analysis",
            description:
                "After you finish or skip the tour, choose the active load case and moment case, then use " +
                "<b>Run Analysis</b>. The solver runs in your browser; the calculation steps, reactions " +
                "and diagrams then appear on this page. Press <b>Next</b> to continue this guided tour.",
            side: "bottom",
            align: "end",
        },
    },
    {
        section: "results",
        element: ".topbar .top-actions",
        popover: {
            title: "Save and export your work",
            description:
                "<b>Save Model</b> stores the model in this browser; <b>Export</b> downloads it as a JSON file. " +
                "You can also rename the model here. Save often — clearing browser data removes saved models.",
            side: "bottom",
            align: "end",
        },
    },
    {
        section: "home",
        popover: {
            title: "You are ready to go",
            description:
                "That covers the essentials — finish the tour, then create your first model to get started." +
                "<br><br>Need a refresher? Use the <b>?</b> button at the bottom-right of the screen " +
                "after the tour closes to replay it.",
            doneBtnText: "Finish",
        },
    },
];

/* ---------------------------------------------------------------------
 * First-visit persistence (localStorage)
 * ------------------------------------------------------------------- */

/* localStorage can throw (private mode, blocked cookies, quota). When it is
 * unusable we cannot remember the visit, so we do NOT auto-start (otherwise
 * the tour would reappear on every page load). The "?" button still works. */
function storageUsable() {
    try {
        const probe = "__structuralTourProbe__";
        if (!safeStorageSetItem(probe, "1")) return false;
        safeStorageRemoveItem(probe);
        return true;
    } catch (e) {
        return false;
    }
}

function hasSeen() {
    try {
```

```
        return safeStorageGetItem(STORAGE_KEY) !== null;
    } catch (e) {
        return false;
    }
}

function saveState(status) {
    try {
        safeStorageSetItem(STORAGE_KEY, JSON.stringify({ status: status, updatedAt: new Date().toISOString() }));
    } catch (e) {
        /* ignore — see storageUsable() */
    }
}

function resetSeen() {
    try {
        safeStorageRemoveItem(STORAGE_KEY);
    } catch (e) {
        /* ignore */
    }
}

/* ------------------------------------------------------------------------
 * Helpers
 * ------------------------------------------------------------------------ */
const tour = {
    driver: null,
    active: false,
    navigating: false,
    origin: null, // { section, x, y } — where the user was when the tour began
};

function frames(count) {
    return new Promise(function (resolve) {
        (function tick(n) {
            if (n <= 0) resolve();
            else requestAnimationFrame(function () { tick(n - 1); });
        })(count);
    });
}

function activeSectionId() {
    const el = document.querySelector(".container > .section.active");
    return el ? el.id : "home";
}

/* Show an app section using the application's own navigation functions. */
function showSection(id) {
    if (!id || activeSectionId() === id) return;
    try {
        if (id === "home" && typeof goHome === "function") {
            goHome();
        } else if (typeof switchSection === "function") {
            switchSection(id, document.querySelector('.nav-btn[data-section="' + id + '"]'));
        }
    } catch (error) {
        console.error("[tour] Could not switch to section:", id, error);
    }
}

/* Driver un-marks the previous target only when its 400 ms transition ends, so
 * clicking Next faster than that leaves the highlight class / ARIA attributes on an
 * earlier (possibly now hidden) element until the tour closes. Clear them first. */
function clearStaleHighlights() {
    document.querySelectorAll(".driver-active-element").forEach(function (el) {
        el.classList.remove("driver-active-element", "driver-no-interaction");
        el.removeAttribute("aria-haspopup");
        el.removeAttribute("aria-expanded");
        el.removeAttribute("aria-controls");
    });
}

function isStepAvailable(step) {
    if (!step.element) return true;
    const el = document.querySelector(step.element);
    return !!el && el.getClientRects().length > 0;
}

/* ------------------------------------------------------------------------
 * Navigation
 * ------------------------------------------------------------------------ */

/* Show step `index`, switching section first. Steps whose target is missing
 * or hidden are skipped in the direction of travel. `fallback` is the step
 * that was on screen (used if nothing before the start is available). */
async function goToStep(index, direction, fallback) {
    let i = index;
```

```
    while (i >= 0 && i < STEPS.length) {
        showSection(STEPS[i].section);
        await frames(2); // let the section render before measuring it
        if (!tour.active) return; // closed while we were waiting
        if (isStepAvailable(STEPS[i])) {
            clearStaleHighlights();
            tour.driver.drive(i);
            return;
        }
        i += direction;
    }
    if (direction > 0) {
        finish("completed");
    } else if (typeof fallback === "number") {
        showSection(STEPS[fallback].section);
        await frames(2);
        if (tour.active) {
            clearStaleHighlights();
            tour.driver.drive(fallback);
        }
    }
}

async function navigate(index, direction, fallback) {
    if (tour.navigating) return; // ignore rapid double clicks / key repeat
    tour.navigating = true;
    try {
        await goToStep(index, direction, fallback);
    } finally {
        tour.navigating = false;
    }
}

function finish(outcome) {
    if (!tour.active) return; // already closed (also guards re-entrancy)
    tour.active = false;
    saveState(outcome); // "completed" | "skipped"
    try {
        tour.driver.destroy();
    } catch (error) {
        console.error("[tour] destroy failed", error);
    }
    // Put the user back where they were before the tour moved them around.
    const origin = tour.origin;
    tour.origin = null;
    if (origin) {
        showSection(origin.section);
        frames(2).then(function () { window.scrollTo(origin.x, origin.y); });
    }
}

/* Adds the "Skip tutorial" button to the popover (runs on every step). */
function decoratePopover(popover) {
    const footer = popover.footer;
    if (!footer.querySelector(".sa-tour-skip")) {
        const skip = document.createElement("button");
        skip.type = "button";
        skip.className = "sa-tour-skip";
        skip.textContent = "Skip tutorial";
        skip.addEventListener("click", function () { finish("skipped"); });
        footer.appendChild(skip);
    }
    // Driver focuses the first control it finds (the × button). Move focus to
    // "Next" so Enter advances the tour; wait one frame so ours runs last.
    requestAnimationFrame(function () {
        if (tour.active && popover.nextButton) popover.nextButton.focus({ preventScroll: true });
    });
}

/* ---------------------------------------------------------------------
 * Public: start
 * ------------------------------------------------------------------- */
function start() {
    if (tour.active) return false;

    const lib = window.driver && window.driver.js && window.driver.js.driver;
    if (typeof lib !== "function") {
        console.error("[tour] Driver.js is not loaded; check the <script> tag in index.html.");
        return false;
    }
    if (!document.querySelector(".app")) return false;

    tour.origin = { section: activeSectionId(), x: window.scrollX, y: window.scrollY };
    tour.active = true;
    saveState("started"); // consumed as soon as it shows, so a reload mid-tour will not repeat it

    const reduceMotion = !!(window.matchMedia && window.matchMedia("(prefers-reduced-motion: reduce)").matches);
```

```
    tour.driver = lib({
        // The steps are driven by navigate() so each one can switch sections first.
        steps: STEPS.map(function (s) { return { element: s.element, popover: s.popover }; }),
        popoverClass: "sa-tour",
        animate: !reduceMotion,
        smoothScroll: false,
        overlayColor: "#0f172a",
        overlayOpacity: 0.62,
        stagePadding: 8,
        stageRadius: 10,
        popoverOffset: 14,
        showProgress: true,
        progressText: "Step {{current}} of {{total}}",
        nextBtnText: "Next",
        prevBtnText: "Back",
        doneBtnText: "Finish",
        allowClose: true, // × button and Esc
        allowKeyboardControl: true, // ← / → / Esc
        disableActiveInteraction: true, // highlighted controls are not clickable mid-tour
        overlayClickBehavior: "none", // Driver ignores any value but "close" / "nextStep"; stops accidental dismissal

        onPopoverRender: decoratePopover,

        onNextClick: function () {
            const i = tour.driver.getActiveIndex();
            if (typeof i !== "number") return;
            if (i >= STEPS.length - 1) finish("completed");
            else navigate(i + 1, 1, i);
        },
        onPrevClick: function () {
            const i = tour.driver.getActiveIndex();
            if (typeof i !== "number" || i <= 0) return;
            navigate(i - 1, -1, i);
        },
        // Fires for ×, Esc and anything else that closes the tour early.
        onDestroyStarted: function () {
            finish("skipped");
        },
    });

    navigate(0, 1, null);
    return true;
}

/* ------------------------------------------------------------------------
 * Permanent "replay" button + first-visit auto start
 * ------------------------------------------------------------------------ */
function createHelpButton() {
    if (document.getElementById("tourHelpButton")) return;
    const btn = document.createElement("button");
    btn.id = "tourHelpButton";
    btn.type = "button";
    btn.className = "tour-help-btn";
    btn.setAttribute("aria-label", "Replay the product tour");
    btn.title = "Replay the product tour";
    btn.innerHTML = '<span class="tour-help-icon" aria-hidden="true">?</span><span class="tour-help-label">Take the tour</span>';
    btn.addEventListener("click", function () { start(); });
    document.body.appendChild(btn);
}

/* Visible "Replay Tutorial" entries in the app's own UI: the Home header (the only
 * navigation visible on Home) and the bottom of the sidebar (workspace tabs). */
function createReplayButtons() {
    const homeActions = document.querySelector("#home .home-standalone-head .home-standalone-actions");
    if (homeActions && !document.getElementById("tourReplayHome")) {
        const b = document.createElement("button");
        b.id = "tourReplayHome";
        b.type = "button";
        b.className = "btn btn-secondary";
        b.textContent = "? Replay Tutorial";
        b.addEventListener("click", function () { start(); });
        homeActions.insertBefore(b, homeActions.firstChild);
    }
    const sidebarBottom = document.querySelector(".sidebar-bottom");
    if (sidebarBottom && !document.getElementById("tourReplaySidebar")) {
        const b = document.createElement("button");
        b.id = "tourReplaySidebar";
        b.type = "button";
        b.className = "sidebar-btn";
        b.textContent = "Replay Tutorial";
        b.addEventListener("click", function () { start(); });
        sidebarBottom.appendChild(b);
    }
}

function autoStartIfFirstVisit() {
    if (tour.active || hasSeen() || !storageUsable()) return;
```

```
        const modal = document.getElementById("newModelModal");
        if (modal && !modal.hidden) return; // another dialog is already open
        start();
    }

    function boot() {
        createHelpButton();
        createReplayButtons();
        const go = function () { setTimeout(autoStartIfFirstVisit, AUTO_START_DELAY_MS); };
        if (document.readyState === "complete") go();
        else window.addEventListener("load", go, { once: true });
    }

    window.StructuralTour = {
        start: start,
        isActive: function () { return tour.active; },
        hasSeen: hasSeen,
        resetSeen: resetSeen,
    };

    // main.js initialises the app on DOMContentLoaded; this listener is registered later, so it runs after it.
    if (document.readyState === "loading") document.addEventListener("DOMContentLoaded", boot);
    else boot();
})();
```

43. service-worker.js

```
"use strict";

/*
 * Offline app shell for GitHub Pages (including the /Structure-Analysis/ path).
 * Keep this list synchronized with index.html and the web app's local assets.
 * Increment CACHE_NAME for each offline-shell release so a failed install cannot
 * partially overwrite the currently active release cache.
 * No model data or calculation results are stored in Cache Storage.
 */
const CACHE_NAME = "structure-analysis-offline-v2";
const CACHE_PREFIX = "structure-analysis-offline-";

const CORE_ASSETS = [
    "index.html",
    "manifest.webmanifest",
    "assets/icons/structure-analysis.svg",
    "assets/css/main.css",
    "assets/js/main.js",
    "assets/js/offline.js",
    "assets/vendor/driver/driver.css",
    "assets/vendor/driver/driver.js.iife.js",
    "assets/vendor/jspdf.umd.min.js",
    "assets/vendor/pdf-fonts.js",
    "assets/vendor/svg2pdf.umd.min.js",
    "modules/dashboard/dashboard.css",
    "modules/dashboard/dashboard.js",
    "modules/elements/elements.js",
    "modules/home/home.css",
    "modules/home/home.js",
    "modules/load-case/load-case.js",
    "modules/load/load.js",
    "modules/moment-case/moment-case.js",
    "modules/moment/moment.js",
    "modules/nodes/nodes.js",
    "modules/results/diagrams/diagrams.css",
    "modules/results/diagrams/diagrams.js",
    "modules/results/results.css",
    "modules/results/results.js",
    "modules/results/step-1/step-1.js",
    "modules/results/step-2/step-2.js",
    "modules/results/step-3/step-3.js",
    "modules/results/step-4/step-4.js",
    "modules/results/step-5/step-5.js",
    "modules/results/step-6/step-6.js",
    "modules/results/step-7/step-7.js",
    "modules/results/step-8/step-8.css",
    "modules/results/step-8/step-8.js",
    "modules/supports/supports.js",
    "modules/tour/tour.css",
    "modules/tour/tour.js"
];

function assetUrl(path) {
    return new URL(path, self.registration.scope);
}

function canonicalRequest(request) {
    const url = new URL(request.url);
    url.search = "";
    url.hash = "";
    return new Request(url.href, { method: "GET" });
}

async function cacheCoreAssets() {
    const cache = await caches.open(CACHE_NAME);
    await Promise.all(CORE_ASSETS.map(async (path) => {
        const url = assetUrl(path);
        const response = await fetch(new Request(url.href, { cache: "reload" }));
        if (!response || !response.ok) {
            throw new Error("Unable to cache required offline asset: " + path);
        }
        await cache.put(new Request(url.href), response.clone());
    }));
}

async function getOfflineStatus() {
    const cache = await caches.open(CACHE_NAME);
    let cachedCount = 0;
    for (const path of CORE_ASSETS) {
        const cached = await cache.match(new Request(assetUrl(path).href), { ignoreSearch: true });
        if (cached) cachedCount += 1;
    }
    return {
        type: "OFFLINE_STATUS",
        ready: cachedCount === CORE_ASSETS.length,
```

```
        cached: cachedCount,
        total: CORE_ASSETS.length
    };
}

self.addEventListener("install", (event) => {
    event.waitUntil((async () => {
        // Install only succeeds when every required app-shell file is available.
        // The currently active cache is retained if this installation fails.
        await cacheCoreAssets();
        await self.skipWaiting();
    })());
});

self.addEventListener("activate", (event) => {
    event.waitUntil((async () => {
        const cacheNames = await caches.keys();
        await Promise.all(cacheNames
            .filter((name) => name.startsWith(CACHE_PREFIX) && name !== CACHE_NAME)
            .map((name) => caches.delete(name)));
        await self.clients.claim();
    })());
});

self.addEventListener("message", (event) => {
    if (event.data?.type !== "GET_OFFLINE_STATUS") return;
    event.waitUntil((async () => {
        const status = await getOfflineStatus();
        if (event.source && typeof event.source.postMessage === "function") {
            event.source.postMessage(status);
        }
    })());
});

self.addEventListener("fetch", (event) => {
    const request = event.request;
    if (!request || request.method !== "GET") return;

    const requestUrl = new URL(request.url);
    // Never attempt to cache external requests or anything outside this app.
    if (requestUrl.origin !== self.location.origin ||
        !requestUrl.pathname.startsWith(new URL(self.registration.scope).pathname)) {
        return;
    }

    if (request.mode === "navigate") {
        event.respondWith((async () => {
            try {
                const response = await fetch(request);
                if (response && response.ok) {
                    const cache = await caches.open(CACHE_NAME);
                    await cache.put(new Request(assetUrl("index.html").href), response.clone());
                }
                return response;
            } catch (error) {
                const cache = await caches.open(CACHE_NAME);
                const cachedPage = await cache.match(new Request(assetUrl("index.html").href));
                if (cachedPage) return cachedPage;
                return new Response(
                    "<!doctype html><html lang=\"en\"><meta charset=\"utf-8\"><meta name=\"viewport\" content=\"width=device-width,initial-scale=1\"><title>Offline</
title><body><h1>Structure Analysis is not cached yet</h1><p>Connect to the internet and reload this page once to prepare offline access.</p></body></html>",
                    { status: 503, headers: { "Content-Type": "text/html; charset=utf-8" } }
                );
            }
        })());
        return;
    }

    event.respondWith((async () => {
        const cache = await caches.open(CACHE_NAME);
        try {
            // Network-first keeps the hosted files fresh while online.
            const response = await fetch(request);
            if (response && response.ok && response.type !== "opaque") {
                await cache.put(canonicalRequest(request), response.clone());
            }
            return response;
        } catch (error) {
            // Strip query strings to support the app's cache-busted script URLs.
            const cached = await cache.match(request, { ignoreSearch: true });
            if (cached) return cached;
            return new Response(
                "This resource is not in the offline cache yet. Reconnect and reload the app to cache it.",
                { status: 503, headers: { "Content-Type": "text/plain; charset=utf-8" } }
            );
        }
    })());
```

```
});
```

44. tests/grid-batch-pdf-cases.test.cjs

```
"use strict";

const test = require("node:test");
const assert = require("node:assert/strict");
const fs = require("node:fs");
const path = require("node:path");

const root = path.resolve(__dirname, "..");
const homeSource = fs.readFileSync(path.join(root, "modules/home/home.js"), "utf8");
const start = homeSource.indexOf("function chooseUnambiguousBatchCases(modelData) {");
const end = homeSource.indexOf("// runAnalysisCore uses a mode-specific success message", start);
assert.notEqual(start, -1, "batch case selector should exist");
assert.notEqual(end, -1, "batch case selector end marker should exist");
const helperSource = homeSource.slice(start, end);
const buildHelpers = new Function(
    "normalizeLegacySelfWeightModel",
    helperSource + "\nreturn { chooseUnambiguousBatchCases, chooseBatchCaseCombinations };"
);
const { chooseBatchCaseCombinations } = buildHelpers((model) => model);

const successStart = homeSource.indexOf("function isBatchAnalysisSuccessMessage(message) {");
const successEnd = homeSource.indexOf("// Loads a model's input data into the working model", successStart);
assert.notEqual(successStart, -1, "silent batch success recognizer should exist");
assert.notEqual(successEnd, -1, "silent batch success recognizer end marker should exist");
const isBatchAnalysisSuccessMessage = new Function(
    homeSource.slice(successStart, successEnd) + "\nreturn isBatchAnalysisSuccessMessage;"
)();

function gridModel() {
    return {
        analysisMode: "grid",
        elements: [{ id: "E1" }, { id: "E2" }],
        loadCases: [
            { id: "LC1", type: "Dead" },
            { id: "LC2", type: "Live" },
            { id: "SW1", type: "Self Weight", memberSelfWeights: { E1: 2, E2: 0 } }
        ],
        loads: [
            { id: "L1", loadCaseId: "LC1" },
            { id: "L2", loadCaseId: "LC2" }
        ],
        momentCases: [{ id: "MC1" }, { id: "MC2" }],
        moments: [
            { id: "M1", momentCaseId: "MC1" },
            { id: "M2", momentCaseId: "MC2" }
        ]
    };
}

test("Grid batch PDF plans every applicable load/torsion case combination", () => {
    const result = chooseBatchCaseCombinations(gridModel());
    assert.equal(result.ok, true);
    assert.equal(result.choices.length, 6);
    assert.deepEqual(result.choices.map(({ loadCaseId, momentCaseId }) => [loadCaseId, momentCaseId]), [
        ["LC1", "MC1"], ["LC1", "MC2"],
        ["LC2", "MC1"], ["LC2", "MC2"],
        ["SW1", "MC1"], ["SW1", "MC2"]
    ]);
});

test("Grid batch PDF does not duplicate a Self Weight case that also has explicit loads", () => {
    const model = gridModel();
    model.loads.push({ id: "L3", loadCaseId: "SW1" });
    const result = chooseBatchCaseCombinations(model);
    assert.equal(result.ok, true);
    assert.equal(result.choices.length, 6);
    assert.equal(result.choices.filter((choice) => choice.loadCaseId === "SW1").length, 2);
});

test("Grid batch PDF keeps the one-case model path to one report", () => {
    const model = gridModel();
    model.loadCases = [{ id: "LC1", type: "Dead", memberSelfWeights: {} }];
    model.loads = [{ id: "L1", loadCaseId: "LC1" }];
    model.momentCases = [{ id: "MC1" }];
    model.moments = [{ id: "M1", momentCaseId: "MC1" }];
    const result = chooseBatchCaseCombinations(model);
    assert.equal(result.ok, true);
    assert.deepEqual(result.choices.map(({ loadCaseId, momentCaseId }) => [loadCaseId, momentCaseId]), [["LC1", "MC1"]]);
});

test("Plane-model batch PDF preserves the existing ambiguity safeguard", () => {
    const model = gridModel();
    model.analysisMode = "plane";
    const result = chooseBatchCaseCombinations(model);
    assert.equal(result.ok, false);
```

```
    assert.match(result.message, /cannot safely guess between multiple load cases/);
});

test("Grid batch PDF includes applicable self-weight when there are no explicit loads", () => {
    const model = gridModel();
    model.loads = [];
    model.loadCases = [
        { id: "LC1", type: "Dead" },
        { id: "SW1", type: "Self Weight", memberSelfWeights: { E1: 2, E2: 0 } }
    ];
    model.moments = [];
    const result = chooseBatchCaseCombinations(model);
    assert.equal(result.ok, true);
    assert.deepEqual(result.choices.map(({ loadCaseId, momentCaseId }) => [loadCaseId, momentCaseId]), [["SW1", ""]]);
});

test("silent batch export recognizes both plane and Grid success messages", () => {
    assert.equal(isBatchAnalysisSuccessMessage("Analysis completed successfully using the LPW-3 Stiffness Method approach."), true);
    assert.equal(isBatchAnalysisSuccessMessage("Grid analysis completed successfully using the Grid Stiffness Method."), true);
    assert.equal(isBatchAnalysisSuccessMessage("Analysis results were computed, but the global equilibrium check failed."), false);
});


test("Grid batch PDF refuses to use zero for a Self Weight element that was never configured", () => {
    const model = gridModel();
    model.loadCases = [{ id: "SW1", type: "Self Weight", memberSelfWeights: { E1: 2 } }];
    model.loads = [];
    model.moments = [];
    const result = chooseBatchCaseCombinations(model);
    assert.equal(result.ok, false);
    assert.match(result.message, /Self Weight case "SW1".*element "E2"/);
});

test("Plane batch PDF also refuses incomplete per-element Self Weight data", () => {
    const model = gridModel();
    model.analysisMode = "plane";
    model.loadCases = [{ id: "SW1", type: "Self Weight", memberSelfWeights: { E1: 2 } }];
    model.loads = [];
    model.moments = [];
    const result = chooseBatchCaseCombinations(model);
    assert.equal(result.ok, false);
    assert.match(result.message, /Self Weight case "SW1".*element "E2"/);
});

test("Batch PDF flags a missing Self Weight even when all configured weights are zero", () => {
    for (const mode of ["plane", "grid"]) {
        const model = gridModel();
        model.analysisMode = mode;
        model.loadCases = [{ id: "SW1", type: "Self Weight", memberSelfWeights: { E1: 0 } }];
        model.loads = [];
        model.moments = [];
        const result = chooseBatchCaseCombinations(model);
        assert.equal(result.ok, false, mode);
        assert.match(result.message, /Self Weight case "SW1".*element "E2"/);
    }
});
```

45. tests/non-calculation-regressions.test.cjs

```
"use strict";

const test = require("node:test");
const assert = require("node:assert/strict");
const fs = require("node:fs");
const path = require("node:path");

const root = path.resolve(__dirname, "..");
const mainSource = fs.readFileSync(path.join(root, "assets/js/main.js"), "utf8");
const resultsForDiagramLabelsSource = fs.readFileSync(path.join(root, "modules/results/results.js"), "utf8");
const step3ThermalSource = fs.readFileSync(path.join(root, "modules/results/step-3/step-3.js"), "utf8");
const loadModuleSource = fs.readFileSync(path.join(root, "modules/load/load.js"), "utf8");
const step4Source = fs.readFileSync(path.join(root, "modules/results/step-4/step-4.js"), "utf8");
const diagramsForDiagramLabelsSource = fs.readFileSync(path.join(root, "modules/results/diagrams/diagrams.js"), "utf8");
const indexSource = fs.readFileSync(path.join(root, "index.html"), "utf8");

function section(source, startMarker, endMarker) {
    const start = source.indexOf(startMarker);
    const end = source.indexOf(endMarker, start);
    assert.notEqual(start, -1, "missing source marker: " + startMarker);
    assert.notEqual(end, -1, "missing end marker: " + endMarker);
    return source.slice(start, end).trim();
}

const builtInSupportTemplates = [
    { id: "FIXED", restraints: { RFx: true, RFy: true, RFz: true, RMx: true, RMy: true, RMz: true }, displacement: { dx: 0, dy: 0, dz: 0, tx: 0, ty: 0, tz: 0 } },
    { id: "ROLLER", restraints: { RFx: false, RFy: true, RFz: false, RMx: false, RMy: false, RMz: false }, displacement: { dx: 0, dy: 0, dz: 0, tx: 0, ty: 0, tz: 0 } },
    { id: "HINGED", restraints: { RFx: true, RFy: true, RFz: false, RMx: false, RMy: false, RMz: false }, displacement: { dx: 0, dy: 0, dz: 0, tx: 0, ty: 0, tz: 0 } }
];

const validatorSource = section(mainSource, "function validatePersistedInputModel(source) {", "\nfunction resetAllEditForms()");
const thermalActionSource = section(step3ThermalSource, "function calculatePlaneThermalFixedEndActions(", "\nfunction renderStep3(");
const calculatePlaneThermalFixedEndActions = new Function("getElementEInternal",
    thermalActionSource + "\nreturn calculatePlaneThermalFixedEndActions;"
)((template) => Number(template.E_kN_m2));
const thermalVectorAssemblySource = section(step3ThermalSource,
    "function assembleThermalEquivalentNodalActions(", "\n// Return the six local fixed-end actions");
const assembleThermalEquivalentNodalActions = new Function("Matrix",
    thermalVectorAssemblySource + "\nreturn assembleThermalEquivalentNodalActions;"
)({
    mul: (A, B) => A.map((row) => [row.reduce((sum, value, index) => sum + value * B[index][0], 0)])
});
const globalLoadAssemblySource = section(step4Source,
    "function assembleGlobalLoadVector(", "\nfunction solveStep4(");
const assembleGlobalLoadVector = new Function("Matrix",
    globalLoadAssemblySource + "\nreturn assembleGlobalLoadVector;"
)({
    add: (A, B) => A.map((row, index) => [row[0] + B[index][0]])
});
const validate = new Function("AppConfig", "normalizeLegacySelfWeightModel", "normalizeLegacySupportMovementAssignments",
    validatorSource + "\nreturn validatePersistedInputModel;"
)({
    persistedModelKeys: ["nodes", "elementTemplates", "elements", "supportTemplates", "supportAssignments", "loadCases", "loads", "momentCases", "moments"],
    builtInSupportTemplates
}, (value) => value, (value) => value);

function baseModel(mode) {
    const grid = mode === "grid";
    return {
        analysisMode: mode || "plane",
        nodes: [{ id: "N1", x: 0, y: 0, z: 0 }, { id: "N2", x: 4, y: 0, z: 0 }],
        elementTemplates: [grid ? { id: "T1", I: 1, E_kN_m2: 1_000_000, G_GPa: 1, J: 1 } : { id: "T1", I: 1, E_kN_m2: 1_000_000, A: 1 }],
        elements: [{ id: "E1", templateId: "T1", startNode: "N1", endNode: "N2" }],
        supportTemplates: [], supportAssignments: [],
        loadCases: [{ id: "LC1", type: "Dead" }], loads: [],
        momentCases: [{ id: "MC1", type: "Dead" }], moments: []
    };
}

test("Plane import accepts blank ignored Mx/My when active Mz is valid", () => {
    const model = baseModel("plane");
    model.moments = [{ id: "M1", momentCaseId: "MC1", type: "Nodal Moment", nodeId: "N1", Mx: "", My: "", Mz: 5 }];
    assert.equal(validate(model), true);
});

test("Grid import still requires active Mx/My components", () => {
    const model = baseModel("grid");
    model.moments = [{ id: "M1", momentCaseId: "MC1", type: "Nodal Moment", nodeId: "N1", Mx: "", My: 5, Mz: 0 }];
    assert.throws(() => validate(model), /Mx must be a finite number/);
});

test("UDL import accepts missing end distance and end intensities using UI defaults", () => {
    const model = baseModel("plane");
    model.loads = [{ id: "L1", loadCaseId: "LC1", type: "Element UDL", elementId: "E1", distance: 1, distanceEnd: "", Fx: 0, Fy: -2 }];
    assert.equal(validate(model), true);
```

```
});

test("import rejects numeric references instead of accepting string-coercion matches", () => {
    const model = baseModel("plane");
    model.elements[0].startNode = 1;
    assert.throws(() => validate(model), /References must match an existing string ID exactly/);
});

test("custom supports cannot shadow reserved built-in support IDs", () => {
    const model = baseModel("plane");
    model.supportTemplates = [{
        id: "FIXED", description: "Custom", type: "custom", builtIn: false,
        restraints: { RFx: false, RFy: false, RFz: false, RMx: false, RMy: false, RMz: false },
        displacement: { dx: 0, dy: 0, dz: 0, tx: 0, ty: 0, tz: 0 }
    }];
    assert.throws(() => validate(model), /reserved for a built-in support/);
});

test("import rejects unsafe __proto__ entity IDs", () => {
    const model = baseModel("plane");
    model.nodes[0].id = "__proto__";
    assert.throws(() => validate(model), /reserved ID "__proto__"/);
});

test("malformed saved-library data is preserved and a save is refused", () => {
    const getLibrarySource = section(mainSource, "function getModelLibrary() {", "\nfunction getModelLibraryReadFailureMessage()");
    const readMessageSource = section(mainSource, "function getModelLibraryReadFailureMessage() {", "\nfunction setModelLibrary(");
    const saveSource = section(mainSource, "function saveCurrentModelToLibrary(silent = false) {", "\nfunction saveModel()");
    const harnessCode = [
        "let lastModelLibraryReadFailed = false;",
        "let lastModelLibraryWriteSucceeded = true;",
        "let lastModelSaveStorageFailed = false;",
        "let currentModelLibraryId = '';",
        "const stored = { structuralModels: rawValue };",
        "let writeCount = 0;",
        "const safeStorageGetItem = (key) => Object.prototype.hasOwnProperty.call(stored, key) ? stored[key] : null;",
        "const safeStorageSetItem = (key, value) => { writeCount++; stored[key] = String(value); return true; };",
        "const safeStorageRemoveItem = (key) => { writeCount++; delete stored[key]; return true; };",
        "const normalizeModelLibraryRecords = (source) => ({ library: source, changed: false });",
        "const normalizeLegacySelfWeightModel = (value) => value;",
        "const modelNameKey = (value) => String(value || '').replace(/\\s+/g, '').toLowerCase();",
        "const getPersistedInputModel = () => ({});",
        "const getModelName = () => 'Test Model';",
        "const findDuplicateModel = () => null;",
        "const createModelLibraryId = () => 'test-id';",
        "const setModelLibrary = (value) => { writeCount++; stored.structuralModels = JSON.stringify(value); lastModelLibraryWriteSucceeded = true; return value; };",
        "const renderModelLibrary = () => {};",
        "const alert = () => {};",
        "const console = { warn: () => {} };",
        "const document = { getElementById: () => null };",
        getLibrarySource,
        readMessageSource,
        saveSource,
        "return { save: () => saveCurrentModelToLibrary(true), state: () => ({ readFailed: lastModelLibraryReadFailed, saveFailed: lastModelSaveStorageFailed, writeCount, rawValue: stored.
structuralModels }) };"
    ].join("\n");
    const factory = new Function("rawValue", harnessCode);
    const badValues = ["{ definitely not valid JSON", JSON.stringify({ brokenRecord: null })];
    for (const rawBad of badValues) {
        const harness = factory(rawBad);
        assert.equal(harness.save(), false);
        const state = harness.state();
        assert.equal(state.readFailed, true);
        assert.equal(state.saveFailed, true);
        assert.equal(state.writeCount, 0);
        assert.equal(state.rawValue, rawBad);
    }
});

test("application JavaScript files parse", () => {
    const files = ["assets/js/main.js", "modules/home/home.js", "modules/nodes/nodes.js", "modules/elements/elements.js", "modules/load-case/load-case.js", "modules/load/load.js", "modules/
moment-case/moment-case.js", "modules/moment/moment.js", "modules/results/results.js", "modules/results/diagrams/diagrams.js"];
    for (const relativePath of files) {
        const source = fs.readFileSync(path.join(root, relativePath), "utf8");
        assert.doesNotThrow(() => new Function(source), relativePath + " should parse");
    }
});


test("import rejects nodes that share the same X/Y coordinates", () => {
    const model = baseModel("plane");
    model.nodes.push({ id: "N3", x: 4, y: 0, z: 0 });
    assert.throws(() => validate(model), /share the same X\/Y coordinates/);
});

test("next ID preserves the existing numeric suffix width", () => {
    const nextIdSource = section(mainSource, "function getNextId(currentId) {", "\nfunction val(");
```

```
    const getNextId = new Function(nextIdSource + "\nreturn getNextId;")();
    assert.equal(getNextId("N09"), "N10");
    assert.equal(getNextId("N009"), "N010");
    assert.equal(getNextId("E99"), "E100");
});

test("live node validation rejects duplicate coordinates while allowing the edited node's own coordinates", () => {
    const nodeValidationSource = section(mainSource, "function validateNodeInput(id, x, y, z, editingId = null) {", "\n//
==========================================================================\n// 4. MATRIX ALGEBRA UTILITIES");
    const model = { nodes: [{ id: "N1", x: 0, y: 0 }, { id: "N2", x: 4, y: 0 }] };
    const hasId = (items, id, editingId = null) => items.some((item) => item.id === id && item.id !== editingId);
    const requireFinite = (value, label) => Number.isFinite(Number(value)) ? null : `${label} must be finite.`;
    const validateNodeInput = new Function("model", "hasId", "requireFinite", nodeValidationSource + "\nreturn validateNodeInput;")(model, hasId, requireFinite);
    assert.match(validateNodeInput("N3", 4, 0, 0), /unique X\/Y coordinates/);
    assert.equal(validateNodeInput("N2", 4, 0, 0, "N2"), null);
});

test("moving a node cannot leave point loads, UDLs, or member moments outside the new element length", () => {
    const nodesSource = fs.readFileSync(path.join(root, "modules/nodes/nodes.js"), "utf8");
    const movementSource = section(nodesSource, "function findNodeMoveDistanceViolation(nodeId, x, y) {", "\nfunction updateNode(");
    const model = {
        nodes: [{ id: "N1", x: 0, y: 0 }, { id: "N2", x: 4, y: 0 }],
        elements: [{ id: "E1", startNode: "N1", endNode: "N2" }],
        loads: [{ id: "L1", type: "Element Point Load", elementId: "E1", distance: 3 }],
        moments: []
    };
    const findViolation = new Function("model", movementSource + "\nreturn findNodeMoveDistanceViolation;")(model);
    assert.match(findViolation("N2", 2, 0), /load "L1"/);
    assert.equal(findViolation("N2", 5, 0), null);
    model.loads = [{ id: "L2", type: "Element UDL", elementId: "E1", distance: 1, distanceEnd: 3.5 }];
    assert.match(findViolation("N2", 3, 0), /UDL "L2"/);
    model.loads = [];
    model.moments = [{ id: "M1", type: "Element Moment", elementId: "E1", distance: 3.5 }];
    assert.match(findViolation("N2", 3, 0), /moment\/torsion "M1"/);
});

test("missing self-weight values are detected; explicitly entered zero is valid", () => {
    const helperSource = section(mainSource, "function getMissingSelfWeightElement(loadCase, elements = model.elements) {", "\nfunction getModelName()");
    const model = { elements: [{ id: "E1" }, { id: "E2" }] };
    const getMissing = new Function("model", helperSource + "\nreturn getMissingSelfWeightElement;")(model);
    const loadCase = { id: "SW1", type: "Self Weight", memberSelfWeights: { E1: 2 } };
    assert.equal(getMissing(loadCase).id, "E2");
    loadCase.memberSelfWeights.E2 = 0;
    assert.equal(getMissing(loadCase), null);
    assert.equal(getMissing({ type: "Dead", memberSelfWeights: {} }), null);
});

test("changing a Self Weight load case to another type clears stale member weights", () => {
    const loadCaseSource = fs.readFileSync(path.join(root, "modules/load-case/load-case.js"), "utf8");
    const updateSource = section(loadCaseSource, "function updateLoadCase(existingId, data) {", "\ndocument.getElementById(\"loadCaseForm\")");
    const loadCase = { id: "SW1", type: "Self Weight", description: "Weights", memberSelfWeights: { E1: 3 } };
    const model = { loadCases: [loadCase], loads: [] };
    let activeLoadCaseId = "SW1";
    let invalidated = false;
    const update = new Function("model", "activeLoadCaseId", "markAnalysisInvalidated", updateSource + "\nreturn updateLoadCase;")(
        model,
        activeLoadCaseId,
        () => { invalidated = true; }
    );
    assert.equal(update("SW1", { id: "LC1", type: "Dead", description: "Dead loads" }), true);
    assert.equal(Object.prototype.hasOwnProperty.call(loadCase, "memberSelfWeights"), false);
    assert.equal(model.loadCases[0].id, "LC1");
    assert.equal(invalidated, true);
});

test("navigation no longer rerenders every model table and diagram", () => {
    const switchSource = section(mainSource, "function switchSection(id, btnEl) {", "\nfunction goHome()");
    assert.doesNotMatch(switchSource, /refreshAll\s*\(/);
});

test("Clear Model resets analysis mode and item deletion resets the matching edit form", () => {
    const clearSource = section(mainSource, "function clearModel() {", "\n// ============================================================================\n// 7. NAVIGATION");
    assert.match(clearSource, /model\.analysisMode\s*=\s*"plane"/);
    const deleteSource = section(mainSource, "function deleteItem(collection, encodedValue, property) {", "\nfunction refreshAll()");
    for (const identifier of ["editingNodeId", "editingElementTemplateId", "editingElementId", "editingSupportTemplateId", "editingSupportAssignmentId", "editingLoadCaseId", "editingLoadId",
 "editingMomentCaseId", "editingMomentId"]) {
        assert.match(deleteSource, new RegExp(identifier));
    }
});

test("single-model import offers replacing without saving and does not abort on Cancel", () => {
    const importSource = section(mainSource, "function importModel(event) {", "\nfunction resetActiveModelAfterSavedModelDeletion()");
    assert.match(importSource, /Choose OK to save the current model first, or Cancel to replace it without saving/);
    assert.match(importSource, /if \(saveBeforeReplace\)\s*\{/);
    assert.doesNotMatch(importSource, /if\s*\(!saveBeforeReplace\)\s*return/);
});
test("saving no longer writes a second full-model copy to legacy storage", () => {
```

```
    const saveSource = section(mainSource, "function saveCurrentModelToLibrary(silent = false) {", "\nfunction saveModel()");
    assert.doesNotMatch(saveSource, /safeStorageSetItem\("structuralInputModel"/);
    assert.match(saveSource, /safeStorageRemoveItem\("structuralInputModel"\)/);
    assert.match(saveSource, /lastModelLibraryWriteSucceeded\s*&&/);
});

test("Plane import ignores blank Fz when the active Fx/Fy values are valid", () => {
    const model = baseModel("plane");
    model.loads = [{ id: "L1", loadCaseId: "LC1", type: "Nodal Load", nodeId: "N1", Fx: 1, Fy: -2, Fz: "" }];
    assert.equal(validate(model), true);
});

test("storage failure is reported as failure and never as a successful save", () => {
    const getLibrarySource = section(mainSource, "function getModelLibrary() {", "\nfunction getModelLibraryReadFailureMessage()");
    const readMessageSource = section(mainSource, "function getModelLibraryReadFailureMessage() {", "\nfunction setModelLibrary(");
    const saveSource = section(mainSource, "function saveCurrentModelToLibrary(silent = false) {", "\nfunction saveModel()");
    const harnessCode = [
        "let lastModelLibraryWriteSucceeded = true;",
        "let lastModelSaveStorageFailed = false;",
        "let lastModelLibraryReadFailed = false;",
        "let currentModelLibraryId = '';",
        "const stored = { structuralModels: '{}' };",
        "const alerts = [];",
        "const safeStorageGetItem = (key) => Object.prototype.hasOwnProperty.call(stored, key) ? stored[key] : null;",
        "const safeStorageSetItem = (key, value) => false;",
        "const safeStorageRemoveItem = (key) => { delete stored[key]; return true; };",
        "const normalizeModelLibraryRecords = (source) => ({ library: source, changed: false });",
        "const normalizeLegacySelfWeightModel = (value) => value;",
        "const modelNameKey = (value) => String(value || '').replace(/\\s+/g, '').toLowerCase();",
        "const getPersistedInputModel = () => ({ nodes: [{ id: 'N1' }] });",
        "const getModelName = () => 'Test Model';",
        "const findDuplicateModel = () => null;",
        "const createModelLibraryId = () => 'test-id';",
        "const setModelLibrary = (value) => { lastModelLibraryWriteSucceeded = safeStorageSetItem('structuralModels', JSON.stringify(value)); return value; };",
        "const getAvailableModelName = (name) => name;",
        "const renderModelLibrary = () => {};",
        "const alert = (message) => alerts.push(String(message));",
        "const console = { warn: () => {} };",
        "const document = { getElementById: () => ({ value: 'Test Model' }) };",
        getLibrarySource,
        readMessageSource,
        saveSource,
        "return { save: () => saveCurrentModelToLibrary(false), state: () => ({ failed: lastModelSaveStorageFailed, alerts, stored }) };"
    ].join("\n");
    const harness = new Function(harnessCode)();
    assert.equal(harness.save(), false);
    const state = harness.state();
    assert.equal(state.failed, true);
    assert.ok(state.alerts.some((message) => /NOT confirmed saved/.test(message)));
    assert.ok(state.alerts.every((message) => !/saved in browser storage/.test(message)));
});


test("Grid My diagram is described as torsion in selectors, legends, and PDF output", () => {
    assert.match(mainSource, /bmd: "Torsion \(My\)"/);
    assert.doesNotMatch(mainSource, /bmd: "Bending Moment \(My\)"/);
    assert.match(mainSource, /afd: "Torsion \(Mx\)", sfd: "Vertical Shear \(Fz\)", bmd: "Torsion \(My\)"/);
    assert.match(mainSource, /bmd: "BMD"/, "the plane-model selector must retain its original label");

    assert.match(resultsForDiagramLabelsSource, /bmd: "Torsion Diagram \(My\)"/);
    assert.match(resultsForDiagramLabelsSource, /Step 8: Grid Torsion \(Mx, My\) and Vertical Shear \(Fz\) Diagrams/);
    assert.match(resultsForDiagramLabelsSource, /bmd: "Bending Moment Diagram \(BMD\)"/, "plane-model PDF naming must remain unchanged");

    assert.match(diagramsForDiagramLabelsSource, /'Grid My torsion diagram'/);
    assert.match(diagramsForDiagramLabelsSource, /'Positive My torsion'/);
    assert.match(diagramsForDiagramLabelsSource, /'Negative My torsion'/);
    assert.match(diagramsForDiagramLabelsSource, /'Grid torsion about Y'/);
    assert.match(diagramsForDiagramLabelsSource, /Mx \/ My — reaction torsions \(kN·m\)/);
});


test("Grid diagram labels refresh at fullscreen open and corrected scripts bypass stale cache", () => {
    const helperSource = section(mainSource, "function syncDiagramModeLabels() {", "\nfunction syncAnalysisModeUI() {");
    const makeSelect = () => ({
        options: ["afd", "sfd", "bmd", "displacement"].map((value) => ({ value, textContent: value }))
    });
    const selects = {
        resultsDiagramMode: makeSelect(),
        fsModeSelect: makeSelect()
    };
    const fakeDocument = { getElementById: (id) => selects[id] || null };

    const syncGridLabels = new Function("isGridAnalysisMode", "document",
        helperSource + "\nreturn syncDiagramModeLabels;"
    )(() => true, fakeDocument);
    syncGridLabels();
    for (const select of Object.values(selects)) {
```

```
        assert.equal(select.options.find((option) => option.value === "afd").textContent, "Torsion (Mx)");
        assert.equal(select.options.find((option) => option.value === "sfd").textContent, "Vertical Shear (Fz)");
        assert.equal(select.options.find((option) => option.value === "bmd").textContent, "Torsion (My)");
    }

    const openSource = section(diagramsForDiagramLabelsSource, "function openFullScreen(source) {", "\nfunction changeFSMode(mode) {");
    assert.match(openSource, /syncDiagramModeLabels\(\)/);
    assert.ok(openSource.indexOf("syncDiagramModeLabels()") < openSource.indexOf('getElementById("fsModeSelect").value'));

    const versionByFile = {
        "assets/js/main.js": "v=20261012-temperature-loads-v1",
        "modules/elements/elements.js": "v=20261012-temperature-loads-v1",
        "modules/load/load.js": "v=20261012-temperature-loads-v2-edit-fields",
        "modules/results/step-1/step-1.js": "v=20261012-temperature-loads-v1",
        "modules/results/step-2/step-2.js": "v=20261012-element-E-kn-m2",
        "modules/results/step-3/step-3.js": "v=20261012-thermal-AT-v1",
        "modules/results/step-4/step-4.js": "v=20261012-thermal-AT-v1",
        "modules/results/diagrams/diagrams.js": "v=20261012-element-E-kn-m2",
        "modules/results/results.js": "v=20261011-grid-my-torsion-refresh"
    };
    for (const [file, version] of Object.entries(versionByFile)) {
        assert.ok(indexSource.includes(file + "?" + version), "missing cache-busted script reference for " + file);
    }
});


test("uniform thermal action uses EA alpha delta-T and Ignore X suppresses it", () => {
    const template = { E_kN_m2: 200_000_000, A: 0.02, I: 0.00008, alpha: 12e-6, thermalDepth: 0.3 };
    const load = { type: "Uniform Temperature Change", temperatureDelta: 30 };
    const actions = calculatePlaneThermalFixedEndActions(template, load, false);
    assert.equal(actions[0], 1440);
    assert.equal(actions[3], -1440);
    assert.deepEqual(actions.slice(1, 3), [0, 0]);
    assert.deepEqual(calculatePlaneThermalFixedEndActions(template, load, true), [0, 0, 0, 0, 0, 0]);
});

test("temperature gradient creates equal-and-opposite fixed-end moments", () => {
    const template = { E_kN_m2: 200_000_000, A: 0.02, I: 0.00008, alpha: 12e-6, thermalDepth: 0.3 };
    const actions = calculatePlaneThermalFixedEndActions(template, {
        type: "Temperature Gradient", temperatureT1: 0, temperatureT2: 30
    }, true);
    assert.ok(Math.abs(actions[2] + 19.2) < 1e-10);
    assert.ok(Math.abs(actions[5] - 19.2) < 1e-10);
    assert.deepEqual([actions[0], actions[1], actions[3], actions[4]], [0, 0, 0, 0]);
});

test("temperature gradient with equal face temperatures produces no thermal moment", () => {
    const template = { E_kN_m2: 200_000_000, I: 0.00008, alpha: 12e-6, thermalDepth: 0.3 };
    const actions = calculatePlaneThermalFixedEndActions(template, {
        type: "Temperature Gradient", temperatureT1: 20, temperatureT2: 20
    }, false);
    assert.equal(Math.abs(actions[2]), 0);
    assert.equal(Math.abs(actions[5]), 0);
});

test("import accepts a valid uniform thermal load on an explicit Beam", () => {
    const model = baseModel("plane");
    model.elementTemplates[0] = { id: "T1", memberType: "beam", I: 1, E_kN_m2: 200_000_000, A: 0.02, alpha: 12e-6, thermalDepth: 0.3 };
    model.loads = [{ id: "TL1", loadCaseId: "LC1", type: "Uniform Temperature Change", elementId: "E1", temperatureDelta: 30 }];
    assert.equal(validate(model), true);
});

test("import rejects thermal loads without a positive expansion coefficient", () => {
    const model = baseModel("plane");
    model.elementTemplates[0] = { id: "T1", memberType: "frame", I: 1, E_kN_m2: 200_000_000, A: 0.02, alpha: 0, thermalDepth: 0.3 };
    model.loads = [{ id: "TL1", loadCaseId: "LC1", type: "Uniform Temperature Change", elementId: "E1", temperatureDelta: 30 }];
    assert.throws(() => validate(model), /alpha must be greater than 0/);
});

test("import accepts a valid temperature gradient and validates section depth", () => {
    const model = baseModel("plane");
    model.elementTemplates[0] = { id: "T1", memberType: "frame", I: 0.00008, E_kN_m2: 200_000_000, A: 0.02, alpha: 12e-6, thermalDepth: 0.3 };
    model.loads = [{ id: "TL1", loadCaseId: "LC1", type: "Temperature Gradient", elementId: "E1", temperatureT1: 30, temperatureT2: 10 }];
    assert.equal(validate(model), true);
    model.elementTemplates[0].thermalDepth = 0;
    assert.throws(() => validate(model), /section depth d must be greater than 0/);
});

test("Grid import rejects plane-member temperature loads", () => {
    const model = baseModel("grid");
    model.loads = [{ id: "TL1", loadCaseId: "LC1", type: "Uniform Temperature Change", elementId: "E1", temperatureDelta: 30 }];
    assert.throws(() => validate(model), /only supported for Beam \/ Truss \/ Frame models/);
});


test("thermal member actions are transformed into a separate global AT vector", () => {
    const template = { E_kN_m2: 200_000_000, A: 0.02, I: 0.00008, alpha: 12e-6, thermalDepth: 0.3 };
```

```
    const thermalAML = calculatePlaneThermalFixedEndActions(template, {
        type: "Uniform Temperature Change", temperatureDelta: 30
    }, false);
    const AT = Array.from({ length: 6 }, () => [0]);
    const identity = Array.from({ length: 6 }, (_, row) =>
        Array.from({ length: 6 }, (_, col) => row === col ? 1 : 0));
    const thermalAMS = assembleThermalEquivalentNodalActions(AT, identity, thermalAML.map((value) => [value]), [0, 1, 2, 3, 4, 5]);

    assert.equal(thermalAMS[0][0], 1440);
    assert.equal(thermalAMS[3][0], -1440);
    assert.deepEqual(AT.map((row) => row[0]), [-1440, 0, 0, 1440, 0, 0]);
    // [AT] is a dedicated vector; this assembly helper never mutates [AE].
    assert.match(step3ThermalSource, /return \{ SJ, AE, AJ, AT, html \}/);
    assert.match(step3ThermalSource, /AE\[idxs\[r\]\]\[0\] -= mechanicalAMS\[r\]\[0\]/);
});

test("Step 4 assembles AT exactly once into AC with the mechanical and prescribed vectors", () => {
    const AE = [[1], [2]];
    const AJ = [[3], [4]];
    const AD = [[5], [6]];
    const AT = [[7], [8]];
    const result = assembleGlobalLoadVector(AE, AJ, AD, AT);

    assert.deepEqual(result.physicalAC, [[4], [6]]);
    assert.deepEqual(result.AC, [[16], [20]]);
    assert.match(step4Source, /\[AC\] = \[AE\] \+ \[AJ\] \+ \[AD\] \+ \[AT\]/);
    assert.match(step4Source, /\{ m: AT, t: "\[AT\] \(Thermal Actions\)"/);
});

test("Grid Step 3 supplies a zero AT vector and does not change Grid loading", () => {
    assert.match(step3ThermalSource, /Grid models do not currently accept temperature loads, so \[AT\] is zero/);
    assert.match(step3ThermalSource, /return \{ SJ, AE, AJ, AT: Matrix\.zeros\(AE\.length, 1\), html \}/);
});

test("editing a saved temperature-gradient load restores and reveals both T1/T2 inputs", () => {
    assert.match(loadModuleSource, /id="loadTemperatureT1"/);
    assert.match(loadModuleSource, /id="loadTemperatureT2"/);
    const editSource = section(loadModuleSource, "function editLoad(encodedId) {", "function renderLoads()");
    const restoredType = editSource.indexOf('document.getElementById("loadType").value = load.type;');
    const restoredT1 = editSource.indexOf('document.getElementById("loadTemperatureT1").value = load.temperatureT1 ?? 0;');
    const restoredT2 = editSource.indexOf('document.getElementById("loadTemperatureT2").value = load.temperatureT2 ?? 0;');
    const refreshFields = editSource.indexOf("toggleLoadTarget();");
    assert.ok(restoredType >= 0 && restoredT1 > restoredType && restoredT2 > restoredT1,
        "editLoad should restore load type and both stored face temperatures");
    assert.ok(refreshFields > restoredT2,
        "editLoad must refresh conditional visibility after restoring the thermal inputs");
});
```

46. tests/offline-pwa.test.cjs

```
"use strict";

const test = require("node:test");
const assert = require("node:assert/strict");
const fs = require("node:fs");
const path = require("node:path");
const vm = require("node:vm");

const root = path.resolve(__dirname, "..");
const indexSource = fs.readFileSync(path.join(root, "index.html"), "utf8");
const workerSource = fs.readFileSync(path.join(root, "service-worker.js"), "utf8");
const manifestSource = fs.readFileSync(path.join(root, "manifest.webmanifest"), "utf8");
const offlineUiSource = fs.readFileSync(path.join(root, "assets/js/offline.js"), "utf8");

const coreAssetsMatch = workerSource.match(/const CORE_ASSETS = \[([\s\S]*?)\];/);
assert.ok(coreAssetsMatch, "Service Worker must declare its precached core assets");
const coreAssets = JSON.parse("[" + coreAssetsMatch[1] + "]");

test("PWA manifest is valid and scoped to the deployed application", () => {
    const manifest = JSON.parse(manifestSource);
    assert.equal(manifest.start_url, "./");
    assert.equal(manifest.scope, "./");
    assert.equal(manifest.display, "standalone");
    assert.ok(Array.isArray(manifest.icons) && manifest.icons.length > 0);
    for (const icon of manifest.icons) {
        assert.ok(fs.existsSync(path.join(root, icon.src)), "missing manifest icon: " + icon.src);
        assert.ok(coreAssets.includes(icon.src), "manifest icon must be precached: " + icon.src);
    }
    assert.match(indexSource, /rel="manifest" href="manifest\.webmanifest"/);
});

test("every precached asset exists and all local index resources are included", () => {
    assert.equal(new Set(coreAssets).size, coreAssets.length, "duplicate precache entries are not expected");
    for (const asset of coreAssets) {
        assert.ok(fs.existsSync(path.join(root, asset)), "missing precache asset: " + asset);
    }
    const references = Array.from(indexSource.matchAll(/(?:src|href)="([^"]+)"/g), (match) => match[1]);
    const localReferences = references
        .filter((value) => !/^(?:[a-z]+:|\/\/|#|data:)/i.test(value))
        .map((value) => value.split("?")[0].replace(/^\.\//, ""));
    for (const localPath of localReferences) {
        if (!localPath) continue;
        assert.ok(coreAssets.includes(localPath), "local index resource is missing from precache list: " + localPath);
    }
    assert.match(indexSource, /assets\/js\/offline\.js\?/);
    assert.match(indexSource, /service-worker|manifest\.webmanifest/);
});

test("Grid Step 2 displays computed member stiffness terms with consistent units", () => {
    const step2Source = fs.readFileSync(path.join(root, "modules/results/step-2/step-2.js"), "utf8");
    assert.match(step2Source, /m = GJ\/L \(kN·m\)/);
    assert.match(step2Source, /a = 12EI\/L³ \(kN\/m\)/);
    assert.match(step2Source, /b = 6EI\/L² \(kN\)/);
    assert.match(step2Source, /c = 4EI\/L \(kN·m\)/);
    assert.match(step2Source, /Number\(member\.G_GPa\) \* 1e6/);
    assert.match(step2Source, /const mTerm = valid \? \(G \* J\) \/ L/);
    assert.match(step2Source, /const aTerm = valid \? \(12 \* E \* I\) \/ Math\.pow\(L, 3\)/);
    assert.match(step2Source, /const bTerm = valid \? \(6 \* E \* I\) \/ Math\.pow\(L, 2\)/);
    assert.match(step2Source, /const cTerm = valid \? \(4 \* E \* I\) \/ L/);
});

test("offline support registers silently and adds no visible readiness UI", () => {
    assert.ok(offlineUiSource.includes('new URL("../../", document.currentScript.src)'));
    assert.ok(offlineUiSource.includes("navigator.serviceWorker.register(workerUrl.href")));
    assert.match(offlineUiSource, /serviceWorker/);
    assert.doesNotMatch(offlineUiSource, /createElement|offlineReadinessStatus|Offline ready|Preparing offline access/);
    const mainCss = fs.readFileSync(path.join(root, "assets/css/main.css"), "utf8");
    assert.ok(!mainCss.includes(".offline-readiness"));
});

function createWorkerHarness(fetchImpl) {
    const listeners = new Map();
    const cacheStores = new Map();
    let skipWaitingCalled = false;
    let claimCalled = false;

    function urlOf(value) {
        return new URL(typeof value === "string" ? value : value.url);
    }
    function requestKey(value, ignoreSearch) {
        const url = urlOf(value);
        if (ignoreSearch) {
            url.search = "";
            url.hash = "";
        }
```

```
        return url.href;
    }

    const caches = {
        async open(name) {
            if (!cacheStores.has(name)) cacheStores.set(name, new Map());
            const store = cacheStores.get(name);
            return {
                async put(request, response) {
                    store.set(requestKey(request, false), response.clone());
                },
                async match(request, options = {}) {
                    const exact = store.get(requestKey(request, false));
                    if (exact) return exact.clone();
                    if (options.ignoreSearch) {
                        const base = requestKey(request, true);
                        for (const [key, response] of store.entries()) {
                            if (requestKey(key, true) === base) return response.clone();
                        }
                    }
                    return undefined;
                },
                async keys() {
                    return Array.from(store.keys(), (key) => new Request(key));
                }
            };
        },
        async keys() {
            return Array.from(cacheStores.keys());
        },
        async delete(name) {
            return cacheStores.delete(name);
        },
        async match(request, options) {
            for (const name of cacheStores.keys()) {
                const cache = await this.open(name);
                const found = await cache.match(request, options);
                if (found) return found;
            }
            return undefined;
        }
    };

    const clients = {
        async claim() { claimCalled = true; },
        async matchAll() { return []; }
    };
    const self = {
        location: new URL("https://example.test/Structure-Analysis/service-worker.js"),
        registration: { scope: "https://example.test/Structure-Analysis/" },
        clients,
        addEventListener(type, handler) { listeners.set(type, handler); },
        async skipWaiting() { skipWaitingCalled = true; }
    };

    vm.runInNewContext(workerSource, {
        self, caches, fetch: fetchImpl, URL, Request, Response, Promise, Set
    }, { filename: "service-worker.js" });

    return { listeners, caches, cacheStores, get skipWaitingCalled() { return skipWaitingCalled; },
        get claimCalled() { return claimCalled; } };
}

test("Service Worker precaches all declared app assets before activating", async () => {
    const requests = [];
    const harness = createWorkerHarness(async (request) => {
        requests.push(new URL(request.url).pathname);
        return new Response("cached " + new URL(request.url).pathname, { status: 200 });
    });
    let installPromise;
    harness.listeners.get("install")({ waitUntil(promise) { installPromise = promise; } });
    await installPromise;

    const cacheNames = await harness.caches.keys();
    assert.deepEqual(cacheNames, ["structure-analysis-offline-v2"]);
    const cache = await harness.caches.open(cacheNames[0]);
    const keys = await cache.keys();
    assert.equal(keys.length, coreAssets.length);
    assert.equal(requests.length, coreAssets.length);
    assert.equal(harness.skipWaitingCalled, true);
});

test("a failed new precache leaves the active app cache unchanged", async () => {
    const failingPath = "/Structure-Analysis/modules/elements/elements.js";
    const harness = createWorkerHarness(async (request) => {
        const url = new URL(request.url);
        if (url.pathname === failingPath) throw new Error("simulated precache failure");
```

```
        return new Response("new app asset " + url.pathname, { status: 200 });
    });

    const previousCache = await harness.caches.open("structure-analysis-offline-v1");
    const previousIndex = new URL("index.html", "https://example.test/Structure-Analysis/");
    await previousCache.put(new Request(previousIndex.href), new Response("previous app shell", { status: 200 }));

    let installPromise;
    harness.listeners.get("install")({ waitUntil(promise) { installPromise = promise; } });
    await assert.rejects(installPromise);

    const preservedResponse = await previousCache.match(new Request(previousIndex.href));
    assert.ok(preservedResponse, "the previous app shell should remain cached");
    assert.equal(await preservedResponse.text(), "previous app shell");
    assert.equal(harness.skipWaitingCalled, false, "a failed precache must not activate the new worker");
});

test("Service Worker preserves cached app and query-busted assets when network is unavailable", async () => {
    let online = true;
    const harness = createWorkerHarness(async (request) => {
        if (!online) throw new Error("simulated offline network");
        const url = new URL(request.url);
        return new Response("online " + url.pathname, { status: 200 });
    });
    let installPromise;
    harness.listeners.get("install")({ waitUntil(promise) { installPromise = promise; } });
    await installPromise;
    let activatePromise;
    harness.listeners.get("activate")({ waitUntil(promise) { activatePromise = promise; } });
    await activatePromise;
    assert.equal(harness.claimCalled, true);

    online = false;
    let navigationResponse;
    await harness.listeners.get("fetch")({
        request: { url: "https://example.test/Structure-Analysis/", method: "GET", mode: "navigate" },
        respondWith(promise) { navigationResponse = promise; }
    });
    const page = await navigationResponse;
    assert.equal(page.status, 200);
    assert.ok((await page.text()).includes("online /Structure-Analysis/index.html"));

    let assetResponse;
    await harness.listeners.get("fetch")({
        request: { url: "https://example.test/Structure-Analysis/assets/js/main.js?v=20261013", method: "GET", mode: "cors" },
        respondWith(promise) { assetResponse = promise; }
    });
    const asset = await assetResponse;
    assert.equal(asset.status, 200);
    assert.ok((await asset.text()).includes("online /Structure-Analysis/assets/js/main.js"));
});
```

47. tests/prescribed-support-movement.test.cjs

```
"use strict";

const test = require("node:test");
const assert = require("node:assert/strict");
const fs = require("node:fs");
const path = require("node:path");

const root = path.resolve(__dirname, "..");
const mainSource = fs.readFileSync(path.join(root, "assets/js/main.js"), "utf8");
const supportsSource = fs.readFileSync(path.join(root, "modules/supports/supports.js"), "utf8");
const step1Source = fs.readFileSync(path.join(root, "modules/results/step-1/step-1.js"), "utf8");
const step2Source = fs.readFileSync(path.join(root, "modules/results/step-2/step-2.js"), "utf8");
const step3Source = fs.readFileSync(path.join(root, "modules/results/step-3/step-3.js"), "utf8");
const step4Source = fs.readFileSync(path.join(root, "modules/results/step-4/step-4.js"), "utf8");
const step5Source = fs.readFileSync(path.join(root, "modules/results/step-5/step-5.js"), "utf8");
const step6Source = fs.readFileSync(path.join(root, "modules/results/step-6/step-6.js"), "utf8");
const step7Source = fs.readFileSync(path.join(root, "modules/results/step-7/step-7.js"), "utf8");
const elementSource = fs.readFileSync(path.join(root, "modules/elements/elements.js"), "utf8");

function section(source, startMarker, endMarker) {
    const start = source.indexOf(startMarker);
    const end = source.indexOf(endMarker, start);
    assert.notEqual(start, -1, "missing source marker: " + startMarker);
    assert.notEqual(end, -1, "missing end marker: " + endMarker);
    return source.slice(start, end).trim();
}

function near(actual, expected, tolerance = 1e-10) {
    assert.ok(Math.abs(actual - expected) <= tolerance * Math.max(1, Math.abs(expected)),
        "expected " + actual + " to be within tolerance of " + expected);
}

const movementHelpersSource = section(
    mainSource,
    "const SUPPORT_MOVEMENT_FIELDS =",
    "\n// ================================================================================\n// 6. PERSISTENCE"
);
const movementHelpers = new Function(
    "AppConfig",
    "isGridAnalysisMode",
    movementHelpersSource + "\nreturn { normalizeLegacySupportMovementAssignments, getSupportAssignmentPrescribedDisplacements, supportTemplateRestrainsMovement };"
)({
    builtInSupportTemplates: [
        { id: "FIXED", restraints: { RFx: true, RFy: true, RFz: true, RMx: true, RMy: true, RMz: true }, displacement: { dx: 0, dy: 0, dz: 0, tx: 0, ty: 0, tz: 0 } },
        { id: "ROLLER", restraints: { RFx: false, RFy: true, RFz: false, RMx: false, RMy: false, RMz: false }, displacement: { dx: 0, dy: 0, dz: 0, tx: 0, ty: 0, tz: 0 } },
        { id: "HINGED", restraints: { RFx: true, RFy: true, RFz: false, RMx: false, RMy: false, RMz: false }, displacement: { dx: 0, dy: 0, dz: 0, tx: 0, ty: 0, tz: 0 } }
    ]
}, () => false);

function matrixApi() {
    return {
        zeros(rows, cols) {
            return Array.from({ length: rows }, () => new Array(cols).fill(0));
        },
        mul(A, B) {
            return A.map((row) => B[0].map((_, col) =>
                row.reduce((sum, value, k) => sum + value * B[k][col], 0)
            ));
        }
    };
}

const buildAD = new Function(
    "Matrix",
    section(step4Source, "function buildPrescribedMovementActionVector(", "\n// Step 4:") +
    "\nreturn buildPrescribedMovementActionVector;"
)(matrixApi());

const reconstructActualD = new Function(
    "Matrix",
    section(step6Source, "function reconstructActualDisplacementVector(", "\n// Step 6:") +
    "\nreturn reconstructActualDisplacementVector;"
)(matrixApi());

function beamBendingStiffness(E, I, L) {
    const a = 12 * E * I / (L ** 3);
    const b = 6 * E * I / (L ** 2);
    const c = 4 * E * I / L;
    const d = 2 * E * I / L;
    return [
        [a, b, -a, b],
        [b, c, -b, d],
        [-a, -b, a, -b],
        [b, d, -b, c]
    ];
```

```
}

const movementUnitHelpersSource = section(
    supportsSource,
    "// ---- Prescribed movement unit conversion ----",
    "\n// ---- Support Assignments (apply a support template to a node) ----"
);
const movementUnitHelpers = new Function(
    movementUnitHelpersSource + "\nreturn { supportMovementInputToModelUnits, supportMovementModelToInputUnits };"
)();

const elementEHelperSource = section(mainSource, "function getElementEInternal(tmpl) {", "\nfunction migrateUnitSystem()");
const getElementEInternal = new Function(elementEHelperSource + "\nreturn getElementEInternal;")();

test("support translations are entered in mm but kept internally in metres; rotations stay in radians", () => {
    for (const field of ["dx", "dy", "dz"]) {
        near(movementUnitHelpers.supportMovementInputToModelUnits(field, 5), 0.005);
        near(movementUnitHelpers.supportMovementModelToInputUnits(field, 0.005), 5);
    }
    near(movementUnitHelpers.supportMovementInputToModelUnits("tx", 0.01), 0.01);
    near(movementUnitHelpers.supportMovementModelToInputUnits("tz", -0.02), -0.02);
    assert.ok(supportsSource.includes('id="assignmentDispX" type="text" inputmode="decimal" placeholder="0" value="0"><span class="input-unit">mm</span>'));
    assert.ok(supportsSource.includes('id="assignmentDispY" type="text" inputmode="decimal" placeholder="0" value="0"><span class="input-unit">mm</span>'));
    assert.ok(supportsSource.includes('id="assignmentDispZ" type="text" inputmode="decimal" placeholder="0" value="0"><span class="input-unit">mm</span>'));
    assert.ok(step4Source.includes("[AD] entries are kN at translational DOFs and kN·m at rotational DOFs"));
});

test("E is entered, stored, and displayed in kN/m² while old saved units still migrate correctly", () => {
    // Canonical E is already kN/m²; legacy GPa is converted exactly once.
    near(getElementEInternal({ E_kN_m2: 30_000_000 }), 30_000_000);
    near(getElementEInternal({ E_GPa: 30 }), 30_000_000);
    near(getElementEInternal({ E: 30_000_000_000 }), 30_000_000);
    assert.ok(elementSource.includes("I (m■)"));
    assert.ok(elementSource.includes("E (kN/m²)"));
    assert.ok(elementSource.includes('id="elementE" type="text" inputmode="decimal" placeholder="1" value="1"><span class="input-unit">kN/m²</span>'));
    assert.ok(elementSource.includes("E_kN_m2"));
    assert.ok(step2Source.includes("E (kN/m²)"));
    assert.ok(step2Source.includes("formatNum(m.E_kN_m2)"));
    assert.ok(step2Source.includes("formatNum(E)"));
    assert.ok(step3Source.includes("let E = getElementEInternal(tmpl)"));
    assert.ok(step3Source.includes("I = tmpl.I"));
    assert.ok(mainSource.includes("if (hasValue(template.E_kN_m2)) return Number(template.E_kN_m2)"));
    assert.ok(mainSource.includes("Number(template.E_GPa) * 1e6"));
    assert.ok(mainSource.includes('elementTemplates: ["id", "memberType", "I", "E_kN_m2", "G_GPa", "J", "A", "alpha", "thermalDepth"]'));
    assert.ok(mainSource.includes("if (hasValue(item.E_GPa)) item.E_kN_m2 = Number(item.E_GPa) * 1e6"));
    assert.ok(mainSource.includes("E_kN_m2: p.E, G_GPa: p.G / 1e6"));
});

test("legacy element E values migrate to canonical kN/m² once without scaling I", () => {
    const migrationSource = section(
        mainSource,
        "function migrateUnitSystem() {",
        "\n\n\n// ============================================================================" + "\n// INPUT VALIDATION HELPERS"
    );
    const oldAndNewModel = {
        elementTemplates: [
            { id: "GPA", E_GPa: 30, I: 0.004 },
            { id: "LEGACY", E: 30_000_000_000, I: 0.004 },
            { id: "CANONICAL", E_kN_m2: 25_000_000, E_GPa: 99, E: 1, I: 0.008 }
        ]
    };
    const migrate = new Function("model", migrationSource + "\nreturn migrateUnitSystem;")(oldAndNewModel);
    migrate();
    assert.equal(oldAndNewModel.elementTemplates[0].E_kN_m2, 30_000_000);
    assert.equal(oldAndNewModel.elementTemplates[1].E_kN_m2, 30_000_000);
    assert.equal(oldAndNewModel.elementTemplates[2].E_kN_m2, 25_000_000);
    assert.equal(oldAndNewModel.elementTemplates[0].I, 0.004);
    assert.equal(oldAndNewModel.elementTemplates[2].I, 0.008);
    for (const template of oldAndNewModel.elementTemplates) {
        assert.equal(Object.hasOwn(template, "E_GPa"), false);
        assert.equal(Object.hasOwn(template, "E"), false);
    }
});

test("AD from a prescribed settlement produces 12EIδ/L³ shears and 6EIδ/L² end actions", () => {
    const E = 30000;
    const I = 0.004;
    const L = 4;
    const delta = 0.015;
    const K = beamBendingStiffness(E, I, L);
    const { Dp, AD } = buildAD(K, [0, 0, delta, 0]);

    near(Dp[2][0], delta);
    near(AD[0][0], 12 * E * I * delta / (L ** 3));
    near(AD[1][0], 6 * E * I * delta / (L ** 2));
    near(AD[2][0], -12 * E * I * delta / (L ** 3));
    near(AD[3][0], 6 * E * I * delta / (L ** 2));
```

```
});

test("AD from a prescribed end rotation uses the corresponding EI/L and EI/L² terms", () => {
    const E = 210000;
    const I = 8e-5;
    const L = 3.5;
    const theta = 0.012;
    const K = beamBendingStiffness(E, I, L);
    const { Dp, AD } = buildAD(K, [0, 0, 0, theta]);

    near(Dp[3][0], theta);
    near(AD[0][0], -6 * E * I * theta / (L ** 2));
    near(AD[1][0], -2 * E * I * theta / L);
    near(AD[2][0], 6 * E * I * theta / (L ** 2));
    near(AD[3][0], -4 * E * I * theta / L);
});


test("settlement equivalent actions recover the same free rotation and support reactions as direct prescribed-displacement stiffness", () => {
    const E = 30000;
    const I = 0.004;
    const L = 4;
    const delta = 0.015;
    const K = beamBendingStiffness(E, I, L);
    const { Dp, AD } = buildAD(K, [0, 0, 0, delta, 0]);
    const physicalLoad = [[0], [0], [0], [0]];
    const AC = physicalLoad.map((row, i) => [row[0] + AD[i][0]]);

    // Left end fixed, right-end vertical movement prescribed, right rotation free.
    const freeDof = 3;
    const DF = [[AC[freeDof][0] / K[freeDof][freeDof]]];
    const D = reconstructActualD(4, Dp.map((row) => row[0]), [freeDof], DF);

    // Incremental free-DOF equilibrium: K_ff * DF = AC_f.
    let freeResidual = 0;
    for (let j = 0; j < 4; j++) freeResidual += K[freeDof][j] * D[j][0];
    near(freeResidual, 0);
    near(D[2][0], delta);

    // Reactions from the AD-based system must equal K*D at restrained DOFs,
    // with no imposed movement counted a second time.
    for (const i of [0, 1, 2]) {
        let directReaction = 0;
        for (let j = 0; j < 4; j++) directReaction += K[i][j] * D[j][0];
        const solvedReaction = -AC[i][0] + K[i][freeDof] * DF[0][0];
        near(solvedReaction, directReaction);
    }

    // The right end is free to rotate, so its recovered end moment is zero.
    let rightEndMoment = 0;
    for (let j = 0; j < 4; j++) rightEndMoment += K[3][j] * D[j][0];
    near(rightEndMoment, 0);
});

test("actual displacement recovery adds prescribed movement after solving free increments", () => {
    const Dp = [0, 0, 0.02, 0.1];
    const DF = [[0.03], [-0.04]];
    const D = reconstructActualD(4, Dp, [0, 1], DF);
    assert.deepEqual(D, [[0.03], [-0.04], [0.02], [0.1]]);
});

test("legacy template movements migrate to every assignment and are then removed from templates", () => {
    const source = {
        analysisMode: "plane",
        supportTemplates: [{
            id: "S1",
            restraints: { RFx: true, RFy: true, RFz: false, RMx: false, RMy: false, RMz: true },
            displacement: { dx: 0.001, dy: -0.02, dz: 0, tx: 0, ty: 0, tz: 0.015 }
        }],
        supportAssignments: [
            { id: "SA1", supportId: "S1", nodeId: "N1" },
            { id: "SA2", supportId: "S1", nodeId: "N2" }
        ]
    };
    const migrated = movementHelpers.normalizeLegacySupportMovementAssignments(source);
    assert.deepEqual(migrated.supportAssignments[0].prescribedDisplacements, {
        dx: 0.001, dy: -0.02, dz: 0, tx: 0, ty: 0, tz: 0.015
    });
    assert.deepEqual(migrated.supportAssignments[1].prescribedDisplacements, migrated.supportAssignments[0].prescribedDisplacements);

    const cleaned = movementHelpers.normalizeLegacySupportMovementAssignments(migrated, { clearTemplateValues: true });
    assert.deepEqual(cleaned.supportTemplates[0].displacement, { dx: 0, dy: 0, dz: 0, tx: 0, ty: 0, tz: 0 });
    assert.equal(cleaned.supportAssignments[1].prescribedDisplacements.dy, -0.02);
});

test("explicit assignment movement overrides legacy template movement, including explicit zero", () => {
    const assignment = {
```

```
        id: "SA1", supportId: "S1", nodeId: "N1",
        prescribedDisplacements: { dx: 0, dy: 0.025, dz: 0, tx: 0, ty: 0, tz: 0 }
    };
    const template = {
        id: "S1",
        restraints: { RFx: true, RFy: true, RFz: false, RMx: false, RMy: false, RMz: true },
        displacement: { dx: 0.5, dy: 0, dz: 0, tx: 0, ty: 0, tz: 0 }
    };
    const value = movementHelpers.getSupportAssignmentPrescribedDisplacements(assignment, template);
    assert.equal(value.dx, 0);
    assert.equal(value.dy, 0.025);
});

test("Grid settlement values are permitted on the built-in vertical support", () => {
    assert.equal(movementHelpers.supportTemplateRestrainsMovement({ id: "ROLLER", restraints: { RFy: true } }, "dz", true), true);
    assert.equal(movementHelpers.supportTemplateRestrainsMovement({ id: "ROLLER", restraints: { RFy: true } }, "tx", true), false);
});

test("prescribed movements are entered and persisted per support assignment, not in template fields", () => {
    assert.match(supportsSource, /id="assignmentDispY"/);
    assert.match(supportsSource, /id="assignmentRotZ"/);
    assert.match(supportsSource, /id="assignmentDispZ"/);
    assert.match(supportsSource, /id="assignmentRotX"/);
    assert.match(supportsSource, /const data = \{ id, supportId, nodeId, prescribedDisplacements \}/);
    assert.doesNotMatch(supportsSource, /document\.getElementById\("dispX"\)/);
    assert.doesNotMatch(supportsSource, /for="dispX"/);
    assert.match(mainSource, /supportAssignments: \["id", "supportId", "nodeId", "prescribedDisplacements"\]/);
    assert.match(mainSource, /normalizeLegacySupportMovementAssignments/);
    assert.match(step1Source, /getSupportAssignmentPrescribedDisplacements\(sa, t\)/);
    assert.match(mainSource, /const movement = assignment\.prescribedDisplacements/);
});

test("Step 4 assembles AC from AE, AJ, and AD; restrained incremental displacement is zero", () => {
    assert.match(step4Source, /const physicalAC = Matrix\.add\(AE, AJ\)/);
    assert.match(step4Source, /let AC = Matrix\.add\(physicalAC, AD\)/);
    assert.match(step4Source, /return \{ Dp, AD \}/);
    assert.match(step4Source, /return \{ AC, physicalAC, AD, Dp, globalLabels, structurallyRelevantDof, html \}/);
    assert.match(step5Source, /DR\[i\]\[0\] = 0/);
    assert.match(step5Source, /prescribedDR/);
});

test("Step 6 uses actual total displacement for Step 7 member end actions", () => {
    assert.match(step6Source, /reconstructActualDisplacementVector\(totalDof, prescribedDisp, freeIndices, DF\)/);
    assert.match(step7Source, /const \{ DF, D \} = step6/);
    assert.match(step7Source, /Matrix\.mul\(el\.RT, DS\)/);
    assert.doesNotMatch(step7Source, /SRR_DR/);
});
```